

Company. In 1971, Gert's husband, who was then running the company, died suddenly of a heart attack. The company was in the midst of an aggressive expansion, which had taken its sales above \$1 million for the first time but which had also left the company financially stressed. Gert took over the small, struggling company with help from her son Tim, who was then a senior at the University of Oregon. Somehow, they kept the company afloat. Today, Columbia has more than 4,000 employees and annual sales in excess of \$1 billion. Its brands include Columbia, Mountain Hardwear, Sorel, and Montrail. Gert still heads up the Board of Directors, and Tim is the company's President and CEO.

Columbia doesn't just focus on financial success. The company is very committed to corporate, social, and environmental responsibility. For example, several of its factories

have participated in a project to increase health awareness of female factory workers in developing countries. Columbia was also a founding member of the Sustainable Apparel Coalition, which is a group that strives to reduce the environmental and social impact of the apparel industry. In addition, it monitors all of the independent factories that produce its products to ensure that they comply with the company's Standards of Manufacturing Practices. These standards address issues including forced labor, child labor, harassment, wages and benefits, health and safety, and the environment.

Employers such as Columbia Sportswear generally assume that managers in all areas of the company are "financially literate." To help prepare you for that, in this textbook you will learn how to read and prepare financial statements, and how to use basic tools to evaluate financial results.

*The **Chapter Outline** presents the chapter's topics and subtopics, as well as practice opportunities.*

Chapter Outline

LEARNING OBJECTIVES

LO 1 Identify the activities and users associated with accounting.	• Three activities • Who uses accounting data	DO IT! 1 Basic Concepts
LO 2 Explain the building blocks of accounting: ethics, principles, and assumptions.	• Ethics • GAAP • Measurement principles • Assumptions	DO IT! 2 Building Blocks of Accounting
LO 3 State the accounting equation, and define its components.	• Assets • Liabilities • Owner's equity	DO IT! 3 Owner's Equity Effects
LO 4 Analyze the effects of business transactions on the accounting equation.	• Accounting transactions • Transaction analysis • Summary of transactions	DO IT! 4 Tabular Analysis
LO 5 Describe the four financial statements and how they are prepared.	• Income statement • Owner's equity statement • Balance sheet • Statement of cash flows	DO IT! 5 Financial Statement Items

Go to the Review and Practice section at the end of the chapter for a review of key concepts and practice applications with solutions.

Visit WileyPLUS with ORION for additional tutorials and practice opportunities.

Accounting Activities and Users

LEARNING OBJECTIVE 1

Identify the activities and users associated with accounting.

What consistently ranks as one of the top career opportunities in business? What frequently rates among the most popular majors on campus? What was the undergraduate degree chosen by **Nike** founder Phil Knight, **Home Depot** co-founder Arthur Blank, former acting director of the **Federal Bureau of Investigation (FBI)** Thomas Pickard, and numerous members of Congress? Accounting.¹ Why did these people choose accounting? They wanted to understand what was happening financially to their organizations. Accounting is the financial information system that provides these insights. In short, to understand your organization, you have to know the numbers.

Accounting consists of three basic activities—it **identifies**, **records**, and **communicates** the economic events of an organization to interested users. Let's take a closer look at these three activities.

Essential terms are printed in blue when they first appear, and are defined in the end-of-chapter Glossary Review.

Three Activities

As a starting point to the accounting process, a company **identifies** the **economic events relevant to its business**. Examples of economic events are the sale of snack chips by **PepsiCo**, the provision of cell phone services by **AT&T**, and the payment of wages by **Facebook**.

Once a company like PepsiCo identifies economic events, it **records** those events in order to provide a history of its financial activities. Recording consists of keeping a **systematic, chronological diary of events**, measured in dollars and cents. In recording, PepsiCo also classifies and summarizes economic events.

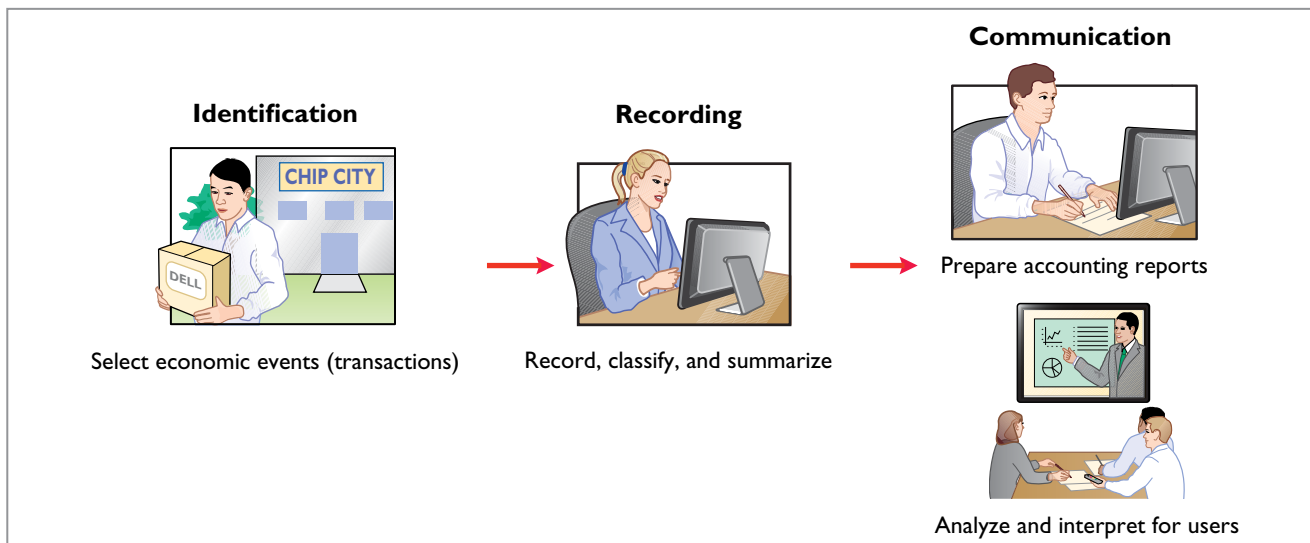
Finally, PepsiCo **communicates** the collected information to interested users by means of **accounting reports**. The most common of these reports are called **financial statements**. To make the reported financial information meaningful, PepsiCo reports the recorded data in a standardized way. It accumulates information resulting from similar transactions. For example, PepsiCo accumulates all sales transactions over a certain period of time and reports the data as one amount in the company's financial statements. Such data are said to be reported **in the aggregate**. By presenting the recorded data in the aggregate, the accounting process simplifies a multitude of transactions and makes a series of activities understandable and meaningful.

A vital element in communicating economic events is the accountant's ability to **analyze and interpret** the reported information. Analysis involves use of ratios, percentages, graphs, and charts to highlight significant financial trends and relationships. Interpretation involves **explaining the uses, meaning, and limitations of reported data**. Appendices A–E show the financial statements of **Apple Inc.**, **PepsiCo Inc.**, **The Coca-Cola Company**, **Amazon.com, Inc.**, and **Wal-Mart Stores, Inc.**, respectively. (In addition, in the *A Look at IFRS* section at the end of each chapter, the French company **Louis Vuitton Moët Hennessy** is analyzed.) We refer to these statements at various places throughout the textbook. At this point, these financial statements probably strike you as complex and confusing. By the end of this course, you'll be surprised at your ability to understand, analyze, and interpret them.

¹ The appendix to this chapter describes job opportunities for accounting majors and explains why accounting is such a popular major.

Illustration 1.1 summarizes the activities of the accounting process.

ILLUSTRATION 1.1 The activities of the accounting process



You should understand that the accounting process **includes** the bookkeeping function. **Bookkeeping** usually involves **only** the recording of economic events. It is therefore just one part of the accounting process. In total, accounting involves **the entire process of identifying, recording, and communicating economic events.**²

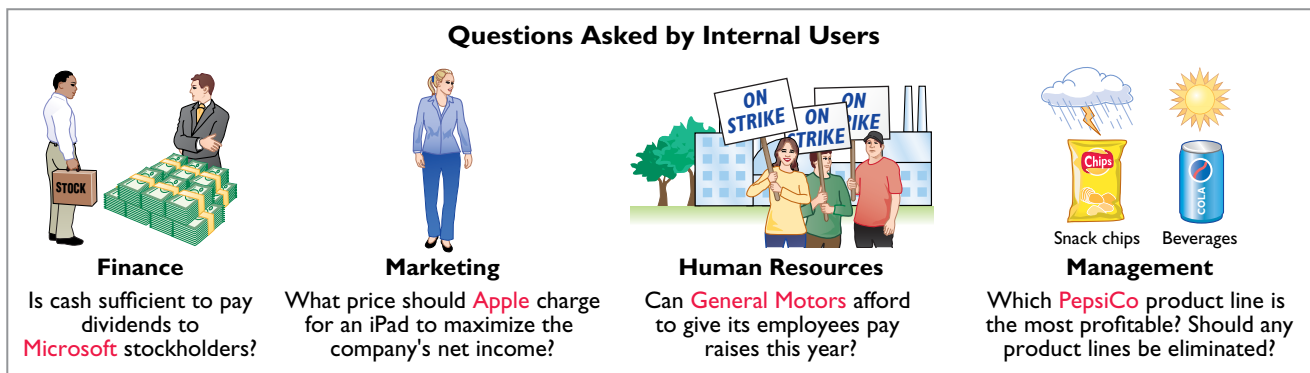
Who Uses Accounting Data

The financial information that users need depends upon the kinds of decisions they make. There are two broad groups of users of financial information: internal users and external users.

Internal Users

Internal users of accounting information are the managers who plan, organize, and run a business. These include marketing managers, production supervisors, finance directors, and company officers. In running a business, internal users must answer many important questions, as shown in **Illustration 1.2**.

ILLUSTRATION 1.2 Questions that internal users ask



²The origins of accounting are generally attributed to the work of Luca Pacioli, an Italian Renaissance mathematician. Pacioli was a close friend and tutor to Leonardo da Vinci and a contemporary of Christopher Columbus. In his 1494 text *Summa de Arithmetica, Geometria, Proportione et Proportionalitate*, Pacioli described a system to ensure that financial information was recorded efficiently and accurately.

To answer these and other questions, internal users need detailed information on a timely basis. **Managerial accounting** provides internal reports to help users make decisions about their companies. Examples are financial comparisons of operating alternatives, projections of income from new sales campaigns, and forecasts of cash needs for the next year.

Accounting Across the Organization Clif Bar & Company



© Dan Moore/iStockphoto

Owning a Piece of the Bar

The original Clif Bar® energy bar was created in 1990 after six months of experimentation by Gary Erickson and his mother in her kitchen. Today, the company has almost 300 employees and is considered one of the leading Landor's Breakaway Brands®. One of **Clif Bar & Company's** proudest moments was the creation of an employee stock own-

ership plan (ESOP) in 2010. This plan gives its employees 20% ownership of the company. The ESOP also resulted in Clif Bar enacting an open-book management program, including the commitment to educate all employee-owners about its finances. Armed with basic accounting knowledge, employees are more aware of the financial impact of their actions, which leads to better decisions.

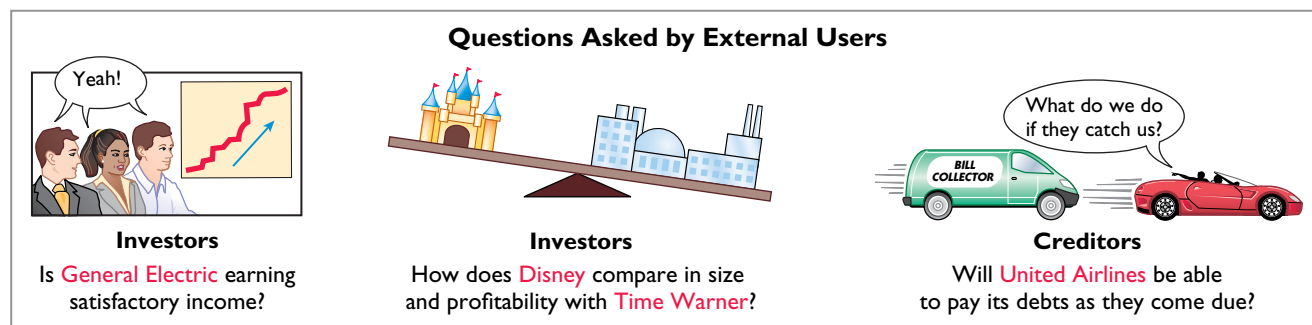
What are the benefits to the company and to the employees of making the financial statements available to all employees? (Go to WileyPLUS for this answer and additional questions.)

Accounting Across the Organization boxes demonstrate applications of accounting information in various business functions.

External Users

External users are individuals and organizations outside a company who want financial information about the company. The two most common types of external users are investors and creditors. **Investors** (owners) use accounting information to decide whether to buy, hold, or sell ownership shares of a company. **Creditors** (such as suppliers and bankers) use accounting information to evaluate the risks of granting credit or lending money. **Illustration 1.3** shows some questions that investors and creditors may ask.

ILLUSTRATION 1.3 Questions that external users ask



Financial accounting answers these questions. It provides economic and financial information for investors, creditors, and other external users. The information needs of external users vary considerably. **Taxing authorities**, such as the Internal Revenue Service, want to know whether the company complies with tax laws. **Regulatory agencies**, such as the Securities and Exchange Commission or the Federal Trade Commission, want to know whether the company is operating within prescribed rules. **Customers** are interested in whether a company like **Tesla Motors** will continue to honor product warranties and support its product lines. **Labor unions**, such as the **Major League Baseball Players Association**, want to know whether the owners have the ability to pay increased wages and benefits.

The **DO IT!** exercises ask you to put newly acquired knowledge to work. They outline the **Action Plan** necessary to complete the exercise, and they show a **Solution**.

DO IT! 1 | Basic Concepts

Indicate whether each of the five statements presented below is true or false. If false, indicate how to correct the statement.

1. The three steps in the accounting process are identification, recording, and communication.
2. Bookkeeping encompasses all steps in the accounting process.
3. Accountants prepare, but do not interpret, financial reports.
4. The two most common types of external users are investors and company officers.
5. Managerial accounting activities focus on reports for internal users.

Solution

1. True 2. False. Bookkeeping involves only the recording step. 3. False. Accountants analyze and interpret information in reports as part of the communication step. 4. False. The two most common types of external users are investors and creditors. 5. True.

Related exercise material: **DO IT! 1.1, E1.1, and E1.2.**

ACTION PLAN

- Review the basic concepts discussed.
- Develop an understanding of the key terms used.

The Building Blocks of Accounting

LEARNING OBJECTIVE 2

Explain the building blocks of accounting: ethics, principles, and assumptions.

A doctor follows certain protocols in treating a patient's illness. An architect follows certain structural guidelines in designing a building. Similarly, an accountant follows certain standards in reporting financial information. These standards are based on specific principles and assumptions. For these standards to work, however, a fundamental business concept must be present—ethical behavior.

Ethics in Financial Reporting

People won't gamble in a casino if they think it is "rigged." Similarly, people won't play the stock market if they think prices are rigged. At one time, the financial press was full of articles about financial scandals at **Enron**, **WorldCom**, **HealthSouth**, and **AIG**. As more scandals came to light, a mistrust of financial reporting in general seemed to be developing. One article in the *Wall Street Journal* noted that "repeated disclosures about questionable accounting practices have bruised investors' faith in the reliability of earnings reports, which in turn has sent stock prices tumbling." Imagine trying to carry on a business or invest money if you could not depend on the financial statements to be honestly prepared. Information would have no credibility. There is no doubt that a sound, well-functioning economy depends on accurate and dependable financial reporting.

United States regulators and lawmakers were very concerned that the economy would suffer if investors lost confidence in corporate accounting because of unethical financial reporting. In response, Congress passed the **Sarbanes-Oxley Act (SOX)** to reduce unethical corporate behavior and decrease the likelihood of future corporate scandals (see **Ethics Note**). As a result of SOX, top management must now certify the accuracy of financial information. In addition, penalties for fraudulent financial activity are much more severe. Also, SOX increased the independence requirements of the outside auditors who review the accuracy of corporate financial statements and increased the oversight role of boards of directors.

ETHICS NOTE

Circus-founder P.T. Barnum is alleged to have said, "Trust everyone, but cut the deck." What Sarbanes-Oxley does is to provide measures that (like cutting the deck of playing cards) help ensure that fraud will not occur.

Ethics Notes help sensitize you to some of the ethical issues in accounting.

The standards of conduct by which actions are judged as right or wrong, honest or dishonest, fair or not fair, are **ethics**. Effective financial reporting depends on sound ethical behavior. To sensitize you to ethical situations in business and to give you practice at solving ethical dilemmas, we address ethics in a number of ways in this textbook:

1. A number of the *Feature Stories* and other parts of the textbook discuss the central importance of ethical behavior to financial reporting.
2. *Ethics Insight* boxes and marginal *Ethics Notes* highlight ethics situations and issues in actual business settings.
3. Many of the *People, Planet, and Profit Insight* boxes focus on ethical issues that companies face in measuring and reporting social and environmental issues.
4. At the end of the chapter, an *Ethics Case* simulates a business situation and asks you to put yourself in the position of a decision-maker in that case.

When analyzing these various ethics cases and your own ethical experiences, you should apply the three steps outlined in **Illustration 1.4**.

ILLUSTRATION 1.4 Steps in analyzing ethics cases and situations



1. Recognize an ethical situation and the ethical issues involved.

Use your personal ethics to identify ethical situations and issues. Some businesses and professional organizations provide written codes of ethics for guidance in some business situations.

2. Identify and analyze the principal elements in the situation.

Identify the **stakeholders**—persons or groups who may be harmed or benefited. Ask the question: What are the responsibilities and obligations of the parties involved?

3. Identify the alternatives, and weigh the impact of each alternative on various stakeholders.

Select the most ethical alternative, considering all the consequences. Sometimes there will be one right answer. Other situations involve more than one right solution; these situations require an evaluation of each and a selection of the best alternative.

Insight boxes provide examples of business situations from various perspectives—ethics, investor, international, and corporate social responsibility. Guideline answers to the critical thinking questions as well as additional questions are available in **WileyPLUS**.

Ethics Insight Dewey & LeBoeuf LLP



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I Felt the Pressure—Would You?

“I felt the pressure.” That’s what some of the employees of the now-defunct law firm of **Dewey & LeBoeuf LLP** indicated when they helped to overstate revenue and use accounting tricks to hide losses and cover up cash shortages. These employees worked for the former finance director and former chief financial officer (CFO) of the firm. Here are some of their comments:

- “I was instructed by the CFO to create invoices, knowing they would not be sent to clients. When I created these invoices, I knew that it was inappropriate.”
- “I intentionally gave the auditors incorrect information in the course of the audit.”

What happened here is that a small group of lower-level employees over a period of years carried out the instructions of their bosses. Their bosses, however, seemed to have no concern as evidenced by various e-mails with one another in which they referred to their financial manipulations as accounting tricks, cooking the books, and fake income.

Source: Ashby Jones, “Guilty Pleas of Dewey Staff Detail the Alleged Fraud,” *Wall Street Journal* (March 28, 2014).

Why did these employees lie, and what do you believe should be their penalty for these lies? (Go to WileyPLUS for this answer and additional questions.)

INTERNATIONAL NOTE

Over 115 countries use international standards (called IFRS). For example, all companies in the European Union follow IFRS. The differences between U.S. and international standards are not generally significant.

International Notes highlight differences between U.S. and international accounting standards.

Generally Accepted Accounting Principles

The accounting profession has developed standards that are generally accepted and universally practiced. This common set of standards is called **generally accepted accounting principles (GAAP)**. These standards indicate how to report economic events.

The primary accounting standard-setting body in the United States is the **Financial Accounting Standards Board (FASB)**. The **Securities and Exchange Commission (SEC)** is the agency of the U.S. government that oversees U.S. financial markets and accounting standard-setting bodies. The SEC relies on the FASB to develop accounting standards, which public companies must follow. Many countries outside of the United States have adopted the accounting standards issued by the **International Accounting Standards Board (IASB)**. These standards are called **International Financial Reporting Standards (IFRS)** (see **International Note**).

As markets become more global, it is often desirable to compare the results of companies from different countries that report using different accounting standards. In order to increase comparability, in recent years the two standard-setting bodies have made efforts to reduce the differences between U.S. GAAP and IFRS. This process is referred to as **convergence**. As a result of these convergence efforts, someday there may be a single set of high-quality accounting standards that are used by companies around the world. Because convergence is such an important issue, we highlight any major differences between GAAP and IFRS in *International Notes* (as shown in the margin here) and provide a more in-depth discussion in the *A Look at IFRS* section at the end of each chapter.

International Insight



Toru-Hanai-Pool/Getty Images, Inc.

The Korean Discount

If you think that accounting standards don't matter, consider these events in South Korea. For many years, international investors complained that the financial reports of South Korean companies were inadequate and inaccurate. Accounting practices there often resulted in huge differences between stated revenues and actual revenues. Because investors did not have faith in the accuracy of the numbers, they were unwilling to pay as much for the shares of these companies relative to shares of comparable companies in different countries. This difference in share price was often referred to as the "Korean discount."

In response, Korean regulators decided that companies would have to comply with international accounting standards. This change was motivated by a desire to "make the country's businesses more transparent" in order to build investor confidence and spur economic growth. Many other Asian countries, including China, India, Japan, and Hong Kong, have also decided either to adopt international standards or to create standards that are based on the international standards.

Source: Evan Ramstad, "End to 'Korea Discount'?" *Wall Street Journal* (March 16, 2007).

What is meant by the phrase "make the country's businesses more transparent"? Why would increasing transparency spur economic growth? (Go to WileyPLUS for this answer and additional questions.)

HELPFUL HINT

Relevance and faithful representation are two primary qualities that make accounting information useful for decision-making.

Helpful Hints further clarify concepts being discussed.

Measurement Principles

GAAP generally uses one of two measurement principles, the historical cost principle or the fair value principle. Selection of which principle to follow generally relates to trade-offs between relevance and faithful representation (see **Helpful Hint**). **Relevance** means that financial information is capable of making a difference in a decision. **Faithful representation** means that the numbers and descriptions match what really existed or happened—they are factual.

Historical Cost Principle

The **historical cost principle** (or cost principle) dictates that companies record assets at their cost. This is true not only at the time the asset is purchased, but also over the time the asset is held. For example, if **Best Buy** purchases land for \$300,000, the company initially reports

it in its accounting records at \$300,000. But what does Best Buy do if, by the end of the next year, the fair value of the land has increased to \$400,000? Under the historical cost principle, it continues to report the land at \$300,000.

Fair Value Principle

The **fair value principle** states that assets and liabilities should be reported at fair value (the price received to sell an asset or settle a liability). Fair value information may be more useful than historical cost for certain types of assets and liabilities. For example, certain investment securities are reported at fair value because market price information is usually readily available for these types of assets. In determining which measurement principle to use, companies weigh the factual nature of cost figures versus the relevance of fair value. In general, most companies choose to use cost. Only in situations where assets are actively traded, such as investment securities, do companies apply the fair value principle extensively.

Assumptions

Assumptions provide a foundation for the accounting process. Two main assumptions are the **monetary unit assumption** and the **economic entity assumption**.

Monetary Unit Assumption

The **monetary unit assumption** requires that companies include in the accounting records only transaction data that can be expressed in money terms. This assumption enables accounting to quantify (measure) economic events. The monetary unit assumption is vital to applying the historical cost principle.

This assumption prevents the inclusion of some relevant information in the accounting records. For example, the health of a company's owner, the quality of service, and the morale of employees are not included. The reason: Companies cannot quantify this information in money terms. Though this information is important, companies record only events that can be measured in money.

Economic Entity Assumption

An economic entity can be any organization or unit in society. It may be a company (such as **Crocs, Inc.**), a governmental unit (the state of Ohio), a municipality (Seattle), a school district (St. Louis District 48), or a church (Southern Baptist). The **economic entity assumption** requires that the activities of the entity be kept separate and distinct from the activities of its owner and all other economic entities (see **Ethics Note**). To illustrate, Sally Rider, owner of Sally's Boutique, must keep her personal living costs separate from the expenses of the business. Similarly, **J. Crew** and **Gap Inc.** are segregated into separate economic entities for accounting purposes.

Proprietorship. A business owned by one person is generally a **proprietorship**. The owner is often the manager/operator of the business. Small service-type businesses (plumbing companies, beauty salons, and auto repair shops), farms, and small retail stores (antique shops, clothing stores, and used-book stores) are often proprietorships. **Usually, only a relatively small amount of money (capital) is necessary to start in business as a proprietorship. The owner (proprietor) receives any profits, suffers any losses, and is personally liable for all debts of the business.** There is no legal distinction between the business as an economic unit and the owner, but the accounting records of the business activities are kept separate from the personal records and activities of the owner.

Partnership. A business owned by two or more persons associated as partners is a **partnership**. In most respects a partnership is like a proprietorship except that more than one owner is involved. Typically, a partnership agreement (written or oral) sets forth such terms as initial investment, duties of each partner, division of net income (or net loss), and settlement to

ETHICS NOTE

The importance of the economic entity assumption is illustrated by scandals involving **Adelphia**. In this case, senior company employees entered into transactions that blurred the line between the employees' financial interests and those of the company. For example, Adelphia guaranteed over \$2 billion of loans to the founding family.

be made upon death or withdrawal of a partner. Each partner generally has unlimited personal liability for the debts of the partnership. **Like a proprietorship, for accounting purposes the partnership transactions must be kept separate from the personal activities of the partners.** Partnerships are often used to organize retail and service-type businesses, including professional practices (lawyers, doctors, architects, and certified public accountants).

Corporation. A business organized as a separate legal entity under state corporation law and having ownership divided into transferable shares of stock is a **corporation**. The holders of the shares (stockholders) **enjoy limited liability**; that is, they are not personally liable for the debts of the corporate entity. Stockholders **may transfer all or part of their ownership shares to other investors at any time** (i.e., sell their shares). The ease with which ownership can change adds to the attractiveness of investing in a corporation. Because ownership can be transferred without dissolving the corporation, the corporation **enjoys an unlimited life**.

Although the combined number of proprietorships and partnerships in the United States is more than five times the number of corporations, the revenue produced by corporations is eight times greater. Most of the largest companies in the United States—for example, **Exxon-Mobil, Ford, Wal-Mart Stores, Inc., Citigroup**, and **Apple**—are corporations.

Accounting Across the Organization



Josef Volavka/iStockphoto

Spinning the Career Wheel

How will the study of accounting help you? A working knowledge of accounting is desirable for virtually every field of business. Some examples of how accounting is used in business careers include:

General management: Managers at **Ford Motors**, Massachusetts General Hospital, California State University—Fullerton, a **McDonald's** franchise, and a **Trek** bike shop all need to understand accounting data in order to make wise business decisions.

Marketing: Marketing specialists at **Procter & Gamble** must be sensitive to costs and benefits, which accounting helps them quantify

and understand. Making a sale is meaningless unless it is a profitable sale.

Finance: Do you want to be a banker for **Citicorp**, an investment analyst for **Goldman Sachs**, or a stock broker for **Merrill Lynch**? These fields rely heavily on accounting knowledge to analyze financial statements. In fact, it is difficult to get a good job in a finance function without two or three courses in accounting.

Real estate: Are you interested in being a real estate broker for **Prudential Real Estate**? Because a third party—the bank—is almost always involved in financing a real estate transaction, brokers must understand the numbers involved: Can the buyer afford to make the payments to the bank? Does the cash flow from an industrial property justify the purchase price? What are the tax benefits of the purchase?

How might accounting help you? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 2 | Building Blocks of Accounting

Indicate whether each of the five statements presented below is true or false. If false, indicate how to correct the statement.

1. Congress passed the Sarbanes-Oxley Act to reduce unethical behavior and decrease the likelihood of future corporate scandals.
2. The primary accounting standard-setting body in the United States is the Financial Accounting Standards Board (FASB).
3. The historical cost principle dictates that companies record assets at their cost. In later periods, however, the fair value of the asset must be used if fair value is higher than its cost.
4. Relevance means that financial information matches what really happened; the information is factual.
5. A business owner's personal expenses must be separated from expenses of the business to comply with accounting's economic entity assumption.

ACTION PLAN

- Review the discussion of ethics and financial reporting standards.
- Develop an understanding of the key terms used.

Solution

1. True. 2. True. 3. False. The historical cost principle dictates that companies record assets at their cost. Under the historical cost principle, the company must also use cost in later periods. 4. False. Faithful representation, not relevance, means that financial information matches what really happened; the information is factual. 5. True.

Related exercise material: **DO IT! 1.2, E1.3, and E1.4.**

The Accounting Equation

LEARNING OBJECTIVE 3

State the accounting equation, and define its components.

The two basic elements of a business are what it owns and what it owes. **Assets** are the resources a business owns. For example, **Alphabet Inc.** has total assets of approximately \$167.5 billion. Liabilities and owner's equity are the rights or claims against these resources. Thus, Alphabet has \$167.5 billion of claims against its \$167.5 billion of assets. Claims of those to whom the company owes money (creditors) are called **liabilities**. Claims of owners are called **owner's equity**. Alphabet has liabilities of \$28.5 billion and owners' equity of \$139.0 billion.

We can express the relationship of assets, liabilities, and owner's equity as an equation, as shown in **Illustration 1.5**.

Assets	=	Liabilities	+	Owner's Equity
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ILLUSTRATION 1.5

The basic accounting equation

This relationship is the **basic accounting equation**. Assets must equal the sum of liabilities and owner's equity. Liabilities appear before owner's equity in the basic accounting equation because they are paid first if a business is liquidated.

The accounting equation applies to all **economic entities** regardless of size, nature of business, or form of business organization. It applies to a small proprietorship such as a corner grocery store as well as to a giant corporation such as **PepsiCo**. The equation provides the **underlying framework** for recording and summarizing economic events.

Let's look in more detail at the categories in the basic accounting equation.

Assets

As noted above, **assets** are resources a business owns. The business uses its assets in carrying out such activities as production and sales. The common characteristic possessed by all assets is **the capacity to provide future services or benefits**. In a business, that service potential or future economic benefit eventually results in cash inflows (receipts). For example, consider Campus Pizza, a local restaurant. It owns a delivery truck that provides economic benefits from delivering pizzas. Other assets of Campus Pizza are tables, chairs, jukebox, cash register, oven, tableware, and, of course, cash.

Liabilities

Liabilities are claims against assets—that is, existing debts and obligations. Businesses of all sizes usually borrow money and purchase merchandise on credit. These economic activities result in payables of various sorts:

- Campus Pizza, for instance, purchases cheese, sausage, flour, and beverages on credit from suppliers. These obligations are called **accounts payable**.

- Campus Pizza also has a **note payable** to First National Bank for the money borrowed to purchase the delivery truck.
- Campus Pizza may also have **salaries and wages payable** to employees and **sales and real estate taxes payable** to the local government.

All of these persons or entities to whom Campus Pizza owes money are its **creditors**.

Creditors may legally force the liquidation of a business that does not pay its debts. In that case, the law requires that creditor claims be paid **before** ownership claims.

Owner's Equity

HELPFUL HINT

In some places, we use the term “owner’s equity” and in others we use “owners’ equity.” *Owner’s* (singular, possessive) refers to one owner (the case with a sole proprietorship). *Owners’* (plural, possessive) refers to multiple owners (the case with partnerships or corporations).

The ownership claim on total assets is **owner’s equity** (see **Helpful Hint**). It is equal to total assets minus total liabilities. Here is why: The assets of a business are claimed by either creditors or owners. To find out what belongs to owners, we subtract the creditors’ claims (the liabilities) from assets. The remainder is the owner’s claim on the assets—the owner’s equity. Since the claims of creditors must be paid **before** ownership claims, owner’s equity is often referred to as **residual equity**.

Increases in Owner’s Equity

In a proprietorship, owner’s investments and revenues increase owner’s equity.

Investments by Owner. **Investments by owner** are the assets the owner puts into the business. These investments increase owner’s equity. They are recorded in a category called **owner’s capital**.

Revenues. **Revenues** are the **increases in assets or decreases in liabilities resulting from the sale of goods or the performance of services in the normal course of business**. Revenues usually result in an increase in an asset. They may arise from different sources and are called different names depending on the nature of the business. Campus Pizza, for instance, has two categories of sales revenues—pizza sales and beverage sales.

Decreases in Owner’s Equity

In a proprietorship, owner’s drawings and expenses decrease owner’s equity.

Drawings. An owner may withdraw cash or other assets for personal use. We use a separate classification called **drawings** to determine the total withdrawals for each accounting period. **Drawings decrease owner’s equity**. They are recorded in a category called owner’s drawings.

Expenses. **Expenses** are the cost of assets consumed or services used in the process of generating revenue. They are **decreases in owner’s equity that result from operating the business**. For example, Campus Pizza recognizes the following expenses: cost of ingredients (meat, flour, cheese, tomato paste, mushrooms, etc.); cost of beverages; salaries and wages expense; utilities expense (electric, gas, and water expense); delivery expense (gasoline, repairs, licenses, etc.); supplies expense (napkins, detergents, aprons, etc.); rent expense; interest expense; and property tax expense.

In summary, owner’s equity is increased by an owner’s investments and by revenues from business operations. Owner’s equity is decreased by an owner’s withdrawals of assets and by expenses. **Illustration 1.6** expands the basic accounting equation by showing the items that comprise owner’s equity. This format is referred to as the **expanded accounting equation**.

ILLUSTRATION 1.6 Expanded accounting equation

Basic Equation	Assets	=	Liabilities	+	Owner's Equity		
Expanded Equation	Assets	=	Liabilities	+	Owner's Capital	−	Owner's Drawings + Revenues − Expenses

DO IT! 3 | Owner's Equity Effects

Classify the following items as investment by owner (I), owner's drawings (D), revenues (R), or expenses (E). Then indicate whether each item increases or decreases owner's equity.

1. Rent Expense.
2. Service Revenue.
3. Drawings.
4. Salaries and Wages Expense.

Solution

1. Rent Expense is an expense (E); it decreases owner's equity. 2. Service Revenue is revenue (R); it increases owner's equity. 3. Drawings is owner's drawings (D); it decreases owner's equity. 4. Salaries and Wages Expense is an expense (E); it decreases owner's equity.

Related exercise material: BE1.1, BE1.2, BE1.3, BE1.4, BE1.5, BE1.6, DO IT! 1.3, and E1.5.

ACTION PLAN

- Understand the sources of revenue.
- Understand what causes expenses.
- Review the rules for changes in owner's equity.
- Recognize that drawings are withdrawals of cash or other assets from the business for personal use.

Analyzing Business Transactions

LEARNING OBJECTIVE 4

Analyze the effects of business transactions on the accounting equation.



The system of collecting and processing transaction data and communicating financial information to decision-makers is known as the **accounting information system**. Factors that shape an accounting information system include the nature of the company's business, the types of transactions, the size of the company, the volume of data, and the information demands of management and others.

Most businesses use computerized accounting systems—sometimes referred to as electronic data processing (EDP) systems. These systems handle all the steps involved in the recording process, from initial data entry to preparation of the financial statements. In order to remain competitive, companies continually improve their accounting systems to provide accurate and timely data for decision-making. For example, in a recent annual report, **Tootsie Roll** stated, “We also invested in additional processing and data storage hardware during the year. We view information technology as a key strategic tool, and are committed to deploying leading edge technology in this area.” In addition, many companies have upgraded their accounting information systems in response to the requirements of Sarbanes-Oxley.

Accounting information systems rely on a process referred to as **the accounting cycle**. As you can see from the graphic above, the accounting cycle begins with the analysis of business transactions and ends with the preparation of a post-closing trial balance. We explain each of the steps, starting in this chapter and continuing in Chapters 2–4.

*This **accounting cycle graphic** illustrates the steps companies follow each period to record transactions and eventually prepare financial statements.*

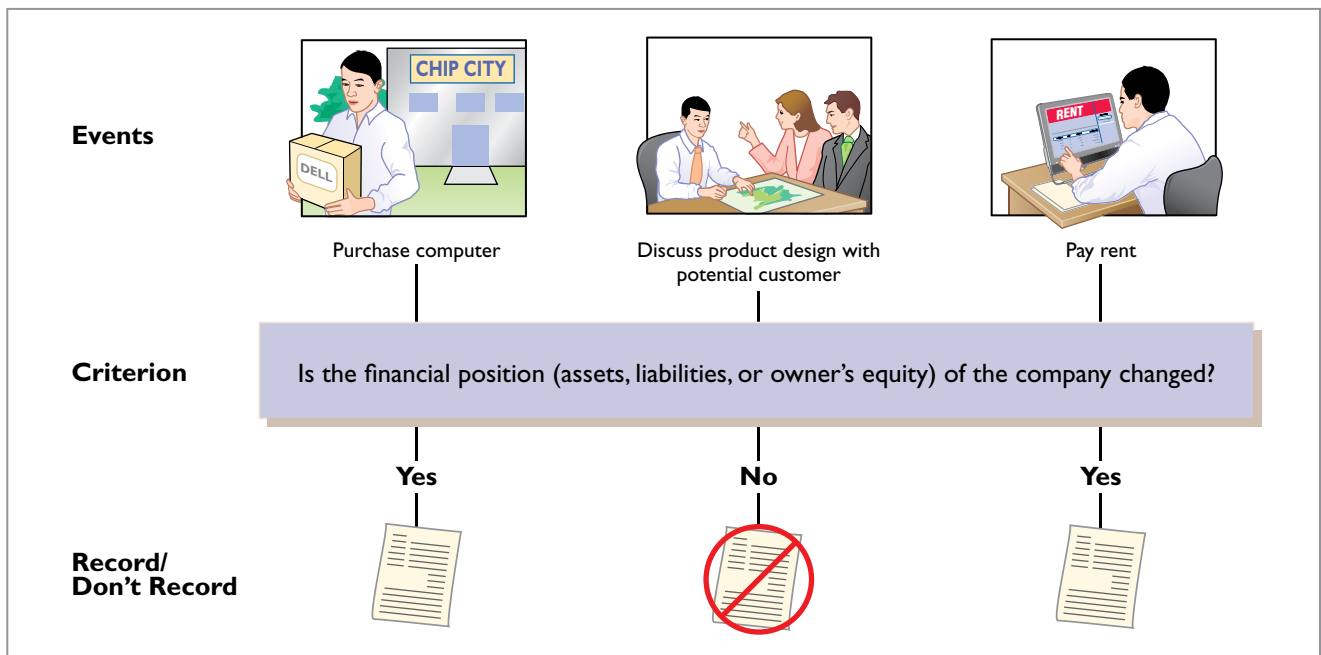
In this textbook, in order to emphasize the underlying concepts and principles, we focus on a manual accounting system. The accounting concepts and principles do not change whether a system is computerized or manual.

Accounting Transactions

Transactions (business transactions) are a business's economic events recorded by accountants. Transactions may be external or internal. **External transactions** involve economic events between the company and some outside enterprise. For example, Campus Pizza's purchase of cooking equipment from a supplier, payment of monthly rent to the landlord, and sale of pizzas to customers are external transactions. **Internal transactions** are economic events that occur entirely within one company. The use of cooking and cleaning supplies are internal transactions for Campus Pizza.

Companies carry on many activities that do not represent business transactions. Examples are hiring employees, responding to e-mails, talking with customers, and placing merchandise orders. Some of these activities may lead to business transactions. Employees will earn wages, and suppliers will deliver ordered merchandise. The company must analyze each event to find out if it affects the components of the accounting equation. If it does, the company will record the transaction. **Illustration 1.7** demonstrates the transaction identification process.

ILLUSTRATION 1.7 Transaction identification process



Each transaction must have a dual effect on the accounting equation. For example, if an asset is increased, there must be a corresponding:

- Decrease in another asset, *or*
- Increase in a specific liability, *or*
- Increase in owner's equity.

HELPFUL HINT

Study these transactions until you are sure you understand them. They are not difficult, but understanding them is important to your success in this course. The ability to analyze transactions in terms of the basic accounting equation is essential in accounting.

Two or more items could be affected. For example, as one asset is increased \$10,000, another asset could decrease \$6,000 and a liability could increase \$4,000. Any change in a liability or ownership claim is subject to similar analysis.

Transaction Analysis

To demonstrate how to analyze transactions in terms of the accounting equation, we will review the business activities of Softbyte, a smartphone app development company (see **Helpful Hint**). Softbyte is the creation of Ray Neal, an entrepreneur who wants to create focused apps

that inspire and engage users of all ages. Ray was encouraged to start his own business after the success of “FoodAlert,” a customizable app he developed that tracks the daily location of local food trucks. The following business transactions occur during Softbyte’s first month of operations.

Transaction (1). Investment of Cash by Owner. Ray Neal starts a smartphone app development company which he names Softbyte. On September 1, 2020, he invests \$15,000 cash in the business. This transaction results in an equal increase in assets and owner’s equity.

Basic Analysis	The asset Cash increases \$15,000, and owner’s equity (specifically Owner’s Capital) increases \$15,000.				
Equation Analysis	<u>Assets</u>	=	<u>Liabilities</u>	+	<u>Owner’s Equity</u>
	Cash	=			Owner’s Capital
	(1) +\$15,000	=			+\$15,000 Initial investment

Observe that the equality of the accounting equation has been maintained. Note that the investments by the owner do not represent revenues, and they are excluded in determining net income. Therefore, it is necessary to make clear that the increase is an investment (increasing Owner’s Capital) rather than revenue.

Transaction (2). Purchase of Equipment for Cash. Softbyte purchases computer equipment for \$7,000 cash. This transaction results in an equal increase and decrease in total assets, though the composition of assets changes.

Basic Analysis	The asset Cash decreases \$7,000, and the asset Equipment increases \$7,000.				
Equation Analysis	<u>Assets</u>	=	<u>Liabilities</u>	+	<u>Owner’s Equity</u>
	Cash + Equipment	=			Owner’s Capital
	\$15,000				\$15,000
	(2) −\$7,000 +\$7,000				
	<u>\$ 8,000 + \$7,000</u>	=			<u>\$15,000</u>
	\$15,000				

Observe that total assets are still \$15,000. Owner’s equity also remains at \$15,000, the amount of Ray Neal’s original investment.

Transaction (3). Purchase of Supplies on Credit. Softbyte purchases headsets (and other computer accessories expected to last several months) for \$1,600 from Mobile Solutions. Mobile Solutions agrees to allow Softbyte to pay this bill in October. This transaction is a purchase on account (a credit purchase). Assets increase because of the expected future benefits of using the headsets and computer accessories, and liabilities increase by the amount due to Mobile Solutions.

The two sides of the equation still balance at \$17,800. Owner's equity decreases when Softbyte incurs the expense. Expenses are not always paid in cash at the time they are incurred. When Softbyte pays at a later date, the liability Accounts Payable will decrease, and the asset Cash will decrease [see Transaction (8)]. The cost of advertising is an expense (rather than an asset) because the company has **used** the benefits. Advertising Expense is included in determining net income.

Transaction (6). Services Performed for Cash and Credit. Softbyte performs \$3,500 of app development services for customers. The company receives cash of \$1,500 from customers, and it bills the balance of \$2,000 on account. This transaction results in an equal increase in assets and owner's equity.

Basic
Analysis

Three specific items are affected: The asset Cash increases \$1,500, the asset Accounts Receivable increases \$2,000, and owner's equity increases \$3,500 due to Service Revenue.

Equation
Analysis

Assets				=	Liabilities	+	Owner's Equity		
Cash	Accounts Receivable	Supplies	Equipment	=	Accounts Payable	+	Owner's Capital	Revenues	Expenses
\$9,200		\$1,600	\$7,000	=	\$1,850	+	\$15,000	\$1,200	\$250
(6) +1,500	+2,000			=				+3,500	
\$10,700	\$2,000	\$1,600	\$7,000	=	\$1,850	+	\$15,000	\$4,700	-\$250
\$21,300					\$21,300				

Service Revenue

Softbyte recognizes \$3,500 in revenue when it performs the service. In exchange for this service, it received \$1,500 in Cash and Accounts Receivable of \$2,000. This Accounts Receivable represents customers' promises to pay \$2,000 to Softbyte in the future. When it later receives collections on account, Softbyte will increase Cash and will decrease Accounts Receivable [see Transaction (9)].

Transaction (7). Payment of Expenses. Softbyte pays the following expenses in cash for September: office rent \$600, salaries and wages of employees \$900, and utilities \$200. These payments result in an equal decrease in assets and owner's equity.

Basic
Analysis

The asset Cash decreases \$1,700, and owner's equity decreases \$1,700 due to the specific expense categories (Rent Expense, Salaries and Wages Expense, and Utilities Expense).

Equation
Analysis

Assets				=	Liabilities	+	Owner's Equity		
Cash	Accounts Receivable	Supplies	Equipment	=	Accounts Payable	+	Owner's Capital	Revenues	Expenses
\$10,700	\$2,000	\$1,600	\$7,000	=	\$1,850	+	\$15,000	\$4,700	\$250
(7) -1,700				=					
\$9,000	\$2,000	\$1,600	\$7,000	=	\$1,850	+	\$15,000	\$4,700	-\$1,950
\$19,600					\$19,600				

-600 Rent Expense
 -900 Sal. and Wages Exp.
 -200 Utilities Exp.

The two sides of the equation now balance at \$19,600. Three lines in the analysis indicate the different types of expenses that have been incurred.

Transaction (8). Payment of Accounts Payable. Softbyte pays its \$250 *Daily News* bill in cash. The company previously [in Transaction (5)] recorded the bill as an increase in Accounts Payable and a decrease in owner's equity.

Observe that the effect of a cash withdrawal by the owner is the opposite of the effect of an investment by the owner. **Owner's drawings are not expenses.** Expenses are incurred for the purpose of earning revenue. Drawings do not generate revenue. They are a **disinvestment**. Like owner's investment, the company excludes owner's drawings in determining net income.

Summary of Transactions

Illustration 1.8 summarizes the September transactions of Softbyte to show their cumulative effect on the basic accounting equation. It also indicates the transaction number and the specific effects of each transaction.

Illustration 1.8 demonstrates some significant facts:

- Each transaction is analyzed in terms of its effect on:
 - The three components of the basic accounting equation (assets, liabilities, and owner's equity).
 - Specific types (kinds) of items within each component (such as the asset Cash).
- The two sides of the equation must always be equal.
- The Owner's Capital, Owner's Drawings, Revenues, and Expenses columns indicate the causes of each change in the owner's claim on assets.

ILLUSTRATION 1.8 Tabular summary of Softbyte transactions

Trans- action	Assets				=	Liabilities	+	Owner's Equity										
	Cash	+	Accounts Receivable	+	Supplies	+	Equipment	=	Accounts Payable	+	Owner's Capital	-	Owner's Drawings	+	Rev.	-	Exp.	
(1)	+\$15,000									+	\$15,000							Initial invest.
(2)	-7,000						+\$7,000											
(3)					+\$1,600				+\$1,600									
(4)	+1,200														+\$1,200			Service Revenue
(5)									+250								-\$250	Adver. Expense
(6)	+1,500		+\$2,000												+3,500			Service Revenue
(7)	-600																-600	Rent Expense
	-900																-900	Sal/Wages Exp.
	-200																-200	Utilities Expense
(8)	-250								-250									
(9)	+600		-600															
(10)	-1,300														-\$1,300			Drawings
	<u>\$ 8,050</u>	+	<u>\$1,400</u>	+	<u>\$1,600</u>	+	<u>\$7,000</u>	=	<u>\$1,600</u>	+	<u>\$15,000</u>	-	<u>\$1,300</u>	+	<u>\$4,700</u>	-	<u>\$1,950</u>	
	\$18,050								\$18,050									

There! You made it through your first transaction analysis. If you feel a bit shaky on any of the transactions, it might be a good idea at this point to get up, take a short break, and come back again for a brief (10- to 15-minute) review of the transactions, to make sure you understand them before you go on to the next section.

DO IT! 4 | Tabular Analysis

Transactions made by Virmari & Co., a public accounting firm, for the month of August are shown below. Prepare a tabular analysis which shows the effects of these transactions on the expanded accounting equation, similar to that shown in Illustration 1.8.

- The owner invested \$25,000 cash in the business.
- The company purchased \$7,000 of office equipment on credit.
- The company received \$8,000 cash in exchange for services performed.
- The company paid \$850 for this month's rent.
- The owner withdrew \$1,000 cash for personal use.

ACTION PLAN

- Analyze the effects of each transaction on the accounting equation.
- Use appropriate category names (not descriptions).
- Keep the accounting equation in balance.

Solution

Trans- action	Assets		=	Liabilities	+	Owner's Equity			
	Cash	+ Equipment	=	Accounts Payable	+	Owner's Capital	- Owner's Drawings	+ Revenues	- Expenses
(1)	+\$25,000					+\$25,000			
(2)		+\$7,000		+\$7,000					
(3)	+8,000							+\$8,000	
(4)	-850								-\$850
(5)	-1,000					-\$1,000			
	<u>\$31,150</u>	+ <u>\$7,000</u>	=	<u>\$7,000</u>	+ <u>\$25,000</u>	- <u>\$1,000</u>	+ <u>\$8,000</u>	- <u>\$850</u>	
	\$38,150			\$38,150					

Service Revenue
Rent Expense
Drawings

Related exercise material: BE1.7, BE1.9, DO IT! 1.4, E1.6, E1.7, and E1.8.

The Four Financial Statements

LEARNING OBJECTIVE 5

Describe the four financial statements and how they are prepared.

HELPFUL HINT

The income statement, owner's equity statement, and statement of cash flows are all for a *period* of time, whereas the balance sheet is for a *point* in time.

Companies prepare four financial statements from the summarized accounting data (see **Helpful Hint**):

1. An **income statement** presents the revenues and expenses and resulting net income or net loss for a specific period of time.
2. An **owner's equity statement** summarizes the changes in owner's equity for a specific period of time.
3. A **balance sheet** reports the assets, liabilities, and owner's equity at a specific date.
4. A **statement of cash flows** summarizes information about the cash inflows (receipts) and outflows (payments) for a specific period of time.

These statements provide relevant financial data for internal and external users. **Illustration 1.9** shows the financial statements of Softbyte.

Note that the statements shown in Illustration 1.9 are interrelated:

1. Net income of \$2,750 on the **income statement** is added to the beginning balance of owner's capital in the **owner's equity statement**.
2. Owner's capital of \$16,450 at the end of the reporting period shown in the **owner's equity statement** is reported on the **balance sheet**.
3. Cash of \$8,050 on the **balance sheet** is reported on the **statement of cash flows**.

Also, explanatory notes and supporting schedules are an integral part of every set of financial statements. We illustrate these notes and schedules in later chapters of this textbook.

Be sure to carefully examine the format and content of each statement in Illustration 1.9. We describe the essential features of each in the following sections.

ALTERNATIVE TERMINOLOGY

The income statement is sometimes referred to as the statement of operations, earnings statement, or profit and loss statement.

Income Statement

Alternative Terminology notes introduce synonymous terms you may come across in practice.

The income statement reports the revenues and expenses for a specific period of time (see **Alternative Terminology**). (In Softbyte's case, this is "For the Month Ended September 30, 2020.") Softbyte's income statement is prepared from the data appearing in the owner's equity columns of Illustration 1.8.

ILLUSTRATION 1.9**Financial statements and their interrelationships**

The heading of each statement identifies the company, the type of statement, and the specific date or time period covered by the statement.

Note that final sums are double-underlined, and negative amounts (in the statement of cash flows) are presented in parentheses.

The arrows show the interrelationships of the four financial statements.

1. Net income is computed first and is needed to determine the ending balance in owner's equity.
2. The ending balance in owner's equity is needed in preparing the balance sheet.
3. The cash shown on the balance sheet is needed in preparing the statement of cash flows.

Softbyte
Income Statement
For the Month Ended September 30, 2020

Revenues		
Service revenue		\$ 4,700
Expenses		
Salaries and wages expense	\$900	
Rent expense	600	
Advertising expense	250	
Utilities expense	200	
Total expenses		<u>1,950</u>
Net income		<u><u>\$ 2,750</u></u>

Softbyte
Owner's Equity Statement
For the Month Ended September 30, 2020

Owner's capital, September 1		\$ -0-
Add: Investments	\$15,000	
Net income	<u>2,750</u>	<u>17,750</u>
		17,750
Less: Drawings		<u>1,300</u>
Owner's capital, September 30		<u><u>\$ 16,450</u></u>

Softbyte
Balance Sheet
September 30, 2020

<u>Assets</u>		
Cash		<u>\$ 8,050</u>
Accounts receivable		1,400
Supplies		1,600
Equipment		<u>7,000</u>
Total assets		<u><u>\$ 18,050</u></u>
<u>Liabilities and Owner's Equity</u>		
Liabilities		
Accounts payable		\$ 1,600
Owner's equity		
Owner's capital		<u>16,450</u>
Total liabilities and owner's equity		<u><u>\$ 18,050</u></u>

Softbyte
Statement of Cash Flows
For the Month Ended September 30, 2020

Cash flows from operating activities		
Cash receipts from revenues		\$ 3,300
Cash payments for expenses		<u>(1,950)</u>
Net cash provided by operating activities		1,350
Cash flows from investing activities		
Purchase of equipment		(7,000)
Cash flows from financing activities		
Investments by owner	\$15,000	
Drawings by owner	<u>(1,300)</u>	<u>13,700</u>
Net increase in cash		8,050
Cash at the beginning of the period		<u>0</u>
Cash at the end of the period		<u><u>\$ 8,050</u></u>

The income statement lists revenues first, followed by expenses. Finally the statement shows net income (or net loss). **Net income** results when revenues exceed expenses. A **net loss** occurs when expenses exceed revenues.

Although practice varies, we have chosen in our illustrations and homework solutions to list expenses in order of magnitude. (We will consider alternative formats for the income statement in later chapters.)

Note that the income statement does **not** include investment and withdrawal transactions between the owner and the business in measuring net income. For example, as explained earlier, Ray Neal's withdrawal of cash from Softbyte was not regarded as a business expense.

Owner's Equity Statement

The owner's equity statement reports the changes in owner's equity for a specific period of time. The time period is the same as that covered by the income statement (in Softbyte's case, this is "For the Month Ended September 30, 2020"). Data for the preparation of the owner's equity statement come from the owner's equity columns of the tabular summary (Illustration 1.8) and from the income statement. The first line of the statement shows the beginning owner's equity amount (which was zero at the start of the business). Then come the owner's investments, net income (or loss), and the owner's drawings. This statement indicates **why** owner's equity has increased or decreased during the period.

What if Softbyte had reported a net loss in its first month? Let's assume that during the month of September 2020, Softbyte lost \$10,000. **Illustration 1.10** shows the presentation of a net loss in the owner's equity statement.

ILLUSTRATION 1.10

Presentation of net loss

Softbyte Owner's Equity Statement For the Month Ended September 30, 2020		
Owner's capital, September 1		\$ —0—
Add: Investments		15,000
		<u>15,000</u>
Less: Drawings	\$ 1,300	
Net loss	10,000	<u>11,300</u>
Owner's capital, September 30		<u>\$ 3,700</u>

If the owner makes any additional investments, the company reports them in the owner's equity statement as investments.

Balance Sheet

Softbyte's balance sheet reports the assets, liabilities, and owner's equity at a specific date (in Softbyte's case, September 30, 2020). The company prepares the balance sheet from the column headings of the tabular summary (Illustration 1.8) and the month-end data shown in its last line.

Observe that the balance sheet lists assets at the top, followed by liabilities and owner's equity. Total assets must equal total liabilities and owner's equity. Softbyte reports only one liability—accounts payable—in its balance sheet. In most cases, there will be more than one liability. When two or more liabilities are involved, a customary way of listing is as shown in **Illustration 1.11**.

ILLUSTRATION 1.11

Presentation of liabilities

	Liabilities	
Notes payable		\$10,000
Accounts payable		63,000
Salaries and wages payable		<u>18,000</u>
Total liabilities		\$91,000

The balance sheet is a snapshot of the company's financial condition at a specific moment in time (usually the month-end or year-end).

Statement of Cash Flows

The primary purpose of a statement of cash flows is to provide financial information about the cash receipts and cash payments of a company for a specific period of time (see **Helpful Hint**). To help investors, creditors, and others in their analysis of a company's cash position, the statement of cash flows reports the cash effects of a company's operating, investing, and financing activities. In addition, the statement shows the net increase or decrease in cash during the period, and the amount of cash at the end of the period.

Reporting the sources, uses, and change in cash is useful because investors, creditors, and others want to know what is happening to a company's most liquid resource. The statement of cash flows provides answers to the following simple but important questions.

1. Where did cash come from during the period?
2. What was cash used for during the period?
3. What was the change in the cash balance during the period?

As shown in Softbyte's statement of cash flows in Illustration 1.9, cash increased \$8,050 during the period. Net cash provided by operating activities increased cash \$1,350 (cash receipts from revenue less cash payments for expenses). Cash flow from investing activities decreased cash \$7,000 (purchase of equipment). And cash flow from financing activities increased cash \$13,700 (investment by owner less drawings by owner). At this time, you need not be concerned with how these amounts are determined. Chapter 17 will examine the statement of cash flows in detail.

HELPFUL HINT

The statement of cash flows helps users determine if the company generates enough cash from operations to fund its investing and financing activities.

People, Planet, and Profit Insight



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Beyond Financial Statements

Should we expand our financial statements beyond the income statement, owner's equity statement, balance sheet, and statement of cash flows? Some believe we should take into account ecological and social performance, in addition to financial results, in evaluating a company. The argument is that a company's responsibility lies with anyone who is influenced by its actions. In other words, a company should be interested in benefiting many different parties, instead of only maximizing stockholders' interests.

A socially responsible business does not exploit or endanger any group of individuals. It follows fair trade practices, provides safe environments for workers, and bears responsibility for environmental damage. Granted, measurement of these factors is difficult. How to report this information is also controversial. But many interesting and useful efforts are underway. Throughout this textbook, we provide additional insights into how companies are attempting to meet the challenge of measuring and reporting their contributions to society, as well as their financial results, to stockholders.

Why might a company's stockholders be interested in its environmental and social performance? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 5 | Financial Statement Items

Presented below is selected information related to Flanagan Company at December 31, 2020. Flanagan reports financial information monthly.

Equipment	\$10,000	Utilities Expense	\$ 4,000
Cash	8,000	Accounts Receivable	9,000
Service Revenue	36,000	Salaries and Wages Expense	7,000
Rent Expense	11,000	Notes Payable	16,500
Accounts Payable	2,000	Owner's Drawings	5,000

- a. Determine the total assets of Flanagan Company at December 31, 2020.
- b. Determine the net income that Flanagan Company reported for December 2020.
- c. Determine the owner's equity of Flanagan Company at December 31, 2020.

ACTION PLAN

- Remember the basic accounting equation: assets must equal liabilities plus owner's equity.
- Review previous financial statements to determine how total assets, net income, and owner's equity are computed.

Solution

- a. The total assets are \$27,000, comprised of Cash \$8,000, Accounts Receivable \$9,000, and Equipment \$10,000.
- b. Net income is \$14,000, computed as follows.

Revenues		
Service revenue		\$36,000
Expenses		
Rent expense	\$11,000	
Salaries and wages expense	7,000	
Utilities expense	<u>4,000</u>	
Total expenses		<u>22,000</u>
Net income		<u>\$14,000</u>

- c. The ending owner's equity of Flanagan Company is \$8,500. By rewriting the accounting equation, we can compute owner's equity as assets minus liabilities, as follows.

Total assets [as computed in (a)]		\$27,000
Less: Liabilities		
Notes payable	\$16,500	
Accounts payable	<u>2,000</u>	<u>18,500</u>
Owner's equity		<u>\$ 8,500</u>

Note that it is not possible to determine the company's owner's equity in any other way because the beginning total for owner's equity is not provided.

Related exercise material: **BE1.10, BE1.11, DO IT! 1.5, E1.9, E1.10, E1.11, E1.12, E1.13, E1.14, E1.15, E1.16, E1.17, and E1.18**

Appendix 1A**Career Opportunities in Accounting****LEARNING OBJECTIVE *6**

Explain the career opportunities in accounting.

Why is accounting such a popular major and career choice? First, there are a lot of jobs. In many cities in recent years, the demand for accountants exceeded the supply. Not only are there a lot of jobs, but there are a wide array of opportunities. As one accounting organization observed, "accounting is one degree with 360 degrees of opportunity."

Accounting is also hot because it is obvious that accounting matters. Interest in accounting has increased, ironically, because of the attention caused by the accounting failures of companies such as **Enron** and **WorldCom**. These widely publicized scandals revealed the important role that accounting plays in society. Most people want to make a difference, and an accounting career provides many opportunities to contribute to society. Finally, the Sarbanes-Oxley Act (SOX) significantly increased the accounting and internal control requirements for corporations. This dramatically increased demand for professionals with accounting training.

Accountants are in such demand that it is not uncommon for accounting students to have accepted a job offer a year before graduation. As the following discussion reveals, the job options of people with accounting degrees are virtually unlimited.

Public Accounting

Individuals in **public accounting** offer expert service to the general public, in much the same way that doctors serve patients and lawyers serve clients. A major portion of public accounting

involves **auditing**. In auditing, a certified public accountant (CPA) examines company financial statements and provides an opinion as to how accurately the financial statements present the company's results and financial position. Analysts, investors, and creditors rely heavily on these "audit opinions," which CPAs have the exclusive authority to issue.

Taxation is another major area of public accounting. The work that tax specialists perform includes tax advice and planning, preparing tax returns, and representing clients before governmental agencies such as the Internal Revenue Service.

A third area in public accounting is **management consulting**. It ranges from installing basic accounting software or highly complex enterprise resource planning systems, to performing support services for major marketing projects and merger and acquisition activities.

Many CPAs are entrepreneurs. They form small- or medium-sized practices that frequently specialize in tax or consulting services.

Private Accounting

Instead of working in public accounting, you might choose to be an employee of a for-profit company such as **Starbucks**, **Alphabet**, or **PepsiCo**. In **managerial (or private) accounting**, you would be involved in activities such as cost accounting (finding the cost of producing specific products), budgeting, accounting information system design and support, and tax planning and preparation. You might also be a member of your company's internal audit team. In response to SOX, the internal auditors' job of reviewing the company's operations to ensure compliance with company policies and to increase efficiency has taken on increased importance.

Alternatively, many accountants work for not-for-profit organizations such as the **Red Cross** or the **Bill and Melinda Gates Foundation**, or for museums, libraries, or performing arts organizations.

Governmental Accounting

Another option is to pursue one of the many accounting opportunities in governmental agencies. For example, the Internal Revenue Service (IRS), Federal Bureau of Investigation (FBI), and the Securities and Exchange Commission (SEC) all employ accountants. The FBI has a stated goal that at least 15% of its new agents should be CPAs. There is also a very high demand for accounting educators at public colleges and universities and in state and local governments.

Forensic Accounting

Forensic accounting uses accounting, auditing, and investigative skills to conduct investigations into theft and fraud. It is listed among the top 20 career paths of the future. The job of forensic accountants is to catch the perpetrators of the estimated \$600 billion per year of theft and fraud occurring at U.S. companies. This includes tracing money-laundering and identity-theft activities as well as tax evasion. Insurance companies hire forensic accountants to detect frauds such as arson, and law offices employ forensic accountants to identify marital assets in divorces. Forensic accountants often have FBI, IRS, or similar government experience.

"Show Me the Money"

How much can a new accountant make? Take a look at the average salaries for college graduates in public and private accounting listed in **Illustration 1A.1**. Keep in mind if you also have a CPA license, you'll make 10–15% more when you start out.

ILLUSTRATION 1A.1

Salary estimates for jobs in public and corporate accounting

Employer	Jr. Level (0–3 yrs.)	Sr. Level (4–6 yrs.)
Public accounting (large firm)	\$63,250–\$83,250	\$78,500–\$106,500
Public accounting (small firm)	\$51,500–\$60,500	\$63,750–\$ 81,500
Corporate accounting (large company)	\$53,750–\$69,500	\$68,750–\$ 87,750
Corporate accounting (small company)	\$45,000–\$59,000	\$57,500–\$ 70,000

Illustration 1A.2 offers some examples of upper-level salaries for managers in corporate accounting. Note that geographic region, experience, education, CPA certification, and company size each play a role in determining salary.

ILLUSTRATION 1A.2

Upper-level management salaries in corporate accounting

Position	Large Company	Small to Medium Company
Chief financial officer	\$207,000–\$465,750	\$105,250–\$208,750
Corporate controller	\$140,000–\$224,750	\$ 92,000–\$161,250
Tax manager	\$112,000–\$158,250	\$ 88,000–\$124,750

The Review and Practice section provides opportunities for students to review key concepts and terms as well as complete multiple-choice questions, exercises, and a comprehensive problem. Detailed solutions are also included.

Review and Practice

Learning Objectives Review

1 Identify the activities and users associated with accounting.

Accounting is an information system that identifies, records, and communicates the economic events of an organization to interested users. The major users and uses of accounting are as follows. (a) Management uses accounting information to plan, organize, and run the business. (b) Investors (owners) decide whether to buy, hold, or sell their financial interests on the basis of accounting data. (c) Creditors (suppliers and bankers) evaluate the risks of granting credit or lending money on the basis of accounting information. Other groups that use accounting information are taxing authorities, regulatory agencies, customers, and labor unions.

2 Explain the building blocks of accounting: ethics, principles, and assumptions.

Ethics are the standards of conduct by which actions are judged as right or wrong. Effective financial reporting depends on sound ethical behavior. Generally accepted accounting principles are a common set of standards used by accountants. The primary accounting standard-setting body in the United States is the Financial Accounting Standards Board. The monetary unit assumption requires that companies include in the accounting records only transaction data that can be expressed in terms of money. The economic entity assumption requires

that the activities of each economic entity be kept separate from the activities of its owner(s) and other economic entities.

3 State the accounting equation, and define its components.

The basic accounting equation is:

Assets = Liabilities + Owner’s Equity

Assets are resources a business owns. Liabilities are creditorship claims on total assets. Owner’s equity is the ownership claim on total assets. The expanded accounting equation is:

Assets = Liabilities + Owner’s Capital – Owner’s Drawings + Revenues – Expenses

Investments by owners (assets the owner puts into the business) are recorded in a category called owner’s capital. Owner’s drawings are the withdrawal of assets by the owner for personal use. Revenues are the gross increase in owner’s equity from business activities for the purpose of earning income. Expenses are the costs of assets consumed or services used in the process of earning revenue.

4 Analyze the effects of business transactions on the accounting equation.

Each business transaction must have a dual effect on the accounting equation. For example, if an individual asset increases, there must be

a corresponding (1) decrease in another asset, (2) increase in a specific liability, or (3) increase in owner's equity.

5 Describe the four financial statements and how they are prepared.

An income statement presents the revenues and expenses, and resulting net income or net loss, for a specific period of time. An owner's equity statement summarizes the changes in owner's equity for a specific period of time. A balance sheet reports the assets, liabilities,

and owner's equity at a specific date. A statement of cash flows summarizes information about the cash inflows (receipts) and outflows (payments) for a specific period of time.

*6 Explain the career opportunities in accounting.

Accounting offers many different jobs in fields such as public and private accounting, governmental, and forensic accounting. Accounting is a popular major because there are many different types of jobs, with unlimited potential for career advancement.

Glossary Review

Accounting The information system that identifies, records, and communicates the economic events of an organization to interested users. (p. 1-3).

Accounting information system The system of collecting and processing transaction data and communicating financial information to decision-makers. (p. 1-13)

Assets Resources a business owns. (p. 1-11).

***Auditing** The examination of financial statements by a certified public accountant in order to express an opinion as to the fairness of presentation. (p. 1-25).

Balance sheet A financial statement that reports the assets, liabilities, and owner's equity at a specific date. (p. 1-20).

Basic accounting equation $\text{Assets} = \text{Liabilities} + \text{Owner's equity}$. (p. 1-11).

Bookkeeping A part of the accounting process that involves only the recording of economic events. (p. 1-4).

Convergence The process of reducing the differences between U.S. GAAP and IFRS. (p. 1-8).

Corporation A business organized as a separate legal entity under state corporation law, having ownership divided into transferable shares of stock. (p. 1-10).

Drawings Withdrawal of cash or other assets from an unincorporated business for the personal use of the owner(s). (p. 1-12).

Economic entity assumption An assumption that requires that the activities of the entity be kept separate and distinct from the activities of its owner and all other economic entities. (p. 1-9).

Ethics The standards of conduct by which actions are judged as right or wrong, honest or dishonest, fair or not fair. (p. 1-7).

Expanded accounting equation $\text{Assets} = \text{Liabilities} + \text{Owner's capital} - \text{Owner's drawings} + \text{Revenues} - \text{Expenses}$. (p. 1-12).

Expenses The cost of assets consumed or services used in the process of generating revenue. (p. 1-12).

Fair value principle An accounting principle stating that assets and liabilities should be reported at fair value (the price received to sell an asset or settle a liability). (p. 1-9).

Faithful representation Numbers and descriptions match what really existed or happened—they are factual. (p. 1-8).

Financial accounting The field of accounting that provides economic and financial information for investors, creditors, and other external users. (p. 1-5).

Financial Accounting Standards Board (FASB) A private organization that establishes generally accepted accounting principles (GAAP) in the United States. (p. 1-8).

***Forensic accounting** An area of accounting that uses accounting, auditing, and investigative skills to conduct investigations into theft and fraud. (p. 1-25).

Generally accepted accounting principles (GAAP) Common standards that indicate how to report economic events. (p. 1-8).

Historical cost principle An accounting principle that states that companies should record assets at their cost. (p. 1-8).

Income statement A financial statement that presents the revenues and expenses and resulting net income or net loss of a company for a specific period of time. (p. 1-20).

International Accounting Standards Board (IASB) An accounting standard-setting body that issues standards adopted by many countries outside of the United States. (p. 1-8).

International Financial Reporting Standards (IFRS) International accounting standards set by the International Accounting Standards Board (IASB). (p. 1-8).

Investments by owner The assets an owner puts into the business. (p. 1-12).

Liabilities Creditor claims against total assets. (p. 1-11).

***Management consulting** An area of public accounting ranging from development of accounting and computer systems to support services for marketing projects and merger and acquisition activities. (p. 1-25).

Managerial accounting The field of accounting that provides internal reports to help users make decisions about their companies. (p. 1-5).

Monetary unit assumption An assumption stating that companies include in the accounting records only transaction data that can be expressed in terms of money. (p. 1-9).

Net income The amount by which revenues exceed expenses. (p. 1-22).

Net loss The amount by which expenses exceed revenues. (p. 1-22).

Owner's equity The ownership claim on total assets. (p. 1-12).

Owner's equity statement A financial statement that summarizes the changes in owner's equity for a specific period of time. (p. 1-20).

Partnership A business owned by two or more persons associated as partners. (p. 1-9).

Proprietorship A business owned by one person. (p. 1-9).

***Public accounting** An area of accounting in which the accountant offers expert service to the general public. (p. 1-24).

Relevance Financial information that is capable of making a difference in a decision. (p. 1-8).

Revenues The increases in assets or decreases in liabilities resulting from the sale of goods or the performance of services in the normal course of business. (p. 1-12).

Sarbanes-Oxley Act (SOX) Law passed by Congress intended to reduce unethical corporate behavior. (p. 1-6).

Securities and Exchange Commission (SEC) A governmental agency that oversees U.S. financial markets and accounting standard-setting bodies. (p. 1-8).

Statement of cash flows A financial statement that summarizes information about the cash inflows (receipts) and cash outflows (payments) for a specific period of time. (p. 1-20).

***Taxation** An area of public accounting involving tax advice, tax planning, preparing tax returns, and representing clients before governmental agencies. (p. 1-25).

Transactions The economic events of a business that are recorded by accountants. (p. 1-14).

Practice Multiple-Choice Questions

1. (LO 1) Which of the following is **not** a step in the accounting process?

- a. Identification.
- b. Economic entity.
- c. Recording.
- d. Communication.

2. (LO 1) Which of the following statements about users of accounting information is **incorrect**?

- a. Management is an internal user.
- b. Taxing authorities are external users.
- c. Present creditors are external users.
- d. Regulatory authorities are internal users.

3. (LO 2) The historical cost principle states that:

- a. assets should be initially recorded at cost and adjusted when the fair value changes.
- b. activities of an entity are to be kept separate and distinct from its owner.
- c. assets should be recorded at their cost.
- d. only transaction data capable of being expressed in terms of money be included in the accounting records.

4. (LO 2) Which of the following statements about basic assumptions is **correct**?

- a. Basic assumptions are the same as accounting principles.
- b. The economic entity assumption states that there should be a particular unit of accountability.
- c. The monetary unit assumption enables accounting to measure employee morale.
- d. Partnerships are not economic entities.

5. (LO 2) The three types of business entities are:

- a. proprietorships, small businesses, and partnerships.
- b. proprietorships, partnerships, and corporations.
- c. proprietorships, partnerships, and large businesses.
- d. financial, manufacturing, and service companies.

6. (LO 3) Net income will result during a time period when:

- a. assets exceed liabilities.
- b. assets exceed revenues.
- c. expenses exceed revenues.
- d. revenues exceed expenses.

7. (LO 3) As of December 31, 2020, Kent Company has assets of \$3,500 and owner's equity of \$2,000. What are the liabilities for Kent Company as of December 31, 2020?

- a. \$1,500.
- b. \$1,000.
- c. \$2,500.
- d. \$2,000.

8. (LO 4) Performing services on account will have the following effects on the components of the basic accounting equation:

- a. increase assets and decrease owner's equity.
- b. increase assets and increase owner's equity.
- c. increase assets and increase liabilities.
- d. increase liabilities and increase owner's equity.

9. (LO 4) Which of the following events is **not** recorded in the accounting records?

- a. Equipment is purchased on account.
- b. An employee is terminated.
- c. A cash investment is made into the business.
- d. The owner withdraws cash for personal use.

10. (LO 4) During 2020, Bruske Company's assets decreased \$50,000 and its liabilities decreased \$50,000. Its owner's equity therefore:

- a. increased \$50,000.
- b. decreased \$50,000.
- c. decreased \$100,000.
- d. did not change.

11. (LO 4) Payment of an account payable affects the components of the accounting equation in the following way.

- a. Decreases owner's equity and decreases liabilities.
- b. Increases assets and decreases liabilities.
- c. Decreases assets and increases owner's equity.
- d. Decreases assets and decreases liabilities.

12. (LO 5) Which of the following statements is **false**?

- a. A statement of cash flows summarizes information about the cash inflows (receipts) and outflows (payments) for a specific period of time.
- b. A balance sheet reports the assets, liabilities, and owner's equity at a specific date.
- c. An income statement presents the revenues, expenses, changes in owner's equity, and resulting net income or net loss for a specific period of time.
- d. An owner's equity statement summarizes the changes in owner's equity for a specific period of time.

13. (LO 5) On the last day of the period, Alan Cesska Company buys a \$900 machine on credit. This transaction will affect the:

- a. income statement only.

- b. balance sheet only.
- c. income statement and owner's equity statement only.
- d. income statement, owner's equity statement, and balance sheet.

14. (LO 5) The financial statement that reports assets, liabilities, and owner's equity is the:

- a. income statement.
- b. owner's equity statement.

- c. balance sheet.
- d. statement of cash flows.

***15. (LO 6)** Services performed by a public accountant include:

- a. auditing, taxation, and management consulting.
- b. auditing, budgeting, and management consulting.
- c. auditing, budgeting, and cost accounting.
- d. auditing, budgeting, and management consulting.

Solutions

1. b. Economic entity is not one of the steps in the accounting process. The other choices are true because (a) identification is the first step in the accounting process, (c) recording is the second step in the accounting process, and (d) communication is the third and final step in the accounting process.

2. d. Regulatory authorities are external, not internal, users of accounting information. The other choices are true statements.

3. c. The historical cost principle states that assets should be recorded at their cost. The other choices are incorrect because (a) the historical cost principle does not say that assets should be adjusted for changes in fair value, (b) describes the economic entity assumption, and (d) describes the monetary unit assumption.

4. b. The economic entity assumption states that there should be a particular unit of accountability. The other choices are incorrect because (a) basic assumptions are not the same as accounting principles, (c) the monetary unit assumption allows accounting to measure economic events, and (d) partnerships are economic entities.

5. b. Proprietorships, partnerships, and corporations are the three types of business entities. Choices (a) and (c) are incorrect because small and large businesses only denote the sizes of businesses. Choice (d) is incorrect because financial, manufacturing, and service companies are types of businesses, not business entities.

6. d. Net income results when revenues exceed expenses. The other choices are incorrect because (a) assets and liabilities are not used in the computation of net income; (b) revenues, not assets, are included in the computation of net income; and (c) when expenses exceed revenues, a net loss results.

7. a. Using a variation of the basic accounting equation, $\text{Assets} - \text{Owner's equity} = \text{Liabilities}$, $\$3,500 - \$2,000 = \$1,500$. Therefore, choices (b) \$1,000, (c) \$2,500, and (d) \$2,000 are incorrect.

8. b. When services are performed on account, assets are increased and owner's equity is increased. The other choices are incorrect because when services are performed on account (a) owner's equity is

increased, not decreased; (c) liabilities are not affected; and (d) owner's equity is increased and liabilities are not affected.

9. b. If an employee is terminated, this represents an activity of a company, not a business transaction. Assets, liabilities, and owner's equity are not affected. Thus, there is no effect on the accounting equation. The other choices are incorrect because they are all recorded: (a) when equipment is purchased on account, both assets and liabilities increase; (c) when a cash investment is made into a business, both assets and owner's equity increase; and (d) when an owner withdraws cash for personal use, both assets and owner's equity decrease.

10. d. In this situation, owner's equity does not change because only assets and liabilities decreased \$50,000. Therefore, the other choices are incorrect.

11. d. Payment of an account payable results in an equal decrease of assets (cash) and liabilities (accounts payable). The other choices are incorrect because payment of an account payable (a) does not affect owner's equity, (b) does not increase assets, and (c) does not affect owner's equity.

12. c. An income statement represents the revenues, expenses, and the resulting net income or net loss for a specific period of time but not the changes in owner's equity. The other choices are true statements.

13. b. This transaction will cause assets to increase by \$900 and liabilities to increase by \$900. The other choices are incorrect because this transaction (a) will have no effect on the income statement, (c) will have no effect on the income statement or the owner's equity statement, and (d) will affect the balance sheet but not the income statement or the owner's equity statement.

14. c. The balance sheet is the statement that reports assets, liabilities and owner's equity. The other choices are incorrect because (a) the income statement reports revenues and expenses, (b) the owner's equity statement reports details about owner's equity, and (d) the statement of cash flows reports inflows and outflows of cash.

***15. a.** Auditing, taxation, and management consulting are all services performed by public accountants. The other choices are incorrect because public accountants do not perform budgeting or cost accounting.

Practice Brief Exercises

1. (LO 3) At the beginning of the year, Ortiz Company had total assets of \$900,000 and total liabilities of \$440,000. Answer the following questions.

Use basic accounting equation.

- a. If total assets decreased \$100,000 during the year and total liabilities increased \$80,000 during the year, what is the amount of owner's equity at the end of the year?
- b. During the year, total liabilities decreased \$100,000 and owner's equity increased \$200,000. What is the amount of total assets at the end of the year?

- c. If total assets increased \$50,000 during the year and owner's equity increased \$60,000 during the year, what is the amount of total liabilities at the end of the year?

Solution

1. a. $(\$900,000 - \$440,000) - \$100,000 - \$80,000 = \$280,000$ owner's equity
 b. $\$900,000 - \$100,000 + \$200,000 = \$1,000,000$ total assets
 c. $\$440,000 - \$60,000 + \$50,000 = \$430,000$ total liabilities

Determine effect of transactions on basic accounting equation

2. (LO 4) Presented below are three business transactions. List the letters (a), (b), and (c) with columns for assets, liabilities and owner's equity. For each column, indicate whether the transactions increased (+), decreased (−), or had no effect (NE) on assets, liabilities, and owner's equity.

- a. Purchased equipment on account.
 b. Withdrawal of cash by owner.
 c. Paid expenses in cash.

Solution

	<u>Assets</u>	<u>Liabilities</u>	<u>Owner's Equity</u>
2. a.	+	+	NE
b.	−	NE	−
c.	−	NE	−

Determine effect of transactions on basic accounting equation.

3. (LO 4) Follow the same format as in **Practice Brief Exercise 2**. Determine the effect on assets, liabilities, and owner's equity of the following three transactions.

- a. Performed accounting services for clients for cash.
 b. Borrowed cash from a bank on a note payable.
 c. Paid cash for rent for the month.

Solution

	<u>Assets</u>	<u>Liabilities</u>	<u>Owner's Equity</u>
3. a.	+	NE	+
b.	+	+	NE
c.	−	NE	−

Determine where items appear on financial statements.

4. (LO 5) **Financial Statement** Indicate whether the following items would appear on the income statement (IS), balance sheet (BS), or owner's equity statement (OE).

- a. Owner's capital.
 b. Cash.
 c. Salaries and wages expense.
 d. Service revenue.
 e. Accounts payable.

Solution

4. a. Owner's capital	<u>OE, BS</u>
b. Cash	<u>BS</u>
c. Salaries and wages expense	<u>IS</u>
d. Service revenue	<u>IS</u>
e. Accounts payable	<u>BS</u>

5. (LO 5) Financial Statement Presented below in alphabetical order are balance sheet items for Feagler Company at December 31, 2020. Carole Feagler is the owner of the company. Prepare a balance sheet following the format of Illustration 1.9.

Prepare a balance sheet.

Accounts receivable	\$12,500
Cash	38,000
Notes payable	40,000
Owner's capital	10,500

Solution

5.

Feagler Company Balance Sheet December 31, 2020

<u>Assets</u>	
Cash	\$38,000
Accounts receivable	<u>12,500</u>
Total assets	<u>\$50,500</u>
<u>Liabilities and Owner's Equity</u>	
Liabilities	
Notes payable	\$40,000
Owner's equity	
Owner's capital	<u>10,500</u>
Total liabilities and owner's equity	<u>\$50,500</u>

Practice Exercises

1. (LO 3, 4) Selected transactions for Fabulous Flora Company are listed below.

Analyze the effect of transactions.

- Made cash investment to start business.
- Purchased equipment on account.
- Paid salaries.
- Billed customers for services performed.
- Received cash from customers billed in (4).
- Withdrew cash for owner's personal use.
- Incurred advertising expense on account.
- Purchased additional equipment for cash.
- Received cash from customers when service was performed.

Instructions

List the numbers of the above transactions and describe the effect of each transaction on assets, liabilities, and owner's equity. For example, the first answer is: (1) Increase in assets and increase in owner's equity.

Solution

1. Increase in assets and increase in owner's equity.
2. Increase in assets and increase in liabilities.
3. Decrease in assets and decrease in owner's equity.
4. Increase in assets and increase in owner's equity.
5. Increase in assets and decrease in assets.
6. Decrease in assets and decrease in owner's equity.

7. Increase in liabilities and decrease in owner's equity.
8. Increase in assets and decrease in assets.
9. Increase in assets and increase in owner's equity.

Analyze the effect of transactions on assets, liabilities, and owner's equity.

2. (LO 3, 4) Alma's Payroll Services Company entered into the following transactions during May 2020.

1. Purchased computers for \$15,000 from Bytes of Data on account.
2. Paid \$3,000 cash for May rent on storage space.
3. Received \$12,000 cash from customers for contracts billed in April.
4. Performed payroll services for Magic Construction Company for \$2,500 cash.
5. Paid Northern Ohio Power Co. \$7,000 cash for energy usage in May.
6. Alma invested an additional \$25,000 in the business.
7. Paid Bytes of Data for the computers purchased in (1) above.
8. Incurred advertising expense for May of \$900 on account.

Instructions

Indicate with the appropriate letter whether each of the transactions above results in:

- a. an increase in assets and a decrease in assets.
- b. an increase in assets and an increase in owner's equity.
- c. an increase in assets and an increase in liabilities.
- d. a decrease in assets and a decrease in owner's equity.
- e. a decrease in assets and a decrease in liabilities.
- f. an increase in liabilities and a decrease in owner's equity.
- g. an increase in owner's equity and a decrease in liabilities.

Solution

- | | | | |
|---------|------|------|------|
| 2. 1. c | 3. a | 5. d | 7. e |
| 2. d | 4. b | 6. b | 8. f |

Practice Problem

Prepare a tabular presentation and financial statements.

(LO 4, 5) Joan Robinson opens her own law office on July 1, 2020. During the first month of operations, the following transactions occurred.

1. Joan invested \$11,000 in cash in the law practice.
2. Paid \$800 for July rent on office space.
3. Purchased equipment on account \$3,000.
4. Performed legal services for clients for cash \$1,500.
5. Borrowed \$700 cash from a bank on a note payable.
6. Performed legal services for client on account \$2,000.
7. Paid monthly expenses: salaries and wages \$500, utilities \$300, and advertising \$100.
8. Joan withdrew \$1,000 cash for personal use.

Instructions

- a. Prepare a tabular summary of the transactions.
- b. **Financial Statement** Prepare the income statement, owner's equity statement, and balance sheet at July 31, 2020, for Joan Robinson, Attorney.

Solution

Trans- action	Assets				=	Liabilities			+	Owner's Equity							
	Cash	+	Accounts Receivable	+	Equipment	=	Notes Payable	+	Accounts Payable	+	Owner's Capital	-	Owner's Drawings	+	Revenues	-	Expenses
1.	+\$11,000					=					+\$11,000						
2.	-800																-\$800
3.					+\$3,000	=			+\$3,000								
4.	+1,500														+\$1,500		
5.	+700						+\$700										
6.			+\$2,000												+\$2,000		
7.	-500																-500
	-300																-300
	-100																-100
8.	-1,000												-\$1,000				
	<u>\$10,500</u>	+	<u>\$2,000</u>	+	<u>\$3,000</u>	=	<u>\$700</u>	+	<u>\$3,000</u>	+	<u>\$11,000</u>	-	<u>\$1,000</u>	+	<u>\$3,500</u>	-	<u>\$1,700</u>
	<u>\$15,500</u>						<u>\$15,500</u>										

b.

Joan Robinson, Attorney
Income Statement
For the Month Ended July 31, 2020

Revenues		
Service revenue		\$3,500
Expenses		
Rent expense	\$800	
Salaries and wages expense	500	
Utilities expense	300	
Advertising expense	100	
Total expenses		<u>1,700</u>
Net income		<u>\$1,800</u>

Joan Robinson, Attorney
Owner's Equity Statement
For the Month Ended July 31, 2020

Owner's capital, July 1		\$ 0
Add: Investments	\$11,000	
Net income	<u>1,800</u>	<u>12,800</u>
		12,800
Less: Drawings		<u>1,000</u>
Owner's capital, July 31		<u>\$11,800</u>

Joan Robinson, Attorney
Balance Sheet
July 31, 2020

<u>Assets</u>		
Cash		\$10,500
Accounts receivable		2,000
Equipment		<u>3,000</u>
Total assets		<u>\$15,500</u>
<u>Liabilities and Owner's Equity</u>		
Liabilities		
Notes payable	\$ 700	
Accounts payable	<u>3,000</u>	
Total liabilities		3,700
Owner's equity		
Owner's capital		<u>11,800</u>
Total liabilities and owner's equity		<u>\$15,500</u>

WileyPLUS

Brief Exercises, DO IT! Exercises, Exercises, Problems, and many additional resources are available for practice in WileyPLUS.

Note: All asterisked Questions, Exercises, and Problems relate to material in the appendix to the chapter.

Questions

1. "Accounting is ingrained in our society and it is vital to our economic system." Do you agree? Explain.
2. Identify and describe the steps in the accounting process.
3. (a) Who are internal users of accounting data? (b) How does accounting provide relevant data to these users?
4. What uses of financial accounting information are made by (a) investors and (b) creditors?
5. "Bookkeeping and accounting are the same." Do you agree? Explain.
6. Benton Travel Agency purchased land for \$90,000 cash on December 10, 2020. At December 31, 2020, the land's value has increased to \$92,000. What amount should be reported for land on Benton's balance sheet at December 31, 2020? Explain.
7. What is the monetary unit assumption?
8. What is the economic entity assumption?
9. What are the three basic forms of business organizations for profit-oriented enterprises?
10. Helen Rupp is the owner of a successful printing shop. Recently, her business has been increasing, and Helen has been thinking about changing the organization of her business from a proprietorship to a corporation. Discuss some of the advantages Helen would enjoy if she were to incorporate her business.
11. What is the basic accounting equation?
12. a. Define the terms assets, liabilities, and owner's equity.
b. What items affect owner's equity?
13. Which of the following items are liabilities of Siebers Jewelry Stores?
 - a. Cash.
 - b. Accounts payable.
 - c. Owner's drawings.
 - d. Accounts receivable.
 - e. Supplies.
 - f. Equipment.
 - g. Salaries and wages payable.
 - h. Service revenue.
 - i. Rent expense.
14. Can a business enter into a transaction in which only the left side of the basic accounting equation is affected? If so, give an example.
15. Are the following events recorded in the accounting records? Explain your answer in each case.
 - a. The owner of the company dies.
 - b. Supplies are purchased on account.
 - c. An employee is fired.
 - d. The owner of the business withdraws cash from the business for personal use.
16. Indicate how the following business transactions affect the basic accounting equation.
 - a. Paid cash for janitorial services.
 - b. Purchased equipment for cash.
 - c. Invested cash in the business.
 - d. Paid accounts payable in full.
17. Listed below are some items found in the financial statements of Tony Gruber Co. Indicate in which financial statement(s) the following items would appear.
 - a. Service revenue.
 - b. Equipment.
 - c. Advertising expense.
 - d. Accounts receivable.
 - e. Owner's capital.
 - f. Salaries and wages payable.
18. In February 2020, Ola Gott invested an additional \$12,000 in her business, Gott's Pharmacy, which is organized as a proprietorship. Gott's accountant, Sal Costa, recorded this receipt as an increase in cash and revenues. Is this treatment appropriate? Why or why not?
19. "A company's net income appears directly on the income statement and the owner's equity statement, and it is included indirectly in the company's balance sheet." Do you agree? Explain.
20. Bayler Enterprises had a capital balance of \$186,000 at the beginning of the period. At the end of the accounting period, the capital balance was \$189,000.
 - a. Assuming no additional investment or withdrawals during the period, what is the net income for the period?
 - b. Assuming an additional investment of \$13,000 but no withdrawals during the period, what is the net income for the period?
21. Summarized operations for Bayles Co. for the month of July are as follows.

Revenues recognized: for cash \$20,000; on account \$70,000.

Expenses incurred: for cash \$26,000; on account \$40,000.

Indicate for Bayles Co. (a) the total revenues, (b) the total expenses, and (c) net income for the month of July.
22. The basic accounting equation is $\text{Assets} = \text{Liabilities} + \text{Owner's equity}$. Replacing the words in that equation with dollar amounts, what is **Apple's** accounting equation at September 26, 2015? (*Hint:* Owner's equity is equivalent to shareholders' equity.)

Brief Exercises

BE1.1 (LO 3) Presented below is the basic accounting equation. Determine the missing amounts.

Use basic accounting equation.

<u>Assets</u>	=	<u>Liabilities</u>	+	<u>Owner's Equity</u>
a. \$90,000		\$50,000		?
b. ?		\$44,000		\$70,000
c. \$94,000		?		\$53,000

BE1.2 (LO 3) Given the accounting equation, answer each of the following questions.

Use basic accounting equation.

- The liabilities of Berber Company are \$120,000 and the owner's equity is \$230,000. What is the amount of Berber Company's total assets?
- The total assets of Berber Company are \$190,000 and its owner's equity is \$89,000. What is the amount of its total liabilities?
- The total assets of Berber Company are \$900,000 and its liabilities are equal to one-half of its total assets. What is the amount of Berber Company's owner's equity?

BE1.3 (LO 3) At the beginning of the year, Gilles Company had total assets of \$800,000 and total liabilities of \$300,000. Answer the following questions.

Use basic accounting equation.

- If total assets increased \$150,000 during the year and total liabilities decreased \$60,000, what is the amount of owner's equity at the end of the year?
- During the year, total liabilities increased \$100,000 and owner's equity decreased \$70,000. What is the amount of total assets at the end of the year?
- If total assets decreased \$80,000 and owner's equity increased \$120,000 during the year, what is the amount of total liabilities at the end of the year?

BE1.4 (LO 3) Use the expanded accounting equation to answer each of the following questions.

Solve expanded accounting equation.

- The liabilities of Platt Company are \$90,000. Owner's capital is \$150,000; drawings are \$40,000; revenues, \$450,000; and expenses, \$340,000. What is the amount of Platt Company's total assets?
- The total assets of Sierra Company are \$57,000. Owner's capital is \$35,000; drawings are \$7,000; revenues, \$52,000; and expenses, \$35,000. What is the amount of the company's total liabilities?
- The total assets of Birch Co. are \$660,000 and its liabilities are equal to two-thirds of its total assets. What is the amount of Birch Co.'s owner's equity?

BE1.5 (LO 3) Indicate whether each of the following items is an asset (A), liability (L), or part of owner's equity (OE).

Identify assets, liabilities, and owner's equity.

_____ a. Accounts receivable	_____ d. Supplies
_____ b. Salaries and wages payable	_____ e. Owner's capital
_____ c. Equipment	_____ f. Notes payable

BE1.6 (LO 3) Classify each of the following items as owner's drawings (D), revenue (R), or expense (E).

Classify items affecting owner's equity.

_____ a. Advertising expense	_____ e. Owner's drawings
_____ b. Service revenue	_____ f. Rent revenue
_____ c. Insurance expense	_____ g. Utilities expense
_____ d. Salaries and wages expense	

BE1.7 (LO 4) Presented below are three business transactions. On a sheet of paper, list the letters (a), (b), and (c) with columns for assets, liabilities, and owner's equity. For each column, indicate whether the transactions increased (+), decreased (−), or had no effect (NE) on assets, liabilities, and owner's equity.

Determine effect of transactions on basic accounting equation.

- Purchased supplies on account.
- Received cash for performing a service.
- Paid expenses in cash.

BE1.8 (LO 4) Follow the same format as in BE1.7. Determine the effect on assets, liabilities, and owner's equity of the following three transactions.

Determine effect of transactions on basic accounting equation.

- Invested cash in the business.
- Withdrawal of cash by owner.
- Received cash from a customer who had previously been billed for services performed.

Determine effect of transactions on basic owner's equity.

BE1.9 (LO 4) Presented below are three transactions. Mark each transaction as affecting owner's investment (I), owner's drawings (D), revenue (R), expense (E), or not affecting owner's equity (NOE).

- _____ a. Received cash for services performed
- _____ b. Paid cash to purchase equipment
- _____ c. Paid employee salaries

Prepare a balance sheet.

BE1.10 (LO 5) Financial Statement In alphabetical order below are balance sheet items for Smyth Company at December 31, 2020. Kathy Smyth is the owner of Smyth Company. Prepare a balance sheet, following the format of Illustration 1.9.

Accounts payable	\$90,000
Accounts receivable	62,500
Cash	49,000
Owner's capital	21,500

Determine where items appear on financial statements.

BE1.11 (LO 5) Financial Statement Indicate whether the following items would appear on the income statement (IS), balance sheet (BS), or owner's equity statement (OE).

- | | |
|------------------------------|--------------------------|
| _____ a. Notes payable | _____ d. Cash |
| _____ b. Advertising expense | _____ e. Service revenue |
| _____ c. Owner's capital | |

DO IT! Exercises

Review basic concepts.

DO IT! 1.1 (LO 1) Indicate whether each of the five statements presented below is true or false.

- The three steps in the accounting process are identification, recording, and examination.
- The accounting process includes the bookkeeping function.
- Managerial accounting provides reports to help investors and creditors evaluate a company.
- The two most common types of external users are investors and creditors.
- Internal users include human resources managers.

Identify building blocks of accounting.

DO IT! 1.2 (LO 2) Indicate whether each of the five statements presented below is true or false.

- Congress passed the Sarbanes-Oxley Act to ensure that investors invest only in companies that will be profitable.
- The standards of conduct by which actions are judged as loyal or disloyal are ethics.
- The primary accounting standard-setting body in the United States is the Securities and Exchange Commission (SEC).
- The historical cost principle dictates that companies record assets at their cost and continue to report them at their cost over the time the assets are held.
- The monetary unit assumption requires that companies record only transactions that can be measured in money.

Evaluate effects of transactions on owner's equity.

DO IT! 1.3 (LO 3) Classify the following items as investment by owner (I), owner's drawings (D), revenues (R), or expenses (E). Then indicate whether each item increases or decreases owner's equity.

- | | |
|------------------|--|
| 1. Drawings. | 3. Advertising expense. |
| 2. Rent revenue. | 4. Owner puts personal assets into the business. |

Prepare tabular analysis.

DO IT! 1.4 (LO 4) Transactions made by A. Marti and Co., a law firm, for the month of March are shown below. Prepare a tabular analysis which shows the effects of these transactions on the expanded accounting equation, similar to that shown in Illustration 1.8.

- The company performed \$20,000 of services for customers, on credit.
- The company received \$20,000 in cash from customers who had been billed for services (in transaction 1).
- The company received a bill for \$3,200 of advertising but will not pay it until a later date.
- A. Marti withdrew \$2,500 cash from the business for personal use.

DO IT! 1.5 (LO 5) Presented below is selected information related to Kirby Company at December 31, 2020. Kirby reports financial information monthly.

Determine specific amounts on the financial statements.

Accounts Payable	\$ 3,000	Salaries and Wages Expense	\$16,500
Cash	6,500	Notes Payable	25,000
Advertising Expense	6,000	Rent Expense	10,500
Service Revenue	53,500	Accounts Receivable	13,500
Equipment	29,000	Owner's Drawings	7,500

- Determine the total assets of Kirby Company at December 31, 2020.
- Determine the net income that Kirby Company reported for December 2020.
- Determine the owner's equity of Kirby Company at December 31, 2020.

Exercises

E1.1 (LO 1) Genesis Company performs the following accounting tasks during the year.

Classify the three activities of accounting.

- _____ Analyzing and interpreting information.
- _____ Classifying economic events.
- _____ Explaining uses, meaning, and limitations of data.
- _____ Keeping a systematic chronological diary of events.
- _____ Measuring events in dollars and cents.
- _____ Preparing accounting reports.
- _____ Reporting information in a standard format.
- _____ Selecting economic activities relevant to the company.
- _____ Summarizing economic events.

Accounting is “an information system that **identifies**, **records**, and **communicates** the economic events of an organization to interested users.”

Instructions

Categorize the accounting tasks performed by Genesis as relating to either the identification (I), recording (R), or communication (C) aspects of accounting.

E1.2 (LO 1) a. The following are users of financial statements.

Identify users of accounting information.

- | | |
|--------------------------------|--|
| _____ Customers | _____ Securities and Exchange Commission |
| _____ Internal Revenue Service | _____ Store manager |
| _____ Labor unions | _____ Suppliers |
| _____ Marketing manager | _____ Vice president of finance |
| _____ Production supervisor | |

Instructions

Identify the users as being either **external users** or **internal users**.

- b.** The following questions could be asked by an internal user or an external user.

- _____ Can we afford to give our employees a pay raise?
- _____ Did the company earn a satisfactory income?
- _____ Do we need to borrow in the near future?
- _____ How does the company's profitability compare to other companies?
- _____ What does it cost us to manufacture each unit produced?
- _____ Which product should we emphasize?
- _____ Will the company be able to pay its short-term debts?

Discuss ethics and the historical cost principle.

Instructions

Identify each of the questions as being more likely asked by an **internal user** or an **external user**.

E1.3 (LO 2) Angela Duffy, president of Duffy Company, has instructed Jana Barth, the head of the accounting department for Duffy Company, to report the company's land in the company's accounting reports at its fair value of \$170,000 instead of its cost of \$100,000. Duffy says, "Showing the land at \$170,000 will make our company look like a better investment when we try to attract new investors next month."

Instructions

Explain the ethical situation involved for Jana Barth, identifying the stakeholders and the alternatives.

Use accounting concepts.

E1.4 (LO 2) The following situations involve accounting principles and assumptions.

1. Sosa Company owns buildings that are worth substantially more than they originally cost. In an effort to provide more relevant information, Sosa reports the buildings at fair value in its accounting reports.
2. Mays Company includes in its accounting records only transaction data that can be expressed in terms of money.
3. Curt Russell, owner of Curt's Photography, records his personal living costs as expenses of the business.

Instructions

For each of the three situations, say if the accounting method used is correct or incorrect. If correct, identify which principle or assumption supports the method used. If incorrect, identify which principle or assumption has been violated.

Classify accounts as assets, liabilities, and owner's equity.

E1.5 (LO 3) Diehl Cleaners has the following balance sheet items.

Accounts payable	Accounts receivable
Cash	Notes payable
Equipment	Salaries and wages payable
Supplies	Owner's capital

Instructions

Classify each item as an asset, liability, or owner's equity.

Analyze the effect of transactions.

E1.6 (LO 4) Selected transactions for Poway Landscaping Company are listed below.

1. Made cash investment to start business.
2. Paid monthly rent.
3. Purchased equipment on account.
4. Billed customers for services performed.
5. Withdrew cash for owner's personal use.
6. Received cash from customers billed in (4).
7. Incurred advertising expense on account.
8. Purchased additional equipment for cash.
9. Received cash from customers when service was performed.

Instructions

List the numbers of the above transactions and describe the effect of each transaction on assets, liabilities, and owner's equity. For example, the first answer is: (1) Increase in assets and increase in owner's equity.

Analyze the effect of transactions on assets, liabilities, and owner's equity.

E1.7 (LO 4) Falske Computer Timeshare Company entered into the following transactions during May 2020.

1. Purchased computers for \$20,000 from Digital Equipment on account.
2. Paid \$4,000 cash for May rent on storage space.
3. Received \$17,000 cash from customers for contracts billed in April.
4. Performed computer services for Viking Construction Company for \$4,000 cash.
5. Paid Tri-State Power Co. \$11,000 cash for energy usage in May.

6. Falske invested an additional \$29,000 in the business.
7. Paid Digital Equipment for the computers purchased in (1) above.
8. Incurred advertising expense for May of \$1,200 on account.

Instructions

Indicate with the appropriate letter whether each of the transactions above results in:

- a. An increase in assets and a decrease in assets.
- b. An increase in assets and an increase in owner's equity.
- c. An increase in assets and an increase in liabilities.
- d. A decrease in assets and a decrease in owner's equity.
- e. A decrease in assets and a decrease in liabilities.
- f. An increase in liabilities and a decrease in owner's equity.
- g. An increase in owner's equity and a decrease in liabilities.

E1.8 (LO 4) Writing An analysis of the transactions made by Peat Deloitte & Co., a certified public accounting firm, for the month of August is shown below. The expenses were \$560 for rent, \$4,800 for salaries and wages, and \$400 for utilities.

Analyze transactions and compute net income.

	Cash	+	Accounts Receivable	+	Supplies	+	Equipment	=	Accounts Payable	+	Owner's Capital	-	Owner's Drawings	+	Revenues	-	Expenses
1.	+\$15,000										+\$15,000						
2.	-2,000						+\$5,000		+\$3,000								
3.	-750				+\$750												
4.	+4,000		+\$4,500												+\$8,500		
5.	-1,500								-1,500								
6.	-2,000												-\$2,000				
7.	-560																-\$560
8.	+450		-450														
9.	-4,800																-4,800
10.									+400								-400

Instructions

- a. Describe each transaction that occurred for the month.
- b. Determine how much owner's equity increased for the month.
- c. Compute the amount of net income for the month.

E1.9 (LO 5) Financial Statement An analysis of transactions for Peat DeLoitte & Co. was presented in E1.8.

Prepare financial statements.

Instructions

Prepare an income statement and an owner's equity statement for August and a balance sheet at August 31, 2020. Assume that August is the company's first month of business.

E1.10 (LO 5) Hawke Company had the following assets and liabilities on the dates indicated.

Determine net income (or loss).

<u>December 31</u>	<u>Total Assets</u>	<u>Total Liabilities</u>
2019	\$400,000	\$250,000
2020	\$460,000	\$300,000
2021	\$590,000	\$400,000

Hawke began business on January 1, 2019, with an investment of \$100,000.

Instructions

From an analysis of the change in owner's equity during the year, compute the net income (or loss) for:

- a. 2019, assuming Hawke's drawings were \$12,000 for the year.
- b. 2020, assuming Hawke made an additional investment of \$34,000 and had no drawings in 2020.
- c. 2021, assuming Hawke made an additional investment of \$12,000 and had drawings of \$25,000 in 2021.

Analyze financial statements items.

E1.11 (LO 5) Two items are omitted from each of the following summaries of balance sheet and income statement data for two proprietorships for the year 2020, Greene's Goods and Solar Enterprises.

	<u>Greene's Goods</u>	<u>Solar Enterprises</u>
Beginning of year:		
Total assets	\$110,000	\$129,000
Total liabilities	85,000	(c)
Total owner's equity	(a)	80,000
End of year:		
Total assets	160,000	180,000
Total liabilities	120,000	50,000
Total owner's equity	40,000	130,000
Changes during year in owner's equity:		
Additional investment	(b)	25,000
Drawings	37,000	(d)
Total revenues	220,000	100,000
Total expenses	175,000	60,000

Instructions

Determine the missing amounts.

Prepare income statement and owner's equity statement.

E1.12 (LO 5) Financial Statement The following information relates to Fleete Co. for the year 2020.

Owner's capital, January 1, 2020	\$42,000	Advertising expense	\$ 1,800
Owner's drawings during 2020	6,000	Rent expense	10,400
Service revenue	63,600	Utilities expense	3,100
Salaries and wages expense	30,200		

Instructions

After analyzing the data, prepare an income statement and an owner's equity statement for the year ending December 31, 2020.

Correct an incorrectly prepared balance sheet.

E1.13 (LO 5) Financial Statement Abby Roland is the bookkeeper for Cheng Company. Abby has been trying to determine the correct balance sheet for Cheng Company. Cheng's balance sheet is shown below.

Cheng Company Balance Sheet December 31, 2020			
Assets		Liabilities	
Cash	\$15,000	Accounts payable	\$21,000
Supplies	8,000	Accounts receivable	(6,500)
Equipment	46,000	Owner's capital	67,500
Owner's drawings	13,000	Total liabilities and	
Total assets	<u>\$82,000</u>	owner's equity	<u>\$82,000</u>

Instructions

Prepare a correct balance sheet.

Compute net income and prepare a balance sheet.

E1.14 (LO 5) Financial Statement Saira Morrow is the sole owner of Buena Vista Park, a public camping ground near the Crater Lake National Recreation Area. Saira has compiled the following financial information as of December 31, 2020.

Revenues during 2020—camping fees	\$140,000	Fair value of equipment	\$140,000
Revenues during 2020—general store	65,000	Notes payable	60,000
Accounts payable	11,000	Expenses during 2020	160,000
Cash on hand	23,000	Accounts receivable	17,500
Original cost of equipment	115,500		

Instructions

- Determine Saira Morrow's net income from Buena Vista Park for 2020.
- Prepare a balance sheet for Buena Vista Park as of December 31, 2020.

E1.15 (LO 5) Financial Statement Presented below is financial information related to the 2020 operations of Sea Legs Cruise Company.

Prepare an income statement.

Maintenance and repairs expense	\$ 95,000
Utilities expense	13,000
Salaries and wages expense	142,000
Advertising expense	24,500
Ticket revenue	410,000

Instructions

Prepare the 2020 income statement for Sea Legs Cruise Company.

E1.16 (LO 5) Financial Statement Presented below is information related to the sole proprietorship of Helen Archer, attorney.

Prepare an owner's equity statement.

Legal service revenue—2020	\$330,000
Total expenses—2020	211,000
Assets, January 1, 2020	98,000
Liabilities, January 1, 2020	62,000
Assets, December 31, 2020	168,000
Liabilities, December 31, 2020	100,000
Drawings—2020	?

Instructions

Prepare the 2020 owner's equity statement for Helen Archer's legal practice.

E1.17 (LO 5) Financial Statement This information is for Paulo Company for the year ended December 31, 2020.

Prepare a cash flow statement.

Cash received from revenues from customers	\$600,000
Cash received from investment by owner	280,000
Cash paid for new equipment	115,000
Cash drawings by owner paid	18,000
Cash paid for expenses	430,000
Cash balance 1/1/20	30,000

Instructions

Prepare the 2020 statement of cash flows for Paulo Company.

E1.18 (LO 5) The statement of cash flows classifies each transaction as an operating activity, an investing activity, or a financing activity. Operating activities are the types of activities the company performs to generate profits. Investing activities include the purchase of long-lived assets such as equipment or the purchase of investment securities. Financing activities are borrowing money, investment by owner, and cash drawings by owner.

Identify cash flow activities.

Presented below are the following transactions.

- Owner invested \$20,000 cash.
- Issued note payable for \$12,000 cash.
- Purchased office equipment for \$11,000 cash.
- Received \$15,000 cash for services performed.
- Paid \$1,000 cash for rent.
- Paid \$600 cash drawings to owner.
- Paid \$5,700 cash for salaries.

Instructions

Classify each of these transactions as operating, investing, or financing activities.

Problems: Set A

Analyze transactions and compute net income.



Check figures provide a key number to let you know you are on the right track.

a. Total assets \$20,800

b. Net income \$6,200

Analyze transactions and prepare income statement, owner's equity statement, and balance sheet.



a. Total assets \$16,500

b. Net income \$3,130
Ending capital \$11,230

Prepare income statement, owner's equity statement, and balance sheet.

P1.1A (LO 3, 4) On April 1, Julie Spengel established Spengel's Travel Agency. The following transactions were completed during the month.

1. Invested \$15,000 cash to start the agency.
2. Paid \$600 cash for April office rent.
3. Purchased equipment for \$3,000 cash.
4. Incurred \$700 of advertising costs in the *Chicago Tribune*, on account.
5. Paid \$900 cash for office supplies.
6. Performed services worth \$10,000: \$3,000 cash is received from customers, and the balance of \$7,000 is billed to customers on account.
7. Withdrew \$600 cash for personal use.
8. Paid *Chicago Tribune* \$500 of the amount due in transaction (4).
9. Paid employees' salaries \$2,500.
10. Received \$4,000 in cash from customers who have previously been billed in transaction (6).

Instructions

- a. Prepare a tabular analysis of the transactions using the following column headings: Cash, Accounts Receivable, Supplies, Equipment, Accounts Payable, Owner's Capital, Owner's Drawings, Revenues, and Expenses.
- b. From an analysis of the owner's equity columns, compute the net income or net loss for April.

P1.2A (LO 4, 5) Financial Statement Sonya Jared opened a law office on July 1, 2020. On July 31, the balance sheet showed Cash \$5,000, Accounts Receivable \$1,500, Supplies \$500, Equipment \$6,000, Accounts Payable \$4,200, and Owner's Capital \$8,800. During August, the following transactions occurred.

1. Collected \$1,200 of accounts receivable.
2. Paid \$2,800 cash on accounts payable.
3. Recognized revenue of \$7,500 of which \$4,000 is collected in cash and the balance is due in September.
4. Purchased additional equipment for \$2,000, paying \$400 in cash and the balance on account.
5. Paid salaries \$2,800, rent for August \$900, and advertising expenses \$400.
6. Withdrew \$700 in cash for personal use.
7. Received \$2,000 from Standard Federal Bank—money borrowed on a note payable.
8. Incurred utility expenses for month on account \$270.

Instructions

- a. Prepare a tabular analysis of the August transactions beginning with July 31 balances. The column headings should be as follows: Cash + Accounts Receivable + Supplies + Equipment = Notes Payable + Accounts Payable + Owner's Capital – Owner's Drawings + Revenues – Expenses.
- b. Prepare an income statement for August, an owner's equity statement for August, and a balance sheet at August 31.

P1.3A (LO 5) Financial Statement On June 1, Cindy Godfrey started Divine Designs Co., a company that provides craft opportunities, by investing \$12,000 cash in the business. Following are the assets and liabilities of the company at June 30 and the revenues and expenses for the month of June.

Cash	\$10,150	Service Revenue	\$6,500
Accounts Receivable	2,800	Advertising Expense	500
Supplies	2,000	Rent Expense	1,600
Equipment	10,000	Gasoline Expense	200
Notes Payable	9,000	Utilities Expense	150
Accounts Payable	1,200		

Cindy made no additional investment in June but withdrew \$1,300 in cash for personal use during the month.

Instructions

- Prepare an income statement and owner's equity statement for the month of June and a balance sheet at June 30, 2020.
- Prepare an income statement and owner's equity statement for June assuming the following data are not included above: (1) \$900 of services were performed and billed but not collected at June 30, and (2) \$150 of gasoline expense was incurred but not paid.

- Net income \$4,050
Owner's equity \$14,750
Total assets \$24,950
- Owner's equity \$15,500

P1.4A (LO 4, 5) Financial Statement Maisie Taft started her own consulting firm, Maisie Consulting, on May 1, 2020. The following transactions occurred during the month of May.

Analyze transactions and prepare financial statements.

- May
- Maisie invested \$7,000 cash in the business.
 - Paid \$900 for office rent for the month.
 - Purchased \$800 of supplies on account.
 - Paid \$125 to advertise in the *County News*.
 - Received \$4,000 cash for services performed.
 - Withdrew \$1,000 cash for personal use.
 - Performed \$6,400 of services on account.
 - Paid \$2,500 for employee salaries.
 - Made a partial payment of \$600 for the supplies purchased on account on May 3.
 - Received a cash payment of \$4,000 for services performed on account on May 15.
 - Borrowed \$5,000 from the bank on a note payable.
 - Purchased equipment for \$4,200 on account.
 - Paid \$275 for utilities.

Instructions

- Show the effects of the previous transactions on the accounting equation using the following format.

- Total assets \$22,000

Assets					Liabilities		Owner's Equity			
Accounts					Notes	Accounts	Owner's	Owner's		
Date	Cash	+ Receivable	+ Supplies	+ Equipment	= Payable	+ Payable	+ Capital	- Drawings	+ Revenues	- Expenses

- Prepare an income statement for the month of May.
- Prepare a balance sheet at May 31, 2020.

- Net income \$6,600
- Cash \$14,600

P1.5A (LO 4, 5) Writing Financial Statement Financial statement information about four different companies is as follows.

Determine financial statement amounts and prepare owner's equity statement.

	Alpha Company	Beta Company	Psi Company	Omega Company
January 1, 2020				
Assets	\$ 80,000	\$ 90,000	(g)	\$150,000
Liabilities	41,000	(d)	80,000	(j)
Owner's equity	(a)	40,000	49,000	90,000
December 31, 2020				
Assets	(b)	112,000	170,000	(k)
Liabilities	60,000	72,000	(h)	100,000
Owner's equity	50,000	(e)	82,000	151,000
Owner's equity changes in year				
Additional investment	(c)	8,000	10,000	15,000
Drawings	15,000	(f)	12,000	10,000
Total revenues	350,000	410,000	(i)	500,000
Total expenses	333,000	385,000	350,000	(l)

Instructions

- Determine the missing amounts. (*Hint:* For example, to solve for (a), Assets – Liabilities = Owner's equity = \$39,000.)
- Prepare the owner's equity statement for Alpha Company.
- Write a memorandum explaining the sequence for preparing financial statements and the interrelationship of the owner's equity statement to the income statement and balance sheet.



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The Recording Process

Chapter Preview

In Chapter 1, we analyzed business transactions in terms of the accounting equation, and we presented the cumulative effects of these transactions in tabular form. Imagine a company like **MF Global** (as in the following Feature Story) using the same tabular format as Softbyte to keep track of its transactions. In a single day, MF Global engaged in thousands of business transactions. To record each transaction this way would be impractical, expensive, and unnecessary. Instead, companies use a set of procedures and records to keep track of transaction data more easily. This chapter introduces and illustrates these basic procedures and records.

Feature Story

Accidents Happen

How organized are you financially? Take a short quiz. Answer yes or no to each question:

- Does your wallet contain so many cash machine receipts that you've been declared a walking fire hazard?
- Do you wait until your debit card is denied before checking the status of your funds?

- Was Aaron Rodgers (the quarterback for the **Green Bay Packers**) playing high school football the last time you verified the accuracy of your bank account?

If you think it is hard to keep track of the many transactions that make up your life, imagine how difficult it is for a big corporation to do so. Not only that, but now consider how important it is for a large company to have good accounting records, especially if it has control of your life savings. **MF Global Holdings Ltd** was such a company. As a large investment broker, it held billions of dollars of investments for clients. If

you had your life savings invested at MF Global, you might be slightly displeased if you heard this from one of its representatives: “You know, I kind of remember an account for someone with a name like yours—now what did we do with that?”

Unfortunately, that is almost exactly what happened to MF Global’s clients shortly before it filed for bankruptcy. During the days immediately following the bankruptcy filing, regulators and auditors struggled to piece things together. In the words of one regulator, “Their books are a disaster . . . we’re trying to figure out what numbers are real numbers.” One company that considered buying an interest in MF Global walked away from

the deal because it “couldn’t get a sense of what was on the balance sheet.” That company said the information that should have been instantly available instead took days to produce.

It now appears that MF Global did not properly segregate customer accounts from company accounts. And, because of its sloppy recordkeeping, customers were not protected when the company had financial troubles. Total customer losses were approximately \$1 billion. As you can see, accounting matters!

Source: S. Patterson and A. Lucchetti, “Inside the Hunt for MF Global Cash,” *Wall Street Journal Online* (November 11, 2011).

Chapter Outline

LEARNING OBJECTIVES

LO 1 Describe how accounts, debits, and credits are used to record business transactions.	• The account • Debits and credits • Summary of debit/credit rules	DO IT! 1 Normal Account Balances
LO 2 Indicate how a journal is used in the recording process.	• The recording process • The journal	DO IT! 2 Recording Business Activities
LO 3 Explain how a ledger and posting help in the recording process.	• The ledger • Posting • Chart of accounts • The recording process illustrated • Summary illustration of journalizing and posting	DO IT! 3 Posting
LO 4 Prepare a trial balance.	• Limitations of a trial balance • Locating errors • Dollar signs and underlining	DO IT! 4 Trial Balance
Go to the Review and Practice section at the end of the chapter for a review of key concepts and practice applications with solutions. Visit WileyPLUS with ORION for additional tutorials and practice opportunities.		

Accounts, Debits, and Credits

LEARNING OBJECTIVE 1

Describe how accounts, debits, and credits are used to record business transactions.

The Account

An **account** is an individual accounting record of increases and decreases in a specific asset, liability, or owner’s equity item. For example, Softbyte (the company discussed in Chapter 1) would have separate accounts for Cash, Accounts Receivable, Accounts Payable, Service

INTERNATIONAL NOTE

Rules for accounting for specific events sometimes differ across countries. Despite the differences, the double-entry accounting system is the basis of accounting systems worldwide.

Debit and Credit Procedure

In Chapter 1, you learned the effect of a transaction on the basic accounting equation. Remember that each transaction must affect two or more accounts to keep the basic accounting equation in balance. In other words, for each transaction, debits must equal credits. The equality of debits and credits provides the basis for the **double-entry system** of recording transactions (see **International Note**).

Under the double-entry system, the dual (two-sided) effect of each transaction is recorded in appropriate accounts. This system provides a logical method for recording transactions and also helps ensure the accuracy of the recorded amounts as well as the detection of errors. If every transaction is recorded with equal debits and credits, the sum of all the debits to the accounts must equal the sum of all the credits.

The double-entry system for determining the equality of the accounting equation is much more efficient than the plus/minus procedure used in Chapter 1. The following discussion illustrates debit and credit procedures in the double-entry system.

Dr./Cr. Procedures for Assets and Liabilities

In Illustration 2.2 for Softbyte, increases in Cash—an asset—are entered on the left side, and decreases in Cash are entered on the right side. We know that both sides of the basic equation (Assets = Liabilities + Owner’s Equity) must be equal. It therefore follows that increases and decreases in liabilities have to be recorded **opposite from** increases and decreases in assets. Thus, increases in liabilities are entered on the right or credit side, and decreases in liabilities are entered on the left or debit side. The effects that debits and credits have on assets and liabilities are summarized in **Illustration 2.3**.

ILLUSTRATION 2.3

Debit and credit effects—assets and liabilities

Debits	Credits
Increase assets	Decrease assets
Decrease liabilities	Increase liabilities

Asset accounts normally show debit balances. That is, debits to a specific asset account should exceed credits to that account. Likewise, **liability accounts normally show credit balances.** That is, credits to a liability account should exceed debits to that account. The **normal balance** of an account is on the side where an increase in the account is recorded. **Illustration 2.4** shows the normal balances for assets and liabilities.

ILLUSTRATION 2.4

Normal balances—assets and liabilities

Assets		Liabilities	
Debit for increase	Credit for decrease	Debit for decrease	Credit for increase
Normal balance			Normal balance

Knowing the normal balance in an account may help you trace errors. For example, a credit balance in an asset account such as Land or a debit balance in a liability account such as Salaries and Wages Payable usually indicates an error. Occasionally, though, an abnormal balance may be correct. The Cash account, for example, will have a credit balance when a company has overdrawn its bank balance by spending more than it has in its account.

Dr./Cr. Procedures for Owner’s Equity

As Chapter 1 indicated, owner’s investments and revenues increase owner’s equity. Owner’s drawings and expenses decrease owner’s equity. Companies keep accounts for each of these types of transactions.

Owner's Capital. Investments by owners are credited to the Owner's Capital account. Credits increase this account, and debits decrease it. When an owner invests cash in the business, the company debits (increases) Cash and credits (increases) Owner's Capital. When the owner's investment in the business is reduced, Owner's Capital is debited (decreased).

Illustration 2.5 shows the rules of debit and credit for the Owner's Capital account.

Debits	Credits
Decrease Owner's Capital	Increase Owner's Capital

ILLUSTRATION 2.5

Debit and credit effects—
Owner's Capital

We can diagram the normal balance in Owner's Capital as shown in **Illustration 2.6**.

Owner's Capital	
Debit for decrease	Credit for increase
	Normal balance

ILLUSTRATION 2.6

Normal balance—Owner's
Capital

Owner's Drawings. An owner may withdraw cash or other assets for personal use. Withdrawals could be debited directly to Owner's Capital to indicate a decrease in owner's equity. However, it is preferable to use a separate account, called Owner's Drawings. This separate account makes it easier to determine total withdrawals for each accounting period. Owner's Drawings is increased by debits and decreased by credits. Normally, the drawings account will have a debit balance.

Illustration 2.7 shows the rules of debit and credit for the Owner's Drawings account.

Debits	Credits
Increase Owner's Drawings	Decrease Owner's Drawings

ILLUSTRATION 2.7

Debit and credit effects—
Owner's Drawings

We can diagram the normal balance as shown in **Illustration 2.8**.

Owner's Drawings	
Debit for increase	Credit for decrease
Normal balance	

ILLUSTRATION 2.8

Normal balance—Owner's
Drawings

The Owner's Drawings account decreases owner's equity. It is not an income statement account like revenues and expenses.

Revenues and Expenses. The purpose of earning revenues is to benefit the owner(s) of the business. When a company recognizes revenues, owner's equity increases. Therefore, **the effect of debits and credits on revenue accounts is the same as their effect on Owner's Capital.** That is, revenue accounts are increased by credits and decreased by debits (see **Helpful Hint**).

Expenses have the opposite effect. Expenses decrease owner's equity. Since expenses decrease net income and revenues increase it, it is logical that the increase and decrease sides of expense accounts should be the opposite of revenue accounts. Thus, expense accounts are increased by debits and decreased by credits. **Illustration 2.9** shows the rules of debits and credits for revenues and expenses.

HELPFUL HINT

Because revenues increase owner's equity, a revenue account has the same debit/credit rules as the Owner's Capital account. Expenses have the opposite effect.

DO IT! 1 | Normal Account Balances

Kate Browne has just rented space in a shopping mall. In this space, she will open a hair salon to be called “Hair It Is.” A friend has advised Kate to set up a double-entry set of accounting records in which to record all of her business transactions.

Identify the balance sheet accounts that Kate will likely need to record the transactions needed to open her business. Indicate whether the normal balance of each account is a debit or a credit.

Solution

Kate would likely need the following accounts in which to record the transactions necessary to ready her hair salon for opening day:

Cash (debit balance)	If she borrows money: Notes Payable (credit balance)
Equipment (debit balance)	Owner's Capital (credit balance)
Supplies (debit balance)	
Accounts Payable (credit balance)	

Related exercise material: **BE2.1, BE2.2, DO IT! 2.1, E2.1, E2.2, and E2.4.**

ACTION PLAN

- Determine the types of accounts needed. Kate will need asset accounts for each different type of asset she invests in the business and liability accounts for any debts she incurs.
- Understand the types of owner's equity accounts. Only Owner's Capital will be needed when Kate begins the business. Other owner's equity accounts will be needed later.

The Journal

LEARNING OBJECTIVE 2

Indicate how a journal is used in the recording process.

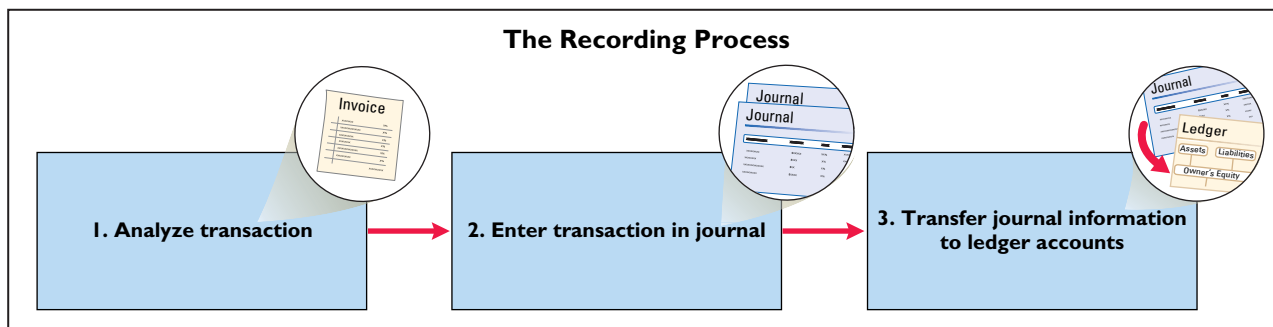


The Recording Process

Although it is possible to enter transaction information directly into the accounts, few businesses do so. Practically every business uses the basic steps shown in **Illustration 2.12** in the recording process (an integral part of the accounting cycle):

1. Analyze each transaction in terms of its effect on the accounts.
2. Enter the transaction information in a journal.
3. Transfer the journal information to the appropriate accounts in the ledger.

ILLUSTRATION 2.12 The recording process



The steps in the recording process occur repeatedly. In Chapter 1, we illustrated the first step, the analysis of transactions, and will give further examples in this and later chapters. The other two steps in the recording process are explained in the next sections.

The Journal

Companies initially record transactions in chronological order (the order in which they occur). Thus, the **journal** is referred to as the book of original entry. For each transaction, the journal shows the debit and credit effects on specific accounts.

Companies may use various kinds of journals, but every company has the most basic form of journal, a **general journal**. Typically, a general journal has spaces for dates, account titles and explanations, references, and two amount columns. See the format of the journal in Illustration 2.13. *Whenever we use the term “journal” in this textbook, we mean the general journal unless we specify otherwise.*

The journal makes several significant contributions to the recording process:

1. It discloses in one place the **complete effects of a transaction**.
2. It provides a **chronological record** of transactions.
3. It helps to **prevent or locate errors** because the debit and credit amounts for each entry can be easily compared.

Journalizing

Entering transaction data in the journal is known as **journalizing**. Companies make separate journal entries for each transaction. A complete entry consists of (1) the date of the transaction, (2) the accounts and amounts to be debited and credited, and (3) a brief explanation of the transaction.

Illustration 2.13 shows the technique of journalizing, using the first two transactions of Softbyte. Recall that on September 1, Ray Neal invested \$15,000 cash in the business, and Softbyte purchased computer equipment for \$7,000 cash. The number J1 indicates that these two entries are recorded on the first page of the journal. **Illustration 2.13** shows the standard form of journal entries for these two transactions. (The boxed numbers correspond to explanations in the list below the illustration.)

ILLUSTRATION 2.13

Technique of journalizing

GENERAL JOURNAL				J1
Date	Account Titles and Explanation	Ref.	Debit	Credit
2020				
Sept. 1	2 Cash	5	15,000	
1	3 Owner's Capital			15,000
	4 (Owner's investment of cash in business)			
1	Equipment		7,000	
	Cash			7,000
	(Purchase of equipment for cash)			

- 1** The date of the transaction is entered in the Date column.
- 2** The debit account title (that is, the account to be debited) is entered first at the extreme left margin of the column headed “Account Titles and Explanation,” and the amount of the debit is recorded in the Debit column.
- 3** The credit account title (that is, the account to be credited) is indented and entered on the next line in the column headed “Account Titles and Explanation,” and the amount of the credit is recorded in the Credit column.
- 4** A brief explanation of the transaction appears on the line below the credit account title. A space is left between journal entries. The blank space separates individual journal entries and makes the entire journal easier to read.
- 5** The column titled Ref. (which stands for Reference) is left blank when the journal entry is made. This column is used later when the journal entries are transferred to the individual accounts.

It is important to use correct and specific account titles in journalizing. Erroneous account titles lead to incorrect financial statements. However, some flexibility exists initially in selecting account titles. The main criterion is that each title must appropriately describe the content of the account. Once a company chooses the specific title to use, it should record under that account title all later transactions involving the account. *In homework problems, you should use specific account titles when they are given.* When account titles are not given, you may select account titles that identify the nature and content of each account. The account titles used in journalizing should not contain explanations such as Cash Paid or Cash Received.

Simple and Compound Entries

Some entries involve only two accounts, one debit and one credit. (See, for example, the entries in Illustration 2.13.) This type of entry is called a **simple entry**. Some transactions, however, require more than two accounts in journalizing. An entry that requires three or more accounts is a **compound entry**. To illustrate, assume that on July 1, Butler Company purchases a delivery truck costing \$14,000. It pays \$8,000 cash now and agrees to pay the remaining \$6,000 on account (to be paid later). **Illustration 2.14** shows the compound entry.

GENERAL JOURNAL				J1
Date	Account Titles and Explanation	Ref.	Debit	Credit
2020 July 1	Equipment Cash Accounts Payable (Purchased truck for cash with balance on account)		14,000	8,000 6,000

ILLUSTRATION 2.14

Compound journal entry

In a compound entry, the standard format requires that all debits be listed before the credits.

Accounting Across the Organization Hain Celestial Group



Keith Homan/
Shutterstock

It Starts with the Transaction

Recording financial transactions in a company's records should be straightforward. If a company determines that a transaction involves revenue, it records revenue. If it has an expense, then it records an expense. However, sometimes this is difficult to do. For example, for more than a year, **Hain Celestial Group** (an organic food company) did not provide income information to investors and regulators. The reason was that the company discovered revenue irregularities and said it could not release financial results until it determined when and how to record revenue for certain transactions.

When Hain missed four deadlines for reporting earnings information, the food company suffered a 34% drop in its stock price. As one analyst noted, it was hard to fathom why a seemingly simple revenue recognition issue took one year to resolve.

In other situations, outright fraud may occur. For example, regulators charged **Obsidian Energy** for fraudulently moving

millions of dollars in expenses from operating expenses to capital expenditure accounts. By understating reported operating expenses, Obsidian made it appear that it was efficiently managing its costs as well as increasing its income.

These examples demonstrate that “getting the basic transaction right” is the foundation for relevant and reliable financial statements. Starting with an incorrect or inappropriate transaction leads to distortions in the financial statements.

Sources: Shawn Tully, “The Mystery of Hain Celestial’s Accounting,” *Fortune.com* (August 20, 2016); and Kelly Cryderman, “U.S. Charges Obsidian, Formerly Penn West, with Accounting Fraud,” *The Globe and Mail* (June 28, 2017).

Why is it important for companies to record financial transactions completely and accurately? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 2 | Recording Business Activities

Kate Browne engaged in the following activities in establishing her salon, Hair It Is:

- 1. Opened a bank account in the name of Hair It Is and deposited \$20,000 of her own money in this account as her initial investment.
- 2. Purchased equipment on account (to be paid in 30 days) for a total cost of \$4,800.
- 3. Interviewed three people for the position of hair stylist.

Prepare the journal entries to record the transactions.

Solution

The three activities would be recorded as follows.

- | | | |
|--|--------|--------|
| 1. Cash | 20,000 | |
| Owner's Capital | | 20,000 |
| (Owner's investment of cash in business) | | |
| 2. Equipment | 4,800 | |
| Accounts Payable | | 4,800 |
| (Purchase of equipment on account) | | |
| 3. No entry because no transaction has occurred. | | |

Related exercise material: BE2.3, BE2.4, BE2.5, BE2.6, DO IT! 2.2, E2.3, E2.5, E2.6, E2.7, E2.8, and E2.9.

ACTION PLAN

- Understand which activities need to be recorded and which do not. Any that affect assets, liabilities, or owner's equity should be recorded in a journal.
- Analyze the effects of transactions on asset, liability, and owner's equity accounts.

The Ledger and Posting

LEARNING OBJECTIVE 3

Explain how a ledger and posting help in the recording process.



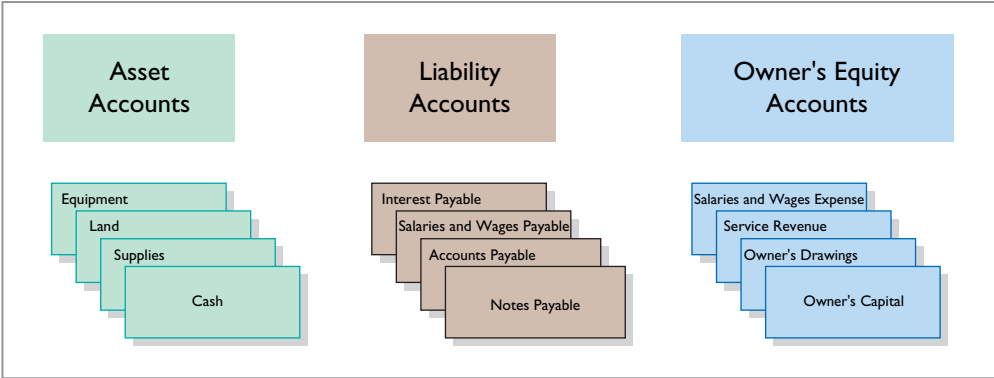
The Ledger

The entire group of accounts maintained by a company is the **ledger**. The ledger provides the balance in each of the accounts as well as keeps track of changes in these balances.

Companies may use various kinds of ledgers, but every company has a general ledger. A **general ledger** contains all the asset, liability, and owner's equity accounts, as shown in **Illustration 2.15**. Whenever we use the term "ledger" in this textbook, we are referring to the general ledger unless we specify otherwise.

ILLUSTRATION 2.15

The general ledger, which contains all of a company's accounts



Companies arrange the ledger in the sequence in which they present the accounts in the financial statements, beginning with the balance sheet accounts. First in order are the asset accounts, followed by liability accounts, owner's capital, owner's drawings, revenues, and expenses. Each account is numbered for easier identification.

The ledger provides the balance in each of the accounts. For example, the Cash account shows the amount of cash available to meet current obligations. The Accounts Receivable account shows amounts due from customers. Accounts Payable shows amounts owed to creditors.

Ethics Insight Credit Suisse Group



© Nuno Silva/iStockphoto

A Convenient Overstatement

Sometimes a company's investment securities suffer a permanent decline in value below their original cost. When this occurs, the company is supposed to reduce the recorded value of the securities on its balance sheet ("write them down" in common financial lingo) and record a loss. It appears, however, that during the financial crisis of 2008, employees at some financial institutions chose to look the other way as the value of their investments skidded.

A number of Wall Street traders that worked for the investment bank **Credit Suisse Group** were charged with intentionally overstating the value of securities that had suffered declines of approximately \$2.85 billion. One reason that they may have been reluctant to record the losses is out of fear that the company's shareholders and clients would panic if they saw the magnitude of the losses. However, personal self-interest might have been equally to blame—the bonuses of the traders were tied to the value of the investment securities.

Source: S. Pulliam, J. Eaglesham, and M. Siconolfi, "U.S. Plans Changes on Bond Fraud," *Wall Street Journal Online* (February 1, 2012).

What incentives might employees have had to overstate the value of these investment securities on the company's financial statements? (Go to WileyPLUS for this answer and additional questions.)

Standard Form of Account

The simple T-account form used in accounting textbooks is often very useful for illustration purposes. However, in practice, the account forms used in ledgers are much more structured. **Illustration 2.16** shows a typical form, using assumed data from a cash account.

CASH					NO. 101
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
June 1			25,000		25,000
2				8,000	17,000
3			4,200		21,200
9			7,500		28,700
17				11,700	17,000
20				250	16,750
30				7,300	9,450

ILLUSTRATION 2.16

Three-column form of account

This format is called the **three-column form of account**. It has three money columns—debit, credit, and balance. The balance in the account is determined after each transaction. Companies use the explanation space and reference columns to provide special information about the transaction.

Posting

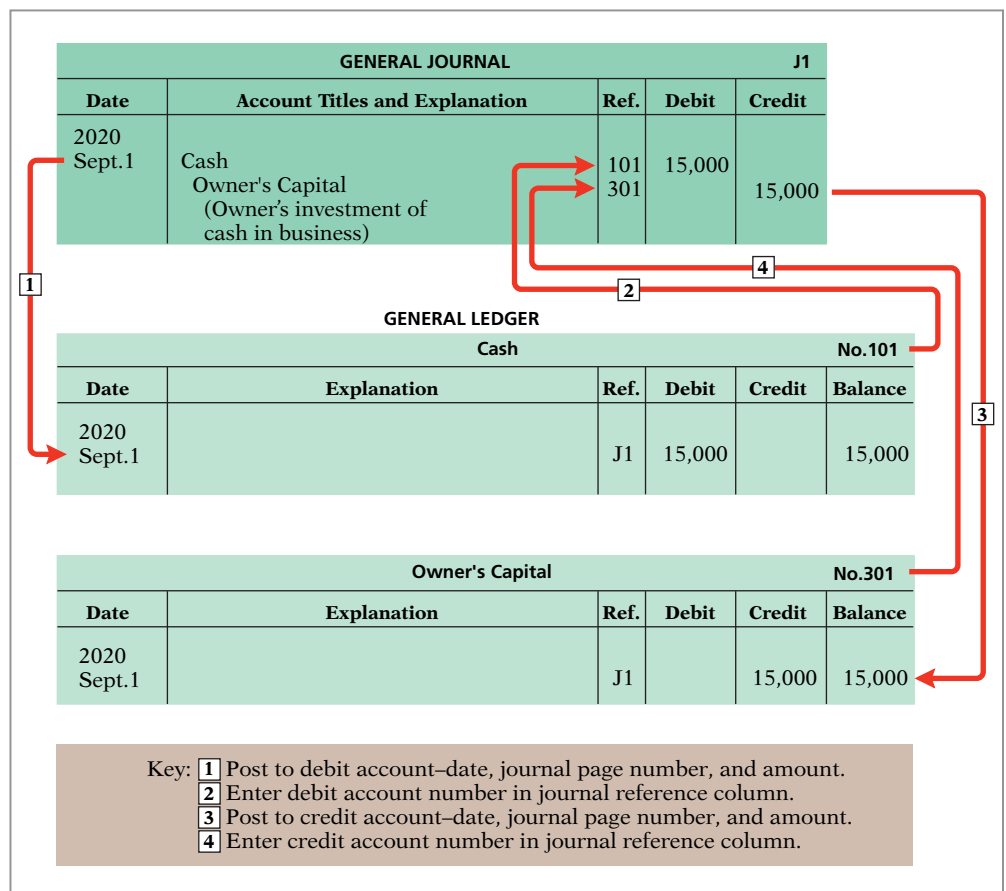
The procedure of transferring journal entries to the ledger accounts is called **posting**. This phase of the recording process accumulates the effects of journalized transactions into the individual accounts. Posting involves the following steps.

1. In the **ledger**, in the appropriate columns of the account(s) debited, enter the date, journal page, and debit amount shown in the journal.
2. In the reference column of the **journal**, write the account number to which the debit amount was posted.
3. In the **ledger**, in the appropriate columns of the account(s) credited, enter the date, journal page, and credit amount shown in the journal.
4. In the reference column of the **journal**, write the account number to which the credit amount was posted.

Illustration 2.17 shows these four steps using Softbyte's first journal entry. The boxed numbers indicate the sequence of the steps.

ILLUSTRATION 2.17

Posting a journal entry



Posting should be performed in chronological order. That is, the company should post all the debits and credits of one journal entry before proceeding to the next journal entry. Postings should be made on a timely basis to ensure that the ledger is up-to-date. *In homework problems, you can journalize all transactions before posting any of the journal entries.*

The reference column of a ledger account indicates the journal page from which the transaction was posted. (After the last entry has been posted, the accountant should scan the reference column **in the journal**, to confirm that all postings have been made.) The explanation space of the ledger account is used infrequently because an explanation already appears in the journal.

Chart of Accounts

The number and type of accounts differ for each company. The number of accounts depends on the amount of detail management desires. For example, the management of one company may want a single account for all types of utility expense. Another may keep separate expense accounts for each type of utility, such as gas, electricity, and water. Similarly, a small company like Softbyte will have fewer accounts than a corporate giant like **Dell**. Softbyte may be able to manage and report its activities in 20 to 30 accounts, while Dell may require thousands of accounts to keep track of its worldwide activities.

Most companies have a **chart of accounts**. This chart lists the accounts and the account numbers that identify their location in the ledger. The numbering system that identifies the accounts usually starts with the balance sheet accounts and follows with the income statement accounts.

In this and the next two chapters, we explain the accounting for Pioneer Advertising (a service company). Accounts 101–199 indicate asset accounts; 200–299 indicate liabilities; 301–350 indicate owner’s equity accounts; 400–499, revenues; 601–799, expenses; 800–899, other revenues; and 900–999, other expenses. **Illustration 2.18** shows Pioneer’s chart of accounts. Accounts listed in red are used in this chapter; accounts shown in black are explained in later chapters.

Pioneer Advertising Chart of Accounts	
Assets	Owner’s Equity
101 Cash	301 Owner’s Capital
112 Accounts Receivable	306 Owner’s Drawings
126 Supplies	350 Income Summary
130 Prepaid Insurance	
157 Equipment	Revenues
158 Accumulated Depreciation—Equipment	400 Service Revenue
Liabilities	Expenses
200 Notes Payable	631 Supplies Expense
201 Accounts Payable	711 Depreciation Expense
209 Unearned Service Revenue	722 Insurance Expense
212 Salaries and Wages Payable	726 Salaries and Wages Expense
230 Interest Payable	729 Rent Expense
	732 Utilities Expense
	905 Interest Expense

ILLUSTRATION 2.18

Chart of accounts

You will notice that there are gaps in the numbering system of the chart of accounts for Pioneer. Companies leave gaps to permit the insertion of new accounts as needed during the life of the business.

The Recording Process Illustrated

Illustrations 2.19 through **2.28** show the basic steps in the recording process, using the October transactions of Pioneer Advertising (see **Helpful Hint**). Pioneer’s accounting period is a month. In these illustrations, a basic analysis, an equation analysis, and a debit-credit analysis precede the journal entry and posting of each transaction. For simplicity, we use the T-account form to show the posting instead of the standard account form.

Study these transaction analyses carefully. **The purpose of transaction analysis is first to identify the type of account involved, and then to determine whether to make a debit**

HELPFUL HINT

The Accounting Cycle Tutorial in WileyPLUS provides an interactive presentation of the accounting cycle using these transaction analyses.

or a credit to the account. You should always perform this type of analysis before preparing a journal entry. Doing so will help you understand the journal entries discussed in this chapter as well as more complex journal entries in later chapters (see **Helpful Hint**).

ILLUSTRATION 2.19**Investment of cash by owner**

Cash flow analyses show the impact of each transaction on cash.

Cash Flows

+10,000

**HELPFUL HINT**

Follow these steps:

1. Determine what type of account is involved.
2. Determine what items increased or decreased and by how much.
3. Translate the increases and decreases into debits and credits.

Transaction

On October 1, C. R. Byrd invests \$10,000 cash in an advertising company called Pioneer Advertising.

Basic Analysis

The asset Cash increases \$10,000; owner's equity (specifically, Owner's Capital) increases \$10,000.

Equation Analysis

<u>Assets</u>	=	<u>Liabilities</u>	+	<u>Owner's Equity</u>
Cash	=			Owner's Capital
+\$10,000				+\$10,000

Debit–Credit Analysis

Debits increase assets: debit Cash \$10,000.
Credits increase owner's equity: credit Owner's Capital \$10,000.

Journal Entry

Oct. 1	Cash	101	10,000	
	Owner's Capital (Owner's investment of cash in business)	301		10,000

Posting

<u>Cash</u> 101		<u>Owner's Capital</u> 301	
Oct. 1	10,000		Oct. 1 10,000

ILLUSTRATION 2.20**Purchase of office equipment****Cash Flows**

no effect

Transaction

On October 1, Pioneer purchases office equipment costing \$5,000 by signing a 3-month, 12%, \$5,000 note payable.

Basic Analysis

The asset Equipment increases \$5,000; the liability Notes Payable increases \$5,000.

Equation Analysis

<u>Assets</u>	=	<u>Liabilities</u>	+	<u>Owner's Equity</u>
Equipment	=	Notes Payable		
+\$5,000		+\$5,000		

Debit–Credit Analysis

Debits increase assets: debit Equipment \$5,000.
Credits increase liabilities: credit Notes Payable \$5,000.

Journal Entry

Oct. 1	Equipment	157	5,000	
	Notes Payable (Issued 3-month, 12% note for office equipment)	200		5,000

Posting

<u>Equipment</u> 157		<u>Notes Payable</u> 200	
Oct. 1	5,000		Oct. 1 5,000

ILLUSTRATION 2.21**Receipt of cash for future service**

Many liabilities have the word “payable” in their title. But, note that Unearned Service Revenue is considered a liability even though the word *payable* is not used.

Cash Flows
+1,200

**Transaction**

On October 2, Pioneer receives a \$1,200 cash advance from R. Knox, a client, for advertising services that are expected to be completed by December 31.

Basic Analysis

The asset Cash increases \$1,200; the liability Unearned Service Revenue increases \$1,200 because the service has not been performed yet. That is, when Pioneer receives an advance payment, it should record an unearned revenue (a liability) in order to recognize the obligation that exists.

Equation Analysis

Assets	=	Liabilities	+	Owner's Equity
Cash	=	Unearned Service Revenue		
+\$1,200		+\$1,200		

Debit–Credit Analysis

Debits increase assets: debit Cash \$1,200.
Credits increase liabilities: credit Unearned Service Revenue \$1,200.

Journal Entry

Oct. 2	Cash	101	1,200	
	Unearned Service Revenue (Received cash from R. Knox for future services)	209		1,200

Posting

Cash		101	Unearned Service Revenue		209
Oct. 1	10,000			Oct. 2	1,200
2	1,200				

Transaction

On October 3, Pioneer pays office rent for October in cash, \$900.

Basic Analysis

The expense account Rent Expense increases \$900 because the payment pertains only to the current month; the asset Cash decreases \$900.

Equation Analysis

Assets	=	Liabilities	+	Owner's Equity
Cash	=			Rent Expense
-\$900				-\$900

Debit–Credit Analysis

Debits increase expenses: debit Rent Expense \$900.
Credits decrease assets: credit Cash \$900.

Journal Entry

Oct. 3	Rent Expense	729	900	
	Cash (Paid cash for October office rent)	101		900

Posting

Cash		101	Rent Expense		729
Oct. 1	10,000		Oct. 3	900	
2	1,200				

ILLUSTRATION 2.22**Payment of monthly rent**

Cash Flows
-900



ILLUSTRATION 2.23**Payment for insurance****Cash Flows**

-600

**Transaction**

On October 4, Pioneer pays \$600 for a one-year insurance policy that will expire next year on September 30.

Basic Analysis

The asset Prepaid Insurance increases \$600 because the payment extends to more than the current month; the asset Cash decreases \$600. Payments of expenses that will benefit more than one accounting period are prepaid expenses or prepayments. When a company makes a payment, it debits an asset account in order to show the service or benefit that will be received in the future.

Equation Analysis

Assets		=	Liabilities	+	Owner's Equity
Cash	+ Prepaid Insurance				
-\$600	+\$600				

Debit–Credit Analysis

Debits increase assets: debit Prepaid Insurance \$600.
Credits decrease assets: credit Cash \$600.

Journal Entry

Oct. 4	Prepaid Insurance	130	600	
	Cash	101		600
	(Paid one-year policy; effective date October 1)			

Posting

Cash		101	Prepaid Insurance		130
Oct. 1	10,000		Oct. 4	600	
2	1,200				
		900			
		4			
		600			

ILLUSTRATION 2.24**Purchase of supplies on credit****Cash Flows**

no effect

Transaction

On October 5, Pioneer purchases an estimated 3-month supply of advertising materials on account from Aero Supply for \$2,500.

Basic Analysis

The asset Supplies increases \$2,500; the liability Accounts Payable increases \$2,500.

Equation Analysis

Assets		=	Liabilities	+	Owner's Equity
Supplies	=	Accounts Payable			
+\$2,500		+\$2,500			

Debit–Credit Analysis

Debits increase assets: debit Supplies \$2,500.
Credits increase liabilities: credit Accounts Payable \$2,500.

Journal Entry

Oct. 5	Supplies	126	2,500	
	Accounts Payable	201		2,500
	(Purchased supplies on account from Aero Supply)			

Posting

Supplies		126	Accounts Payable		201
Oct. 5	2,500		Oct. 5	2,500	

Event

On October 9, Pioneer hires four employees to begin work on October 15. Each employee is to receive a weekly salary of \$500 for a 5-day work week, payable every 2 weeks—first payment made on October 26.

Basic Analysis

A business transaction has not occurred. There is only an agreement between the employer and the employees to enter into a business transaction beginning on October 15. Thus, a debit–credit analysis is not needed because there is no accounting entry (see October 26 transaction for first entry).

ILLUSTRATION 2.25**Hiring of employees****Cash Flows****no effect****Transaction**

On October 20, C. R. Byrd withdraws \$500 cash for personal use.

Basic Analysis

The owner's equity account Owner's Drawings increases \$500; the asset Cash decreases \$500.

Equation Analysis

<u>Assets</u>	=	<u>Liabilities</u>	+	<u>Owner's Equity</u>
Cash	=			Owner's Drawings
-\$500				-\$500

Debit–Credit Analysis

Debits increase drawings: debit Owner's Drawings \$500.
Credits decrease assets: credit Cash \$500.

Journal Entry

Oct. 20	Owner's Drawings	306	500	
	Cash	101		500
	(Withdraw cash for personal use)			

Posting

Cash 101		Owner's Drawings 306	
Oct. 1	10,000	Oct. 3	900
2	1,200	4	600
		20	500

ILLUSTRATION 2.26**Withdrawal of cash by owner**
Cash Flows
 -500


ILLUSTRATION 2.27**Payment of salaries****Cash Flows**

-4,000

**Transaction**

On October 26, Pioneer owes employee salaries of \$4,000 and pays them in cash (see October 9 event).

Basic Analysis

The expense account Salaries and Wages Expense increases \$4,000; the asset Cash decreases \$4,000.

Equation Analysis

Assets	=	Liabilities	+	Owner's Equity
Cash	=			Salaries and Wages Expense
-\$4,000				-\$4,000

Debit–Credit Analysis

Debits increase expenses: debit Salaries and Wages Expense \$4,000.
Credits decrease assets: credit Cash \$4,000.

Journal Entry

Oct. 26	Salaries and Wages Expense Cash (Paid salaries to date)	726 101	4,000	4,000
---------	---	------------	-------	-------

Posting

Cash				Salaries and Wages Expense	
			101		726
Oct. 1	10,000	Oct. 3	900	Oct. 26	4,000
2	1,200	4	600		
		20	500		
		26	4,000		

ILLUSTRATION 2.28**Receipt of cash for services performed****Cash Flows**

+10,000

**Transaction**

On October 31, Pioneer receives \$10,000 in cash from Copa Company for advertising services performed in October.

Basic Analysis

The asset Cash increases \$10,000; the revenue account Service Revenue increases \$10,000.

Equation Analysis

Assets	=	Liabilities	+	Owner's Equity
Cash	=			Service Revenue
+\$10,000				+\$10,000

Debit–Credit Analysis

Debits increase assets: debit Cash \$10,000.
Credits increase revenues: credit Service Revenue \$10,000.

Journal Entry

Oct. 31	Cash Service Revenue (Received cash for services performed)	101 400	10,000	10,000
---------	---	------------	--------	--------

Posting

Cash				Service Revenue	
			101		400
Oct. 1	10,000	Oct. 3	900		
2	1,200	4	600		
31	10,000	20	500	Oct. 31	10,000
		26	4,000		

Summary Illustration of Journalizing and Posting

Illustration 2.29 shows the journal for Pioneer Advertising for October.

GENERAL JOURNAL				PAGE J1
Date	Account Titles and Explanation	Ref.	Debit	Credit
2020				
Oct. 1	Cash	101	10,000	
	Owner's Capital	301		10,000
	(Owner's investment of cash in business)			
1	Equipment	157	5,000	
	Notes Payable	200		5,000
	(Issued 3-month, 12% note for office equipment)			
2	Cash	101	1,200	
	Unearned Service Revenue	209		1,200
	(Received cash from R. Knox for future services)			
3	Rent Expense	729	900	
	Cash	101		900
	(Paid cash for October office rent)			
4	Prepaid Insurance	130	600	
	Cash	101		600
	(Paid one-year policy; effective date October 1)			
5	Supplies	126	2,500	
	Accounts Payable	201		2,500
	(Purchased supplies on account from Aero Supply)			
20	Owner's Drawings	306	500	
	Cash	101		500
	(Withdrew cash for personal use)			
26	Salaries and Wages Expense	726	4,000	
	Cash	101		4,000
	(Paid salaries to date)			
31	Cash	101	10,000	
	Service Revenue	400		10,000
	(Received cash for services performed)			

ILLUSTRATION 2.29

General journal entries

Illustration 2.30 shows the ledger, with all balances in red.

ILLUSTRATION 2.30 General ledger

GENERAL LEDGER											
Cash No. 101						Accounts Payable No. 201					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 1		J1	10,000		10,000	Oct. 5		J1		2,500	2,500
2		J1	1,200		11,200						
3		J1		900	10,300						
4		J1		600	9,700						
20		J1		500	9,200						
26		J1		4,000	5,200						
31		J1	10,000		15,200						
Supplies No. 126						Unearned Service Revenue No. 209					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 5		J1	2,500		2,500	Oct. 2		J1		1,200	1,200
Prepaid Insurance No. 130						Owner's Capital No. 301					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 4		J1	600		600	Oct. 1		J1		10,000	10,000
Equipment No. 157						Owner's Drawings No. 306					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 1		J1	5,000		5,000	Oct. 20		J1	500		500
Notes Payable No. 200						Service Revenue No. 400					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 1		J1		5,000	5,000	Oct. 31		J1		10,000	10,000
						Salaries and Wages Expense No. 726					
						Date	Explanation	Ref.	Debit	Credit	Balance
						2020					
						Oct. 26		J1	4,000		4,000
						Rent Expense No. 729					
						Date	Explanation	Ref.	Debit	Credit	Balance
						2020					
						Oct. 3		J1	900		900

DO IT! 3 | Posting

Kate Browne recorded the following transactions in a general journal during the month of March.

Mar. 4	Cash	2,280	
	Service Revenue		2,280
15	Salaries and Wages Expense	400	
	Cash		400
19	Utilities Expense	92	
	Cash		92

Post these entries to the Cash account of the general ledger to determine its ending balance. The beginning balance of Cash on March 1 was \$600.

Solution

Cash			
3/1 Bal.	600	3/15	400
3/4	2,280	3/19	92
3/31 Bal.	2,388		

Related exercise material: **BE2.7, BE2.8, DO IT! 2.3, E2.11, and E2.14.**

ACTION PLAN

- Recall that posting involves transferring the journalized debits and credits to specific accounts in the ledger.
- Determine the ending balance by netting the total debits and credits.

The Trial Balance

LEARNING OBJECTIVE 4

Prepare a trial balance.



A **trial balance** is a list of accounts and their balances at a given time. Companies usually prepare a trial balance at the end of an accounting period. They list accounts in the order in which they appear in the ledger. Debit balances appear in the left column and credit balances in the right column. The totals of the two columns must equal.

The trial balance proves the mathematical equality of debits and credits after posting. Under the double-entry system, this equality occurs when the sum of the debit account balances equals the sum of the credit account balances. **A trial balance may also uncover errors in journalizing and posting.** For example, a trial balance may well have detected the error at **MF Global** discussed in the Feature Story. **In addition, a trial balance is useful in the preparation of financial statements,** as we will explain in the next two chapters.

The steps for preparing a trial balance are:

1. List the account titles and their balances in the appropriate debit or credit column.
2. Total the debit and credit columns.
3. Verify the equality of the two columns.

Illustration 2.31 shows the trial balance prepared from Pioneer Advertising's ledger. Note that the total debits equal the total credits.

Pioneer Advertising Trial Balance October 31, 2020

	Debit	Credit
Cash	\$15,200	
Supplies	2,500	
Prepaid Insurance	600	
Equipment	5,000	
Notes Payable		\$ 5,000
Accounts Payable		2,500
Unearned Service Revenue		1,200
Owner's Capital		10,000
Owner's Drawings	500	
Service Revenue		10,000
Salaries and Wages Expense	4,000	
Rent Expense	900	
	<u>\$28,700</u>	<u>\$28,700</u>

ILLUSTRATION 2.31

A Trial balance

Note that the order of presentation in the trial balance is:

Assets
Liabilities
Owner's equity
Revenues
Expenses

A trial balance is a necessary checkpoint for uncovering certain types of errors. For example, if only the debit portion of a journal entry has been posted, the trial balance would bring this error to light.

Limitations of a Trial Balance

ETHICS NOTE

An **error** is the result of an unintentional mistake; it is neither ethical nor unethical. An **irregularity** is an intentional misstatement, which is viewed as unethical.

A trial balance does not guarantee freedom from recording errors, however (see **Ethics Note**). Numerous errors may exist even though the totals of the trial balance columns agree. For example, the trial balance may balance even when:

1. A transaction is not journalized.
2. A correct journal entry is not posted.
3. A journal entry is posted twice.
4. Incorrect accounts are used in journalizing or posting.
5. Offsetting errors are made in recording the amount of a transaction.

As long as equal debits and credits are posted, even to the wrong account or in the wrong amount, the total debits will equal the total credits. **The trial balance does not prove that the company has recorded all transactions or that the ledger is correct.**

Locating Errors

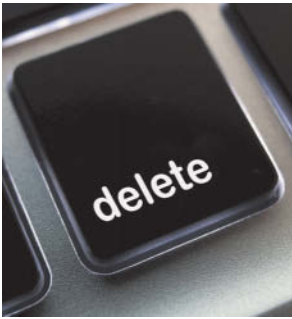
Errors in a trial balance generally result from mathematical mistakes, incorrect postings, or simply transcribing data incorrectly. What do you do if you are faced with a trial balance that does not balance? First, determine the amount of the difference between the two columns of the trial balance. After this amount is known, the following steps are often helpful:

1. If the error is \$1, \$10, \$100, or \$1,000, re-add the trial balance columns and recompute the account balances.
2. If the error is divisible by 2, scan the trial balance to see whether a balance equal to half the error has been entered in the wrong column.
3. If the error is divisible by 9, retrace the account balances on the trial balance to see whether they are incorrectly copied from the ledger. For example, if a balance was \$12 and it was listed as \$21, a \$9 error has been made. Reversing the order of numbers is called a **transposition error**.
4. If the error is not divisible by 2 or 9, scan the ledger to see whether an account balance in the amount of the error has been omitted from the trial balance, and scan the journal to see whether a posting of that amount has been omitted.

Dollar Signs and Underlining

Note that dollar signs do not appear in journals or ledgers. Dollar signs are typically used only in the trial balance and the financial statements. Generally, a dollar sign is shown only for the first item in the column and for the total of that column. A single line (a totaling rule) is placed under the column of figures to be added or subtracted. Total amounts are double-underlined to indicate they are final sums.

Investor Insight Fannie Mae



Enviromatic/iStockphoto

Why Accuracy Matters

While most companies record transactions very carefully, the reality is that mistakes still happen. For example, bank regulators fined **Bank One Corporation** (now **Chase**) \$1.8 million because they felt that the unreliability of the bank's accounting system caused it to violate regulatory requirements.

Also, in recent years **Fannie Mae**, the government-chartered mortgage association, announced a series of large accounting errors. These announcements caused alarm among investors, regulators, and politicians because they fear that the errors may suggest larger, undetected problems. This is important because

the home-mortgage market depends on Fannie Mae to buy hundreds of billions of dollars of mortgages each year from banks, thus enabling the banks to issue new mortgages.

Finally, before a major overhaul of its accounting system, the financial records of **Waste Management Inc.** were in such disarray that of the company's 57,000 employees, 10,000 were receiving pay slips that were in error.

The Sarbanes-Oxley Act was created to minimize the occurrence of errors like these by increasing every employee's responsibility for accurate financial reporting.

In order for these companies to prepare and issue financial statements, their accounting equations (debits and credits) must have been in balance at year-end. How could these errors or misstatements have occurred? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 4 | Trial Balance

The following accounts come from the ledger of SnowGo Company at December 31, 2020.

157	Equipment	\$88,000	301	Owner's Capital	\$20,000
306	Owner's Drawings	8,000	212	Salaries and Wages	
201	Accounts Payable	22,000		Payable	2,000
726	Salaries and Wages		200	Notes Payable (due in 3 months)	19,000
	Expense	42,000	732	Utilities Expense	3,000
112	Accounts Receivable	4,000	130	Prepaid Insurance	6,000
400	Service Revenue	95,000	101	Cash	7,000

Prepare a trial balance in good form.

Solution

SnowGo Company
Trial Balance
December 31, 2020

	Debit	Credit
Cash	\$ 7,000	
Accounts Receivable	4,000	
Prepaid Insurance	6,000	
Equipment	88,000	
Notes Payable		\$ 19,000
Accounts Payable		22,000
Salaries and Wages Payable		2,000
Owner's Capital		20,000
Owner's Drawings	8,000	
Service Revenue		95,000
Utilities Expense	3,000	
Salaries and Wages Expense	42,000	
	\$158,000	\$158,000

ACTION PLAN

- Determine normal balances and list accounts in the order they appear in the ledger.
- Accounts with debit balances appear in the left column, and those with credit balances in the right column.
- Total the debit and credit columns to prove equality.

Review and Practice

Learning Objectives Review

1 Describe how accounts, debits, and credits are used to record business transactions.

An account is a record of increases and decreases in specific asset, liability, and owner's equity items. The terms debit and credit are synonymous with left and right. Assets, drawings, and expenses are increased by debits and decreased by credits. Liabilities, owner's capital, and revenues are increased by credits and decreased by debits.

2 Indicate how a journal is used in the recording process.

The basic steps in the recording process are (a) analyze each transaction for its effects on the accounts, (b) enter the transaction information in a journal, and (c) transfer the journal information to the appropriate accounts in the ledger.

The initial accounting record of a transaction is entered in a journal before the data are entered in the accounts. A journal (a) discloses in one place the complete effects of a transaction, (b) provides a chronological record of transactions, and (c) prevents

or locates errors because the debit and credit amounts for each entry can be easily compared.

3 Explain how a ledger and posting help in the recording process.

The ledger is the entire group of accounts maintained by a company. The ledger provides the balance in each of the accounts as well as keeps track of changes in these balances. Posting is the transfer of journal entries to the ledger accounts. This phase of the recording process accumulates the effects of journalized transactions in the individual accounts.

4 Prepare a trial balance.

A trial balance is a list of accounts and their balances at a given time. Its primary purpose is to prove the equality of debits and credits after posting. A trial balance also uncovers errors in journalizing and posting and is useful in preparing financial statements.

Glossary Review

Account A record of increases and decreases in specific asset, liability, or owner's equity items. (p. 2-2).

Chart of accounts A list of accounts and the account numbers that identify their location in the ledger. (p. 2-13).

Compound entry A journal entry that involves three or more accounts. (p. 2-9).

Credit The right side of an account. (p. 2-3).

Debit The left side of an account. (p. 2-3).

Double-entry system A system that records in appropriate accounts the dual effect of each transaction. (p. 2-4).

General journal The most basic form of journal. (p. 2-8).

General ledger A ledger that contains all asset, liability, and owner's equity accounts. (p. 2-10).

Journal An accounting record in which transactions are initially recorded in chronological order. (p. 2-8).

Journalizing The entering of transaction data in the journal. (p. 2-8).

Ledger The entire group of accounts maintained by a company. (p. 2-10).

Normal balance An account balance on the side where an increase in the account is recorded. (p. 2-4).

Posting The procedure of transferring journal entries to the ledger accounts. (p. 2-12).

Simple entry A journal entry that involves only two accounts. (p. 2-9).

T-account The basic form of an account, consisting of (1) a title, (2) a left or debit side, and (3) a right or credit side. (p. 2-3).

Three-column form of account A form with columns for debit, credit, and balance amounts in an account. (p. 2-11).

Trial balance A list of accounts and their balances at a given time. (p. 2-21).

Practice Multiple-Choice Questions

1. (LO 1) Which of the following statements about an account is true?

- a. The right side of an account is the debit, or increase, side.
- b. An account is an individual accounting record of increases and decreases in specific asset, liability, and owner's equity items.

- c. There are separate accounts for specific assets and liabilities but only one account for owner's equity items.
- d. The left side of an account is the credit, or decrease, side.

2. (LO 1) Debits:
 - a. increase both assets and liabilities.
 - b. decrease both assets and liabilities.
 - c. increase assets and decrease liabilities.
 - d. decrease assets and increase liabilities.
3. (LO 1) A revenue account:
 - a. is increased by debits.
 - b. is decreased by credits.
 - c. has a normal balance of a debit.
 - d. is increased by credits.
4. (LO 1) Accounts that normally have debit balances are:
 - a. assets, expenses, and revenues.
 - b. assets, expenses, and owner's capital.
 - c. assets, liabilities, and owner's drawings.
 - d. assets, owner's drawings, and expenses.
5. (LO 1) The expanded accounting equation is:
 - a. $\text{Assets} + \text{Liabilities} = \text{Owner's Capital} + \text{Owner's Drawings} + \text{Revenues} + \text{Expenses}$.
 - b. $\text{Assets} = \text{Liabilities} + \text{Owner's Capital} + \text{Owner's Drawings} + \text{Revenues} - \text{Expenses}$.
 - c. $\text{Assets} = \text{Liabilities} - \text{Owner's Capital} - \text{Owner's Drawings} - \text{Revenues} - \text{Expenses}$.
 - d. $\text{Assets} = \text{Liabilities} + \text{Owner's Capital} - \text{Owner's Drawings} + \text{Revenues} - \text{Expenses}$.
6. (LO 2) Which of the following is **not** part of the recording process?
 - a. Analyzing transactions.
 - b. Preparing an income statement.
 - c. Entering transactions in a journal.
 - d. Posting journal entries.
7. (LO 2) Which of the following statements about a journal is **false**?
 - a. It is not a book of original entry.
 - b. It provides a chronological record of transactions.
 - c. It helps to locate errors because the debit and credit amounts for each entry can be readily compared.
 - d. It discloses in one place the complete effect of a transaction.
8. (LO 2) The purchase of supplies on account should result in:
 - a. a debit to Supplies Expense and a credit to Cash.
 - b. a debit to Supplies Expense and a credit to Accounts Payable.
 - c. a debit to Supplies and a credit to Accounts Payable.
 - d. a debit to Supplies and a credit to Accounts Receivable.
9. (LO 3) The order of the accounts in the ledger is:
 - a. assets, revenues, expenses, liabilities, owner's capital, owner's drawings,
 - b. assets, liabilities, owner's capital, owner's drawings, revenues, expenses.
 - c. owner's capital, assets, revenues, expenses, liabilities, owner's drawings.
 - d. revenues, assets, expenses, liabilities, owner's capital, owner's drawings.
10. (LO 3) A ledger:
 - a. contains only asset and liability accounts.
 - b. should show accounts in alphabetical order.
 - c. is a collection of the entire group of accounts maintained by a company.
 - d. is a book of original entry.
11. (LO 3) Posting:
 - a. normally occurs before journalizing.
 - b. transfers ledger transaction data to the journal.
 - c. is an optional step in the recording process.
 - d. transfers journal entries to ledger accounts.
12. (LO 3) Before posting a payment of \$5,000, the Accounts Payable of Senator Company had a normal balance of \$16,000. The balance after posting this transaction was:
 - a. \$21,000.
 - b. \$5,000.
 - c. \$11,000.
 - d. Cannot be determined.
13. (LO 4) A trial balance:
 - a. is a list of accounts with their balances at a given time.
 - b. proves the journalized transactions are correct.
 - c. will not balance if a correct journal entry is posted twice.
 - d. proves that all transactions have been recorded.
14. (LO 4) A trial balance will not balance if:
 - a. a correct journal entry is posted twice.
 - b. the purchase of supplies on account is debited to Supplies and credited to Cash.
 - c. a \$100 cash drawing by the owner is debited to Owner's Drawings for \$1,000 and credited to Cash for \$100.
 - d. a \$450 payment on account is debited to Accounts Payable for \$45 and credited to Cash for \$45.
15. (LO 4) The trial balance of Jeong Company had accounts with the following normal balances: Cash \$5,000, Service Revenue \$85,000, Salaries and Wages Payable \$4,000, Salaries and Wages Expense \$40,000, Rent Expense \$10,000, Owner's Capital \$42,000, Owner's Drawings \$15,000, and Equipment \$61,000. In preparing a trial balance, the total in the debit column is:
 - a. \$131,000.
 - b. \$216,000.
 - c. \$91,000.
 - d. \$116,000.

Solutions

1. b. An account is an individual accounting record of increases and decreases in specific asset, liability, and owner's equity items. The other choices are incorrect because (a) the right side of the account is the credit side, not the debit side, and can be the increase or

the decrease side, depending on the specific classification account; (c) there are also separate accounts for different owner's equity items; and (d) the left side of the account is the debit side, not the credit side, and can be either the decrease or the increase side, depending on the specific classification account.

2. c. Debits increase assets but they decrease liabilities. The other choices are incorrect because debits (a) decrease, not increase, liabilities; (b) increase, not decrease, assets; and (d) increase, not decrease, assets and decrease, not increase, liabilities.

3. d. A revenue account is increased by credits. The other choices are incorrect because a revenue account (a) is increased by credits, not debits; (b) is decreased by debits, not credits; and (c) has a normal balance of a credit, not a debit.

4. d. Assets, owner's drawings, and expenses all have normal debit balances. The other choices are incorrect because (a) revenues have normal credit balances, (b) owner's capital has a normal credit balance, and (c) liabilities have normal credit balances.

5. d. The expanded accounting equation is $\text{Assets} = \text{Liabilities} + \text{Owner's Capital} - \text{Owner's Drawings} + \text{Revenue} - \text{Expenses}$. The other choices are incorrect because (a) both Owner's Drawings and Expenses must be subtracted, not added, and Liabilities should be added to the right side of equation, not to the left side; (b) Owner's Drawings must be subtracted, not added; and (c) Owner's Capital and Revenues must be added, not subtracted.

6. b. Preparing an income statement is not part of the recording process. Choices (a) analyzing transactions, (c) entering transactions in a journal, and (d) posting journal entries are all part of the recording process.

7. a. The journal is a book of original entry. The other choices are all true statements.

8. c. The purchase of supplies on account results in a debit to Supplies and a credit to Accounts Payable. The other choices are incorrect because the purchase of supplies on account results in (a) a debit to Supplies, not Supplies Expense, and a credit to Accounts Payable, not Cash; (b) a debit to Supplies, not Supplies Expense; and (d) a credit to Accounts Payable, not Accounts Receivable.

9. b. The correct order of the accounts in the ledger is assets, liabilities, owner's capital, owner's drawing, revenues, expenses. The other choices are incorrect because they do not reflect this order. The order of the accounts in the ledger is (1) balance sheet accounts: assets, liabilities, and owner's equity accounts (owner's capital and owner's drawings); and then (2) income statement accounts: revenues and expenses.

10. c. A ledger is a collection of all the accounts maintained by a company. The other choices are incorrect because a ledger (a) contains all account types—assets, liabilities, and owner's equity—not just assets and liability accounts; (b) usually shows accounts in account number order, not alphabetical order; and (d) is not a book of original entry because entries made in the ledger come from the journals (the books of original entry).

11. d. Posting transfers journal entries to ledger accounts. The other choices are incorrect because posting (a) occurs after journalizing, (b) transfers journal transaction data to the ledger; and (c) is not an optional step in the recording process.

12. c. The balance is \$11,000 (\$16,000 normal balance – \$5,000 payment), not (a) \$21,000 or (b) \$5,000. Choice (d) is incorrect because the balance can be determined.

13. a. A trial balance is a list of accounts with their balances at a given time. The other choices are incorrect because (b) the trial balance does not prove that journalized transactions are mathematically correct; (c) if a journal entry is posted twice, the trial balance will still balance; and (d) the trial balance does not prove that all transactions have been recorded.

14. c. The trial balance will not balance in this case because the debit of \$1,000 to Owner's Drawings is not equal to the credit of \$100 to Cash. The other choices are incorrect because (a) if a correct journal entry is posted twice, the trial balance will still balance; (b) if the purchase of supplies on account is debited to Supplies and credited to Cash, Cash and Accounts Payable will be understated but the trial balance will still balance; and (d) since the debit and credit amounts are the same, the trial balance will still balance but both Accounts Payable and Cash will be overstated.

15. a. The total debit column = \$5,000 (Cash) + \$40,000 (Salaries and Wages Expense) + \$10,000 (Rent Expense) + \$15,000 (Owner's Drawings) + \$61,000 (Equipment) = \$131,000. The normal balance for Assets, Expenses, and Owner's Drawings is a debit. The other choices are incorrect because (b) revenue of \$85,000 should not be included in the total of \$216,000 and its normal balance is a credit; (c) the total of \$91,000 is missing the Salaries and Wages Expense of \$40,000, which has a normal balance of a debit; and (d) the total of \$116,000 is missing the Owner's Drawings of \$15,000, which has a normal balance of a debit.

Practice Brief Exercises

Identify accounts to be debited and credited.

1. (LO 1) Transactions for the Warren Potter Company for the month of May are presented below. Identify the accounts to be debited and credited for each transaction.

- May 1 Warren Potter invested \$22,000 in the business.
 6 Paid office rent of \$900.
 12 Performed consulting services and billed client \$4,400.
 18 Purchased equipment on account for \$1,200.

Solution

1.	Account Debited	Account Credited
May 1	Cash	Owner's Capital
6	Rent Expense	Cash
12	Accounts Receivable	Service Revenue
18	Equipment	Accounts Payable

Journalize transactions.

2. (LO 2) Using the data from **Practice Brief Exercise 1**, journalize the transactions (omit explanations).

Solution

2. May	1	Cash	22,000	
		Owner's Capital		22,000
	6	Rent Expense	900	
		Cash		900
	12	Accounts Receivable	4,400	
		Service Revenue		4,400
	18	Equipment	1,200	
		Accounts Payable		1,200

3. (LO 3) Selected transactions for Carlos Santana Company are presented in journal form below. Post the transactions to T-accounts. Make one T-account for each and determine each account's ending balance.

Post journal entries to T-accounts.

J1				
Date	Account Titles and Explanation	Ref.	Debit	Credit
June 6	Cash		22,000	
	Owner's Capital (Owner's investment of cash in business)			22,000
13	Accounts Receivable		8,200	
	Service Revenue (Billed for services performed)			8,200
14	Cash		3,700	
	Accounts Receivable (Received cash in payment of account)			3,700

Solution

Cash			Accounts Receivable		
6/6	22,000		6/13	8,200	6/14 3,700
6/14	3,700			Bal. 4,500	
	Bal. 25,700				
Service Revenue			Owner's Capital		
	6/13	8,200		6/6	22,000
		Bal. 8,200			Bal. 22,000

4. (LO 3) Selected journal entries for Carlos Santana Company are presented in **Practice Brief Exercise 3**. Post the transactions using the standard form of account.

Post journal entries to standard form of account.

Solution**4. Cash**

Date	Explanation	Ref.	Debit	Credit	Balance
June 6		J1	22,000		22,000
14		J1	3,700		25,700

Accounts Receivable

Date	Explanation	Ref.	Debit	Credit	Balance
June 13		J1	8,200		8,200
14		J1		3,700	4,500

Service Revenue

Date	Explanation	Ref.	Debit	Credit	Balance
June 13		J1		8,200	8,200

Owner's Capital

Date	Explanation	Ref.	Debit	Credit	Balance
June 6		J1		22,000	22,000

Prepare a trial balance.

5. (LO 4) From the ledger accounts below, prepare a trial balance for Bundy Company at December 31, 2020. List the accounts in the order shown in the textbook. All account balances are normal.

Accounts Receivable	\$10,000	Salaries and Wages Expense	\$ 2,300
Supplies	4,100	Rent Expense	1,200
Accounts Payable	3,500	Owner's Capital	10,200
Owner's Drawing	1,100	Cash	6,000
Service Revenue	11,000		

Solution

5.

Bundy Company Trial Balance December 31, 2020		
	Debit	Credit
Cash	\$ 6,000	
Accounts Receivable	10,000	
Supplies	4,100	
Accounts Payable		\$ 3,500
Owner's Capital		10,200
Owner's Drawings	1,100	
Service Revenue		11,000
Salaries and Wages Expense	2,300	
Rent Expense	1,200	
	<u>\$24,700</u>	<u>\$24,700</u>

Practice Exercises

Analyze and journalize transactions.

1. (LO 2) Presented below is information related to Hammond Real Estate Agency.

- Oct. 1 Lia Berge begins business as a real estate agent with a cash investment of \$30,000.
 2 Paid rent, \$700, on office space.
 3 Purchases office equipment for \$2,800, on account.
 6 Sells a house and lot for Hal Smith; bills Hal Smith \$4,400 for realty services performed.
 27 Pays \$1,100 on the balance related to the transaction of October 3.
 30 Receives bill for October utilities, \$130 (not paid at this time).

Instructions

Journalize the transactions. (You may omit explanations)

Solution

1. GENERAL JOURNAL				
Date	Account Titles and Explanation	Ref.	Debit	Credit
Oct. 1	Cash Owner's Capital		30,000	30,000
2	Rent Expense Cash		700	700
3	Equipment Accounts Payable		2,800	2,800
6	Accounts Receivable Service Revenue		4,400	4,400
27	Accounts Payable Cash		1,100	1,100
30	Utilities Expense Accounts Payable		130	130

2. (LO 2, 4) The T-accounts below summarize the ledger of Depot Company at the end of the first month of operations.

Journalize transactions from account data and prepare a trial balance.

Cash No. 101	
4/1 16,000	4/15 700
4/12 1,200	4/25 1,600
4/29 900	
4/30 1,600	

Accounts Receivable No. 112	
4/7 2,900	4/29 900

Supplies No. 126	
4/4 1,900	

Accounts Payable No. 201	
4/25 1,600	4/4 1,900

Unearned Service Revenue No. 209	
	4/30 1,600

Owner's Capital No. 301	
	4/1 16,000

Service Revenue No. 400	
	4/7 2,900
	4/12 1,200

Salaries and Wages Expense No. 726	
4/15 700	

Instructions

- Prepare the complete general journal (including explanations) from which the postings to Cash were made.
- Prepare a trial balance at April 30, 2020.

Solution

2. a. GENERAL JOURNAL

Date	Account Titles and Explanation	Ref.	Debit	Credit
Apr. 1	Cash Owner's Capital (Owner's investment of cash in business)		16,000	16,000
4	Supplies Accounts Payable (Purchased supplies on account)		1,900	1,900
7	Accounts Receivable Service Revenue (Billed customers for services performed)		2,900	2,900
12	Cash Service Revenue (Received cash for services performed)		1,200	1,200
15	Salaries and Wages Expense Cash (Paid salaries to date)		700	700
25	Accounts Payable Cash (Paid creditors on account)		1,600	1,600
29	Cash Accounts Receivable (Received cash in payment of account)		900	900
30	Cash Unearned Service Revenue (Received cash for future services)		1,600	1,600

b.

Depot Company Trial Balance April 30, 2020

	Debit	Credit
Cash	\$17,400	
Accounts Receivable	2,000	
Supplies	1,900	

Accounts Payable		\$ 300
Unearned Service Revenue		1,600
Owner's Capital		16,000
Service Revenue		4,100
Salaries and Wages Expense	700	
	<u>\$22,000</u>	<u>\$22,000</u>

Practice Problem

Journalize transactions, post, and prepare a trial balance.

(LO 1, 2, 3, 4) Bob Sample opened the Campus Laundromat on September 1, 2020. During the first month of operations, the following transactions occurred.

- Sept. 1 Bob invested \$20,000 cash in the business.
 2 The company paid \$1,000 cash for store rent for September.
 3 Purchased washers and dryers for \$25,000, paying \$10,000 in cash and signing a \$15,000, 6-month, 12% note payable.
 4 Paid \$1,200 for a one-year accident insurance policy.
 10 Received a bill from the *Daily News* for online advertising of the opening of the laundromat \$200.
 20 Bob withdrew \$700 cash for personal use.
 30 The company determined that cash receipts for laundry services for the month were \$6,200.

The chart of accounts for the company is the same as that for Pioneer Advertising plus No. 610 Advertising Expense.

Instructions

- Journalize the September transactions. (Use J1 for the journal page number.)
- Open ledger accounts and post the September transactions.
- Prepare a trial balance at September 30, 2020.

Solution

a. GENERAL JOURNAL J1				
Date	Account Titles and Explanation	Ref.	Debit	Credit
2020				
Sept. 1	Cash	101	20,000	
	Owner's Capital	301		20,000
	(Owner's investment of cash in business)			
2	Rent Expense	729	1,000	
	Cash	101		1,000
	(Paid September rent)			
3	Equipment	157	25,000	
	Cash	101		10,000
	Notes Payable	200		15,000
	(Purchased laundry equipment for cash and 6-month, 12% note payable)			
4	Prepaid Insurance	130	1,200	
	Cash	101		1,200
	(Paid one-year insurance policy)			
10	Advertising Expense	610	200	
	Accounts Payable	201		200
	(Received bill from <i>Daily News</i> for advertising)			
20	Owner's Drawings	306	700	
	Cash	101		700
	(Withdrew cash for personal use)			
30	Cash	101	6,200	
	Service Revenue	400		6,200
	(Received cash for services performed)			

b.

GENERAL LEDGER

Cash No. 101						Accounts Payable No. 201					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Sept. 1		J1	20,000		20,000	Sept. 10		J1		200	200
2		J1		1,000	19,000	Owner's Capital No. 301					
3		J1		10,000	9,000	Date	Explanation	Ref.	Debit	Credit	Balance
4		J1		1,200	7,800	2020					
20		J1		700	7,100	Sept. 1		J1		20,000	20,000
30		J1	6,200		13,300	Owner's Drawings No. 306					
Prepaid Insurance No. 130						Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Sept. 4		J1	1,200		1,200	Sept. 20		J1	700		700
Equipment No. 157						Service Revenue No. 400					
2020						Date	Explanation	Ref.	Debit	Credit	Balance
Sept. 3		J1	25,000		25,000	2020					
						Sept. 30		J1		6,200	6,200
Notes Payable No. 200						Advertising Expense No. 610					
2020						Date	Explanation	Ref.	Debit	Credit	Balance
Sept. 3		J1		15,000	15,000	2020					
						Sept. 10		J1	200		200
						Rent Expense No. 729					
						Date	Explanation	Ref.	Debit	Credit	Balance
						2020					
						Sept. 2		J1	1,000		1,000

c.

Campus Laundromat Trial Balance September 30, 2020		
	Debit	Credit
Cash	\$13,300	
Prepaid Insurance	1,200	
Equipment	25,000	
Notes Payable		\$15,000
Accounts Payable		200
Owner's Capital		20,000
Owner's Drawings	700	
Service Revenue		6,200
Advertising Expense	200	
Rent Expense	1,000	
	<u>\$41,400</u>	<u>\$41,400</u>

WileyPLUS

Brief Exercises, DO IT! Exercises, Exercises, Problems, and many additional resources are available for practice in WileyPLUS.

Questions

1. Describe the parts of a T-account.
2. “The terms debit and credit mean increase and decrease, respectively.” Do you agree? Explain.
3. Pete Harcourt, a fellow student, contends that the double-entry system means each transaction must be recorded twice. Is Pete correct? Explain.
4. Melissa Estes, a beginning accounting student, believes debit balances are favorable and credit balances are unfavorable. Is Melissa correct? Discuss.
5. State the rules of debit and credit as applied to (a) asset accounts, (b) liability accounts, and (c) the owner’s equity accounts (revenue, expenses, owner’s drawings, and owner’s capital).
6. What is the normal balance for each of the following accounts? (a) Accounts Receivable. (b) Cash. (c) Owner’s Drawings. (d) Accounts Payable. (e) Service Revenue. (f) Salaries and Wages Expense. (g) Owner’s Capital.
7. Indicate whether each of the following accounts is an asset, a liability, or an owner’s equity account and whether it has a normal debit or credit balance: (a) Accounts Receivable, (b) Accounts Payable, (c) Equipment, (d) Owner’s Drawings, and (e) Supplies.
8. For the following transactions, indicate the account debited and the account credited.
 - a. Supplies are purchased on account.
 - b. Cash is received on signing a note payable.
 - c. Employees are paid salaries in cash.
9. Indicate whether the following accounts generally will have (a) debit entries only, (b) credit entries only, or (c) both debit and credit entries.

1. Cash.	5. Salaries and Wages Expense.
2. Accounts Receivable.	6. Service Revenue.
3. Owner’s Drawings.	
4. Accounts Payable.	
10. What are the basic steps in the recording process?
11. What are the advantages of using a journal in the recording process?
12. a. When entering a transaction in the journal, should the debit or credit be written first?
b. Which should be indented, the debit or credit?
13. Describe a compound entry, and provide an example.
14. a. Should business transaction debits and credits be recorded directly in the ledger accounts?
b. What are the advantages of first recording transactions in the journal and then posting to the ledger?
15. The account number is entered as the last step in posting the amounts from the journal to the ledger. What is the advantage of this step?
16. Journalize the following business transactions.
 - a. Wes Lee invests \$7,000 cash in the business.
 - b. Insurance of \$800 is paid for the year.
 - c. Supplies of \$2,000 are purchased on account.
 - d. Cash of \$8,500 is received for services performed.
17. a. What is a ledger?
b. What is a chart of accounts and why is it important?
18. What is a trial balance and what are its purposes?
19. Victor Grimm is confused about how accounting information flows through the accounting system. He believes the flow of information is as follows.
 - a. Debits and credits posted to the ledger.
 - b. Business transaction occurs.
 - c. Information entered in the journal.
 - d. Financial statements are prepared.
 - e. Trial balance is prepared.

Is Victor correct? If not, indicate to Victor the proper flow of the information.
20. Two students are discussing the use of a trial balance. They wonder whether the following errors, each considered separately, would prevent the trial balance from balancing.
 - a. The bookkeeper debited Cash for \$600 and credited Salaries and Wages Expense for \$600 for payment of wages.
 - b. Cash collected on account was debited to Cash for \$800 and Service Revenue was credited for \$80.

What would you tell them?
21. What are the normal balances for **Apple’s** Cash, Accounts Payable, and Interest Expense accounts?

Brief Exercises

Indicate debit and credit effects and normal balance.

BE2.1 (LO 1) For each of the following accounts, indicate the effects of (a) a debit and (b) a credit on the accounts and (c) the normal balance of the account.

- | | |
|-------------------------|-------------------------|
| 1. Accounts Payable. | 4. Accounts Receivable. |
| 2. Advertising Expense. | 5. Owner’s Capital. |
| 3. Service Revenue. | 6. Owner’s Drawings. |

BE2.2 (LO 1) Transactions for the Oleg Thorn Company for the month of June are presented below. Identify the accounts to be debited and credited for each transaction.

- June 1 Oleg Thorn invests \$5,000 cash in a small welding business of which he is the sole proprietor.
 2 Purchases equipment on account for \$3,600.
 3 \$800 cash is paid to landlord for June rent.
 12 Sends a bill to K. Johnsen for \$400 after completing welding work done on account.

Identify accounts to be debited and credited.

BE2.3 (LO 2) Using the data in BE2.2, journalize the transactions. (You may omit explanations.)

Journalize transactions.

BE2.4 (LO 2) Writing Shea Jonas, a fellow student, is unclear about the basic steps in the recording process. Identify and briefly explain the steps in the order in which they occur.

Identify and explain steps in recording process.

BE2.5 (LO 2) M. Gonzales has the following transactions during August of the current year. Indicate (a) the effect on the accounting equation and (b) the debit-credit analysis illustrated in the textbook.

Indicate basic and debit-credit analysis.

- Aug. 1 Opens an office as a financial advisor, investing \$9,000 in cash.
 4 Pays insurance in advance for 6 months, \$2,100 cash.
 16 Receives \$3,600 from clients for services performed.
 27 Pays secretary \$1,000 salary.

BE2.6 (LO 2) Using the data in BE2.5, journalize the transactions. (You may omit explanations.)

Journalize transactions.

BE2.7 (LO 3) Selected transactions for the Walt Bryce Company are presented in journal form below. Post the transactions to T-accounts. Make one T-account for each item and determine each account's ending balance.

Post journal entries to T-accounts.

J1				
Date	Account Titles and Explanation	Ref.	Debit	Credit
May 5	Accounts Receivable Service Revenue (Billed for services performed)		5,400	5,400
12	Cash Accounts Receivable (Received cash in payment of account)		4,200	4,200
15	Cash Service Revenue (Received cash for services performed)		3,000	3,000

BE2.8 (LO 3) Selected journal entries for the Walt Bryce Company are presented in BE2.7. Post the transactions using the standard form of account.

Post journal entries to standard form of account.

BE2.9 (LO 4) From the ledger balances given below, prepare a trial balance for the Amaro Company at June 30, 2020. List the accounts in the order shown in the textbook. All account balances are normal.

Prepare a trial balance.

Accounts Payable \$8,100, Cash \$5,800, Owner's Capital \$15,000, Owner's Drawings \$1,200, Equipment \$17,000, Service Revenue \$10,000, Accounts Receivable \$3,000, Salaries and Wages Expense \$5,100, and Rent Expense \$1,000.

BE2.10 (LO 4) An inexperienced bookkeeper prepared the following trial balance. Prepare a correct trial balance, assuming all account balances are normal.

Prepare a correct trial balance.

**Shaushank Company
 Trial Balance
 December 31, 2020**

	<u>Debit</u>	<u>Credit</u>
Cash	\$10,600	
Prepaid Insurance		\$ 3,500
Accounts Payable		3,000
Unearned Service Revenue	2,200	
Owner's Capital		9,000
Owner's Drawings		4,500
Service Revenue		25,400
Salaries and Wages Expense	18,600	
Rent Expense		2,400
	<u>\$31,400</u>	<u>\$47,800</u>

DO IT! Exercises

Identify normal balances.

DO IT! 2.1 (LO 1) Tom Rast has just rented space in a strip mall. In this space, he will open a photography studio, to be called “Picture This!” A friend has advised Tom to set up a double-entry set of accounting records in which to record all of his business transactions.

Identify the balance sheet accounts that Tom will likely need to record the transactions needed to open his business. Indicate whether the normal balance of each account is a debit or credit.

Record business activities.

DO IT! 2.2 (LO 2) Ron Tost engaged in the following activities in establishing his photography studio, Shutter Bug:

1. Opened a bank account in the name of Shutter Bug and deposited \$6,500 of his own money into this account as his initial investment.
2. Purchased photography supplies at a total cost of \$1,200. The business paid \$400 in cash and the balance is on account.
3. Obtained estimates on the cost of photography equipment from three different manufacturers.

Prepare the journal entries to record the transactions. (You may omit explanations.)

Post transactions.

DO IT! 2.3 (LO 3) Tom Rast recorded the following transactions during the month of April.

April 3	Cash	3,400	
	Service Revenue		3,400
April 16	Rent Expense	700	
	Cash		700
April 20	Salaries and Wages Expense	250	
	Cash		250

Post these entries to the Cash T-account of the general ledger to determine the ending balance in cash. The beginning balance in cash on April 1 was \$1,600.

Prepare a trial balance.

DO IT! 2.4 (LO 4) The following accounts are taken from the ledger of Macon Company at December 31, 2020.

200	Notes Payable	\$20,000	101	Cash	\$ 6,000
301	Owner's Capital	28,000	126	Supplies	7,000
157	Equipment	80,000	729	Rent Expense	4,000
306	Owner's Drawings	9,000	212	Salaries and Wages Payable	3,000
726	Salaries and Wages Expense	38,000	201	Accounts Payable	11,000
400	Service Revenue	90,000	112	Accounts Receivable	8,000

Prepare a trial balance in good form.

Exercises

Analyze statements about accounting and the recording process.

E2.1 (LO 1) Kim Yi has prepared the following list of statements about accounts.

1. An account is an accounting record of either a specific asset or a specific liability.
2. An account shows only increases, not decreases, in the item it relates to.
3. Some items, such as Cash and Accounts Receivable, are combined into one account.
4. An account has a left, or credit side, and a right, or debit side.
5. A simple form of an account consisting of just the account title, the left side, and the right side, is called a T-account.

Instructions

Identify each statement as true or false. If false, indicate how to correct the statement.

E2.2 (LO 1) Selected transactions for M. Acosta, an interior decorator, in her first month of business, are as follows.

Identify debits, credits, and normal balances.

- Jan. 2 Invested \$10,000 cash in business.
 3 Purchased used car for \$3,000 cash for use in business.
 9 Purchased supplies on account for \$600.
 11 Billed customers \$2,400 for services performed.
 16 Paid \$350 cash for advertising.
 20 Received \$900 cash from customers billed on January 11.
 23 Paid creditor \$300 cash on balance owed.
 28 Withdrew \$1,000 cash for personal use by owner.

Instructions

For each transaction, indicate the following.

- The basic type of account debited and credited (asset, liability, owner's equity).
- The specific account debited and credited (Cash, Rent Expense, Service Revenue, etc.).
- Whether the specific account is increased or decreased.
- The normal balance of the specific account.

Use the following format, in which the January 2 transaction is given as an example.

Account Debited					Account Credited			
Date	(a) Basic Type	(b) Specific Account	(c) Effect	(d) Normal Balance	(a) Basic Type	(b) Specific Account	(c) Effect	(d) Normal Balance
Jan. 2	Asset	Cash	Increase	Debit	Owner's Equity	Owner's Capital	Increase	Credit

E2.3 (LO 2) Data for M. Acosta, interior decorator, are presented in E2.2.

Journalize transactions.

Instructions

Journalize the transactions using journal page J1. (You may omit explanations.)

E2.4 (LO 1) The following information relates to Royale Real Estate Agency.

Analyze transactions and determine their effect on accounts.

- Oct. 1 James Royale begins business as a real estate agent with a cash investment of \$17,000.
 2 Hires an administrative assistant.
 3 Purchases office furniture for \$1,900, on account.
 6 Sells a house and lot for C. Rouse; bills C. Rouse \$3,800 for realty services performed.
 27 Pays \$1,300 on the balance related to the transaction of October 3.
 30 Pays the administrative assistant \$2,500 in salary for October.

Instructions

Prepare the debit-credit analysis for each transaction as illustrated in the textbook.

E2.5 (LO 2) Transaction data for Royale Real Estate Agency are presented in E2.4.

Journalize transactions.

Instructions

Journalize the transactions. (You may omit explanations.)

E2.6 (LO 1, 2) Lennon Industries had the following transactions.

Analyze transactions and journalize.

- Borrowed \$5,000 from the bank by signing a note.
- Paid \$3,900 cash for a computer.
- Purchased \$650 of supplies on account.

Instructions

- Indicate what accounts are increased and decreased by each transaction.
- Journalize each transaction. (Omit explanations.)

Analyze transactions and journalize.

E2.7 (LO 1, 2) Halladay Enterprises had the following selected transactions.

1. Bo Halladay invested \$4,000 cash in the business.
2. Paid office rent of \$840.
3. Performed consulting services and billed a client \$5,200.
4. Bo Halladay withdrew \$750 cash for personal use.

Instructions

- a. Indicate the effect each transaction has on the accounting equation (Assets = Liabilities + Owner's Equity), using plus and minus signs.
- b. Journalize each transaction. (Omit explanations.)

Journalize a series of transactions.

E2.8 (LO 2) Selected transactions for Sophie's Dog Care are as follows during the month of March.

- | | | |
|-------|----|--|
| March | 1 | Paid monthly rent of \$1,200. |
| | 3 | Performed services for \$160 on account. |
| | 5 | Performed services for cash of \$75. |
| | 8 | Purchased equipment for \$600. The company paid cash of \$90 and the balance was on account. |
| | 12 | Received cash from customers billed on March 3. |
| | 14 | Paid salaries and wages to employees of \$525. |
| | 22 | Paid utilities of \$72. |
| | 24 | Borrowed \$1,500 from Grafton State Bank by signing a note. |
| | 27 | Paid \$220 to repair service for plumbing repairs. |
| | 28 | Paid balance amount owed from equipment purchase on March 8. |
| | 30 | Paid \$1,800 for six months of insurance. |

Instructions

Journalize the transactions. (Omit explanations.)

Record journal entries.

E2.9 (LO 2) On April 1, Adventures Travel Agency began operations. The following transactions were completed during the month.

1. Owner invested \$24,000 in the business.
2. Obtained a bank loan for \$7,000 by issuing a note payable.
3. Paid \$11,000 cash to buy equipment.
4. Paid \$1,200 cash for April office rent.
5. Paid \$1,450 for supplies.
6. Purchased \$600 of advertising in the *Daily Herald*, on account.
7. Performed services for \$18,000: cash of \$2,000 was received from customers, and the balance of \$16,000 was billed to customers on account.
8. Cash withdrawal of \$400 by owner for personal use.
9. Paid the utility bill for the month, \$2,000.
10. Paid *Daily Herald* the amount due in transaction (6).
11. Paid \$40 of interest on the bank loan obtained in transaction (2).
12. Paid employees' salaries and wages, \$6,400.
13. Received \$12,000 cash from customers billed in transaction (7).

Instructions

Journalize the transactions. (Omit explanations.)

Analyze statements about the ledger.

E2.10 (LO 3) Alma Ortiz has prepared the following list of statements about the general ledger.

1. The general ledger contains all the asset and liability accounts but no owner's equity accounts.
2. The general ledger is sometimes referred to as simply the ledger.
3. The accounts in the general ledger are arranged in alphabetical order.
4. Each account in the general ledger is numbered for easier identification.
5. The general ledger is a book of original entry.

Instructions

Identify each statement as true or false. If false, indicate how to correct the statement.

E2.11 (LO 3, 4) Selected transactions from the journal of June Feldman, investment broker, are presented below.

Post journal entries and prepare a trial balance.

Date	Account Titles and Explanation	Ref.	Debit	Credit
Aug. 1	Cash Owner's Capital (Owner's investment of cash in business)		5,000	5,000
10	Cash Service Revenue (Received cash for services performed)		2,600	2,600
12	Equipment Cash Notes Payable (Purchased equipment for cash and notes payable)		5,000	2,300 2,700
25	Accounts Receivable Service Revenue (Billed clients for services performed)		1,700	1,700
31	Cash Accounts Receivable (Receipt of cash on account)		900	900

Instructions

- Post the transactions to T-accounts.
- Prepare a trial balance at August 31, 2020.

E2.12 (LO 2, 4) The T-accounts below summarize the ledger of Negrete Landscaping Company at the end of the first month of operations.

Journalize transactions from account data and prepare a trial balance.

Cash		No. 101	
4/1	14,000	4/15	1,300
4/12	900	4/25	1,500
4/29	400		
4/30	1,000		

Accounts Receivable		No. 112	
4/7	3,000	4/29	400

Supplies		No. 126	
4/4	1,800		

Accounts Payable		No. 201	
4/25	1,500	4/4	1,800

Unearned Service Revenue		No. 209	
	4/30		1,000

Owner's Capital		No. 301	
	4/1		14,000

Service Revenue		No. 400	
	4/7		3,000
	4/12		900

Salaries and Wages Expense		No. 726	
4/15	1,300		

Instructions

- Prepare the complete general journal (including explanations) from which the postings to Cash were made.
- Prepare a trial balance at April 30, 2020.

Journalize transactions from account data and prepare a trial balance.

E2.13 (LO 2, 4) Presented below is the ledger for Shumway Co.

Cash		No. 101
10/1	3,000	10/4 400
10/10	750	10/12 1,500
10/10	4,000	10/15 350
10/20	500	10/30 300
10/25	2,000	10/31 500

Accounts Receivable		No. 112
10/6	800	10/20 500
10/20	940	

Supplies		No. 126
10/4	400	

Equipment		No. 157
10/3	2,000	

Notes Payable		No. 200
	10/10	4,000

Accounts Payable		No. 201
10/12	1,500	10/3 2,000

Owner's Capital		No. 301
	10/1	3,000
	10/25	2,000

Owner's Drawings		No. 306
10/30	300	

Service Revenue		No. 400
	10/6	800
	10/10	750
	10/20	940

Salaries and Wages Expense		No. 726
10/31	500	

Rent Expense		No. 729
10/15	350	

Instructions

- Reproduce the journal entries for the transactions that occurred on October 1, 10, and 20, and provide explanations for each.
- Determine the October 31 balance for each of the accounts above, and prepare a trial balance at October 31, 2020.

Prepare journal entries and post using standard account form.

E2.14 (LO 2, 3) Selected transactions for Bonnie Donne Company during its first month in business are presented below.

- Sept. 1 Invested \$10,000 cash in the business.
 5 Purchased equipment for \$14,000 paying \$4,000 in cash and the balance on account.
 25 Paid \$3,000 cash on balance owed for equipment.
 30 Withdrew \$900 cash for personal use.

Donne's chart of accounts shows No. 101 Cash, No. 157 Equipment, No. 201 Accounts Payable, No. 301 Owner's Capital, and No. 306 Owner's Drawings.

Instructions

- Journalize the transactions on page J1 of the journal. (Omit explanations.)
- Post the transactions using the standard account form.

Analyze errors and their effects on trial balance.

E2.15 (LO 4) The bookkeeper for J.L. Kang Equipment Repair made a number of errors in journalizing and posting, as described below.

- A credit posting of \$525 to Accounts Receivable was omitted.
- A debit posting of \$750 for Prepaid Insurance was debited to Insurance Expense.
- A collection from a customer of \$100 in payment of its account owed was journalized and posted as a debit to Cash \$100 and a credit to Service Revenue \$100.
- A credit posting of \$415 to Property Taxes Payable was made twice.
- A cash purchase of supplies for \$250 was journalized and posted as a debit to Supplies \$25 and a credit to Cash \$25.
- A debit of \$625 to Advertising Expense was posted as \$652.

Instructions

For each error:

- Indicate whether the trial balance will balance.
- If the trial balance will not balance, indicate the amount of the difference.
- Indicate the trial balance column that will have the larger total.

Consider each error separately. Use the following form, in which error (1) is given as an example.

Error	(a) In Balance	(b) Difference	(c) Larger Column
(1)	No	\$525	debit

E2.16 (LO 4) The accounts in the ledger of Prompt Delivery Service contain the following balances on July 31, 2020.

Prepare a trial balance.

Accounts Receivable	\$ 7,640	Prepaid Insurance	\$ 1,968
Accounts Payable	8,394	Maintenance and Repairs Expense	961
Cash	?	Service Revenue	10,610
Equipment	45,360	Owner's Drawings	700
Gasoline Expense	758	Owner's Capital	38,000
Utilities Expense	523	Salaries and Wages Expense	4,428
Notes Payable	17,000	Salaries and Wages Payable	815

Instructions

Prepare a trial balance with the accounts arranged as illustrated in the chapter and fill in the missing amount for Cash.

E2.17 (LO 2, 3, 4) Beyers Security Company provides security services. Selected transactions for Beyers are presented below.

Journalize transactions, post transactions to T-accounts, and prepare trial balance.

- Oct. 1 Invested \$66,000 cash in the business.
 2 Hired part-time security consultant. Salary will be \$2,000 per month. First day of work will be October 15.
 4 Paid one month of rent for building for \$2,000.
 7 Purchased equipment for \$18,000, paying \$4,000 cash and the balance on account.
 8 Paid \$500 for advertising.
 10 Received bill for equipment repair cost of \$390.
 12 Provided security services for event for \$3,200 on account.
 16 Purchased supplies for \$410 on account.
 21 Paid balance due from October 7 purchase of equipment.
 24 Received and paid utility bill for \$148.
 27 Received payment from customer for October 12 services performed.
 31 Paid employee salaries and wages of \$5,100.

Instructions

- Journalize the transactions. Do not provide explanations.
- Post the transactions to T-accounts.
- Prepare a trial balance at October 31, 2020. (*Hint: Compute ending balances of T-accounts first.*)

Problems: Set A

P2.1A (LO 1, 2) Holz Disc Golf Course was opened on March 1 by Ian Holz. The following selected events and transactions occurred during March.

Journalize a series of transactions.

- Mar. 1 Invested \$20,000 cash in the business.
 3 Purchased Rainbow Golf Land for \$15,000 cash. The price consists of land \$12,000, shed \$2,000, and equipment \$1,000. (Make one compound entry.)
 5 Paid advertising expenses of \$900.
 6 Paid cash \$600 for a one-year insurance policy.
 10 Purchased golf discs and other equipment for \$1,050 from Stevenson Company payable in 30 days.



- Mar. 18 Received \$1,100 in cash for golf fees (Holz records golf fees as service revenue).
 19 Sold 150 coupon books for \$10 each. Each book contains 4 coupons that enable the holder to play one round of disc golf.
 25 Withdrew \$800 cash for personal use.
 30 Paid salaries of \$250.
 30 Paid Stevenson Company in full.
 31 Received \$2,700 cash for golf fees.

Holz Disc Golf uses the following accounts: Cash, Prepaid Insurance, Land, Buildings, Equipment, Accounts Payable, Unearned Service Revenue, Owner's Capital, Owner's Drawings, Service Revenue, Advertising Expense, and Salaries and Wages Expense.

Instructions

Journalize the March transactions.

Journalize transactions, post, and prepare a trial balance.



P2.2A (LO 1, 2, 3, 4) Vera Ernst is a licensed dentist. During the first month of the operation of her business, the following events and transactions occurred.

- April 1 Invested \$20,000 cash in her business.
 1 Hired a secretary-receptionist at a salary of \$700 per week payable monthly.
 2 Paid office rent for the month \$1,500.
 3 Purchased dental supplies on account from Dazzle Company \$4,000.
 10 Performed dental services and billed insurance companies \$5,100.
 11 Received \$1,000 cash advance from Leah Mataruka for an implant.
 20 Received \$2,100 cash for services performed from Michael Santos.
 30 Paid secretary-receptionist for the month \$2,800.
 30 Paid \$2,600 to Dazzle for accounts payable due.

Vera uses the following chart of accounts: No. 101 Cash, No. 112 Accounts Receivable, No. 126 Supplies, No. 201 Accounts Payable, No. 209 Unearned Service Revenue, No. 301 Owner's Capital, No. 400 Service Revenue, No. 726 Salaries and Wages Expense, and No. 729 Rent Expense.

Instructions

- Journalize the transactions.
- Post to the ledger accounts.
- Prepare a trial balance on April 30, 2020.

c. Trial balance totals \$29,600

Journalize transactions, post, and prepare a trial balance.

P2.3A (LO 1, 2, 3, 4) Maquoketa Services was formed on May 1, 2020. The following transactions took place during the first month.

Transactions on May 1:

- Jay Bradford invested \$40,000 cash in the company, as its sole owner.
- Hired two employees to work in the warehouse. They will each be paid a salary of \$3,050 per month.
- Signed a 2-year rental agreement on a warehouse; paid \$24,000 cash in advance for the first year.
- Purchased furniture and equipment costing \$30,000. A cash payment of \$10,000 was made immediately; the remainder will be paid in 6 months.
- Paid \$1,800 cash for a one-year insurance policy on the furniture and equipment.

Transactions during the remainder of the month:

- Purchased basic office supplies for \$420 cash.
- Purchased more office supplies for \$1,500 on account.
- Total revenues earned were \$20,000—\$8,000 cash and \$12,000 on account.
- Paid \$400 to suppliers for accounts payable due.
- Received \$3,000 from customers in payment of accounts receivable.
- Received utility bills in the amount of \$380, to be paid next month.
- Paid the monthly salaries of the two employees, totaling \$6,100.

Instructions

- Prepare journal entries to record each of the events listed. (Omit explanations.)
- Post the journal entries to T-accounts.
- Prepare a trial balance as of May 31, 2020.

c. Trial balance totals \$81,480

P2.4A (LO 4) The following trial balance of Sergei Asbrecht Co. does not balance.

Prepare a correct trial balance.



Sergei Asbrecht Co.
Trial Balance
June 30, 2020

	Debit	Credit
Cash		\$ 3,140
Accounts Receivable	\$ 2,812	
Supplies	1,200	
Equipment	2,600	
Accounts Payable		3,666
Unearned Service Revenue	1,100	
Owner's Capital		8,000
Owner's Drawings	800	
Service Revenue		2,480
Salaries and Wages Expense	3,200	
Utilities Expense	810	
	\$12,522	\$17,286

Each of the listed accounts has a normal balance per the general ledger. An examination of the ledger and journal reveals the following errors.

1. Cash received from a customer in payment of its account was debited for \$850, and Accounts Receivable was credited for the same amount. The actual collection was for \$580.
2. The purchase of a computer on account for \$710 was recorded as a debit to Supplies for \$710 and a credit to Accounts Payable for \$710.
3. Services were performed on account for a client for \$980. Accounts Receivable was debited for \$980, and Service Revenue was credited for \$98.
4. A debit posting to Salaries and Wages Expense of \$900 was omitted.
5. A payment of a balance due for \$306 was credited to Cash for \$306 and credited to Accounts Payable for \$360.
6. The withdrawal of \$600 cash for Sergei's personal use was debited to Salaries and Wages Expense for \$600 and credited to Cash for \$600.

Instructions

Prepare a correct trial balance. (*Hint:* It helps to prepare the correct journal entry for the transaction described and compare it to the mistake made.)

Trial balance totals \$15,462

P2.5A (LO 1, 2, 3, 4) The Starr Theater, owned by Meg Vargo, will begin operations in March. The Starr will be unique in that it will show only triple features of sequential theme movies. As of March 1, the ledger of Starr showed No. 101 Cash \$3,000, No. 140 Land \$24,000, No. 145 Buildings (concession stand, projection room, ticket booth, and screen) \$10,000, No. 157 Equipment \$10,000, No. 201 Accounts Payable \$7,000, and No. 301 Owner's Capital \$40,000. During the month of March, the following events and transactions occurred.

Journalize transactions, post, and prepare a trial balance.



- Mar. 2 Rented the three *Indiana Jones* movies to be shown for the first 3 weeks of March. The film rental was \$3,500; \$1,500 was paid in cash and \$2,000 will be paid on March 10.
- 3 Ordered the *Lord of the Rings* movies to be shown the last 10 days of March. It will cost \$200 per night.
- 9 Received \$4,300 cash from admissions.
- 10 Paid balance due on *Indiana Jones* movies rental and \$2,100 on March 1 accounts payable.
- 11 Starr Theater contracted with Adam Ladd to operate the concession stand. Ladd is to pay 15% of gross concession receipts, payable monthly, for the rental of the concession stand.
- 12 Paid advertising expenses \$900.
- 20 Received \$5,000 cash from customers for admissions.
- 20 Received the *Lord of the Rings* movies and paid the rental fee of \$2,000.
- 31 Paid salaries of \$3,100.
- 31 Received statement from Adam Ladd showing gross receipts from concessions of \$6,000 and the balance due to Starr Theater of \$900 ($\$6,000 \times 15\%$) for March. Ladd paid one-half the balance due and will remit the remainder on April 5.
- 31 Received \$9,000 cash from customers for admissions.

In addition to the accounts identified above, the chart of accounts includes No. 112 Accounts Receivable, No. 400 Service Revenue, No. 429 Rent Revenue, No. 610 Advertising Expense, No. 726 Salaries and Wages Expense, and No. 729 Rent Expense.

Instructions

- a. Enter the beginning balances in the ledger. Insert a check mark (✓) in the reference column of the ledger for the beginning balance.
 - b. Journalize the March transactions. Starr records admission revenue as service revenue, rental of the concession stand as rent revenue, and film rental expense as rent expense.
 - c. Post the March journal entries to the ledger. Assume that all entries are posted from page 1 of the journal.
 - d. Trial balance totals \$64,100
- d. Prepare a trial balance on March 31, 2020.

Continuing Case

Cookie Creations

(Note: This is a continuation of the Cookie Creations case from Chapter 1.)

CC2 After researching the different forms of business organization, Natalie Koebel decides to operate “Cookie Creations” as a proprietorship. She then starts the process of getting the business running. In November 2019, the following activities take place.



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- Nov. 8 Natalie cashes her U.S. Savings Bonds and receives \$520, which she deposits in her personal bank account.
- 8 She opens a bank account under the name “Cookie Creations” and transfers \$500 from her personal account to the new account.
- 11 Natalie pays \$65 for advertising.
- 13 She buys baking supplies, such as flour, sugar, butter, and chocolate chips, for \$125 cash. (Hint: Use Supplies account.)
- 14 Natalie starts to gather some baking equipment to take with her when teaching the cookie classes. She has an excellent top-of-the-line food processor and mixer that originally cost her \$750. Natalie decides to start using it only in her new business. She estimates that the equipment is currently worth \$300. She invests the equipment in the business.
- 16 Natalie realizes that her initial cash investment is not enough. Her grandmother lends her \$2,000 cash, for which Natalie signs a note payable in the name of the business. Natalie deposits the money in the business bank account. (Hint: The note does not have to be repaid for 24 months. As a result, the note payable should be reported in the accounts as the last liability and also on the balance sheet as the last liability.)
- 17 She buys more baking equipment for \$900 cash.
- 20 She teaches her first class and collects \$125 cash.
- 25 Natalie books a second class for December 4 for \$150. She receives \$30 cash in advance as a down payment.
- 30 Natalie pays \$1,320 for a one-year insurance policy that will expire on December 1, 2020.

Instructions

- a. Prepare journal entries to record the November transactions.
- b. Post the journal entries to general ledger accounts.
- c. Prepare a trial balance at November 30.

Ethics Cases

EC2.1 Ellynn Kole is the assistant chief accountant at Doman Company, a manufacturer of computer chips and cellular phones. The company presently has total sales of \$20 million. It is the end of the first quarter. Ellynn is hurriedly trying to prepare a trial balance so that quarterly financial statements can be prepared and released to management and the regulatory agencies. The total credits on the trial balance exceed the debits by \$1,000. In order to meet the 4 p.m. deadline, Ellynn decides to force the debits and credits into balance by adding the amount of the difference to the Equipment account. She chooses

A Look at IFRS

LEARNING OBJECTIVE 5

Compare the procedures for the recording process under GAAP and IFRS.

International companies use the same set of procedures and records to keep track of transaction data. Thus, the material in Chapter 2 dealing with the account, general rules of debit and credit, and steps in the recording process—the journal, ledger, and chart of accounts—is the same under both GAAP and IFRS.

Key Points

Following are the key similarities and differences between GAAP and IFRS as related to the recording process.

Similarities

- Transaction analysis is the same under IFRS and GAAP.
- Both the IASB and the FASB go beyond the basic definitions provided in the textbook for the key elements of financial statements, that is assets, liabilities, equity, revenue, and expenses. The implications of the expanded definitions are discussed in more advanced accounting courses.
- As shown in the textbook, dollar signs are typically used only in the trial balance and the financial statements. The same practice is followed under IFRS, using the currency of the country where the reporting company is headquartered.
- A trial balance under IFRS follows the same format as shown in the textbook.

Differences

- IFRS relies less on historical cost and more on fair value than do FASB standards.
- Internal controls are a system of checks and balances designed to prevent and detect fraud and errors. While most public U.S. companies have these systems in place, many non-U.S. companies have never completely documented the controls nor had an independent auditors attest to their effectiveness.

Looking to the Future

The basic recording process shown in this textbook is followed by companies across the globe. It is unlikely to change in the future. The definitional structure of assets, liabilities, equity, revenues, and expenses may change over time as the IASB and FASB evaluate their overall conceptual framework for establishing accounting standards.

IFRS Practice

IFRS Self-Test Questions

1. Which statement is **correct** regarding IFRS?
 - a. IFRS reverses the rules of debits and credits, that is, debits are on the right and credits are on the left.
 - b. IFRS uses the same process for recording transactions as GAAP.
 - c. The chart of accounts under IFRS is different because revenues follow assets.
 - d. None of the above statements are correct.
2. The expanded accounting equation under IFRS is as follows:
 - a. $\text{Assets} = \text{Liabilities} + \text{Owner's Capital} + \text{Owner's Drawings} + \text{Revenues} - \text{Expenses}$.
 - b. $\text{Assets} + \text{Liabilities} = \text{Owner's Capital} + \text{Owner's Drawings} + \text{Revenues} - \text{Expenses}$.
 - c. $\text{Assets} = \text{Liabilities} + \text{Owner's Capital} - \text{Owner's Drawings} + \text{Revenues} - \text{Expenses}$.
 - d. $\text{Assets} = \text{Liabilities} + \text{Owner's Capital} + \text{Owner's Drawings} - \text{Revenues} - \text{Expenses}$.
3. A trial balance:
 - a. is the same under IFRS and GAAP.
 - b. proves that transactions are recorded correctly.
 - c. proves that all transactions have been recorded.
 - d. will not balance if a correct journal entry is posted twice.
4. One difference between IFRS and GAAP is that:
 - a. GAAP uses accrual-accounting concepts and IFRS uses primarily the cash basis of accounting.
 - b. IFRS uses a different posting process than GAAP.



Rudy Archuleta/Redux Pictures

Adjusting the Accounts

Chapter Preview

In Chapter 1, you learned a neat little formula: $\text{Net income} = \text{Revenues} - \text{Expenses}$. In Chapter 2, you learned some rules for recording revenue and expense transactions. Guess what? Things are not really that nice and neat. In fact, it is often difficult for companies to determine in what time period they should report some revenues and expenses. In other words, in measuring net income, timing is everything.

Feature Story

Keeping Track of Groupons

Who doesn't like buying things at a discount? That's why it's not surprising that three years after it started as a company, **Groupon, Inc.** was estimated to be worth \$16 billion. This translates into an average increase in value of almost \$15 million per day.

Now consider that Groupon had previously been estimated to be worth even more than that. What happened? Well, accounting regulators and investors began to question the way that Groupon had accounted for some of its transactions. Groupon sells coupons ("groupons"), so how hard can it be

to accurately account for that? It turns out that accounting for coupons is not as easy as you might think.

First, consider what happens when Groupon makes a sale. Suppose it sells agroupon for \$30 for Highrise Hamburgers. When it receives the \$30 from the customer, it must turn over half of that amount (\$15) to Highrise Hamburgers. So should Groupon record revenue for the full \$30 or just \$15? Until recently, Groupon recorded the full \$30. But, in response to an SEC ruling on the issue, Groupon now records revenue of \$15 instead.

A second issue is a matter of timing. When should Groupon record this \$15 revenue? Should it record the revenue when it sells thegroupon, or must it wait until the customer uses thegroupon at Highrise Hamburgers? You can find the answer to

this question in the notes to Groupon’s financial statements. It recognizes the revenue once “the number of customers who purchase the daily deal exceeds the predetermined threshold, the Groupon has been electronically delivered to the purchaser and a listing of Groupons sold has been made available to the merchant.”

The accounting becomes even more complicated when you consider the company’s loyalty programs. Groupon offers free or discounted groupons to its subscribers for doing things such as referring new customers or participating in promotions. These groupons are to be used for future purchases, yet the company must record the expense at the time the customer receives the

groupon. The cost of these programs is huge for Groupon, so the timing of this expense can definitely affect its reported income.

Finally, Groupon, like all other companies, relies on many estimates in its financial reporting. For example, Groupon reports that “estimates are utilized for, but not limited to, stock-based compensation, income taxes, valuation of acquired goodwill and intangible assets, customer refunds, contingent liabilities and the depreciable lives of fixed assets.” It notes that “actual results could differ materially from those estimates.” So, next time you use a coupon, think about what that means for the company’s accountants!

Chapter Outline

LEARNING OBJECTIVES

LO 1 Explain the accrual basis of accounting and the reasons for adjusting entries.	• Fiscal and calendar years • Accrual- vs. cash-basis accounting • Recognizing revenues and expenses • Need for adjusting entries • Types of adjusting entries	DO IT! 1 Timing Concepts
LO 2 Prepare adjusting entries for deferrals.	• Prepaid expenses • Unearned revenues	DO IT! 2 Adjusting Entries for Deferrals
LO 3 Prepare adjusting entries for accruals.	• Accrued revenues • Accrued expenses • Summary of basic relationships	DO IT! 3 Adjusting Entries for Accruals
LO 4 Describe the nature and purpose of an adjusted trial balance.	• Preparing the adjusted trial balance • Preparing financial statements	DO IT! 4 Trial Balance
Go to the Review and Practice section at the end of the chapter for a review of key concepts and practice applications with solutions. Visit WileyPLUS with ORION for additional tutorials and practice opportunities.		

Accrual-Basis Accounting and Adjusting Entries

LEARNING OBJECTIVE 1

Explain the accrual basis of accounting and the reasons for adjusting entries.

If we could wait to prepare financial statements until a company ended its operations, no adjustments would be needed. At that point, we could easily determine its final balance sheet and the amount of lifetime income it earned.

However, most companies need feedback about how well they are performing during a period of time. For example, management usually wants monthly financial statements. The Internal Revenue Service requires all businesses to file annual tax returns. Therefore, **accountants**

divide the economic life of a business into artificial time periods. This convenient assumption is referred to as the **time period assumption** (see **Alternative Terminology**).

Many business transactions affect more than one of these arbitrary time periods. For example, the airplanes purchased by **Southwest Airlines** five years ago are still in use today. It would not make sense to expense the full cost of the airplanes at the time of purchase because they will be used for many subsequent periods. Instead, companies must therefore allocate the costs to the periods of use (how much of the cost of an airplane contributed to operations this year?).

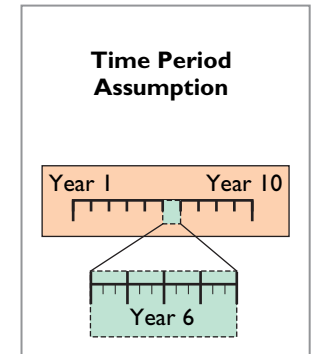
Fiscal and Calendar Years

Both small and large companies prepare financial statements periodically in order to assess their financial condition and results of operations. **Accounting time periods are generally a month, a quarter, or a year.** Monthly and quarterly time periods are called **interim periods**. Most large companies must prepare both quarterly and annual financial statements.

An accounting time period that is one year in length is a **fiscal year**. A fiscal year usually begins with the first day of a month and ends 12 months later on the last day of a month. Many businesses use the **calendar year** (January 1 to December 31) as their accounting period. Some do not. Companies whose fiscal year differs from the calendar year include **Delta Air Lines**, June 30, and **The Walt Disney Company**, September 30. Sometimes a company's year-end will vary from year to year. For example, **PepsiCo's** fiscal year ends on the Friday closest to December 31, which was December 27 in 2014 and December 26 in 2015.

ALTERNATIVE TERMINOLOGY

The time period assumption is also called the *periodicity assumption*.



Accrual- versus Cash-Basis Accounting

What you will learn in this chapter is **accrual-basis accounting**. Under the accrual basis, companies record transactions that change a company's financial statements **in the periods in which the events occur**. For example, using the accrual basis to determine net income means companies recognize revenues when they perform services (rather than when they receive cash). It also means recognizing expenses when incurred (rather than when paid).

An alternative to the accrual basis is the cash basis. Under **cash-basis accounting**, companies record revenue at the time they receive cash. They record an expense at the time they pay out cash. The cash basis seems appealing due to its simplicity, but it often produces misleading financial statements. For example, it fails to record revenue for a company that has performed services but has not yet received payment. As a result, the cash basis may not recognize revenue in the period that a performance obligation is satisfied.

Accrual-basis accounting is therefore in accordance with generally accepted accounting principles (GAAP). Individuals and some small companies, however, do use cash-basis accounting. The cash basis is justified for small businesses because they often have few receivables and payables. Medium and large companies use accrual-basis accounting.

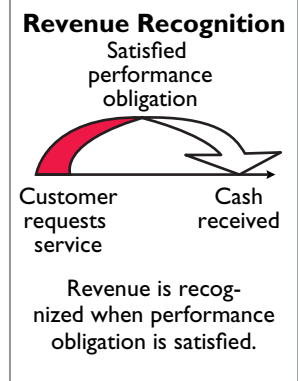
Recognizing Revenues and Expenses

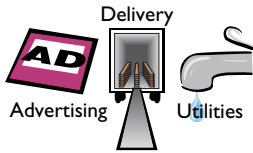
It can be difficult to determine when to report revenues and expenses. The revenue recognition principle and the expense recognition principle help in this task.

Revenue Recognition Principle

When a company agrees to perform a service or sell a product to a customer, it has a **performance obligation**. When the company meets this performance obligation, it recognizes revenue. The **revenue recognition principle** therefore requires that companies recognize revenue in the accounting period in which the performance obligation is satisfied. A company satisfies its performance obligation by performing a service or providing a good to a customer.

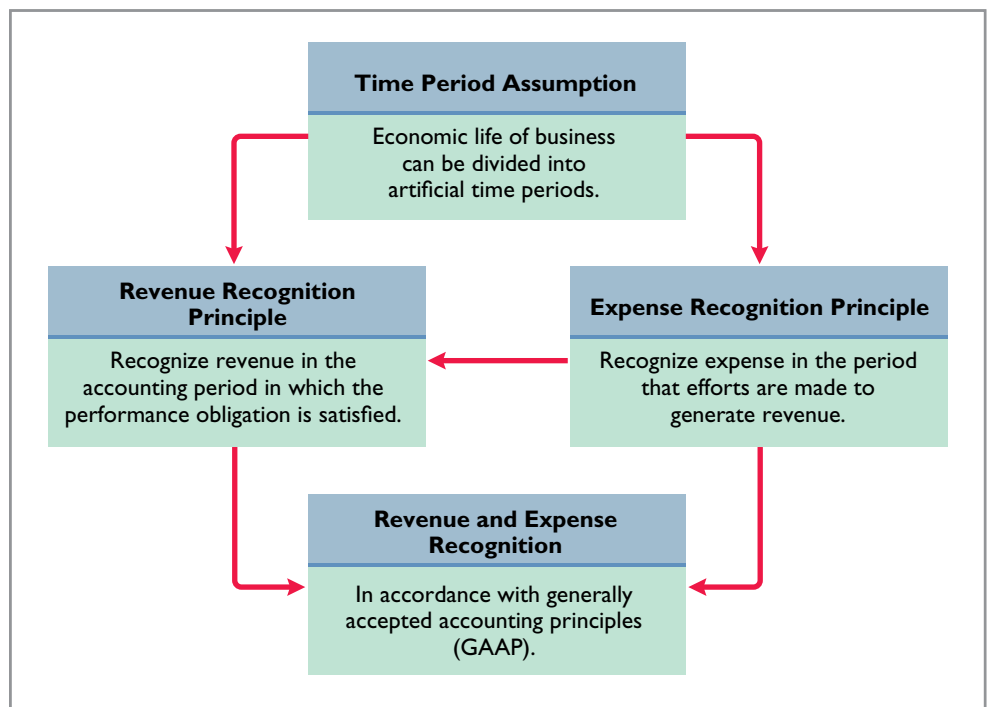
To illustrate, assume that Dave's Dry Cleaning cleans clothing on June 30 but customers do not claim and pay for their clothes until the first week of July. Dave's should record revenue in June when it performed the service (satisfied the performance obligation) rather than in July when it received the cash. At June 30, Dave's would report a receivable on its balance sheet and revenue in its income statement for the service performed.



Expense RecognitionEfforts generated
revenueExpense is recognized
when efforts are made to
generate revenue.**Expense Recognition Principle**

Accountants follow a simple rule in recognizing expenses: “Let the expenses follow the revenues.” Thus, expense recognition is tied to revenue recognition. In the dry cleaning example, this means that Dave’s should report the salary expense incurred in performing the June 30 cleaning service in the same period in which it recognizes the service revenue. The critical issue in expense recognition is when the expense makes its contribution to revenue. This may or may not be the same period in which the expense is paid. If Dave’s does not pay the salary incurred on June 30 until July, it would report salaries payable on its June 30 balance sheet.

This practice of expense recognition is referred to as the **expense recognition principle**. It requires that companies recognize expenses in the period in which they make efforts (consume assets or incur liabilities) to generate revenue. The term matching is sometimes used in expense recognition to indicate the relationship between the effort expended and the revenue generated. **Illustration 3.1** summarizes the revenue and expense recognition principles.

ILLUSTRATION 3.1 GAAP relationships in revenue and expense recognition**Investor Insight** Apple Inc.

PhotoAlto/James Hardy/Getty Images, Inc.

Reporting Revenue Accurately

Until recently, electronics manufacturer **Apple** was required to spread the revenues from iPhone sales over the two year period following the sale of the phone. Accounting standards required this because Apple was obligated to provide software updates after the phone was sold. Since Apple had service obligations after the initial date of sale, it was forced to spread the revenue over a two-year period.

As a result, the rapid growth of iPhone sales was not fully reflected in the revenue amounts reported in Apple’s income statement. A new accounting standard now enables Apples to report much more of its iPhone revenue at the point of sale. It was estimated that under the new rule revenues would have been about 17% higher and earnings per share almost 50% higher.

In the past, why was it argued that Apple should spread the recognition of iPhone revenue over a two-year period, rather than recording it upfront? (Go to WileyPLUS for this answer and additional questions.)

The Need for Adjusting Entries

In order for revenues to be recorded in the period in which services are performed and for expenses to be recognized in the period in which they are incurred, companies make adjusting entries. **Adjusting entries ensure that the revenue recognition and expense recognition principles are followed** (see **International Note**).

Adjusting entries are necessary because the **trial balance**—the first pulling together of the transaction data—may not contain up-to-date and complete data. This is true for several reasons:

1. Some events are not recorded daily because it is not efficient to do so. Examples are the use of supplies and the earning of wages by employees.
2. Some costs are not recorded during the accounting period because these costs expire with the passage of time rather than as a result of recurring daily transactions. Examples are charges related to the use of buildings and equipment, rent, and insurance.
3. Some items may be unrecorded. An example is a utility service bill that will not be received until the next accounting period.

Adjusting entries are required every time a company prepares financial statements. The company analyzes each account in the trial balance to determine whether it is complete and up-to-date for financial statement purposes. **Every adjusting entry will include one income statement account and one balance sheet account.**

INTERNATIONAL NOTE

Internal controls are a system of checks and balances designed to detect and prevent fraud and errors. The Sarbanes-Oxley Act requires U.S. companies to enhance their systems of internal control. However, many foreign companies do not have to meet strict internal control requirements. Some U.S. companies believe that this gives foreign firms an unfair advantage because developing and maintaining internal controls can be very expensive.

Types of Adjusting Entries

Adjusting entries are classified as either **deferrals** or **accruals**. As **Illustration 3.2** shows, each of these classes has two subcategories.

Deferrals:

1. **Prepaid expenses:** Expenses paid in cash before they are used or consumed.
2. **Unearned revenues:** Cash received before services are performed.

Accruals:

1. **Accrued revenues:** Revenues for services performed but not yet received in cash or recorded.
2. **Accrued expenses:** Expenses incurred but not yet paid in cash or recorded.

ILLUSTRATION 3.2

Categories of adjusting entries

Subsequent sections give examples of each type of adjustment. Each example is based on the October 31 trial balance of Pioneer Advertising from Chapter 2, reproduced in **Illustration 3.3**.

Pioneer Advertising Trial Balance October 31, 2020

	Debit	Credit
Cash	\$15,200	
Supplies	2,500	
Prepaid Insurance	600	
Equipment	5,000	
Notes Payable		\$ 5,000
Accounts Payable		2,500
Unearned Service Revenue		1,200
Owner's Capital		10,000
Owner's Drawings	500	
Service Revenue		10,000
Salaries and Wages Expense	4,000	
Rent Expense	900	
	\$28,700	\$28,700

ILLUSTRATION 3.3

Trial balance

We assume that Pioneer uses an accounting period of one month. Thus, monthly adjusting entries are made. The entries are dated October 31.

DO IT! 1 | Timing Concepts

Below is a list of timing concepts in the left column, with a description of the concept in the right column. There are more descriptions provided than concepts. Match the description to the concept.

- | | |
|--|--|
| 1. ____ Accrual-basis accounting. | (a) Monthly and quarterly time periods. |
| 2. ____ Calendar year. | (b) Efforts (expenses) should be recognized in the period in which a company uses assets or incurs liabilities to generate results (revenues). |
| 3. ____ Time period assumption. | (c) Accountants divide the economic life of a business into artificial time periods. |
| 4. ____ Expense recognition principle. | (d) Companies record revenues when they receive cash and record expenses when they pay out cash. |
| | (e) An accounting time period that starts on January 1 and ends on December 31. |
| | (f) Companies record transactions in the period in which the events occur. |

ACTION PLAN

- Review the definitions of the timing concepts in the Glossary Review section.
- Study carefully the revenue recognition principle, the expense recognition principle, and the time period assumption.

Solution

1. f 2. e 3. c 4. b

Related exercise material: BE3.1, BE3.2, DO IT! 3.1, E3.1, E3.2, and E3.3.

Adjusting Entries for Deferrals

LEARNING OBJECTIVE 2

Prepare adjusting entries for deferrals.



To defer means to postpone or delay. **Deferrals** are expenses or revenues that are recognized at a date later than the point when cash was originally exchanged. The two types of deferrals are prepaid expenses and unearned revenues.

Prepaid Expenses

When companies record payments of expenses that will benefit more than one accounting period, they record an asset called **prepaid expenses** or **prepayments**. When expenses are prepaid, an asset account is increased (debited) to show the service or benefit that the company will receive in the future. Examples of common prepayments are insurance, supplies, advertising, and rent. In addition, companies make prepayments when they purchase buildings and equipment.

Prepaid expenses are costs that expire either with the passage of time (e.g., rent and insurance) **or through use** (e.g., supplies). The expiration of these costs does not require daily entries, which would be impractical and unnecessary. Accordingly, companies postpone the recognition of such cost expirations until they prepare financial statements. At each statement date, they make adjusting entries to record the expenses applicable to the current accounting period and to show the remaining amounts in the asset accounts.

Prior to adjustment, assets are overstated and expenses are understated. Therefore, as shown in **Illustration 3.4**, an adjusting entry for prepaid expenses results in an increase (a debit) to an expense account and a decrease (a credit) to an asset account.

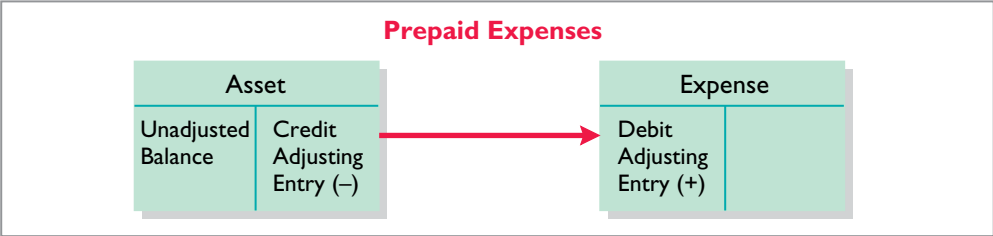


ILLUSTRATION 3.4
Adjusting entries for prepaid expenses

Let’s look in more detail at some specific types of prepaid expenses, beginning with supplies.

Supplies

The purchase of supplies, such as paper and envelopes, results in an increase (a debit) to an asset account. During the accounting period, the company uses supplies. Rather than record supplies expense as the supplies are used, companies recognize supplies expense at the **end** of the accounting period. At the end of the accounting period, the company counts the remaining supplies. As shown in Illustration 3.5, the difference between the unadjusted balance in the Supplies (asset) account and the actual cost of supplies on hand represents the supplies used (an expense) for that period.

Recall from Chapter 2 that Pioneer Advertising purchased supplies costing \$2,500 on October 5. Pioneer recorded the purchase by increasing (debiting) the asset Supplies. This account shows a balance of \$2,500 in the October 31 trial balance. An inventory count at the close of business on October 31 reveals that \$1,000 of supplies are still on hand. Thus, the cost of supplies used is \$1,500 (\$2,500 – \$1,000). This use of supplies decreases an asset, Supplies. It also decreases owner’s equity by increasing an expense account, Supplies Expense. This is shown in **Illustration 3.5**.

Supplies

Oct. 5



Supplies purchased;
record asset

Oct. 31



Supplies used;
record supplies expense

ILLUSTRATION 3.5 Adjustment for supplies

Basic Analysis

Equation Analysis

Debit–Credit Analysis

Journal Entry

Posting

The expense Supplies Expense is increased \$1,500; the asset Supplies is decreased \$1,500.

(1)

<u>Assets</u>	=	<u>Liabilities</u>	+	<u>Owner’s Equity</u>
Supplies	=			Supplies Expense
–\$1,500	=			–\$1,500

Debits increase expenses: debit Supplies Expense \$1,500.
Credits decrease assets: credit Supplies \$1,500.

Oct. 31	Supplies Expense	631	1,500	
	Supplies	126		1,500
	(To record supplies used)			

Supplies		126	Supplies Expense		631
Oct. 5	2,500	Oct. 31	Adj. 1,500	Oct. 31	Adj. 1,500
Oct. 31	Bal. 1,000			Oct. 31	Bal. 1,500

After adjustment, the asset account Supplies shows a balance of \$1,000, which is equal to the cost of supplies on hand at the statement date. In addition, Supplies Expense shows a balance of \$1,500, which equals the cost of supplies used in October. **If Pioneer does not make the adjusting entry, October expenses are understated and net income is overstated by \$1,500. Moreover, both assets and owner's equity will be overstated by \$1,500 on the October 31 balance sheet.**

Insurance

Oct. 4



Insurance purchased;
record asset

Insurance Policy			
Oct \$50	Nov \$50	Dec \$50	Jan \$50
Feb \$50	March \$50	April \$50	May \$50
June \$50	July \$50	Aug \$50	Sept \$50
1 YEAR \$600			

Oct. 31

Insurance expired;
record insurance expense

Insurance

Companies purchase insurance to protect themselves from losses due to fire, theft, and unforeseen events. Insurance must be paid in advance, often for multiple months. The cost of insurance (premiums) paid in advance is recorded as an increase (debit) in the asset account Prepaid Insurance. At the financial statement date, companies increase (debit) Insurance Expense and decrease (credit) Prepaid Insurance for the cost of insurance that has expired during the period.

On October 4, Pioneer Advertising paid \$600 for a one-year fire insurance policy. Coverage began on October 1. Pioneer recorded the payment by increasing (debiting) Prepaid Insurance. This account shows a balance of \$600 in the October 31 trial balance. Insurance of \$50 ($\$600 \div 12$) expires each month. The expiration of prepaid insurance decreases an asset, Prepaid Insurance. It also decreases owner's equity by increasing an expense account, Insurance Expense.

As shown in **Illustration 3.6**, the asset Prepaid Insurance shows a balance of \$550, which represents the unexpired cost for the remaining 11 months of coverage. At the same time, the balance in Insurance Expense equals the insurance cost that expired in October. **If Pioneer does not make this adjustment, October expenses are understated by \$50 and net income is overstated by \$50. Moreover, both assets and owner's equity will be overstated by \$50 on the October 31 balance sheet.**

ILLUSTRATION 3.6 Adjustment for insurance

Basic Analysis	The expense Insurance Expense is increased \$50; the asset Prepaid Insurance is decreased \$50.				
Equation Analysis	(2)	<u>Assets</u> Prepaid Insurance -\$50	=	<u>Liabilities</u> 	<u>+</u> <u>Owner's Equity</u> Insurance Expense -\$50
Debit–Credit Analysis	Debits increase expenses: debit Insurance Expense \$50. Credits decrease assets: credit Prepaid Insurance \$50.				
Journal Entry	Oct. 31	Insurance Expense Prepaid Insurance (To record insurance expired)	722 130	50 50	
Posting	Prepaid Insurance130 Oct. 4600 Oct. 31Bal. 550		Insurance Expense722 Oct. 31Adj. 50 Oct. 31Bal. 50		

Depreciation

A company typically owns a variety of assets that have long lives, such as buildings, equipment, and motor vehicles. The period of service is referred to as the **useful life** of the asset. Because a building is expected to be of service for many years, it is recorded as an asset, rather than an expense, on the date it is acquired. As explained in Chapter 1, companies record such assets **at cost**, as required by the historical cost principle. To follow the expense recognition principle, companies allocate a portion of this cost as an expense during each period of the

asset’s useful life. **Depreciation** is the process of allocating the cost of an asset to expense over its useful life.

Need for Adjustment. The acquisition of long-lived assets is essentially a long-term prepayment for the use of an asset. An adjusting entry for depreciation is needed to recognize the cost that has been used (an expense) during the period and to report the unused cost (an asset) at the end of the period. One very important point to understand: **Depreciation is an allocation concept, not a valuation concept.** That is, depreciation **allocates an asset’s cost to the periods in which it is used. Depreciation does not attempt to report the actual change in the value of the asset.**

For Pioneer Advertising, assume that depreciation on the equipment is \$480 a year, or \$40 per month. As shown in **Illustration 3.7**, rather than decrease (credit) the asset account directly, Pioneer instead credits Accumulated Depreciation—Equipment. Accumulated Depreciation is called a **contra asset account**. Such an account is offset against an asset account on the balance sheet (see **Helpful Hint**). Thus, the Accumulated Depreciation—Equipment account offsets the asset Equipment. **This account keeps track of the total amount of depreciation expense taken over the life of the asset.** To keep the accounting equation in balance, Pioneer decreases owner’s equity by increasing an expense account, Depreciation Expense.

ILLUSTRATION 3.7 Adjustment for depreciation

Basic Analysis

The expense Depreciation Expense is increased \$40; the contra asset Accumulated Depreciation—Equipment is increased \$40.

Equation Analysis

Assets	=	Liabilities	+	Owner's Equity
Accumulated Depreciation—Equipment	=			Depreciation Expense
–\$40	=			–\$40

Debit–Credit Analysis

Debits increase expenses: debit Depreciation Expense \$40.

Credits increase contra assets: credit Accumulated Depreciation—Equipment \$40.

Journal Entry

Oct. 31	Depreciation Expense	711	40	
	Accumulated Depreciation—Equipment	158		40
	(To record monthly depreciation)			

Posting

Equipment		157
Oct. 1	5,000	
Oct. 31	Bal. 5,000	

Accumulated Depreciation—Equipment		158
	Oct. 31	Adj. 40
	Oct. 31	Bal. 40

Depreciation Expense		711
	Oct. 31	Adj. 40
	Oct. 31	Bal. 40

Depreciation

Oct. 1



Equipment purchased; record asset

Equipment			
Oct	Nov	Dec	Jan
\$40	\$40	\$40	\$40
Feb	March	April	May
\$40	\$40	\$40	\$40
June	July	Aug	Sept
\$40	\$40	\$40	\$40
Depreciation = \$480/year			

Oct. 31

Depreciation recognized; record depreciation expense

HELPFUL HINT

All contra accounts have increases, decreases, and normal balances opposite to the account to which they relate.

The balance in the Accumulated Depreciation—Equipment account will increase \$40 each month, and the balance in Equipment remains \$5,000.

Statement Presentation. As indicated, Accumulated Depreciation—Equipment is a contra asset account. It is offset against Equipment on the balance sheet. The normal balance of a contra asset account is a credit. A theoretical alternative to using a contra asset account would be to decrease (credit) the asset account by the amount of depreciation each period. But using the contra account is preferable for a simple reason: It discloses **both** the original cost of the equipment **and** the total cost that has been expensed to date. Thus, in the balance sheet, Pioneer deducts Accumulated Depreciation—Equipment from the related asset account, as shown in **Illustration 3.8**.

ILLUSTRATION 3.8

Balance sheet presentation of accumulated depreciation

Equipment	\$5,000
Less: Accumulated depreciation—equipment	40
	\$4,960

ALTERNATIVE TERMINOLOGY

Book value is also referred to as *carrying value*.

Book value is the difference between the cost of any depreciable asset and its related accumulated depreciation (see **Alternative Terminology**). In Illustration 3.8, the book value of the equipment at the balance sheet date is \$4,960. The book value and the fair value of the asset are generally two different values. As noted earlier, **the purpose of depreciation is not valuation but a means of cost allocation**.

Depreciation expense identifies the portion of an asset’s cost that expired during the period (in this case, in October). The accounting equation shows that **without this adjusting entry, total assets, total owner’s equity, and net income are overstated by \$40 and depreciation expense is understated by \$40**.

Illustration 3.9 summarizes the accounting for prepaid expenses.

ILLUSTRATION 3.9

Accounting for prepaid expenses

Accounting for Prepaid Expenses			
Examples	Reason for Adjustment	Accounts Before Adjustment	Adjusting Entry
Insurance, supplies, advertising, rent, depreciation	Prepaid expenses originally recorded in asset accounts have been used.	Assets overstated. Expenses understated.	Dr. Expenses Cr. Assets or Contra Assets

Unearned Revenues

Oct. 2



Cash is received in advance; liability is recorded



Oct. 31

Some service has been performed; some revenue is recorded

Unearned Revenues

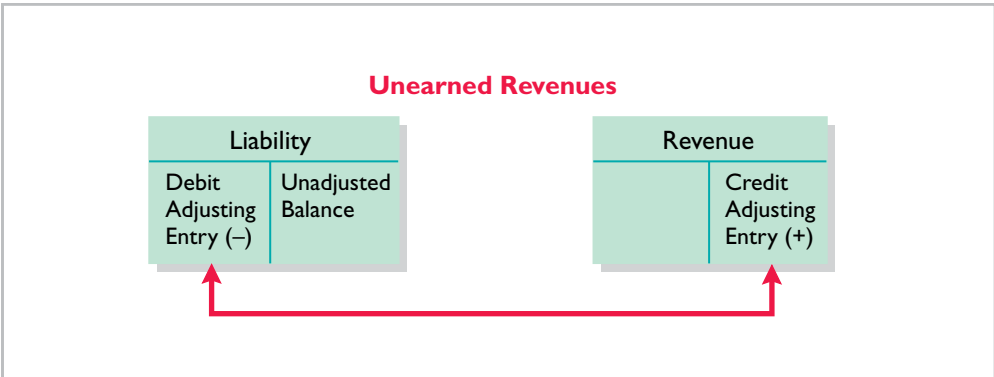
When companies receive cash before services are performed, they record a liability by increasing (crediting) a liability account called **unearned revenues**. In other words, a **company now has a performance obligation** (liability) to transfer a service to one of its customers. Items like rent, magazine subscriptions, and customer deposits for future service may result in unearned revenues. Airlines such as **United**, **Southwest**, and **Delta**, for instance, treat receipts from the sale of tickets as unearned revenue until the flight service is provided.

Unearned revenues are the opposite of prepaid expenses. Indeed, unearned revenue on the books of one company is likely to be a prepaid expense on the books of the company that has made the advance payment. For example, if identical accounting periods are assumed, a landlord will have unearned rent revenue when a tenant has prepaid rent.

When a company receives payment for services to be performed in a future accounting period, it increases (credits) an unearned revenue (a liability) account to recognize the liability that exists. The company subsequently recognizes revenues when it performs the service (see **Illustration 3.10**).

ILLUSTRATION 3.10

Adjusting entries for unearned revenues



During the accounting period, it is not practical to make daily entries as the company performs services. Instead, the company delays recognition of revenue until the adjustment process. Then, the company makes an adjusting entry to record the revenue for services performed during the period and to show the liability that remains at the end of the accounting period. Typically, prior to adjustment, liabilities are overstated and revenues are understated. Therefore, **the adjusting entry for unearned revenues results in a decrease (a debit) to a liability account and an increase (a credit) to a revenue account.**

Pioneer Advertising received \$1,200 on October 2 from R. Knox for advertising services expected to be completed by December 31. Pioneer credited the payment to Unearned Service Revenue. This liability account shows a balance of \$1,200 in the October 31 trial balance. From an evaluation of the services Pioneer performed for Knox during October, the company determines that it should recognize \$400 of revenue in October. The liability (Unearned Service Revenue) is therefore decreased, and owner's equity (Service Revenue) is increased.

As shown in **Illustration 3.11**, the liability Unearned Service Revenue now shows a balance of \$800. That amount represents the remaining advertising services Pioneer is obligated to perform in the future. At the same time, Service Revenue shows total revenue recognized in October of \$10,400. **Without this adjustment, revenues and net income are understated by \$400 in the income statement. Moreover, liabilities will be overstated and owner's equity will be understated by \$400 on the October 31 balance sheet.**

ILLUSTRATION 3.11 Service revenue accounts after adjustment

Basic
Analysis

The liability Unearned Service Revenue is decreased \$400; the revenue Service Revenue is increased \$400.

Equation
Analysis

Assets	=	Liabilities	+	Owner's Equity
		Unearned Service Revenue		Service Revenue
		−\$400		+\$400

Debit–Credit
Analysis

Debits decrease liabilities: debit Unearned Service Revenue \$400.
Credits increase revenues: credit Service Revenue \$400.

Journal
Entry

Oct. 31	Unearned Service Revenue Service Revenue (To record revenue for services performed)	209 400	400	400
---------	---	------------	-----	-----

Posting

Unearned Service Revenue		209
Oct. 31	Adj. 400	
	Oct. 2	1,200
	Oct. 31	Bal. 800

Service Revenue		400
	Oct. 31	10,000
	31	Adj. 400
	Oct. 31	Bal. 10,400

Illustration 3.12 summarizes the accounting for unearned revenues.

Accounting for Unearned Revenues			
Examples	Reason for Adjustment	Accounts Before Adjustment	Adjusting Entry
Rent, magazine subscriptions, customer deposits for future service	Unearned revenues recorded in liability accounts are now recognized as revenue for services performed.	Liabilities overstated. Revenues understated.	Dr. Liabilities Cr. Revenues

ILLUSTRATION 3.12

Accounting for unearned revenues

Accounting Across the Organization Best Buy



© Skip O'Donnell/iStockphoto

Turning Gift Cards into Revenue

Those of you who are marketing majors (and even most of you who are not) know that gift cards are among the hottest marketing tools in merchandising today. Customers purchase gift cards and give them to someone for later use. In a recent year, gift-card sales were expected to exceed \$124 billion.

Although these programs are popular with marketing executives, they create accounting questions. Should

revenue be recorded at the time the gift card is sold, or when it is exercised? How should expired gift cards be accounted for? In a recent balance sheet, **Best Buy** reported unearned revenue related to gift cards of \$427 million.

Source: "2014 Gift Card Sales to Top \$124 Billion, but Growth Slowing," *PR Newswire* (December 10, 2014).

Suppose that Robert Jones purchases a \$100 gift card at Best Buy on December 24, 2019, and gives it to his wife, Mary Jones, on December 25, 2019. On January 3, 2020, Mary uses the card to purchase \$100 worth of CDs. When do you think Best Buy should recognize revenue and why? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 2 | Adjusting Entries for Deferrals

The ledger of Hammond Company, on March 31, 2020, includes these selected accounts before adjusting entries are prepared.

	<u>Debit</u>	<u>Credit</u>
Prepaid Insurance	\$ 3,600	
Supplies	2,800	
Equipment	25,000	
Accumulated Depreciation—Equipment		\$5,000
Unearned Service Revenue		9,200

An analysis of the accounts shows the following.

1. Insurance expires at the rate of \$100 per month.
2. Supplies on hand total \$800.
3. The equipment depreciates \$200 a month.
4. During March, services were performed for \$4,000 of the unearned service revenue reported.

Prepare the adjusting entries for the month of March.

Solution

1. Insurance Expense	100	
Prepaid Insurance		100
(To record insurance expired)		
2. Supplies Expense (\$2,800 – \$800)	2,000	
Supplies		2,000
(To record supplies used)		
3. Depreciation Expense	200	
Accumulated Depreciation—Equipment		200
(To record monthly depreciation)		
4. Unearned Service Revenue	4,000	
Service Revenue		4,000
(To record revenue for services performed)		

Related exercise material: BE3.2, BE3.3, BE3.4, BE3.5, BE3.6, and DO IT! 3.2.

ACTION PLAN

- Make adjusting entries at the end of the period for revenues recognized and expenses incurred in the period.
- Don't forget to make adjusting entries for deferrals. Failure to adjust for deferrals leads to overstatement of the asset or liability and understatement of the related expense or revenue.

Adjusting Entries for Accruals

LEARNING OBJECTIVE 3

Prepare adjusting entries for accruals.



The second category of adjusting entries is **accruals**. Prior to an accrual adjustment, the revenue account (and the related asset account) or the expense account (and the related liability account) are understated. Thus, the adjusting entry for accruals will **increase both a balance sheet and an income statement account**.

Accrued Revenues

Revenues for services performed but not yet recorded at the statement date are **accrued revenues**. Accrued revenues may accumulate (accrue) with the passing of time, as in the case of interest revenue. These are unrecorded because the earning of interest does not involve daily transactions. Companies do not record interest revenue on a daily basis because it is often impractical to do so. Accrued revenues also may result from services that have been performed but not yet billed nor collected, as in the case of commissions and fees. These may be unrecorded because only a portion of the total service has been performed and the clients will not be billed until the service has been completed.

An adjusting entry records the receivable that exists at the balance sheet date and the revenue for the services performed during the period. Prior to adjustment, both assets and revenues are understated. As shown in **Illustration 3.13**, an **adjusting entry for accrued revenues results in an increase (a debit) to an asset account and an increase (a credit) to a revenue account**.

Accrued Revenues



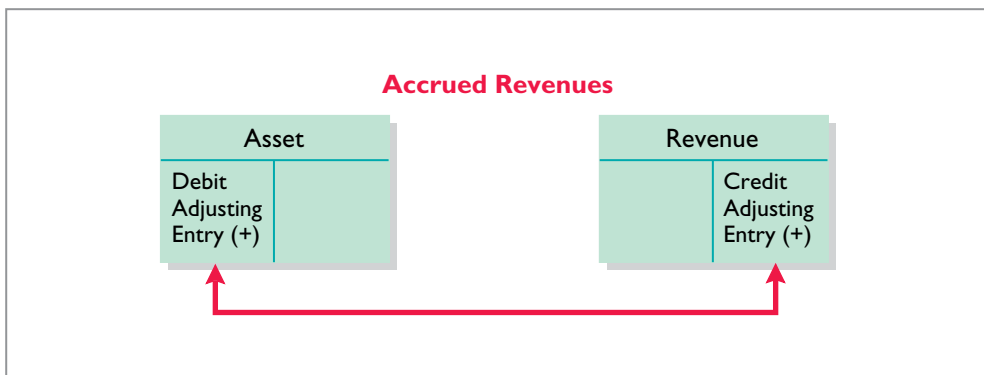
Revenue and receivable are recorded for unbilled services

Nov. 10



Cash is received; receivable is reduced

ILLUSTRATION 3.13 Adjusting entries for accrued revenues



In October, Pioneer Advertising performed services worth \$200 that were not billed to clients on or before October 31. Because these services were not billed, they were not recorded. The accrual of unrecorded service revenue increases an asset account, Accounts Receivable. It also increases owner's equity by increasing a revenue account, Service Revenue, as shown in **Illustration 3.14**.

ILLUSTRATION 3.14 Adjustment for accrued revenue

Basic
Analysis

The asset Accounts Receivable is increased \$200; the revenue Service Revenue is increased \$200.

Equation
Analysis

Assets	=	Liabilities	+	Owner's Equity
Accounts Receivable				Service Revenue
+ \$200				+ \$200

Debit–Credit
Analysis

Debits increase assets: debit Accounts Receivable \$200.
Credits increase revenues: credit Service Revenue \$200.

Journal
Entry

Oct. 31	Accounts Receivable	112	200	
	Service Revenue	400		200
	(To record revenue for services performed)			

Posting

Accounts Receivable			112	Service Revenue			400
Oct. 31	Adj. 200				Oct. 31	10,000	
					31	400	
					31	Adj. 200	
Oct. 31	Bal. 200				Oct. 31	Bal. 10,600	

Equation analyses summarize the effects of transactions on the three elements of the accounting equation, as well as the effect on cash flows.

A	=	L	+	OE
+200				
–200				
Cash Flows				
+200				

The asset Accounts Receivable shows that clients owe Pioneer \$200 at the balance sheet date. The balance of \$10,600 in Service Revenue represents the total revenue for services performed by Pioneer during the month (\$10,000 + \$400 + \$200). **Without the adjusting entry, assets and owner's equity on the balance sheet and revenues and net income on the income statement are understated.**

On November 10, Pioneer receives cash of \$200 for the services performed in October and makes the following entry.

Nov. 10	Cash	200	
	Accounts Receivable		200
	(To record cash collected on account)		

The company records the collection of the receivables by a debit (increase) to Cash and a credit (decrease) to Accounts Receivable.

Illustration 3.15 summarizes the accounting for accrued revenues.

ILLUSTRATION 3.15**Accounting for accrued revenues****ETHICS NOTE**

A report released by **Fannie Mae's** board of directors stated that improper adjusting entries at the mortgage-finance company resulted in delayed recognition of expenses caused by interest rate changes. The motivation for such accounting apparently was the desire to hit earnings estimates.

Accounting for Accrued Revenues

Examples	Reason for Adjustment	Accounts Before Adjustment	Adjusting Entry
Interest, rent, services	Services performed but not yet received in cash or recorded.	Assets understated. Revenues understated.	Dr. Assets Cr. Revenues

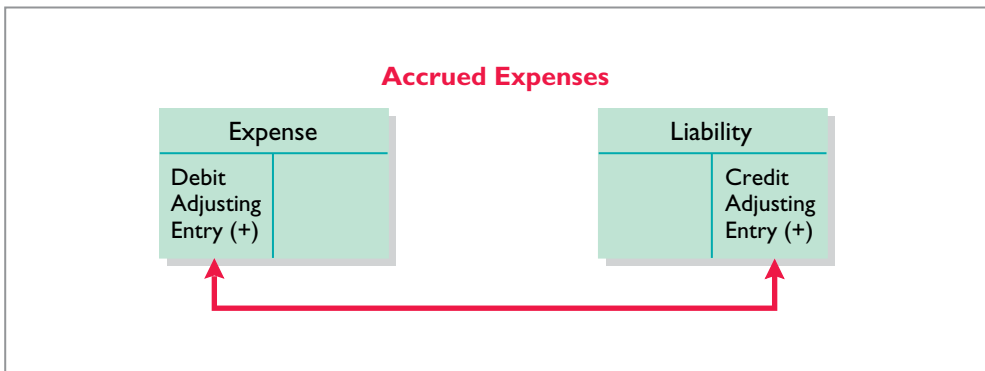
Accrued Expenses

Expenses incurred but not yet paid or recorded at the statement date are called **accrued expenses**. Interest, taxes, and salaries are common examples of accrued expenses.

Companies make adjustments for accrued expenses to record the obligations that exist at the balance sheet date and to recognize the expenses that apply to the current accounting period (see **Ethics Note**). Prior to adjustment, both liabilities and expenses are understated. Therefore, as **Illustration 3.16** shows, **an adjusting entry for accrued expenses results in an increase (a debit) to an expense account and an increase (a credit) to a liability account.**

ILLUSTRATION 3.16

Adjusting entries for accrued expenses



Let's look in more detail at some specific types of accrued expenses, beginning with accrued interest.

Accrued Interest

Pioneer Advertising signed a three-month note payable in the amount of \$5,000 on October 1. The note requires Pioneer to pay interest at an annual rate of 12%.

The amount of the interest recorded is determined by three factors: (1) the face value of the note; (2) the interest rate, which is always expressed as an annual rate; and (3) the length of time the note is outstanding. For Pioneer, the total interest due on the \$5,000 note at its maturity date three months in the future is \$150 ($\$5,000 \times 12\% \times \frac{3}{12}$), or \$50 for one month. **Illustration 3.17** shows the formula for computing interest and its application to Pioneer for the month of October (see **Helpful Hint**).

HELPFUL HINT

In computing interest, we express the time period as a fraction of a year.

Face Value of Note	×	Annual Interest Rate	×	Time in Terms of One Year	=	Interest
\$5,000	×	12%	×	$\frac{1}{12}$	=	\$50

ILLUSTRATION 3.17

Formula for computing interest

As **Illustration 3.18** shows, the accrual of interest at October 31 increases a liability account, Interest Payable. It also decreases owner's equity by increasing an expense account, Interest Expense.

ILLUSTRATION 3.18 Adjustment for accrued interest

Basic
Analysis

The expense Interest Expense is increased \$50; the liability Interest Payable is increased \$50.

Equation
Analysis

$$\begin{array}{rcccl} \text{Assets} & = & \text{Liabilities} & + & \text{Owner's Equity} \\ & & \text{Interest Payable} & & \text{Interest Expense} \\ & & +\$50 & & -\$50 \end{array}$$

Debit–Credit
Analysis

Debits increase expenses: debit Interest Expense \$50.
Credits increase liabilities: credit Interest Payable \$50.

Journal
Entry

Oct. 31	Interest Expense Interest Payable (To record interest on notes payable)	905 230	50	50
---------	---	------------	----	----

Posting

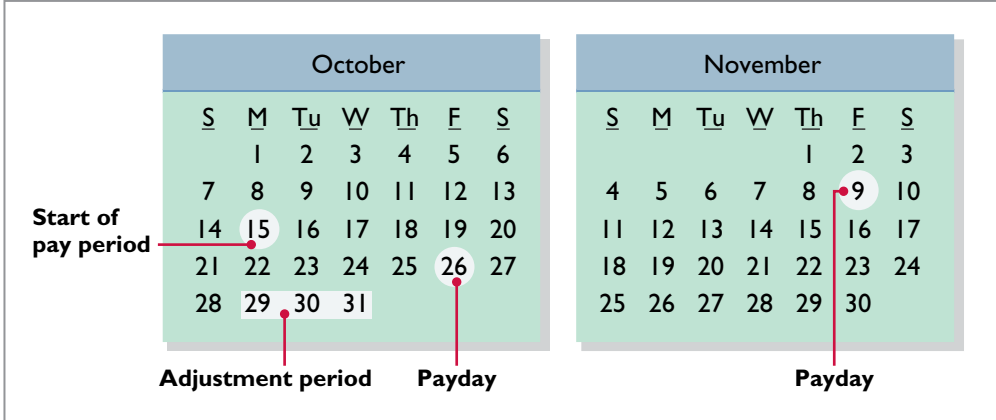
Interest Expense		905	Interest Payable		230
Oct. 31	Adj. 50			Oct. 31	Adj. 50
Oct. 31	Bal. 50			Oct. 31	Bal. 50

Interest Expense shows the interest charges for the month of October. Interest Payable shows the amount of interest the company owes at the statement date. Pioneer will not pay the interest until the note comes due at the end of three months. Companies use the Interest Payable account, instead of crediting Notes Payable, to disclose the two different types of obligations—interest and principal—in the accounts and statements. **Without this adjusting entry, liabilities and interest expense are understated, and net income and owner’s equity are overstated.**

Accrued Salaries and Wages

Companies pay for some types of expenses, such as employee salaries and wages, after the services have been performed. Pioneer Advertising paid salaries and wages on October 26 for its employees’ first two weeks of work. The next payment of salaries will not occur until November 9. As **Illustration 3.19** shows, three working days remain in October (October 29–31).

ILLUSTRATION 3.19
Calendar showing Pioneer’s pay periods



At October 31, the salaries and wages for these three days represent an accrued expense and a related liability to Pioneer. The employees receive total salaries and wages of \$2,000 for a five-day work week, or \$400 per day. Thus, accrued salaries and wages at October 31 are \$1,200 ($\400×3). This accrual increases a liability, Salaries and Wages Payable. It also decreases owner’s equity by increasing an expense account, Salaries and Wages Expense, as shown in **Illustration 3.20**.

ILLUSTRATION 3.20 Adjustment for accrued salaries and wages

Basic Analysis

The expense Salaries and Wages Expense is increased \$1,200; the liability Salaries and Wages Payable is increased \$1,200.

Equation Analysis

Assets	=	Liabilities	+	Owner's Equity
		Salaries and Wages Payable		Salaries and Wages Expense
		+ \$1,200		− \$1,200

Debit–Credit Analysis

Debits increase expenses: debit Salaries and Wages Expense \$1,200.

Credits increase liabilities: credit Salaries and Wages Payable \$1,200.

Journal Entry

Oct. 31	Salaries and Wages Expense Salaries and Wages Payable (To record accrued salaries and wages)	726 212	1,200	1,200
---------	--	------------	-------	-------

Posting

Salaries and Wages Expense		726	Salaries and Wages Payable		212
Oct. 26	4,000			Oct. 31	Adj. 1,200
31	Adj. 1,200				
Oct. 31	Bal. 5,200			Oct. 31	Bal. 1,200

After this adjustment, the balance in Salaries and Wages Expense of \$5,200 (13 days × \$400) is the actual salary and wages expense for October. The balance in Salaries and Wages Payable of \$1,200 is the amount of the liability for salaries and wages Pioneer owes as of October 31. **Without the \$1,200 adjustment for salaries and wages, Pioneer's expenses are understated \$1,200 and its liabilities are understated \$1,200.**

Pioneer pays salaries and wages every two weeks. Consequently, the next payday is November 9, when the company will again pay total salaries and wages of \$4,000. The payment consists of \$1,200 of salaries and wages payable at October 31 plus \$2,800 of salaries and wages expense for November (7 working days, as shown in the November calendar × \$400). Therefore, Pioneer makes the following entry on November 9.

Nov. 9	Salaries and Wages Payable	1,200	
	Salaries and Wages Expense	2,800	
	Cash		4,000
	(To record November 9 payroll)		

A	=	L	+	OE
		-1,200		-2,800
-4,000				
Cash Flows				
-4,000				

This entry eliminates the liability for Salaries and Wages Payable that Pioneer recorded in the October 31 adjusting entry, and it records the proper amount of Salaries and Wages Expense for the period between November 1 and November 9.

Illustration 3.21 summarizes the accounting for accrued expenses.

Accounting for Accrued Expenses

Examples	Reason for Adjustment	Accounts Before Adjustment	Adjusting Entry
Interest, rent, salaries	Expenses have been incurred but not yet paid in cash or recorded.	Expenses understated. Liabilities understated.	Dr. Expenses Cr. Liabilities

ILLUSTRATION 3.21

Accounting for accrued expenses

People, Planet, and Profit Insight



© Nathan Gleave/iStockphoto

Got Junk?

Do you have an old computer or two that you no longer use? How about an old TV that needs replacing? Many people do. Approximately 163,000 computers and televisions become obsolete each day. Yet, in a recent year, only 11% of computers were recycled. It is estimated that 75% of all

computers ever sold are sitting in storage somewhere, waiting to be disposed of. Each of these old TVs and computers is loaded with lead, cadmium, mercury, and other toxic chemicals. If you have one of these electronic gadgets, you have a responsibility, and a probable cost, for disposing of it. Companies have the same problem, but their discarded materials may include lead paint, asbestos, and other toxic chemicals.

What accounting issue might this cause for companies? (Go to WileyPLUS for this answer and additional questions.)

Summary of Basic Relationships

Illustration 3.22 summarizes the four basic types of adjusting entries. Take some time to study and analyze the adjusting entries. Be sure to note that **each adjusting entry affects one balance sheet account and one income statement account.**

ILLUSTRATION 3.22

Summary of adjusting entries

Type of Adjustment

Prepaid expenses

Unearned revenues

Accrued revenues

Accrued expenses

Accounts Before Adjustment

Assets overstated.
Expenses understated.

Liabilities overstated.
Revenues understated.

Assets understated.
Revenues understated.

Expenses understated.
Liabilities understated.

Adjusting Entry

Dr. Expenses
Cr. Assets or
Contra Assets

Dr. Liabilities
Cr. Revenues

Dr. Assets
Cr. Revenues

Dr. Expenses
Cr. Liabilities

Illustrations 3.23 and 3.24 show the journalizing and posting of adjusting entries for Pioneer Advertising on October 31. The ledger identifies all adjustments by the reference J2 because they have been recorded on page 2 of the general journal. The company may insert a center caption “Adjusting Entries” between the last transaction entry and the first adjusting entry in the journal. When you review the general ledger in Illustration 3.24, note that the entries highlighted in red are the adjustments.

ILLUSTRATION 3.23

General journal showing adjusting entries

- (1) Adjusting entries should not involve debits or credits to Cash.
- (2) Evaluate whether the adjustment makes sense. For example, an adjustment to recognize supplies used should increase Supplies Expense.
- (3) Double-check all computations.
- (4) Each adjusting entry affects one balance sheet account and one income statement account.

GENERAL JOURNAL					J2
Date	Account Titles and Explanation	Ref.	Debit	Credit	
2020	Adjusting Entries				
Oct. 31	Supplies Expense	631	1,500		
	Supplies	126		1,500	
	(To record supplies used)				
31	Insurance Expense	722	50		
	Prepaid Insurance	130		50	
	(To record insurance expired)				
31	Depreciation Expense	711	40		
	Accumulated Depreciation—Equipment	158		40	
	(To record monthly depreciation)				
31	Unearned Service Revenue	209	400		
	Service Revenue	400		400	
	(To record revenue for services performed)				
31	Accounts Receivable	112	200		
	Service Revenue	400		200	
	(To record revenue for services performed)				
31	Interest Expense	905	50		
	Interest Payable	230		50	
	(To record interest on notes payable)				
31	Salaries and Wages Expense	726	1,200		
	Salaries and Wages Payable	212		1,200	
	(To record accrued salaries and wages)				

ILLUSTRATION 3.24 General ledger after adjustment

GENERAL LEDGER

Cash					No. 101
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 1		J1	10,000		10,000
2		J1	1,200		11,200
3		J1		900	10,300
4		J1		600	9,700
20		J1		500	9,200
26		J1		4,000	5,200
31		J1	10,000		15,200

Accounts Receivable					No. 112
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31	Adj. entry	J2	200		200

Supplies					No. 126
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 5		J1	2,500		2,500
31	Adj. entry	J2		1,500	1,000

Prepaid Insurance					No. 130
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 4		J1	600		600
31	Adj. entry	J2		50	550

Equipment					No. 157
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 1		J1	5,000		5,000

Accumulated Depreciation—Equipment					No. 158
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31	Adj. entry	J2		40	40

Notes Payable					No. 200
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 1		J1		5,000	5,000

Accounts Payable					No. 201
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 5		J1		2,500	2,500

Unearned Service Revenue					No. 209
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 2		J1		1,200	1,200
31	Adj. entry	J2	400		800

Salaries and Wages Payable					No. 212
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31	Adj. entry	J2		1,200	1,200

Interest Payable					No. 230
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31	Adj. entry	J2		50	50

Owner's Capital					No. 301
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 1		J1		10,000	10,000

Owner's Drawings					No. 306
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 20		J1	500		500

Service Revenue					No. 400
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31		J1		10,000	10,000
31	Adj. entry	J2		400	10,400
31	Adj. entry	J2		200	10,600

Supplies Expense					No. 631
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31	Adj. entry	J2	1,500		1,500

Depreciation Expense					No. 711
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31	Adj. entry	J2	40		40

Insurance Expense					No. 722
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31	Adj. entry	J2	50		50

Salaries and Wages Expense					No. 726
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 26		J1	4,000		4,000
31	Adj. entry	J2	1,200		5,200

Rent Expense					No. 729
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 3		J1	900		900

Interest Expense					No. 905
Date	Explanation	Ref.	Debit	Credit	Balance
2020					
Oct. 31	Adj. entry	J2	50		50

DO IT! 3 | Adjusting Entries for Accruals

Micro Computer Services began operations on August 1, 2020. At the end of August 2020, management prepares monthly financial statements. The following information relates to August.

1. At August 31, the company owed its employees \$800 in salaries and wages that will be paid on September 1.
2. On August 1, the company borrowed \$30,000 from a local bank on a 15-year mortgage. The annual interest rate is 10%.
3. Revenue for services performed but unrecorded for August totaled \$1,100.

Prepare the adjusting entries needed at August 31, 2020.

Solution

1. Salaries and Wages Expense	800	
Salaries and Wages Payable		800
(To record accrued salaries)		
2. Interest Expense	250	
Interest Payable		250
(To record accrued interest:		
$\$30,000 \times 10\% \times \frac{1}{12} = \250)		
3. Accounts Receivable	1,100	
Service Revenue		1,100
(To record revenue for services performed)		

Related exercise material: **BE3.7, DO IT! 3.3, E3.5, E3.6, E3.7, E3.8 and E3.9.**

ACTION PLAN

- Make adjusting entries at the end of the period to recognize revenues for services performed and for expenses incurred.
- Don't forget to make adjusting entries for accruals. Adjusting entries for accruals will increase both a balance sheet and an income statement account.

Adjusted Trial Balance and Financial Statements

LEARNING OBJECTIVE 4

Describe the nature and purpose of an adjusted trial balance.



After a company has journalized and posted all adjusting entries, it prepares another trial balance from the ledger accounts. This trial balance is called an **adjusted trial balance**. It shows the balances of all accounts, including those adjusted, at the end of the accounting period. The purpose of an adjusted trial balance is to **prove the equality** of the total debit balances and the total credit balances in the ledger after all adjustments. Because the accounts contain all data needed for financial statements, the adjusted trial balance is the **primary basis for the preparation of financial statements**.

Preparing the Adjusted Trial Balance

Illustration 3.25 presents the adjusted trial balance for Pioneer Advertising prepared from the ledger accounts in Illustration 3.24. The amounts affected by the adjusting entries are highlighted in red. Compare these amounts to those in the unadjusted trial balance in Illustration 3.3. In this comparison, you will see that there are more accounts in the adjusted trial balance as a result of the adjusting entries made at the end of the month.

ILLUSTRATION 3.25

Adjusted trial balance

Pioneer Advertising Adjusted Trial Balance October 31, 2020		
	Debit	Credit
Cash	\$15,200	
Accounts Receivable	200	
Supplies	1,000	
Prepaid Insurance	550	
Equipment	5,000	
Accumulated Depreciation—Equipment		\$ 40
Notes Payable		5,000
Accounts Payable		2,500
Interest Payable		50
Unearned Service Revenue		800
Salaries and Wages Payable		1,200
Owner's Capital		10,000
Owner's Drawings	500	
Service Revenue		10,600
Salaries and Wages Expense	5,200	
Supplies Expense	1,500	
Rent Expense	900	
Insurance Expense	50	
Interest Expense	50	
Depreciation Expense	40	
	<u>\$30,190</u>	<u>\$30,190</u>

Preparing Financial Statements

Companies can prepare financial statements directly from the adjusted trial balance.

Illustrations 3.26 and **3.27** present the interrelationships of data in the adjusted trial balance and the financial statements.

As Illustration 3.26 shows, companies prepare the income statement from the revenue and expense accounts. Next, they use the owner's capital and drawings accounts and the net income (or net loss) from the income statement to prepare the owner's equity statement.

ILLUSTRATION 3.26 Preparation of the income statement and owner's equity statement from the adjusted trial balance

Pioneer Advertising Adjusted Trial Balance October 31, 2020			Pioneer Advertising Income Statement For the Month Ended October 31, 2020	
Account	Debit	Credit		
Cash	\$15,200		Revenues	
Accounts Receivable	200		Service revenue	\$10,600
Supplies	1,000		Expenses	
Prepaid Insurance	550		Salaries and wages expense	\$5,200
Equipment	5,000		Supplies expense	1,500
Accumulated Depreciation— Equipment		\$ 40	Rent expense	900
Notes Payable		5,000	Insurance expense	50
Accounts Payable		2,500	Interest expense	50
Unearned Service Revenue		800	Depreciation expense	40
Salaries and Wages Payable		1,200	Total expenses	7,740
Interest Payable		50	Net income	\$ 2,860
Owner's Capital		10,000		
Owner's Drawings	500			
Service Revenue		10,600		
Salaries and Wages Expense	5,200			
Supplies Expense	1,500			
Rent Expense	900			
Insurance Expense	50			
Interest Expense	50			
Depreciation Expense	40			
	<u>\$30,190</u>	<u>\$30,190</u>		

Pioneer Advertising Owner's Equity Statement For the Month Ended October 31, 2020	
Owner's capital, October 1	\$ -0-
Add: Investments	10,000
	<u>10,000</u>
Net income	2,860
	<u>12,860</u>
Owner's drawings	500
Owner's capital, October 31	<u>\$12,360</u>

To balance sheet

As Illustration 3.27 shows, companies then prepare the balance sheet from the asset and liability accounts and the ending owner's capital balance as reported in the owner's equity statement.

ILLUSTRATION 3.27 Preparation of the balance sheet from the adjusted trial balance

Pioneer Advertising Adjusted Trial Balance October 31, 2020			Pioneer Advertising Balance Sheet October 31, 2020	
Account	Debit	Credit	<u>Assets</u>	
Cash	\$15,200		Cash	\$15,200
Accounts Receivable	200		Accounts receivable	200
Supplies	1,000		Supplies	1,000
Prepaid Insurance	550		Prepaid insurance	550
Equipment	5,000		Equipment	\$5,000
Accumulated Depreciation—Equipment		\$ 40	Less: Accumulated depreciation—equip.	40
Notes Payable		5,000		4,960
Accounts Payable		2,500	Total assets	\$21,910
Unearned Service Revenue		800	<u>Liabilities and Owner's Equity</u>	
Salaries and Wages Payable		1,200	Liabilities	
Interest Payable		50	Notes payable	\$ 5,000
Owner's Capital		10,000	Accounts payable	2,500
Owner's Drawings	500		Unearned service revenue	800
Service Revenue		10,600	Salaries and wages payable	1,200
Salaries and Wages Expense	5,200		Interest payable	50
Supplies Expense	1,500		Total liabilities	9,550
Rent Expense	900		Owner's equity	
Insurance Expense	50		Owner's capital	12,360
Interest Expense	50		Total liabilities and owner's equity	\$21,910
Depreciation Expense	40			
	<u>\$30,190</u>	<u>\$30,190</u>		

Capital balance at Oct. 31 from owner's equity statement in Illustration 3.26

DO IT! 4 | Trial Balance

Skolnick Co. was organized on April 1, 2020. The company prepares quarterly financial statements. The adjusted trial balance amounts at June 30 are shown below.

	Debit		Credit
Cash	\$ 6,700	Accumulated Depreciation—Equipment	\$ 850
Accounts Receivable	600	Notes Payable	5,000
Prepaid Rent	900	Accounts Payable	1,510
Supplies	1,000	Salaries and Wages Payable	400
Equipment	15,000	Interest Payable	50
Owner's Drawings	600	Unearned Rent Revenue	500
Salaries and Wages Expense	9,400	Owner's Capital	14,000
Rent Expense	1,500	Service Revenue	14,200
Depreciation Expense	850	Rent Revenue	800
Supplies Expense	200		
Utilities Expense	510		
Interest Expense	50		
	<u>\$37,310</u>		<u>\$37,310</u>

- Determine the net income for the quarter April 1 to June 30.
- Determine the total assets and total liabilities at June 30, 2020, for Skolnick Co.
- Determine the amount of owner's capital at June 30, 2020.

ACTION PLAN

- In an adjusted trial balance, all asset, liability, revenue, and expense accounts are properly stated.
- To determine the ending balance in Owner's Capital, add net income and subtract dividends.

Solution

- a. The net income is determined by adding revenues and subtracting expenses. The net income is computed as follows.

Revenues		
Service revenue	\$14,200	
Rent revenue	800	
Total revenues		\$15,000
Expenses		
Salaries and wages expense	9,400	
Rent expense	1,500	
Depreciation expense	850	
Utilities expense	510	
Supplies expense	200	
Interest expense	50	
Total expenses		12,510
Net income		<u>\$ 2,490</u>

- b. Total assets and liabilities are computed as follows.

Assets		Liabilities	
Cash	\$ 6,700	Notes payable	\$5,000
Accounts receivable	600	Accounts payable	1,510
Supplies	1,000	Unearned rent revenue	500
Prepaid rent	900	Salaries and wages payable	400
Equipment	\$15,000	Interest payable	50
Less: Accumulated depreciation—equipment	850		
	<u>14,150</u>		
Total assets	<u>\$23,350</u>	Total liabilities	<u>\$7,460</u>

c. Owner's capital, April 1	\$ 0
Add: Investments	14,000
Net income	2,490
Less: Owner's drawings	600
Owner's capital, June 30	<u>\$15,890</u>

Related exercise material: **BE3.9, BE3.10, DO IT! 3.4, and E3.11.**

Appendix 3A**Alternative Treatment of Deferrals****LEARNING OBJECTIVE *5**

Prepare adjusting entries for the alternative treatment of deferrals.

In discussing adjusting entries for prepaid expenses and unearned revenues, we illustrated transactions for which companies made the initial entries to balance sheet accounts. In the case of prepaid expenses, the company debited the prepayment to an asset account. In the case of unearned revenue, the company credited a liability account to record the cash received.

Some companies use an alternative treatment. (1) When a company prepays an expense, it debits that amount to an expense account. (2) When it receives payment for future services, it credits the amount to a revenue account. In this appendix, we describe the circumstances

that justify such entries and the different adjusting entries that may be required. This alternative treatment of prepaid expenses and unearned revenues has the same effect on the financial statements as the procedures described in the chapter.

Prepaid Expenses

Prepaid expenses become expired costs either through the passage of time (e.g., insurance) or through consumption (e.g., advertising supplies). If at the time of purchase the company expects to consume the supplies before the next financial statement date, **it may choose to debit (increase) an expense account rather than an asset account. This alternative treatment is simply more convenient.**

Assume that Pioneer Advertising expects that it will use before the end of the month all of the supplies purchased on October 5. A debit of \$2,500 to Supplies Expense (rather than to the asset account Supplies) on October 5 will eliminate the need for an adjusting entry on October 31. At October 31, the Supplies Expense account will show a balance of \$2,500, which is the cost of supplies used between October 5 and October 31.

But what if the company does not use all the supplies? For example, what if an inventory of \$1,000 of advertising supplies remains on October 31? Obviously, the company would need to make an adjusting entry. Prior to adjustment, the expense account Supplies Expense is overstated \$1,000, and the asset account Supplies is understated \$1,000. Thus, Pioneer makes the following adjusting entry.

Oct. 31	Supplies	1,000	
	Supplies Expense		1,000
	(To record supplies inventory)		

A	=	L	+	OE
+1,000				
				+1,000 Exp

Cash Flows
no effect

Illustration 3A.1 shows the accounts after the company posts the adjusting entry.

Supplies				Supplies Expense			
10/31	Adj.	1,000		10/5	2,500	10/31	Adj. 1,000
				10/31	Bal. 1,500		

ILLUSTRATION 3A.1

Prepaid expenses accounts after adjustment

After adjustment, the asset account Supplies shows a balance of \$1,000, which is equal to the cost of supplies on hand at October 31. In addition, Supplies Expense shows a balance of \$1,500. This is equal to the cost of supplies used between October 5 and October 31. Without the adjusting entry, expenses are overstated and net income is understated by \$1,000 in the October income statement. Also, both assets and owner's equity are understated by \$1,000 on the October 31 balance sheet.

Illustration 3A.2 compares the entries and accounts for advertising supplies in the two adjustment approaches.

Prepayment Initially Debited to Asset Account (per chapter)				Prepayment Initially Debited to Expense Account (per appendix)			
Oct. 5	Supplies	2,500		Oct. 5	Supplies Expense	2,500	
	Accounts Payable		2,500		Accounts Payable		2,500
Oct. 31	Supplies Expense	1,500		Oct. 31	Supplies	1,000	
	Supplies		1,500		Supplies Expense		1,000

ILLUSTRATION 3A.2

Adjustment approaches—a comparison

After Pioneer posts the entries, the accounts appear as shown in **Illustration 3A.3**.

ILLUSTRATION 3A.3

Comparison of accounts

(per chapter)				(per appendix)			
Supplies				Supplies			
10/5	2,500	10/31	Adj. 1,500	10/31	Adj. 1,000		
10/31	Bal. 1,000						
Supplies Expense				Supplies Expense			
10/31	Adj. 1,500			10/5	2,500	10/31	Adj. 1,000
				10/31	Bal. 1,500		

Note that the account balances under each alternative are the same at October 31: Supplies \$1,000 and Supplies Expense \$1,500.

Unearned Revenues

Unearned revenues are recognized as revenue at the time services are performed. Similar to the case for prepaid expenses, companies may credit (increase) a revenue account when they receive cash for future services.

To illustrate, assume that Pioneer Advertising received \$1,200 for future services on October 2. Pioneer expects to perform the services before October 31.¹ In such a case, the company credits Service Revenue. If Pioneer in fact performs the service before October 31, no adjustment is needed.

However, if at the statement date Pioneer has not performed \$800 of the services, it would make an adjusting entry (see **Helpful Hint**). Without the entry, the revenue account Service Revenue is overstated \$800, and the liability account Unearned Service Revenue is understated \$800. Thus, Pioneer makes the following adjusting entry.

HELPFUL HINT

The required adjusted balances here are Service Revenue \$400 and Unearned Service Revenue \$800.

A	=	L	+	OE
				-800 Rev
		+800		

Cash Flows
no effect

Oct. 31	Service Revenue	800	
	Unearned Service Revenue		800
	(To record unearned service revenue)		

Illustration 3A.4 shows the accounts after Pioneer posts the adjusting entry.

ILLUSTRATION 3A.4

Unearned service revenue accounts after adjustment

Unearned Service Revenue				Service Revenue				
	10/31	Adj.	800	10/31	Adj.	800	10/2	1,200
							10/31	Bal. 400

The liability account Unearned Service Revenue shows a balance of \$800. This equals the services that will be performed in the future. In addition, the balance in Service Revenue equals the services performed in October. Without the adjusting entry, both revenues and net income are overstated by \$800 in the October income statement. Also, liabilities are understated by \$800 and owner's equity is overstated by \$800 on the October 31 balance sheet.

¹This example focuses only on the alternative treatment of unearned revenues. For simplicity, we have ignored the entries to Service Revenue pertaining to the immediate recognition of revenue (\$10,000) and the adjusting entry for accrued revenue (\$200).

Illustration 3A.5 compares the entries and accounts for initially recording unearned service revenue in (1) a liability account or (2) a revenue account.

Unearned Service Revenue Initially Credited to Liability Account (per chapter)			Unearned Service Revenue Initially Credited to Revenue Account (per appendix)		
Oct. 2	Cash	1,200	Oct. 2	Cash	1,200
	Unearned Service Revenue	1,200		Service Revenue	1,200
Oct. 31	Unearned Service Revenue	400	Oct. 31	Service Revenue	800
	Service Revenue	400		Unearned Service Revenue	800

ILLUSTRATION 3A.5

Adjustment approaches—
a comparison

After Pioneer posts the entries, the accounts appear as shown in **Illustration 3A.6**.

(per chapter) Unearned Service Revenue				(per appendix) Unearned Service Revenue					
10/31	Adj.	400	10/2	1,200		10/31	Adj.	800	
			10/31	Bal.	800				
Service Revenue				Service Revenue					
	10/31	Adj.	400	10/31	Adj.	800	10/2	1,200	
							10/31	Bal.	400

ILLUSTRATION 3A.6

Comparison of accounts

Note that the balances in the accounts are the same under the two alternatives: Unearned Service Revenue \$800 and Service Revenue \$400.

Summary of Additional Adjustment Relationships

Illustration 3A.7 provides a summary of basic relationships for deferrals.

ILLUSTRATION 3A.7 Summary of basic relationships for deferrals

Type of Adjustment	Reason for Adjustment	Account Balances before Adjustment	Adjusting Entry
1. Prepaid expenses	(a) Prepaid expenses initially recorded in asset accounts have been used. (b) Prepaid expenses initially recorded in expense accounts have not been used.	Assets overstated. Expenses understated. Assets understated. Expenses overstated.	Dr. Expenses Cr. Assets Dr. Assets Cr. Expenses
2. Unearned revenues	(a) Unearned revenues initially recorded in liability accounts are now recognized as revenue. (b) Unearned revenues initially recorded in revenue accounts are still unearned.	Liabilities overstated. Revenues understated. Liabilities understated. Revenues overstated.	Dr. Liabilities Cr. Revenues Dr. Revenues Cr. Liabilities

Alternative adjusting entries **do not apply** to accrued revenues and accrued expenses because **no entries occur before companies make these types of adjusting entries.**

Appendix 3B

Financial Reporting Concepts

LEARNING OBJECTIVE *6

Discuss financial reporting concepts.

This appendix provides a summary of the concepts in action used in this textbook. In addition, it provides other useful concepts which accountants use as a basis for recording and reporting financial information.

Qualities of Useful Information

Recently, the FASB completed the first phase of a project in which it developed a conceptual framework to serve as the basis for future accounting standards. The framework begins by stating that the primary objective of financial reporting is to provide financial information that is **useful** to investors and creditors for making decisions about providing capital. Useful information should possess two fundamental qualities, relevance and faithful representation, as shown in **Illustration 3B.1**.

ILLUSTRATION 3B.1

Fundamental qualities of useful information



Relevance Accounting information has **relevance** if it would make a difference in a business decision. Information is considered relevant if it provides information that has **predictive value**, that is, helps provide accurate expectations about the future, and has **confirmatory value**, that is, confirms or corrects prior expectations. **Materiality** is a company-specific aspect of relevance. An item is material when its **size** makes it likely to influence the decision of an investor or creditor.



Faithful Representation **Faithful representation** means that information accurately depicts what really happened. To provide a faithful representation, information must be **complete** (nothing important has been omitted), **neutral** (is not biased toward one position or another), and **free from error**.

Enhancing Qualities

In addition to the two fundamental qualities, the FASB also describes a number of enhancing qualities of useful information. These include **comparability**, **verifiability**, **timeliness**, and **understandability**. In accounting, **comparability** results when different companies use the same accounting principles. Another type of comparability is consistency. **Consistency** means that a company uses the same accounting principles and methods from year to year. Information is **verifiable** if independent observers, using the same methods, obtain similar results. For accounting information to have relevance, it must be **timely**. That is, it must be available to decision-makers before it loses its capacity to influence decisions. For example, public companies like **Google** or **Best Buy** provide their annual reports to investors within 60 days of their year-end. Information has the quality of **understandability** if it is presented in a clear and concise fashion, so that reasonably informed users of that information can interpret it and comprehend its meaning.

Assumptions in Financial Reporting

To develop accounting standards, the FASB relies on some key assumptions, as shown in **Illustration 3B.2**. These include assumptions about the monetary unit, economic entity, time period, and going concern.

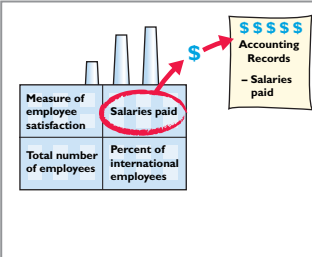
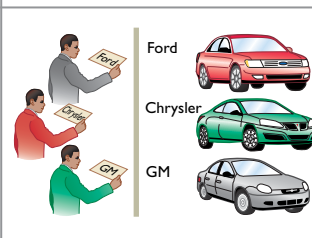
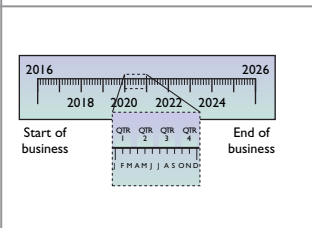
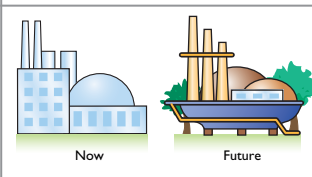
	Monetary Unit Assumption The monetary unit assumption requires that only those things that can be expressed in money are included in the accounting records. This means that certain important information needed by investors, creditors, and managers, such as customer satisfaction, is not reported in the financial statements. This assumption relies on the monetary unit remaining relatively stable in value.
	Economic Entity Assumption The economic entity assumption states that every economic entity can be separately identified and accounted for. In order to assess a company's performance and financial position accurately, it is important to not blur company transactions with personal transactions (especially those of its managers) or transactions of other companies.
	Time Period Assumption Notice that the income statement, retained earnings statement, and statement of cash flows all cover periods of one year, and the balance sheet is prepared at the end of each year. The time period assumption states that the life of a business can be divided into artificial time periods and that useful reports covering those periods can be prepared for the business.
	Going Concern Assumption The going concern assumption states that the business will remain in operation for the foreseeable future. Of course, many businesses do fail, but in general it is reasonable to assume that the business will continue operating.

ILLUSTRATION 3B.2**Key assumptions in financial reporting**

Principles in Financial Reporting

Measurement Principles

GAAP generally uses one of two measurement principles, the historical cost principle or the fair value principle. Selection of which principle to follow generally relates to trade-offs between relevance and faithful representation.

Historical Cost Principle. The **historical cost principle** (or cost principle, discussed in Chapter 1) dictates that companies record assets at their cost. This is true not only at the time the asset is purchased but also over the time the asset is held. For example, if land that was purchased for \$30,000 increases in value to \$40,000, it continues to be reported at \$30,000.

Fair Value Principle. The **fair value principle** (discussed in Chapter 1) indicates that assets and liabilities should be reported at fair value (the price received to sell an asset or settle a liability). Fair value information may be more useful than historical cost for certain types of assets and liabilities. For example, certain investment securities are reported at fair value because market price information is often readily available for these types of assets. In choosing between cost and fair value, two qualities that make accounting information useful for decision-making are used—relevance and faithful representation. In determining which measurement principle to use, the factual nature of cost figures are weighed versus the relevance of fair value. In general, most assets follow the historical cost principle because fair values may not be representationally faithful. Only in situations where assets are actively traded, such as investment securities, is the fair value principle applied.

Revenue Recognition Principle

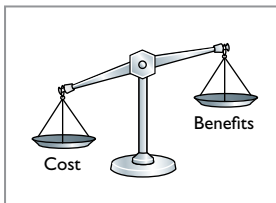
The **revenue recognition principle** requires that companies recognize revenue in the accounting period in which the performance obligation is satisfied. As discussed earlier in the chapter, in a service company, revenue is recognized at the time the service is performed. In a merchandising company, the performance obligation is generally satisfied when the goods transfer from the seller to the buyer (discussed in Chapter 5). At this point, the sales transaction is complete and the sales price established.

Expense Recognition Principle

The **expense recognition principle** (discussed earlier in the chapter) dictates that companies recognize expense in the period in which they make efforts to generate revenue. Thus, expenses follow revenues.

Full Disclosure Principle

The **full disclosure principle** (discussed in Chapter 11) requires that companies disclose all circumstances and events that would make a difference to financial statement users. If an important item cannot reasonably be reported directly in one of the four types of financial statements, then it should be discussed in notes that accompany the statements.



Cost Constraint

Providing information is costly. In deciding whether companies should be required to provide a certain type of information, accounting standard-setters consider the **cost constraint**. It weighs the cost that companies will incur to provide the information against the benefit that financial statement users will gain from having the information available.

Review and Practice

Learning Objectives Review

1 Explain the accrual basis of accounting and the reasons for adjusting entries.

The time period assumption assumes that the economic life of a business is divided into artificial time periods. Accrual-basis accounting means that companies record events that change a company's financial statements in the periods in which those events occur, rather than in the periods in which the company receives or pays cash.

Companies make adjusting entries at the end of an accounting period. Such entries ensure that companies recognize revenues in the period in which the performance obligation is satisfied and recognize expenses in the period in which they are incurred. The major types of adjusting entries are deferrals (prepaid expenses and unearned revenues) and accruals (accrued revenues and accrued expenses).

2 Prepare adjusting entries for deferrals.

Deferrals are either prepaid expenses or unearned revenues. Companies make adjusting entries for deferrals to record the portion of the prepayment that represents the expense incurred or the revenue for services performed in the current accounting period.

3 Prepare adjusting entries for accruals.

Accruals are either accrued revenues or accrued expenses. Companies make adjusting entries for accruals to record revenues for services performed and expenses incurred in the current accounting period that have not been recognized through daily entries.

4 Describe the nature and purpose of an adjusted trial balance.

An adjusted trial balance shows the balances of all accounts, including those that have been adjusted, at the end of an accounting period. Its purpose is to prove the equality of the total debit balances and total credit balances in the ledger after all adjustments.

***5 Prepare adjusting entries for the alternative treatment of deferrals.**

Companies may initially debit prepayments to an expense account. Likewise, they may credit unearned revenues to a revenue account. At the end of the period, these accounts may be overstated. The adjusting entries for prepaid expenses are a debit to an asset account and a credit to an expense account. Adjusting entries for unearned revenues are a debit to a revenue account and a credit to a liability account.

***6 Discuss financial reporting concepts.**

To be judged useful, information should have the primary characteristics of relevance and faithful representation. In addition, it

should be comparable, consistent, verifiable, timely, and understandable.

The **monetary unit assumption** requires that companies include in the accounting records only transaction data that can be expressed in terms of money. The **economic entity assumption** states that economic events can be identified with a particular unit of accountability. The **time period assumption** states that the economic life of a business can be divided into artificial time periods and that meaningful accounting reports can be prepared for each period. The **going concern assumption** states that the company will continue in operation long enough to carry out its existing objectives and commitments.

The **historical cost principle** states that companies should record assets at their cost. The **fair value principle** indicates that assets and liabilities should be reported at fair value. The **revenue recognition principle** requires that companies recognize revenue in the accounting period in which the performance obligation is satisfied. The **expense recognition principle** dictates that efforts (expenses) be matched with results (revenues). The **full disclosure principle** requires that companies disclose circumstances and events that matter to financial statement users.

The **cost constraint** weighs the cost that companies incur to provide a type of information against its benefits to financial statement users.

Glossary Review

Accrual-basis accounting Accounting basis in which companies record transactions that change a company's financial statements in the periods in which the events occur. (p. 3-3).

Accruals Adjusting entries for either accrued revenues or accrued expenses. (p. 3-5).

Accrued expenses Expenses incurred but not yet paid in cash or recorded. (p. 3-14).

Accrued revenues Revenues for services performed but not yet received in cash or recorded. (p. 3-13).

Adjusted trial balance A list of accounts and their balances after the company has made all adjustments. (p. 3-20).

Adjusting entries Entries made at the end of an accounting period to ensure that companies follow the revenue recognition and expense recognition principles. (p. 3-5).

Book value The difference between the cost of a depreciable asset and its related accumulated depreciation. (p. 3-10).

Calendar year An accounting period that extends from January 1 to December 31. (p. 3-3).

Cash-basis accounting Accounting basis in which companies record revenue when they receive cash and an expense when they pay out cash. (p. 3-3).

***Comparability** Ability to compare the accounting information of different companies because they use the same accounting principles. (p. 3-28).

***Consistency** Use of the same accounting principles and methods from year to year within a company. (p. 3-28).

Contra asset account An account offset against an asset account on the balance sheet. (p. 3-9).

***Cost constraint** Constraint that weighs the cost that companies will incur to provide the information against the benefit that financial statement users will gain from having the information available. (p. 3-30).

Deferrals Adjusting entries for either prepaid expenses or unearned revenues. (p. 3-5).

Depreciation The process of allocating the cost of an asset to expense over its useful life. (p. 3-9).

***Economic entity assumption** An assumption that every economic entity can be separately identified and accounted for. (p. 3-29).

Expense recognition principle (matching principle) The principle that companies recognize expense in the period in which they make efforts (consume assets or incur liabilities) to generate revenue. (pp. 3-4, 3-30).

***Fair value principle** Assets and liabilities should be reported at fair value (the price received to sell an asset or settle a liability). (p. 3-29).

***Faithful representation** Information that accurately depicts what really happened. (p. 3-28).

Fiscal year An accounting period that is one year in length. (p. 3-3).

***Full disclosure principle** Accounting principle that dictates that companies disclose circumstances and events that make a difference to financial statement users. (p. 3-30).

***Going concern assumption** The assumption that the company will continue in operation for the foreseeable future. (p. 3-29).

***Historical cost principle** An accounting principle that states that companies should record assets at their cost. (p. 3-29).

Interim periods Monthly or quarterly accounting time periods. (p. 3-3).

***Materiality** A company-specific aspect of relevance. An item is material when its size makes it likely to influence the decision of an investor or creditor. (p. 3-28).

***Monetary unit assumption** An assumption that requires that only those things that can be expressed in money are included in the accounting records. (p. 3-29).

Prepaid expenses (prepayments) Expenses paid in cash before they are used or consumed. (p. 3-6).

***Relevance** The quality of information that indicates the information makes a difference in a decision. (p. 3-28).

Revenue recognition principle The principle that companies recognize revenue in the accounting period in which the performance obligation is satisfied. (pp. 3-3, 3-30).

***Timely** Information that is available to decision-makers before it loses its capacity to influence decisions. (p. 3-28).

Time period assumption An assumption that accountants can divide the economic life of a business into artificial time periods. (pp. 3-3, 3-29).

***Understandability** Information presented in a clear and concise fashion so that users can interpret it and comprehend its meaning. (p. 3-28).

Unearned revenues A liability recorded for cash received before services are performed. (p. 3-10).

Useful life The length of service of a long-lived asset. (p. 3-8).

***Verifiable** The quality of information that occurs when independent observers, using the same methods, obtain similar results. (p. 3-28).

Practice Multiple-Choice Questions

1. (LO 1) The revenue recognition principle states that:

- a. revenue should be recognized in the accounting period in which a performance obligation is satisfied.
- b. expenses should be matched with revenues.
- c. the economic life of a business can be divided into artificial time periods.
- d. the fiscal year should correspond with the calendar year.

2. (LO 1) The time period assumption states that:

- a. companies must wait until the calendar year is completed to prepare financial statements.
- b. companies use the fiscal year to report financial information.
- c. the economic life of a business can be divided into artificial time periods.
- d. companies record information in the time period in which the events occur.

3. (LO 1) Which of the following statements about the accrual basis of accounting is **false**?

- a. Events that change a company's financial statements are recorded in the periods in which the events occur.
- b. Revenue is recognized in the period in which services are performed.
- c. This basis is in accordance with generally accepted accounting principles.
- d. Revenue is recorded only when cash is received, and expense is recorded only when cash is paid.

4. (LO 1) The principle or assumption dictating that efforts (expenses) should be recognized in the period in which a company consumes assets or incurs liabilities to generate revenue is the:

- a. expense recognition principle.
- b. cost assumption.
- c. time period assumption.
- d. revenue recognition principle.

5. (LO 1) Adjusting entries are made to ensure that:

- a. expenses are recognized in the period in which they are incurred.
- b. revenues are recorded in the period in which services are performed.
- c. balance sheet and income statement accounts have correct balances at the end of an accounting period.
- d. All the responses above are correct.

6. (LO 1) Each of the following is a major type (or category) of adjusting entries **except**:

- a. prepaid expenses.
- b. accrued revenues.
- c. accrued expenses.
- d. recognized revenues.

7. (LO 2) The trial balance shows Supplies \$1,350 and Supplies Expense \$0. If \$600 of supplies are on hand at the end of the period, the adjusting entry is:

a. Supplies	600	
Supplies Expense		600
b. Supplies	750	
Supplies Expense		750
c. Supplies Expense	750	
Supplies		750
d. Supplies Expense	600	
Supplies		600

8. (LO 2) Adjustments for prepaid expenses:

- a. decrease assets and increase revenues.
- b. decrease expenses and increase assets.
- c. decrease assets and increase expenses.
- d. decrease revenues and increase assets.

9. (LO 2) Accumulated Depreciation is:

- a. a contra asset account.
- b. an expense account.
- c. an owner's equity account.
- d. a liability account.

10. (LO 2) Rivera Company computes depreciation on delivery equipment at \$1,000 for the month of June. The adjusting entry to record this depreciation is as follows.

a. Depreciation Expense	1,000	
Accumulated Depreciation— Rivera Company		1,000
b. Depreciation Expense	1,000	
Equipment		1,000
c. Depreciation Expense	1,000	
Accumulated Depreciation— Equipment		1,000
d. Equipment Expense	1,000	
Accumulated Depreciation— Equipment		1,000

11. (LO 2) Adjustments for unearned revenues:

- a. decrease liabilities and increase revenues.
- b. have an assets-and-revenues-account relationship.
- c. increase assets and increase revenues.
- d. decrease revenues and decrease assets.

12. (LO 3) Adjustments for accrued revenues:

- a. have a liabilities-and-revenues-account relationship.
- b. have an assets-and-revenues-account relationship.
- c. decrease assets and revenues.
- d. decrease liabilities and increase revenues.

13. (LO 3) Anika Wilson earned a salary of \$400 for the last week of September. She will be paid on October 1. The adjusting entry for Anika's employer at September 30 is:

- a. No entry is required.
- b. Salaries and Wages Expense 400
 Salaries and Wages Payable 400
- c. Salaries and Wages Expense 400
 Cash 400
- d. Salaries and Wages Payable 400
 Cash 400

14. (LO 4) Which of the following statements is **incorrect** concerning the adjusted trial balance?

- a. An adjusted trial balance proves the equality of the total debit balances and the total credit balances in the ledger after all adjustments are made.
- b. The adjusted trial balance provides the primary basis for the preparation of financial statements.
- c. The adjusted trial balance lists the account balances segregated by assets and liabilities.

d. The adjusted trial balance is prepared after the adjusting entries have been journalized and posted.

***15. (LO 5)** The trial balance shows Supplies \$0 and Supplies Expense \$1,500. If \$800 of supplies are on hand at the end of the period, the adjusting entry is:

- a. debit Supplies \$800 and credit Supplies Expense \$800.
- b. debit Supplies Expense \$800 and credit Supplies \$800.
- c. debit Supplies \$700 and credit Supplies Expense \$700.
- d. debit Supplies Expense \$700 and credit Supplies \$700.

***16. (LO 6)** Neutrality is an ingredient of:

	<u>Faithful Representation</u>	<u>Relevance</u>
a.	Yes	Yes
b.	No	No
c.	Yes	No
d.	No	Yes

***17. (LO 6)** Which item is a constraint in financial accounting?

- a. Comparability. c. Cost.
- b. Materiality. d. Consistency.

Solutions

1. a. Revenue should be recognized in the accounting period in which a performance obligation is satisfied. The other choices are incorrect because (b) defines the expense recognition principle, (c) describes the time period assumption, and (d) a company's fiscal year does not need to correspond with the calendar year.

2. c. The economic life of a business can be divided into artificial time periods. The other choices are incorrect because (a) companies report their activities on a more frequent basis and not necessarily based on a calendar year; (b) companies report financial information more frequently than annually, such as monthly or quarterly, in order to evaluate results of operations; and (d) describes accrual-basis accounting.

3. d. Under the accrual basis of accounting, revenue is recognized when the performance obligation is satisfied, not when cash is received, and expense is recognized when incurred, not when cash is paid. The other choices are all true statements.

4. a. The expense recognition principle dictates that companies recognize expense in the period in which they make efforts to generate revenue. The other choices are incorrect because (b) there is no cost assumption, but the historical cost principle states that assets should be recorded at their cost; (c) the time period assumption states that the economic life of a business can be divided into artificial time periods; and (d) the revenue recognition principle indicates that revenue should be recognized in the accounting period in which a performance obligation is satisfied.

5. d. Adjusting entries are made for the reasons noted in choices (a), (b), and (c). The other choices are true statements, but (d) is the best answer.

6. d. Unearned revenues, not recognized revenues, is one of the major categories of adjusting entries. The other choices all list one of the major categories of adjusting entries.

7. c. Debiting Supplies Expense for \$750 and crediting Supplies for \$750 (\$1,350 – \$600) will decrease Supplies and increase Supplies Expense. The other choices are incorrect because (a) will increase

Supplies and decrease Supplies Expense and also for the wrong amounts, (b) will increase Supplies and decrease Supplies Expense, and (d) will cause Supplies to have an incorrect balance of \$750 (\$1,350 – \$600) and Supplies Expense to have an incorrect balance of \$600 (\$0 + \$600).

8. c. Adjustments for prepaid expenses decrease assets and increase expenses. The other choices are incorrect because an adjusting entry for prepaid expenses (a) increases expenses, not revenues; (b) increases, not decreases, expenses and decreases, not increases, assets; and (d) increases expenses, not decreases, revenues and decreases, not increases, assets.

9. a. Accumulated Depreciation is a contra asset account; it is offset against an asset account on the balance sheet. The other choices are incorrect because Accumulated Depreciation is not (b) an expense account nor located on the income statement, (c) an owner's equity account, or (d) a liability account.

10. c. The adjusting entry is to debit Depreciation Expense and credit Accumulated Depreciation—Equipment. The other choices are incorrect because (a) the contra asset account title includes the asset being depreciated, not the company name; (b) the credit should be to the contra asset account, not directly to the asset; and (d) the debit for this entry should be Depreciation Expense, not Equipment Expense.

11. a. Adjustments for unearned revenues will consist of a debit (decrease) to unearned revenues (a liability) and a credit (increase) to a revenue account. Choices (b), (c), and (d) are incorrect because adjustments for unearned revenues will increase revenues but will have no effect on assets.

12. b. Adjustments for accrued revenues will have an assets-and-revenues-account relationship. Choices (a) and (d) are incorrect because adjustments for accrued revenues have no effect on liabilities. Choice (c) is incorrect because these adjustments will increase, not decrease, both assets and revenues.

13. b. The adjusting entry should be to debit Salaries and Wages Expense for \$400 and credit Salaries and Wages Payable for \$400.

The other choices are incorrect because (a) if an adjusting entry is not made, the amount of money owed (liability) that is shown on the balance sheet will be understated and the amount of salaries and wages expense will also be understated; (c) the credit account is incorrect as adjusting entries never affect cash; and (d) the debit account should be Salaries and Wages Expense and the credit account should be Salaries and Wages Payable. Adjusting entries never affect cash.

14. c. The accounts on the trial balance can be segregated by the balance in the account—either debit or credit—not whether they are assets or liabilities. All accounts in the ledger are included in the adjusted trial balance, not just assets and liabilities. The other choices are all true statements.

***15. a.** This adjusting entry correctly states the Supplies account at \$800 ($\$0 + \800) and the Supplies Expense account at \$700

($\$1,500 - \800). The other choices are incorrect because (b) will cause the Supplies account to have a credit balance (assets have a normal debit balance) and the Supplies Expense account to be stated at \$2,300, which is too high; (c) will result in a \$700 balance in the Supplies account (\$100 too low) and an \$800 balance in the Supplies Expense account (\$100 too high); and (d) will cause the Supplies account to have a credit balance (assets have a normal debit balance) and the Supplies Expense account to be stated at \$2,200, which is too high.

***16. c.** Neutrality is one of the enhancing qualities that makes information more representationally faithful, not relevant. Therefore, choices (a), (b), and (d) are incorrect.

***17. c.** Cost is a constraint in financial accounting. The other choices are all enhancing qualities of useful information.

Practice Brief Exercises

Indicate why adjusting entries are needed.

1. (LO 1) The ledger of Dey Company includes the following accounts. Explain why each account may need adjustment.

- | | |
|------------------------------|--------------------------------|
| a. Supplies. | c. Salaries and Wages Payable. |
| b. Unearned Service Revenue. | d. Interest Payable. |

Solution

1. a. Supplies: to recognize supplies used during the period.
- b. Unearned Service Revenue: to record revenue generated for services performed.
- c. Salaries and Wages Payable: to recognize salaries and wages accrual to employees at the end of a reporting period.
- d. Interest Payable: to recognize interest accrued but unpaid on notes payable.

Prepare adjusting entry for depreciation.

2. (LO 2) Financial Statement At the end of its first year, the trial balance of Denton Company shows Equipment of \$40,000 and zero balances in Accumulated Depreciation—Equipment and Depreciation Expense. Depreciation for the year is estimated to be \$8,000. Prepare the adjusting entry for depreciation at December 31, post the adjustments to T-accounts, and indicate the balance sheet presentation of the equipment at December 31.

Solution

2. Dec. 31

Depreciation Expense		8,000	
Accumulated Depreciation—Equipment			8,000

<u>Depreciation Expense</u> 12/31 8,000	<u>Accum. Depreciation—Equipment</u> 12/31 8,000
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Balance Sheet:

Equipment	\$40,000	
Less: Accumulated Depreciation—Equipment	8,000	\$32,000

Prepare adjusting entries for accruals.

3. (LO 3) You are asked to prepare the following accrued adjusting entries at December 31.

1. Services performed but not recorded are \$4,200.
2. Utility expenses incurred but not paid are \$660.
3. Salaries and wages earned by employees of \$3,000 are unpaid.

Use the following account titles: Accounts Payable, Accounts Receivable, Service Revenue, Salaries and Wages Expense, Salaries and Wages Payable, and Utility Expense.

Solution

3. Dec. 31	Accounts Receivable	4,200	
	Service Revenue		4,200
31	Utility Expense	660	
	Accounts Payable		660
31	Salaries and Wage Expense	3,000	
	Salaries and Wages Payable		3,000

4. (LO 1, 2, 3) The trial balance for Blair Company includes the following balance sheet accounts. Identify the accounts that may require adjustment. For each account that requires adjustment, indicate (a) the type of adjusting entry (prepaid expense, unearned revenue, accrued revenue, or accrued expense) and (b) the related account in the adjusting entry.

Analyze accounts in an unadjusted trial balance.

Accounts Receivable	Interest Payable
Supplies	Unearned Service Revenue
Prepaid Insurance	

Solution

4. Account	Type of Adjustment	Related Account
Accounts Receivable	Accrued Revenue	Service Revenue
Supplies	Prepaid Expense	Supplies Expense
Prepaid Insurance	Prepaid Expense	Insurance Expense
Interest Payable	Accrued Expense	Interest Expense
Unearned Service Revenue	Unearned Revenue	Service Revenue

5. (LO 4) Financial Statement The adjusted trial balance of Harmony Company includes the following accounts at December 31, 2020: Cash \$12,000, Owner's Capital \$22,000, Owner's Drawing \$3,000, Service Revenue \$41,000, Rent Expense \$900, Salaries and Wages Expense \$6,000, Supplies Expense \$700, and Depreciation Expense \$1,800. Prepare an income statement for the year.

Prepare an income statement from an adjusted trial balance.

Solution

Harmony Company Income Statement For the Year Ended December 31, 2020		
Revenues		
Service revenue		\$41,000
Expenses		
Salaries and wages expense	\$6,000	
Rent expense	900	
Depreciation expense	1,800	
Supplies expense	700	
Total expenses		9,400
Net income		<u>\$31,600</u>

Practice Exercises

1. (LO 2, 3) Evan Watts, D.D.S., opened a dental practice on January 1, 2020. During the first month of operations, the following transactions occurred.

Prepare adjusting entries.

- Watts performed services for patients totaling \$2,400. These services have not yet been recorded.
- Utility expenses incurred but not paid prior to January 31 totaled \$400.
- Purchased dental equipment on January 1 for \$80,000, paying \$20,000 in cash and signing a \$60,000, 3-year note payable. The equipment depreciates \$500 per month. Interest is \$600 per month.
- Purchased a one-year malpractice insurance policy on January 1 for \$12,000.
- Purchased \$2,600 of dental supplies. On January 31, determined that \$900 of supplies were on hand.

Instructions

Prepare the adjusting entries on January 31. Account titles are Accumulated Depreciation—Equipment, Depreciation Expense, Service Revenue, Accounts Receivable, Insurance Expense, Interest Expense, Interest Payable, Prepaid Insurance, Supplies, Supplies Expense, Utilities Expense, and Utilities Payable.

Solution**1.**

Jan. 31	Accounts Receivable	2,400	
	Service Revenue		2,400
	Utilities Expense	400	
	Utilities Payable		400
	Depreciation Expense	500	
	Accumulated Depreciation—Equipment		500
	Interest Expense	600	
	Interest Payable		600
	Insurance Expense (\$12,000 ÷ 12)	1,000	
	Prepaid Insurance		1,000
	Supplies Expense (\$2,600 – \$900)	1,700	
	Supplies		1,700

Prepare correct income statement.

2. (LO 2, 3, 4) Financial Statement The income statement of Venden Co. for the month of July shows net income of \$4,000 based on Service Revenue \$8,700, Salaries and Wages Expense \$2,500, Supplies Expense \$1,700, and Utilities Expense \$500. In reviewing the statement, you discover the following.

- Insurance expired during July of \$700 was omitted.
- Supplies expense includes \$250 of supplies that are still on hand at July 31.
- Depreciation on equipment of \$300 was omitted.
- Accrued but unpaid wages at July 31 of \$400 were not included.
- Services performed but unrecorded totaled \$650.

Instructions

Prepare a correct income statement for July 2020.

Solution**2.**

Venden Co. Income Statement For the Month Ended July 31, 2020			
Revenues			
Service revenue (\$8,700 + \$650)			\$9,350
Expenses			
Salaries and wages expense (\$2,500 + \$400)	\$2,900		
Supplies expense (\$1,700 – \$250)	1,450		
Utilities expense	500		
Insurance expense	700		
Depreciation expense	300		
Total expenses			<u>5,850</u>
Net income			<u><u>\$3,500</u></u>

Practice Problem

(LO 2, 3) The Green Thumb Lawn Care Company began operations on April 1. At April 30, the trial balance shows the following balances for selected accounts.

Prepare adjusting entries from selected data.

Prepaid Insurance	\$ 3,600
Equipment	28,000
Notes Payable	20,000
Unearned Service Revenue	4,200
Service Revenue	1,800

Analysis reveals the following additional data.

1. Prepaid insurance is the cost of a 2-year insurance policy, effective April 1.
2. Depreciation on the equipment is \$500 per month.
3. The note payable is dated April 1. It is a 6-month, 12% note.
4. Seven customers paid for the company's 6-month lawn service package of \$600 beginning in April. The company performed services for these customers in April.
5. Lawn services performed for other customers but not recorded at April 30 totaled \$1,500.

Instructions

Prepare the adjusting entries for the month of April. Show computations.

Solution

GENERAL JOURNAL				J1
Date	Account Titles and Explanation	Ref.	Debit	Credit
	<u>Adjusting Entries</u>			
Apr. 30	Insurance Expense Prepaid Insurance (To record insurance expired: $\$3,600 \div 24 = \150 per month)		150	150
30	Depreciation Expense Accumulated Depreciation—Equipment (To record monthly depreciation)		500	500
30	Interest Expense Interest Payable (To record interest on notes payable: $\$20,000 \times 12\% \times 1/12 = \200)		200	200
30	Unearned Service Revenue Service Revenue (To record revenue for services performed: $\$600 \div 6 = \100 ; $\$100$ per month $\times 7 = \$700$)		700	700
30	Accounts Receivable Service Revenue (To record revenue for services performed)		1,500	1,500

WileyPLUS

Brief Exercises, DO IT! Exercises, Exercises, Problems, and many additional resources are available for practice in WileyPLUS.

Note: All asterisked Questions, Exercises, and Problems relate to material in the appendices to the chapter.

Questions

1. **a.** How does the time period assumption affect an accountant's analysis of business transactions?
b. Explain the terms fiscal year, calendar year, and interim periods.
2. Define two generally accepted accounting principles that relate to adjusting the accounts.
3. Susan Hardy, a lawyer, accepts a legal engagement in March, performs the work in April, and is paid in May. If Hardy's law firm prepares monthly financial statements, when should it recognize revenue from this engagement? Why?
4. Why do accrual-basis financial statements provide more useful information than cash-basis statements?
5. In completing the engagement in Question 3, Hardy pays no costs in March, \$2,000 in April, and \$2,500 in May (incurred in April). How much expense should the firm deduct from revenues in the month when it recognizes the revenue? Why?
6. "Adjusting entries are required by the historical cost principle of accounting." Do you agree? Explain.
7. Why may a trial balance not contain up-to-date and complete financial information?
8. Distinguish between the two categories of adjusting entries, and identify the types of adjustments applicable to each category.
9. What is the debit/credit effect of a prepaid expense adjusting entry?
10. "Depreciation is a valuation process that results in the reporting of the fair value of the asset." Do you agree? Explain.
11. Explain the differences between depreciation expense and accumulated depreciation.
12. G. Phillips Company purchased equipment for \$18,000. By the current balance sheet date, \$6,000 had been depreciated. Indicate the balance sheet presentation of the data.
13. What is the debit/credit effect of an unearned revenue adjusting entry?
14. Whistler Corp. performed services for a customer but has not yet recorded payment, nor has it recorded any entry related to the work. Which of the following accounts are involved in the adjusting entry: (a) asset, (b) liability, (c) revenue, or (d) expense? For the accounts selected, indicate whether they would be debited or credited in the entry.
15. A company fails to recognize an expense incurred but not paid. Indicate which of the following accounts is debited and which is credited in the adjusting entry: (a) asset, (b) liability, (c) revenue, or (d) expense.
16. A company makes an accrued revenue adjusting entry for \$900 and an accrued expense adjusting entry for \$700. How much was net income understated prior to these entries? Explain.
17. On January 9, a company pays \$5,000 for salaries and wages of which \$2,000 was reported as Salaries and Wages Payable on December 31. Give the entry to record the payment.
18. For each of the following items before adjustment, indicate the type of adjusting entry (prepaid expense, unearned revenue, accrued revenue, or accrued expense) that is needed to correct the misstatement. If an item could result in more than one type of adjusting entry, indicate each of the types.

a. Assets are understated.	d. Expenses are understated.
b. Liabilities are overstated.	e. Assets are overstated.
c. Liabilities are understated.	f. Revenue is understated.
19. One-half of the adjusting entry is given below. Indicate the account title for the other half of the entry.

a. Salaries and Wages Expense is debited.
b. Depreciation Expense is debited.
c. Interest Payable is credited.
d. Supplies is credited.
e. Accounts Receivable is debited.
f. Unearned Service Revenue is debited.
20. "An adjusting entry may affect more than one balance sheet or income statement account." Do you agree? Why or why not?
21. Why is it possible to prepare financial statements directly from an adjusted trial balance?
- *22. Dashan Company debits Supplies Expense for all purchases of supplies and credits Rent Revenue for all advanced rentals. For each type of adjustment, give the adjusting entry.
- *23. **a.** What is the primary objective of financial reporting?
b. Identify the characteristics of useful accounting information.
- *24. Dan Fineman, the president of King Company, is pleased. King substantially increased its net income in 2020 while keeping its unit inventory relatively the same. Howard Gross, chief accountant, cautions Dan, however. Gross says that since King changed its method of inventory valuation, there is a consistency problem and it is difficult to determine whether King is better off. Is Gross correct? Why or why not?
- *25. What is the distinction between comparability and consistency?
- *26. Describe the constraint inherent in the presentation of accounting information.
- *27. Quinn Becker is president of Better Books. She has no accounting background. Becker cannot understand why fair value is not used as the basis for all accounting measurement and reporting. Discuss.
- *28. What is the economic entity assumption? Give an example of its violation.

Brief Exercises

Indicate why adjusting entries are needed.

BE3.1 (LO 1) The ledger of Althukair Company includes the following accounts. Explain why each account may require adjustment.

- | | |
|---------------------------------|-------------------------------------|
| a. Prepaid Insurance. | c. Unearned Service Revenue. |
| b. Depreciation Expense. | d. Interest Payable. |

BE3.2 (LO 1, 2) Gee Company accumulates the following adjustment data at December 31. Indicate (a) the type of adjustment (prepaid expense, accrued revenue, and so on), and (b) the status of accounts before adjustment (for example, “assets understated and revenues understated”).

1. Supplies of \$150 are on hand.
2. Services performed but not recorded total \$900.
3. Interest of \$200 has accumulated on a note payable.
4. Rent collected in advance totaling \$850 has been earned.

Identify the major types of adjusting entries.

BE3.3 (LO 2) Schramel Advertising Company’s trial balance at December 31 shows Supplies \$6,700 and Supplies Expense \$0. On December 31, there are \$2,100 of supplies on hand. Prepare the adjusting entry at December 31, and using T-accounts, enter the balances in the accounts, post the adjusting entry, and indicate the adjusted balance in each account.

Prepare adjusting entry for supplies.

BE3.4 (LO 2) At the end of its first year, the trial balance of Wolowitz Company shows Equipment \$30,000 and zero balances in Accumulated Depreciation—Equipment and Depreciation Expense. Depreciation for the year is estimated to be \$3,750. Prepare the annual adjusting entry for depreciation at December 31, post the adjustments to T-accounts, and indicate the balance sheet presentation of the equipment at December 31.

Prepare adjusting entry for depreciation.

BE3.5 (LO 2) On July 1, 2019, Major Co. pays \$15,120 to Mesa Insurance Co. for a 4-year insurance contract. Both companies have fiscal years ending December 31. For Major Co., journalize and post the entry on July 1 and the annual adjusting entry on December 31.

Prepare adjusting entry for prepaid expense.

BE3.6 (LO 2) Using the data in BE3.5, journalize and post the entry on July 1 and the adjusting entry on December 31 for Mesa Insurance Co. Mesa uses the accounts Unearned Service Revenue and Service Revenue.

Prepare adjusting entry for unearned revenue.

BE3.7 (LO 3) The bookkeeper for Abdul Company asks you to prepare the following accrued adjusting entries at December 31.

1. Interest on notes payable of \$400 is accrued.
2. Services performed but not recorded total \$2,300.
3. Salaries earned by employees of \$900 have not been recorded.

Prepare adjusting entries for accruals.

Use the following account titles: Service Revenue, Accounts Receivable, Interest Expense, Interest Payable, Salaries and Wages Expense, and Salaries and Wages Payable.

BE3.8 (LO 1, 2, 3) The trial balance of Beowulf Company includes the following balance sheet accounts, which may require adjustment. For each account that requires adjustment, indicate (a) the type of adjusting entry (prepaid expense, unearned revenue, accrued revenue, or accrued expense) and (b) the related account in the adjusting entry.

Analyze accounts in an unadjusted trial balance.

Accounts Receivable	Interest Payable
Prepaid Insurance	Unearned Service Revenue
Accumulated Depreciation—Equipment	

BE3.9 (LO 4) Financial Statement The adjusted trial balance of Miller Company at December 31, 2020, includes the following accounts: Owner’s Capital \$16,400, Owner’s Drawings \$7,000, Service Revenue \$39,000, Salaries and Wages Expense \$16,000, Insurance Expense \$2,000, Rent Expense \$4,000, Supplies Expense \$1,500, and Depreciation Expense \$1,300. Prepare an income statement for the year.

Prepare an income statement from an adjusted trial balance.

BE3.10 (LO 4) Financial Statement Partial adjusted trial balance data for Miller Company is presented in BE3.9. The balance in Owner’s Capital is the balance as of January 1. Prepare an owner’s equity statement for the year assuming net income is \$14,200 for the year.

Prepare an owner’s equity statement from an adjusted trial balance.

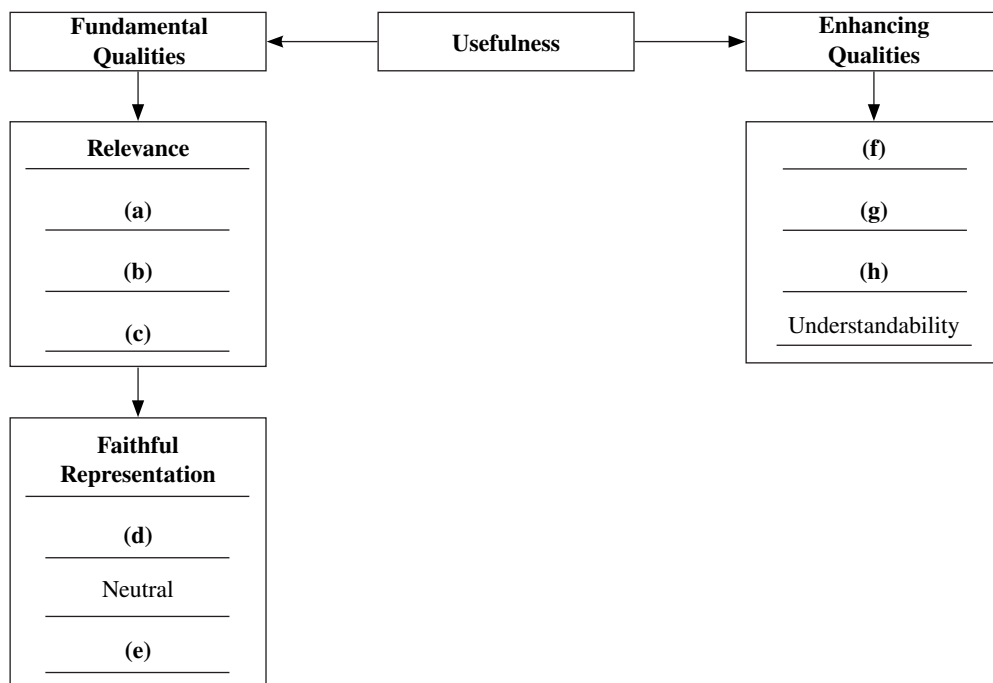
***BE3.11 (LO 5)** Eckholm Company records all prepayments in income statement accounts. At April 30, the trial balance shows Supplies Expense \$2,800, Service Revenue \$9,200, and zero balances in related

Prepare adjusting entries under alternative treatment of deferrals.

balance sheet accounts. Prepare the adjusting entries at April 30 assuming (a) \$400 of supplies on hand and (b) \$3,000 of service revenue should be reported as unearned.

Identify characteristics of useful information.

***BE3.12 (LO 6)** The accompanying chart shows the qualitative characteristics of useful accounting information. Fill in the blanks.



Identify characteristics of useful information.

***BE3.13 (LO 6)** Given the characteristics of useful accounting information, complete each of the following statements.

- For information to be _____, it should have predictive value, confirmatory value, and be material.
- _____ is the quality of information that gives assurance that the information accurately depicts what really happened.
- _____ means using the same accounting principles and methods from year to year within a company.

Identify characteristics of useful information.

***BE3.14 (LO 6)** Here are some qualitative characteristics of useful accounting information:

- Predictive value.
- Neutral.
- Verifiable.
- Timely.

Match each qualitative characteristic to one of the following statements.

- Accounting information should help provide accurate expectations about future events.
- Accounting information cannot be selected, prepared, or presented to favor one set of interested users over another.
- The quality of information that occurs when independent observers, using the same methods, obtain similar results.
- Accounting information must be available to decision-makers before it loses its capacity to influence their decisions.

Define full disclosure principle.

***BE3.15 (LO 6)** Select the response that completes the following statement correctly. The full disclosure principle dictates that:

- financial statements should disclose all assets at their cost.
- financial statements should disclose only those events that can be measured in currency.
- financial statements should disclose all events and circumstances that would matter to users of financial statements.
- financial statements should not be relied on unless an auditor has expressed an unqualified opinion on them.

DO IT! Exercises

DO IT! 3.1 (LO 1) Below is a list of concepts in the left column, with a description of the concept in the right column. There are more descriptions provided than concepts. Match the description of the concept to the concept.

- | | |
|--|---|
| 1. ____ Cash-basis accounting. | a. Monthly and quarterly time periods. |
| 2. ____ Fiscal year. | b. Accountants divide the economic life of a business into artificial time periods. |
| 3. ____ Revenue recognition principle. | c. Efforts (expenses) should be recognized in the period in which a company uses assets or incurs liabilities to generate accomplishments (revenues). |
| 4. ____ Expense recognition principle. | d. Companies record revenues when they receive cash and record expenses when they pay out cash. |
| | e. An accounting time period that is one year in length. |
| | f. An accounting time period that starts on January 1 and ends on December 31. |
| | g. Companies record transactions in the period in which the events occur. |
| | h. Recognize revenue in the accounting period in which a performance obligation is satisfied. |

Identify timing concepts.

DO IT! 3.2 (LO 2) The ledger of Milton, Inc. on March 31, 2020, includes the following selected accounts before adjusting entries.

Prepare adjusting entries for deferrals.

	<u>Debit</u>	<u>Credit</u>
Prepaid Insurance	\$ 2,400	
Supplies	2,500	
Equipment	30,000	
Unearned Service Revenue		\$9,000

An analysis of the accounts shows the following.

- Insurance expires at the rate of \$400 per month.
- Supplies on hand total \$1,600.
- The equipment depreciates \$480 per month.
- During March, services were performed for two-fifths of the unearned service revenue.

Prepare the adjusting entries for the month of March.

DO IT! 3.3 (LO 3) Fiske Computer Services began operations in July 2020. At the end of the month, the company prepares monthly financial statements. It has the following information for the month.

Prepare adjusting entries for accruals.

- At July 31, the company owed employees \$1,300 in salaries that the company will pay in August.
- On July 1, the company borrowed \$20,000 from a local bank on a 10-year note. The annual interest rate is 6%.
- Service revenue unrecorded in July totaled \$2,400.

Prepare the adjusting entries needed at July 31, 2020.

DO IT! 3.4 (LO 4) Financial Statement Ming Company was organized on April 1, 2020. The company prepares quarterly financial statements. The adjusted trial balance amounts at June 30 are as follows.

Calculate amounts from trial balance.

	<u>Debit</u>		<u>Credit</u>
Cash	\$ 5,360	Accumulated Depreciation—	
Accounts Receivable	580	Equipment	\$ 700
Prepaid Rent	1,120	Notes Payable	4,000
Supplies	920	Accounts Payable	790
Equipment	12,000	Salaries and Wages Payable	300
Owner's Drawings	500	Interest Payable	40
Salaries and Wages Expense	7,400	Unearned Rent Revenue	400
Rent Expense	1,200	Owner's Capital	11,200
Depreciation Expense	700	Service Revenue	11,360
Supplies Expense	160	Rent Revenue	1,600
Utilities Expense	410		<u>\$30,390</u>
Interest Expense	40		
	<u>\$30,390</u>		

- a. Determine the net income for the quarter April 1 to June 30.
- b. Determine the total assets and total liabilities at June 30, 2020, for Ming Company.
- c. Determine the amount that appears for Owner's Capital at June 30, 2020.

Exercises

Explain the time period assumption.

E3.1 (LO 1) Chloe Davis has prepared the following list of statements about the time period assumption.

1. Adjusting entries would not be necessary if a company's life were not divided into artificial time periods.
2. The IRS requires companies to file annual tax returns.
3. Accountants divide the economic life of a business into artificial time periods, but each transaction affects only one of these periods.
4. Accounting time periods are generally a month, a quarter, or a year.
5. A time period lasting one year is called an interim period.
6. All fiscal years are calendar years, but not all calendar years are fiscal years.

Instructions

Identify each statement as true or false. If false, indicate how to correct the statement.

Distinguish between cash and accrual basis of accounting.

E3.2 (LO 1) Writing On numerous occasions, proposals have surfaced to put the federal government on the accrual basis of accounting. This is no small issue. If this basis were used, it would mean that billions in unrecorded liabilities would have to be booked, and the federal deficit would increase substantially.

Instructions

- a. What is the difference between accrual-basis accounting and cash-basis accounting?
- b. Why would politicians prefer the cash basis over the accrual basis?
- c. Write a letter to your senator explaining why the federal government should adopt the accrual basis of accounting.

Compute cash and accrual accounting income.

E3.3 (LO 1) Carillo Industries collected \$108,000 from customers in 2020. Of the amount collected, \$25,000 was for services performed in 2019. In addition, Carillo performed services worth \$36,000 in 2020, which will not be collected until 2021.

Carillo Industries also paid \$72,000 for expenses in 2020. Of the amount paid, \$30,000 was for expenses incurred on account in 2019. In addition, Carillo incurred \$42,000 of expenses in 2020, which will not be paid until 2021.

Instructions

- Compute 2020 cash-basis net income.
- Compute 2020 accrual-basis net income.

E3.4 (LO 1, 2, 3) Luong Corporation encounters the following situations:

Identify the type of adjusting entry needed.

1. Luong collects \$1,300 from a customer in 2020 for services to be performed in 2021.
2. Luong incurs utility expense which is not yet paid in cash or recorded.
3. Luong's employees worked 3 days in 2020 but will not be paid until 2021.
4. Luong performs services for customers but has not yet received cash or recorded the transaction.
5. Luong paid \$2,800 rent on December 1 for the 4 months starting December 1.
6. Luong received cash for future services and recorded a liability until the service was performed.
7. Luong performed consulting services for a client in December 2020. On December 31, it had not billed the client for services provided of \$1,200.
8. Luong paid cash for an expense and recorded an asset until the item was used up.
9. Luong purchased \$900 of supplies in 2020; at year-end, \$400 of supplies remain unused.
10. Luong purchased equipment on January 1, 2020; the equipment will be used for 5 years.
11. Luong borrowed \$12,000 on October 1, 2020, signing an 8% one-year note payable.

Instructions

Identify what type of adjusting entry (prepaid expense, unearned revenue, accrued expense, or accrued revenue) is needed in each situation at December 31, 2020.

E3.5 (LO 2, 3) Devin Wolf Company has the following balances in selected accounts on December 31, 2020.

Prepare adjusting entries from selected data.

Accounts Receivable	\$ -0-
Accumulated Depreciation—Equipment	-0-
Equipment	7,000
Interest Payable	-0-
Notes Payable	10,000
Prepaid Insurance	2,100
Salaries and Wages Payable	-0-
Supplies	2,450
Unearned Service Revenue	32,000

All the accounts have normal balances. The information below has been gathered at December 31, 2020.

1. Devin Wolf Company borrowed \$10,000 by signing a 9%, one-year note on September 1, 2020.
2. A count of supplies on December 31, 2020, indicates that supplies of \$900 are on hand.
3. Depreciation on the equipment for 2020 is \$1,000.
4. Devin Wolf Company paid \$2,100 for 12 months of insurance coverage on June 1, 2020.
5. On December 1, 2020, Devin Wolf collected \$32,000 for consulting services to be performed from December 1, 2020, through March 31, 2021.
6. Devin Wolf performed consulting services for a client in December 2020. The client will be billed \$4,200.
7. Devin Wolf Company pays its employees total salaries of \$9,000 every Monday for the preceding 5-day week (Monday through Friday). On Monday, December 29, employees were paid for the week ending December 26. All employees worked the last 3 days of 2020.

Instructions

Prepare adjusting entries for the seven items described above.

E3.6 (LO 2, 3, 4) Mendoza Company accumulates the following adjustment data at December 31.

Identify types of adjustments and account relationships.

1. Services performed but unbilled total \$3,000.
2. Supplies of \$300 have been used.
3. Utility expenses of \$552 are unpaid.
4. Services performed of \$260 collected in advance.
5. Salaries of \$800 are unpaid.
6. Prepaid insurance totaling \$350 has expired.

Instructions

For each of the above items indicate the following.

- The type of adjustment (prepaid expense, unearned revenue, accrued revenue, or accrued expense).
- The status of accounts before adjustment (overstatement or understatement).

Prepare adjusting entries from selected account data.

E3.7 (LO 2, 3) The ledger of Passehl Rental Agency on March 31 of the current year includes the selected accounts, shown below, before adjusting entries have been prepared.

	<u>Debit</u>	<u>Credit</u>
Prepaid Insurance	\$ 3,600	
Supplies	2,800	
Equipment	25,000	
Accumulated Depreciation—Equipment		\$ 8,400
Notes Payable		20,000
Unearned Rent Revenue		10,200
Rent Revenue		60,000
Interest Expense	—0—	
Salaries and Wages Expense	14,000	

An analysis of the accounts shows the following.

- The equipment depreciates \$400 per month.
- One-third of the unearned rent revenue was earned during the quarter.
- Interest of \$500 is accrued on the notes payable.
- Supplies on hand total \$750.
- Insurance expires at the rate of \$300 per month.

Instructions

Prepare the adjusting entries at March 31, assuming that adjusting entries are made **quarterly**. Additional accounts are Depreciation Expense, Insurance Expense, Interest Payable, and Supplies Expense.

Prepare adjusting entries.

E3.8 (LO 2, 3) Lorena Manzone, D.D.S., opened a dental practice on January 1, 2020. During the first month of operations, the following transactions occurred.

- Performed services for patients who had dental plan insurance. At January 31, \$785 of such services were performed but not yet recorded.
- Utility expenses incurred but not paid prior to January 31 totaled \$650.
- Purchased dental equipment on January 1 for \$80,000, paying \$30,000 in cash and signing a \$50,000, 3-year note payable. The equipment depreciates \$400 per month. Interest is \$500 per month.
- Purchased a one-year malpractice insurance policy on January 1 for \$24,000.
- Purchased \$1,600 of dental supplies. On January 31, determined that \$400 of supplies were on hand.

Instructions

Prepare the adjusting entries on January 31. Account titles are Accumulated Depreciation—Equipment, Depreciation Expense, Service Revenue, Accounts Receivable, Insurance Expense, Interest Expense, Interest Payable, Prepaid Insurance, Supplies, Supplies Expense, Utilities Expense, and Utilities Payable.

Prepare adjusting entries.

E3.9 (LO 2, 3) The trial balance for Pioneer Advertising is shown in Illustration 3.3. Instead of the adjusting entries shown in the textbook at October 31, assume the following adjustment data.

- Supplies on hand at October 31 total \$500.
- Expired insurance for the month is \$120.
- Depreciation for the month is \$50.
- Services related to unearned service revenue in October worth \$600 were performed.
- Services performed but not recorded at October 31 are \$360.
- Interest accrued at October 31 is \$95.
- Accrued salaries at October 31 are \$1,625.

Instructions

Prepare the adjusting entries for the items above.

E3.10 (LO 1, 2, 3, 4) Financial Statement The income statement of Lundeen Co. for the month of July shows net income of \$1,400 based on Service Revenue \$5,500, Salaries and Wages Expense \$2,300, Supplies Expense \$1,200, and Utilities Expense \$600. In reviewing the statement, you discover the following.

1. Insurance expired during July of \$500 was omitted.
2. Supplies expense includes \$250 of supplies that are still on hand at July 31.
3. Depreciation on equipment of \$150 was omitted.
4. Accrued but unpaid salaries and wages at July 31 of \$400 were not included.
5. Services performed but unrecorded totaled \$650.

Instructions

Prepare a correct income statement for July 2020.

Prepare correct income statement.



E3.11 (LO 1, 2, 3, 4) A partial adjusted trial balance of Frangesch Company at January 31, 2020, shows the following.

Analyze adjusted data.



Frangesch Company
Adjusted Trial Balance
January 31, 2020

	<u>Debit</u>	<u>Credit</u>
Supplies	\$ 850	
Prepaid Insurance	2,400	
Salaries and Wages Payable		\$ 920
Unearned Service Revenue		750
Supplies Expense	950	
Insurance Expense	400	
Salaries and Wages Expense	2,900	
Service Revenue		2,000

Instructions

Answer the following questions, assuming the year begins January 1.

- a. If the amount in Supplies Expense is the January 31 adjusting entry and \$1,000 of supplies was purchased in January, what was the balance in Supplies on January 1?
- b. If the amount in Insurance Expense is the January 31 adjusting entry and the original insurance premium was for one year, what was the total premium and when was the policy purchased?
- c. If \$3,800 of salaries was paid in January, what was the balance in Salaries and Wages Payable at December 31, 2019?

E3.12 (LO 2, 3) Selected accounts of Shannon Company are shown as follows.

Journalize basic transactions and adjusting entries.

Supplies Expense			
7/31	900		
Supplies			
7/1 Bal.	1,100	7/31	900
7/10	650		
Salaries and Wages Payable			
		7/31	1,200
Accounts Receivable			
7/31	500		
Unearned Service Revenue			
7/31	1,150	7/1 Bal.	1,500
		7/20	1,000
Salaries and Wages Expense			
7/15	1,200		
7/31	1,200		
Service Revenue			
		7/14	2,200
		7/31	1,150
		7/31	500

Instructions

After analyzing the accounts, journalize (a) the July transactions and (b) the adjusting entries that were made on July 31. (*Hint:* July transactions were for cash.)

Prepare adjusting entries from selected account data.

E3.13 (LO 2, 3) The ledger of Armour Lake Lumber Supply on July 31, 2020, includes the selected accounts below before adjusting entries have been prepared.

	<u>Debit</u>	<u>Credit</u>
Notes Payable		\$ 20,000
Supplies	\$ 24,000	
Prepaid Rent	3,600	
Buildings	250,000	
Accumulated Depreciation—Buildings		140,000
Unearned Service Revenue		11,500

An analysis of the company's accounts shows the following.

1. The notes payable pays interest at a rate of 6% per year.
2. Supplies on hand at the end of the month totaled \$18,600.
3. The balance in Prepaid Rent represents 4 months of rent costs.
4. Employees were owed \$3,100 related to unpaid salaries and wages.
5. Depreciation on buildings is \$6,000 per year.
6. During the month, the company satisfied obligations worth \$4,700 related to the Unearned Services Revenue.
7. Unpaid maintenance and repairs costs were \$2,300.

Instructions

Prepare the adjusting entries at July 31 assuming that adjusting entries are made monthly. Use additional accounts as needed.

Determine effect of adjusting entries.

E3.14 (LO 2, 3) On December 31, 2020, Waters Company prepared an income statement and balance sheet, but failed to take into account three adjusting entries. The balance sheet showed total assets \$150,000, total liabilities \$70,000, and owner's equity \$80,000. The incorrect income statement showed net income of \$70,000.

The data for the three adjusting entries were:

1. Salaries and wages amounting to \$10,000 for the last 2 days in December were not paid and not recorded. The next payroll will be in January.
2. Rent payments of \$8,000 was received for two months in advance on December 1. The entire amount was credited to Unearned Rent Revenue when paid.
3. Depreciation expense for 2020 is \$9,000.

Instructions

Complete the following table to correct the financial statement amounts shown (indicate deductions with parentheses).

<u>Item</u>	<u>Net Income</u>	<u>Total Assets</u>	<u>Total Liabilities</u>	<u>Owner's Equity</u>
Incorrect balances	\$70,000	\$150,000	\$70,000	\$80,000
Effects of:				
Salaries and Wages	_____	_____	_____	_____
Rent Revenue	_____	_____	_____	_____
Depreciation	_____	_____	_____	_____
Correct balances	=====	=====	=====	=====

Prepare and post transaction and adjusting entries for prepayments.

E3.15 (LO 2) Action Quest Games Co. adjusts its accounts annually. The following information is available for the year ended December 31, 2020.

1. Purchased a 1-year insurance policy on June 1 for \$1,800 cash.
2. Paid \$6,500 on August 31 for 5 months' rent in advance.
3. On September 4, received \$3,600 cash in advance from a company to sponsor a game each month for a total of 9 months for the most improved students at a local school.
4. Signed a contract for cleaning services starting December 1 for \$1,000 per month. Paid for the first 2 months on November 30. (*Hint:* Use the account Prepaid Cleaning to record prepayments.)
5. On December 5, received \$1,500 in advance from a gaming club. Determined that on December 31, \$475 of these games had not yet been played.

Instructions

- For each of the above transactions, prepare the journal entry to record the initial transaction.
- For each of the above transactions, prepare the adjusting journal entry that is required on December 31, (Hint: Use the account Service Revenue for item 3 and Repairs and Maintenance Expense for item 4.)
- Post the journal entries in parts (a) and (b) to T-accounts and determine the final balance in each account balance. (Note: Posting to the Cash account is not required.)

E3.16 (LO 3) Greenock Company has the following information available for accruals for the year ended December 31, 2020. The company adjusts its accounts annually.

Prepare adjusting and subsequent entries for accruals.

- The December utility bill for \$425 was unrecorded on December 31. Greenock paid the bill on January 11.
- Greenock is open 7 days a week and employees are paid a total of \$3,500 every Monday for a 7-day (Monday–Sunday) workweek. December 31 is a Thursday, so employees will have worked 4 days (Monday, December 28–Thursday, December 31) that they have not been paid for by year-end. Employees will be paid next on January 4.
- Greenock signed a \$48,000, 5% bank loan on November 1, 2019, due in 2 years. Interest is payable on the first day of each following month.
- Greenock receives a fee from Pizza Shop next door for all pizzas sold to customers using Greenock's facility. The amount owed for December is \$300, which Pizza Shop will pay on January 4. (Hint: Use the Service Revenue account.)
- Greenock rented some of its unused warehouse space to a client for \$6,000 a month, payable the first day of the following month. It received the rent for the month of December on January 2.

Instructions

- For each situation, prepare the adjusting entry required at December 31. (Round all calculations to the nearest dollar.)
- For each situation, prepare the journal entry to record the subsequent cash transaction in 2021.

E3.17 (LO 2, 3, 4) The trial balances before and after adjustment for Renfro Company at the end of its fiscal year are presented below.

Prepare adjusting entries from analysis of trial balances.

Renfro Company
Trial Balance
August 31, 2020

	Before Adjustment		After Adjustment	
	Dr.	Cr.	Dr.	Cr.
Cash	\$10,400		\$10,400	
Accounts Receivable	8,800		11,200	
Supplies	2,300		700	
Prepaid Insurance	4,000		2,500	
Equipment	14,000		14,000	
Accumulated Depreciation—Equipment		\$ 3,600		\$ 4,500
Accounts Payable		5,800		5,800
Salaries and Wages Payable		—0—		1,100
Unearned Rent Revenue		1,500		400
Owner's Capital		15,600		15,600
Service Revenue		34,000		36,400
Rent Revenue		11,000		12,100
Salaries and Wages Expense	17,000		18,100	
Supplies Expense	—0—		1,600	
Rent Expense	15,000		15,000	
Insurance Expense	—0—		1,500	
Depreciation Expense	—0—		900	
	<u>\$71,500</u>	<u>\$71,500</u>	<u>\$75,900</u>	<u>\$75,900</u>

Instructions

Prepare the adjusting entries that were made.

Prepare financial statements from adjusted trial balance.

Record transactions on accrual basis; convert revenue to cash receipts.

E3.18 (LO 4) Financial Statement The adjusted trial balance for Renfro Company is given in E3.17.

Instructions

Prepare the income and owner's equity statements for the year and the balance sheet at August 31.

E3.19 (LO 2, 3) The following data are taken from the comparative balance sheets of Bundies Billiards Club, which prepares its financial statements using the accrual basis of accounting.

<u>December 31</u>	<u>2020</u>	<u>2019</u>
Accounts receivable from members	\$16,000	\$ 8,000
Unearned service revenue	17,000	25,000

Members are billed based upon their use of the club's facilities. Unearned service revenues arise from the sale of gift certificates, which members can apply to their future use of club facilities. The 2020 income statement for the club showed that service revenue of \$161,000 was earned during the year.

Instructions

(Hint: You will probably find it helpful to use T-accounts to analyze these data.)

- a. Prepare journal entries for each of the following events that took place during 2020.
 1. Accounts receivable from 2019 were all collected.
 2. Gift certificates outstanding at the end of 2019 were all redeemed.
 3. An additional \$38,000 worth of gift certificates were sold during 2020. A portion of these was used by the recipients during the year; the remainder was still outstanding at the end of 2020.
 4. Services performed for members for 2020 were billed to members.
 5. Accounts receivable for 2020 (i.e., those billed in item [4] above) were partially collected.
- b. Determine the amount of cash received by the club, with respect to member services, during 2020.

Journalize adjusting entries.

***E3.20 (LO 5)** Bob Zeller Company has the following balances in selected accounts on December 31, 2020.

Service Revenue	\$40,000
Insurance Expense	2,400
Supplies Expense	2,450

All the accounts have normal balances. Bob Zeller Company debits prepayments to expense accounts when paid, and credits unearned revenues to revenue accounts when received. The following information below has been gathered at December 31, 2020.

1. Bob Zeller Company paid \$2,400 for 12 months of insurance coverage on June 1, 2020.
2. On December 1, 2020, Bob Zeller Company collected \$40,000 for consulting services to be performed from December 1, 2020, through March 31, 2021.
3. A count of supplies on December 31, 2020, indicates that supplies of \$600 are on hand.

Instructions

Prepare the adjusting entries needed at December 31, 2020.

Journalize transactions and adjusting entries.

***E3.21 (LO 5)** At Sekon Company, prepayments are debited to expense when paid, and unearned revenues are credited to revenue when cash is received. During January of the current year, the following transactions occurred.

- Jan. 2 Paid \$1,920 for fire insurance protection for the year.
 10 Paid \$1,700 for supplies.
 15 Received \$6,100 for services to be performed in the future.

On January 31, it is determined that \$2,100 of the services were performed and that there are \$650 of supplies on hand.

Instructions

- a. Journalize and post the January transactions. (Use T-accounts.)
- b. Journalize and post the adjusting entries at January 31.
- c. Determine the ending balance in each of the accounts.

***E3.22 (LO 6)** Presented below are the assumptions and principles discussed in this chapter.

- | | |
|-------------------------------|--------------------------------|
| 1. Full disclosure principle. | 4. Time period assumption. |
| 2. Going concern assumption. | 5. Historical cost principle. |
| 3. Monetary unit assumption. | 6. Economic entity assumption. |

Identify accounting assumptions and principles.

Instructions

Identify by number the accounting assumption or principle that is described below. Do not use a number more than once.

- _____ a. Is the rationale for why plant assets are not reported at liquidation value. (*Note:* Do not use the historical cost principle.)
- _____ b. Indicates that personal and business recordkeeping should be separately maintained.
- _____ c. Assumes that the monetary unit is the “measuring stick” used to report on financial performance.
- _____ d. Separates financial information into time periods for reporting purposes.
- _____ e. Measurement basis used when a reliable estimate of fair value is not available.
- _____ f. Dictates that companies should disclose all circumstances and events that make a difference to financial statement users.

***E3.23 (LO 6)** Weber Co. had three major business transactions during 2020.

- Reported at its fair value of \$260,000 merchandise inventory with a cost of \$208,000.
- The president of Weber Co., Austin Weber, purchased a truck for personal use and charged it to his expense account.
- Weber Co. wanted to make its 2020 income look better, so it added 2 more weeks to the year (a 54-week year). Previous years were 52 weeks.

Identify the assumption or principle that has been violated.

Instructions

In each situation, identify the assumption or principle that has been violated, if any, and discuss what the company should have done.

***E3.24 (LO 6)** The following characteristics, assumptions, principles, or constraint guide the FASB when it creates accounting standards.

Identify financial accounting concepts and principles.

Relevance	Expense recognition principle
Faithful representation	Time period assumption
Comparability	Going concern assumption
Consistency	Historical cost principle
Monetary unit assumption	Full disclosure principle
Economic entity assumption	Materiality

Match each item above with a description below.

- _____ Ability to easily evaluate one company's results relative to another's.
- _____ Requirement that a company will continue to operate for the foreseeable future.
- _____ The judgment concerning whether an item's size is large enough to matter to decision-makers.
- _____ The reporting of all information that would make a difference to financial statement users.
- _____ The practice of preparing financial statements at regular intervals.
- _____ The quality of information that indicates the information makes a difference in a decision.
- _____ A belief that items should be reported on the balance sheet at the price that was paid to acquire them.
- _____ A company's use of the same accounting principles and methods from year to year.
- _____ Tracing accounting events to particular companies.
- _____ The desire to minimize bias in financial statements.
- _____ Reporting only those things that can be measured in monetary units.
- _____ Dictates that efforts (expenses) be recognized in the period in which a company generates results (revenues).

Comment on the objective and qualitative characteristics of accounting information.

***E3.25 (LO 6) Writing** Speyeware International Inc., headquartered in Vancouver, Canada, specializes in Internet safety and computer security products for both the home and commercial markets. In a recent balance sheet, it reported a deficit of US\$5,678,288. It has reported only net losses since its inception. In spite of these losses, Speyeware's shares of stock have traded anywhere from a high of \$3.70 to a low of \$0.32 on the Canadian Venture Exchange.

Speyeware's financial statements have historically been prepared in Canadian dollars. Recently, the company adopted the U.S. dollar as its reporting currency.

Instructions

- What is the objective of financial reporting? How does this objective meet or not meet Speyeware's investors' needs?
- Why would investors want to buy Speyeware's shares if the company has consistently reported losses over the last few years? Include in your answer an assessment of the relevance of the information reported on Speyeware's financial statements.
- Comment on how the change in reporting information from Canadian dollars to U.S. dollars likely affected the readers of Speyeware's financial statements. Include in your answer an assessment of the comparability of the information.

Comment on the objective and qualitative characteristics of financial reporting.

***E3.26 (LO 6) Writing** A friend of yours, Mindy Gare, recently completed an undergraduate degree in science and has just started working with a biotechnology company. Mindy tells you that the owners of the business are trying to secure new sources of financing which are needed in order for the company to proceed with development of a new healthcare product. Mindy said that her boss told her that the company must put together a report to present to potential investors.

Mindy thought that the company should include in this package the detailed scientific findings related to the Phase I clinical trials for this product. She said, "I know that the biotech industry sometimes has only a 10% success rate with new products, but if we report all the scientific findings, everyone will see what a sure success this is going to be! The president was talking about the importance of following some set of accounting principles. Why do we need to look at some accounting rules? What they need to realize is that we have scientific results that are quite encouraging, some of the most talented employees around, and the start of some really great customer relationships. We haven't made any sales yet, but we will. We just need the funds to get through all the clinical testing and get government approval for our product. Then these investors will be quite happy that they bought in to our company early!"

Instructions

- What is accounting information?
- Comment on how Mindy's suggestions for what should be reported to prospective investors conforms to the qualitative characteristics of accounting information. Do you think that the things that Mindy wants to include in the information for investors will conform to financial reporting guidelines?

Problems: Set A

Prepare adjusting entries, post to ledger accounts, and prepare an adjusted trial balance.

P3.1A (LO 2, 3, 4) Logan Krause started her own consulting firm, Krause Consulting, on May 1, 2020. The trial balance at May 31 is as follows.

Krause Consulting Trial Balance May 31, 2020

Account Number		Debit	Credit
101	Cash	\$ 4,500	
112	Accounts Receivable	6,000	
126	Supplies	1,900	
130	Prepaid Insurance	3,600	
149	Equipment	11,400	
201	Accounts Payable		\$ 4,500
209	Unearned Service Revenue		2,000
301	Owner's Capital		18,700
400	Service Revenue		9,500
726	Salaries and Wages Expense	6,400	
729	Rent Expense	900	
		<u>\$34,700</u>	<u>\$34,700</u>

In addition to those accounts listed on the trial balance, the chart of accounts for Krause Consulting also contains the following accounts and account numbers: No. 150 Accumulated Depreciation—Equipment, No. 212 Salaries and Wages Payable, No. 631 Supplies Expense, No. 717 Depreciation Expense, No. 722 Insurance Expense, and No. 732 Utilities Expense.

Other data:

1. \$900 of supplies have been used during the month.
2. Utilities expense incurred but not paid on May 31, 2020, \$250.
3. The insurance policy is for 2 years.
4. \$400 of the balance in the unearned service revenue account remains unearned at the end of the month.
5. May 31 is a Wednesday, and employees are paid on Fridays. Krause Consulting has two employees, who are paid \$920 each for a 5-day work week.
6. The office furniture has a 5-year life with no salvage value. It is being depreciated at \$190 per month for 60 months.
7. Invoices representing \$1,700 of services performed during the month have not been recorded as of May 31.

Instructions

- a. Prepare the adjusting entries for the month of May. Use J4 as the page number for your journal.
- b. Post the adjusting entries to the ledger accounts. Enter the totals from the trial balance as beginning account balances and place a check mark in the posting reference column.
- c. Prepare an adjusted trial balance at May 31, 2020.

P3.2A (LO 2, 3, 4) Financial Statement Hank's Hotel opened for business on May 1, 2020. Its trial balance before adjustment on May 31 is as follows.

c. Adj. trial balance \$37,944

Prepare adjusting entries, post, and prepare adjusted trial balance and financial statements.



Hank's Hotel Trial Balance May 31, 2020

Account Number		Debit	Credit
101	Cash	\$ 3,400	
126	Supplies	2,080	
130	Prepaid Insurance	2,400	
140	Land	12,000	
141	Buildings	60,000	
149	Equipment	15,000	
201	Accounts Payable		\$ 4,700
208	Unearned Rent Revenue		3,300
275	Mortgage Payable		40,000
301	Owner's Capital		41,380
429	Rent Revenue		10,300
610	Advertising Expense	600	
726	Salaries and Wages Expense	3,300	
732	Utilities Expense	900	
		<u>\$99,680</u>	<u>\$99,680</u>

In addition to those accounts listed on the trial balance, the chart of accounts for Hank's Hotel also contains the following accounts and account numbers: No. 142 Accumulated Depreciation—Buildings, No. 150 Accumulated Depreciation—Equipment, No. 212 Salaries and Wages Payable, No. 230 Interest Payable, No. 619 Depreciation Expense, No. 631 Supplies Expense, No. 718 Interest Expense, and No. 722 Insurance Expense.

Other data:

1. Prepaid insurance is a 1-year policy starting May 1, 2020.
2. A count of supplies shows \$750 of unused supplies on May 31.
3. Annual depreciation is \$3,600 on the buildings and \$1,500 on equipment.
4. The mortgage interest rate is 6%. (The mortgage was taken out on May 1.)
5. Two-thirds of the unearned rent revenue has been earned.
6. Salaries of \$750 are accrued and unpaid at May 31.

Instructions

- c. Adj. trial balance \$101,055
- d. Net income \$4,795
Ending capital \$46,175
Total assets \$92,925

Prepare adjusting entries and financial statements.

- Journalize the adjusting entries on May 31.
- Prepare a ledger using the three-column form of account. Enter the trial balance amounts and post the adjusting entries. (Use J1 as the posting reference.)
- Prepare an adjusted trial balance on May 31.
- Prepare an income statement and an owner's equity statement for the month of May and a balance sheet at May 31.

P3.3A (LO 2, 3, 4) Financial Statement Alena Co. was organized on July 1, 2020. Quarterly financial statements are prepared. The unadjusted and adjusted trial balances as of September 30 are shown below.

Alena Co.
Trial Balance
September 30, 2020

	Unadjusted		Adjusted	
	Dr.	Cr.	Dr.	Cr.
Cash	\$ 8,700		\$ 8,700	
Accounts Receivable	10,400		11,500	
Supplies	1,500		650	
Prepaid Rent	2,200		500	
Equipment	18,000		18,000	
Accumulated Depreciation—Equipment		\$ —0—		\$ 700
Notes Payable		10,000		10,000
Accounts Payable		2,500		2,500
Salaries and Wages Payable		—0—		725
Interest Payable		—0—		100
Unearned Rent Revenue		1,900		450
Owner's Capital		22,000		22,000
Owner's Drawings	1,600		1,600	
Service Revenue		16,000		17,100
Rent Revenue		1,410		2,860
Salaries and Wages Expense	8,000		8,725	
Rent Expense	1,900		3,600	
Depreciation Expense			700	
Supplies Expense			850	
Utilities Expense	1,510		1,510	
Interest Expense			100	
	<u>\$53,810</u>	<u>\$53,810</u>	<u>\$56,435</u>	<u>\$56,435</u>

Instructions

- b. Net income \$4,475
Ending capital \$24,875
Total assets \$38,650

Prepare adjusting entries.

- Journalize the adjusting entries that were made.
- Prepare an income statement and an owner's equity statement for the 3 months ending September 30 and a balance sheet at September 30.
- If the note bears interest at 12%, how many months has it been outstanding?

P3.4A (LO 2, 3) A review of the ledger of Gina Company at December 31, 2020, produces the following data pertaining to the preparation of annual adjusting entries.

- Insurance expense \$4,840
- Rent revenue \$75,500

- Prepaid Insurance \$10,440. The company has separate insurance policies on its buildings and its motor vehicles. Policy B4564 on the building was purchased on April 1, 2019, for \$7,920. The policy has a term of 3 years. Policy A2958 on the vehicles was purchased on January 1, 2020, for \$4,400. This policy has a term of 2 years.
- Unearned Rent Revenue \$429,000. The company began subleasing office space in its new building on November 1. At December 31, the company had the following rental contracts that are paid in full for the entire term of the lease.

Date	Term (in months)	Monthly Rent	Number of Leases
Nov. 1	9	\$5,000	5
Dec. 1	6	\$8,500	3

3. Notes Payable \$120,000. This balance consists of a note for 9 months at an annual interest rate of 6%, dated November 1.
4. Salaries and Wages Payable \$0. There are eight salaried employees. Salaries are paid every Friday for the current week. Five employees receive a salary of \$700 each per week, and three employees earn \$500 each per week. Assume December 31 is a Tuesday. Employees do not work weekends. All employees worked the last 2 days of December.

3. Interest expense \$1,200

4. Salaries and wages expense \$2,000

Instructions

Prepare the adjusting entries at December 31, 2020.

P3.5A (LO 2, 3, 4) Financial Statement On November 1, 2020, the account balances of Hamm Equipment Repair were as follows.

Journalize transactions and follow through accounting cycle to preparation of financial statements.

No.		Debit	No.		Credit
101	Cash	\$ 2,400	154	Accumulated Depreciation—Equipment	\$ 2,000
112	Accounts Receivable	4,250	201	Accounts Payable	2,600
126	Supplies	1,800	209	Unearned Service Revenue	1,200
153	Equipment	12,000	212	Salaries and Wages Payable	700
			301	Owner's Capital	13,950
		<u>\$20,450</u>			<u>\$20,450</u>



During November, the following summary transactions were completed.

- Nov. 8 Paid \$1,700 for salaries due employees, of which \$700 is for October salaries.
 10 Received \$3,620 cash from customers on account.
 12 Received \$3,100 cash for services performed in November.
 15 Purchased equipment on account \$2,000.
 17 Purchased supplies on account \$700.
 20 Paid creditors on account \$2,700.
 22 Paid November rent \$400.
 25 Paid salaries \$1,700.
 27 Performed services on account and billed customers for these services \$2,200.
 29 Received \$600 from customers for future service.

Adjustment data consist of:

1. Supplies on hand \$1,400.
2. Accrued salaries payable \$350.
3. Depreciation for the month is \$200.
4. Services related to unearned service revenue of \$1,220 were performed.

Instructions

- a. Enter the November 1 balances in the ledger accounts.
- b. Journalize the November transactions.
- c. Post to the ledger accounts. Use J1 for the posting reference. Use the following additional accounts: No. 407 Service Revenue, No. 615 Depreciation Expense, No. 631 Supplies Expense, No. 726 Salaries and Wages Expense, and No. 729 Rent Expense.
- d. Prepare a trial balance at November 30.
- e. Journalize and post adjusting entries.
- f. Prepare an adjusted trial balance.
- g. Prepare an income statement and an owner's equity statement for November and a balance sheet at November 30.

d. Trial balance \$25,650

f. Adj. trial balance \$26,200

g. Net income \$1,770;
 Ending capital \$15,720
 Total assets \$19,250

***P3.6A (LO 2, 3, 4, 5) Financial Statement** Gabriel's Graphics Company was organized on January 1, 2020, by Gabriel Mediña. At the end of the first 6 months of operations, the trial balance contained the following accounts.

Prepare adjusting entries, adjusted trial balance, and financial statements using appendix.

	<u>Debit</u>		<u>Credit</u>
Cash	\$ 8,600	Notes Payable	\$ 20,000
Accounts Receivable	14,000	Accounts Payable	9,000
Equipment	45,000	Owner's Capital	22,000
Insurance Expense	2,700	Sales Revenue	52,100
Salaries and Wages Expense	30,000	Service Revenue	6,000
Supplies Expense	3,700		
Advertising Expense	1,900		
Rent Expense	1,500		
Utilities Expense	1,700		
	<u>\$109,100</u>		<u>\$109,100</u>

Analysis reveals the following additional data.

1. The \$3,700 balance in Supplies Expense represents supplies purchased in January. At June 30, \$1,300 of supplies are on hand.
2. The note payable was issued on February 1. It is a 6%, 6-month note.
3. The balance in Insurance Expense is the premium on a one-year policy, dated April 1, 2020.
4. Service revenues are credited to revenue when received. At June 30, services revenue of \$1,300 are unearned.
5. Revenue for services performed but unrecorded at June 30 totals \$2,000.
6. Depreciation is \$2,250 per year.

Instructions

- a. Journalize the adjusting entries at June 30. (Assume adjustments are recorded every 6 months.)
- b. Prepare an adjusted trial balance.
- c. Prepare an income statement and owner's equity statement for the 6 months ended June 30 and a balance sheet at June 30.

b. Adj. trial balance \$112,725
 c. Net income \$19,000
 Ending capital \$41,000
 Total assets \$71,800

Continuing Case

Cookie Creations

(Note: This is a continuation of the Cookie Creations case from Chapters 1 and 2. Use the information from the previous chapters and follow the instructions below using the general ledger accounts you have already prepared.)

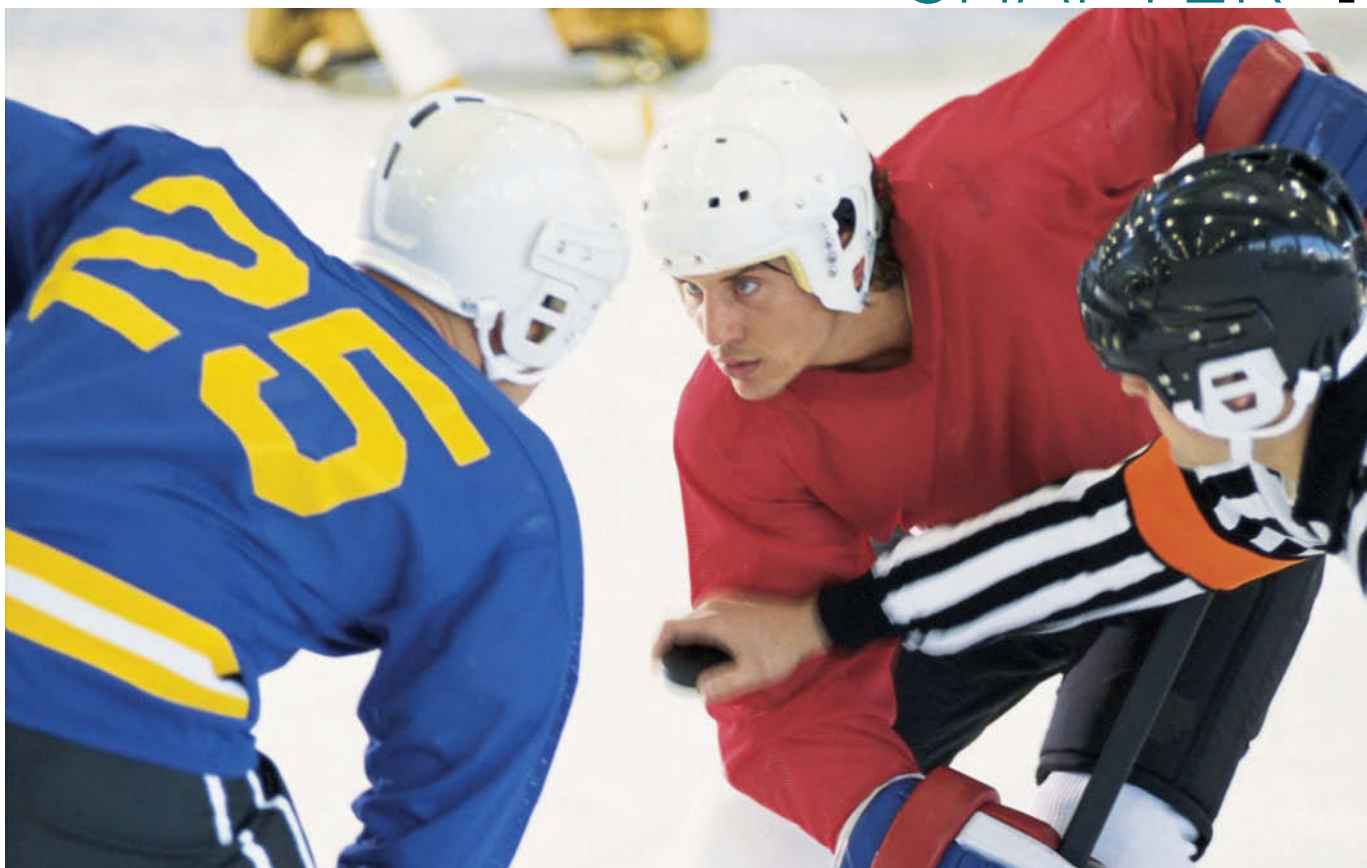
CC3 It is the end of November and Natalie has been in touch with her grandmother. Her grandmother asked Natalie how well things went in her first month of business. Natalie, too, would like to know if she has been profitable or not during November. Natalie realizes that in order to determine Cookie Creations' income, she must first make adjustments.

Natalie puts together the following additional information.

1. A count reveals that \$35 of baking supplies were used during November.
2. Natalie estimates that all of her baking equipment will have a useful life of 5 years or 60 months. (Assume Natalie decides to record a full month's worth of depreciation, regardless of when the equipment was obtained by the business.)
3. Natalie's grandmother has decided to charge interest of 6% on the note payable extended on November 16. The loan plus interest is to be repaid in 24 months. (Assume that half a month of interest accrued during November.)
4. On November 30, a friend of Natalie's asks her to teach a class at the neighborhood school. Natalie agrees and teaches a group of 35 first-grade students how to make gingerbread cookies. The next



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Completing the Accounting Cycle

Chapter Preview

As the following Feature Story highlights, at **Rhino Foods, Inc.**, financial statements help employees understand what is happening in the business. In Chapter 3, we prepared financial statements directly from the adjusted trial balance. However, with so many details involved in the end-of-period accounting procedures, it is easy to make errors. One way to minimize errors in the records and to simplify the end-of-period procedures is to use a worksheet.

In this chapter, we will explain the role of the worksheet in accounting. We also will study the remaining steps in the accounting cycle, especially the closing process, again using Pioneer Advertising as an example. Then, we will consider correcting entries and classified balance sheets.

Feature Story

Everyone Likes to Win

When Ted Castle was a hockey coach at the University of Vermont, his players were self-motivated by their desire to win. Hockey was a game you usually either won or lost. But at **Rhino Foods, Inc.**, a bakery-foods company he founded in Burlington, Vermont, he discovered that manufacturing-line workers were

not so self-motivated. Ted thought, what if he turned the food-making business into a game, with rules, strategies, and trophies?

In a game, knowing the score is all-important. Ted felt that only if the employees know the score—know exactly how the business is doing daily, weekly, monthly—could he turn food-making into a game. But Rhino is a closely held, family-owned business, and its financial statements and profits were confidential. Ted wondered, should he open Rhino's books to the employees?

A consultant put Ted’s concerns in perspective when he said, “Imagine you’re playing touch football. You play for an hour or two, and the whole time I’m sitting there with a book, keeping score. All of a sudden I blow the whistle, and I say, ‘OK, that’s it. Everybody go home.’ I close my book and walk away. How would you feel?” Ted opened his books and revealed the financial statements to his employees.

The next step was to teach employees the rules and strategies of how to “win” at making food. The first lesson: “Your opponent at Rhino is expenses. You must cut and control expenses.” Ted and his staff distilled those lessons into daily

scorecards—production reports and income statements—that keep Rhino’s employees up-to-date on the game. At noon each day, Ted posts the previous day’s results at the entrance to the production room. Everyone checks whether they made or lost money on what they produced the day before. And it’s not just an academic exercise: There’s a bonus check for each employee at the end of every four-week “game” that meets profitability guidelines.

Rhino has flourished since the first game. Employment has increased from 20 to 130 people, while both revenues and profits have grown dramatically.

Chapter Outline

LEARNING OBJECTIVES

LO 1 Prepare a worksheet.	• Steps in preparing a worksheet • Preparing financial statements from a worksheet • Preparing adjusting entries from a worksheet	DO IT! 1 Worksheet
LO 2 Prepare closing entries and a post-closing trial balance.	• Preparing closing entries • Posting closing entries • Preparing a post-closing trial balance	DO IT! 2 Closing Entries
LO 3 Explain the steps in the accounting cycle and how to prepare correcting entries.	• Summary of the accounting cycle • Reversing entries • Correcting entries	DO IT! 3 Correcting Entries
LO 4 Identify the sections of a classified balance sheet.	• Current assets • Long-term investments • Property, plant, and equipment • Intangible assets • Current liabilities • Long-term liabilities • Owner’s equity	DO IT! 4 Balance Sheet Classifications
Go to the Review and Practice section at the end of the chapter for a review of key concepts and practice applications with solutions. Visit WileyPLUS with ORION for additional tutorials and practice opportunities.		

The Worksheet

LEARNING OBJECTIVE 1

Prepare a worksheet.

A **worksheet** is a multiple-column form used in the adjustment process and in preparing financial statements. As its name suggests, the worksheet is a working tool. **It is not a**

permanent accounting record. It is neither a journal nor a part of the general ledger. The worksheet is merely a device used in preparing adjusting entries and the financial statements. Companies generally computerize worksheets using an electronic spreadsheet program such as **Microsoft Excel**.

Illustration 4.1 shows the basic form of a worksheet and the five steps for preparing it. Each step is performed in sequence. **The use of a worksheet is optional.** When a company chooses to use one, it prepares financial statements directly from the worksheet. It enters the adjustments in the worksheet columns and then journalizes and posts the adjustments after it has prepared the financial statements. Thus, worksheets make it possible to provide the financial statements to management and other interested parties at an earlier date.

Steps in Preparing a Worksheet

We will use the October 31 trial balance and adjustment data of Pioneer Advertising from Chapter 3 to illustrate how to prepare a worksheet. In the following pages, we describe and then demonstrate each step of the process.

Worksheet											
P18 fx											
	A	B	C	D	E	F	G	H	I	J	K
1	Worksheet										
2											
3											
4		Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
5	Account Titles	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
6											
7											
8											
9											
10											
11											
12											
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14											
15											
16											
17											
18											
19											
20											
21											
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25											

- 1 Prepare a trial balance on the worksheet.
- 2 Enter adjustment data.
- 3 Enter adjusted balances.
- 4 Extend adjusted balances to appropriate statement columns.
- 5 Total the statement columns, compute net income (or net loss), and complete worksheet.

ILLUSTRATION 4.1

Form and procedure for a worksheet

Step 1

Prepare a Trial Balance on the Worksheet

The first step in preparing a worksheet is to enter all ledger accounts with balances in the account titles column and then enter debit and credit amounts from the ledger in the trial balance columns. **Illustration 4.2** shows the worksheet trial balance for Pioneer Advertising. This trial balance is the same one that appears in Illustration 2.31 and Illustration 3.3.

ILLUSTRATION 4.2 Preparing a trial balance

Pioneer Advertising											
Home Insert Page Layout Formulas Data Review View											
P18 fx											
	A	B	C	D	E	F	G	H	I	J	K
1	Pioneer Advertising Worksheet For the Month Ended October 31, 2020										
2											
3											
4											
5		Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
6											
7	Account Titles	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
8	Cash	15,200									
9	Supplies	2,500									
10	Prepaid Insurance	600									
11	Equipment	5,000									
12	Notes Payable		5,000								
13	Accounts Payable		2,500								
14	Unearned Service Revenue		1,200								
15	Owner's Capital		10,000								
16	Owner's Drawings	500									
17	Service Revenue		10,000								
18	Salaries and Wages Expense	4,000									
19	Rent Expense	900									
20	Totals	28,700	28,700								
21											
22											
23											
24											
25											
26											
27											
28											
29											
30											
31											
32											
33											
34											
35											
36											

Include all accounts with balances from ledger.

Trial balance amounts come directly from ledger accounts.

Step 2**Enter the Adjustments in the Adjustments Columns**

As shown in **Illustration 4.3**, the second step when using a worksheet is to enter all adjustments in the adjustments columns. In entering the adjustments, use applicable trial balance accounts. If additional accounts are needed, insert them on the lines immediately below the trial balance totals. A different letter identifies the debit and credit for each adjusting entry. The term used to describe this process is **keying**. **Companies do not journalize the adjustments until after they complete the worksheet and prepare the financial statements.**

The adjustments for Pioneer Advertising are the same as the adjustments in Illustration 3.23. They are keyed in the adjustments columns of the worksheet as follows.

- a. Pioneer debits an additional account, Supplies Expense, \$1,500 for the cost of supplies used, and credits Supplies \$1,500.
- b. Pioneer debits an additional account, Insurance Expense, \$50 for the insurance that has expired, and credits Prepaid Insurance \$50.
- c. The company needs two additional depreciation accounts. It debits Depreciation Expense \$40 for the month's depreciation, and credits Accumulated Depreciation—Equipment \$40.
- d. Pioneer debits Unearned Service Revenue \$400 for services performed, and credits Service Revenue \$400.
- e. Pioneer debits an additional account, Accounts Receivable, \$200 for services performed but not billed, and credits Service Revenue \$200.
- f. The company needs two additional accounts relating to interest. It debits Interest Expense \$50 for accrued interest, and credits Interest Payable \$50.
- g. Pioneer debits Salaries and Wages Expense \$1,200 for accrued salaries, and credits an additional account, Salaries and Wages Payable, \$1,200.

After Pioneer has entered all the adjustments, the adjustments columns are totaled to prove their equality.

ILLUSTRATION 4.3 Entering the adjustments in the adjustments columns

Pioneer Advertising										
Pioneer Advertising Worksheet For the Month Ended October 31, 2020										
Account Titles	Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
Cash	15,200									
Supplies	2,500			(a) 1,500						
Prepaid Insurance	600			(b) 50						
Equipment	5,000									
Notes Payable		5,000								
Accounts Payable		2,500								
Unearned Service Revenue		1,200	(d) 400							
Owner's Capital		10,000								
Owner's Drawings	500									
Service Revenue		10,000		(d) 400						
				(e) 200						
Salaries and Wages Expense	4,000		(g) 1,200							
Rent Expense	900									
Totals	28,700	28,700								
Supplies Expense			(a) 1,500							
Insurance Expense			(b) 50							
Accum. Depreciation— Equipment				(c) 40						
Depreciation Expense			(c) 40							
Accounts Receivable			(e) 200							
Interest Expense			(f) 50							
Interest Payable				(f) 50						
Salaries and Wages Payable				(g) 1,200						
Totals			3,440	3,440						

Add additional accounts as needed to complete the adjustments:

- (a) Supplies Used.
- (b) Insurance Expired.
- (c) Depreciation Expensed.
- (d) Service Revenue Recognized.
- (e) Service Revenue Accrued.
- (f) Interest Accrued.
- (g) Salaries Accrued.

Enter adjustment amounts in appropriate columns, and use letters to cross-reference the debit and credit adjustments.

Total adjustments columns and check for equality.

Step 3

Enter Adjusted Balances in the Adjusted Trial Balance Columns

As shown in **Illustration 4.4**, Pioneer Advertising next determines the adjusted balance of an account by combining the amounts entered in the first four columns of the worksheet for each account. For example, the Prepaid Insurance account in the trial balance columns has a \$600 debit balance and a \$50 credit in the adjustments columns. The result is a \$550 debit balance recorded in the adjusted trial balance columns. **For each account, the amount in the adjusted trial balance columns is the balance that will appear in the ledger after journalizing and posting the adjusting entries.** The balances in these columns are the same as those in the adjusted trial balance in Illustration 3.25.

After Pioneer has entered all account balances in the adjusted trial balance columns, the columns are totaled to prove their equality. If the column totals do not agree, the financial statement columns will not balance and the financial statements will be incorrect.

ILLUSTRATION 4.4 Entering adjusted balances in the adjusted trial balance columns

Pioneer Advertising											
Home Insert Page Layout Formulas Data Review View											
P18 fx											
	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
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36											

Pioneer Advertising											
Worksheet											
For the Month Ended October 31, 2020											
	Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet		
Account Titles	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	
Cash	15,200				15,200						
Supplies	2,500			(a) 1,500	1,000						
Prepaid Insurance	600			(b) 50	550						
Equipment	5,000				5,000						
Notes Payable		5,000				5,000					
Accounts Payable		2,500				2,500					
Unearned Service Revenue		1,200	(d) 400			800					
Owner's Capital		10,000				10,000					
Owner's Drawings	500				500						
Service Revenue		10,000		(d) 400		10,600					
				(e) 200							
Salaries and Wages Expense	4,000		(g) 1,200		5,200						
Rent Expense	900				900						
Totals	28,700	28,700									
Supplies Expense			(a) 1,500		1,500						
Insurance Expense			(b) 50		50						
Accum. Depreciation— Equipment				(c) 40		40					
Depreciation Expense			(c) 40		40						
Accounts Receivable			(e) 200		200						
Interest Expense			(f) 50		50						
Interest Payable				(f) 50		50					
Salaries and Wages Payable				(g) 1,200		1,200					
Totals			3,440	3,440	30,190	30,190					

Combine trial balance amounts with adjustment amounts to obtain the adjusted trial balance.
Total adjusted trial balance columns and check for equality.

Step 4

Extend Adjusted Trial Balance Amounts to Appropriate Financial Statement Columns

HELPFUL HINT

Every adjusted trial balance amount must be extended to one of the four statement columns.

As shown in **Illustration 4.5**, the fourth step is to extend adjusted trial balance amounts to the income statement and balance sheet columns of the worksheet (see **Helpful Hint**). Pioneer Advertising enters balance sheet accounts in the appropriate balance sheet debit and credit columns. For instance, it enters Cash in the balance sheet debit column, and Notes Payable in the balance sheet credit column. Pioneer extends Accumulated Depreciation—Equipment to the balance sheet credit column. The reason is that accumulated depreciation is a contra asset account with a credit balance.

Because the worksheet does not have columns for the owner's equity statement, Pioneer extends the balance in owner's capital to the balance sheet credit column. In addition, it extends the balance in owner's drawings to the balance sheet debit column because it is an owner's equity account with a debit balance.

The company enters the expense and revenue accounts such as Salaries and Wages Expense and Service Revenue in the appropriate income statement columns.

ILLUSTRATION 4.5 Extending the adjusted trial balance amounts to appropriate financial statement columns

Pioneer Advertising										
Worksheet										
For the Month Ended October 31, 2020										
Account Titles	Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
Cash	15,200				15,200				15,200	
Supplies	2,500			(a) 1,500	1,000				1,000	
Prepaid Insurance	600			(b) 50	550				550	
Equipment	5,000				5,000				5,000	
Notes Payable		5,000				5,000				5,000
Accounts Payable		2,500				2,500				2,500
Unearned Service Revenue		1,200	(d) 400			800				800
Owner's Capital		10,000				10,000				10,000
Owner's Drawings	500				500				500	
Service Revenue		10,000		(d) 400		10,600	10,600			
				(e) 200						
Salaries and Wages Expense	4,000		(g) 1,200		5,200		5,200			
Rent Expense	900				900		900			
Totals	28,700	28,700								
Supplies Expense			(a) 1,500		1,500		1,500			
Insurance Expense			(b) 50		50		50			
Accum. Depreciation—Equipment				(c) 40		40				40
Depreciation Expense			(c) 40		40		40			
Accounts Receivable			(e) 200		200				200	
Interest Expense			(f) 50		50		50			
Interest Payable				(f) 50		50				50
Salaries and Wages Payable				(g) 1,200		1,200				1,200
Totals			3,440	3,440	30,190	30,190				

Extend all revenue and expense account balances to the income statement columns.

Extend all asset and liability account balances, as well as owner's capital and drawings account balances, to the balance sheet columns.

Step 5

Total the Statement Columns, Compute the Net Income (or Net Loss), and Complete the Worksheet

As shown in **Illustration 4.6**, Pioneer Advertising must now total each of the financial statement columns. The net income or loss for the period is the difference between the totals of the two income statement columns. If total credits exceed total debits, the result is net income. In such a case, the company inserts the words “Net Income” in the account titles space. It then enters the amount in the income statement debit column and the balance sheet credit column. **The debit amount balances the income statement columns; the credit amount balances the balance sheet columns.** In addition, the credit in the balance sheet column indicates the increase in owner’s equity resulting from net income.

What if total debits in the income statement columns exceed total credits? In that case, Pioneer has a net loss. It enters the amount of the net loss in the income statement credit column and the balance sheet debit column.

After entering the net income or net loss, Pioneer determines new column totals. The totals shown in the debit and credit income statement columns will match. So will the totals shown in the debit and credit balance sheet columns. If either the income statement columns or the balance sheet columns are not equal after the net income or net loss has been entered, there is an error in the worksheet.

ILLUSTRATION 4.6 Computing net income or net loss and completing the worksheet

Pioneer Advertising											
Home Insert Page Layout Formulas Data Review View											
	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5											
6											
7	Account Titles	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
8	Cash	15,200				15,200				15,200	
9	Supplies	2,500			(a) 1,500	1,000				1,000	
10	Prepaid Insurance	600			(b) 50	550				550	
11	Equipment	5,000				5,000				5,000	
12	Notes Payable		5,000				5,000				5,000
13	Accounts Payable		2,500				2,500				2,500
14	Unearned Service Revenue		1,200	(d) 400			800				800
15	Owner's Capital		10,000				10,000				10,000
16	Owner's Drawings	500				500				500	
17	Service Revenue		10,000		(d) 400		10,600		10,600		
18					(e) 200						
19	Salaries and Wages Expense	4,000		(g) 1,200		5,200		5,200			
20	Rent Expense	900				900		900			
21	Totals	28,700	28,700								
22											
23											
24	Supplies Expense			(a) 1,500		1,500		1,500			
25	Insurance Expense			(b) 50		50		50			
26	Accum. Depreciation—										
27	Equipment				(c) 40		40				40
28	Depreciation Expense			(c) 40		40		40			
29	Accounts Receivable			(e) 200		200				200	
30	Interest Expense			(f) 50		50		50			
31	Interest Payable				(f) 50		50				50
32	Salaries and Wages Payable				(g) 1,200		1,200				1,200
33	Totals			3,440	3,440	30,190	30,190	7,740	10,600	22,450	19,590
34											
35	Net Income							2,860			2,860
36	Totals							10,600	10,600	22,450	22,450

The difference between the totals of the two income statement columns determines net income or net loss.

Net income is extended to the credit column of the balance sheet columns. (Net loss would be extended to the debit column.)

Preparing Financial Statements from a Worksheet

After a company has completed a worksheet, it has at hand all the data required for preparation of financial statements. The income statement is prepared from the income statement columns. The owner's equity statement and balance sheet are prepared from the balance sheet columns. **Illustration 4.7** shows the financial statements prepared from Pioneer Advertising's

ILLUSTRATION 4.7

Financial statements from a worksheet

Pioneer Advertising Income Statement For the Month Ended October 31, 2020			
Revenues			
Service revenue			\$10,600
Expenses			
Salaries and wages expense	\$5,200		
Supplies expense	1,500		
Rent expense	900		
Insurance expense	50		
Interest expense	50		
Depreciation expense	40		
Total expenses		<u>7,740</u>	
Net income			<u>\$ 2,860</u>

Pioneer Advertising Owner's Equity Statement For the Month Ended October 31, 2020			
Owner's capital, October 1			\$ -0-
Add: Investments	\$10,000		
Net income	<u>2,860</u>		<u>12,860</u>
			12,860
Less: Drawings			<u>500</u>
Owner's capital, October 31			<u>\$12,360</u>

Pioneer Advertising Balance Sheet October 31, 2020			
<u>Assets</u>			
Cash			\$15,200
Accounts receivable			200
Supplies			1,000
Prepaid insurance			550
Equipment	\$5,000		
Less: Accumulated depreciation—equipment	<u>40</u>		<u>4,960</u>
Total assets			<u>\$21,910</u>
<u>Liabilities and Owner's Equity</u>			
Liabilities			
Notes payable	\$5,000		
Accounts payable	2,500		
Interest payable	50		
Unearned service revenue	800		
Salaries and wages payable	<u>1,200</u>		
Total liabilities			\$ 9,550
Owner's equity			
Owner's capital			<u>12,360</u>
Total liabilities and owner's equity			<u>\$21,910</u>

worksheet. At this point, the company has not journalized or posted adjusting entries. Therefore, ledger balances for some accounts are not the same as the financial statement amounts.

The amount shown for owner's capital on the worksheet is the account balance **before considering drawings and net income (or loss)**. When the owner has made no additional investments of capital during the period, this worksheet amount for owner's capital is the balance at the beginning of the period.

Using a worksheet, companies can prepare financial statements before they journalize and post adjusting entries. **However, the completed worksheet is not a substitute for formal financial statements.** The format of the data in the financial statement columns of the worksheet is not the same as the format of the financial statements. **A worksheet is essentially a working tool of the accountant;** companies do not distribute it to management and other parties.

Preparing Adjusting Entries from a Worksheet

A worksheet is not a journal, and it cannot be used as a basis for posting to ledger accounts. To adjust the accounts, the company must journalize the adjustments and post them to the ledger. **The adjusting entries are prepared from the adjustments columns of the worksheet.** The reference letters in the adjustments columns and the explanations of the adjustments at the bottom of the worksheet help identify the adjusting entries (see **Helpful Hint**). The journalizing and posting of adjusting entries **follows** the preparation of financial statements when a worksheet is used. The adjusting entries on October 31 for Pioneer Advertising are the same as those shown in Illustration 3.23.

HELPFUL HINT

Note that writing the explanation to the adjustment at the bottom of the worksheet is not required.

DO IT! 1 | Worksheet

Susan Elbe is preparing a worksheet. Explain to Susan how she should extend the following adjusted trial balance accounts to the financial statement columns of the worksheet.

Cash	Owner's Drawings
Accumulated Depreciation—Equipment	Service Revenue
Accounts Payable	Salaries and Wages Expense

Solution

Income statement debit column—Salaries and Wages Expense
 Income statement credit column—Service Revenue
 Balance sheet debit column—Cash; Owner's Drawings
 Balance sheet credit column—Accumulated Depreciation—Equipment; Accounts Payable

Related exercise material: **BE4.1, BE4.2, BE4.3, DO IT! 4.1, E4.1, E4.2, E4.3, E4.5, and E4.6.**

ACTION PLAN

- **Balance sheet:** Extend assets to debit column. Extend liabilities to credit column. Extend contra assets to credit column. Extend drawings account to debit column.
- **Income statement:** Extend expenses to debit column. Extend revenues to credit column.

Closing the Books

LEARNING OBJECTIVE 2

Prepare closing entries and a post-closing trial balance.

ANALYZE

JOURNALIZE

POST

TRIAL
BALANCEADJUSTING
ENTRIESADJUSTED
TRIAL
BALANCEPREPARE
FINANCIAL
STATEMENTSJournalize and
post closing
entriesPrepare a
post-closing
trial balance

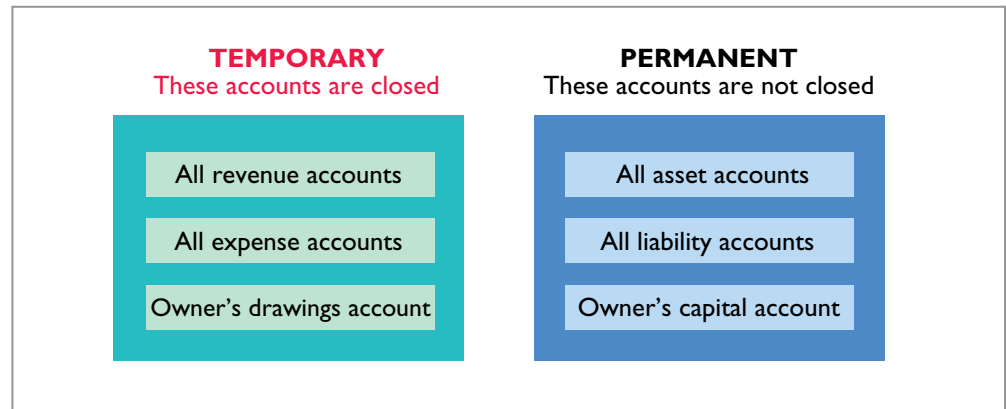
At the end of the accounting period, the company makes the accounts ready for the next period. This is called **closing the books**. In closing the books, the company distinguishes between temporary and permanent accounts.

**ALTERNATIVE
TERMINOLOGY**

Temporary accounts are sometimes called *nominal accounts*, and permanent accounts are sometimes called *real accounts*.

Temporary accounts relate only to a given accounting period. They include all income statement accounts and the owner's drawings account. **The company closes all temporary accounts at the end of the period.**

In contrast, **permanent accounts** relate to one or more future accounting periods. They consist of all balance sheet accounts, including the owner's capital account. **Permanent accounts are not closed from period to period.** Instead, the company carries forward the balances of permanent accounts into the next accounting period. **Illustration 4.8** identifies the accounts in each category (see **Alternative Terminology**).

ILLUSTRATION 4.8**Temporary versus permanent accounts**

Preparing Closing Entries

At the end of the accounting period, the company transfers temporary account balances to the permanent owner's equity account, Owner's Capital, by means of closing entries.¹

Closing entries formally recognize in the ledger the transfer of net income (or net loss) and owner's drawings to owner's capital. The owner's equity statement shows the results of these entries. **Closing entries also produce a zero balance in each temporary account.** The temporary accounts are then ready to accumulate data in the next accounting period separate from the data of prior periods. Permanent accounts are not closed.

Journalizing and posting closing entries is a required step in the accounting cycle (see Illustration 4.15). The company performs this step after it has prepared financial statements. In contrast to the steps in the cycle that you have already studied, companies generally journalize and post closing entries **only at the end of the annual accounting period**. Thus, all temporary accounts will contain data for the entire accounting period.

In preparing closing entries, companies could close each income statement account directly to owner's capital. However, to do so would result in excessive detail in the permanent Owner's Capital account. Instead, companies close the revenue and expense accounts to another temporary account, **Income Summary**, and then transfer the resulting net income or net loss from this account to owner's capital.

Companies **record closing entries in the general journal**. A center caption, Closing Entries, inserted in the journal between the last adjusting entry and the first closing entry, identifies these entries. Then the company posts the closing entries to the ledger accounts.

Companies generally prepare closing entries directly from the adjusted balances in the ledger. They could prepare separate closing entries for each nominal account, but the following four entries accomplish the desired result more efficiently:

1. Debit each revenue account for its balance, and credit Income Summary for total revenues.

¹We explain closing entries for a partnership and for a corporation in Chapters 12 and 13, respectively.

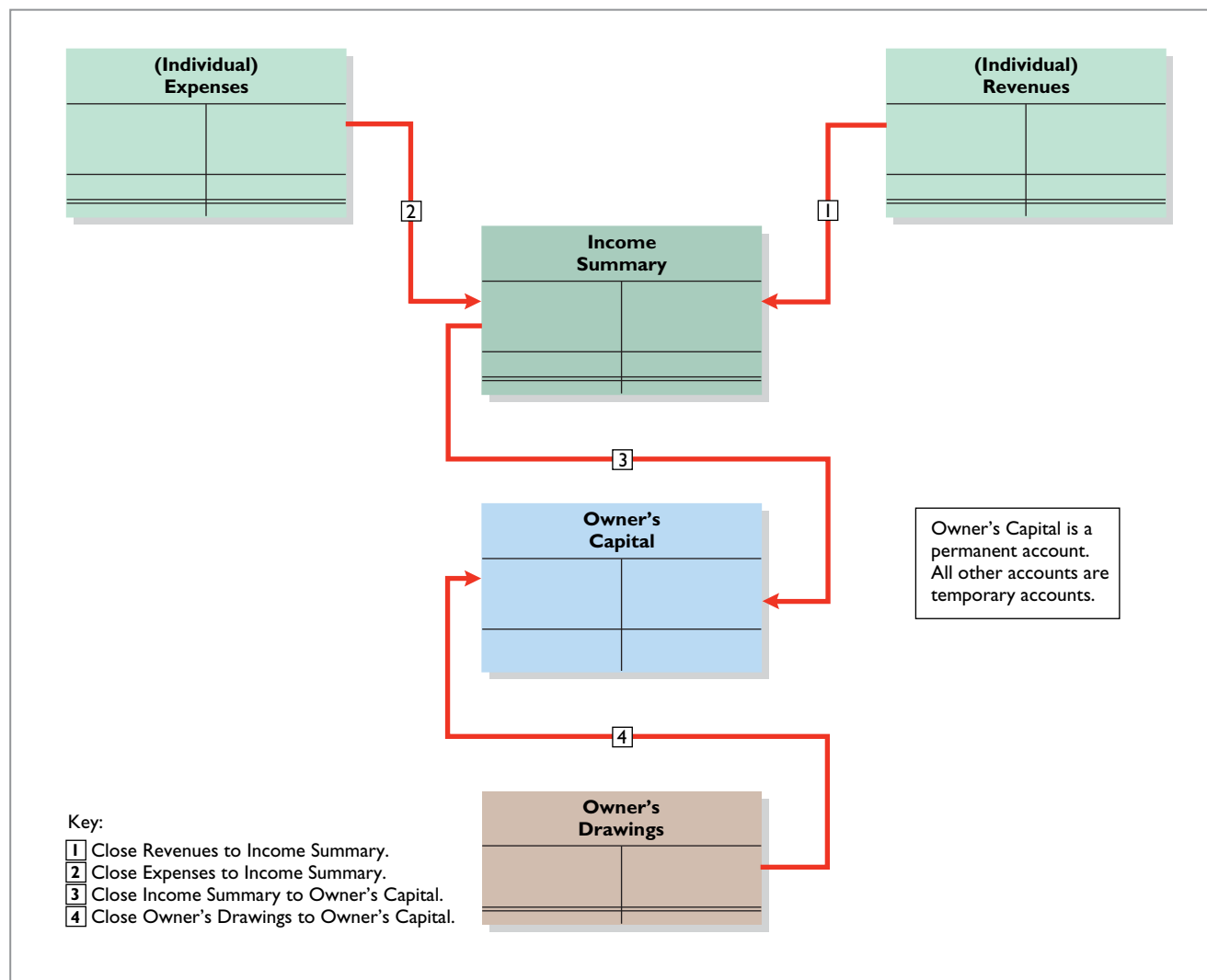
2. Debit Income Summary for total expenses, and credit each expense account for its balance.
3. Debit Income Summary and credit Owner's Capital for the amount of net income.
4. Debit Owner's Capital for the balance in the Owner's Drawings account, and credit Owner's Drawings for the same amount (see **Helpful Hint**).

HELPFUL HINT

Owner's Drawings is closed directly to Owner's Capital and not to Income Summary. Owner's Drawings is not an expense.

Illustration 4.9 presents a diagram of the closing process. In it, the boxed numbers refer to the four entries required in the closing process.

ILLUSTRATION 4.9 Diagram of closing process—proprietorship



If there were a net loss (because expenses exceeded revenues), entry 3 in Illustration 4.9 would be reversed: there would be a credit to Income Summary and a debit to Owner's Capital.

Closing Entries Illustrated

In practice, companies generally prepare closing entries only at the end of the annual accounting period. However, to illustrate the journalizing and posting of closing entries, we will assume that Pioneer Advertising closes its books monthly. **Illustration 4.10** shows the closing entries at October 31. (The numbers in parentheses before each entry correspond to the four entries diagrammed in Illustration 4.9.)

ILLUSTRATION 4.10

Closing entries journalized

GENERAL JOURNAL				J3
Date	Account Titles and Explanation	Ref.	Debit	Credit
	Closing Entries			
2020	(1)			
Oct. 31	Service Revenue	400	10,600	
	Income Summary	350		10,600
	(To close revenue account)			
	(2)			
31	Income Summary	350	7,740	
	Supplies Expense	631		1,500
	Depreciation Expense	711		40
	Insurance Expense	722		50
	Salaries and Wages Expense	726		5,200
	Rent Expense	729		900
	Interest Expense	905		50
	(To close expense accounts)			
	(3)			
31	Income Summary	350	2,860	
	Owner's Capital	301		2,860
	(To close net income to capital)			
	(4)			
31	Owner's Capital	301	500	
	Owner's Drawings	306		500
	(To close drawings to capital)			

Note that the amounts for Income Summary in entries (1) and (2) are the totals of the income statement credit and debit columns, respectively, in the worksheet.

A couple of cautions in preparing closing entries. (1) Avoid unintentionally doubling the revenue and expense balances rather than zeroing them. (2) Do not close Owner's Drawings through the Income Summary account. **Owner's Drawings is not an expense, and it is not a factor in determining net income.**

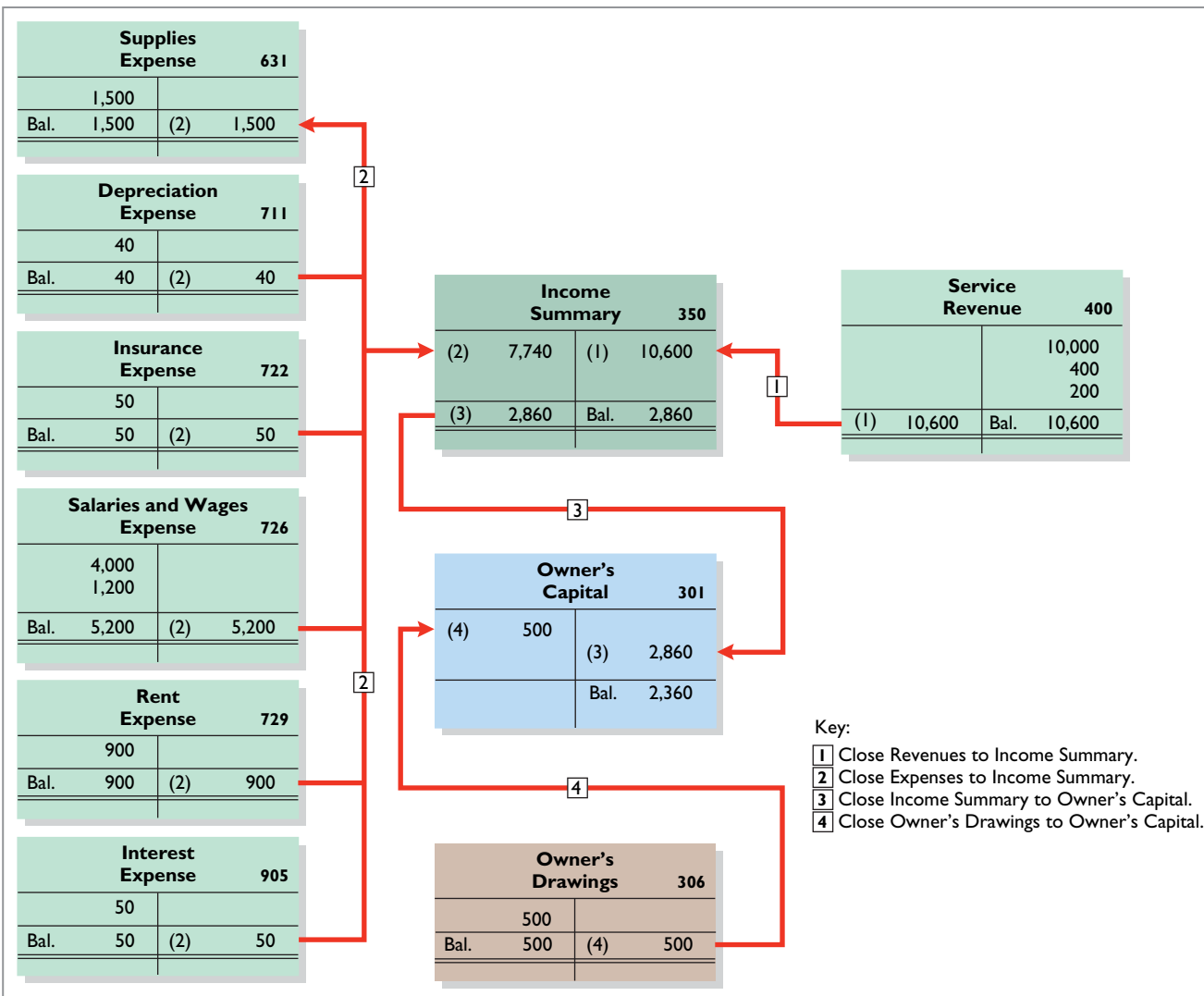
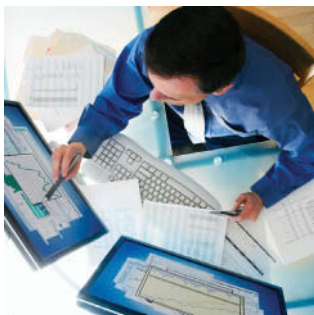
Posting Closing Entries

Illustration 4.11 shows the posting of the closing entries and the underlining (ruling) of the accounts. Note that all temporary accounts have zero balances after posting the closing entries. In addition, notice that the balance in owner's capital (Owner's Capital) represents the total equity of the owner at the end of the accounting period. This balance is shown on the balance sheet and is the ending capital reported on the owner's equity statement, as shown in Illustration 4.7. Pioneer Advertising uses the Income Summary account only in closing. It does not journalize and post entries to this account during the year (see **Helpful Hint**).

As part of the closing process, Pioneer totals, balances, and double-underlines its temporary accounts—revenues, expenses, and Owner's Drawings, as shown in T-account form in Illustration 4.11. It does not close its permanent accounts—assets, liabilities, and Owner's Capital. Instead, Pioneer draws a single underline beneath the current-period entries for the permanent accounts. The account balance is then entered below the single underline and is carried forward to the next period (for example, see Owner's Capital).

HELPFUL HINT

The balance in Income Summary before it is closed must equal the net income or net loss for the period.

ILLUSTRATION 4.11 Posting of closing entries**Accounting Across the Organization** Cisco Systems

© Steve Cole/iStockphoto

Performing the Virtual Close

Technology has dramatically shortened the closing process. Recent surveys have reported that the average company now takes only six to seven days to close, rather than the previous 20 days. But a few companies do much better. **Cisco Systems** can perform a “virtual close”—closing within 24 hours on any day

in the quarter. The same is true at **Lockheed Martin Corp.**, which improved its closing time by 85% in just the last few years. Not very

long ago, it took 14 to 16 days. Managers at these companies emphasize that this increased speed has not reduced the accuracy and completeness of the data.

This is not just showing off. Knowing exactly where you are financially all of the time allows the company to respond faster than its competitors. It also means that the hundreds of people who used to spend 10 to 20 days a quarter tracking transactions can now be more usefully employed on things such as mining data for business intelligence to find new business opportunities.

Source: “Reporting Practices: Few Do It All,” *Financial Executive* (November 2003), p. 11.

Who else benefits from a shorter closing process? (Go to WileyPLUS for this answer and additional questions.)

Preparing a Post-Closing Trial Balance

After Pioneer Advertising has journalized and posted all closing entries, it prepares another trial balance, called a **post-closing trial balance**, from the ledger. The post-closing trial balance lists permanent accounts and their balances after the journalizing and posting of closing entries. The purpose of the post-closing trial balance is **to prove the equality of the permanent account balances carried forward into the next accounting period**. Since all temporary accounts will have zero balances, **the post-closing trial balance will contain only permanent—balance sheet—accounts**.

Illustration 4.12 shows the post-closing trial balance for Pioneer Advertising.

ILLUSTRATION 4.12

Post-closing trial balance

Pioneer Advertising Post-Closing Trial Balance October 31, 2020		
	Debit	Credit
Cash	\$15,200	
Accounts Receivable	200	
Supplies	1,000	
Prepaid Insurance	550	
Equipment	5,000	
Accumulated Depreciation—Equipment		\$ 40
Notes Payable		5,000
Accounts Payable		2,500
Unearned Service Revenue		800
Salaries and Wages Payable		1,200
Interest Payable		50
Owner's Capital		12,360
	<u>\$21,950</u>	<u>\$21,950</u>

Pioneer prepares the post-closing trial balance from the permanent accounts in the ledger.

Illustration 4.13 shows the permanent accounts in Pioneer's general ledger.

ILLUSTRATION 4.13 General ledger, permanent accounts**(Permanent Accounts Only)**

GENERAL LEDGER											
Cash No. 101						Accounts Payable No. 201					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 1		J1	10,000		10,000	Oct. 5		J1		2,500	2,500
2		J1	1,200		11,200						
3		J1		900	10,300						
4		J1		600	9,700						
20		J1		500	9,200						
26		J1		4,000	5,200						
31		J1	10,000		15,200						
Accounts Receivable No. 112						Unearned Service Revenue No. 209					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 31	Adj. entry	J2	200		200	Oct. 2		J1		1,200	1,200
						31	Adj. entry	J2	400		800
Supplies No. 126						Salaries and Wages Payable No. 212					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 31	Adj. entry	J2	200		200	Oct. 31	Adj. entry	J2		1,200	1,200
Prepaid Insurance No. 130						Interest Payable No. 230					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 5		J1	2,500		2,500	Oct. 31	Adj. entry	J2		50	50
31	Adj. entry	J2		1,500	1,000						
Equipment No. 157						Owner's Capital No. 301					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 1		J1	5,000		5,000	Oct. 1		J1		10,000	10,000
						31	Closing entry	J3		2,860	12,860
						31	Closing entry	J3	500		12,360
Accumulated Depreciation—Equipment No. 158											
Date	Explanation	Ref.	Debit	Credit	Balance						
2020											
Oct. 31	Adj. entry	J2		40	40						
Notes Payable No. 200											
Date	Explanation	Ref.	Debit	Credit	Balance						
2020											
Oct. 1		J1		5,000	5,000						

Note: The permanent accounts for Pioneer Advertising are shown here. Illustration 4.14 shows the temporary accounts. Both permanent and temporary accounts are part of the general ledger. They are segregated here to aid in learning.

A post-closing trial balance provides evidence that the company has properly journalized and posted the closing entries. It also shows that the accounting equation is in balance at the end of the accounting period. However, like the trial balance, it does not prove that Pioneer has recorded all transactions or that the ledger is correct. For example, the post-closing trial balance still will balance even if a transaction is not journalized and posted or if a transaction is journalized and posted twice.

The remaining accounts in the general ledger are temporary accounts, shown in [Illustration 4.14](#). After Pioneer correctly posts the closing entries, each temporary account has a zero balance. These accounts are double-underlined to finalize the closing process.

ILLUSTRATION 4.14 General ledger, temporary accounts

(Temporary Accounts Only)

GENERAL LEDGER											
Owner's Drawings No. 306						Insurance Expense No. 722					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 20		J1	500		500	Oct. 31	Adj. entry	J2	50		50
31	Closing entry	J3		500	-0-	31	Closing entry	J3		50	-0-
Income Summary No. 350						Salaries and Wages Expense No. 726					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 31	Closing entry	J3		10,600	10,600	Oct. 26		J1	4,000		4,000
31	Closing entry	J3	7,740		2,860	31	Adj. entry	J2	1,200		5,200
31	Closing entry	J3	2,860		-0-	31	Closing entry	J3		5,200	-0-
Service Revenue No. 400						Rent Expense No. 729					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
2020						2020					
Oct. 31		J1		10,000	10,000	Oct. 3		J1	900		900
31	Adj. entry	J2		400	10,400	31	Closing entry	J3		900	-0-
31	Adj. entry	J2		200	10,600						
31	Closing entry	J3	10,600		-0-	Interest Expense No. 905					
Supplies Expense No. 631						Date	Explanation	Ref.	Debit	Credit	Balance
Date	Explanation	Ref.	Debit	Credit	Balance	2020					
2020						Oct. 31	Adj. entry	J2	50		50
Oct. 31	Adj. entry	J2	1,500		1,500	31	Closing entry	J3		50	-0-
31	Closing entry	J3		1,500	-0-						
Depreciation Expense No. 711						<i>Note:</i> The temporary accounts for Pioneer Advertising are shown here. Illustration 4.13 shows the permanent accounts. Both permanent and temporary accounts are part of the general ledger. They are segregated here to aid in learning.					
Date	Explanation	Ref.	Debit	Credit	Balance						
2020											
Oct. 31	Adj. entry	J2	40		40						
31	Closing entry	J3		40	-0-						

DO IT! 2 | Closing Entries

Hancock Company has the following balances in selected accounts of its adjusted trial balance.

Accounts Payable	\$27,000	Owner's Drawings	\$15,000
Service Revenue	98,000	Owner's Capital	42,000
Rent Expense	22,000	Accounts Receivable	38,000
Salaries and Wages Expense	51,000	Supplies Expense	7,000

Prepare the closing entries at December 31.

ACTION PLAN

- Close revenue and expense accounts to Income Summary.
- Close Income Summary to Owner's Capital.
- Close Owner's Drawings to Owner's Capital.

Solution

Dec. 31	Service Revenue	98,000	
	Income Summary		98,000
	(To close revenue account to Income Summary)		
31	Income Summary	80,000	
	Salaries and Wages Expense		51,000
	Rent Expense		22,000
	Supplies Expense		7,000
	(To close expense accounts to Income Summary)		

31	Income Summary (\$98,000 – \$80,000)	18,000	
	Owner's Capital		18,000
	(To close net income to owner's capital)		
31	Owner's Capital	15,000	
	Owner's Drawings		15,000
	(To close owner's drawings to owner's capital)		

Related exercise material: **BE4.4, BE4.5, BE4.6, DO IT! 4.2, E4.4, E4.7, E4.8, and E4.11.**

The Accounting Cycle and Correcting Entries

LEARNING OBJECTIVE 3

Explain the steps in the accounting cycle and how to prepare correcting entries.

Summary of the Accounting Cycle

Illustration 4.15 summarizes the steps in the accounting cycle. You can see that the cycle begins with the analysis of business transactions and ends with the preparation of a post-closing trial balance. Companies perform the steps in the cycle in sequence and repeat them in each accounting period.

Steps 1–3 may occur daily during the accounting period. Companies perform Steps 4–7 on a periodic basis, such as monthly, quarterly, or annually. Steps 8 and 9—closing entries and a post-closing trial balance—usually take place only at the end of a company's **annual** accounting period.

There are also two **optional steps** in the accounting cycle. As you have seen, companies may use a worksheet in preparing adjusting entries and financial statements. In addition, they may use reversing entries, as explained below.

Reversing Entries—An Optional Step

Some accountants prefer to reverse certain adjusting entries by making a **reversing entry** at the beginning of the next accounting period. A reversing entry is the exact opposite of the adjusting entry made in the previous period. **Use of reversing entries is an optional book-keeping procedure; it is not a required step in the accounting cycle.** Accordingly, we have chosen to cover this topic in an appendix at the end of the chapter.

Correcting Entries—An Avoidable Step

Unfortunately, errors may occur in the recording process. Companies should correct errors, **as soon as they discover them**, by journalizing and posting **correcting entries**. If the accounting records are free of errors, no correcting entries are needed.

You should recognize several differences between correcting entries and adjusting entries. First, adjusting entries are an integral part of the accounting cycle. Correcting entries, on the other hand, are unnecessary if the records are error-free. Second, companies journalize and post adjustments **only at the end of an accounting period**. In contrast, companies make correcting entries **whenever they discover an error** (see **Ethics Note**). Finally, adjusting entries always affect at least one balance sheet account and one income statement account. In

ETHICS NOTE

When companies find errors in previously released income statements, they restate those numbers. Perhaps because of the increased scrutiny caused by Sarbanes-Oxley, in a recent year companies filed a record 1,195 restatements

Required steps in the accounting cycle

Equation Analysis

Assets					=	Liabilities		Owner's Equity	
Cash	Receivable	Supplies	Equipment			Payable	Capital	Revenues	Expenses
\$9,200		\$1,600	\$7,000	=		\$1,850	\$15,000	\$1,200	\$250
(6) <u>+1,500</u>	<u>+2,000</u>							<u>+3,500</u>	
\$10,700	\$2,000	\$1,600	\$7,000	=		\$1,850	\$15,000	\$4,700	\$250
		\$21,300					\$21,300		

Pioneer Advertising
Post-Closing Trial Balance
October 31, 2020

	Debit	Credit
Cash	\$15,200	
Accounts Receivable	200	
Supplies	1,000	
Prepaid Insurance	550	
Equipment	5,000	
Accumulated Depreciation—Equipment		\$ 400
Notes Payable		5,000
Accounts Payable		2,500
Unearned Service Revenue		800
Salaries and Wages Payable		1,200

GENERAL JOURNAL

GENERAL JOURNAL			PAGE J1	
Date	Account Titles and Explanation	Ref.	Debit	Credit
2020				
Oct. 1	Cash	101	10,000	
	Owner's Capital	301		10,000
	(Owner's investment of cash in business)			
1	Equipment	157	5,000	
	Notes Payable	200		5,000
	(Issued 3-month, 12% note for office equipment)			
2	Cash	101	1,200	
	Unearned Service Revenue	209		1,200
	(Received cash from R. Knox for future services)			

g Entries

GENERAL JOURNAL					J3
Date	Account Titles and Explanation	Ref.	Debit	Credit	
	Closing Entries				
	(1)				
2020	Service Revenue	400	10,600		
Oct. 31	Income Summary	350		10,600	
	(To close revenue account)				
	(2)				
31	Income Summary	350	7,740		
	Supplies Expense	631		1,500	
	Depreciation Expense	711		40	
	Insurance Expense	722		50	
	Salaries and Wages Expense	726		5,200	
	Rent Expense	729		900	

GENERAL LEDGER

[illegible]

Pioneer Advertising
Income Statement
For the Month Ended October 31, 2020

Revenue

Service

Pioneer Advertising Owner's Equity Statement For the Month Ended October 31, 2020	
--	--

Pioneer Advertising Balance Sheet October 31, 2020	
<u>Assets</u>	
Cash	\$15,200
Accounts receivable	200
Supplies	1,000
Prepaid insurance	750

**Pioneer Advertising
Trial Balance
October 31, 2020**

	Debit	Credit
Cash	\$15,200	
Supplies	2,500	
Prepaid Insurance	600	
Equipment	5,000	
Notes Payable		\$ 5,000
Accounts Payable		2,500
Unearned Service Revenue		1,200
Owner's Capital		10,000
Owner's Drawings	500	
Service Revenue		10,000

**Pioneer Advertising
Adjusted Trial Balance
October 31, 2020**

	<u>Debit</u>	<u>Credit</u>
Cash	\$15,200	
Accounts Receivable	200	
Supplies	1,000	
Prepaid Insurance	550	
Equipment	5,000	
Accumulated Depreciation—Equipment		\$ 40
Notes Payable		5,000
Accounts Payable		2,500
Interest Payable		50
Unearned Service Revenue		800
Salaries and Wages Payable		1,200

GENERAL JOURNAL

[illegible]

contrast, correcting entries may involve any combination of accounts in need of correction.
Correcting entries must be posted before closing entries.

To determine the correcting entry, it is useful to compare the incorrect entry with the correct entry. Doing so helps identify the accounts and amounts that should—and should not—be corrected. After comparison, the accountant makes an entry to correct the accounts. The following two cases for Mercato Co. illustrate this approach.

Case 1

On May 10, Mercato Co. journalized and posted a \$50 cash collection on account from a customer as a debit to Cash \$50 and a credit to Service Revenue \$50. The company discovered the error on May 20, when the customer paid the remaining balance in full (see **Illustration 4.16**).

Incorrect Entry (May 10)			Correct Entry (May 10)		
Cash	50		Cash	50	
Service Revenue		50	Accounts Receivable		50

ILLUSTRATION 4.16

Comparison of entries

Comparison of the incorrect entry with the correct entry reveals that the debit to Cash \$50 is correct. However, the \$50 credit to Service Revenue should have been credited to Accounts Receivable. As a result, both Service Revenue and Accounts Receivable are overstated in the ledger. Mercato makes the correcting entry shown in **Illustration 4.17**.

Correcting Entry		
May 20	Service Revenue	50
	Accounts Receivable	50
	(To correct entry of May 10)	

ILLUSTRATION 4.17

Correcting entry

A	=	L	+	OE
-50				-50 Rev

Cash Flows

no effect

Case 2

On May 18, Mercato purchased on account equipment costing \$450. The transaction was journalized and posted as a debit to Equipment \$45 and a credit to Accounts Payable \$45. The error was discovered on June 3, when Mercato received the monthly statement for May from the creditor (see **Illustration 4.18**).

Incorrect Entry (May 18)			Correct Entry (May 18)		
Equipment	45		Equipment	450	
Accounts Payable		45	Accounts Payable		450

ILLUSTRATION 4.18

Comparison of entries

Comparison of the two entries shows that two accounts are incorrect. Equipment is understated \$405, and Accounts Payable is understated \$405. Mercato makes the correcting entry shown in **Illustration 4.19**.

Correcting Entry		
June 3	Equipment	405
	Accounts Payable	405
	(To correct entry of May 18)	

ILLUSTRATION 4.19

Correcting entry

A	=	L	+	OE
+405				+405

Cash Flows

no effect

Instead of preparing a correcting entry, **it is possible to reverse the incorrect entry and then prepare the correct entry.** This approach will result in more entries and postings than a correcting entry, but it will accomplish the desired result.

Accounting Across the Organization Yale Express



© Christian Lagereek/iStockphoto

Lost in Transportation

Yale Express, a short-haul trucking firm, turned over much of its cargo to local truckers to complete deliveries. Yale collected the entire delivery charge. When billed by the local trucker, Yale sent payment for the final phase to the local trucker. Yale used a cutoff period of 20 days into the next accounting period in making its adjusting entries for accrued liabilities. That is, it waited 20 days to receive the local truckers' bills to determine the amount of the unpaid but incurred delivery charges as of the balance sheet date.

On the other hand, **Republic Carloading**, a nationwide, long-distance freight forwarder, frequently did not receive transportation bills from truckers to whom it passed on cargo until months after the year-end. In making its year-end adjusting entries, Republic waited for months in order to include all of these outstanding transportation bills.

When Yale Express merged with Republic Carloading, Yale's vice president employed the 20-day cutoff procedure for both firms. As a result, millions of dollars of Republic's accrued transportation bills went unrecorded. When the company detected the error and made correcting entries, these and other errors changed a reported profit of \$1.14 million into a loss of \$1.88 million!

What might Yale Express's vice president have done to produce more accurate financial statements without waiting months for Republic's outstanding transportation bills? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 3 | Correcting Entries

Sanchez Company discovered the following errors made in January 2020.

1. A payment of Salaries and Wages Expense of \$600 was debited to Supplies and credited to Cash, both for \$600.
2. A collection of \$3,000 from a client on account was debited to Cash \$200 and credited to Service Revenue \$200.
3. The purchase of supplies on account for \$860 was debited to Supplies \$680 and credited to Accounts Payable \$680.

Correct the errors without reversing the incorrect entry.

Solution

1. Salaries and Wages Expense	600	
Supplies		600
2. Service Revenue	200	
Cash	2,800	
Accounts Receivable		3,000
3. Supplies (\$860 – \$680)	180	
Accounts Payable		180

Related exercise material: **BE4.9, DO IT! 4.3, E4.12, and E4.13.**

ACTION PLAN

- Compare the incorrect entry with correct entry.
- After comparison, make an entry to correct the accounts.

Classified Balance Sheet

LEARNING OBJECTIVE 4

Identify the sections of a classified balance sheet.

The balance sheet presents a snapshot of a company's financial position at a point in time. To improve users' understanding of a company's financial position, companies often use a classified balance sheet. A **classified balance sheet** groups together similar assets and similar liabilities, using a number of standard classifications and sections. This is useful because items within a group have similar economic characteristics. A classified balance sheet generally contains the standard classifications listed in **Illustration 4.20**.

Assets	Liabilities and Owner's Equity
Current assets	Current liabilities
Long-term investments	Long-term liabilities
Property, plant, and equipment	Owner's (Stockholders') equity
Intangible assets	

ILLUSTRATION 4.20

Standard balance sheet classifications

These groupings help financial statement readers determine such things as (1) whether the company has enough assets to pay its debts as they come due, and (2) the claims of short- and long-term creditors on the company's total assets. Many of these groupings can be seen in the balance sheet of Franklin Company shown in **Illustration 4.21** (see **Helpful Hint**). In the sections that follow, we explain each of these groupings.

HELPFUL HINT

Recall that the basic accounting equation is $\text{Assets} = \text{Liabilities} + \text{Owner's Equity}$.

Franklin Company

Balance Sheet

October 31, 2020

<u>Assets</u>			
Current assets			
Cash		\$ 6,600	
Debt investments		2,000	
Accounts receivable		7,000	
Notes receivable		1,000	
Inventory		3,000	
Supplies		2,100	
Prepaid insurance		400	
Total current assets			\$22,100
Long-term investments			
Stock investments		5,200	
Investment in real estate		2,000	7,200
Property, plant, and equipment			
Land		10,000	
Equipment	\$24,000		
Less: Accumulated depreciation—equipment	5,000	19,000	29,000
Intangible assets			
Patents			3,100
Total assets			\$61,400

ILLUSTRATION 4.21

Classified balance sheet

ILLUSTRATION 4.21*(continued)*

		<u>Liabilities and Owner's Equity</u>
Current liabilities		
Notes payable	\$11,000	
Accounts payable	2,100	
Unearned service revenue	900	
Salaries and wages payable	1,600	
Interest payable	450	
Total current liabilities		\$16,050
Long-term liabilities		
Mortgage payable	10,000	
Notes payable	1,300	
Total long-term liabilities		11,300
Total liabilities		27,350
Owner's equity		
Owner's capital		34,050
Total liabilities and owner's equity		\$61,400

Current Assets

Current assets are assets that a company expects to convert to cash or use up within one year or its operating cycle, whichever is longer. In Illustration 4.21, Franklin Company had current assets of \$22,100. For most businesses, the cutoff for classification as current assets is one year from the balance sheet date. For example, accounts receivable are current assets because the company will collect them and convert them to cash within one year. Supplies is a current asset because the company expects to use them up in operations within one year.

Some companies use a period longer than one year to classify assets and liabilities as current because they have an operating cycle longer than one year. The **operating cycle** of a company is the average time that it takes to purchase inventory, sell it on account, and then collect cash from customers. For most businesses, this cycle takes less than a year so they use a one-year cutoff. But for some businesses, such as vineyards or airplane manufacturers, this period may be longer than a year. **Except where noted, we will assume that companies use one year to determine whether an asset or liability is current or long-term.**

Common types of current assets are (1) cash, (2) investments (such as short-term U.S. government securities), (3) receivables (notes receivable, accounts receivable, and interest receivable), (4) inventories, and (5) prepaid expenses (supplies and insurance). **On the balance sheet, companies usually list these items in the order in which they expect to convert them into cash.** Illustration 4.22 presents the current assets of **Southwest Airlines Co.**

ILLUSTRATION 4.22

Current assets section

Real World		Southwest Airlines Co.
		Balance Sheet (partial)
		(in millions)
Current assets		
Cash and cash equivalents	\$1,355	
Short-term investments	1,797	
Accounts receivable	419	
Inventories	467	
Prepaid expenses and other current assets	418	
Total current assets	\$4,456	

As we explain later in the chapter, a company's current assets are important in assessing its short-term debt-paying ability.

Long-Term Investments

Long-term investments are generally (1) investments in stocks and bonds of other companies that are normally held for many years, (2) long-term assets such as land or buildings that a company is not currently using in its operating activities, and (3) long-term notes receivable (see **Alternative Terminology**). In Illustration 4.21, Franklin Company reported total long-term investments of \$7,200 on its balance sheet. **Alphabet Inc.** reported long-term investments in its balance sheet, as shown in **Illustration 4.23**.

ALTERNATIVE TERMINOLOGY

Long-term investments are often referred to simply as *investments*.

Real World Alphabet Inc. Balance Sheet (partial) (in millions)	
Long-term investments	
Non-marketable equity investments	\$1,469

ILLUSTRATION 4.23

Long-term investments section

Property, Plant, and Equipment

Property, plant, and equipment are assets with relatively long useful lives that a company is currently using in operating the business. This category includes land, buildings, machinery and equipment, delivery equipment, and furniture (see **Alternative Terminology**). In Illustration 4.21, Franklin Company reported property, plant, and equipment of \$29,000.

Depreciation is the practice of allocating the cost of assets to a number of years. Companies do this by systematically assigning a portion of an asset's cost as an expense each year (rather than expensing the full purchase price in the year of purchase). The assets that the company depreciates are reported on the balance sheet at cost less accumulated depreciation. The **accumulated depreciation** account shows the total amount of depreciation that the company has expensed thus far in the asset's life. In Illustration 4.21, Franklin Company reported accumulated depreciation of \$5,000. **Illustration 4.24** presents the property, plant, and equipment of **Tesla Motors, Inc.**

ALTERNATIVE TERMINOLOGY

Property, plant, and equipment is sometimes called *fixed assets* or *plant assets*.

Real World Tesla Motors, Inc. Balance Sheet (partial) (in thousands)	
Property, plant, and equipment	
Machinery, equipment and office furniture	\$ 322,394
Tooling	230,385
Leasehold improvements	94,763
Building and building improvements	67,707
Land	45,020
Computer equipment and software	42,073
Construction in progress	76,294
	878,636
Less: Accumulated depreciation	140,142
Total	\$ 738,494

ILLUSTRATION 4.24

Property, plant, and equipment section

Intangible Assets

Many companies have long-lived assets that do not have physical substance yet often are very valuable. We call these assets **intangible assets** (see **Helpful Hint**). One significant intangible asset is goodwill. Others include patents, copyrights, and trademarks or trade names that give

HELPFUL HINT

Sometimes intangible assets are reported under a broader heading called "*Other assets*."

the company **exclusive right** of use for a specified period of time. In Illustration 4.21, Franklin Company reported intangible assets of \$3,100. **Illustration 4.25** shows the intangible assets of media and theme-park giant **The Walt Disney Company** in a recent year.

ILLUSTRATION 4.25**Intangible assets section**

Real World The Walt Disney Company Balance Sheet (partial) (in millions)	
Intangible assets and goodwill	
Character/franchise intangibles and copyrights	\$ 5,830
Other amortizable intangible assets	903
Accumulated amortization	(1,204)
Net amortizable intangible assets	5,529
FCC licenses	667
Trademarks	1,218
Other indefinite lived intangible assets	20
	7,434
Goodwill	27,881
	<u>\$35,315</u>

People, Planet, and Profit Insight



© Gehring/iStockphoto

Regaining Goodwill

After falling to unforeseen lows amidst scandals, recalls, and economic crises, the American public's positive perception of the reputation of corporate America is on the rise. Overall corporate reputation is experiencing rehabilitation as the American public gives high marks overall to corporate America, specific industries, and the largest number of individual companies in a dozen years. This is according to the findings of a *Harris Interactive*

RQ Study, which measures the reputations of the 60 most visible companies in the United States.

The survey focuses on six reputational dimensions that influence reputation and consumer behavior. Four of these dimensions, along

with the five corporations that ranked highest within each, are as follows.

- **Social Responsibility:** (1) Whole Foods Market, (2) Johnson & Johnson, (3) Google, (4) The Walt Disney Company, (5) Procter & Gamble Co.
- **Emotional Appeal:** (1) Johnson & Johnson, (2) Amazon.com, (3) UPS, (4) General Mills, (5) Kraft Foods
- **Financial Performance:** (1) Google, (2) Berkshire Hathaway, (3) Apple, (4) Intel, (5) The Walt Disney Company
- **Products and Services:** (1) Intel Corporation, (2) 3M Company, (3) Johnson & Johnson, (4) Google, (5) Procter & Gamble Co.

Source: www.harrisinteractive.com.

Name two industries today which are probably rated low on the reputational characteristics of “being trusted” and “having high ethical standards.” (Go to WileyPLUS for this answer and additional questions.)

ETHICS NOTE

A company that has more current assets than current liabilities can increase the ratio of current assets to current liabilities by using cash to pay off some current liabilities. This gives the appearance of being more liquid. Do you think this move is ethical?

Current Liabilities

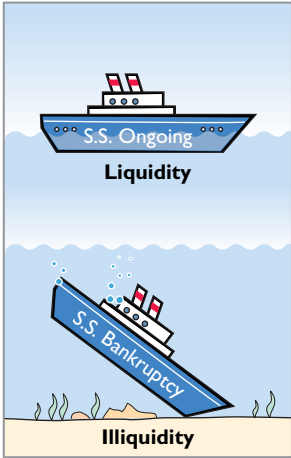
In the liabilities and owner's equity section of the balance sheet, the first grouping is current liabilities. **Current liabilities** are obligations that the company is to pay within the coming year or its operating cycle, whichever is longer (see **Ethics Note**). Common examples are accounts payable, salaries and wages payable, notes payable, interest payable, and income taxes payable. Also included as current liabilities are current maturities of long-term obligations—payments to be made within the next year on long-term obligations. In Illustration 4.21, Franklin Company reported five different types of current liabilities, for a total of

\$16,050. **Illustration 4.26** shows the current liabilities section adapted from the balance sheet of **Marcus Corporation**.

Real World	
Marcus Corporation	
Balance Sheet (partial)	
(in thousands)	
Current liabilities	
Notes payable	\$ 239
Accounts payable	24,242
Current maturities of long-term debt	57,250
Other current liabilities	27,477
Income taxes payable	11,215
Salaries and wages payable	6,720
Total current liabilities	\$127,143

ILLUSTRATION 4.26

Current liabilities section



Users of financial statements look closely at the relationship between current assets and current liabilities. This relationship is important in evaluating a company’s **liquidity**—its ability to pay obligations expected to be due within the next year. When current assets exceed current liabilities, the likelihood for paying the liabilities is favorable. When the reverse is true, short-term creditors may not be paid, and the company may ultimately be forced into bankruptcy.

Accounting Across the Organization REL Consultancy Group



© Jorge Salcedo/iStockphoto

Can a Company Be Too Liquid?

There actually is a point where a company can be too liquid—that is, it can have too much working capital (current assets less current liabilities). While it is important to be liquid enough to be able to pay short-term bills as they come due, a company does not want to tie up its cash in extra inventory or receivables that are not earning the company money.

By one estimate from the **REL Consultancy Group**, the

thousand largest U.S. companies have on their books cumulative excess working capital of \$764 billion. Based on this figure, companies could have reduced debt by 36% or increased net income by 9%. Given that managers throughout a company are interested in improving profitability, it is clear that they should have an eye toward managing working capital. They need to aim for a “Goldilocks solution”—not too much, not too little, but just right.

Source: K. Richardson, “Companies Fall Behind in Cash Management,” *Wall Street Journal* (June 19, 2007).

What can various company managers do to ensure that working capital is managed efficiently to maximize net income? (Go to WileyPLUS for this answer and additional questions.)

Long-Term Liabilities

Long-term liabilities are obligations that a company expects to pay **after** one year. Liabilities in this category include bonds payable, mortgages payable, long-term notes payable, lease liabilities, and pension liabilities. Many companies report long-term debt maturing after one year as a single amount in the balance sheet and show the details of the debt in notes that accompany the financial statements. Others list the various types of long-term liabilities. In Illustration 4.21, Franklin Company reported long-term liabilities of \$11,300. **Illustration 4.27** shows the long-term liabilities that **Nike, Inc.** reported in its balance sheet in a recent year.

ILLUSTRATION 4.27

Long-term liabilities section

Real World Nike, Inc. Balance Sheet (partial) (in millions)	
Long-term liabilities	
Bonds payable	\$1,106
Notes payable	51
Deferred income taxes and other	<u>1,544</u>
Total long-term liabilities	\$2,701

Owner's Equity

The content of the owner's equity section varies with the form of business organization. In a proprietorship, there is one capital account. In a partnership, there is a capital account for each partner. Corporations divide owners' equity into two accounts—Common Stock (sometimes referred to as Capital Stock) and Retained Earnings. Corporations record stockholders' investments in the company by debiting an asset account and crediting the Common Stock account. They record in the Retained Earnings account income retained for use in the business. Corporations combine the Common Stock and Retained Earnings accounts and report them on the balance sheet as **stockholders' equity**. (We discuss these corporation accounts in later chapters.) **Nordstrom, Inc.** recently reported its stockholders' equity section as shown in **Illustration 4.28**.

ILLUSTRATION 4.28

Stockholders' equity section

Real World Nordstrom, Inc. Balance Sheet (partial) (in thousands)	
Stockholders' equity	
Common stock, 271,331 shares	\$ 685,934
Retained earnings	<u>1,406,747</u>
Total stockholders' equity	\$2,092,681

DO IT! 4 | Balance Sheet Classifications

The following accounts were taken from the financial statements of Callahan Company.

_____ Salaries and wages payable	_____ Stock investments (long-term)
_____ Service revenue	_____ Equipment
_____ Interest payable	_____ Accumulated depreciation—
_____ Goodwill	_____ equipment
_____ Debt investments (short-term)	_____ Depreciation expense
_____ Mortgage payable (due in 3 years)	_____ Owner's capital
	_____ Unearned service revenue

Match each of the following to its proper balance sheet classification, shown below. If the item would not appear on a balance sheet, use "NA."

Current assets (CA)	Current liabilities (CL)
Long-term investments (LTI)	Long-term liabilities (LTL)
Property, plant, and equipment (PPE)	Owner's equity (OE)
Intangible assets (IA)	

Solution

__CL__ Salaries and wages payable	__LTI__ Stock investments (long-term)
__NA__ Service revenue	__PPE__ Equipment
__CL__ Interest payable	__PPE__ Accumulated depreciation—
__IA__ Goodwill	_____ equipment
__CA__ Debt investments (short-term)	__NA__ Depreciation expense
__LTL__ Mortgage payable (due in	__OE__ Owner's capital
3 years)	__CL__ Unearned service revenue

Related exercise material: **BE4.11**, **DO IT! 4.4**, **E4.14**, **E4.15**, **E4.16**, and **E4.17**.

ACTION PLAN

- Analyze whether each financial statement item is an asset, liability, or owner's equity.
- Determine if asset and liability items are short-term or long-term.

Appendix 4A

Reversing Entries

LEARNING OBJECTIVE *5

Prepare reversing entries.

After preparing the financial statements and closing the books, it is often helpful to reverse some of the adjusting entries before recording the regular transactions of the next period. Such entries are **reversing entries**. Companies make a **reversing entry at the beginning of the next accounting period**. Each reversing entry is the **exact opposite of the adjusting entry made in the previous period**. The recording of reversing entries is an **optional step** in the accounting cycle.

The purpose of reversing entries is to simplify the recording of a subsequent transaction related to an adjusting entry. For example, in Chapter 3, the payment of salaries after an adjusting entry resulted in two debits: one to Salaries and Wages Payable and the other to Salaries and Wages Expense. With reversing entries, the company can debit the entire subsequent payment to Salaries and Wages Expense. **The use of reversing entries does not change the amounts reported in the financial statements.** What it does is simplify the recording of subsequent transactions.

Reversing Entries Example

Companies most often use reversing entries to reverse two types of adjusting entries: accrued revenues and accrued expenses. To illustrate the optional use of reversing entries for accrued expenses, we will use the salaries expense transactions for Pioneer Advertising as illustrated in Chapters 2, 3, and 4. The transaction and adjustment data are as follows.

1. October 26 (initial salary entry): Pioneer pays \$4,000 of salaries and wages earned between October 15 and October 26.
2. October 31 (adjusting entry): Salaries and wages earned between October 29 and October 31 are \$1,200. The company will pay these in the November 9 payroll.
3. November 9 (subsequent salary entry): Salaries and wages paid are \$4,000. Of this amount, \$1,200 applied to accrued salaries and wages payable and \$2,800 was earned between November 1 and November 9.

Illustration 4A.1 shows the entries with and without reversing entries.

ILLUSTRATION 4A.1 Comparative entries—without and with reversing

Without Reversing Entries (per chapter)				With Reversing Entries (per appendix)			
<u>Initial Salary Entry</u>				<u>Initial Salary Entry</u>			
Oct. 26	Salaries and Wages Expense Cash	4,000	4,000	Oct. 26 (Same entry)			
<u>Adjusting Entry</u>				<u>Adjusting Entry</u>			
Oct. 31	Salaries and Wages Expense Salaries and Wages Payable	1,200	1,200	Oct. 31 (Same entry)			
<u>Closing Entry</u>				<u>Closing Entry</u>			
Oct. 31	Income Summary Salaries and Wages Expense	5,200	5,200	Oct. 31 (Same entry)			
<u>Reversing Entry</u>				<u>Reversing Entry</u>			
Nov. 1 No reversing entry is made.				Nov. 1 Salaries and Wages Payable 1,200 1,200 Salaries and Wages Expense			
<u>Subsequent Salary Entry</u>				<u>Subsequent Salary Entry</u>			
Nov. 9	Salaries and Wages Payable Salaries and Wages Expense Cash	1,200 2,800	4,000	Nov. 9 Salaries and Wages Expense 4,000 4,000 Cash			

The first three entries are the same whether or not Pioneer uses reversing entries. The last two entries are different. The November 1 **reversing entry** eliminates the \$1,200 balance in Salaries and Wages Payable created by the October 31 adjusting entry. The reversing entry also creates a \$1,200 credit balance in the Salaries and Wages Expense account. As you know, it is unusual for an expense account to have a credit balance. The balance is correct in this instance, though, because it anticipates that the entire amount of the first salaries and wages payment in the new accounting period will be debited to Salaries and Wages Expense. This debit will eliminate the credit balance. The resulting debit balance in the expense account will equal the salaries and wages expense incurred in the new accounting period (\$2,800 in this example).

If Pioneer makes reversing entries, it can debit all cash payments of expenses to the expense account. This means that on November 9 (and every payday) Pioneer can debit Salaries and Wages Expense for the amount paid, without regard to any accrued salaries and wages payable. Being able to make the **same entry each time** simplifies the recording process. The company can record subsequent transactions as if the related adjusting entry had never been made.

Illustration 4A.2 shows the posting of the entries with reversing entries.

ILLUSTRATION 4A.2 Postings with reversing entries

Salaries and Wages Expense				Salaries and Wages Payable			
10/26	Paid	4,000		11/1	Reversing	1,200	
31	Adjusting	1,200					10/31
		5,200	10/31	Closing			Adjusting
			5,200				1,200
11/9	Paid	4,000	11/1	Reversing	1,200		

A company can also use reversing entries for accrued revenue adjusting entries. For Pioneer, the adjusting entry was Accounts Receivable (Dr.) \$200 and Service Revenue (Cr.) \$200. Thus, the reversing entry on November 1 is:

A	=	L	+	OE
				-200 Rev
+200				
Cash Flows				
no effect				

Nov. 1	Service Revenue	200	
	Accounts Receivable		200
	(To reverse October 31 adjusting entry)		

When Pioneer collects the accrued service revenue, it debits Cash and credits Service Revenue.

Review and Practice

Learning Objectives Review

- 1 Prepare a worksheet.

The steps in preparing a worksheet are as follows. (a) Prepare a trial balance on the worksheet. (b) Enter the adjustments in the adjustments columns. (c) Enter adjusted balances in the adjusted trial balance columns. (d) Extend adjusted trial balance amounts to appropriate financial statement columns. (e) Total the statement columns, compute net income (or net loss), and complete the worksheet.
- 2 Prepare closing entries and a post-closing trial balance.

Closing the books occurs at the end of an accounting period. The process is to journalize and post closing entries and then underline and balance all accounts. In closing the books, companies make separate entries to close revenues and expenses to Income Summary, Income Summary to Owner's Capital, and Owner's Drawings to Owner's Capital. Only temporary accounts are closed. A post-closing trial

balance contains the balances in permanent accounts that are carried forward to the next accounting period. The purpose of this trial balance is to prove the equality of these balances.

3 Explain the steps in the accounting cycle and how to prepare correcting entries.

The required steps in the accounting cycle are (1) analyze business transactions, (2) journalize the transactions, (3) post to ledger accounts, (4) prepare a trial balance, (5) journalize and post adjusting entries, (6) prepare an adjusted trial balance, (7) prepare financial statements, (8) journalize and post closing entries, and (9) prepare a post-closing trial balance.

One way to determine the correcting entry is to compare the incorrect entry with the correct entry. After comparison, the company makes a correcting entry to correct the accounts. An alternative to a correcting entry is to reverse the incorrect entry and then prepare the correct entry.

4 Identify the sections of a classified balance sheet.

A classified balance sheet categorizes assets as current assets; long-term investments; property, plant, and equipment; and intangibles. Liabilities are classified as either current or long-term. There is also an owner's (owners') equity section, which varies with the form of business organization.

*5 Prepare reversing entries.

Reversing entries are the opposite of the adjusting entries made in the preceding period. Some companies choose to make reversing entries at the beginning of a new accounting period to simplify the recording of later transactions related to the adjusting entries. In most cases, only accrued adjusting entries are reversed.

Glossary Review

Classified balance sheet A balance sheet that contains standard classifications or sections. (p. 4-23).

Closing entries Entries made at the end of an accounting period to transfer the balances of temporary accounts to a permanent owner's equity account, Owner's Capital. (p. 4-12).

Correcting entries Entries to correct errors made in recording transactions. (p. 4-19).

Current assets Assets that a company expects to convert to cash or use up within one year or its operating cycle, whichever is longer. (p. 4-24).

Current liabilities Obligations that a company expects to pay within the coming year or its operating cycle, whichever is longer. (p. 4-26).

Income Summary A temporary account used in closing revenue and expense accounts. (p. 4-12).

Intangible assets Long-lived assets that do not have physical substance. (p. 4-25).

Liquidity The ability of a company to pay obligations expected to be due within the next year. (p. 4-27).

Long-term investments Generally, (1) investments in stocks and bonds of other companies that companies normally hold for many years; (2) long-term assets, such as land and buildings, not currently being used in operating activities; and (3) long-term notes receivable. (p. 4-25).

Long-term liabilities Obligations that a company expects to pay after one year. (p. 4-27).

Operating cycle The average time that it takes to purchase inventory, sell it on account, and then collect cash from customers. (p. 4-24).

Permanent (real) accounts Accounts that relate to one or more future accounting periods. Consist of all balance sheet accounts. Balances are carried forward to the next accounting period. (p. 4-12).

Post-closing trial balance A list of permanent accounts and their balances after a company has journalized and posted closing entries. (p. 4-16).

Property, plant, and equipment Assets with relatively long useful lives and currently being used in operations. (p. 4-25).

Reversing entry An entry, made at the beginning of the next accounting period that is the exact opposite of the adjusting entry made in the previous period. (p. 4-19).

Stockholders' equity The ownership claim of shareholders on total assets. It is to a corporation what owner's equity is to a proprietorship. (p. 4-28).

Temporary (nominal) accounts Accounts that relate only to a given accounting period. Consist of all income statement accounts and owner's drawings account. All temporary accounts are closed at end of the accounting period. (p. 4-12).

Worksheet A multiple-column form that may be used in making adjustments and in preparing financial statements. (p. 4-2).

Practice Multiple-Choice Questions

1. (LO 1) Which of the following statements is **incorrect** concerning the worksheet?

- a. The worksheet is essentially a working tool of the accountant.
- b. The worksheet is distributed to management and other interested parties.
- c. The worksheet cannot be used as a basis for posting to ledger accounts.
- d. Financial statements can be prepared directly from the worksheet before journalizing and posting the adjusting entries.

2. (LO 1) In a worksheet, net income is entered in the following columns:

- a. income statement (Dr) and balance sheet (Dr).
- b. income statement (Cr) and balance sheet (Dr).
- c. income statement (Dr) and balance sheet (Cr).
- d. income statement (Cr) and balance sheet (Cr).

3. (LO 1) In the unadjusted trial balance of its worksheet for the year ended December 31, 2020, Knox Company reported Equipment of \$120,000. The year-end adjusting entries require an adjustment of

\$15,000 for depreciation expense for the equipment. After the adjusted trial balance is completed, what amount should be shown in the financial statement columns?

- A debit of \$105,000 for Equipment in the balance sheet column.
 - A credit of \$15,000 for Depreciation Expense in the income statement column.
 - A debit of \$120,000 for Equipment in the balance sheet column.
 - A debit of \$15,000 for Accumulated Depreciation—Equipment in the balance sheet column.
4. (LO 2) An account that will have a zero balance after closing entries have been journalized and posted is:
- Service Revenue.
 - Supplies.
 - Prepaid Insurance.
 - Accumulated Depreciation—Equipment.
5. (LO 2) When a net loss has occurred, Income Summary is:
- debited and Owner's Capital is credited.
 - credited and Owner's Capital is debited.
 - debited and Owner's Drawings is credited.
 - credited and Owner's Drawings is debited.
6. (LO 2) The closing process involves separate entries to close (1) expenses, (2) drawings, (3) revenues, and (4) income summary. The correct sequencing of the entries is:
- (4), (3), (2), (1).
 - (1), (2), (3), (4).
 - (3), (1), (4), (2).
 - (3), (2), (1), (4).
7. (LO 2) Which types of accounts will appear in the post-closing trial balance?
- Permanent (real) accounts.
 - Temporary (nominal) accounts.
 - Accounts shown in the income statement columns of a worksheet.
 - None of these answer choices is correct.
8. (LO 3) All of the following are required steps in the accounting cycle **except**:
- journalizing and posting closing entries.
 - preparing financial statements.
 - journalizing the transactions.
 - preparing a worksheet.
9. (LO 3) The proper order of the following steps in the accounting cycle is:
- prepare unadjusted trial balance, journalize transactions, post to ledger accounts, journalize and post adjusting entries.
 - journalize transactions, prepare unadjusted trial balance, post to ledger accounts, journalize and post adjusting entries.
 - journalize transactions, post to ledger accounts, prepare unadjusted trial balance, journalize and post adjusting entries.

- prepare unadjusted trial balance, journalize and post adjusting entries, journalize transactions, post to ledger accounts.

10. (LO 3) When Lopez Company purchased supplies worth \$600, it incorrectly recorded a credit to Supplies for \$6,000 and a debit to Cash for \$6,000. Before correcting this error:

- Cash is overstated and Supplies is overstated.
- Cash is understated and Supplies is understated.
- Cash is understated and Supplies is overstated.
- Cash is overstated and Supplies is understated.

11. (LO 3) Cash of \$100 received at the time the service was performed was journalized and posted as a debit to Cash \$100 and a credit to Accounts Receivable \$100. Assuming the incorrect entry is not reversed, the correcting entry is:

- debit Service Revenue \$100 and credit Accounts Receivable \$100.
- debit Accounts Receivable \$100 and credit Service Revenue \$100.
- debit Cash \$100 and credit Service Revenue \$100.
- debit Accounts Receivable \$100 and credit Cash \$100.

12. (LO 4) The correct order of presentation in a classified balance sheet for the following current assets is:

- accounts receivable, cash, prepaid insurance, inventory.
- cash, inventory, accounts receivable, prepaid insurance.
- cash, accounts receivable, inventory, prepaid insurance.
- inventory, cash, accounts receivable, prepaid insurance.

13. (LO 4) A company has purchased a tract of land. It expects to build a production plant on the land in approximately 5 years. During the 5 years before construction, the land will be idle. The land should be reported as:

- property, plant, and equipment.
- land expense.
- a long-term investment.
- an intangible asset.

14. (LO 4) In a classified balance sheet, assets are usually classified using the following categories:

- current assets; long-term assets; property, plant, and equipment; and intangible assets.
- current assets; long-term investments; property, plant, and equipment; and tangible assets.
- current assets; long-term investments; tangible assets; and intangible assets.
- current assets; long-term investments; property, plant, and equipment; and intangible assets.

15. (LO 4) Current assets are listed:

- by expected conversion to cash.
- by importance.
- by longevity.
- alphabetically.

*16. (LO 5) On December 31, Harvey Kohl Company correctly made an adjusting entry to recognize \$2,000 of accrued salaries payable. On January 8 of the next year, total salaries of \$3,800 were paid. Assuming

the correct reversing entry was made on January 1, the entry on January 8 will result in a credit to Cash \$3,800 and the following debit(s):

- a. Salaries and Wages Payable \$1,800 and Salaries and Wages Expense \$2,000.

- b. Salaries and Wages Payable \$2,000 and Salaries and Wages Expense \$1,800.

- c. Salaries and Wages Expense \$3,800.

- d. Salaries and Wages Payable \$3,800.

Solutions

1. **b.** The worksheet is a working tool of the accountant; it is not distributed to management and other interested parties. The other choices are all true statements.

2. **c.** Net income is entered in the Dr column of the income statement and the Cr column of the balance sheet. The other choices are incorrect because net income is entered in the (a) Cr (not Dr) column of the balance sheet, (b) Dr (not Cr) column of the income statement and in the Cr (not Dr) column of the balance sheet, and (d) Dr (not Cr) column of the income statement.

3. **c.** A debit of \$120,000 for Equipment would appear in the balance sheet column. The other choices are incorrect because (a) Equipment, less accumulated depreciation of \$15,000, would total \$105,000 under assets on the balance sheet, not on the worksheet; (b) a debit, not credit, for Depreciation Expense would appear in the income statement column; and (d) a credit, not debit, of \$15,000 for Accumulated Depreciation—Equipment would appear in the balance sheet column.

4. **a.** The Service Revenue account will have a zero balance after closing entries have been journalized and posted because it is a temporary account. The other choices are incorrect because (b) Supplies, (c) Prepaid Insurance, and (d) Accumulated Depreciation—Equipment are all permanent accounts and therefore not closed in the closing process.

5. **b.** The effect of a net loss is a credit to Income Summary and a debit to Owner's Capital. The other choices are incorrect because (a) Income Summary is credited, not debited, and Owner's Capital is debited, not credited; (c) Income Summary is credited, not debited, and Owner's Drawings is not affected; and (d) Owner's Capital, not Owner's Drawings, is debited.

6. **c.** The correct order is (3) revenues, (1) expenses, (4) income summary, and (2) drawings. Therefore, choices (a), (b), and (d) are incorrect.

7. **a.** Permanent accounts appear in the post-closing trial balance. The other choices are incorrect because (b) temporary accounts and (c) income statement accounts are closed to a zero balance and are therefore not included in the post-closing trial balance. Choice (d) is wrong as there is only one correct answer for this question.

8. **d.** Preparing a worksheet is not a required step in the accounting cycle. The other choices are all required steps in the accounting cycle.

9. **c.** The proper order of the steps in the accounting cycle is (1) journalize transactions, (2) post to ledger accounts, (3) prepare unadjusted trial balance, and (4) journalize and post adjusting entries. Therefore, choices (a), (b), and (d) are incorrect.

10. **d.** This entry causes Cash to be overstated and Supplies to be understated. Supplies should have been debited (increasing supplies) and Cash should have been credited (decreasing cash). The other choices are incorrect because (a) Supplies is understated, not overstated; (b) Cash is overstated, not understated; and (c) Cash is overstated, not understated, and Supplies is understated, not overstated.

11. **b.** The correcting entry is to debit Accounts Receivable \$100 and credit Service Revenue \$100. The other choices are incorrect because (a) Service Revenue should be credited, not debited, and Accounts Receivable should be debited, not credited; (c) Service Revenue should be credited for \$100, and Cash should not be included in the correcting entry as it was recorded properly; and (d) Accounts Receivable should be debited for \$100 and Cash should not be included in the correcting entry as it was recorded properly.

12. **c.** Companies list current assets on balance sheet in the order of liquidity: cash, accounts receivable, inventory, and prepaid insurance. Therefore, choices (a), (b), and (d) are incorrect.

13. **c.** Long-term investments include long-term assets such as land that a company is not currently using in its operating activities. The other choices are incorrect because (a) land would be reported as property, plant, and equipment only if it is being currently used in the business; (b) land is an asset, not an expense; and (d) land has physical substance and thus is a tangible property.

14. **d.** These are the categories usually used in a classified balance sheet. The other choices are incorrect because the categories (a) "long-term assets" and (b) and (c) "tangible assets" are generally not used.

15. **a.** Current assets are listed in order of their liquidity, not (b) by importance, (c) by longevity, or (d) alphabetically.

*16. **c.** The use of reversing entries simplifies the recording of the first payroll following the end of the year by eliminating the need to make an entry to the Salaries and Wages Payable account. The other choices are incorrect because (a) Salaries and Wages Payable is not part of the payroll entry on January 8, and the debit to Salaries and Wages Expense should be for \$3,800, not \$2,000; and (b) and (d) the Salaries and Wages Expense account, not the Salaries and Wages Payable account, should be debited.

Practice Brief Exercises

1. **(LO 2)** The ledger of Quintana Company contains the following balances: Owner's Capital \$40,000, Owner's Drawings \$3,000, Service Revenue \$65,000, Salaries and Wages Expense \$39,000, and Maintenance and Repairs Expense \$9,000. Prepare the closing entries at December 31.

Prepare closing entries from ledger balances.

Solution

1.	Dec. 31	Service Revenue	65,000	
		Income Summary		65,000
	31	Income Summary	48,000	
		Salaries and Wages Expense		39,000
		Maintenance and Repairs Expense		9,000
	31	Income Summary	17,000	
		Owner's Capital		17,000
	31	Owner's Capital	3,000	
		Owner's Drawing		3,000

Prepare correcting entries.

2. (LO 3) At Shields Company, the following errors were discovered after the transactions had been journalized and posted. Prepare the correcting entries.

- A cash sale to a customer for \$780 was recorded as a debit to Cash \$780 and a credit to Accounts Receivable \$780.
- The purchase of equipment on account for \$1,730 was recorded as a debit to Equipment \$1,370 and a credit to Accounts Payable \$1,370.

Solution

2. a.	Accounts Receivable	780	
	Service Revenue		780
b.	Equipment (\$1,730 – \$1,370)	360	
	Accounts Payable		360

Prepare the current assets section of a balance sheet.

3. (LO 4) **Financial Statement** The balance sheet debit column of the worksheet for Miguel Company includes the following accounts: Accounts Receivable \$25,000, Prepaid Insurance \$7,000, Cash \$8,000, Supplies \$11,000, and Stock Investments (short-term) \$14,000. Prepare the current assets section of the balance sheet, listing the accounts in proper sequence.

Solution

3.

Miguel Company	
Partial Balance Sheet	
Current assets	
Cash	\$ 8,000
Stock investments	14,000
Accounts receivable	25,000
Supplies	11,000
Prepaid insurance	<u>7,000</u>
Total current assets	<u>\$65,000</u>

Practice Exercises

Journalize and post closing entries, and prepare a post-closing trial balance.

1. (LO 2) Hercules Company ended its fiscal year on August 31, 2020. The company's adjusted trial balance as of the end of its fiscal year is as shown below.

Hercules Company
Adjusted Trial Balance
August 31, 2020

No.	Account Titles	Debit	Credit
101	Cash	\$10,900	
112	Accounts Receivable	6,200	
157	Equipment	10,600	
167	Accumulated Depr.—Equip.		\$ 5,400
201	Accounts Payable		2,800
208	Unearned Rent Revenue		1,200
301	Owner's Capital		31,700
306	Owner's Drawings	12,000	
404	Service Revenue		42,400
429	Rent Revenue		6,100
711	Depreciation Expense	2,700	
720	Salaries and Wages Expense	37,100	
732	Utilities Expense	10,100	
		<u>\$89,600</u>	<u>\$89,600</u>

Instructions

- a. Prepare the closing entries using page J15 in a general journal.
- b. Post to Owner's Capital and No. 350 Income Summary accounts. (Use the three-column form.)
- c. Prepare a post-closing trial balance at August 31, 2020.

Solution

1. a.

GENERAL JOURNAL

J15

Date	Account Titles	Ref.	Debit	Credit
Aug. 31	Service Revenue	404	42,400	
	Rent Revenue	429	6,100	
	Income Summary	350		48,500
	(To close revenue accounts)			
31	Income Summary	350	49,900	
	Salaries and Wages Expense	720		37,100
	Utilities Expense	732		10,100
	Depreciation Expense	711		2,700
	(To close expense accounts)			
31	Owner's Capital (\$49,900 – \$48,500)	301	1,400	
	Income Summary	350		1,400
	(To close net loss to capital)			
31	Owner's Capital	301	12,000	
	Owner's Drawings	306		12,000
	(To close drawings to capital)			

b.

Owner's Capital

No. 301

Date	Explanation	Ref.	Debit	Credit	Balance
Aug. 31	Balance				31,700
31	Close net loss	J15	1,400		30,300
31	Close drawings	J15	12,000		18,300

Income Summary

No. 350

Date	Explanation	Ref.	Debit	Credit	Balance
Aug. 31	Close revenue	J15		48,500	48,500
31	Close expenses	J15	49,000		(1,400)
31	Close net loss	J15		1,400	0

c.

Hercules Company Post-Closing Trial Balance August 31, 2020		
	Debit	Credit
Cash	\$10,900	
Accounts Receivable	6,200	
Equipment	10,600	
Accumulated Depreciation—Equipment		\$ 5,400
Accounts Payable		2,800
Unearned Rent Revenue		1,200
Owner's Capital		18,300
	<u>\$27,700</u>	<u>\$27,700</u>

Prepare financial statements.

2. (LO 4) Financial Statement The adjusted trial balance for Hercules Company is presented in **Practice Exercise 1**.

Instructions

- Prepare an income statement and an owner's equity statement for the year ended August 31, 2020. Hercules did not make any capital investments during the year.
- Prepare a classified balance sheet at August 31, 2020.

Solution**2. a.**

Hercules Company Income Statement For the Year Ended August 31, 2020		
Revenues		
Service revenue	\$42,400	
Rent revenue	<u>6,100</u>	
Total revenues		\$48,500
Expenses		
Salaries and wages expense	37,100	
Utilities expense	10,100	
Depreciation expense	<u>2,700</u>	
Total expenses		<u>49,900</u>
Net loss		<u>\$ (1,400)</u>

Hercules Company Owner's Equity Statement For the Year Ended August 31, 2020		
Owner's capital, September 1, 2019		\$ 31,700
Less: Net loss	\$ 1,400	
Owner's drawings	<u>12,000</u>	<u>13,400</u>
Owner's capital, August 31, 2020		<u>\$ 18,300</u>

b.

Hercules Company		
Balance Sheet		
August 31, 2020		
<u>Assets</u>		
Current assets		
Cash	\$10,900	
Accounts receivable	<u>6,200</u>	
Total current assets		\$17,100
Property, plant, and equipment		
Equipment	10,600	
Less: Accumulated depreciation—equip.	<u>5,400</u>	<u>5,200</u>
Total assets		<u>\$22,300</u>
<u>Liabilities and Owner's Equity</u>		
Current liabilities		
Accounts payable	\$ 2,800	
Unearned rent revenue	<u>1,200</u>	
Total current liabilities		\$ 4,000
Owner's equity		
Owner's capital		<u>18,300</u>
Total liabilities and owner's equity		<u>\$22,300</u>

Practice Problem

(LO 1, 2, 4) Financial Statement At the end of its first month of operations, Pampered Pet Service has the following unadjusted trial balance.

Prepare worksheet and classified balance sheet, and journalize closing entries.

Pampered Pet Service		
August 31, 2020		
Trial Balance		
	<u>Debit</u>	<u>Credit</u>
Cash	\$ 5,400	
Accounts Receivable	2,800	
Supplies	1,300	
Prepaid Insurance	2,400	
Equipment	60,000	
Notes Payable		\$40,000
Accounts Payable		2,400
Owner's Capital		30,000
Owner's Drawings	1,000	
Service Revenue		4,900
Salaries and Wages Expense	3,200	
Utilities Expense	800	
Advertising Expense	<u>400</u>	
	<u>\$77,300</u>	<u>\$77,300</u>

Other data:

- Insurance expires at the rate of \$200 per month.
- \$1,000 of supplies are on hand at August 31.
- Monthly depreciation on the equipment is \$900.
- Interest of \$500 on the notes payable has accrued during August.

Instructions

- Prepare a worksheet.
- Prepare a classified balance sheet assuming \$35,000 of the notes payable are long-term.
- Journalize the closing entries.

Solution

a. Pampered Pet Service
Worksheet for the Month Ended August 31, 2020

Account Titles	Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
Cash	5,400				5,400				5,400	
Accounts Receivable	2,800				2,800				2,800	
Supplies	1,300			(b) 300	1,000				1,000	
Prepaid Insurance	2,400			(a) 200	2,200				2,200	
Equipment	60,000				60,000				60,000	
Notes Payable		40,000				40,000				40,000
Accounts Payable		2,400				2,400				2,400
Owner's Capital		30,000				30,000				30,000
Owner's Drawings	1,000				1,000				1,000	
Service Revenue		4,900				4,900	4,900			
Salaries and Wages Expense	3,200				3,200		3,200			
Utilities Expense	800				800		800			
Advertising Expense	400				400		400			
Totals	<u>77,300</u>	<u>77,300</u>								
Insurance Expense			(a) 200		200		200			
Supplies Expense			(b) 300		300		300			
Depreciation Expense			(c) 900		900		900			
Accumulated Depreciation—Equipment				(c) 900		900				900
Interest Expense			(d) 500		500		500			
Interest Payable				(d) 500		500				500
Totals			<u>1,900</u>	<u>1,900</u>	<u>78,700</u>	<u>78,700</u>	<u>6,300</u>	<u>4,900</u>	<u>72,400</u>	<u>73,800</u>
Net Loss								1,400	1,400	
Totals							<u>6,300</u>	<u>6,300</u>	<u>73,800</u>	<u>73,800</u>

Explanation: (a) insurance expired, (b) supplies used, (c) depreciation expensed, and (d) interest accrued.

b.

Pampered Pet Service
Balance Sheet
August 31, 2020

<u>Assets</u>	
Current assets	
Cash	\$ 5,400
Accounts receivable	2,800
Supplies	1,000
Prepaid insurance	2,200
Total current assets	<u>11,400</u>
Property, plant, and equipment	
Equipment	60,000
Less: Accumulated depreciation—equipment	<u>900</u>
Total assets	<u>\$70,500</u>

Liabilities and Owner's Equity

Current liabilities		
Notes payable	\$ 5,000	
Accounts payable	2,400	
Interest payable	<u>500</u>	
Total current liabilities		\$ 7,900
Long-term liabilities		
Notes payable		<u>35,000</u>
Total liabilities		42,900
Owner's equity		
Owner's capital		<u>27,600*</u>
Total liabilities and owner's equity		<u>\$70,500</u>

*Owner's capital \$30,000 less drawings \$1,000 and net loss \$1,400.

c.

Aug. 31	Service Revenue	4,900	
	Income Summary		4,900
	(To close revenue account)		
31	Income Summary	6,300	
	Salaries and Wages Expense		3,200
	Depreciation Expense		900
	Utilities Expense		800
	Interest Expense		500
	Advertising Expense		400
	Supplies Expense		300
	Insurance Expense		200
	(To close expense accounts)		
31	Owner's Capital (\$6,300 – \$4,900)	1,400	
	Income Summary		1,400
	(To close net loss to capital)		
31	Owner's Capital	1,000	
	Owner's Drawings		1,000
	(To close drawings to capital)		

WileyPLUS

Brief Exercises, DO IT! Exercises, Exercises, Problems, and many additional resources are available for practice in WileyPLUS.

Note: All asterisked Questions, Exercises, and Problems relate to material in the appendix to the chapter.

Questions

1. "A worksheet is a permanent accounting record and its use is required in the accounting cycle." Do you agree? Explain.
2. Explain the purpose of the worksheet.
3. What is the relationship, if any, between the amount shown in the adjusted trial balance column for an account and that account's ledger balance?
4. If a company's revenues are \$125,000 and its expenses are \$113,000, in which financial statement columns of the worksheet will the net income of \$12,000 appear? When expenses exceed revenues, in which columns will the difference appear?
5. Why is it necessary to prepare formal financial statements if all of the data are in the statement columns of the worksheet?
6. Identify the account(s) debited and credited in each of the four closing entries, assuming the company has net income for the year.
7. Describe the nature of the Income Summary account and identify the types of summary data that may be posted to this account.

8. What are the content and purpose of a post-closing trial balance?
9. Which of the following accounts would not appear in the post-closing trial balance? Interest Payable, Equipment, Depreciation Expense, Owner's Drawings, Unearned Service Revenue, Accumulated Depreciation—Equipment, and Service Revenue.
10. Distinguish between a reversing entry and an adjusting entry. Are reversing entries required?
11. Indicate, in the sequence in which they are made, the three required steps in the accounting cycle that involve journalizing.
12. Identify, in the sequence in which they are prepared, the three trial balances that are used in the accounting cycles.
13. How do correcting entries differ from adjusting entries?
14. What standard classifications are used in preparing a classified balance sheet?
15. What is meant by the term "operating cycle?"
16. Define current assets. What basis is used for arranging individual items within the current assets section?
17. Distinguish between long-term investments and property, plant, and equipment.
18. (a) What is the term used to describe the owner's equity section of a corporation? (b) Identify the two owners' equity accounts in a corporation and indicate the purpose of each.
19. Using **Apple's** annual report, determine its current liabilities at September 27, 2014, and September 26, 2015. Were current liabilities higher or lower than current assets in these two years?
- *20. Cigale Company prepares reversing entries. If the adjusting entry for interest payable is reversed, what type of an account balance, if any, will there be in Interest Payable and Interest Expense after the reversing entry is posted?
- *21. At December 31, accrued salaries payable totaled \$3,500. On January 10, total salaries of \$8,000 are paid. (a) Assume that reversing entries are made at January 1. Give the January 10 entry, and indicate the Salaries and Wages Expense account balance after the entry is posted. (b) Repeat part (a) assuming reversing entries are not made.

Brief Exercises

List the steps in preparing a worksheet.

BE4.1 (LO 1) The steps in using a worksheet are presented in random order below. List the steps in the proper order by placing numbers 1–5 in the blank spaces.

- a. _____ Prepare a trial balance on the worksheet.
- b. _____ Enter adjusted balances.
- c. _____ Extend adjusted balances to appropriate statement columns.
- d. _____ Total the statement columns, compute net income (loss), and complete the worksheet.
- e. _____ Enter adjustment data.

Prepare partial worksheet.

BE4.2 (LO 1) The ledger of Walters Company includes the following unadjusted balances: Prepaid Insurance \$3,000, Service Revenue \$60,000, and Salaries and Wages Expense \$25,000. Adjusting entries are required for (a) expired insurance \$1,800, (b) services performed \$1,100 are billed and not yet collected, and (c) accrued salaries payable \$900. Enter the unadjusted balances and adjustments into a worksheet and complete the worksheet for all accounts. (*Note:* You will need to add the following accounts: Accounts Receivable, Salaries and Wages Payable, and Insurance Expense.)

Identify worksheet columns for selected accounts.

BE4.3 (LO 1) The following selected accounts appear in the adjusted trial balance columns of the worksheet for Ashram Company: Accumulated Depreciation, Depreciation Expense, Owner's Capital, Owner's Drawings, Service Revenue, Supplies, and Accounts Payable. Indicate the financial statement column (income statement Dr., balance sheet Cr., etc.) to which each balance should be extended.

Prepare closing entries from ledger balances.

BE4.4 (LO 2) The ledger of Pitt Company contains the following balances: Owner's Capital \$30,000, Owner's Drawings \$2,000, Service Revenue \$58,000, Salaries and Wages Expense \$39,000, and Supplies Expense \$7,000. Prepare the closing entries at December 31.

Post closing entries; underline and balance T-accounts.

BE4.5 (LO 2) Using the data in BE4.4, enter the balances in T-accounts, post the closing entries, and underline and balance the accounts.

Journalize and post closing entries using the three-column form of account.

BE4.6 (LO 2) The income statement for Salt Creek Golf Club for the month ending July 31 shows Service Revenue \$17,800, Salaries and Wages Expense \$9,600, Maintenance and Repairs Expense \$2,500, and Net Income \$5,700. Prepare the entries to close the revenue and expense accounts. Post the entries to the revenue and expense accounts, and complete the closing process for these accounts using the three-column form of account.

Identify post-closing trial balance accounts.

BE4.7 (LO 2) Using the data in BE4.3, identify the accounts that would be included in a post-closing trial balance.

BE4.8 (LO 3) The steps in the accounting cycle are listed in random order below. List the steps in proper sequence, assuming no worksheet is prepared, by placing numbers 1–9 in the blank spaces.

- _____ Prepare a trial balance.
- _____ Journalize the transactions.
- _____ Journalize and post closing entries.
- _____ Prepare financial statements.
- _____ Journalize and post adjusting entries.
- _____ Post to ledger accounts.
- _____ Prepare a post-closing trial balance.
- _____ Prepare an adjusted trial balance.
- _____ Analyze business transactions.

List the required steps in the accounting cycle in sequence.

BE4.9 (LO 3) At Raymond Company, the following errors were discovered after the transactions had been journalized and posted. Prepare the correcting entries.

Prepare correcting entries.

1. A collection on account from a customer for \$870 was recorded as a debit to Cash \$870 and a credit to Service Revenue \$870.
2. The purchase of store supplies on account for \$1,510 was recorded as a debit to Supplies \$1,150 and a credit to Accounts Payable \$1,150.

BE4.10 (LO 4) Financial Statement The balance sheet debit column of the worksheet for Jolie Company includes the following accounts: Accounts Receivable \$12,500, Prepaid Insurance \$4,500, Cash \$4,100, Supplies \$5,200, and Debt Investments (short-term) \$7,600. Prepare the current assets section of the balance sheet, listing the accounts in proper sequence.

Prepare the current assets section of a balance sheet.

BE4.11 (LO 4) The following are the major balance sheet classifications:

Classify accounts on balance sheet.

Current assets (CA)	Current liabilities (CL)
Long-term investments (LTI)	Long-term liabilities (LTL)
Property, plant, and equipment (PPE)	Owner's equity (OE)
Intangible assets (IA)	

Match each of the following accounts to its proper balance sheet classification.

_____ Accounts payable	_____ Income taxes payable
_____ Accounts receivable	_____ Debt investments (long-term)
_____ Accumulated depreciation—buildings	_____ Land
_____ Buildings	_____ Inventory
_____ Cash	_____ Patents
_____ Copyrights	_____ Supplies

***BE4.12 (LO 5)** At October 31, Miras Company made an accrued expense adjusting entry of \$3,300 for salaries. Prepare the reversing entry on November 1, and indicate the balances in Salaries and Wages Payable and Salaries and Wages Expense after posting the reversing entry.

Prepare reversing entries.

DO IT! Exercises

DO IT! 4.1 (LO 1) Jordan Carr is preparing a worksheet. Explain to Jordan how he should extend the following adjusted trial balance accounts to the financial statement columns of the worksheet.

Prepare a worksheet.

Service Revenue	Accounts Receivable
Notes Payable	Accumulated Depreciation
Owner's Capital	Utilities Expense

DO IT! 4.2 (LO2) Paloma Company shows the following balances in selected accounts of its adjusted trial balance.

Prepare closing entries.

Supplies	\$32,000	Service Revenue	\$108,000
Supplies Expense	6,000	Salaries and Wages Expense	40,000
Accounts Receivable	12,000	Utilities Expense	8,000
Owner's Drawings	22,000	Rent Expense	18,000
Owner's Capital	70,000		

Prepare the closing entries at December 31.

Prepare correcting entries.

DO IT! 4.3 (LO 3) Hanson Company has an inexperienced accountant. During the first month on the job, the accountant made the following errors in journalizing transactions. All entries were posted as made.

1. The purchase of supplies for \$650 cash was debited to Equipment \$210 and credited to Cash \$210.
2. A \$500 withdrawal of cash for B. Hanson's personal use was debited to Salaries and Wages Expense \$900 and credited to Cash \$900.
3. A payment on account of \$820 to a creditor was debited to Accounts Payable \$280 and credited to Cash \$280.

Prepare the correcting entries.

Match accounts to balance sheet classifications.

DO IT! 4.4 (LO 4) The following accounts were taken from the financial statements of Giles Company.

_____ Interest revenue	_____ Owner's capital
_____ Utilities payable	_____ Accumulated depreciation—equipment
_____ Accounts payable	_____ Equipment
_____ Supplies	_____ Salaries and wages expense
_____ Bonds payable	_____ Debt investments (long-term)
_____ Goodwill	_____ Unearned rent revenue

Match each of the accounts to its proper balance sheet classification, as shown below. If the item would not appear on a balance sheet, use "NA."

Current assets (CA)	Current liabilities (CL)
Long-term investments (LTI)	Long-term liabilities (LTL)
Property, plant, and equipment (PPE)	Owner's equity (OE)
Intangible assets (IA)	

Exercises

Complete the worksheet.

E4.1 (LO 1, 3) The trial balance columns of the worksheet for Dixon Company at June 30, 2020, are as follows.

Dixon Company
Worksheet
For the Month Ended June 30, 2020

Account Titles	Trial Balance	
	Dr.	Cr.
Cash	2,320	
Accounts Receivable	2,440	
Supplies	1,880	
Accounts Payable		1,120
Unearned Service Revenue		240
Owner's Capital		3,600
Service Revenue		2,400
Salaries and Wages Expense	560	
Miscellaneous Expense	160	
	<u>7,360</u>	<u>7,360</u>

Other data:

1. A physical count reveals \$500 of supplies on hand.
2. \$100 of the unearned revenue is still unearned at month-end.
3. Accrued salaries are \$210.

Instructions

Enter the trial balance on a worksheet and complete the worksheet.

E4.2 (LO 1) The adjusted trial balance columns of the worksheet for Auburn Company are as follows.

Complete the worksheet.



Auburn Company
Worksheet (partial)
For the Month Ended April 30, 2020

Account Titles	Adjusted Trial Balance		Income Statement		Balance Sheet	
	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
Cash	10,000					
Accounts Receivable	7,840					
Prepaid Rent	2,280					
Equipment	23,050					
Accumulated Depreciation—Equip.		4,900				
Notes Payable		5,700				
Accounts Payable		4,920				
Owner's Capital		27,960				
Owner's Drawings	3,650					
Service Revenue		15,590				
Salaries and Wages Expense	10,840					
Rent Expense	760					
Depreciation Expense	650					
Interest Expense	57					
Interest Payable		57				
Totals	59,127	59,127				

Instructions

Complete the worksheet.

E4.3 (LO 1, 4) Financial Statement Worksheet data for Auburn Company are presented in E4.2. The owner did not make any additional investments in the business in April.

Prepare financial statements from worksheet.



Instructions

Prepare an income statement, an owner's equity statement, and a classified balance sheet.

E4.4 (LO 2) Worksheet data for Auburn Company are presented in E4.2.

Journalize and post closing entries and prepare a post-closing trial balance.

Instructions

- Journalize the closing entries at April 30.
- Post the closing entries to Income Summary and Owner's Capital. Use T-accounts.
- Prepare a post-closing trial balance at April 30.

E4.5 (LO 1) The adjustments columns of the worksheet for Becker Company are shown below.

Prepare adjusting entries from a worksheet, and extend balances to worksheet columns.

Account Titles	Adjustments	
	Debit	Credit
Accounts Receivable	1,100	
Prepaid Insurance		300
Accumulated Depreciation—Equipment		900
Salaries and Wages Payable		500
Service Revenue		1,100
Salaries and Wages Expense	500	
Insurance Expense	300	
Depreciation Expense	900	
	2,800	2,800

Instructions

- Prepare the adjusting entries.
- Assuming the adjusted trial balance amount for each account is normal, indicate the financial statement column to which each balance should be extended.

Derive adjusting entries from worksheet data.

E4.6 (LO 1) Selected worksheet data for Bonita Company are presented below.

Account Titles	Trial Balance		Adjusted Trial Balance	
	Dr.	Cr.	Dr.	Cr.
Accounts Receivable	?		34,000	
Prepaid Insurance	26,000		20,000	
Supplies	7,000		?	
Accumulated Depreciation—Equipment		12,000		?
Salaries and Wages Payable		?		5,600
Service Revenue		88,000		97,000
Insurance Expense			?	
Depreciation Expense			10,000	
Supplies Expense			4,500	
Salaries and Wages Expense	?		49,000	

Instructions

- Fill in the missing amounts.
- Prepare the adjusting entries that were made.

Prepare closing entries, and prepare a post-closing trial balance.

E4.7 (LO 2) Victoria Lee Company had the following adjusted trial balance.

Victoria Lee Company
Adjusted Trial Balance
For the Month Ended June 30, 2020

Account Titles	Adjusted Trial Balance	
	Debit	Credit
Cash	\$ 3,712	
Accounts Receivable	3,904	
Supplies	480	
Accounts Payable		\$ 1,382
Unearned Service Revenue		160
Owner's Capital		5,760
Owner's Drawings	550	
Service Revenue		4,300
Salaries and Wages Expense	1,260	
Miscellaneous Expense	256	
Supplies Expense	1,900	
Salaries and Wages Payable		460
	<u>\$12,062</u>	<u>\$12,062</u>

Instructions

- Prepare closing entries at June 30, 2020.
- Prepare a post-closing trial balance.

Journalize and post closing entries, and prepare a post-closing trial balance.

E4.8 (LO 2) Phoebe Company ended its fiscal year on July 31, 2020. The company's adjusted trial balance as of the end of its fiscal year is as follows.

Phoebe Company
Adjusted Trial Balance
July 31, 2020

No.	Account Titles	Debit	Credit
101	Cash	\$ 9,840	
112	Accounts Receivable	8,780	
157	Equipment	15,900	
158	Accumulated Depreciation—Equip.		\$ 7,400
201	Accounts Payable		4,220
208	Unearned Rent Revenue		1,800
301	Owner's Capital		45,200
306	Owner's Drawings	16,000	
400	Service Revenue		64,000
429	Rent Revenue		6,500
711	Depreciation Expense	8,000	
726	Salaries and Wages Expense	55,700	
732	Utilities Expense	14,900	
		<u>\$129,120</u>	<u>\$129,120</u>

Instructions

- Prepare the closing entries using page J15.
- Post to Owner's Capital and No. 350 Income Summary accounts. (Use the three-column form.)
- Prepare a post-closing trial balance at July 31.

E4.9 (LO 4) Financial Statement The adjusted trial balance for Phoebe Company is presented in E4.8.

Prepare financial statements.

Instructions

- Prepare an income statement and an owner's equity statement for the year. Phoebe did not make any capital investments during the year.
- Prepare a classified balance sheet at July 31.

E4.10 (LO 3) Ray Louis has prepared the following list of statements about the accounting cycle.

Answer questions related to the accounting cycle.

- "Journalize the transactions" is the first step in the accounting cycle.
- Reversing entries are a required step in the accounting cycle.
- Correcting entries do not have to be part of the accounting cycle.
- If a worksheet is prepared, some steps of the accounting cycle are incorporated into the worksheet.
- The accounting cycle begins with the analysis of business transactions and ends with the preparation of a post-closing trial balance.
- All steps of the accounting cycle occur daily during the accounting period.
- The step of "post to the ledger accounts" occurs before the step of "journalize the transactions."
- Closing entries must be prepared before financial statements can be prepared.

Instructions

Identify each statement as true or false. If false, indicate how to correct the statement.

E4.11 (LO 2) Selected accounts for Tamora's Salon are presented below. All June 30 postings are from closing entries.

Prepare closing entries.

Salaries and Wages Expense				Service Revenue				Owner's Capital			
6/10	3,200				6/15	9,700		6/30	2,500	6/1	12,000
6/28	5,600				6/24	8,400				6/30	5,000
Bal.	8,800	6/30	8,800	6/30	18,100	Bal.	18,100			Bal.	14,500
Supplies Expense				Rent Expense				Owner's Drawings			
6/12	600			6/1	3,000			6/13	1,000		
6/24	700			Bal.	3,000	6/30	3,000	6/25	1,500		
Bal.	1,300	6/30	1,300					Bal.	2,500	6/30	2,500

Instructions

- Prepare the closing entries that were made.
- Post the closing entries to Income Summary.

Prepare correcting entries.

E4.12 (LO 3) Blair Natt Company discovered the following errors made in January 2020.

- A payment of Salaries and Wages Expense of \$700 was debited to Equipment and credited to Cash, both for \$700.
- A collection of \$1,500 from a client on account was debited to Cash \$150 and credited to Service Revenue \$150.
- The purchase of equipment on account for \$670 was debited to Equipment \$760 and credited to Accounts Payable \$760.

Instructions

- Correct the errors by reversing the incorrect entry and preparing the correct entry.
- Correct the errors without reversing the incorrect entry.

Prepare correcting entries.

E4.13 (LO 3) Patel Company has an inexperienced accountant. During the first 2 weeks on the job, the accountant made the following errors in journalizing transactions. All entries were posted as made.

- A payment on account of \$750 to a creditor was debited to Accounts Payable \$570 and credited to Cash \$570.
- The purchase of supplies on account for \$560 was debited to Equipment \$56 and credited to Accounts Payable \$56.
- A \$500 withdrawal of cash for N. Patel's personal use was debited to Salaries and Wages Expense \$500 and credited to Cash \$500.

Instructions

Prepare the correcting entries.

Prepare a classified balance sheet.

E4.14 (LO 4) Writing Financial Statement The adjusted trial balance for Carter Bowling Alley at December 31, 2020, contains the following accounts.

	Debit		Credit
Buildings	\$128,800	Owner's Capital	\$113,000
Accounts Receivable	14,520	Accumulated Depreciation—Buildings	42,600
Prepaid Insurance	4,680	Accounts Payable	12,300
Cash	18,040	Notes Payable	97,780
Equipment	62,400	Accumulated Depreciation—Equipment	18,720
Land	65,000	Interest Payable	3,800
Insurance Expense	780	Service Revenue	17,180
Depreciation Expense	7,360		\$305,380
Interest Expense	3,800		
	<u>\$305,380</u>		

Instructions

- Prepare a classified balance sheet; assume that \$30,000 of the note payable will be paid in 2021.
- Comment on the liquidity of the company.

Classify accounts on balance sheet.

E4.15 (LO 4) The following are the major balance sheet classifications.

Current assets (CA)	Current liabilities (CL)
Long-term investments (LTI)	Long-term liabilities (LTL)
Property, plant, and equipment (PPE)	Owner's equity (OE)
Intangible assets (IA)	

Instructions

Classify each of the following accounts taken from Faust Company's balance sheet.

_____ Accounts payable	_____ Accumulated depreciation—equipment
_____ Accounts receivable	_____ Buildings
_____ Cash	_____ Land (in use)
_____ Owner's capital	_____ Notes payable (due in 2 years)
_____ Patents	_____ Supplies
_____ Salaries and wages payable	_____ Equipment
_____ Inventory	_____ Prepaid expenses
_____ Stock investments (to be sold in 7 months)	

E4.16 (LO 4) Financial Statement The following items were taken from the financial statements of P. Jimenez Company. (All amounts are in thousands.)

Prepare a classified balance sheet.

Long-term debt	\$ 1,000	Accumulated depreciation—equipment	\$ 5,655
Prepaid insurance	650	Accounts payable	1,214
Equipment	11,500	Notes payable (due after 2021)	400
Stock investments (long-term)	264	Owner's capital	12,955
Debt investments (short-term)	3,690	Accounts receivable	1,696
Notes payable (due in 2021)	500	Inventory	1,256
Cash	2,668		

Instructions

Prepare a classified balance sheet in good form as of December 31, 2020.

E4.17 (LO 4) Financial Statement These financial statement items are for Basten Company at year-end, July 31, 2020.

Prepare financial statements.

Salaries and wages payable	\$ 2,080	Notes payable (long-term)	\$ 1,800
Salaries and wages expense	48,700	Cash	14,200
Utilities expense	22,600	Accounts receivable	9,780
Equipment	34,400	Accumulated depreciation—equipment	6,000
Accounts payable	4,100	Owner's drawings	3,000
Service revenue	63,000	Depreciation expense	4,000
Rent revenue	8,500	Owner's capital (beginning of the year)	51,200

Instructions

- Prepare an income statement and an owner's equity statement for the year. The owner did not make any new investments during the year.
- Prepare a classified balance sheet at July 31.

***E4.18 (LO 5)** Krantz Company pays salaries of \$15,000 every Monday for the preceding 5-day week (Monday through Friday). Assume December 31 falls on a Tuesday, so Krantz's employees have worked 2 days without being paid at the end of the fiscal year.

Use reversing entries.

Instructions

- Assume the company does not use reversing entries. Prepare the December 31 adjusting entry and the entry on Monday, January 6, when Krantz pays the payroll.
- Assume the company does use reversing entries. Prepare the December 31 adjusting entry, the January 1 reversing entry, and the entry on Monday, January 6, when Krantz pays the payroll.

***E4.19 (LO 2, 5)** On December 31, the adjusted trial balance of Shihata Employment Agency shows the following selected data.

Prepare closing and reversing entries.

Accounts Receivable	\$24,500	Service Revenue	\$92,500
Interest Expense	7,700	Interest Payable	2,200

Analysis shows that adjusting entries were made to (1) accrue \$5,000 of service revenue and (2) accrue \$2,200 interest expense.

Instructions

- Prepare the closing entries for the temporary accounts shown above at December 31.
- Prepare the reversing entries on January 1.
- Post the entries in (a) and (b). Underline and balance the accounts. (Use T-accounts.)
- Prepare the entries to record (1) the collection of the accrued revenue on January 10 and (2) the payment of all interest due (\$3,000) on January 15.
- Post the entries in (d) to the temporary accounts.

Problems: Set A

P4.1A (LO 1, 2, 4) Financial Statement The trial balance columns of the worksheet for Warren Roofing at March 31, 2020, are as follows.

Prepare a worksheet, financial statements, and adjusting and closing entries.



Warren Roofing
Worksheet
For the Month Ended March 31, 2020

Account Titles	Trial Balance	
	Dr.	Cr.
Cash	4,500	
Accounts Receivable	3,200	
Supplies	2,000	
Equipment	11,000	
Accumulated Depreciation—Equipment		1,250
Accounts Payable		2,500
Unearned Service Revenue		550
Owner's Capital		12,900
Owner's Drawings	1,100	
Service Revenue		6,300
Salaries and Wages Expense	1,300	
Miscellaneous Expense	400	
	<u>23,500</u>	<u>23,500</u>

Other data:

1. A physical count reveals only \$480 of roofing supplies on hand.
2. Depreciation for March is \$250.
3. Unearned revenue amounted to \$260 at March 31.
4. Accrued salaries are \$700.

Instructions

- a. Adjusted trial balance
\$24,450
- b. Net income \$2,420
Total assets \$17,680

- a. Enter the trial balance on a worksheet and complete the worksheet.
- b. Prepare an income statement and owner's equity statement for the month of March and a classified balance sheet at March 31. T. Warren made an additional investment in the business of \$10,000 in March.
- c. Journalize the adjusting entries from the adjustments columns of the worksheet.
- d. Journalize the closing entries from the financial statement columns of the worksheet.

Complete worksheet; prepare financial statements, closing entries, and post-closing trial balance.



P4.2A (LO 1, 2, 4) Financial Statement The adjusted trial balance columns of the worksheet for Nguyen Company, owned by C. Nguyen, are as follows.

Nguyen Company
Worksheet
For the Year Ended December 31, 2020

Account No.	Account Titles	Adjusted Trial Balance	
		Dr.	Cr.
101	Cash	5,300	
112	Accounts Receivable	10,800	
126	Supplies	1,500	
130	Prepaid Insurance	2,000	
157	Equipment	27,000	
158	Accumulated Depreciation—Equipment		5,600
200	Notes Payable		15,000
201	Accounts Payable		6,100
212	Salaries and Wages Payable		3,600
230	Interest Payable		600
301	Owner's Capital		13,000
306	Owner's Drawings	7,600	
400	Service Revenue		61,000
610	Advertising Expense	9,000	
631	Supplies Expense	4,000	
711	Depreciation Expense	5,600	
722	Insurance Expense	3,500	
726	Salaries and Wages Expense	28,000	
905	Interest Expense	600	
	Totals	<u>104,900</u>	<u>104,900</u>

Instructions

- Complete the worksheet by extending the balances to the financial statement columns.
- Prepare an income statement, owner's equity statement, and a classified balance sheet. (*Note:* \$4,000 of the notes payable become due in 2021.) C. Nguyen did not make any additional investments in the business during the year.
- Prepare the closing entries. Use J14 for the journal page.
- Post the closing entries. Use the three-column form of account. Income Summary is No. 350.
- Prepare a post-closing trial balance.

- Net income \$10,300
- Current assets \$19,600
Current liabilities \$14,300
- Post-closing trial balance
\$46,600

P4.3A (LO 1, 2, 4) Financial Statement The completed financial statement columns of the worksheet for Bray Company are shown as follows.

Prepare financial statements, closing entries, and post-closing trial balance.

Bray Company
Worksheet
For the Year Ended December 31, 2020

Account No.	Account Titles	Income Statement		Balance Sheet	
		Dr.	Cr.	Dr.	Cr.
101	Cash			8,800	
112	Accounts Receivable			10,800	
130	Prepaid Insurance			2,800	
157	Equipment			24,000	
158	Accumulated Depreciation—Equip.				4,200
201	Accounts Payable				9,000
212	Salaries and Wages Payable				2,400
301	Owner's Capital				19,500
306	Owner's Drawings			11,000	
400	Service Revenue		60,000		
622	Maintenance and Repairs Expense	1,700			
711	Depreciation Expense	2,800			
722	Insurance Expense	1,800			
726	Salaries and Wages Expense	30,000			
732	Utilities Expense	1,400			
	Totals	37,700	60,000	57,400	35,100
	Net Income	22,300			22,300
		<u>60,000</u>	<u>60,000</u>	<u>57,400</u>	<u>57,400</u>

Instructions

- Prepare an income statement, an owner's equity statement, and a classified balance sheet.
- Prepare the closing entries. L. Bray did not make any additional investments during the year.
- Post the closing entries and underline and balance the accounts. (Use T-accounts.) Income Summary is account No. 350.
- Prepare a post-closing trial balance.

- Ending capital \$30,800
Total current assets \$22,400

P4.4A (LO 1, 2, 4) Financial Statement Rusthe Management Services began business on January 1, 2020, with a capital investment of \$120,000. The company manages condominiums for owners (Service Revenue) and rents space in its own office building (Rent Revenue). The trial balance and adjusted trial balance columns of the worksheet at the end of the first year are as follows.

- Post-closing trial balance
\$46,400

Complete worksheet; prepare classified balance sheet, entries, and post-closing trial balance.

Rusthe Management Services
Worksheet
For the Year Ended December 31, 2020

Account Titles	Trial Balance		Adjusted Trial Balance	
	Dr.	Cr.	Dr.	Cr.
Cash	13,800		13,800	
Accounts Receivable	28,300		28,300	
Prepaid Insurance	3,600		2,100	
Land	67,000		67,000	
Buildings	127,000		127,000	

Account Titles	Trial Balance		Adjusted Trial Balance	
	Dr.	Cr.	Dr.	Cr.
Equipment	59,000		59,000	
Accounts Payable		12,500		12,500
Unearned Rent Revenue		6,000		1,000
Mortgage Payable		120,000		120,000
Owner's Capital		144,000		144,000
Owner's Drawings	22,000		22,000	
Service Revenue		90,700		90,700
Rent Revenue		29,000		34,000
Salaries and Wages Expense	42,000		42,000	
Advertising Expense	20,500		20,500	
Utilities Expense	19,000		19,000	
Totals	<u>402,200</u>	<u>402,200</u>		
Insurance Expense			1,500	
Depreciation Expense			6,600	
Accumulated Depreciation—Buildings				3,000
Accumulated Depreciation—Equipment				3,600
Interest Expense			10,000	
Interest Payable				10,000
Totals			<u>418,800</u>	<u>418,800</u>

Instructions

- a. Net income \$25,100
b. Total current assets \$44,200

- a. Prepare a complete worksheet.
b. Prepare a classified balance sheet. (*Note:* \$45,000 of the mortgage note payable is due for payment next year.)
c. Journalize the adjusting entries.
d. Journalize the closing entries.
e. Prepare a post-closing trial balance.

- e. Post-closing trial balance
\$297,200

Complete all steps in accounting cycle.



P4.5A (LO 1, 2, 4) Anya Clark opened Anya's Cleaning Service on July 1, 2020. During July, the following transactions were completed.

- July 1 Anya invested \$20,000 cash in the business.
1 Purchased used truck for \$12,000, paying \$4,000 cash and the balance on account.
3 Purchased cleaning supplies for \$2,100 on account.
5 Paid \$1,800 cash on a 1-year insurance policy effective July 1.
12 Billed customers \$4,500 for cleaning services.
18 Paid \$1,500 cash on amount owed on truck and \$1,400 on amount owed on cleaning supplies.
20 Paid \$2,800 cash for employee salaries.
21 Collected \$3,400 cash from customers billed on July 12.
25 Billed customers \$6,000 for cleaning services.
31 Paid \$350 for the monthly gasoline bill for the truck.
31 Withdraw \$5,600 cash for personal use.

The chart of accounts for Anya's Cleaning Service contains the following accounts: No. 101 Cash, No. 112 Accounts Receivable, No. 126 Supplies, No. 130 Prepaid Insurance, No. 157 Equipment, No. 158 Accumulated Depreciation—Equipment, No. 201 Accounts Payable, No. 212 Salaries and Wages Payable, No. 301 Owner's Capital, No. 306 Owner's Drawings, No. 350 Income Summary, No. 400 Service Revenue, No. 631 Supplies Expense, No. 633 Gasoline Expense, No. 711 Depreciation Expense, No. 722 Insurance Expense, and No. 726 Salaries and Wages Expense.

Instructions

- b. Trial balance \$37,700
c. Adjusted trial balance
\$41,900

- a. Journalize and post the July transactions. Use page J1 for the journal and the three-column form of account.
b. Prepare a trial balance at July 31 on a worksheet.
c. Enter the following adjustments on the worksheet and complete the worksheet.
1. Unbilled and uncollected revenue for services performed at July 31 were \$2,700.
2. Depreciation on equipment for the month was \$500.

3. One-twelfth of the insurance expired.
4. An inventory count shows \$600 of cleaning supplies on hand at July 31.
5. Accrued but unpaid employee salaries were \$1,000.
- d. Prepare the income statement and owner's equity statement for July and a classified balance sheet at July 31.
- e. Journalize and post adjusting entries. Use page J2 for the journal.
- f. Journalize and post closing entries and complete the closing process. Use page J3 for the journal.
- g. Prepare a post-closing trial balance at July 31.

d. Net income \$6,900
Total assets \$29,500

g. Post-closing trial balance
\$30,000

P4.6A (LO 3) Horace Culpepper, CPA, was retained by Pulsar Cable to prepare financial statements for April 2020. Horace accumulated all the ledger balances per Pulsar's records and found the following.

Analyze errors and prepare correcting entries and trial balance.

**Pulsar Cable
Trial Balance
April 30, 2020**

	<u>Debit</u>	<u>Credit</u>
Cash	\$ 4,100	
Accounts Receivable	3,200	
Supplies	800	
Equipment	10,800	
Accumulated Depreciation—Equip.		\$ 1,350
Accounts Payable		2,100
Salaries and Wages Payable		700
Unearned Service Revenue		890
Owner's Capital		12,900
Service Revenue		5,650
Salaries and Wages Expense	3,300	
Advertising Expense	600	
Miscellaneous Expense	290	
Depreciation Expense	500	
	<u>\$23,590</u>	<u>\$23,590</u>

Horace Culpepper then reviewed the records and found the following errors.

1. Cash received from a customer on account was recorded as \$950 instead of \$590.
2. A payment of \$75 for advertising expense was entered as a debit to Miscellaneous Expense \$75 and a credit to Cash \$75.
3. The first salary payment in April was for \$1,900, which included \$900 of salaries payable on March 31. The payment was recorded as a debit to Salaries and Wages Expense \$1,900 and a credit to Cash \$1,900. (No reversing entries were made on April 1.)
4. The purchase on account of a printer costing \$510 was recorded as a debit to Supplies and a credit to Accounts Payable for \$510.
5. A cash payment of repair expense on equipment for \$96 was recorded as a debit to Equipment \$69 and a credit to Cash \$69.

Instructions

- a. Prepare an analysis of each error showing (1) the incorrect entry, (2) the correct entry, and (3) the correcting entry. Items 4 and 5 occurred on April 30, 2020.
- b. Prepare a correct trial balance.

b. Trial balance \$22,890

Continuing Case

Cookie Creations

(Note: This is a continuation of the Cookie Creations case from Chapters 1 through 3.)

CC4 Natalie had a very busy December. At the end of the month, after journalizing and posting the December transactions and adjusting entries, Natalie prepared the following adjusted trial balance.



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**Cookie Creations
Adjusted Trial Balance
December 31, 2019**

	<u>Debit</u>	<u>Credit</u>
Cash	\$1,180	
Accounts Receivable	875	
Supplies	350	
Prepaid Insurance	1,210	
Equipment	1,200	
Accumulated Depreciation—Equipment		\$ 40
Accounts Payable		75
Salaries and Wages Payable		56
Interest Payable		15
Unearned Service Revenue		300
Notes Payable		2,000
Owner's Capital		800
Owner's Drawings	500	
Service Revenue		4,515
Salaries and Wages Expense	1,006	
Utilities Expense	125	
Advertising Expense	165	
Supplies Expense	1,025	
Depreciation Expense	40	
Insurance Expense	110	
Interest Expense	15	
	<u>\$7,801</u>	<u>\$7,801</u>

Instructions

Using the information in the adjusted trial balance, do the following.

- Prepare an income statement and an owner's equity statement for the 2 months ended December 31, 2019, and a classified balance sheet at December 31, 2019. The note payable has a stated interest rate of 6%, and the principal and interest are due on November 16, 2021.
- Natalie has decided that her year-end will be December 31, 2019. Prepare and post closing entries as of December 31, 2019.
- Prepare a post-closing trial balance.

Ethics Case

EC4 As the controller of Take No Prisoners Perfume Company, you discover a misstatement that overstated net income in the prior year's financial statements. The misleading financial statements appear in the company's annual report which was issued to banks and other creditors less than a month ago. After much thought about the consequences of telling the president, Mike Flanary, about this misstatement, you gather your courage to inform him. Mike says, "Hey! What they don't know won't hurt them. But, just so we set the record straight, we'll adjust this year's financial statements for last year's misstatement. We can absorb that misstatement better in this year than in last year anyway! Just don't make such a mistake again."

Instructions

- Who are the stakeholders in this situation?
- What are the ethical issues in this situation?
- What would you do as a controller in this situation?

Comprehensive Accounting Cycle Review

Complete all steps in
accounting cycle.



ACR4.1 Financial Statement Mike Greenberg opened Kleene Window Washing Co. on July 1, 2020. During July, the following transactions were completed.

- July 1 Owner invested \$12,000 cash in the company.
 1 Purchased used truck for \$8,000, paying \$2,000 cash and the balance on account.
 3 Purchased cleaning supplies for \$900 on account.
 5 Paid \$1,800 cash on a 1-year insurance policy effective July 1.
 12 Billed customers \$3,700 for cleaning services performed.
 18 Paid \$1,000 cash on amount owed on truck and \$500 on amount owed on cleaning supplies.
 20 Paid \$2,000 cash for employee salaries.
 21 Collected \$1,600 cash from customers billed on July 12.
 25 Billed customers \$2,500 for cleaning services performed.
 31 Paid \$290 for maintenance of the truck during month.
 31 Owner withdrew \$600 cash from the company.

The chart of accounts for Kleene Window Washing contains the following accounts: Cash, Accounts Receivable, Supplies, Prepaid Insurance, Equipment, Accumulated Depreciation—Equipment, Accounts Payable, Salaries and Wages Payable, Owner's Capital, Owner's Drawings, Income Summary, Service Revenue, Maintenance and Repairs Expense, Supplies Expense, Depreciation Expense, Insurance Expense, and Salaries and Wages Expense.

Instructions

- Journalize the July transactions.
- Post to the ledger accounts. (Use T-accounts.)
- Prepare a trial balance at July 31.
- Journalize the following adjustments.
 - Services performed but unbilled and uncollected at July 31 were \$1,700.
 - Depreciation on equipment for the month was \$180.
 - One-twelfth of the insurance expired.
 - A count shows \$320 of cleaning supplies on hand at July 31.
 - Accrued but unpaid employee salaries were \$400.
- Post adjusting entries to the T-accounts.
- Prepare an adjusted trial balance. f. Cash \$5,410
- Prepare the income statement and an owner's equity statement for July and a classified balance sheet at July 31. g. Tot. assets \$21,500
- Journalize and post closing entries and complete the closing process.
- Prepare a post-closing trial balance at July 31.

ACR4.2 Financial Statement Lars Linken opened Lars Cleaners on March 1, 2020. During March, the following transactions were completed.

- Mar. 1 Owner invested \$15,000 cash in the company.
 1 Borrowed \$6,000 cash by signing a 6-month, 6%, \$6,000 note payable. Interest will be paid the first day of each subsequent month.
 1 Purchased used truck for \$8,000 cash.
 2 Paid \$1,500 cash to cover rent from March 1 through May 31.
 3 Paid \$2,400 cash on a 6-month insurance policy effective March 1.
 6 Purchased cleaning supplies for \$2,000 on account.
 14 Billed customers \$3,700 for cleaning services performed.
 18 Paid \$500 on amount owed on cleaning supplies.
 20 Paid \$1,750 cash for employee salaries.
 21 Collected \$1,600 cash from customers billed on March 14.
 28 Billed customers \$4,200 for cleaning services performed.
 31 Paid \$350 for gas and oil used in truck during month (use Maintenance and Repairs Expense).
 31 Owner withdrew \$900 cash from the company.

Complete all steps in accounting cycle.



The chart of accounts for Lars Cleaners contains the following accounts: Cash, Accounts Receivable, Supplies, Prepaid Insurance, Prepaid Rent, Equipment, Accumulated Depreciation—Equipment, Accounts Payable, Salaries and Wages Payable, Notes Payable, Interest Payable, Owner's Capital, Owner's Drawings, Income Summary, Service Revenue, Maintenance and Repairs Expense, Supplies Expense, Depreciation Expense, Insurance Expense, Salaries and Wages Expense, Rent Expense, and Interest Expense.

Instructions

- a. Journalize the March transactions.
- b. Post to the ledger accounts. (Use T-accounts.)
- c. Prepare a trial balance at March 31.
- d. Journalize the following adjustments.
 1. Services performed but unbilled and uncollected at March 31 were \$200.
 2. Depreciation on equipment for the month was \$250.
 3. One-sixth of the insurance expired.
 4. An inventory count shows \$280 of cleaning supplies on hand at March 31.
 5. Accrued but unpaid employee salaries were \$1,080.
 6. One month of the prepaid rent has expired.
 7. One month of interest expense related to the note payable has accrued and will be paid April 1. (Hint: Use the formula from Illustration 3.17 to compute interest.)
- e. Post adjusting entries to the T-accounts.
- f. Prepare an adjusted trial balance.
- g. Prepare the income statement and an owner's equity statement for March and a classified balance sheet at March 31.
- h. Journalize and post closing entries and complete the closing process.
- i. Prepare a post-closing trial balance at March 31.

f. Tot. adj. trial balance
\$31,960

g. Tot. assets \$24,730

Journalize transactions and follow through accounting cycle to preparation of financial statements.

**ACR4.3 Financial Statement**

On August 1, 2020, the following were the account balances of B&B Repair Services.

	Debit		Credit
Cash	\$ 6,040	Accumulated Depreciation—Equipment	\$ 600
Accounts Receivable	2,910	Accounts Payable	2,300
Notes Receivable	4,000	Unearned Service Revenue	1,260
Supplies	1,030	Salaries and Wages Payable	1,420
Equipment	10,000	Owner's Capital	18,400
	<u>\$23,980</u>		<u>\$23,980</u>

During August, the following summary transactions were completed.

- Aug. 1 Paid \$400 cash for advertising in local newspapers. Advertising flyers will be included with newspapers delivered during August and September.
- 3 Paid August rent \$380.
- 5 Received \$1,200 cash from customers in payment on account.
- 10 Paid \$3,120 for salaries due employees, of which \$1,700 is for August and \$1,420 is for July salaries payable.
- 12 Received \$2,800 cash for services performed in August.
- 15 Purchased store equipment on account \$2,000.
- 20 Paid creditors \$2,000 of accounts payable due.
- 22 Purchased supplies on account \$800.
- 25 Paid \$2,900 cash for employees' salaries.
- 27 Billed customers \$3,760 for services performed.
- 29 Received \$780 from customers for services to be performed in the future.

Adjustment data:

1. A count shows supplies on hand of \$960.
2. Accrued but unpaid employees' salaries are \$1,540.
3. Depreciation on equipment for the month is \$320.
4. Services were performed to satisfy \$800 of unearned service revenue.
5. One month's worth of advertising services has been received.
6. One month of interest revenue related to the \$4,000 note receivable has accrued. The 4-month note has a 6% annual interest rate. (Hint: Use the formula from Illustration 3.17 to compute interest.)

Instructions

- a. Enter the August 1 balances in the ledger accounts. (Use T-accounts.)
- b. Journalize the August transactions.

- c. Post to the ledger accounts. B&B's chart of accounts includes Prepaid Advertising, Interest Receivable, Service Revenue, Interest Revenue, Advertising Expense, Depreciation Expense, Supplies Expense, Salaries and Wages Expense, and Rent Expense.
- d. Prepare a trial balance at August 31.
- e. Journalize and post adjusting entries.
- f. Prepare an adjusted trial balance.
- g. Prepare an income statement and an owner's equity statement for August and a classified balance sheet at August 31.
- h. Journalize and post closing entries and complete the closing process.
- i. Prepare a post-closing trial balance at August 31.

f. Cash \$2,020
 Tot. adj. trial balance \$32,580
 g. Net loss \$530

ACR4.4 Financial Statement At June 30, 2020, the end of its most recent fiscal year, Green River Computer Consultants' post-closing trial balance was as follows:

Record and post transaction, adjusting, and closing journal entries; prepare adjusted trial balance and financial statements.

	<u>Debit</u>	<u>Credit</u>
Cash	\$5,230	
Accounts receivable	1,200	
Supplies	690	
Accounts payable		\$ 400
Unearned service revenue		1,120
Owner's capital		5,600
	<u>\$7,120</u>	<u>\$7,120</u>

The company underwent a major expansion in July. New staff was hired and more financing was obtained. Green River conducted the following transactions during July 2020, and adjusts its accounts monthly.

- July 1 Purchased equipment, paying \$4,000 cash and signing a 2-year note payable for \$20,000. The equipment has a 4-year useful life. The note has a 6% interest rate which is payable on the first day of each following month.
- 2 Owner invested \$50,000 cash in the company.
- 3 Paid \$3,600 cash for a 12-month insurance policy effective July 1.
- 3 Paid the first 2 (July and August 2020) months' rent for an annual lease of office space for \$4,000 per month.
- 6 Paid \$3,800 for supplies.
- 9 Visited client offices and agreed on the terms of a consulting project. Green River will bill the client, Connor Productions, on the 20th of each month for services performed.
- 10 Collected \$1,200 cash on account from Milani Brothers. This client was billed in June when Green River performed the service.
- 13 Performed services for Fitzgerald Enterprises. This client paid \$1,120 in advance last month. All services relating to this payment are now completed.
- 14 Paid \$400 cash for a utility bill. This related to June utilities that were accrued at the end of June.
- 16 Met with a new client, Thunder Bay Technologies. Received \$12,000 cash in advance for future services to be performed.
- 18 Paid semi-monthly salaries for \$11,000.
- 20 Performed services worth \$28,000 on account and billed customers.
- 20 Received a bill for \$2,200 for advertising services received during July. The amount is not due until August 15.
- 23 Performed the first phase of the project for Thunder Bay Technologies. Recognized \$10,000 of revenue from the cash advance received July 16.
- 27 Received \$15,000 cash from customers billed on July 20.

Adjustment data:

- 1. Adjustment of prepaid insurance.
- 2. Adjustment of prepaid rent.
- 3. Supplies used, \$1,250.
- 4. Equipment depreciation, \$500 per month.
- 5. Accrual of interest on note payable. (*Hint: Use the formula from Illustration 3.17 to compute interest.*)
- 6. Salaries for the second half of July, \$11,000, to be paid on August 1.



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Accounting for Merchandising Operations

Chapter Preview

Merchandising is one of the largest and most influential industries in the United States. It is likely that a number of you will work for a merchandiser. Therefore, understanding the financial statements of merchandising companies is important. In this chapter, you will learn the basics about reporting merchandising transactions. In addition, you will learn how to prepare and analyze a commonly used form of the income statement—the multiple-step income statement.

Feature Story

Buy Now, Vote Later

Have you ever shopped for outdoor gear at an **REI (Recreational Equipment Incorporated)** store? If so, you might have been surprised if a salesclerk asked if you were a member. A member? What do you mean a member? REI is a consumer cooperative, or “co-op” for short. To figure out what that means, consider this quote from the company’s annual stewardship report:

As a cooperative, the Company is owned by its members. Each member is entitled to one vote in the election of the Company’s Board of Directors. Since January 1, 2008, the nonrefundable, nontransferable, one-time membership fee has been 20 dollars. As of December 31, 2010, there were approximately 10.8 million members.

Voting rights? Now that's something you don't get from shopping at **Wal-Mart**. REI members get other benefits as well, including sharing in the company's profits through a dividend at the end of the year. The more you spend, the bigger your dividend.

Since REI is a co-op, you might also wonder whether management's incentives might be a little different than at other stores. Management is still concerned about making a profit as it ensures the long-term viability of the company. REI's members also want the company to be run efficiently, so that prices remain low. In order for its members to evaluate just how well management is doing, REI publishes an audited annual report, just like publicly traded companies do.

How well is this business model working for REI? Well, it has consistently been rated as one of the best places to work

in the United States by *Fortune* magazine. It is one of only five companies named each year since the list was created in 1998. Also, REI had sustainable business practices long before social responsibility became popular at other companies. The CEO's stewardship report states "we reduced the absolute amount of energy we use despite opening four new stores and growing our business; we grew the amount of FSC-certified paper we use to 58.4 percent of our total paper footprint—including our cash register receipt paper; we facilitated 2.2 million volunteer hours and we provided \$3.7 million to more than 330 conservation and recreation nonprofits."

So, while REI, like other retailers, closely monitors its financial results, it also strives to succeed in other areas. And, with over 10 million votes at stake, REI's management knows that it has to deliver.

Chapter Outline

LEARNING OBJECTIVES

LO 1 Describe merchandising operations and inventory systems.	• Operating cycles • Flow of costs	DO IT! 1 Merchandising Operations and Inventory Systems
LO 2 Record purchases under a perpetual inventory system.	• Freight costs • Purchase returns and allowances • Purchase discounts • Summary of purchasing transactions	DO IT! 2 Purchase Transactions
LO 3 Record sales under a perpetual inventory system.	• Sales returns and allowances • Sales discounts	DO IT! 3 Sales Transactions
LO 4 Apply the steps in the accounting cycle to a merchandising company.	• Adjusting entries • Closing entries • Summary of merchandising entries	DO IT! 4 Closing Entries
LO 5 Prepare a multiple-step income statement and a comprehensive income statement.	• Multiple-step income statement • Single-step income statement • Comprehensive income statement • Classified balance sheet	DO IT! 5 Multiple-Step Income Statement

Go to the Review and Practice section at the end of the chapter for a review of key concepts and practice applications with solutions.
Visit WileyPLUS with ORION for additional tutorials and practice opportunities.

Merchandising Operations and Inventory Systems

LEARNING OBJECTIVE 1

Describe merchandising operations and inventory systems.

REI, **Wal-Mart**, and **Amazon.com** are called merchandising companies because they buy and sell merchandise rather than perform services as their primary source of revenue. Merchandising companies that purchase and sell directly to consumers are called **retailers**. Merchandising companies that sell to retailers are known as **wholesalers**. For example, retailer **Walgreens** might buy goods from wholesaler **McKesson**. Retailer **Office Depot** might buy office supplies from wholesaler **United Stationers**. The primary source of revenue for merchandising companies is the sale of merchandise, often referred to simply as **sales revenue** or **sales**. A merchandising company has two categories of expenses: cost of goods sold and operating expenses.

Cost of goods sold is the total cost of merchandise sold during the period. This expense is directly related to the revenue recognized from the sale of goods. **Illustration 5.1** shows the income measurement process for a merchandising company. The items in the two blue boxes are unique to a merchandising company; they are not used by a service company.

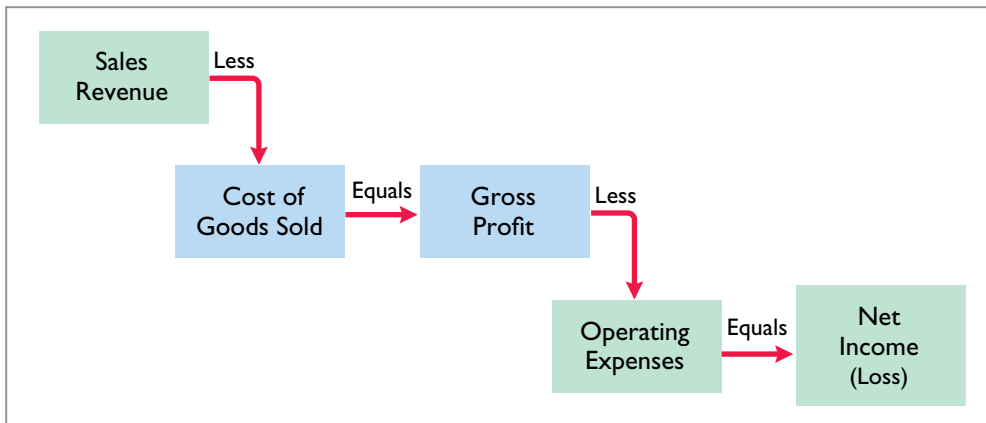


ILLUSTRATION 5.1

Income measurement process for a merchandising company

Operating Cycles

The operating cycle of a merchandising company ordinarily is longer than that of a service company. The purchase of merchandise inventory and its eventual sale lengthen the cycle. **Illustration 5.2** shows the operating cycle of a service company.

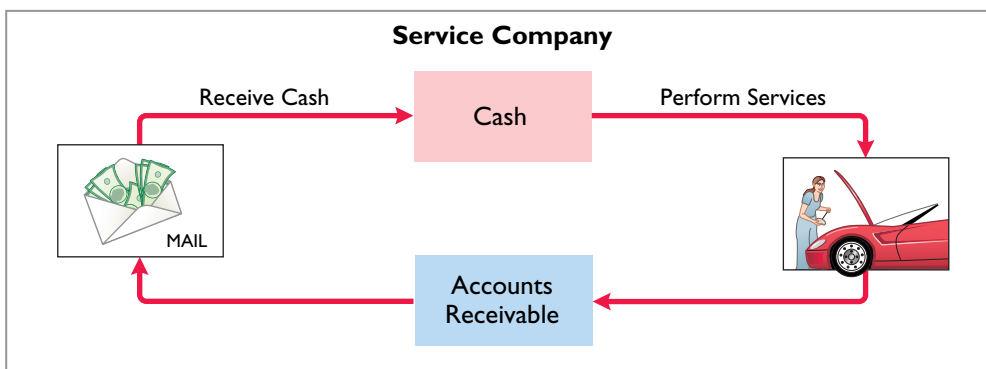


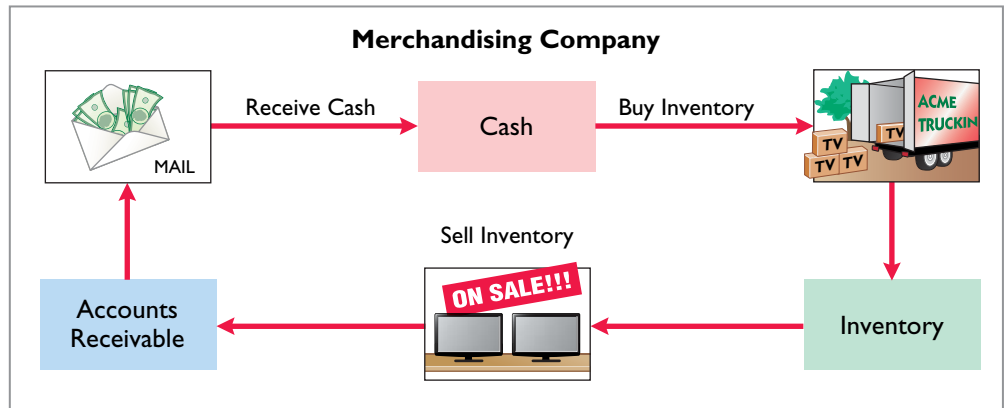
ILLUSTRATION 5.2

Operating cycle for a service company

Illustration 5.3 shows the operating cycle of a merchandising company.

ILLUSTRATION 5.3

Operating cycle for a merchandising company



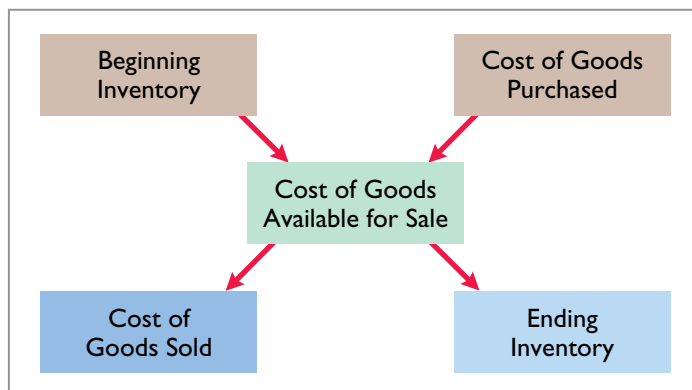
Note that the added asset account for a merchandising company is the Inventory account. Companies report inventory as a current asset on the balance sheet.

Flow of Costs

The flow of costs for a merchandising company is as follows. Beginning inventory plus the cost of goods purchased is the cost of goods available for sale. As goods are sold, they are assigned to cost of goods sold. Those goods that are not sold by the end of the accounting period represent ending inventory. **Illustration 5.4** describes these relationships. Companies use one of two systems to account for inventory: a **perpetual inventory system** or a **periodic inventory system**.

ILLUSTRATION 5.4

Flow of costs



Perpetual System

HELPFUL HINT

Even under perpetual inventory systems, companies take a physical inventory count. This is done as a control procedure to verify inventory levels, in order to detect theft or “shrinkage.”

In a **perpetual inventory system**, companies keep detailed records of the cost of each inventory purchase and sale (see **Helpful Hint**). These records continuously—perpetually—show the inventory that should be on hand for every item. For example, a **Ford** dealership has separate inventory records for each automobile, truck, and van on its lot and showroom floor. Similarly, a **Kroger** grocery store uses bar codes and optical scanners to keep a daily running record of every box of cereal and every jar of jelly that it buys and sells. Under a perpetual inventory system, a company determines the cost of goods sold **each time a sale occurs**.

Periodic System

In a **periodic inventory system**, companies do not keep detailed inventory records of the goods on hand throughout the period. Instead, they determine the cost of goods sold **only at the end of the accounting period**—that is, periodically. At that point, the company takes a physical inventory count to determine the cost of goods on hand.

To determine the cost of goods sold under a periodic inventory system, the following steps are necessary:

1. Determine the cost of goods on hand at the beginning of the accounting period.
2. Add to it the cost of goods purchased.
3. Subtract the cost of goods on hand as determined by the physical inventory count at the end of the accounting period.

Illustration 5.5 compares the sequence of activities and the timing of the cost of goods sold computation under the two inventory systems.

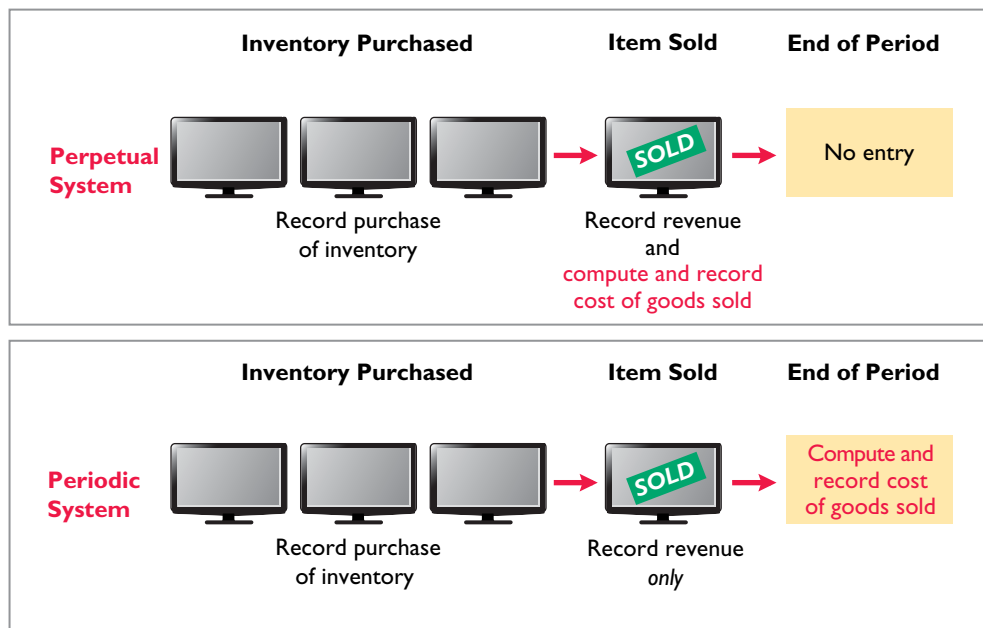


ILLUSTRATION 5.5

Comparing perpetual and periodic inventory systems

Advantages of the Perpetual System

Companies that sell merchandise with high unit values, such as automobiles, furniture, and major home appliances, have traditionally used perpetual systems. The growing use of computers and electronic scanners has enabled many more companies to install perpetual inventory systems. The perpetual inventory system is so named because the accounting records continuously—perpetually—show the quantity and cost of the inventory that should be on hand at any time.

A perpetual inventory system provides better control over inventories than a periodic system. Since the inventory records show the quantities that should be on hand, the company can count the goods at any time to see whether the amount of goods actually on hand agrees with the inventory records. If shortages are uncovered, the company can investigate immediately. Although a perpetual inventory system requires both additional clerical work and expense to maintain the subsidiary records, a computerized system can minimize this cost. Much of **Amazon.com**'s success is attributed to its sophisticated inventory system.

Some businesses find it either unnecessary or uneconomical to invest in a sophisticated, computerized perpetual inventory system such as Amazon's. Many small merchandising businesses now use basic accounting software which provides some of the essential benefits of a

perpetual inventory system. Also, managers of some small businesses still find that they can control their merchandise and manage day-to-day operations using a periodic inventory system.

Because of the widespread use of the perpetual inventory system, we illustrate it in this chapter. We discuss and illustrate the periodic system in Appendix 5B.

Investor Insight Morrow Snowboards, Inc.



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Improving Stock Appeal

Investors are often eager to invest in a company that has a hot new product. However, when snowboard-maker **Morrow Snowboards, Inc.** issued shares of stock to the public for the first time, some investors expressed reluctance to invest in Morrow because of a number of accounting control problems. To reduce investor concerns, Morrow implemented a

perpetual inventory system to improve its control over inventory. In addition, the company stated that it would perform a physical inventory count every quarter until it felt that its perpetual inventory system was reliable.

If a perpetual system keeps track of inventory on a daily basis, why do companies ever need to do a physical count? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 1 | Merchandising Operations and Inventory Systems

Indicate whether the following statements are true or false. If false, indicate how to correct the statement.

1. The primary source of revenue for a merchandising company results from performing services for customers.
2. The operating cycle of a service company is usually shorter than that of a merchandising company.
3. Sales revenue less cost of goods sold equals gross profit.
4. Ending inventory plus the cost of goods purchased equals cost of goods available for sale.

Solution

1. False. The primary source of revenue for a service company results from performing services for customers. 2. True. 3. True. 4. False. Beginning inventory plus the cost of goods purchased equals cost of goods available for sale.

Related exercise material: BE5.1, BE5.2, DO IT! 5.1, and E5.1.

ACTION PLAN

- Review merchandising concepts.
- Understand the flow of costs in a merchandising company.

Recording Purchases under a Perpetual System

LEARNING OBJECTIVE 2

Record purchases under a perpetual inventory system.

Companies purchase inventory using cash or credit (on account). They normally record purchases when they receive the goods from the seller. Every purchase should be supported by

business documents that provide written evidence of the transaction. Each cash purchase should be supported by a canceled check or a cash register receipt indicating the items purchased and amounts paid. Companies record cash purchases by an increase in Inventory and a decrease in Cash.

A **purchase invoice** should support each credit purchase. This invoice indicates the total purchase price and other relevant information. However, the purchaser does not prepare a separate purchase invoice. Instead, the purchaser uses as a purchase invoice a copy of the sales invoice sent by the seller. In **Illustration 5.6**, for example, Sauk Stereo (the buyer) uses as a purchase invoice the sales invoice prepared by PW Audio Supply (the seller) (see **Helpful Hint**).

ILLUSTRATION 5.6

Sales invoice used as purchase invoice by Sauk Stereo

HELPFUL HINT

To better understand the contents of this invoice, identify these items:

1. Seller
2. Invoice date
3. Purchaser
4. Salesperson
5. Credit terms
6. Freight terms
7. Goods sold: catalog number, description, quantity, price per unit
8. Total invoice amount

INVOICE NO. 731


PW AUDIO SUPPLY
27 CIRCLE DRIVE
HARDING, MICHIGAN 48281

SOLD TO Firm Name Sauk Stereo

Attention of James Hoover, Purchasing Agent

Address 125 Main Street

Chelsea Illinois 60915
City State Zip

Date 5/4/20	Salesperson Malone	Terms 2/10, n/30	FOB Shipping Point		
Catalog No.	Description	Quantity	Price	Amount	
X572Y9820	Printed Circuit Board-prototype	1	2,300	\$2,300	
A2547Z45	Production Model Circuits	5	300	1,500	
IMPORTANT: ALL RETURNS MUST BE MADE WITHIN 10 DAYS			TOTAL	\$3,800	

Sauk Stereo makes the following journal entry to record its purchase from PW Audio Supply on account. The entry increases (debits) Inventory and increases (credits) Accounts Payable.

May 4	Inventory	3,800	
	Accounts Payable		3,800
	(To record goods purchased on account from PW Audio Supply)		

A	=	L	+	OE
+3,800				
		+3,800		

Cash Flows
no effect

Under the perpetual inventory system, companies record purchases of merchandise for sale in the Inventory account. Thus, **REI** would increase (debit) Inventory for clothing, sporting goods, and anything else purchased for resale to customers.

Not all purchases are debited to Inventory, however. Companies record purchases of assets acquired for use and not for resale, such as supplies, equipment, and similar items, as increases to specific asset accounts rather than to Inventory. For example, to record the purchase

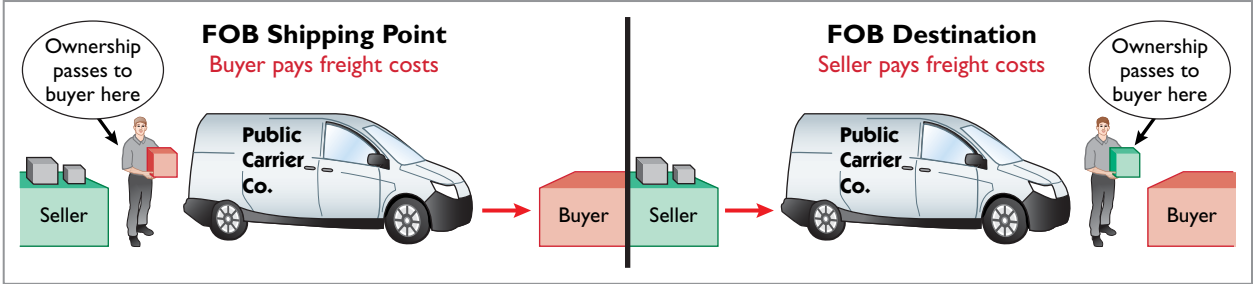
of materials used to make shelf signs or for cash register receipt paper, REI would increase (debit) Supplies.

Freight Costs

The sales agreement should indicate who—the seller or the buyer—is to pay for transporting the goods to the buyer’s place of business. When a common carrier such as a railroad, trucking company, or airline transports the goods, the carrier prepares a freight bill in accord with the sales agreement.

Freight terms are expressed as either FOB shipping point or FOB destination. The letters FOB mean **free on board**. Thus, **FOB shipping point** means that the seller places the goods free on board the carrier, and the buyer pays the freight costs. Conversely, **FOB destination** means that the seller places the goods free on board to the buyer’s place of business, and the seller pays the freight. For example, the sales invoice in Illustration 5.6 indicates FOB shipping point. Thus, the buyer (Sauk Stereo) pays the freight charges. Illustration 5.7 illustrates these shipping terms.

ILLUSTRATION 5.7 Shipping terms



A	=	L	+	OE
+150				
-150				
Cash Flows				
-150				



Thus, any freight costs incurred by the buyer are part of the cost of merchandise purchased. The reason: Inventory cost should include all costs to acquire the inventory, including freight necessary to deliver the goods to the buyer. Companies recognize these costs as cost of goods sold when inventory is sold.

Freight Costs Incurred by the Buyer

When the buyer incurs the transportation costs, these costs are considered part of the cost of purchasing inventory. Therefore, the buyer debits (increases) the Inventory account. For example, if Sauk Stereo (the buyer) pays Public Carrier Co. \$150 for freight charges on May 6, the entry on Sauk Stereo’s books is:

A	=	L	+	OE
-150				-150 Exp
Cash Flows				
-150				



May 6	Inventory	150	
	Cash		150
	(To record payment of freight on goods purchased)		

Freight Costs Incurred by the Seller

In contrast, **freight costs incurred by the seller on outgoing merchandise are an operating expense to the seller**. These costs increase an expense account titled Freight-Out (sometimes called Delivery Expense). For example, if the freight terms on the invoice in Illustration 5.6 had required PW Audio Supply (the seller) to pay the freight charges, the entry by PW Audio Supply would be:

May 4	Freight-Out (or Delivery Expense)	150	
	Cash		150
	(To record payment of freight on goods sold)		

When the seller pays the freight charges, the seller will usually establish a higher invoice price for the goods to cover the shipping expense.

Purchase Returns and Allowances

A purchaser may be dissatisfied with the merchandise received because the goods are damaged or defective, of inferior quality, or do not meet the purchaser’s specifications. In such cases, the purchaser may return the goods to the seller for credit if the sale was made on credit, or for a cash refund if the purchase was for cash. This transaction is known as a **purchase return**. Alternatively, the purchaser may choose to keep the merchandise if the seller is willing to grant an allowance (deduction) from the purchase price. This transaction is known as a **purchase allowance**.

Assume that Sauk Stereo returned goods costing \$300 to PW Audio Supply on May 8. The following entry by Sauk Stereo for the returned merchandise decreases (debits) Accounts Payable and decreases (credits) Inventory.

		A = L + OE	
		-300	
		Cash Flows	
		no effect	

May 8	Accounts Payable	300	
	Inventory		300
	(To record return of goods purchased from PW Audio Supply)		

Because Sauk Stereo increased Inventory when the goods were received, Inventory is decreased when Sauk Stereo returns the goods.

Suppose instead that Sauk Stereo chose to keep the goods after being granted a \$50 allowance (reduction in price). It would reduce (debit) Accounts Payable and reduce (credit) Inventory for \$50.

Purchase Discounts

The credit terms of a purchase on account may permit the buyer to claim a cash discount for prompt payment. The buyer calls this cash discount a **purchase discount**. This incentive offers advantages to both parties. The purchaser saves money, and the seller is able to shorten the operating cycle by converting the accounts receivable into cash.

Credit terms specify the amount of the cash discount and time period in which it is offered. They also indicate the time period in which the purchaser is expected to pay the full invoice price. In the sales invoice in Illustration 5.6, credit terms are 2/10, n/30, which is read “two-ten, net thirty” (see **Helpful Hint**). This means that the buyer may take a 2% cash discount on the invoice price, less (“net of”) any returns or allowances, if payment is made within 10 days of the invoice date (the **discount period**). Otherwise, the invoice price, less any returns or allowances, is due 30 days from the invoice date.

Alternatively, the discount period may extend to a specified number of days following the month in which the sale occurs. For example, 1/10 EOM (end of month) means that a 1% discount is available if the invoice is paid within the first 10 days of the next month.

When the seller elects not to offer a cash discount for prompt payment, credit terms will specify only the maximum time period for paying the balance due. For example, the invoice may state the time period as n/30, n/60, or n/10 EOM. This means, respectively, that the buyer must pay the net amount in 30 days, 60 days, or within the first 10 days of the next month.

When the buyer pays an invoice within the discount period, the amount of the discount decreases Inventory. Why? Because companies record inventory at cost and, by paying within the discount period, the buyer has reduced its cost. To illustrate, assume Sauk Stereo pays the balance due of \$3,500 (gross invoice price of \$3,800 less purchase returns and allowances of \$300) on May 14, the last day of the discount period. Since the terms are 2/10, n/30, the cash discount is \$70 (\$3,500 × 2%), and Sauk Stereo pays \$3,430 (\$3,500 – \$70). The entry Sauk Stereo makes to record its May 14 payment decreases (debits) Accounts Payable by

HELPFUL HINT

The term *net* in “net 30” means the remaining amount due after subtracting any sales returns and allowances and partial payments.

A	=	L	+	OE
		-3,500		
-3,430				
-70				

Cash Flows

-3,430



A	=	L	+	OE
		-3,500		
-3,500				

Cash Flows

-3,500



the amount of the gross invoice price, reduces (credits) Inventory by the \$70 discount, and reduces (credits) Cash by the net amount owed.

May 14	Accounts Payable	3,500	
	Cash		3,430
	Inventory		70
	(To record payment within discount period)		

If Sauk Stereo failed to take the discount and instead made full payment of \$3,500 on June 3, it would debit Accounts Payable and credit Cash for \$3,500 each.

June 3	Accounts Payable	3,500	
	Cash		3,500
	(To record payment with no discount taken)		

A merchandising company usually should take all available discounts. Passing up the discount may be viewed as **paying interest** for use of the money. For example, passing up the discount offered by PW Audio Supply would be comparable to Sauk Stereo paying an interest rate of 2% for the use of \$3,500 for 20 days. This is the equivalent of an annual interest rate of approximately 36.5% ($2\% \times 365/20$). Obviously, it would be better for Sauk Stereo to borrow at prevailing bank interest rates of 6% to 10% than to lose the discount.

Summary of Purchasing Transactions

The following T-account (with transaction descriptions in red) provides a summary of the effect of the previous transactions on Inventory. Sauk Stereo originally purchased \$3,800 worth of inventory on account for resale. It then returned \$300 of goods. It paid \$150 in freight charges, and finally, it received a \$70 discount off the balance owed because it paid within the discount period. This results in a balance in Inventory of \$3,580.

	Inventory				
Purchase	May 4	3,800	May 8	300	Purchase return
Freight-in	6	150	14	70	Purchase discount
Balance		3,580			

DO IT! 2 | Purchase Transactions

On September 5, De La Hoya Company buys merchandise on account from Junot Diaz Company. The purchase price of the goods paid by De La Hoya is \$1,500, and the cost to Diaz Company was \$800. On September 8, De La Hoya returns defective goods with a selling price of \$200. Record the transactions on the books of De La Hoya Company.

Solution

Sept. 5	Inventory	1,500	
	Accounts Payable		1,500
	(To record goods purchased on account)		
8	Accounts Payable	200	
	Inventory		200
	(To record return of defective goods)		

Related exercise material: **BE5.3, BE5.5, DO IT! 5.2, E5.2, E5.3, and E5.4.**

ACTION PLAN

- Purchaser records goods at cost.
- When goods are returned, purchaser reduces Inventory.

Recording Sales Under a Perpetual System

LEARNING OBJECTIVE 3

Record sales under a perpetual inventory system.

In accordance with the revenue recognition principle, companies record sales revenue when the performance obligation is satisfied. Typically, the performance obligation is satisfied when the goods transfer from the seller to the buyer. At this point, the sales transaction is complete and the sales price established.

Sales may be made on credit or for cash. A **business document** should support every sales transaction, to provide written evidence of the sale. **Cash register documents** provide evidence of cash sales. A **sales invoice**, like the one shown in Illustration 5.6, provides support for a credit sale. The original copy of the invoice goes to the customer, and the seller keeps a copy for use in recording the sale. The invoice shows the date of sale, customer name, total sales price, and other relevant information.

The seller makes two entries for each sale. **The first entry records the sale:** The seller increases (debits) Cash (or Accounts Receivable if a credit sale) and also increases (credits) Sales Revenue. **The second entry records the cost of the merchandise sold:** The seller increases (debits) Cost of Goods Sold and also decreases (credits) Inventory for the cost of those goods. As a result, the Inventory account will show at all times the amount of inventory that should be on hand.

To illustrate a credit sales transaction, PW Audio Supply records its May 4 sale of \$3,800 to Sauk Stereo (see Illustration 5.6) as follows (assume the merchandise cost PW Audio Supply \$2,400).

May 4	Accounts Receivable	3,800	
	Sales Revenue		3,800
	(To record credit sale to Sauk Stereo per invoice #731)		
4	Cost of Goods Sold	2,400	
	Inventory		2,400
	(To record cost of merchandise sold on invoice #731 to Sauk Stereo)		

A = L + OE

+3,800

+3,800 Rev

Cash Flows

no effect

A = L + OE

-2,400

-2,400 Exp

Cash Flows

no effect

For internal decision-making purposes, merchandising companies may use more than one sales account. For example, PW Audio Supply may decide to keep separate sales accounts for its sales of TVs, Blu-ray players, and headsets. **REI** might use separate accounts for camping gear, children’s clothing, and ski equipment—or it might have even more narrowly defined accounts. By using separate sales accounts for major product lines, rather than a single combined sales account, company management can more closely monitor sales trends and respond to changes in sales patterns more strategically. For example, if TV sales are increasing while Blu-ray player sales are decreasing, PW Audio Supply might reevaluate both its advertising and pricing policies on these items to ensure they are optimal.

On its income statement presented to outside investors, a merchandising company normally would provide only a single sales figure—the sum of all of its individual sales accounts. This is done for two reasons. First, providing detail on all of its individual sales accounts would add considerable length to its income statement. Second, companies do not want their competitors to know the details of their operating results. However, **Microsoft** recently expanded its disclosure of revenue from three to five types. The reason: The additional categories enabled financial statement users to better evaluate the growth of the company’s consumer and Internet businesses (see **Ethics Note**).

ETHICS NOTE

Many companies are trying to improve the quality of their financial reporting. For example, **General Electric** now provides more detail on its revenues and operating profits.

Anatomy of a Fraud¹

Holly Harmon was a cashier at a national superstore for only a short time when she began stealing merchandise using three methods. Under the first method, her husband or friends took UPC labels from cheaper items and put them on more expensive items. Holly then scanned the goods at the register. Using the second method, Holly scanned an item at the register but then voided the sale and left the merchandise in the shopping cart. A third approach was to put goods into large plastic containers. She scanned the plastic containers but not the goods within them. After Holly quit, a review of past surveillance tapes enabled the store to observe the thefts and to identify the participants.

Total take: \$12,000

The Missing Controls

Human resource controls. A background check would have revealed Holly’s previous criminal record. She would not have been hired as a cashier.

Physical controls. Software can flag high numbers of voided transactions or a high number of sales of low-priced goods. Random comparisons of video records with cash register records can ensure that the goods reported as sold on the register are the same goods that are shown being purchased on the video recording. Finally, employees should be aware that they are being monitored.

Source: Adapted from Wells, *Fraud Casebook* (2007), pp. 251–259.

At the end of “Anatomy of a Fraud” stories, which describe some recent real-world frauds, we discuss the missing control activities that would likely have prevented or uncovered the fraud.

Sales Returns and Allowances

We now look at the “flip side” of purchase returns and allowances, which the seller records as **sales returns and allowances**. These are transactions where the seller either accepts goods back from the buyer (a return) or grants a reduction in the purchase price (an allowance) so the buyer will keep the goods. PW Audio Supply’s entries to record credit for returned goods involve (1) an increase (debit) in Sales Returns and Allowances (a contra account to Sales Revenue) and a decrease (credit) in Accounts Receivable at the \$300 selling price, and (2) an increase (debit) in Inventory (assume a \$140 cost) and a decrease (credit) in Cost of Goods Sold, as shown below (assuming that the goods were not defective).

A	=	L	+	OE
				–300 Rev
–300				
Cash Flows				
no effect				

A	=	L	+	OE
				+140 Exp
+140				
Cash Flows				
no effect				

May 8	Sales Returns and Allowances	300	
	Accounts Receivable		300
	(To record credit granted to Sauk Stereo for returned goods)		
8	Inventory	140	
	Cost of Goods Sold		140
	(To record cost of goods returned)		

If Sauk Stereo returns goods because they are damaged or defective, then PW Audio Supply’s entry to Inventory and Cost of Goods Sold should be for the fair value of the returned goods, rather than their cost. For example, if the returned goods were defective and had a fair value of \$50, PW Audio Supply would debit Inventory for \$50 and credit Cost of Goods Sold for \$50.

¹The “Anatomy of a Fraud” stories in this textbook are adapted from *Fraud Casebook: Lessons from the Bad Side of Business*, edited by Joseph T. Wells (Hoboken, NJ: John Wiley & Sons, Inc., 2007). Used by permission. The names of some of the people and organizations in the stories are fictitious, but the facts in the stories are true.

What happens if the goods are not returned but the seller grants the buyer an allowance by reducing the purchase price? In this case, the seller debits Sales Returns and Allowances and credits Accounts Receivable for the amount of the allowance. An allowance has no impact on Inventory or Cost of Goods Sold.

Sales Returns and Allowances is a **contra revenue account** to Sales Revenue. This means that it is offset against a revenue account on the income statement. The normal balance of Sales Returns and Allowances is a debit. Companies use a contra account, instead of debiting Sales Revenue, to disclose in the accounts and in the income statement the amount of sales returns and allowances. Disclosure of this information is important to management. Excessive returns and allowances may suggest problems—inferior merchandise, inefficiencies in filling orders, errors in billing customers, or delivery or shipment mistakes. Moreover, a decrease (debit) recorded directly to Sales Revenue would obscure the relative importance of sales returns and allowances as a percentage of sales. It also could distort comparisons between total sales in different accounting periods.

At the end of the accounting period, if the company anticipates that sales returns and allowances will be material, the company should make an adjusting entry to estimate the amount of returns. In some industries, such as those relating to the sale of books and periodicals, returns are often material. The accounting for situations where returns must be estimated is addressed in advanced accounting courses.

Accounting Across the Organization Costco Wholesale Corp.



© Jacob Wackerhausen/iStockphoto

The Point of No Returns?

In most industries, sales returns are relatively minor. But returns of consumer electronics can really take a bite out of profits. Recently, the marketing executives at **Costco Wholesale Corp.** faced a difficult decision. Costco has always prided itself on its generous return policy. Most goods have had

an unlimited grace period for returns. However, a new policy requires that certain electronics must be returned within 90 days of their purchase. The reason? The cost of returned products such as high-definition TVs, computers, and iPods cut an estimated 8¢ per share off Costco's earnings per share, which was \$2.30.

Source: Kris Hudson, "Costco Tightens Policy on Returning Electronics," *Wall Street Journal* (February 27, 2007), p. B4.

If a company expects significant returns, what are the implications for revenue recognition? (Go to WileyPLUS for this answer and additional questions.)

Sales Discounts

As mentioned in our discussion of purchase transactions, the seller may offer the customer a cash discount—called by the seller a **sales discount**—for the prompt payment of the balance due. Like a purchase discount, a sales discount is based on the invoice price less returns and allowances, if any. The seller increases (debits) the Sales Discounts account for discounts that are taken. For example, PW Audio Supply makes the following entry to record the cash receipt on May 14 from Sauk Stereo within the discount period.

May 14	Cash	3,430	
	Sales Discounts	70	
	Accounts Receivable		3,500
	(To record collection within 2/10, n/30 discount period from Sauk Stereo)		

A	=	L	+	OE
+3,430				
				-70 Rev
-3,500				

Cash Flows

+3,430

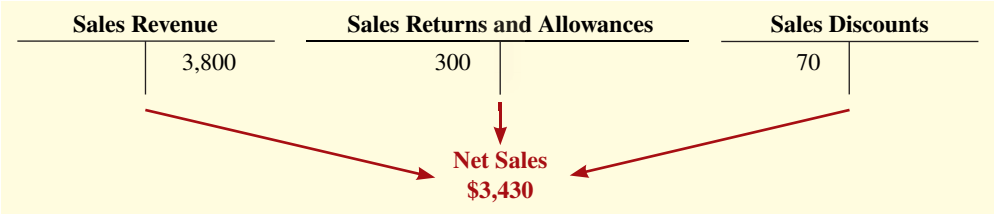


Like Sales Returns and Allowances, Sales Discounts is a **contra revenue account** to Sales Revenue. Its normal balance is a debit. PW Audio Supply uses this account, instead of debiting Sales Revenue, to disclose the amount of cash discounts taken by customers. If Sauk

Stereo does not take the discount, PW Audio Supply increases (debits) Cash for \$3,500 and decreases (credits) Accounts Receivable for the same amount at the date of collection.

At the end of the accounting period, if the amount of potential discounts is material, the company should make an adjusting entry to estimate the discounts. This would not usually be the case for sales discounts but might be necessary for other types of discounts such as volume discounts, which are addressed in more advanced accounting courses.

The following T-accounts summarize the three sales-related transactions and show their combined effect on net sales.



People, Planet, and Profit Insight PepsiCo



Helen Sessions/Alamy

Selling Green

Here is a question an executive of **PepsiCo** was asked: Should PepsiCo market green? The executive indicated that the company should, as he believes it's the No. 1 thing consumers all over the world care about. Here are some of his thoughts on this issue:

"Sun Chips are part of the food business I run. It's a 'healthy snack.' We decided that Sun Chips, if it's a healthy snack, should be made in facilities that have a net-zero footprint. In other words, I want off the electric grid everywhere we make Sun Chips.

We did that. Sun Chips should be made in a facility that puts back more water than it uses. It does that. And we partnered with our

suppliers and came out with the world's first compostable chip package.

Now, there was an issue with this package: It was louder than the New York subway, louder than jet engines taking off. What would a company that's committed to green do: walk away or stay committed? If your people are passionate, they're going to fix it for you as long as you stay committed. Six months later, the compostable bag has half the noise of our current package.

So the view today is: we should market green, we should be proud to do it . . . it has to be a 360-degree process, both internal and external. And if you do that, you can monetize environmental sustainability for the shareholders."

Source: "Four Problems—and Solutions," *Wall Street Journal* (March 7, 2011), p. R2.

What is meant by "monetize environmental sustainability" for shareholders? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 3 Sales Transactions

On September 5, De La Hoya Company buys merchandise on account from Junot Diaz Company. The selling price of the goods is \$1,500, and the cost to Diaz Company was \$800. On September 8, De La Hoya returns defective goods with a selling price of \$200 and a fair value of \$30. Record the transactions on the books of Junot Diaz Company.

Solution

Sept. 5	Accounts Receivable	1,500	
	Sales Revenue		1,500
	(To record credit sale)		
5	Cost of Goods Sold	800	
	Inventory		800
	(To record cost of goods sold on account)		

ACTION PLAN

- Seller records both the sale and the cost of goods sold at the time of the sale.
- When goods are returned, the seller records the return in a contra account, Sales Returns and Allowances, and reduces Accounts Receivable.

8	Sales Returns and Allowances	200	
	Accounts Receivable		200
	(To record credit granted for receipt of returned goods)		
8	Inventory	30	
	Cost of Goods Sold		30
	(To record fair value of goods returned)		

ACTION PLAN (cont.)

- Any goods returned increase Inventory and reduce Cost of Goods Sold. Defective or damaged inventory is recorded at fair value (scrap value).

Related exercise material: **BE5.3, BE5.4, DO IT! 5.3, E5.3, E5.4, and E5.5.**

The Accounting Cycle for a Merchandising Company

LEARNING OBJECTIVE 4

Apply the steps in the accounting cycle to a merchandising company.

Up to this point, we have illustrated the basic entries for transactions relating to purchases and sales in a perpetual inventory system. Now we consider the remaining steps in the accounting cycle for a merchandising company. Each of the required steps described in Chapter 4 for service companies apply to merchandising companies. Appendix 5A to this chapter shows use of a worksheet by a merchandiser (an optional step).

Adjusting Entries

A merchandising company generally has the same types of adjusting entries as a service company. However, a merchandiser using a perpetual system will require one additional adjustment to make the records agree with the actual inventory on hand. Here's why. At the end of each period, for control purposes, a merchandising company that uses a perpetual system will take a physical count of its goods on hand. The company's unadjusted balance in Inventory usually does not agree with the actual amount of inventory on hand. The perpetual inventory records may be incorrect due to recording errors, theft, or waste. Thus, the company needs to adjust the perpetual records to make the recorded inventory amount agree with the inventory on hand. **This involves adjusting Inventory and Cost of Goods Sold.**

For example, suppose that PW Audio Supply has an unadjusted balance of \$40,500 in Inventory. Through a physical count, PW Audio Supply determines that its actual merchandise inventory at December 31 is \$40,000. The company would make an adjusting entry as follows.

Dec. 31	Cost of Goods Sold	500	
	Inventory (\$40,500 – \$40,000)		500
	(To adjust inventory to physical count)		

A	=	L	+	OE
–500				–500 Exp

Cash Flows
no effect

Closing Entries

A merchandising company, like a service company, closes to Income Summary all accounts that affect net income. In journalizing, the company credits all temporary accounts with debit balances, and debits all temporary accounts with credit balances, as the following shows for PW Audio Supply (see **Helpful Hint**). Note that PW Audio Supply closes Cost of Goods Sold to Income Summary.

Dec. 31	Sales Revenue	480,000	
	Income Summary		480,000
	(To close income statement accounts with credit balances)		

HELPFUL HINT

The easiest way to prepare the first two closing entries is to identify the temporary accounts by their balances and then prepare one entry for the credits and one for the debits.

31	Income Summary	450,000	
	Sales Returns and Allowances		12,000
	Sales Discounts		8,000
	Cost of Goods Sold		316,000
	Salaries and Wages Expense		64,000
	Freight-Out		7,000
	Advertising Expense		16,000
	Utilities Expense		17,000
	Depreciation Expense		8,000
	Insurance Expense		2,000
	(To close income statement accounts with debit balances)		
31	Income Summary	30,000	
	Owner's Capital		30,000
	(To close net income to capital)		
31	Owner's Capital	15,000	
	Owner's Drawings		15,000
	(To close drawings to capital)		

After PW Audio Supply has posted the closing entries, all temporary accounts have zero balances. Also, Owner's Capital has a balance that is carried over to the next period.

Summary of Merchandising Entries

Illustration 5.8 summarizes the entries for the merchandising accounts using a perpetual inventory system.

ILLUSTRATION 5.8 Daily recurring and adjusting and closing entries

Transactions		Daily Recurring Entries		Dr.	Cr.
Sales Transactions	Selling merchandise to customers.	Cash or Accounts Receivable		XX	
		Sales Revenue			XX
		Cost of Goods Sold		XX	
		Inventory			XX
	Granting sales returns or allowances to customers.	Sales Returns and Allowances		XX	
Purchase Transactions		Cash or Accounts Receivable			XX
		Inventory		XX	
		Cost of Goods Sold			XX
	Paying freight costs on sales; FOB destination.	Freight-Out		XX	
		Cash			XX
	Receiving payment from customers within discount period.	Cash		XX	
		Sales Discounts		XX	
		Accounts Receivable			XX
Purchase Transactions	Purchasing merchandise for resale.	Inventory		XX	
		Cash or Accounts Payable			XX
	Paying freight costs on merchandise purchased; FOB shipping point.	Inventory		XX	
		Cash			XX
	Receiving purchase returns or allowances from suppliers.	Cash or Accounts Payable		XX	
		Inventory			XX
	Paying suppliers within discount period.	Accounts Payable		XX	
		Inventory			XX
		Cash			XX

ILLUSTRATION 5.8 (Continued)

Events	Adjusting and Closing Entries
Adjust because book amount is higher than the inventory amount determined to be on hand.	Cost of Goods Sold XX Inventory XX
Closing temporary accounts with credit balances.	Sales Revenue XX Income Summary XX
Closing temporary accounts with debit balances.	Income Summary XX Sales Returns and Allowances XX Sales Discounts XX Cost of Goods Sold XX Freight-Out XX Expenses XX

DO IT! 4 | Closing Entries

The trial balance of Celine's Sports Wear Shop at December 31 shows Inventory \$25,000, Sales Revenue \$162,400, Sales Returns and Allowances \$4,800, Sales Discounts \$3,600, Cost of Goods Sold \$110,000, Rent Revenue \$6,000, Freight-Out \$1,800, Rent Expense \$8,800, and Salaries and Wages Expense \$22,000. Prepare the closing entries for the above accounts.

Solution

The two closing entries are:

Dec. 31	Sales Revenue Rent Revenue Income Summary (To close accounts with credit balances)	162,400 6,000 168,400
31	Income Summary Cost of Goods Sold Sales Returns and Allowances Sales Discounts Freight-Out Rent Expense Salaries and Wages Expense (To close accounts with debit balances)	151,000 110,000 4,800 3,600 1,800 8,800 22,000

Related exercise material: **BE5.6, BE5.7, DO IT! 5.4, E5.6, E5.7, and E5.8.**

ACTION PLAN

- Close all temporary accounts with credit balances to Income Summary by debiting these accounts.
- Close all temporary accounts with debit balances, except drawings, to Income Summary by crediting these accounts.

Multiple-Step and Comprehensive Income Statements

LEARNING OBJECTIVE 5

Prepare a multiple-step income statement and a comprehensive income statement.

Merchandising companies widely use the classified balance sheet introduced in Chapter 4 and one of two forms for the income statement. This section explains the use of these financial statements by merchandisers.

Multiple-Step Income Statement

The **multiple-step income statement** is so named because it shows several steps in determining net income. Two of these steps relate to the company’s principal operating activities. A multiple-step statement also distinguishes between **operating** and **nonoperating activities**. Finally, the statement highlights intermediate components of income and shows subgroupings of expenses.

Income Statement Presentation of Sales

The multiple-step income statement begins by presenting **sales revenue**. It then deducts contra revenue accounts—sales returns and allowances, and sales discounts—from sales revenue to arrive at **net sales**. **Illustration 5.9** presents the sales section for PW Audio Supply, using assumed data.

ILLUSTRATION 5.9
 Computation of net sales

PW Audio Supply Income Statement (partial)			
Sales			
Sales revenue			\$480,000
Less: Sales returns and allowances	\$12,000		
Sales discounts	8,000	20,000	
Net sales			\$460,000

ALTERNATIVE
 TERMINOLOGY
 Gross profit is sometimes referred to as *gross margin*.

Gross Profit

From Illustration 5.1, you learned that companies deduct cost of goods sold from sales revenue to determine **gross profit** (see **Alternative Terminology**). For this computation, companies use **net sales** (which takes into consideration Sales Returns and Allowances and Sales Discounts) as the amount of sales revenue. On the basis of the sales data in Illustration 5.9 (net sales of \$460,000) and cost of goods sold under the perpetual inventory system (assume \$316,000), PW Audio Supply’s gross profit is \$144,000, computed as shown in **Illustration 5.10**.

ILLUSTRATION 5.10
 Computation of gross profit

Net sales	\$460,000
Cost of goods sold	316,000
Gross profit	\$144,000

We also can express a company’s gross profit as a percentage, called the **gross profit rate**. To do so, we divide the amount of gross profit by net sales. For PW Audio Supply, the **gross profit rate** is 31.3%, computed as shown in **Illustration 5.11**.

ILLUSTRATION 5.11
 Gross profit rate formula and computation

Gross Profit	÷	Net Sales	=	Gross Profit Rate
\$144,000	÷	\$460,000	=	31.3%

Analysts generally consider the gross profit **rate** to be more useful than the gross profit **amount**. The rate expresses a more meaningful (qualitative) relationship between net sales and gross profit. For example, a gross profit of \$1,000,000 may sound impressive. But if it is the result of a gross profit rate of only 7%, it is not so impressive. The gross profit rate tells how many cents of each sales dollar go to gross profit.

Gross profit represents the **merchandising profit** of a company. It is not a measure of the overall profitability because operating expenses are not yet deducted. But managers and other interested parties closely watch the amount and trend of gross profit. They compare current gross profit with amounts reported in past periods. They also compare the company’s gross profit rate with rates of competitors and with industry averages. Such comparisons provide

information about the effectiveness of a company's purchasing function and the soundness of its pricing policies.

Operating Expenses and Net Income

Operating expenses are the next component in measuring net income for a merchandising company. They are the expenses incurred in the process of earning sales revenue. These expenses are similar in merchandising and service companies. Companies sometimes segregate operating expenses into selling expenses and administrative expenses to provide additional information. Selling expenses include such items as expenses for sales salaries and advertising. Administrative expenses include such items as insurance expense and utility expense.

At PW Audio Supply, operating expenses were \$114,000. The company determines its net income by subtracting operating expenses from gross profit. Thus, net income is \$30,000, as **Illustration 5.12** shows.

Gross profit	\$144,000
Operating expenses	<u>114,000</u>
Net income	<u>\$ 30,000</u>

ILLUSTRATION 5.12

Operating expenses in computing net income

The net income amount is the so-called "bottom line" of a company's income statement.

Nonoperating Activities

Nonoperating activities consist of various revenues and expenses and gains and losses that are unrelated to the company's main line of operations. When nonoperating items are included, the label "**Income from operations**" (or "Operating income") precedes them. This label clearly identifies the results of the company's normal operations, an amount determined by subtracting cost of goods sold and operating expenses from net sales. The results of nonoperating activities are shown in the categories "**Other revenues and gains**" and "**Other expenses and losses**." **Illustration 5.13** lists examples of each.

Other Revenues and Gains

Interest revenue from notes receivable and marketable securities.

Dividend revenue from investments in common stock.

Rent revenue from subleasing a portion of the store.

Gain from the sale of property, plant, and equipment.

Other Expenses and Losses

Interest expense on notes and loans payable.

Casualty losses from recurring causes, such as vandalism and accidents.

Loss from the sale or abandonment of property, plant, and equipment.

Loss from strikes by employees and suppliers.

ILLUSTRATION 5.13

Other items of nonoperating activities

Merchandising companies report the nonoperating activities in the income statement immediately after the company's operating activities. **Illustration 5.14** shows these sections for PW Audio Supply, using assumed data.

The distinction between operating and nonoperating activities is crucial to many external users of financial data. These users view operating income as sustainable and many nonoperating activities as non-recurring (see **Ethics Note**). Therefore, when forecasting next year's income, analysts put the most weight on this year's operating income and less weight on this year's nonoperating activities.

ETHICS NOTE

Companies manage earnings in various ways. **ConAgra Foods** recorded a non-recurring gain for \$186 million from the sale of **Pilgrim's Pride** stock to help meet an earnings projection for the quarter.

ILLUSTRATION 5.14**Multiple-step income statement****Calculation of gross profit****Calculation of income from operations****Results of nonoperating activities**
PW Audio Supply
Income Statement
For the Year Ended December 31, 2020

Sales			
Sales revenue			\$480,000
Less: Sales returns and allowances	\$12,000		
Sales discounts	8,000	20,000	
Net sales			460,000
Cost of goods sold			316,000
Gross profit			144,000
Operating expenses			
Salaries and wages expense	64,000		
Utilities expense	17,000		
Advertising expense	16,000		
Depreciation expense	8,000		
Freight-out	7,000		
Insurance expense	2,000		
Total operating expenses		114,000	
Income from operations			30,000
Other revenues and gains			
Interest revenue	3,000		
Gain on disposal of plant assets	600	3,600	
Other expenses and losses			
Interest expense	1,800		
Casualty loss from vandalism	200	2,000	
Net income			\$ 31,600

Ethics Insight IBM

ImageRite/Getty Images, Inc.

Disclosing More Details

After **Enron**, increased investor criticism and regulator scrutiny forced many companies to improve the clarity of their financial disclosures. For example, **IBM** began providing more detail regarding its “Other gains and losses.” It had previously included these items in its selling, general, and administrative expenses, with little disclosure. For example, previously if IBM

sold off one of its buildings at a gain, it would include this gain in the selling, general and administrative expense line item, thus reducing that expense. This made it appear that the company had done a better job of controlling operating expenses than it actually had.

As another example, when **eBay** sold the remainder of its investment in **Skype** to **Microsoft**, it reported a gain in “Other revenues and gains” of \$1.7 billion. Since eBay’s total income from operations was \$2.4 billion, it was very important that the gain from the Skype sale not be buried in operating income.

Why have investors and analysts demanded more accuracy in isolating “Other gains and losses” from operating items? (Go to WileyPLUS for this answer and additional questions.)

Single-Step Income Statement

Another income statement format is the **single-step income statement**. The statement is so named because only one step—subtracting total expenses from total revenues—is required in determining net income.

In a single-step statement, all data are classified into two categories: (1) **revenues**, which include both operating revenues and other revenues and gains; and (2) **expenses**, which include cost of goods sold, operating expenses, and other expenses and losses. **Illustration 5.15** shows a single-step statement for PW Audio Supply.

PW Audio Supply
Income Statement
For the Year Ended December 31, 2020

Revenues

Net sales	\$460,000
Interest revenue	3,000
Gain on disposal of plant assets	600
Total revenues	463,600

Expenses

Cost of goods sold	\$316,000
Operating expenses	114,000
Interest expense	1,800
Casualty loss from vandalism	200
Total expenses	432,000

Net income	\$ 31,600
-------------------	------------------

ILLUSTRATION 5.15**Single-step income statement**

There are two primary reasons for using the single-step format. (1) A company does not realize any type of profit or income until total revenues exceed total expenses, so it makes sense to divide the statement into these two categories. (2) The format is simpler and easier to read. *For homework problems, however, you should use the single-step format only when specifically instructed to do so.*

Comprehensive Income Statement

Chapter 1 discussed the fair value principle. Accounting standards require companies to adjust the recorded values of certain types of assets and liabilities to their fair values at the end of each reporting period. In some instances, the unrealized gains or losses that result from adjusting recorded amounts to fair value are included in net income. However, in other cases, these unrealized gains and losses are not included in net income. Instead, these excluded items are reported as part of a more inclusive earnings measure, called **comprehensive income**. Examples of such items include certain adjustments to pension plan assets, gains and losses on foreign currency translation, and unrealized gains and losses on certain types of investments. Items that are excluded from net income but included in comprehensive income are either reported in a combined statement of net income and comprehensive income, or in a separate comprehensive income statement. The **comprehensive income statement** presents items that are not included in the determination of net income, referred to as other comprehensive income. **Illustration 5.16** shows how comprehensive income is presented in a separate comprehensive income statement. It assumes that PW Audio Supply had an unrealized gain of \$2,300. *Use this format when preparing your homework.*

PW Audio Supply
Comprehensive Income Statement
For the Year Ended December 31, 2020

Net income	\$31,600
Other comprehensive income	
Unrealized holding gain on investment securities	2,300
Comprehensive income	\$33,900

ILLUSTRATION 5.16**Combined statement of net income and comprehensive income**

Classified Balance Sheet

In the balance sheet, merchandising companies report inventory as a current asset immediately below accounts receivable. Recall from Chapter 4 that companies generally list current asset items in the order of their closeness to cash (liquidity). Inventory is less close to cash than accounts receivable because the goods must first be sold and then collection made from

the customer. **Illustration 5.17** presents the assets section of a classified balance sheet for PW Audio Supply (see **Helpful Hint**).

ILLUSTRATION 5.17 Assets section of a classified balance sheet

HELPFUL HINT

The \$40,000 is the cost of the inventory on hand, not its expected selling price.

PW Audio Supply Balance Sheet (Partial) December 31, 2020			
<u>Assets</u>			
Current assets			
Cash			\$ 9,500
Accounts receivable			16,100
Inventory			40,000
Prepaid insurance			<u>1,800</u>
Total current assets			67,400
Property, plant, and equipment			
Equipment	\$80,000		
Less: Accumulated depreciation—equipment	<u>24,000</u>		<u>56,000</u>
Total assets			<u>\$123,400</u>

DO IT! 5 | Multiple-Step Income Statement

The following information is available for Art Center for the year ended December 31, 2020.

Other revenues and gains	\$ 8,000	Sales revenue	\$462,000
Other expenses and losses	3,000	Operating expenses	187,000
Cost of goods sold	147,000	Sales discounts	20,000
		Other comprehensive loss	10,000

Prepare a multiple-step income statement and comprehensive income statement for Art Center.

ACTION PLAN

- Subtract cost of goods sold from net sales to determine gross profit.
- Subtract operating expenses from gross profit to determine income from operations.
- Add/subtract nonoperating items to determine net income.

Solution

Art Center Income Statement For the Year Ended December 31, 2020			
Sales			
Sales revenue			\$462,000
Sales discounts			<u>20,000</u>
Net sales			442,000
Cost of goods sold			<u>147,000</u>
Gross profit			295,000
Operating expenses			<u>187,000</u>
Income from operations			108,000
Other revenues and gains	\$8,000		
Other expenses and losses	<u>3,000</u>		<u>5,000</u>
Net income			<u>\$113,000</u>

Art Center Comprehensive Income Statement For the Year Ended December 31, 2020			
Net income			\$113,000
Other comprehensive loss			<u>10,000</u>
Comprehensive income			<u>\$103,000</u>

Worksheet for a Merchandising Company

Prepare a worksheet for a merchandising company.

As indicated in Chapter 4, a worksheet enables companies to prepare financial statements before they journalize and post adjusting entries. The steps in preparing a worksheet for a merchandising company are the same as for a service company. **Illustration 5A.1** shows the worksheet for PW Audio Supply (excluding nonoperating items). The unique accounts for a merchandiser using a **perpetual inventory system** are in red.

ILLUSTRATION 5A.1 Worksheet for merchandising company—perpetual inventory system

PW Audio Supply											
Home Insert Page Layout Formulas Data Review View											
P18 fx											
	A	B	C	D	E	F	G	H	I	J	K
1	PW Audio Supply Worksheet For the Year Ended December 31, 2020										
2											
3											
4											
5											
6		Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
7	Accounts	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
8	Cash	9,500				9,500				9,500	
9	Accounts Receivable	16,100				16,100				16,100	
10	Inventory	40,500		(a) 500		40,000				40,000	
11	Prepaid Insurance	3,800		(b) 2,000		1,800				1,800	
12	Equipment	80,000				80,000				80,000	
13	Accumulated Depreciation—Equipment		16,000		(c) 8,000		24,000				24,000
14	Accounts Payable		20,400				20,400				20,400
15	Owner's Capital		83,000				83,000				83,000
16	Owner's Drawings	15,000				15,000				15,000	
17	Sales Revenue		480,000				480,000		480,000		
18	Sales Returns and Allowances	12,000				12,000		12,000			
19	Sales Discounts	8,000				8,000		8,000			
20	Cost of Goods Sold	315,500		(a) 500		316,000		316,000			
21	Freight-Out	7,000				7,000		7,000			
22	Advertising Expense	16,000				16,000		16,000			
23	Salaries and Wages Expense	59,000		(d) 5,000		64,000		64,000			
24	Utilities Expense	17,000				17,000		17,000			
25	Totals	599,400	599,400								
26	Insurance Expense			(b) 2,000		2,000		2,000			
27	Depreciation Expense			(c) 8,000		8,000		8,000			
28	Salaries and Wages Payable				(d) 5,000		5,000				5,000
29	Totals			15,500	15,500	612,400	612,400	450,000	480,000	162,400	132,400
30	Net Income							30,000			30,000
31	Totals							480,000	480,000	162,400	162,400
32											
33	Key: (a) Adjustment to inventory on hand. (b) Insurance expired. (c) Depreciation expense. (d) Salaries and wages accrued.										

Trial Balance Columns

Data for the trial balance come from the ledger balances of PW Audio Supply at December 31. The amount shown for Inventory, \$40,500, is the year-end inventory amount from the perpetual inventory system.

Adjustments Columns

A merchandising company generally has the same types of adjustments as a service company. As you see in the worksheet, adjustments (b), (c), and (d) are for insurance, depreciation, and salaries. Pioneer Advertising, as illustrated in Chapters 3 and 4, also had these adjustments. Adjustment (a) was required to adjust the perpetual inventory carrying amount to the actual count.

After PW Audio Supply enters all adjustments data on the worksheet, it establishes the equality of the adjustments column totals. It then extends the balances in all accounts to the adjusted trial balance columns.

Adjusted Trial Balance

The adjusted trial balance shows the balance of all accounts after adjustment at the end of the accounting period.

Income Statement Columns

Next, the merchandising company transfers the accounts and balances that affect the income statement from the adjusted trial balance columns to the income statement columns. PW Audio Supply shows Sales Revenue of \$480,000 in the credit column. It shows the contra revenue accounts Sales Returns and Allowances \$12,000 and Sales Discounts \$8,000 in the debit column. The difference of \$460,000 is the net sales shown on the income statement (Illustration 5.14).

Finally, the company totals all the credits in the income statement column and compares those totals to the total of the debits in the income statement column. If the credits exceed the debits, the company has net income. PW Audio Supply has net income of \$30,000. If the debits exceed the credits, the company would report a net loss.

Balance Sheet Columns

The major difference between the balance sheets of a service company and a merchandiser is inventory. PW Audio Supply shows the ending inventory amount of \$40,000 in the balance sheet debit column. The information to prepare the owner's equity statement is also found in these columns. That is, the Owner's Capital account is \$83,000. Owner's Drawings are \$15,000. Net income results when the total of the debit column exceeds the total of the credit column in the balance sheet columns. A net loss results when the total of the credits exceeds the total of the debit balances.

Appendix 5B

Periodic Inventory System

LEARNING OBJECTIVE *7

Record purchases and sales under a periodic inventory system.

As described in this chapter, companies may use one of two basic systems of accounting for inventories: (1) the perpetual inventory system or (2) the periodic inventory system. In the chapter, we focused on the characteristics of the perpetual inventory system. In this appendix, we discuss and illustrate the **periodic inventory system**. One key difference between the two systems is the point at which the company computes cost of goods sold. For a visual reminder of this difference, refer back to Illustration 5.5.

Determining Cost of Goods Sold Under a Periodic System

Determining cost of goods sold is different when a periodic inventory system is used rather than a perpetual system. As you have seen, a company using a **perpetual system** makes an entry to record cost of goods sold and to reduce inventory each time a sale is made. A company using a **periodic system** does not determine cost of goods sold until the end of the period. At the end of the period, the company performs a count to determine the ending balance of inventory. It then **calculates cost of goods sold by subtracting ending inventory from the cost of goods available for sale**. Cost of goods available for sale is the sum of beginning inventory plus purchases, as shown in **Illustration 5B.1**.

Beginning Inventory	
+ Cost of Goods Purchased	
Cost of Goods Available for Sale	
– Ending Inventory	
Cost of Goods Sold	

ILLUSTRATION 5B.1

Basic formula for cost of goods sold using the periodic system

Another difference between the two approaches is that the perpetual system directly adjusts the Inventory account for any transaction that affects inventory (such as freight costs, returns, and discounts). The periodic system does not do this. Instead, it creates different accounts for purchases, freight costs, returns, and discounts. These various accounts are shown in **Illustration 5B.2**, which presents the calculation of cost of goods sold for PW Audio Supply using the periodic approach (see **Helpful Hint**).

PW Audio Supply Cost of Goods Sold For the Year Ended December 31, 2020

Cost of goods sold			
Inventory, January 1			\$ 36,000
Purchases		\$325,000	
Less: Purchase returns and allowances	\$10,400		
Purchase discounts	6,800	17,200	
Net purchases		307,800	
Add: Freight-in		12,200	
Cost of goods purchased			320,000
Cost of goods available for sale			356,000
Less: Inventory, December 31			40,000
Cost of goods sold			\$316,000

ILLUSTRATION 5B.2

Cost of goods sold for a merchandiser using a periodic inventory system

HELPFUL HINT

The far right column identifies the primary items that make up cost of goods sold of \$316,000. The middle column explains cost of goods purchased of \$320,000. The left column reports contra purchase items of \$17,200.

Note that the basic elements from Illustration 5B.1 are highlighted in Illustration 5B.2. You will learn more in Chapter 6 about how to determine cost of goods sold using the periodic system.

The use of the periodic inventory system does not affect the form of presentation in the balance sheet. As under the perpetual system, a company reports inventory in the current assets section.

Recording Merchandise Transactions

In a **periodic inventory system**, companies record revenues from the sale of merchandise when sales are made, just as in a perpetual system. Unlike the perpetual system, however, companies **do not attempt on the date of sale to record the cost of the merchandise sold**.

Instead, they take a physical inventory count at the **end of the period** to determine (1) the cost of the merchandise then on hand and (2) the cost of the goods sold during the period. And, **under a periodic system, companies record purchases of merchandise in the Purchases account rather than in the Inventory account.** Purchase returns and allowances, purchase discounts, and freight costs on purchases are recorded in separate accounts.

To illustrate the recording of merchandise transactions under a periodic inventory system, we will use purchase/sales transactions between PW Audio Supply and Sauk Stereo, as illustrated for the perpetual inventory system in this chapter.

Recording Purchases of Merchandise

HELPFUL HINT

Be careful not to debit purchases of equipment or supplies to a Purchases account.

On the basis of the sales invoice (Illustration 5.6) and receipt of the merchandise ordered from PW Audio Supply, Sauk Stereo records the \$3,800 purchase as follows (see **Helpful Hint**).

May 4	Purchases	3,800	
	Accounts Payable		3,800
	(To record goods purchased on account from PW Audio Supply)		

Purchases is a temporary account whose normal balance is a debit.

Freight Costs

When the purchaser directly incurs the freight costs, it debits the account Freight-In (or Transportation-In). For example, if Sauk Stereo pays Public Carrier Co. \$150 for freight charges on its purchase from PW Audio Supply on May 6, the entry on Sauk Stereo's books is as follows.

May 6	Freight-In (Transportation-In)	150	
	Cash		150
	(To record payment of freight on goods purchased)		

Like Purchases, Freight-In is a temporary account whose normal balance is a debit. **Freight-In is part of cost of goods purchased.** The reason is that cost of goods purchased should include any freight charges necessary to bring the goods to the purchaser. Freight costs are not subject to a purchase discount. Purchase discounts apply only to the invoice cost of the merchandise.

Purchase Returns and Allowances

Sauk Stereo returns goods costing \$300 to PW Audio Supply and prepares the following entry to recognize the return.

May 8	Accounts Payable	300	
	Purchase Returns and Allowances		300
	(To record return of goods purchased from PW Audio Supply)		

Purchase Returns and Allowances is a temporary account whose normal balance is a credit.

Purchase Discounts

On May 14, Sauk Stereo pays the balance due on account to PW Audio Supply, taking the 2% cash discount allowed by PW Audio Supply for payment within 10 days. Sauk Stereo records the payment and discount as follows.

May 14	Accounts Payable (\$3,800 – \$300)	3,500	
	Purchase Discounts (\$3,500 × .02)		70
	Cash		3,430
	(To record payment within the discount period)		

Purchase Discounts is a temporary account whose normal balance is a credit.

Recording Sales of Merchandise

The seller, PW Audio Supply, records the sale of \$3,800 of merchandise to Sauk Stereo on May 4 (sales invoice No. 731, Illustration 5.6) as follows.

May 4	Accounts Receivable	3,800	
	Sales Revenue		3,800
	(To record credit sales per invoice #731 to Sauk Stereo)		

Sales Returns and Allowances

To record the returned goods received from Sauk Stereo on May 8, PW Audio Supply records the \$300 sales return as follows.

May 8	Sales Returns and Allowances	300	
	Accounts Receivable		300
	(To record credit granted to Sauk Stereo for returned goods)		

Sales Discounts

On May 14, PW Audio Supply receives payment of \$3,430 on account from Sauk Stereo. PW Audio Supply honors the 2% cash discount and records the payment of Sauk Stereo's account receivable in full as follows.

May 14	Cash	3,430	
	Sales Discounts ($\$3,500 \times .02$)	70	
	Accounts Receivable ($\$3,800 - \300)		3,500
	(To record collection within 2/10, n/30 discount period from Sauk Stereo)		

Comparison of Entries—Perpetual vs. Periodic

Illustration 5B.3 summarizes the periodic inventory entries shown in this appendix and compares them to the perpetual system entries from the chapter. Entries that differ in the two systems are shown in color.

ILLUSTRATION 5B.3 Comparison of entries for perpetual and periodic inventory systems

ENTRIES ON SAUK STEREO'S BOOKS						
Transaction		Perpetual Inventory System		Periodic Inventory System		
May 4	Purchase of merchandise on credit.	Inventory	3,800	Purchases	3,800	
		Accounts Payable	3,800	Accounts Payable		3,800
6	Freight costs on purchases.	Inventory	150	Freight-In	150	
		Cash	150	Cash		150
8	Purchase returns and allowances.	Accounts Payable	300	Accounts Payable	300	
		Inventory	300	Purchase Returns and Allowances		300
14	Payment on account with a discount.	Accounts Payable	3,500	Accounts Payable	3,500	
		Cash	3,430	Cash		3,430
		Inventory	70	Purchase Discounts		70

ILLUSTRATION 5B.3 (Continued)

ENTRIES ON PW AUDIO SUPPLY'S BOOKS						
Transaction		Perpetual Inventory System			Periodic Inventory System	
May 4	Sale of merchandise on credit.	Accounts Receivable	3,800		Accounts Receivable	3,800
		Sales Revenue		3,800	Sales Revenue	3,800
		Cost of Goods Sold	2,400		No entry for cost of goods sold	
8	Return of merchandise sold.	Inventory		2,400		
		Sales Returns and Allowances	300		Sales Returns and Allowances	300
		Accounts Receivable		300	Accounts Receivable	300
14	Cash received on account with a discount.	Inventory	140		No entry	
		Cost of Goods Sold		140		
		Cash	3,430		Cash	3,430
		Sales Discounts	70		Sales Discounts	70
		Accounts Receivable		3,500	Accounts Receivable	3,500

Journalizing and Posting Closing Entries

For a merchandising company, like a service company, all accounts that affect the determination of net income are closed to Income Summary. Data for the preparation of closing entries may be obtained from the income statement columns of the worksheet. In journalizing, all debit column amounts are credited, and all credit column amounts are debited. To close the merchandise inventory in a periodic inventory system:

1. The beginning inventory balance is debited to Income Summary and credited to Inventory.
2. The ending inventory balance is debited to Inventory and credited to Income Summary.

The two entries for PW Audio Supply are as follows.

(1)				
Dec. 31	Income Summary		36,000	
	Inventory			36,000
	(To close beginning inventory)			
(2)				
31	Inventory		40,000	
	Income Summary			40,000
	(To record ending inventory)			

Illustration 5B.4 shows the Inventory and Income Summary accounts after posting.

ILLUSTRATION 5B.4

Posting closing entries for merchandise inventory

Inventory					Income Summary						
1/1	Bal.	36,000	12/31	Close	36,000	12/31	Close	36,000	12/31	Close	40,000
12/31	Close	40,000									
12/31	Bal.	40,000									

HELPFUL HINT

Except for merchandise inventory, the easiest way to prepare the first two closing entries is to identify the temporary accounts by their balances and then prepare one entry for the credits and one for the debits.

Often, the closing of inventory is included with other closing entries, as shown below for PW Audio Supply (see **Helpful Hint**). (Close inventory with other accounts in homework problems unless stated otherwise.)

Dec. 31	Inventory (Dec. 31)	40,000	
	Sales Revenue	480,000	
	Purchase Returns and Allowances	10,400	
	Purchase Discounts	6,800	
	Income Summary		537,200
	(To record ending inventory and close accounts with credit balances)		

31	Income Summary	507,200	
	Inventory (Jan. 1)		36,000
	Sales Returns and Allowances		12,000
	Sales Discounts		8,000
	Purchases	325,000	
	Freight-In	12,200	
	Salaries and Wages Expense	64,000	
	Freight-Out	7,000	
	Advertising Expense	16,000	
	Utilities Expense	17,000	
	Depreciation Expense	8,000	
	Insurance Expense	2,000	
	(To close beginning inventory and other income statement accounts with debit balances)		
31	Income Summary	30,000	
	Owner's Capital		30,000
	(To transfer net income to capital)		
31	Owner's Capital	15,000	
	Owner's Drawings		15,000
	(To close drawings to capital)		

After the closing entries are posted, all temporary accounts have zero balances. In addition, Owner's Capital has a credit balance of \$98,000: beginning balance + net income – drawings (\$83,000 + \$30,000 – \$15,000).

Using a Worksheet

As indicated in Chapter 4, a worksheet enables companies to prepare financial statements before journalizing and posting adjusting entries. The steps in preparing a worksheet for a merchandising company are the same as they are for a service company.

Trial Balance Columns

Data for the trial balance come from the ledger balances of PW Audio Supply at December 31. The amount shown for Inventory, \$36,000, is the beginning inventory amount from the periodic inventory system.

Adjustments Columns

A merchandising company generally has the same types of adjustments as a service company. As you see in the worksheet in **Illustration 5B.5**, adjustments (a), (b), and (c) are for insurance, depreciation, and salaries and wages. These adjustments were also required for Pioneer Advertising, as illustrated in Chapters 3 and 4. The unique accounts for a merchandiser using a **periodic inventory system** are shown in capital red letters. Note, however, that the worksheet excludes nonoperating items.

After all adjustment data are entered on the worksheet, the equality of the adjustment column totals is established. The balances in all accounts are then extended to the adjusted trial balance columns.

Income Statement Columns

Next, PW Audio Supply transfers the accounts and balances that affect the income statement from the adjusted trial balance columns to the income statement columns. The company shows Sales Revenue of \$480,000 in the credit column. It shows the contra revenue accounts, Sales Returns and Allowances of \$12,000 and Sales Discounts of \$8,000 in the debit column. The difference of \$460,000 is the net sales shown on the income statement (Illustration 5.9).

ILLUSTRATION 5B.5 Worksheet for merchandising company—periodic inventory system

PW Audio Supply											
P18 fx											
	A	B	C	D	E	F	G	H	I	J	K
1	PW Audio Supply Worksheet For the Year Ended December 31, 2020										
2											
3											
4											
5											
6		Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
7	Accounts	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
8	Cash	9,500				9,500				9,500	
9	Accounts Receivable	16,100				16,100				16,100	
10	INVENTORY	36,000				36,000		36,000	40,000	40,000	
11	Prepaid Insurance	3,800			(a) 2,000	1,800				1,800	
12	Equipment	80,000				80,000				80,000	
13	Accumulated Depreciation—Equipment		16,000		(b) 8,000		24,000				24,000
14	Accounts Payable		20,400				20,400				20,400
15	Owner's Capital		83,000				83,000				83,000
16	Owner's Drawings	15,000				15,000				15,000	
17	SALES REVENUE		480,000				480,000		480,000		
18	SALES RETURNS AND ALLOWANCES	12,000				12,000		12,000			
19	SALES DISCOUNTS	8,000				8,000		8,000			
20	PURCHASES	325,000				325,000		325,000			
21	PURCHASE RETURNS AND ALLOWANCES		10,400				10,400		10,400		
22	PURCHASE DISCOUNTS		6,800				6,800		6,800		
23	FREIGHT-IN	12,200				12,200		12,200			
24	Freight-Out	7,000				7,000		7,000			
25	Advertising Expense	16,000				16,000		16,000			
26	Salaries and Wages Expense	59,000		(c) 5,000		64,000		64,000			
27	Utilities Expense	17,000				17,000		17,000			
28	Totals	616,600	616,600								
29	Insurance Expense			(a) 2,000		2,000		2,000			
30	Depreciation Expense			(b) 8,000		8,000		8,000			
31	Salaries and Wages Payable				(c) 5,000		5,000				5,000
32	Totals			15,000	15,000	629,600	629,600	507,200	537,200	162,400	132,400
33	Net Income							30,000			30,000
34	Totals							537,200	537,200	162,400	162,400
35											
36											
	Key: (a) Insurance expired. (b) Depreciation expense. (c) Salaries and wages accrued.										

Similarly, Purchases of \$325,000 and Freight-In of \$12,200 are extended to the debit column. The contra purchase accounts, Purchase Returns and Allowances of \$10,400 and Purchase Discounts of \$6,800, are extended to the credit columns.

The worksheet procedures for the Inventory account merit specific comment. The procedures are:

1. The beginning balance, \$36,000, is extended from the adjusted trial balance column to the **income statement debit column**. From there, it can be added in reporting cost of goods available for sale in the income statement.
2. The ending inventory, \$40,000, is added to the worksheet by an **income statement credit and a balance sheet debit**. The credit makes it possible to deduct ending inventory from

the cost of goods available for sale in the income statement to determine cost of goods sold. The debit means the ending inventory can be reported as an asset on the balance sheet.

Illustration 5B.6 shows these two procedures.

	Income Statement		Balance Sheet	
	Dr.	Cr.	Dr.	Cr.
Inventory	(1) 36,000	40,000	(2) 40,000	

ILLUSTRATION 5B.6

Worksheet procedures for inventories

The computation for cost of goods sold, taken from the income statement column in Illustration 5B.5, is as shown in **Illustration on 5B.7** (see **Helpful Hint**).

Debit Column		Credit Column	
Beginning inventory	\$ 36,000	Ending inventory	\$40,000
Purchases	325,000	Purchase returns and allowances	10,400
Freight-in	12,200	Purchase discounts	6,800
Total debits	373,200	Total credits	\$57,200
Less: Total credits	57,200		
Cost of goods sold	\$316,000		

ILLUSTRATION 5B.7

Computation of cost of goods sold from worksheet columns

HELPFUL HINT

In a periodic system, cost of goods sold is a computation—it is not a separate account with a balance.

Finally, PW Audio Supply totals all the credits in the income statement column and compares these totals to the total of the debits in the income statement column. If the credits exceed the debits, the company has net income. PW Audio Supply has net income of \$30,000. If the debits exceed the credits, the company would report a net loss.

Balance Sheet Columns

The major difference between the balance sheets of a service company and a merchandising company is inventory. PW Audio Supply shows ending inventory of \$40,000 in the balance sheet debit column. The information to prepare the owner's equity statement is also found in these columns. That is, the Owner's Capital account is \$83,000. Owner's Drawings are \$15,000. Net income results when the total of the debit column exceeds the total of the credit column in the balance sheet columns. A net loss results when the total of the credits exceeds the total of the debit balances.

Review and Practice

Learning Objectives Review

1 Describe merchandising operations and inventory systems.

Because of inventory, a merchandising company has sales revenue, cost of goods sold, and gross profit. To account for inventory, a merchandising company must choose between a perpetual and a periodic inventory system.

2 Record purchases under a perpetual inventory system.

The company debits the Inventory account for all purchases of merchandise and freight-in, and credits it for purchase discounts and purchase returns and allowances.

3 Record sales under a perpetual inventory system.

When a merchandising company sells inventory, it debits Accounts Receivable (or Cash) and credits Sales Revenue for the **selling price** of the merchandise. At the same time, it debits Cost of Goods Sold and credits Inventory for the **cost** of the inventory items sold. Sales Returns and Allowances and Sales Discounts are debited and are contra revenue accounts.

4 Apply the steps in the accounting cycle to a merchandising company.

Each of the required steps in the accounting cycle for a service company applies to a merchandising company. A worksheet is again an optional step. Under a perpetual inventory system, the company must adjust the Inventory account to agree with the physical count.

5 Prepare a multiple-step income statement and a comprehensive income statement.

A multiple-step income statement shows numerous steps in determining net income, including nonoperating activities sections. A

single-step income statement classifies all data under two categories, revenues or expenses, and determines net income in one step. A comprehensive income statement adds or subtracts any items of other comprehensive income to net income to arrive at comprehensive income.

***6 Prepare a worksheet for a merchandising company.**

The steps in preparing a worksheet for a merchandising company are the same as for a service company. The unique accounts for a merchandiser are Inventory, Sales Revenue, Sales Returns and Allowances, Sales Discounts, and Cost of Goods Sold.

***7 Record purchases and sales under a periodic inventory system.**

In recording purchases under a periodic system, companies must make entries for (a) cash and credit purchases, (b) purchase returns and allowances, (c) purchase discounts, and (d) freight costs. In recording sales, companies must make entries for (a) cash and credit sales, (b) sales returns and allowances, and (c) sales discounts.

Glossary Review

Comprehensive income An income measure that includes gains and losses that are excluded from the determination of net income. (p. 5-21).

Comprehensive income statement A statement that presents items that are not included in the determination of net income, referred to as other comprehensive income. (p. 5-21).

Contra revenue account An account that is offset against a revenue account on the income statement. (p. 5-13).

Cost of goods sold The total cost of merchandise sold during the period. (p. 5-3).

FOB destination Freight terms indicating that the seller places the goods free on board to the buyer's place of business, and the seller pays the freight. (p. 5-8).

FOB shipping point Freight terms indicating that the seller places goods free on board the carrier, and the buyer pays the freight costs. (p. 5-8).

Gross profit The excess of net sales over the cost of goods sold. (p. 5-18).

Gross profit rate Gross profit expressed as a percentage, by dividing the amount of gross profit by net sales. (p. 5-18).

Income from operations Income from a company's principal operating activity; determined by subtracting cost of goods sold and operating expenses from net sales. (p. 5-19).

Multiple-step income statement An income statement that shows several steps in determining net income. (p. 5-18).

Net sales Sales revenue less sales returns and allowances and less sales discounts. (p. 5-18).

Nonoperating activities Various revenues, expenses, gains, and losses that are unrelated to a company's main line of operations. (p. 5-19).

Operating expenses Expenses incurred in the process of earning sales revenue. (p. 5-19).

Other expenses and losses A nonoperating-activities section of the income statement that shows expenses and losses unrelated to the company's main line of operations. (p. 5-19).

Other revenues and gains A nonoperating-activities section of the income statement that shows revenues and gains unrelated to the company's main line of operations. (p. 5-19).

Periodic inventory system An inventory system under which the company does not keep detailed inventory records throughout the accounting period but determines the cost of goods sold only at the end of an accounting period. (p. 5-5).

Perpetual inventory system An inventory system under which the company keeps detailed records of the cost of each inventory purchase and sale, and the records continuously show the inventory that should be on hand. (p. 5-4).

Purchase allowance A deduction made to the selling price of merchandise, granted by the seller so that the buyer will keep the merchandise. (p. 5-9).

Purchase discount A cash discount claimed by a buyer for prompt payment of a balance due. (p. 5-9).

Purchase invoice A document that supports each credit purchase. (p. 5-7).

Purchase return A return of goods from the buyer to the seller for a cash or credit refund. (p. 5-9).

Sales discount A reduction given by a seller for prompt payment of a credit sale. (p. 5-13).

Sales invoice A document that supports each credit sale. (p. 5-11).

Sales returns and allowances Purchase returns and allowances from the seller's perspective. See *Purchase return* and *Purchase allowance*, above. (p. 5-12).

Sales revenue (Sales) The primary source of revenue in a merchandising company. (p. 5-3).

Single-step income statement An income statement that shows only one step in determining net income. (p. 5-20).

Practice Multiple-Choice Questions

1. (LO 1) Gross profit will result if:
 - a. operating expenses are less than net income.
 - b. net sales are greater than operating expenses.
 - c. net sales are greater than cost of goods sold.
 - d. operating expenses are greater than cost of goods sold.
2. (LO 2) Under a perpetual inventory system, when goods are purchased for resale by a company:
 - a. purchases on account are debited to Inventory.
 - b. purchases on account are debited to Purchases.
 - c. purchase returns are debited to Purchase Returns and Allowances.
 - d. freight costs are debited to Freight-Out.
3. (LO 3) The sales accounts that normally have a debit balance are:
 - a. Sales Discounts.
 - b. Sales Returns and Allowances.
 - c. Both (a) and (b).
 - d. Neither (a) nor (b).
4. (LO 3) A credit sale of \$750 is made on June 13, terms 2/10, net/30. A return of \$50 is granted on June 16. The amount received as payment in full on June 23 is:
 - a. \$700.
 - b. \$686.
 - c. \$685.
 - d. \$650.
5. (LO 2) Which of the following accounts will normally appear in the ledger of a merchandising company that uses a perpetual inventory system?
 - a. Purchases.
 - b. Freight-In.
 - c. Cost of Goods Sold.
 - d. Purchase Discounts.
6. (LO 3) To record the sale of goods for cash in a perpetual inventory system:
 - a. only one journal entry is necessary to record cost of goods sold and reduction of inventory.
 - b. only one journal entry is necessary to record the receipt of cash and the sales revenue.
 - c. two journal entries are necessary: one to record the receipt of cash and sales revenue, and one to record the cost of goods sold and reduction of inventory.
 - d. two journal entries are necessary: one to record the receipt of cash and reduction of inventory, and one to record the cost of goods sold and sales revenue.
7. (LO 4) The steps in the accounting cycle for a merchandising company are the same as those in a service company **except**:
 - a. an additional adjusting journal entry for inventory may be needed in a merchandising company.
 - b. closing journal entries are not required for a merchandising company.
 - c. a post-closing trial balance is not required for a merchandising company.
 - d. a multiple-step income statement is required for a merchandising company.
8. (LO 5) The multiple-step income statement for a merchandising company shows each of the following features **except**:
 - a. gross profit.
 - b. cost of goods sold.
 - c. a sales section.
 - d. an investing activities section.
9. (LO 5) If net sales are \$400,000, cost of goods sold is \$310,000, and operating expenses are \$60,000, the gross profit is:
 - a. \$30,000.
 - b. \$90,000.
 - c. \$340,000.
 - d. \$400,000.
10. (LO 5) A single-step income statement:
 - a. reports gross profit.
 - b. does not report cost of goods sold.
 - c. reports sales revenue and "Other revenues and gains" in the revenues section of the income statement.
 - d. reports operating income separately.
11. (LO 5) Which of the following appears on both a single-step and a multiple-step income statement?
 - a. Inventory.
 - b. Gross profit.
 - c. Income from operations.
 - d. Cost of goods sold.
- *12. (LO 6) In a worksheet using a perpetual inventory system, Inventory is shown in the following columns:
 - a. adjusted trial balance debit and balance sheet debit.
 - b. income statement debit and balance sheet debit.
 - c. income statement credit and balance sheet debit.
 - d. income statement credit and adjusted trial balance debit.
- *13. (LO 7) In determining cost of goods sold in a periodic system:
 - a. purchase discounts are deducted from net purchases.
 - b. freight-out is added to net purchases.
 - c. purchase returns and allowances are deducted from net purchases.
 - d. freight-in is added to net purchases.
- *14. (LO 7) If beginning inventory is \$60,000, cost of goods purchased is \$380,000, and ending inventory is \$50,000, cost of goods sold is:
 - a. \$390,000.
 - b. \$370,000.
 - c. \$330,000.
 - d. \$420,000.
- *15. (LO 7) When goods are purchased for resale by a company using a periodic inventory system:
 - a. purchases on account are debited to Inventory.
 - b. purchases on account are debited to Purchases.
 - c. purchase returns are debited to Purchase Returns and Allowances.
 - d. freight costs are debited to Purchases.

Solutions

1. **c.** Gross profit will result if net sales are greater than cost of goods sold. The other choices are incorrect because (a) operating expenses and net income are not used in the computation of gross profit; (b) gross profit results when net sales are greater than cost of goods sold, not operating expenses; and (d) gross profit results when net sales, not operating expenses, are greater than cost of goods sold.
2. **a.** Under a perpetual inventory system, when a company purchases goods for resale, purchases on account are debited to the Inventory account, not (b) Purchases or (c) Purchase Returns and Allowances. Choice (d) is incorrect because freight costs are also debited to the Inventory account, not the Freight-Out account.
3. **c.** Both Sales Discounts and Sales Returns and Allowances normally have a debit balance. Choices (a) and (b) are both correct, but (c) is the better answer. Choice (d) is incorrect as both (a) and (b) are correct.
4. **b.** The full amount of \$686 is paid within 10 days of the purchase $(\$750 - \$50) - [(\$750 - \$50) \times 2\%]$. The other choices are incorrect because (a) does not consider the discount of \$14; (c) the amount of the discount is based upon the amount after the return is granted $(\$700 \times 2\%)$, not the amount before the return of merchandise $(\$750 \times 2\%)$; and (d) does not constitute payment in full on June 23.
5. **c.** The Cost of Goods Sold account normally appears in the ledger of a merchandising company using a perpetual inventory system. The other choices are incorrect because (a) the Purchases account, (b) the Freight-In account, and (d) the Purchase Discounts account all appear in the ledger of a merchandising company that uses a periodic inventory system.
6. **c.** Two journal entries are necessary: one to record the receipt of cash and sales revenue, and one to record the cost of goods sold and reduction of inventory. The other choices are incorrect because (a) only considers the recognition of the expense and ignores the revenue, (b) only considers the recognition of revenue and leaves out the expense or cost of merchandise sold, and (d) the receipt of cash and sales revenue, not reduction of inventory, are paired together, and the cost of goods sold and reduction of inventory, not sales revenue, are paired together.
7. **a.** An additional adjusting journal entry for inventory may be needed in a merchandising company to adjust for a physical inventory count, but it is not needed for a service company. The other choices are incorrect because (b) closing journal entries and (c) a post-closing trial balance are required for both types of companies. Choice (d) is incorrect because while a multiple-step income statement is not required for a merchandising company, it is useful to distinguish income generated from operating the business versus income or loss from non-recurring, nonoperating items.
8. **d.** An investing activities section appears on the statement of cash flows, not on a multiple-step income statement. Choices (a) gross profit, (b) cost of goods sold, and (c) a sales section are all features of a multiple-step income statement.
9. **b.** Gross profit = Net sales (\$400,000) – Cost of goods sold (\$310,000) = \$90,000, not (a) \$30,000, (c) \$340,000, or (d) \$400,000.
10. **c.** Both sales revenue and “Other revenues and gains” are reported in the revenues section of a single-step income statement. The other choices are incorrect because (a) gross profit is not reported on a single-step income statement, (b) cost of goods sold is included in the expenses section of a single-step income statement, and (d) income from operations is not shown separately on a single-step income statement.
11. **d.** Cost of goods sold appears on both a single-step and a multiple-step income statement. The other choices are incorrect because (a) inventory does not appear on either a single-step or a multiple-step income statement and (b) gross profit and (c) income from operations appear on a multiple-step income statement but not on a single-step income statement.
12. **a.** In a worksheet using a perpetual inventory system, inventory is shown in the adjusted trial balance debit column and in the balance sheet debit column. The other choices are incorrect because the Inventory account is not shown in the income statement columns.
- *13. **d.** In determining cost of goods sold in a periodic system, freight-in is added to net purchases. The other choices are incorrect because (a) purchase discounts are deducted from purchases, not net purchases; (b) freight-out is a cost of sales, not a cost of purchases; and (c) purchase returns and allowances are deducted from purchases, not net purchases.
- *14. **a.** Beginning inventory (\$60,000) + Cost of goods purchased (\$380,000) – Ending inventory (\$50,000) = Cost of goods sold (\$390,000), not (b) \$370,000, (c) \$330,000, or (d) \$420,000.
- *15. **b.** Purchases for resale are debited to the Purchases account. The other choices are incorrect because (a) purchases on account are debited to Purchases, not Inventory; (c) Purchase Returns and Allowances are always credited; and (d) freight costs are debited to Freight-In, not Purchases.

Practice Brief Exercises

Compute the missing amounts in determining cost of goods sold.

1. (LO 1) Presented below are the components in determining cost of goods sold for (a) Frazier Company, (b) Todd Company, and (c) Abreu Enterprises. Determine the missing amounts.

	Beginning Inventory	Purchases	Cost of Goods Available for Sale	Ending Inventory	Cost of Goods Sold
a.	\$120,000	\$150,000	?	?	\$160,000
b.	\$ 50,000	?	\$125,000	\$45,000	?
c.	?	\$220,000	\$330,000	\$61,000	?

Solution

1. **a.** Cost of goods available for sale = \$120,000 + \$150,000 = \$270,000
Ending inventory = \$270,000 – \$160,000 = \$110,000

- b. $\text{Purchases} = \$125,000 - \$50,000 = \$75,000$
 $\text{Cost of goods sold} = \$125,000 - \$45,000 = \$80,000$
- c. $\text{Beginning inventory} = \$330,000 - \$220,000 = \$110,000$
 $\text{Cost of goods sold} = \$330,000 - \$61,000 = \$269,000$

2. (LO 2) Prepare the journal entries to record the following transactions on Robertson Company's books using a perpetual inventory system.

Journalize purchase transactions.

- a. On March 2, Melky Company sold \$800,000 of merchandise to Robertson Company, terms 2/10, n/30.
- b. On March 6, Robertson Company returned \$100,000 of the merchandise purchased on March 2.
- c. On March 12, Robertson Company paid the balance due to Melky Company.

Solution

2. a. Inventory	800,000	
Accounts Payable		800,000
b. Accounts Payable	100,000	
Inventory		100,000
c. Accounts Payable (\$800,000 – \$100,000)	700,000	
Inventory (\$700,000 × 2%)		14,000
Cash (\$700,000 – \$14,000)		686,000

3. (LO 4) Cabrera Company has the following account balances: Sales Revenue \$300,000, Sales Returns and Allowances \$10,000, Cost of Goods Sold \$174,000, and Inventory \$50,000. Prepare the entries to record the closing of these items to Income Summary.

Prepare closing entries.

Solution

3. Sales Revenue	300,000	
Income Summary		300,000
Income Summary	184,000	
Cost of Goods Sold		174,000
Sales Returns and Allowances		10,000

4. (LO 5) Assume Yoan Company has the following reported amounts: Sales revenue \$400,000, Sales discounts \$10,000, Cost of goods sold \$234,000, and Operating expenses \$60,000. Compute the following: (a) net sales, (b) gross profit, (c) income from operations, and (d) gross profit rate. (Round to one decimal place.)

Compute net sales, gross profit, income from operations, and gross profit rate.

Solution

- 4. a.** $\text{Net sales} = \$400,000 - \$10,000 = \$390,000$
- b.** $\text{Gross profit} = \$390,000 - \$234,000 = \$156,000$
- c.** $\text{Income from operations} = \$156,000 - \$60,000 = \$96,000$
- d.** $\text{Gross profit rate} = \$156,000 \div \$390,000 = 40\%$

Practice Exercises

1. (LO 2, 3) On June 10, Spinner Company purchased \$10,000 of merchandise on account from Lawrence Company, FOB shipping point, terms 2/10, n/30. Spinner pays the freight costs of \$600 on June 11. Damaged goods totaling \$700 are returned to Lawrence for credit on June 12. The fair value of these goods is \$300. On June 19, Spinner pays Lawrence in full, less the purchase discount. Both companies use a perpetual inventory system.

Prepare purchase and sales entries.

Instructions

- Prepare separate entries for each transaction on the books of Spinner Company.
- Prepare separate entries for each transaction for Lawrence Company. The merchandise purchased by Spinner on June 10 had cost Lawrence \$6,400.

Solution**1. a.**

June 10	Inventory	10,000	
	Accounts Payable		10,000
11	Inventory	600	
	Cash		600
12	Accounts Payable	700	
	Inventory		700
19	Accounts Payable (\$10,000 – \$700)	9,300	
	Inventory (\$9,300 × 2%)		186
	Cash (\$9,300 – \$186)		9,114

b.

June 10	Accounts Receivable	10,000	
	Sales Revenue		10,000
	Cost of Goods Sold	6,400	
	Inventory		6,400
12	Sales Returns and Allowances	700	
	Accounts Receivable		700
	Inventory	300	
	Cost of Goods Sold		300
19	Cash (\$9,300 – \$186)	9,114	
	Sales Discounts (\$9,300 × 2%)	186	
	Accounts Receivable (\$10,000 – \$700)		9,300

Prepare multiple-step and single-step income statements.

- 2. (LO 5) Financial Statement** In its income statement for the year ended December 31, 2020, Sale Company reported the following condensed data.

Interest expense	\$ 50,000	Net sales	\$1,650,000
Operating expenses	590,000	Interest revenue	20,000
Cost of goods sold	902,000	Loss on disposal of equipment	7,000

Instructions

- Prepare a multiple-step income statement.
- Prepare a single-step income statement.

Solution**2. a.**

Sale Company Income Statement For the Year Ended December 31, 2020			
Net sales			\$1,650,000
Cost of goods sold			<u>902,000</u>
Gross profit			748,000
Operating expenses			<u>590,000</u>
Income from operations			158,000
Other revenues and gains			
Interest revenue		\$20,000	
Other expenses and losses			
Interest expense	\$50,000		
Loss on disposal of equipment	<u>7,000</u>	<u>57,000</u>	<u>(37,000)</u>
Net income			<u>\$ 121,000</u>

b.

Sale Company Income Statement For the Year Ended December 31, 2020		
Revenues		
Net sales	\$1,650,000	
Interest revenue	<u>20,000</u>	
Total revenues		\$1,670,000
Expenses		
Cost of goods sold	902,000	
Operating expenses	590,000	
Interest expenses	50,000	
Loss on sale of equipment	<u>7,000</u>	
Total expenses		<u>1,549,000</u>
Net income		<u>\$ 121,000</u>

Practice Problem

(LO 5) Financial Statement The adjusted trial balance columns of Falcetto Company's worksheet for the year ended December 31, 2020, are as follows.

Prepare a multiple-step income statement.

	Debit		Credit
Cash	14,500	Accumulated Depreciation—	
Accounts Receivable	11,100	Equipment	18,000
Inventory	29,000	Notes Payable	25,000
Prepaid Insurance	2,500	Accounts Payable	10,600
Equipment	95,000	Owner's Capital	81,000
Owner's Drawings	12,000	Sales Revenue	536,800
Sales Returns and Allowances	6,700	Interest Revenue	<u>2,500</u>
Sales Discounts	5,000		<u>673,900</u>
Cost of Goods Sold	363,400		
Freight-Out	7,600		
Advertising Expense	12,000		
Salaries and Wages Expense	56,000		
Utilities Expense	18,000		
Rent Expense	24,000		
Depreciation Expense	9,000		
Insurance Expense	4,500		
Interest Expense	<u>3,600</u>		
	<u>673,900</u>		

Instructions

Prepare a multiple-step income statement for Falcetto Company.

Solution

Falcetto Company Income Statement For the Year Ended December 31, 2020		
Sales		
Sales revenue		\$536,800
Less: Sales returns and allowances	\$ 6,700	
Sales discounts	<u>5,000</u>	<u>11,700</u>

Net sales		525,100
Cost of goods sold		<u>363,400</u>
Gross profit		161,700
Operating expenses		
Salaries and wages expense	56,000	
Rent expense	24,000	
Utilities expense	18,000	
Advertising expense	12,000	
Depreciation expense	9,000	
Freight-out	7,600	
Insurance expense	<u>4,500</u>	
Total operating expenses		<u>131,100</u>
Income from operations		30,600
Other revenues and gains		
Interest revenue	2,500	
Other expenses and losses		
Interest expense	<u>3,600</u>	<u>1,100</u>
Net income		<u><u>\$ 29,500</u></u>

WileyPLUS

Brief Exercises, DO IT! Exercises, Exercises, Problems, and many additional resources are available for practice in WileyPLUS.

Note: All asterisked Questions, Exercises, and Problems relate to material in the appendices to the chapter.

Questions

- (a) "The steps in the accounting cycle for a merchandising company are different from the accounting cycle for a service company." Do you agree or disagree? (b) Is the measurement of net income for a merchandising company conceptually the same as for a service company? Explain.
- Why is the normal operating cycle for a merchandising company likely to be longer than for a service company?
- What components of revenues and expenses are different between merchandising and service companies?
- How does income measurement differ between a merchandising and a service company?
- When is cost of goods sold determined in a perpetual inventory system?
- Distinguish between FOB shipping point and FOB destination. Identify the freight terms that will result in a debit to Inventory by the buyer and a debit to Freight-Out by the seller.
- Explain the meaning of the credit terms 2/10, n/30.
- Goods costing \$2,000 are purchased on account on July 15 with credit terms of 2/10, n/30. On July 18, a \$200 credit memo is received from the supplier for damaged goods. Give the journal entry on July 24 to record payment of the balance due within the discount period using a perpetual inventory system.
- Celina Harris believes revenues from credit sales may be recorded before they are collected in cash. Do you agree? Explain.
- (a) What is the primary source document for recording (1) cash sales and (2) credit sales? (b) Using XXs for amounts, give the journal entry for each of the transactions in part (a).
- A credit sale is made on July 10 for \$900, terms 2/10, n/30. On July 12, \$100 of goods are returned for credit. Give the journal entry on July 19 to record the receipt of the balance due within the discount period.
- Explain why the Inventory account will usually require adjustment at year-end.
- Prepare the closing entries for the Sales Revenue account, assuming a balance of \$200,000 and the Cost of Goods Sold account with a \$145,000 balance.
- What merchandising account(s) will appear in the post-closing trial balance?
- Cupery Co. has net sales of \$105,000, cost of goods sold of \$70,000, and operating expenses of \$20,000. What is its gross profit and its gross profit rate?

16. Stefan Page Company reports net sales of \$800,000, gross profit of \$370,000, and net income of \$240,000. What are its operating expenses?
17. Identify the distinguishing features of an income statement for a merchandising company.
18. Identify the sections of a multiple-step income statement that relate to (a) operating activities, and (b) nonoperating activities.
19. How does the single-step form of income statement differ from the multiple-step form?
20. Determine **Apple's** gross profit rate for 2015 and 2014. Indicate whether it increased or decreased from 2014 to 2015.
- *21. Indicate the columns of the worksheet in a perpetual system in which (a) inventory and (b) cost of goods sold will be shown.
- *22. Identify the accounts that are added to or deducted from Purchases in a periodic system to determine the cost of goods purchased. For each account, indicate whether it is added or deducted.
- *23. Goods costing \$3,000 are purchased on account on July 15 with credit terms of 2/10, n/30. On July 18, a \$200 credit was received from the supplier for damaged goods. Give the journal entry on July 24 to record payment of the balance due within the discount period, assuming a periodic inventory system.

Brief Exercises

BE5.1 (LO 1) Presented below are the components in determining cost of goods sold. Determine the missing amounts.

Compute missing amounts in determining cost of goods sold.

	<u>Beginning Inventory</u>	<u>Purchases</u>	<u>Cost of Goods Available for Sale</u>	<u>Ending Inventory</u>	<u>Cost of Goods Sold</u>
a.	\$80,000	\$100,000	?	?	\$120,000
b.	\$50,000	?	\$115,000	\$35,000	?
c.	?	\$110,000	\$160,000	\$29,000	?

BE5.2 (LO 1) Presented below are the components in Veasy Company's income statement. Determine the missing amounts.

Compute missing amounts in determining net income.

	<u>Sales Revenue</u>	<u>Cost of Goods Sold</u>	<u>Gross Profit</u>	<u>Operating Expenses</u>	<u>Net Income</u>
a.	\$ 75,000	?	\$28,000	?	\$ 9,800
b.	\$108,000	\$70,000	?	?	\$29,500
c.	?	\$83,900	\$79,600	\$39,500	?

BE5.3 (LO 2, 3) Cha Company buys merchandise on account from Wirtz Company. The selling price of the goods is \$780, and the cost of the goods is \$470. Both companies use perpetual inventory systems. Journalize the transaction on the books of both companies.

Journalize perpetual inventory entries.

BE5.4 (LO 3) Prepare the journal entries to record the following transactions on Novy Company's books using a perpetual inventory system.

Journalize sales transactions.

- On March 2, Novy Company sold \$900,000 of merchandise on account to Opps Company, terms 2/10, n/30. The cost of the merchandise sold was \$590,000.
- On March 6, Opps Company returned \$90,000 of the merchandise purchased on March 2. The cost of the returned merchandise was \$62,000.
- On March 12, Novy Company received the balance due from Opps Company.

BE5.5 (LO 2) From the information in BE5.4, prepare the journal entries to record these transactions on Opps Company's books under a perpetual inventory system.

Journalize purchase transactions.

BE5.6 (LO 4) At year-end, the perpetual inventory records of Gutierrez Company showed merchandise inventory of \$98,000. The company determined, however, that its actual inventory on hand was \$96,100. Record the necessary adjusting entry.

Prepare adjusting entry for inventory.

BE5.7 (LO 4) Brueser Company has the following account balances: Sales Revenue \$195,000, Sales Discounts \$2,000, Cost of Goods Sold \$117,000, and Inventory \$40,000. Prepare the entries to record the closing of these items to Income Summary.

Prepare closing entries for accounts.

BE5.8 (LO 5) Financial Statement Nelson Company provides the following information for the month ended October 31, 2020: sales on credit \$280,000, cash sales \$95,000, sales discounts \$5,000, and sales returns and allowances \$11,000. Prepare the sales section of the income statement based on this information.

Prepare sales section of income statement.

Contrast presentation in multiple-step and single-step income statements.

Compute net sales, gross profit, income from operations, and gross profit rate.

Identify worksheet columns for selected accounts.

BE5.9 (LO 5) Writing Financial Statement Explain where each of the following items would appear on (1) a multiple-step income statement, and on (2) a single-step income statement: (a) gain on sale of equipment, (b) interest expense, (c) casualty loss from vandalism, and (d) cost of goods sold.

BE5.10 (LO 5) Assume Kupfer Company has the following reported amounts: Sales revenue \$510,000, Sales returns and allowances \$15,000, Cost of goods sold \$330,000, and Operating expenses \$90,000. Compute the following: (a) net sales, (b) gross profit, (c) income from operations, and (d) gross profit rate. (Round to one decimal place.)

***BE5.11 (LO 6)** Presented below is the format of the worksheet using the perpetual inventory system presented in Appendix 5A.

Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.

Indicate where the following items will appear on the worksheet: (a) Cash, (b) Inventory, (c) Sales revenue, and (d) Cost of goods sold.

Example:

Cash: Trial balance debit column; Adjusted trial balance debit column; and Balance sheet debit column.

Compute net purchases and cost of goods purchased.

***BE5.12 (LO 7)** Assume that Morgan Company uses a periodic inventory system and has these account balances: Purchases \$450,000, Purchase Returns and Allowances \$13,000, Purchase Discounts \$9,000, and Freight-In \$18,000. Determine net purchases and cost of goods purchased.

Compute cost of goods sold and gross profit.

***BE5.13 (LO 7)** Assume the same information as in BE5.12 and also that Morgan Company has beginning inventory of \$60,000, ending inventory of \$90,000, and net sales of \$730,000. Determine the amounts to be reported for cost of goods sold and gross profit.

Journalize purchase transactions.

***BE5.14 (LO 7)** Prepare the journal entries to record these transactions on Shabani Company's books using a periodic inventory system.

- On March 2, Shabani Company purchased \$900,000 of merchandise on account from Ballas Company, terms 2/10, n/30.
- On March 6, Shabani Company returned \$110,000 of the merchandise purchased on March 2.
- On March 12, Shabani Company paid the balance due to Ballas Company.

Prepare closing entries for merchandise accounts.

***BE5.15 (LO 7)** T. Corlett Company has the following merchandise account balances: Sales Revenue \$180,000, Sales Discounts \$2,000, Purchases \$120,000, and Purchases Returns and Allowances \$30,000. In addition, it has a beginning inventory of \$40,000 and an ending inventory of \$30,000. Prepare the entries to record the closing of these items to Income Summary using the periodic inventory system.

Identify worksheet columns for selected accounts

***BE5.16 (LO 7)** Presented below is the format of the worksheet using the periodic inventory system presented in Appendix 5B.

Trial Balance		Adjustments		Adjusted Trial Balance		Income Statement		Balance Sheet	
Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.

Indicate where the following items will appear on the worksheet: (a) Cash, (b) Beginning inventory, (c) Accounts payable, and (d) Ending inventory.

Example

Cash: Trial balance debit column; Adjustment trial balance debit column; and Balance sheet debit column.

DO IT! Exercises

Answer general questions about merchandisers.

DO IT! 5.1 (LO 1) Indicate whether the following statements are true or false. If false, indicate how to correct the statement.

- A merchandising company reports gross profit but a service company does not.
- Under a periodic inventory system, a company determines the cost of goods sold each time a sale occurs.

3. A service company is likely to use accounts receivable but a merchandising company is not likely to do so.
4. Under a periodic inventory system, the cost of goods on hand at the beginning of the accounting period plus the cost of goods purchased less the cost of goods on hand at the end of the accounting period equals cost of goods sold.

DO IT! 5.2 (LO 2) On October 5, Wang Company buys merchandise on account from Davis Company. The selling price of the goods is \$4,800, and the cost to Davis Company is \$3,100. On October 8, Wang returns defective goods with a selling price of \$650 and a fair value of \$100. Record the transactions on the books of Wang Company.

Record transactions of purchasing company.

DO IT! 5.3 (LO 3) Assume information similar to that in **DO IT! 5.2**. On October 5, Wang Company buys merchandise on account from Davis Company. The selling price of the goods is \$4,800, and the cost to Davis Company is \$3,100. On October 8, Wang returns defective goods with a selling price of \$650 and a fair value of \$100. Record the transactions on the books of Davis Company.

Record transactions of selling company.

DO IT! 5.4 (LO 4) The trial balance of Beads and Bangles at December 31 shows Inventory \$21,000, Sales Revenue \$156,000, Sales Returns and Allowances \$4,000, Sales Discounts \$3,000, Cost of Goods Sold \$92,400, Interest Revenue \$5,000, Freight-Out \$1,800, Utilities Expense \$7,700, and Salaries and Wages Expense \$19,500. Prepare the closing entries for Beads and Bangles for these accounts.

Prepare closing entries for a merchandising company.

DO IT! 5.5 (LO 5) Financial Statement The following information is available for Berlin Corp. for the year ended -December 31, 2020:

Prepare a multiple-step income statement and comprehensive income statement.

Other revenues and gains	\$ 12,700	Sales revenue	\$592,000
Other expenses and losses	13,300	Operating expenses	186,000
Cost of goods sold	156,000	Sales returns and	
Other comprehensive income	5,400	allowances	40,000

Prepare a multiple-step income statement for Berlin Corp. and a comprehensive income statement.

Exercises

E5.1 (LO 1) Mr. McKenzie has prepared the following list of statements about service companies and merchandisers.

Answer general questions about merchandisers.

1. Measuring net income for a merchandiser is conceptually the same as for a service company.
2. For a merchandiser, sales less operating expenses is called gross profit.
3. For a merchandiser, the primary source of revenues is the sale of inventory.
4. Sales salaries and wages is an example of an operating expense.
5. The operating cycle of a merchandiser is the same as that of a service company.
6. In a perpetual inventory system, no detailed inventory records of goods on hand are maintained.
7. In a periodic inventory system, the cost of goods sold is determined only at the end of the accounting period.
8. A periodic inventory system provides better control over inventories than a perpetual system.

Instructions

Identify each statement as true or false. If false, indicate how to correct the statement.

E5.2 (LO 2) Information related to Kerber Co. is presented below.

Journalize purchase transactions.

1. On April 5, purchased merchandise on account from Wilkes Company for \$23,000, terms 2/10, net/30, FOB shipping point.
2. On April 6, paid freight costs of \$900 on merchandise purchased from Wilkes.
3. On April 7, purchased equipment on account for \$26,000.
4. On April 8, returned damaged merchandise to Wilkes Company and was granted a \$3,000 credit for returned merchandise.
5. On April 15, paid the amount due to Wilkes Company in full.

Instructions

- Prepare the journal entries to record these transactions on the books of Kerber Co. under a perpetual inventory system.
- Assume that Kerber Co. paid the balance due to Wilkes Company on May 4 instead of April 15. Prepare the journal entry to record this payment.

Journalize perpetual inventory entries.

E5.3 (LO 2, 3) On September 1, Nixa Office Supply had an inventory of 30 calculators at a cost of \$18 each. The company uses a perpetual inventory system. During September, the following transactions occurred.

- | | |
|---------|--|
| Sept. 6 | Purchased 90 calculators at \$22 each from York. |
| 9 | Paid freight of \$90 on calculators purchased from York Co. |
| 10 | Returned 3 calculators to York Co. for \$69 cash (including freight) because they did not meet specifications. |
| 12 | Sold 26 calculators costing \$23 (including freight) for \$31 each on account to Sura Book Store, terms n/30. |
| 14 | Granted credit of \$31 to Sura Book Store for the return of one calculator that was not ordered. |
| 20 | Sold 30 calculators costing \$23 for \$32 each on account to Davis Card Shop, terms n/30. |

Instructions

Journalize the September transactions.

Prepare purchase and sales entries.

E5.4 (LO 2, 3) On June 10, Diaz Company purchased \$8,000 of merchandise on account from Taylor Company, FOB shipping point, terms 2/10, n/30. Diaz pays the freight costs of \$400 on June 11. Damaged goods totaling \$300 are returned to Taylor for credit on June 12. The fair value of these goods is \$70. On June 19, Diaz pays Taylor Company in full, less the purchase discount. Both companies use a perpetual inventory system.

Instructions

- Prepare separate entries for each transaction on the books of Diaz Company.
- Prepare separate entries for each transaction for Taylor Company. The merchandise purchased by Diaz on June 10 had cost Taylor \$4,800.

Journalize sales transactions.

E5.5 (LO 3) Presented below are transactions related to R. Humphrey Company.

- On December 3, R. Humphrey Company sold \$570,000 of merchandise on account to Frazier Co., terms 1/10, n/30, FOB destination. R. Humphrey paid \$400 for freight charges. The cost of the merchandise sold was \$350,000.
- On December 8, Frazier Co. was granted an allowance of \$20,000 for merchandise purchased on December 3.
- On December 13, R. Humphrey Company received the balance due from Frazier Co.

Instructions

- Prepare the journal entries to record these transactions on the books of R. Humphrey Company using a perpetual inventory system.
- Assume that R. Humphrey Company received the balance due from Frazier Co. on January 2 of the following year instead of December 13. Prepare the journal entry to record the receipt of payment on January 2.

Prepare sales section and closing entries.

E5.6 (LO 4, 5) Financial Statement The adjusted trial balance of Sang Company shows the following data pertaining to sales at the end of its fiscal year October 31, 2020: Sales Revenue \$820,000, Freight-Out \$16,000, Sales Returns and Allowances \$25,000, and Sales Discounts \$13,000.

Instructions

- Prepare the sales section of the income statement.
- Prepare separate closing entries for (1) sales revenue, and (2) the contra accounts to sales revenue.

Prepare adjusting and closing entries.

E5.7 (LO 4) Tim Jarosz Company had the following account balances at year-end: Cost of Goods Sold \$60,000, Inventory \$15,000, Operating Expenses \$29,000, Sales Revenue \$115,000, Sales Discounts

\$1,200, and Sales Returns and Allowances \$1,700. A physical count of inventory determines that merchandise inventory on hand is \$13,600.

Instructions

- Prepare the adjusting entry necessary as a result of the physical count.
- Prepare closing entries.

E5.8 (LO 4) Presented below is information related to Hoerl Co. for the month of January 2020.

Prepare adjusting and closing entries.

Ending inventory per perpetual records	\$ 21,600	Insurance expense	\$ 12,000
Ending inventory actually on hand	21,000	Rent expense	20,000
Cost of goods sold	218,000	Salaries and wages expense	55,000
Freight-out	7,000	Sales discounts	10,000
		Sales returns and allowances	13,000
		Sales revenue	380,000

Instructions

- Prepare the necessary adjusting entry for inventory.
- Prepare the necessary closing entries.

E5.9 (LO 5) Financial Statement Presented below is information for Kaila Company for the month of March 2020.

Prepare a multiple-step income statement and calculate profitability.

Cost of goods sold	\$215,000	Rent expense	\$ 30,000
Freight-out	7,000	Sales discounts	8,000
Insurance expense	6,000	Sales returns and allowances	13,000
Salaries and wages expense	58,000	Sales revenue	380,000

Instructions

- Prepare a multiple-step income statement.
- Compute the gross profit rate.

E5.10 (LO 5) Financial Statement In its income statement for the year ended December 31, 2020, Anhad Company reported the following condensed data.

Prepare income statements and a comprehensive income statement.

Operating expenses	\$ 725,000	Interest revenue	\$ 28,000
Cost of goods sold	1,289,000	Loss on disposal of plant assets	17,000
Interest expense	70,000	Net sales	2,200,000
		Other comprehensive loss	8,300



Instructions

- Prepare a multiple-step income statement.
- Prepare a single-step income statement.
- Prepare a comprehensive income statement.

E5.11 (LO 2, 3) An inexperienced accountant for Stahr Company made the following errors in recording merchandising transactions.

Prepare correcting entries for sales and purchases.

- A \$210 refund to a customer for faulty merchandise was debited to Sales Revenue \$210 and credited to Cash \$210.
- A \$180 credit purchase of supplies was debited to Inventory \$180 and credited to Cash \$180.
- A \$215 sales discount was debited to Sales Revenue.
- A cash payment of \$20 for freight on merchandise purchases was debited to Freight-Out \$200 and credited to Cash \$200.

Instructions

Prepare separate correcting entries for each error, assuming that the incorrect entry is not reversed. (Omit explanations.)

E5.12 (LO 5) Financial Statement In 2020, Laquen Company had net sales of \$900,000 and cost of goods sold of \$522,000. Operating expenses were \$225,000, and interest expense was \$11,000. Laquen prepares a multiple-step income statement.

Compute various income measures.

Instructions

- Compute Laquen's gross profit.
- Compute the gross profit rate. Why is this rate computed by financial statement users?
- What is Laquen's income from operations and net income?
- If Laquen prepared a single-step income statement, what amount would it report for net income?
- In what section of its classified balance sheet should Laquen report inventory?

Compute missing amounts and compute gross profit rate.

E5.13 (LO 5) Presented below is financial information for two different companies.

	<u>Summer Company</u>	<u>Winter Company</u>
Sales revenue	\$92,000	(d)
Sales returns	(a)	\$ 5,000
Net sales	87,000	102,000
Cost of goods sold	56,000	(e)
Gross profit	(b)	41,500
Operating expenses	15,000	(f)
Net income	(c)	18,000

Instructions

- Determine the missing amounts.
- Determine the gross profit rates. (Round to one decimal place.)

Compute missing amounts.

E5.14 (LO 5) Financial information is presented below for three different companies.

	<u>Hardy Cosmetics</u>	<u>Yee Grocery</u>	<u>Wang Wholesalers</u>
Sales revenue	\$90,000	\$ (e)	\$122,000
Sales returns and allowances	(a)	5,000	12,000
Net sales	86,000	95,000	(i)
Cost of goods sold	56,000	(f)	(j)
Gross profit	(b)	38,000	24,000
Operating expenses	15,000	(g)	18,000
Income from operations	(c)	(h)	(k)
Other expenses and losses	4,000	7,000	(l)
Net income	(d)	11,000	5,000

Instructions

Determine the missing amounts.

Complete worksheet using a perpetual inventory system.

***E5.15 (LO 6)** The following are selected accounts for McPhan Company as reported in the worksheet using a perpetual inventory system at the end of May 2020.

<u>Accounts</u>	<u>Adjusted Trial Balance</u>		<u>Income Statement</u>		<u>Balance Sheet</u>	
	<u>Dr.</u>	<u>Cr.</u>	<u>Dr.</u>	<u>Cr.</u>	<u>Dr.</u>	<u>Cr.</u>
Cash	11,000					
Inventory	76,000					
Sales Revenue		480,000				
Sales Returns and Allowances	10,000					
Sales Discounts	9,000					
Cost of Goods Sold	300,000					

Instructions

Complete the worksheet by extending amounts reported in the adjusted trial balance to the appropriate columns in the worksheet. Do not total individual columns.

Prepare a worksheet using a perpetual inventory system.

***E5.16 (LO 6)** The trial balance columns of the worksheet using a perpetual inventory system for Balistreri Company at June 30, 2020, are as follows.

Balistreri Company
Worksheet
For the Month Ended June 30, 2020

Account Titles	Trial Balance	
	Debit	Credit
Cash	1,920	
Accounts Receivable	2,440	
Inventory	11,640	
Accounts Payable		1,120
Owner's Capital		3,500
Sales Revenue		42,500
Cost of Goods Sold	20,560	
Operating Expenses	10,560	
	<u>47,120</u>	<u>47,120</u>

Other data:

Operating expenses incurred on account, but not yet recorded, total \$1,500.

Instructions

Enter the trial balance on a worksheet and complete the worksheet.

***E5.17 (LO 7)** The trial balance of A. Wiencek Company at the end of its fiscal year, August 31, 2020, includes these accounts: Inventory \$19,500; Purchases \$149,000; Sales Revenue \$190,000; Freight-In \$5,000; Sales Returns and Allowances \$3,000; Freight-Out \$1,000; and Purchase Returns and Allowances \$2,000. The ending inventory is \$23,000.

Prepare cost of goods sold section.

Instructions

Prepare a cost of goods sold section for the year ending August 31 (periodic inventory).

***E5.18 (LO 7)** On January 1, 2020, Brooke Hanson Corporation had inventory of \$50,000. At December 31, 2020, Brooke Hanson had the following account balances.

Compute various income statement items.

Freight-in	\$ 4,000
Purchases	509,000
Purchase discounts	6,000
Purchase returns and allowances	2,000
Sales revenue	840,000
Sales discounts	5,000
Sales returns and allowances	10,000

At December 31, 2020, Brooke Hanson determines that its ending inventory is \$60,000.

Instructions

- Compute Brooke Hanson's 2020 gross profit.
- Compute Brooke Hanson's 2020 operating expenses if net income is \$130,000 and there are no nonoperating activities.

***E5.19 (LO 7)** Below is a series of cost of goods sold sections for companies B, F, L, and R.

Prepare cost of goods sold section.

	B	F	L	R
Beginning inventory	\$ 180	\$ 70	\$1,000	\$ (j)
Purchases	1,620	1,060	(g)	43,590
Purchase returns and allowances	40	(d)	290	(k)
Net purchases	(a)	1,030	6,210	41,090
Freight-in	110	(e)	(h)	2,240
Cost of goods purchased	(b)	1,280	7,940	(l)
Cost of goods available for sale	1,870	1,350	(i)	49,530
Ending inventory	250	(f)	1,450	6,230
Cost of goods sold	(c)	1,230	7,490	43,300

Instruction

Fill in the lettered blanks to complete the cost of goods sold sections.

Journalize purchase transactions.

***E5.20 (LO 7)** This information relates to Nandi Co.

- On April 5, purchased merchandise on account from Dion Company for \$25,000, terms 2/10, net/30, FOB shipping point.
- On April 6, paid freight costs of \$900 on merchandise purchased from Dion Company.
- On April 7, purchased equipment on account for \$30,000.
- On April 8, returned some of April 5 merchandise, which cost \$2,800, to Dion Company.
- On April 15, paid the amount due to Dion Company in full.

Instructions

- Prepare the journal entries to record these transactions on the books of Nandi Co. using a periodic inventory system.
- Assume that Nandi Co. paid the balance due to Dion Company on May 4 instead of April 15. Prepare the journal entry to record this payment.

Journalize purchase transactions.

***E5.21 (LO 7)** Presented below is information related to Chung Co.

- On April 5, purchased merchandise on account from Jose Company for \$21,000, terms 2/10, net/30, FOB shipping point.
- On April 6, paid freight costs of \$800 on merchandise purchased from Jose.
- On April 7, purchased equipment on account from Winker Mfg. Co. for \$26,000.
- On April 8, returned merchandise, which cost \$4,000, to Jose Company.
- On April 15, paid the amount due to Jose Company in full.

Instructions

- Prepare the journal entries to record these transactions on the books of Chung Co. using a periodic inventory system.
- Assume that Chung Co. paid the balance due to Jose Company on May 4 instead of April 15. Prepare the journal entry to record this payment.

Complete worksheet for selected accounts.

***E5.22 (LO 7)** Presented below are selected accounts for T. Swift Company as reported in the worksheet at the end of May 2020. Ending inventory is \$75,000.

Accounts	Adjusted Trial Balance		Income Statement		Balance Sheet	
	Dr.	Cr.	Dr.	Cr.	Dr.	Cr.
Cash	9,000					
Inventory	80,000					
Purchases	240,000					
Purchase Returns and Allowances		30,000				
Sales Revenue		450,000				
Sales Returns and Allowances	10,000					
Sales Discounts	5,000					
Rent Expense	42,000					

Instructions

Complete the worksheet by extending amounts reported in the adjustment trial balance to the appropriate columns in the worksheet. The company uses the periodic inventory system.

Problems: Set A

Journalize purchase and sales transactions under a perpetual inventory system.



P5.1A (LO 2, 3) Kern's Book Warehouse distributes hardcover books to retail stores and extends credit terms of 2/10, n/30 to all of its customers. At the end of May, Kern's inventory consisted of books purchased for \$1,800. During June, the following merchandising transactions occurred.

- June 1 Purchased books on account for \$1,600 from Binsfeld Publishers, FOB destination, terms 2/10, n/30. The appropriate party also made a cash payment of \$50 for the freight on this date.
- 3 Sold books on account to Reading Rainbow for \$2,500. The cost of the books sold was \$1,440.
- 6 Received \$100 credit for books returned to Binsfeld Publishers.
- 9 Paid Binsfeld Publishers in full, less discount.
- 15 Received payment in full from Reading Rainbow.
- 17 Sold books on account to Rapp Books for \$1,800. The cost of the books sold was \$1,080.
- 20 Purchased books on account for \$1,800 from McGinn Publishers, FOB destination, terms 2/15, n/30. The appropriate party also made a cash payment of \$60 for the freight on this date.
- 24 Received payment in full from Rapp Books.
- 26 Paid McGinn Publishers in full, less discount.
- 28 Sold books on account to Baeten Bookstore for \$1,600. The cost of the books sold was \$970.
- 30 Granted Baeten Bookstore \$120 credit for books returned costing \$72.

Kern's Book Warehouse's chart of accounts includes the following: No. 101 Cash, No. 112 Accounts Receivable, No. 120 Inventory, No. 201 Accounts Payable, No. 401 Sales Revenue, No. 412 Sales Returns and Allowances, No. 414 Sales Discounts, and No. 505 Cost of Goods Sold.

Instructions

Journalize the transactions for the month of June for Kern's Book Warehouse using a perpetual inventory system.

P5.2A (LO 2, 3, 5) Financial Statement Renner Hardware Store completed the following merchandising transactions in the month of May. At the beginning of May, the ledger of Renner showed Cash of \$5,000 and Owner's Capital of \$5,000.

Journalize, post, and prepare a partial income statement.



- May 1 Purchased merchandise on account from Braun's Wholesale Supply \$4,200, terms 2/10, n/30.
- 2 Sold merchandise on account \$2,100, terms 1/10, n/30. The cost of the merchandise sold was \$1,300.
- 5 Received credit from Braun's Wholesale Supply for merchandise returned \$300.
- 9 Received collections in full, less discounts, from customers billed on sales of \$2,100 on May 2.
- 10 Paid Braun's Wholesale Supply in full, less discount.
- 11 Purchased supplies for cash \$400.
- 12 Purchased merchandise for cash \$1,400.
- 15 Received refund for poor quality merchandise from supplier on cash purchase \$150.
- 17 Purchased merchandise from Valley Distributors \$1,300, FOB shipping point, terms 2/10, n/30.
- 19 Paid freight on May 17 purchase \$130.
- 24 Sold merchandise for cash \$3,200. The merchandise sold had a cost of \$2,000.
- 25 Purchased merchandise on account from Lumley, Inc. \$620, FOB destination, terms 2/10, n/30.
- 27 Paid Valley Distributors in full, less discount.
- 29 Made refunds to cash customers for defective merchandise \$70. The returned merchandise had a fair value of \$30.
- 31 Sold merchandise on account \$1,000 terms n/30. The cost of the merchandise sold was \$560.

Renner Hardware's chart of accounts includes the following: No. 101 Cash, No. 112 Accounts Receivable, No. 120 Inventory, No. 126 Supplies, No. 201 Accounts Payable, No. 301 Owner's Capital, No. 401 Sales Revenue, No. 412 Sales Returns and Allowances, No. 414 Sales Discounts, and No. 505 Cost of Goods Sold.

Instructions

- Journalize the transactions using a perpetual inventory system.
- Enter the beginning cash and capital balances and post the transactions. (Use J1 for the journal reference.)
- Prepare an income statement through gross profit for the month of May 2020.

c. Gross profit \$2,379

Prepare financial statements and adjusting and closing entries.

P5.3A (LO 4, 5) Financial Statement Big Box Store is located in midtown Madison. During the past several years, net income has been declining because of suburban shopping centers. At the end of the company's fiscal year on November 30, 2020, the following accounts appeared in two of its trial balances.

	Unadjusted	Adjusted		Unadjusted	Adjusted
Accounts Payable	\$ 25,200	\$ 25,200	Notes Payable	\$ 37,000	\$ 37,000
Accounts Receivable	30,500	30,500	Owner's Capital	101,700	101,700
Accumulated Depr.—Equip.	34,000	45,000	Owner's Drawings	10,000	10,000
Cash	26,000	26,000	Prepaid Insurance	10,500	3,500
Cost of Goods Sold	518,000	518,000	Property Tax Expense		2,500
Freight-Out	6,500	6,500	Property Taxes Payable		2,500
Equipment	146,000	146,000	Rent Expense	15,000	15,000
Depreciation Expense		11,000	Salaries and Wages Expense	96,000	96,000
Insurance Expense		7,000	Sales Revenue	720,000	720,000
Interest Expense	6,400	6,400	Sales Commissions Expense	6,500	11,000
Interest Revenue	2,000	2,000	Sales Commissions Payable		4,500
Inventory	32,000	32,000	Sales Returns and Allowances	8,000	8,000
			Utilities Expense	8,500	8,500

Instructions

- a. Net income \$32,100
Owner's capital \$123,800
Total assets \$193,000

- Prepare a multiple-step income statement, an owner's equity statement, and a classified balance sheet. Notes payable are due in 2023.
- Journalize the adjusting entries that were made.
- Journalize the closing entries that are necessary.

Journalize, post, and prepare a trial balance.



P5.4A (LO 2, 3, 4) Financial Statement Yolanda Hagen, a former disc golf star, operates Yolanda's Discorama. At the beginning of the current season on April 1, the ledger of Yolanda's Discorama showed Cash \$1,800, Inventory \$2,500, and Owner's Capital \$4,300. The following transactions were completed during April.

- Apr. 5 Purchased golf discs, bags, and other inventory on account from Mumford Co. \$1,200, FOB shipping point, terms 2/10, n/60.
- 7 Paid freight on the Mumford purchase \$50.
- 9 Received credit from Mumford Co. for merchandise returned \$100.
- 10 Sold merchandise on account for \$900, terms n/30. The merchandise sold had a cost of \$540.
- 12 Purchased disc golf shirts and other accessories on account from Saucer Sportswear \$670, terms 1/10, n/30.
- 14 Paid Mumford Co. in full, less discount.
- 17 Received credit from Saucer Sportswear for merchandise returned \$70.
- 20 Made sales on account for \$610, terms n/30. The cost of the merchandise sold was \$370.
- 21 Paid Saucer Sportswear in full, less discount.
- 27 Granted an allowance to customers for clothing that was flawed \$20.
- 30 Received payments on account from customers \$900.

The chart of accounts for the store includes the following: No. 101 Cash, No. 112 Accounts Receivable, No. 120 Inventory, No. 201 Accounts Payable, No. 301 Owner's Capital, No. 401 Sales Revenue, No. 412 Sales Returns and Allowances, and No. 505 Cost of Goods Sold.

Instructions

- Journalize the April transactions using a perpetual inventory system.
- Enter the beginning balances in the ledger accounts and post the April transactions. (Use J1 for the journal reference.)
- Prepare a trial balance on April 30, 2020.

- c. Total debits \$5,810

Complete accounting cycle beginning with a worksheet.



P5.5A (LO 4, 5, 6) Financial Statement The trial balance of Gaolee Fashion Center contained the following accounts at November 30, the end of the company's fiscal year.

Gaolee Fashion Center
Trial Balance
November 30, 2020

	<u>Debit</u>	<u>Credit</u>
Cash	\$ 20,700	
Accounts Receivable	30,700	
Inventory	44,700	
Supplies	6,200	
Equipment	133,000	
Accumulated Depreciation—Equipment		\$ 28,000
Notes Payable		60,000
Accounts Payable		48,500
Owner's Capital		93,000
Owner's Drawings	12,000	
Sales Revenue		755,200
Sales Returns and Allowances	8,800	
Cost of Goods Sold	497,400	
Salaries and Wages Expense	140,000	
Advertising Expense	24,400	
Utilities Expense	14,000	
Maintenance and Repairs Expense	12,100	
Freight-Out	16,700	
Rent Expense	24,000	
Totals	<u>\$984,700</u>	<u>\$984,700</u>

Adjustment data:

1. Supplies on hand totaled \$2,600.
2. Depreciation is \$11,500 on the equipment.
3. Interest of \$3,800 is accrued on notes payable at November 30.
4. Inventory actually on hand is \$44,400.

Instructions

- a. Enter the trial balance on a worksheet, and complete the worksheet.
- b. Prepare a multiple-step income statement and an owner's equity statement for the year, and a classified balance sheet as of November 30, 2020. Notes payable of \$20,000 are due in January 2021.
- c. Journalize the adjusting entries.
- d. Journalize the closing entries.
- e. Prepare a post-closing trial balance.

- a. Adj. trial balance \$1,000,000
Net loss \$1,400
- b. Gross profit \$248,700
Total assets \$191,900

P5.6A (LO 5, 7) Financial Statement At the end of Donaldson Department Store's fiscal year on November 30, 2020, these accounts appeared in its adjusted trial balance.

Determine cost of goods sold and gross profit under periodic approach.

Freight-In	\$ 7,500
Inventory	40,000
Purchases	585,000
Purchase Discounts	6,300
Purchase Returns and Allowances	2,700
Sales Revenue	1,000,000
Sales Returns and Allowances	20,000

Additional facts:

1. Merchandise inventory on November 30, 2020, is \$52,600.
2. Donaldson Department Store uses a periodic system.

Instructions

Prepare an income statement through gross profit for the year ended November 30, 2020.

Gross profit \$409,100

Calculate missing amounts and assess profitability.

P5.7A (LO 5, 7) Writing Kayla Inc. operates a retail operation that purchases and sells home entertainment products. The company purchases all merchandise inventory on credit and uses a periodic inventory system. The Accounts Payable account is used for recording inventory purchases only; all other current liabilities are accrued in separate accounts. You are provided with the following selected information for the fiscal years 2017 through 2020, inclusive.

	2017	2018	2019	2020
Income Statement Data				
Sales revenue		\$55,000	\$ (e)	\$47,000
Cost of goods sold		(a)	14,800	14,300
Gross profit		38,300	35,200	(i)
Operating expenses		34,900	(f)	28,800
Net income		\$ (b)	\$ 2,500	\$ (j)
Balance Sheet Data				
Inventory	\$7,200	\$ (c)	\$ 8,100	\$ (k)
Accounts payable	3,200	3,600	2,500	(l)
Additional Information				
Purchases of merchandise inventory on account		\$14,200	\$ (g)	\$13,200
Cash payments to suppliers		(d)	(h)	13,600

- c. \$4,700
g. \$18,200
i. \$32,700

Instructions

- Calculate the missing amounts.
- Sales declined over the 3-year fiscal period, 2018–2020. Does that mean that profitability necessarily also declined? Explain, computing the gross profit rate and the profit margin (Net income ÷ Sales revenue) for each fiscal year to help support your answer. (Round to one decimal place.)

Journalize, post, and prepare trial balance and partial income statement using periodic approach.



P5.8A (LO 7) Financial Statement At the beginning of the current season on April 1, the ledger of Gage Pro Shop showed Cash \$3,000, Inventory \$4,000, and Owner's Capital \$7,000. These transactions occurred during April 2020.

- Apr. 5 Purchased golf bags, clubs, and balls on account from Tiger Co. \$1,200, FOB shipping point, terms 2/10, n/60.
7 Paid freight on Tiger Co. purchases \$50.
9 Received credit from Tiger Co. for merchandise returned \$100.
10 Sold merchandise on account to customers \$600, terms n/30.
12 Purchased golf shoes, sweaters, and other accessories on account from Classic Sportswear \$450, terms 1/10, n/30.
14 Paid Tiger Co. in full.
17 Received credit from Classic Sportswear for merchandise returned \$50.
20 Made sales on account to customers \$600, terms n/30.
21 Paid Classic Sportswear in full.
27 Granted credit to customers for clothing that had flaws \$35.
30 Received payments on account from customers \$600.

The chart of accounts for the pro shop includes Cash, Accounts Receivable, Inventory, Accounts Payable, Owner's Capital, Sales Revenue, Sales Returns and Allowances, Purchases, Purchase Returns and Allowances, Purchase Discounts, and Freight-In.

Instructions

- Journalize the April transactions using a periodic inventory system.
- Using T-accounts, enter the beginning balances in the ledger accounts and post the April transactions.
- Prepare a trial balance on April 30, 2020.
- Prepare an income statement through gross profit, assuming merchandise inventory on hand at April 30 is \$4,824.

- c. Tot. trial balance \$8,376
d. Gross profit \$465

Continuing Case

Cookie Creations

(Note: This is a continuation of the Cookie Creations case from Chapters 1 through 4.)

CC5 Because Natalie has had such a successful first few months, she is considering other opportunities to develop her business. One opportunity is the sale of fine European mixers. The owner of Kzinski Supply Company has approached Natalie to become the exclusive U.S. distributor of these fine mixers. The current cost of a mixer is approximately \$525 (U.S.), and Natalie would sell each one for \$1,050. Natalie comes to you for advice on how to account for these mixers.

Go to **WileyPLUS** for complete case details and instructions.



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Ethics Case

EC5 Tiffany Lyons was just hired as the assistant treasurer of Key West Stores. The company is a specialty chain store with nine retail stores concentrated in one metropolitan area. Among other things, the payment of all invoices is centralized in one of the departments Tiffany will manage. Her primary responsibility is to maintain the company's high credit rating by paying all bills when due and to take advantage of all cash discounts.

Jay Barnes, the former assistant treasurer who has been promoted to treasurer, is training Tiffany in her new duties. He instructs Tiffany that she is to continue the practice of preparing all checks "net of discount" and dating the checks the last day of the discount period. "But," Jay continues, "we always hold the checks at least 4 days beyond the discount period before mailing them. That way, we get another 4 days of interest on our money. Most of our creditors need our business and don't complain. And, if they scream about our missing the discount period, we blame it on the mailroom or the post office. We've only lost one discount out of every hundred we take that way. I think everybody does it. By the way, welcome to our team!"

Instructions

- What are the ethical considerations in this case?
- Who are the stakeholders that are harmed or benefitted in this situation?
- Should Tiffany continue the practice started by Jay? Does she have any choice?

Comprehensive Accounting Cycle Review

ACR5 Financial Statement On December 1, 2020, Rodriguez Distributing Company had the following account balances.

	Debit		Credit
Cash	\$ 7,200	Accumulated Depreciation—	
Accounts Receivable	4,600	Equipment	\$ 2,200
Inventory	12,000	Accounts Payable	4,500
Supplies	1,200	Salaries and Wages Payable	1,000
Equipment	22,000	Owner's Capital	39,300
	<u>\$47,000</u>		<u>\$47,000</u>

During December, the company completed the following summary transactions.

- Dec. 6 Paid \$1,600 for salaries and wages due employees, of which \$600 is for December and \$1,000 is for November salaries and wages payable.
- 8 Received \$2,200 cash from customers in payment of account (no discount allowed).
- 10 Sold merchandise for cash \$6,300. The cost of the merchandise sold was \$4,100.
- 13 Purchased merchandise on account from Boehm Co. \$9,000, terms 2/10, n/30.
- 15 Purchased supplies for cash \$2,000.
- 18 Sold merchandise on account \$15,000, terms 3/10, n/30. The cost of the merchandise sold was \$10,000.
- 20 Paid salaries and wages \$1,800.
- 23 Paid Boehm Co. in full, less discount.
- 27 Received collections in full, less discounts, from customers billed on December 18.



James Porter/Workbook/Getty Images, Inc.

Inventories

Chapter Preview

In the previous chapter, we discussed the accounting for merchandise inventory using a perpetual inventory system. In this chapter, we explain the methods used to calculate the cost of inventory on hand at the balance sheet date and the cost of goods sold.

Feature Story

“Where Is That Spare Bulldozer Blade?”

Let’s talk inventory—big, bulldozer-size inventory. **Caterpillar Inc.** is the world’s largest manufacturer of construction and mining equipment, diesel and natural gas engines, and industrial gas turbines. It sells its products in over 200 countries, making it one of the most successful U.S. exporters. More than 70% of its productive assets are located domestically, and nearly 50% of its sales are foreign.

In the past, Caterpillar’s profitability suffered, but today it is very successful. A big part of this turnaround can be attributed to effective management of its inventory. Imagine what it costs Caterpillar to have too many bulldozers sitting around in

inventory—a situation the company definitely wants to avoid. Yet Caterpillar must also make sure it has enough inventory to meet demand.

At one time during a 7-year period, Caterpillar’s sales increased by 100% while its inventory increased by only 50%. To achieve this dramatic reduction in the amount of resources tied up in inventory while continuing to meet customers’ needs, Caterpillar used a two-pronged approach. First, it completed a factory modernization program, which greatly increased its production efficiency. The program reduced by 60% the amount of inventory the company processes at any one time. It also reduced by an incredible 75% the time it takes to manufacture a part.

Second, Caterpillar dramatically improved its parts distribution system. It ships more than 100,000 items daily from its

23 distribution centers strategically located around the world (10 million square feet of warehouse space—remember, we’re talking bulldozers). The company can virtually guarantee that it can get any part to anywhere in the world within 24 hours.

These changes led to record exports, profits, and revenues for Caterpillar. It would seem that things couldn’t be better. But industry analysts, as well as the company’s managers, thought otherwise. In order to maintain Caterpillar’s

position as the industry leader, management began another major overhaul of inventory production and inventory management processes. The goal: to cut the number of repairs in half, increase productivity by 20%, and increase inventory turnover by 40%.

In short, Caterpillar’s ability to manage its inventory has been a key reason for its past success and will very likely play a huge part in its future profitability as well.

Chapter Outline

LEARNING OBJECTIVES

LO 1 Discuss how to classify and determine inventory.	• Classifying inventory • Determining inventory quantities	DO IT! 1 Rules of Ownership
LO 2 Apply inventory cost flow methods and discuss their financial effects.	• Specific identification • Cost flow assumptions • Financial statement and tax effects • Using inventory cost flow methods consistently	DO IT! 2 Cost Flow Methods
LO 3 Indicate the effects of inventory errors on the financial statements.	• Income statement effects • Balance sheet effects	DO IT! 3 Inventory Errors
LO 4 Explain the statement presentation and analysis of inventory.	• Presentation • Lower-of-cost-or-net realizable value • Analysis	DO IT! 4 LCNRV and Inventory Turnover

Go to the Review and Practice section at the end of the chapter for a review of key concepts and practice applications with solutions.
Visit WileyPLUS with ORION for additional tutorials and practice opportunities.

Classifying and Determining Inventory

LEARNING OBJECTIVE 1

Discuss how to classify and determine inventory.

Two important steps in the reporting of inventory at the end of the accounting period are the classification of inventory based on its degree of completeness and the determination of inventory amounts.

Classifying Inventory

How a company classifies its inventory depends on whether the firm is a merchandiser or a manufacturer. In a **merchandising** company, such as those described in Chapter 5, inventory consists of many different items. For example, in a grocery store, canned goods, dairy products, meats, and produce are just a few of the inventory items on hand. These items have two common characteristics: (1) they are owned by the company, and (2) they are in a form ready for sale to customers in the ordinary course of business. Thus, merchandisers need only one inventory classification, **merchandise inventory**, to describe the many different items that make up the total inventory.

In a **manufacturing** company, some inventory may not yet be ready for sale. As a result, manufacturers usually classify inventory into three categories: finished goods, work in process, and raw materials (see **Helpful Hint**). **Finished goods inventory** is manufactured items that are completed and ready for sale. **Work in process** is that portion of manufactured inventory that has been placed into the production process but is not yet complete. **Raw materials** are the basic goods that will be used in production but have not yet been placed into production.

For example, **Caterpillar** classifies earth-moving tractors completed and ready for sale as **finished goods**. It classifies the tractors on the assembly line in various stages of production as **work in process**. The steel, glass, tires, and other components that are on hand waiting to be used in the production of tractors are identified as **raw materials**. **Illustration 6.1** shows an adapted excerpt from Note 7 of Caterpillar's annual report.

HELPFUL HINT

Regardless of the classification, companies report all inventories under Current Assets on the balance sheet.

	December 31	
(millions of dollars)	2016	2015
Raw materials	\$2,102	\$2,467
Work-in-process	1,719	1,857
Finished goods	4,576	5,122
Supplies	217	254
Total inventories	\$8,614	\$9,700

ILLUSTRATION 6.1

Composition of Caterpillar's inventory

By observing the levels and changes in the levels of these three inventory types, financial statement users can gain insight into management's production plans. For example, low levels of raw materials and high levels of finished goods suggest that management believes it has enough inventory on hand and production will be slowing down—perhaps in anticipation of a recession. Conversely, high levels of raw materials and low levels of finished goods probably signal that management is planning to step up production.

Many companies have significantly lowered inventory levels and costs using **just-in-time (JIT) inventory** methods. Under a just-in-time method, companies manufacture or purchase goods only when needed. **Dell** is famous for having developed a system for making computers in response to individual customer requests. Even though it makes each computer to meet each customer's particular specifications, Dell is able to assemble the computer and put it on a truck in less than 48 hours. The success of the JIT system depends on reliable suppliers. By integrating its information systems with those of its suppliers, Dell reduced its inventories to nearly zero. This is a huge advantage in an industry where products become obsolete nearly overnight.

The accounting concepts discussed in this chapter apply to the inventory classifications of both merchandising and manufacturing companies. Our focus here is on merchandise inventory. Additional issues specific to manufacturing companies are discussed as part of managerial accounting.

Accounting Across the Organization Ford



© PeskyMonkey/iStockphoto

A Big Hiccup

JIT can save a company a lot of money, but it isn't without risk. An unexpected disruption in the supply chain can cost a company a lot of money. Japanese automakers experienced just such a disruption when a 6.8-magnitude earthquake caused major damage to the company that produces 50% of their piston rings. The rings themselves cost only \$1.50, but you can't make a car without them. As a result, the automakers were

forced to shut down production for a few days—a loss of tens of thousands of cars.

Similarly, a major snowstorm halted production at the Canadian plants of **Ford**. A Ford spokesperson said, "Because the plants run with just-in-time inventory, we don't have large stockpiles of parts sitting around. When you have a somewhat significant disruption, you can pretty quickly run out of parts."

Sources: Amy Chozick, "A Key Strategy of Japan's Car Makers Backfires," *Wall Street Journal* (July 20, 2007); and Kate Linebaugh, "Canada Military Evacuates Motorists Stranded by Snow," *Wall Street Journal* (December 15, 2010).

What steps might the companies take to avoid such a serious disruption in the future? (Go to WileyPLUS for this answer and additional questions.)

Determining Inventory Quantities

No matter whether they are using a periodic or perpetual inventory system, all companies need to determine inventory quantities at the end of the accounting period. If using a perpetual system, companies take a physical inventory for the following reasons:

1. To check the accuracy of their perpetual inventory records.
2. To determine the amount of inventory lost due to wasted raw materials, shoplifting, or employee theft.

Companies using a periodic inventory system take a physical inventory for **two different purposes**: to determine the inventory on hand at the balance sheet date, and to determine the cost of goods sold for the period.

Determining inventory quantities involves two steps: (1) taking a physical inventory of goods on hand and (2) determining the ownership of goods.

Taking a Physical Inventory

Companies take a physical inventory at the end of the accounting period. Taking a physical inventory involves actually counting, weighing, or measuring each kind of inventory on hand (see **Ethics Note**). In many companies, taking an inventory is a formidable task. Retailers such as **Target**, **True Value Hardware**, or **Home Depot** have thousands of different inventory items. An inventory count is generally more accurate when goods are not being sold or received during the counting. Consequently, companies often "take inventory" when the business is closed or when business is slow. Many retailers close early on a chosen day in January—after the holiday sales and returns, when inventories are at their lowest level—to count inventory. **Wal-Mart Stores, Inc.**, for example, has a year-end of January 31.

ETHICS NOTE

In a famous fraud, a salad oil company filled its storage tanks mostly with water. The oil rose to the top, so auditors thought the tanks were full of oil. The company also said it had more tanks than it really did: It repainted numbers on the tanks to confuse auditors.

Determining Ownership of Goods

One challenge in computing inventory quantities is determining what inventory a company owns. To determine ownership of goods, two questions must be answered: Do all of the goods included in the count belong to the company? Does the company own any goods that were not included in the count?

Ethics Insight Leslie Fay



© Greg Brookes/iStockphoto

Falsifying Inventory to Boost Income

Managers at women's apparel maker **Leslie Fay** were convicted of falsifying inventory records to boost net income in an attempt to increase management bonuses. In another case, executives at **Craig Consumer Electronics** were accused of defrauding lenders by manipulating inventory records. The

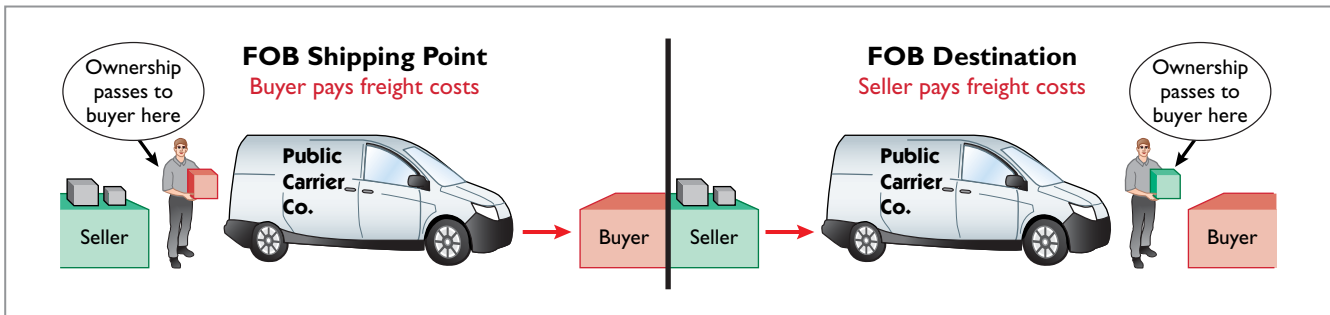
indictment said the company classified "defective goods as new or refurbished" and claimed that it owned certain shipments "from overseas suppliers" when, in fact, Craig either did not own the shipments or the shipments did not exist.

What effect does an overstatement of inventory have on a company's financial statements? (Go to WileyPLUS for this answer and additional questions.)

Goods in Transit. A complication in determining ownership is **goods in transit** (on board a truck, train, ship, or plane) at the end of the period. The company may have purchased goods that have not yet been received, or it may have sold goods that have not yet been delivered. To arrive at an accurate count, the company must determine ownership of these goods.

Goods in transit should be included in the inventory of the company that has legal title to the goods. Legal title is determined by the terms of the sale, as shown in **Illustration 6.2** and described below.

ILLUSTRATION 6.2 Terms of sale



1. When the terms are **FOB (free on board) shipping point**, ownership of the goods passes to the buyer when the public carrier accepts the goods from the seller.
2. When the terms are **FOB destination**, ownership of the goods remains with the seller until the goods reach the buyer.

If goods in transit at the statement date are ignored, inventory quantities may be seriously miscounted. Assume, for example, that Hargrove Company has 20,000 units of inventory on hand on December 31. It also has the following goods in transit:

1. Sales of 1,500 units shipped December 31 FOB destination.
2. Purchases of 2,500 units shipped FOB shipping point by the seller on December 31.

Hargrove has legal title to both the 1,500 units sold and the 2,500 units purchased. If the company ignores the units in transit, it would understate inventory quantities by 4,000 units (1,500 + 2,500).

As we will see later in the chapter, inaccurate inventory counts affect not only the inventory amount shown on the balance sheet but also the cost of goods sold calculation on the income statement.

Consigned Goods. In some lines of business, it is common to hold the goods of other parties and try to sell the goods for them for a fee, but without taking ownership of the goods. These are called **consigned goods**.

For example, you might have a used car that you would like to sell. If you take the item to a dealer, the dealer might be willing to put the car on its lot and charge you a commission if it is sold. Under this agreement, the dealer **would not take ownership** of the car, which would still belong to you. Therefore, if an inventory count were taken, the car would not be included in the dealer's inventory because the dealer does not own it.

Many car, boat, and antique dealers sell goods on consignment to keep their inventory costs down and to avoid the risk of purchasing an item that they will not be able to sell. Today, even some manufacturers are making consignment agreements with their suppliers in order to keep their inventory levels low.

Anatomy of a Fraud

Ted Nickerson, CEO of clock manufacturer Dally Industries, had expensive tastes. To support this habit, Ted took out large loans, which he collateralized with his shares of Dally Industries stock. If the price of Dally's stock fell, he was required to provide the bank with more shares of stock. To achieve target net income figures and thus maintain the stock price, Ted coerced employees in the company to alter inventory figures. Inventory quantities were manipulated by changing the amounts on inventory control tags after the year-end physical inventory count. For example, if a tag said there were 20 units of a particular item, the tag was changed to 220. Similarly, the unit costs that were used to determine the value of ending inventory were increased from, for example, \$125 per unit to \$1,250. Both of these fraudulent changes had the

effect of increasing the amount of reported ending inventory. This reduced cost of goods sold and increased net income.

Total take: \$245,000

The Missing Control

Independent internal verification. The company should have spot-checked its inventory records periodically, verifying that the number of units in the records agreed with the amount on hand and that the unit costs agreed with vendor price sheets.

Source: Adapted from Wells, *Fraud Casebook* (2007), pp. 502–509.

DO IT! 1 | Rules of Ownership

Hasbeen Company completed its inventory count. It arrived at a total inventory value of \$200,000. As a new member of Hasbeen's accounting department, you have been given the information listed below. Discuss how this information affects the reported cost of inventory.

1. Hasbeen included in the inventory goods held on consignment for Falls Co., costing \$15,000.
2. The company did not include in the count purchased goods of \$10,000 which were in transit (terms: FOB shipping point).
3. The company did not include in the count sold inventory with a cost of \$12,000 which was in transit (terms: FOB shipping point).

Solution

The goods of \$15,000 held on consignment should be deducted from the inventory count. The goods of \$10,000 purchased FOB shipping point should be added to the inventory count. Sold goods of \$12,000 which were in transit FOB shipping point should not be included in the ending inventory. Thus, inventory should be carried at \$195,000 (\$200,000 – \$15,000 + \$10,000).

Related exercise material: **BE6.1, BE6.2, DO IT! 6.1, E6.1, E6.2, and E6.3.**

ACTION PLAN

- Apply the rules of ownership to goods held on consignment.
- Apply the rules of ownership to goods in transit.

Inventory Methods and Financial Effects

LEARNING OBJECTIVE 2

Apply inventory cost flow methods and discuss their financial effects.

Inventory is accounted for at cost. Cost includes all expenditures necessary to acquire goods and place them in a condition ready for sale. For example, freight costs incurred to acquire inventory are added to the cost of inventory, but the cost of shipping goods to a customer is a selling expense.

After a company has determined the quantity of units of inventory, it applies unit costs to the quantities to compute the total cost of the inventory and the cost of goods sold. This process can be complicated if a company has purchased inventory items at different times and at different prices.

For example, assume that Crivitz TV Company purchases three identical 50-inch TVs on different dates at costs of \$720, \$750, and \$800. During the year, Crivitz sold two TVs at \$1,200 each. These facts are summarized in [Illustration 6.3](#).

Purchases

February 3	1 TV	at	\$720
March 5	1 TV	at	\$750
May 22	1 TV	at	\$800

Sales

June 1	2 TVs	for	\$2,400 (\$1,200 × 2)
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ILLUSTRATION 6.3

Data for inventory costing example

Cost of goods sold will differ depending on which two TVs the company sold. For example, it might be \$1,470 (\$720 + \$750), or \$1,520 (\$720 + \$800), or \$1,550 (\$750 + \$800). In this section, we discuss alternative costing methods available to Crivitz.

Specific Identification

If Crivitz can positively identify which particular units it sold and which are still in ending inventory, it can use the [specific identification method](#) of inventory costing. For example, if Crivitz sold the TVs it purchased on February 3 and May 22, then its cost of goods sold is \$1,520 (\$720 + \$800), and its ending inventory is \$750 (see [Illustration 6.4](#)). Using this method, companies can accurately determine ending inventory and cost of goods sold.

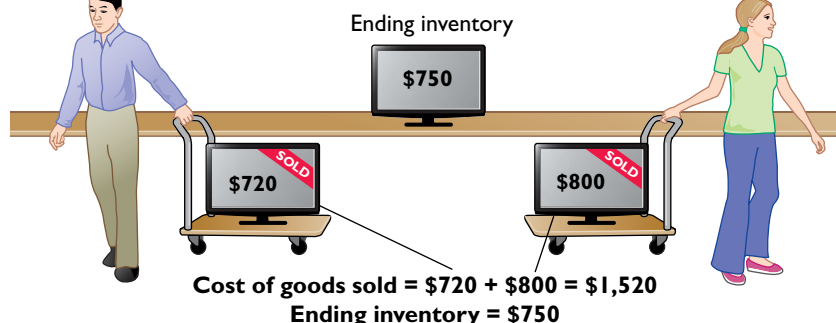


ILLUSTRATION 6.4

Specific identification method

ETHICS NOTE

A major disadvantage of the specific identification method is that management may be able to manipulate net income. For example, it can boost net income by selling units purchased at a low cost, or reduce net income by selling units purchased at a high cost.

Specific identification requires that companies keep records of the original cost of each individual inventory item. Historically, specific identification was possible only when a company sold a limited variety of high-unit-cost items that could be identified clearly from the time of purchase through the time of sale. Examples of such products are cars, pianos, or expensive antiques.

Today, bar coding, electronic product codes, and radio frequency identification make it theoretically possible to do specific identification with nearly any type of product. The reality is, however, that this practice is still relatively rare (see **Ethics Note**). Instead, rather than keep track of the cost of each particular item sold, most companies make assumptions, called **cost flow assumptions**, about which units were sold.

Cost Flow Assumptions

Because specific identification is often impractical, other cost flow methods are permitted. These differ from specific identification in that they **assume** flows of costs that may be unrelated to the physical flow of goods. There are three assumed cost flow methods:

1. First-in, first-out (FIFO)
2. Last-in, first-out (LIFO)
3. Average-cost

There is no accounting requirement that the cost flow assumption be consistent with the physical movement of the goods. Company management selects the appropriate cost flow method.

To demonstrate the three cost flow methods, we will use a **periodic** inventory system. We assume a periodic system because **very few companies use perpetual LIFO, FIFO, or average-cost** to cost their inventory and related cost of goods sold. Instead, companies that use perpetual systems often use an assumed cost (called a standard cost) to record cost of goods sold at the time of sale. Then, at the end of the period when they count their inventory, they **recalculate cost of goods sold using periodic FIFO, LIFO, or average-cost** as shown in this chapter and adjust cost of goods sold to this recalculated number.¹

To illustrate the three inventory cost flow methods, we will use the data for Houston Electronics' Astro condensers, shown in **Illustration 6.5**.

ILLUSTRATION 6.5**Data for Houston Electronics**

Houston Electronics Astro Condensers				
Date	Explanation	Units	Unit Cost	Total Cost
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total units available for sale	1,000		\$12,000
	Units in ending inventory	450		
	Units sold	550		

The cost of goods sold formula in a periodic system is:

Beginning Inventory	+	Cost of Goods Purchased	−	Ending Inventory	=	Cost of Goods Sold
--------------------------------	----------	--	----------	-----------------------------	----------	-----------------------------------

¹Also, some companies use a perpetual system to keep track of units, but they do not make an entry for perpetual cost of goods sold. In addition, firms that employ LIFO tend to use **dollar-value LIFO**, a method discussed in upper-level courses. FIFO periodic and FIFO perpetual give the same result. Therefore, companies should not incur the additional cost to use FIFO perpetual. Few companies use perpetual average-cost because of the added cost of recordkeeping. Finally, for instructional purposes, we believe it is easier to demonstrate the cost flow assumptions under the periodic system, which makes it more pedagogically appropriate.

Houston Electronics had a total of 1,000 units available to sell during the period (beginning inventory plus purchases). The total cost of these 1,000 units is \$12,000, referred to as **cost of goods available for sale**. A physical inventory taken at December 31 determined that there were 450 units in ending inventory. Therefore, Houston sold 550 units (1,000 – 450) during the period. To determine the cost of the 550 units that were sold (the cost of goods sold), we assign a cost to the ending inventory and subtract that value from the cost of goods available for sale. The value assigned to the ending inventory **depends on which cost flow method we use**. No matter which cost flow assumption we use, though, the sum of cost of goods sold plus the cost of the ending inventory must equal the cost of goods available for sale—in this case, \$12,000.

First-In, First-Out (FIFO)

The **first-in, first-out (FIFO) method** assumes that the **earliest goods** purchased are the first to be sold. FIFO often parallels the actual physical flow of merchandise. That is, it generally is good business practice to sell the oldest units first. Under the FIFO method, therefore, the **costs** of the earliest goods purchased are the first to be recognized in determining cost of goods sold. (This does not necessarily mean that the oldest units **are** sold first, but that the costs of the oldest units are **recognized** first. In a bin of picture hangers at the hardware store, for example, no one really knows, nor would it matter, which hangers are sold first.) **Illustration 6.6** shows the allocation of the cost of goods available for sale at Houston Electronics under FIFO (see **Helpful Hint**).

COST OF GOODS AVAILABLE FOR SALE				
Date	Explanation	Units	Unit Cost	Total Cost
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total	1,000		\$12,000

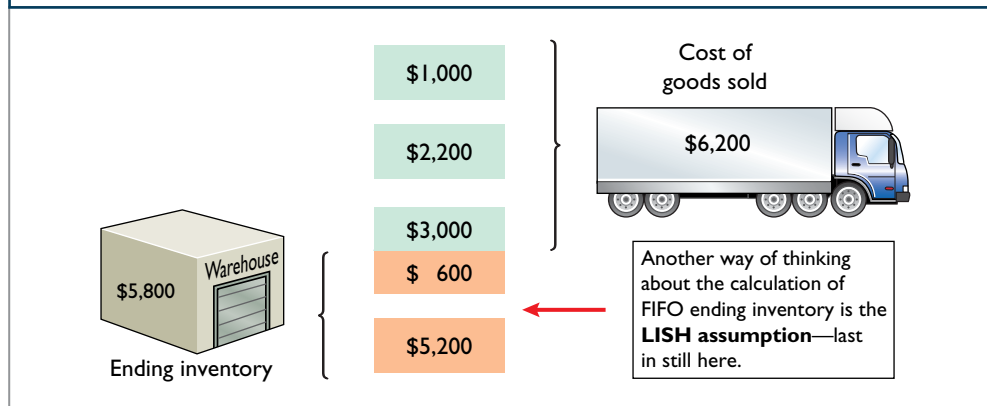
STEP 1: ENDING INVENTORY			STEP 2: COST OF GOODS SOLD		
Date	Units	Unit Cost	Total Cost		
Nov. 27	400	\$13	\$5,200	Cost of goods available for sale	\$12,000
Aug. 24	50	12	600	Less: Ending inventory	5,800
Total	450		\$5,800	Cost of goods sold	\$ 6,200

ILLUSTRATION 6.6

Allocation of costs—FIFO method

HELPFUL HINT

Note the sequencing of the allocation: (1) compute ending inventory, and (2) determine cost of goods sold.



Under FIFO, since it is assumed that the first goods purchased were the first goods sold, ending inventory is based on the prices of the most recent units purchased. That is, **under FIFO, companies obtain the cost of the ending inventory by taking the unit cost**

of the most recent purchase and working backward until all units of inventory have been costed. In this example, Houston Electronics prices the 450 units of ending inventory using the **most recent** prices. The last purchase was 400 units at \$13 on November 27. The remaining 50 units are priced using the unit cost of the second most recent purchase, \$12, on August 24. Next, Houston Electronics calculates cost of goods sold by subtracting the cost of the units **not sold** (ending inventory) from the cost of all goods available for sale.

Illustration 6.7 demonstrates that companies also can calculate cost of goods sold by pricing the 550 units sold using the prices of the first 550 units acquired. Note that of the 300 units purchased on August 24, only 250 units are assumed sold. This agrees with our calculation of the cost of ending inventory, where 50 of these units were assumed unsold and thus included in ending inventory.

ILLUSTRATION 6.7

Proof of cost of goods sold

	Date	Units	Unit Cost	Total Cost
	Jan. 1	100	\$10	\$1,000
	Apr. 15	200	11	2,200
	Aug. 24	250	12	3,000
	Total	550		\$6,200

Last-In, First-Out (LIFO)

The **last-in, first-out (LIFO) method** assumes that the **latest goods** purchased are the first to be sold. LIFO seldom coincides with the actual physical flow of inventory. (Exceptions include goods stored in piles, such as coal or hay, where goods are removed from the top of the pile as they are sold.) Under the LIFO method, the **costs** of the latest goods purchased are the first to be recognized in determining cost of goods sold. **Illustration 6.8** shows the allocation of the cost of goods available for sale at Houston Electronics under LIFO.

ILLUSTRATION 6.8

Allocation of costs—LIFO method

COST OF GOODS AVAILABLE FOR SALE					
Date	Explanation	Units	Unit Cost	Total Cost	
Jan. 1	Beginning inventory	100	\$10	\$ 1,000	
Apr. 15	Purchase	200	11	2,200	
Aug. 24	Purchase	300	12	3,600	
Nov. 27	Purchase	400	13	5,200	
	Total	1,000		\$12,000	
STEP 1: ENDING INVENTORY				STEP 2: COST OF GOODS SOLD	
Date	Units	Unit Cost	Total Cost		
Jan. 1	100	\$10	\$1,000	Cost of goods available for sale	\$12,000
Apr. 15	200	11	2,200	Less: Ending inventory	5,000
Aug. 24	150	12	1,800	Cost of goods sold	\$ 7,000
Total	450		\$5,000		

Under LIFO, since it is assumed that the first goods sold were those that were most recently purchased, ending inventory is based on the prices of the oldest units purchased. That is, **under LIFO, companies obtain the cost of the ending inventory by taking the unit cost of the earliest goods available for sale and working forward until all units of inventory have been costed.** In this example, Houston Electronics prices the 450 units of ending inventory using the **earliest** prices. The first purchase was 100 units at \$10 in the January 1 beginning inventory. Then, 200 units were purchased at \$11. The remaining 150 units needed are priced at \$12 per unit (August 24 purchase). Next, Houston Electronics calculates cost of goods sold by subtracting the cost of the units **not sold** (ending inventory) from the cost of all goods available for sale.

Illustration 6.9 demonstrates that companies also can calculate cost of goods sold by pricing the 550 units sold using the prices of the last 550 units acquired. Note that of the 300 units purchased on August 24, only 150 units are assumed sold. This agrees with our calculation of the cost of ending inventory, where 150 of these units were assumed unsold and thus included in ending inventory.

Date	Units	Unit Cost	Total Cost
Nov. 27	400	\$13	\$5,200
Aug. 24	150	12	1,800
Total	550		\$7,000

ILLUSTRATION 6.9

Proof of cost of goods sold

Under a periodic inventory system, which we are using here, **all goods purchased during the period are assumed to be available for the first sale, regardless of the date of purchase.**

Average-Cost

The **average-cost method** allocates the cost of goods available for sale on the basis of the **weighted-average unit cost** incurred. The average-cost method assumes that goods are similar in nature. **Illustration 6.10** presents the formula and a sample computation of the weighted-average unit cost.

Cost of Goods Available for Sale	÷	Total Units Available for Sale	=	Weighted-Average Unit Cost
\$12,000	÷	1,000	=	\$12

ILLUSTRATION 6.10

Formula for weighted-average unit cost

The company then applies the weighted-average unit cost to the units on hand to determine the cost of the ending inventory. **Illustration 6.11** shows the allocation of the cost of goods available for sale at Houston Electronics using average-cost.

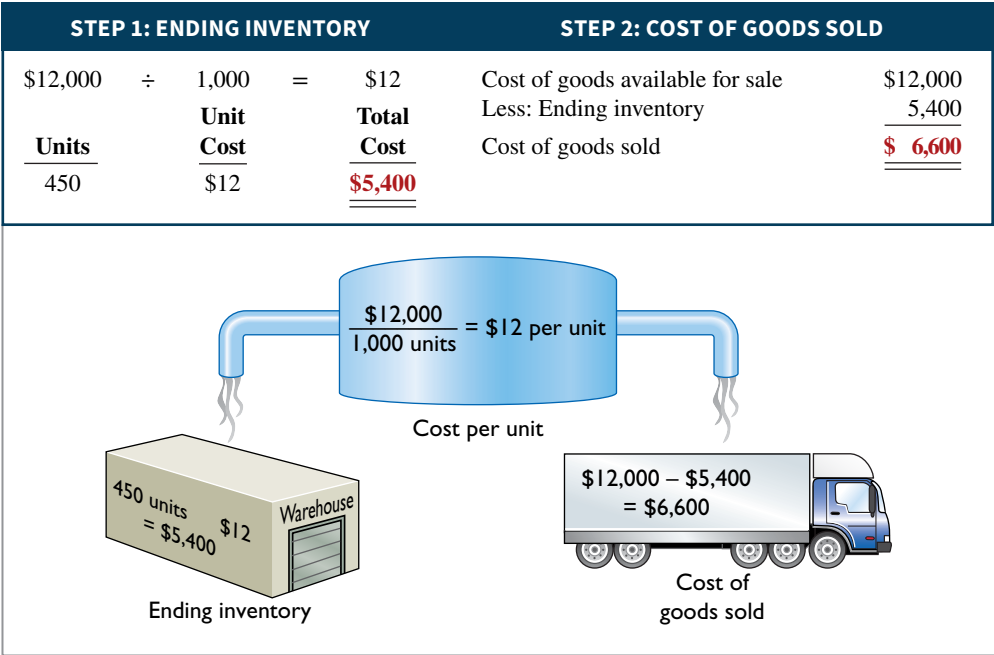
COST OF GOODS AVAILABLE FOR SALE				
Date	Explanation	Units	Unit Cost	Total Cost
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total	1,000		\$12,000

ILLUSTRATION 6.11

Allocation of costs—average-cost method

ILLUSTRATION 6.11

(Continued)



We can verify the cost of goods sold under this method by multiplying the units sold times the weighted-average unit cost ($550 \times \$12 = \$6,600$). Note that this method does **not** use the average of the unit costs. That average is $\$11.50$ ($\$10 + \$11 + \$12 + \$13 = \$46$; $\$46 \div 4$). The average-cost method instead uses the average **weighted by** the quantities purchased at each unit cost.

Financial Statement and Tax Effects of Cost Flow Methods

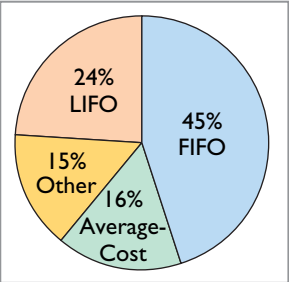


ILLUSTRATION 6.12

Use of cost flow methods in major U.S. companies

Each of the three assumed cost flow methods is acceptable for use. For example, **Reebok International Ltd.** and **Wendy's International** currently use the FIFO method of inventory costing. **Campbell Soup Company**, **Kroger**, and **Walgreen Drugs** use LIFO for part or all of their inventory. **Bristol-Myers Squibb**, **Starbucks**, and **Motorola** use the average-cost method. In fact, a company may also use more than one cost flow method at the same time. **Stanley Black & Decker Manufacturing Company**, for example, uses LIFO for domestic inventories and FIFO for foreign inventories. **Illustration 6.12** shows the use of the three cost flow methods in 500 large U.S. companies.

The reasons companies adopt different inventory cost flow methods are varied, but they usually involve one of three factors: (1) income statement effects, (2) balance sheet effects, or (3) tax effects.

Income Statement Effects

To understand why companies might choose a particular cost flow method, let's examine the effects of the different cost flow assumptions on the financial statements of Houston Electronics. The condensed income statements in **Illustration 6.13** assume that Houston sold its 550 units for \$18,500, had operating expenses of \$9,000, and is subject to an income tax rate of 30%.

Note the cost of goods available for sale (\$12,000) is the same under each of the three inventory cost flow methods. However, the ending inventories and the costs of goods sold are different. This difference is due to the unit costs that the company allocated to cost of goods sold and to ending inventory. Each dollar of difference in ending inventory results in a corresponding dollar difference in income before income taxes. For Houston, an \$800 difference exists between FIFO and LIFO cost of goods sold.

Houston Electronics Condensed Income Statements

ILLUSTRATION 6.13

Comparative effects of cost flow methods

	FIFO	LIFO	Average-Cost
Sales revenue	\$18,500	\$18,500	\$18,500
Beginning inventory	1,000	1,000	1,000
Purchases	11,000	11,000	11,000
Cost of goods available for sale	12,000	12,000	12,000
Ending inventory	5,800	5,000	5,400
Cost of goods sold	6,200	7,000	6,600
Gross profit	12,300	11,500	11,900
Operating expenses	9,000	9,000	9,000
Income before income taxes*	3,300	2,500	2,900
Income tax expense (30%)	990	750	870
Net income	\$ 2,310	\$ 1,750	\$ 2,030

*We are assuming that Houston Electronics is a corporation, and corporations are required to pay income taxes.

In periods of changing prices, the cost flow assumption can have significant impacts both on income and on evaluations of income, such as the following.

1. In a period of inflation, FIFO produces a higher net income because lower unit costs of the first units purchased are matched against revenue.
2. In a period of inflation, LIFO produces a lower net income because higher unit costs of the last goods purchased are matched against revenue.
3. If prices are falling, the results from the use of FIFO and LIFO are reversed. FIFO will report the lowest net income and LIFO the highest.
4. Regardless of whether prices are rising or falling, average-cost produces net income between FIFO and LIFO.

As shown in the Houston example (Illustration 6.13), in a period of rising prices FIFO reports the highest net income (\$2,310) and LIFO the lowest (\$1,750); average-cost falls between these two amounts (\$2,030).

To management, higher net income is an advantage. It causes external users to view the company more favorably. In addition, management bonuses, if based on net income, will be higher. Therefore, when prices are rising (which is usually the case), companies tend to prefer FIFO because it results in higher net income.

Others believe that LIFO presents a more realistic net income number. That is, LIFO matches the more recent costs against current revenues to provide a better measure of net income. During periods of inflation, many challenge the quality of non-LIFO earnings, noting that failing to match current costs against current revenues leads to an understatement of cost of goods sold and an overstatement of net income. As some indicate, net income computed using FIFO creates “**paper or phantom profits**”—that is, earnings that do not really exist.

Balance Sheet Effects

A major advantage of the FIFO method is that in a period of inflation, the costs allocated to ending inventory will approximate their current cost. For example, for Houston Electronics, 400 of the 450 units in the ending inventory are costed under FIFO at the higher November 27 unit cost of \$13.

Conversely, a major shortcoming of the LIFO method is that in a period of inflation, the costs allocated to ending inventory may be significantly understated in terms of current cost. The understatement becomes greater over prolonged periods of inflation if the inventory includes goods purchased in one or more prior accounting periods. For example, **Caterpillar** has used LIFO for more than 50 years. Its balance sheet shows

ending inventory of \$9,700 million. But the inventory's actual current cost if FIFO had been used is \$12,189 million.

Tax Effects

HELPFUL HINT

A tax rule, often referred to as the LIFO conformity rule, requires that if companies use LIFO for tax purposes they must also use it for financial reporting purposes. This means that if a company chooses the LIFO method to reduce its tax bills, it will also have to report lower net income in its financial statements.

We have seen that both inventory on the balance sheet and net income on the income statement are higher when companies use FIFO in a period of inflation. Yet, many companies have selected LIFO. Why? The reason is that LIFO results in the lowest income taxes (because of lower net income) during times of rising prices (see **Helpful Hint**). For example, at Houston Electronics, income taxes are \$750 under LIFO, compared to \$990 under FIFO. The tax savings of \$240 makes more cash available for use in the business.

Using Inventory Cost Flow Methods Consistently

Whatever cost flow method a company chooses, it should use that method consistently from one accounting period to another. This approach is often referred to as the **consistency concept**, which means that a company uses the same accounting principles and methods from year to year. Consistent application enhances the comparability of financial statements over successive time periods. In contrast, using the FIFO method one year and the LIFO method the next year would make it difficult to compare the net incomes of the two years.

Although consistent application is preferred, it does not mean that a company may never change its inventory costing method. When a company adopts a different method, it should disclose in the financial statements the change and its effects on net income. **Illustration 6.14** shows a typical disclosure, using information from recent financial statements of **Quaker Oats** (now a unit of **PepsiCo**).

ILLUSTRATION 6.14

Disclosure of change in cost flow method

Real
World

Quaker Oats Notes to the Financial Statements

Note 1: Effective July 1, the Company adopted the LIFO cost flow assumption for valuing the majority of U.S. Grocery Products inventories. The Company believes that the use of the LIFO method better matches current costs with current revenues. The effect of this change on the current year was to decrease net income by \$16.0 million.

International Insight ExxonMobil Corporation



Bloomberg/Getty Images

Is LIFO Fair?

ExxonMobil Corporation, like many U.S. companies, uses LIFO to value its inventory for financial reporting and tax purposes. In one year, this resulted in a cost of goods sold figure that was \$5.6 billion higher than under FIFO. By increasing cost of goods sold, ExxonMobil reduces net income, which reduces taxes. Critics say that LIFO provides an unfair "tax dodge." As Congress looks for more sources of tax revenue, some lawmakers favor the

elimination of LIFO. Supporters of LIFO argue that the method is conceptually sound because it matches current costs with current revenues. In addition, they point out that this matching provides protection against inflation.

International accounting standards do not allow the use of LIFO. Because of this, the net income of foreign oil companies such as **BP** and **Royal Dutch Shell** are not directly comparable to U.S. companies, which can make analysis difficult.

Source: David Reilly, "Big Oil's Accounting Methods Fuel Criticism," *Wall Street Journal* (August 8, 2006), p. C1.

What are the arguments for and against the use of LIFO? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 2 | Cost Flow Methods

The accounting records of Shumway Ag Implements show the following data.

Beginning inventory	4,000 units at \$ 3
Purchases	6,000 units at \$ 4
Sales	7,000 units at \$12

Determine the cost of goods sold during the period under a periodic inventory system using (a) the FIFO method, (b) the LIFO method, and (c) the average-cost method.

Solution

Cost of goods available for sale = $(4,000 \times \$3) + (6,000 \times \$4) = \$36,000$

Ending inventory = $10,000 - 7,000 = 3,000$ units

a. FIFO: $\$36,000 - (3,000 \times \$4) = \$24,000$

b. LIFO: $\$36,000 - (3,000 \times \$3) = \$27,000$

c. Average cost per unit: $[(4,000 @ \$3) + (6,000 @ \$4)] \div 10,000 = \$3.60$

Average-cost: $\$36,000 - (3,000 \times \$3.60) = \$25,200$

Related exercise material: **BE6.3, BE6.4, BE6.5, BE6.6, DO IT! 6.2, E6.4, E6.5, E6.6, E6.7, E6.8, and E6.9.**

ACTION PLAN

- Understand the periodic inventory system.
- Allocate costs between goods sold and goods on hand (ending inventory) for each cost flow method.
- Compute cost of goods sold for each method.

Effects of Inventory Errors

LEARNING OBJECTIVE 3

Indicate the effects of inventory errors on the financial statements.

Unfortunately, errors occasionally occur in accounting for inventory. In some cases, errors are caused by failure to count or price the inventory correctly. In other cases, errors occur because companies do not properly recognize the transfer of legal title to goods that are in transit. When errors occur, they affect both the income statement and the balance sheet.

Income Statement Effects

The ending inventory of one period automatically becomes the beginning inventory of the next period. Thus, inventory errors affect the computation of cost of goods sold and net income in two periods.

The effects on cost of goods sold can be computed by first entering incorrect data in the formula in **Illustration 6.15** and then substituting the correct data.

Beginning Inventory	+	Cost of Goods Purchased	-	Ending Inventory	=	Cost of Goods Sold
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ILLUSTRATION 6.15

Formula for cost of goods sold

If **beginning** inventory is understated, cost of goods sold will be understated. If **ending** inventory is understated, cost of goods sold will be overstated. **Illustration 6.16** shows the effects of inventory errors on the current year's income statement.

ILLUSTRATION 6.16**Effects of inventory errors on current year's income statement**

<u>When Inventory Error:</u>	<u>Cost of Goods Sold Is:</u>	<u>Net Income Is:</u>
Understates beginning inventory	Understated	Overstated
Overstates beginning inventory	Overstated	Understated
Understates ending inventory	Overstated	Understated
Overstates ending inventory	Understated	Overstated

ETHICS NOTE

Inventory fraud increases during recessions. Such fraud includes pricing inventory at amounts in excess of its actual value, or claiming to have inventory when no inventory exists. Inventory fraud usually overstates ending inventory, thereby understating cost of goods sold and creating higher income.

An error in the ending inventory of the current period will have a **reverse effect on net income of the next accounting period**. **Illustration 6.17** shows this effect. Note that the understatement of ending inventory in 2019 results in an understatement of beginning inventory in 2020 and an overstatement of net income in 2020.

Over the two years, though, total net income is correct because the errors **offset each other**. Notice that total income using incorrect data is \$35,000 (\$22,000 + \$13,000), which is the same as the total income of \$35,000 (\$25,000 + \$10,000) using correct data. Also note in this example that an error in the beginning inventory does not result in a corresponding error in the ending inventory for that period. The correctness of the ending inventory depends entirely on the accuracy of taking and costing the inventory at the balance sheet date under the periodic inventory system (see **Ethics Note**).

ILLUSTRATION 6.17 Effects of inventory errors on two years' income statements

Sample Company Condensed Income Statements					
	2019		2020		
	<u>Incorrect</u>	<u>Correct</u>	<u>Incorrect</u>	<u>Correct</u>	
Sales revenue	\$80,000	\$80,000	\$90,000	\$90,000	
Beginning inventory	\$ 20,000	20,000	\$12,000	\$15,000	
Cost of goods purchased	40,000	40,000	68,000	68,000	
Cost of goods available for sale	60,000	60,000	80,000	83,000	
Ending inventory	12,000	15,000	23,000	23,000	
Cost of goods sold	48,000	45,000	57,000	60,000	
Gross profit	32,000	35,000	33,000	30,000	
Operating expenses	10,000	10,000	20,000	20,000	
Net income	\$22,000	\$25,000	\$13,000	\$10,000	
	\$(3,000)		\$3,000		
	Net income understated		Net income overstated		
The errors cancel. Thus, the combined total income for the 2-year period is correct.					

Balance Sheet Effects

Companies can determine the effect of ending inventory errors on the balance sheet by using the basic accounting equation: Assets = Liabilities + Owner's Equity. Errors in the ending inventory have the effects shown in **Illustration 6.18**.

ILLUSTRATION 6.18**Effects of ending inventory errors on balance sheet**

<u>Ending Inventory Error</u>	<u>Assets</u>	<u>Liabilities</u>	<u>Owner's Equity</u>
Overstated	Overstated	No effect	Overstated
Understated	Understated	No effect	Understated

The effect of an error in ending inventory on the subsequent period was shown in Illustration 6.17. Note that if the error is not corrected, the combined total net income for the two periods would be correct. Thus, total owner's equity reported on the balance sheet at the end of 2020 will also be correct.

DO IT! 3 | Inventory Errors

Visual Company overstated its 2019 ending inventory by \$22,000. Determine the impact this error has on ending inventory, cost of goods sold, and owner's equity in 2019 and 2020.

Solution

	2019	2020
Ending inventory	\$22,000 overstated	No effect
Cost of goods sold	\$22,000 understated	\$22,000 overstated
Owner's equity	\$22,000 overstated	No effect

Related exercise material: BE6.7, DO IT! 6.3, E6.10, and E6.11.

ACTION PLAN

- An ending inventory error in one period will have an equal and opposite effect on cost of goods sold and net income in the next period.
- After two years, the errors have offset each other.

Inventory Statement Presentation and Analysis

LEARNING OBJECTIVE 4

Explain the statement presentation and analysis of inventory.

Presentation

As indicated in Chapter 5, inventory is classified in the balance sheet as a current asset immediately below receivables. In a multiple-step income statement, cost of goods sold is subtracted from net sales. There also should be disclosure of (1) the major inventory classifications, (2) the basis of accounting (cost, or lower-of-cost-or-net realizable value), and (3) the cost method (FIFO, LIFO, or average-cost).

Lower-of-Cost-or-Net Realizable Value

The value of inventory for companies selling high-technology or fashion goods can drop very quickly due to continual changes in technology or fashions. These circumstances sometimes call for inventory valuation methods other than those presented so far. For example, purchasing managers at **Ford** at one time decided to make a large purchase of palladium, a precious metal used in vehicle emission devices. They made this purchase because they feared a future shortage. The shortage did not materialize, and by the end of the year the price of palladium had plummeted. Ford's inventory was then worth \$1 billion less than its original cost. Do you think Ford's inventory should have been stated at cost, in accordance with the historical cost principle, or at its lower net realizable value?

As you probably reasoned, this situation requires a departure from the cost basis of accounting. When the value of inventory is lower than its cost, companies must "write down" the inventory to its net realizable value. This is done by valuing the inventory at the **lower-of-cost-or-net realizable value (LCNRV)** in the period in which the price decline occurs. LCNRV is an example of the accounting concept of **conservatism**, which means

that the best choice among accounting alternatives is the method that is least likely to overstate assets and net income.

Under the LCNRV basis, **net realizable value** refers to the net amount that a company expects to realize (receive) from the sale of inventory. Specifically, net realizable value is the estimated selling price in the normal course of business, less estimated costs to complete and sell.

Companies apply LCNRV to the items in inventory after they have used one of the inventory costing methods (specific identification, FIFO, or average-cost) to determine cost.² To illustrate the application of LCNRV, assume that Ken Tuckie TV has the following lines of merchandise with costs and net realizable values as indicated. LCNRV produces the results shown in **Illustration 6.19**. Note that the amounts shown in the final column are the lower-of-cost-or-net realizable value amounts for each item.

ILLUSTRATION 6.19

Computation of lower-of-cost-or-net realizable value

	Units	Cost per Unit	Net Realizable Value per Unit	Lower-of-Cost-or-Net Realizable Value
Flat-screen TVs	100	\$600	\$550	\$ 55,000 (\$550 × 100)
Satellite radios	500	90	104	45,000 (\$90 × 500)
DVD recorders	850	50	48	40,800 (\$48 × 850)
DVDs	3,000	5	6	15,000 (\$5 × 3,000)
Total inventory				<u>\$155,800</u>

Analysis

The amount of inventory carried by a company has significant economic consequences. And inventory management is a double-edged sword that requires constant attention. On the one hand, management wants to have a great variety and quantity available so that customers have a wide selection and items are always in stock. But, such a policy may incur high carrying costs (e.g., investment, storage, insurance, obsolescence, and damage). On the other hand, low inventory levels lead to stock-outs and lost sales. Common ratios used to manage and evaluate inventory levels are inventory turnover and a related measure, days in inventory.

Inventory turnover measures the number of times on average the inventory is sold during the period. Its purpose is to measure the liquidity of the inventory. The inventory turnover is computed by dividing cost of goods sold by the average inventory during the period. Unless seasonal factors are significant, average inventory can be computed from the beginning and ending inventory balances. For example, **Wal-Mart** reported in its January 31, 2016, annual report a beginning inventory of \$45,141 million, an ending inventory of \$44,469 million, and cost of goods sold for the year ended January 31, 2016, of \$360,984 million. The inventory turnover formula and computation for Wal-Mart are shown in **Illustration 6.20**.

ILLUSTRATION 6.20

Inventory turnover formula and computation for Wal-Mart

Cost of Goods Sold	÷	Average Inventory	=	Inventory Turnover
\$360,984	÷	$\frac{\$44,469 + \$45,141}{2}$	=	8.1 times

A variant of the inventory turnover is **days in inventory**. This measures the average number of days inventory is held. It is calculated as 365 divided by the inventory turnover. For example, Wal-Mart's inventory turnover of 8.1 times divided into 365 is approximately 45.1 days. This is the approximate time that it takes a company to sell the inventory once it arrives at the store.

²Special rules apply to companies that use LIFO, which are discussed in more advanced courses.

There are typical levels of inventory in every industry. Companies that are able to keep their inventory at lower levels and higher turnovers and still satisfy customer needs are the most successful.

Accounting Across the Organization Sony



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Too Many TVs or Too Few?

Financial analysts closely monitor the inventory management practices of companies. For example, some analysts following **Sony** expressed concern because the company built up its inventory of televisions in an attempt to sell 25 million liquid crystal display (LCD) TVs—a 60% increase over the prior year. A year earlier, Sony had cut its inventory

levels so that its quarterly days in inventory was down to 38 days, compared to 61 days for the same quarter a year before that. But in the next year, as a result of its inventory build-up, days in inventory rose to 59 days. Management said that it didn't think that Sony's inventory levels were too high. However, analysts were concerned that the company would have to engage in very heavy discounting in order to sell off its inventory. Analysts noted that the losses from discounting can be "punishing."

Source: Daisuke Wakabayashi, "Sony Pledges to Corral Inventory," *Wall Street Journal Online* (November 2, 2010).

For Sony, what are the advantages and disadvantages of having a low days in inventory measure? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 4 | LCMRV and Inventory Turnover

- a. Tracy Company sells three different types of home heating stoves (gas, wood, and pellet). The cost and net realizable value of its inventory of stoves are as follows.

	Cost	Net Realizable Value
Gas	\$ 84,000	\$ 79,000
Wood	250,000	280,000
Pellet	112,000	101,000

Determine the value of the company's inventory under the lower-of-cost-or-net realizable value approach.

Solution

The lowest value for each inventory type is gas \$79,000, wood \$250,000, and pellet \$101,000. The total inventory value is the sum of these amounts, \$430,000.

- b. Early in 2020, Westmoreland Company switched to a just-in-time inventory system. Its sales revenue, cost of goods sold, and inventory amounts for 2019 and 2020 are shown below.

	2019	2020
Sales revenue	\$2,000,000	\$1,800,000
Cost of goods sold	1,000,000	910,000
Beginning inventory	290,000	210,000
Ending inventory	210,000	50,000

Determine the inventory turnover and days in inventory for 2019 and 2020. Discuss the changes in the amount of inventory, the inventory turnover and days in inventory, and the amount of sales across the two years.

ACTION PLAN

- Determine whether cost or net realizable value is lower for each inventory type.
- Sum the lowest value of each inventory type to determine the total value of inventory.

ACTION PLAN

- To find the inventory turnover, divide cost of goods sold by average inventory.
- To determine days in inventory, divide 365 days by the inventory turnover.

Solution

	<u>2019</u>		<u>2020</u>	
Inventory turnover	$\frac{\$1,000,000}{(\$290,000 + \$210,000)/2} = 4$		$\frac{\$910,000}{(\$210,000 + \$50,000)/2} = 7$	
Days in inventory	$365 \div 4 = 91.3 \text{ days}$		$365 \div 7 = 52.1 \text{ days}$	

ACTION PLAN (CONT.)

- Just-in-time inventory reduces the amount of inventory on hand, which reduces carrying costs. Reducing inventory levels by too much has potential negative implications for sales.

The company experienced a very significant decline in its ending inventory as a result of the just-in-time inventory. This decline improved its inventory turnover and its days in inventory. However, its sales declined by 10%. It is possible that this decline was caused by the dramatic reduction in the amount of inventory that was on hand, which increased the likelihood of “stock-outs.” To determine the optimal inventory level, management must weigh the benefits of reduced inventory against the potential lost sales caused by stock-outs.

Related exercise material: **BE6.8, BE6.9, DO IT! 6.4, E6.12, E6.13, E6.14, and E6.15.**

Appendix 6A**Inventory Methods and the Perpetual System****LEARNING OBJECTIVE *5**

Apply the inventory cost flow methods to perpetual inventory records.

What inventory cost flow methods can companies employ if they use a perpetual inventory system? Simple—they can use any of the inventory cost flow methods described in the chapter. To illustrate the application of the three assumed cost flow methods (FIFO, LIFO, and average-cost), we will use the data shown in **Illustration 6A.1** and in this chapter for Houston Electronics’ Astro condensers.

ILLUSTRATION 6A.1

Inventoriable units and costs

Houston Electronics Astro Condensers					
Date	Explanation	Units	Unit Cost	Total Cost	Balance in Units
1/1	Beginning inventory	100	\$10	\$ 1,000	100
4/15	Purchases	200	11	2,200	300
8/24	Purchases	300	12	3,600	600
9/10	Sale	550			50
11/27	Purchases	400	13	5,200	450
				<u>\$12,000</u>	

First-In, First-Out (FIFO)

Under perpetual FIFO, the company charges to cost of goods sold the cost of the earliest goods on hand **prior to each sale**. Therefore, the cost of goods sold on September 10 consists of the units on hand January 1 and the units purchased April 15 and August 24. **Illustration 6A.2** shows the inventory under a FIFO method perpetual system.

ILLUSTRATION 6A.2

Perpetual system—FIFO

Date	Purchases	Cost of Goods Sold	Balance (in units and cost)	
January 1			(100 @ \$10)	\$1,000
April 15	(200 @ \$11) \$2,200		(100 @ \$10) } (200 @ \$11) }	\$3,200
August 24	(300 @ \$12) \$3,600		(100 @ \$10) } (200 @ \$11) } (300 @ \$12) }	\$6,800
September 10		(100 @ \$10) (200 @ \$11) (250 @ \$12) <u>\$6,200</u>	(50 @ \$12)	\$ 600
November 27	(400 @ \$13) \$5,200		(50 @ \$12) } (400 @ \$13) }	\$5,800

Cost of goods sold

Ending inventory

The ending inventory in this situation is \$5,800, and the cost of goods sold is \$6,200 [(100 @ \$10) + (200 @ \$11) + (250 @ \$12)].

Compare Illustrations 6.6 and 6A.2. You can see that the results under FIFO in a perpetual system are the **same as in a periodic system**. In both cases, the ending inventory is \$5,800 and cost of goods sold is \$6,200. Regardless of the system, the first costs in are the costs assigned to cost of goods sold.

Last-In, First-Out (LIFO)

Under the LIFO method using a perpetual system, the company charges to cost of goods sold the cost of the most recent purchase prior to sale. Therefore, the cost of the goods sold on September 10 consists of all the units from the August 24 and April 15 purchases plus 50 of the units in beginning inventory. **Illustration 6A.3** shows the computation of the ending inventory under the LIFO method.

ILLUSTRATION 6A.3

Perpetual system—LIFO

Date	Purchases	Cost of Goods Sold	Balance (in units and cost)	
January 1			(100 @ \$10)	\$1,000
April 15	(200 @ \$11) \$2,200		(100 @ \$10) } (200 @ \$11) }	\$3,200
August 24	(300 @ \$12) \$3,600		(100 @ \$10) } (200 @ \$11) } (300 @ \$12) }	\$6,800
September 10		(300 @ \$12) (200 @ \$11) (50 @ \$10) <u>\$6,300</u>	(50 @ \$10)	\$ 500
November 27	(400 @ \$13) \$5,200		(50 @ \$10) } (400 @ \$13) }	\$5,700

Cost of goods sold

Ending inventory

The use of LIFO in a perpetual system will usually produce cost allocations that differ from those using LIFO in a periodic system. In a perpetual system, the latest units purchased **prior to each sale** are allocated to cost of goods sold. In contrast, in a periodic system, the latest units purchased **during the period** are allocated to cost of goods sold. Thus, when a purchase is made after the last sale, the LIFO periodic system will apply this purchase to the previous sale.

See Illustration 6.9, which shows the proof that the 400 units at \$13 purchased on November 27 applied to the sale of 550 units on September 10.

Under the LIFO perpetual system in Illustration 6A.3, the 400 units at \$13 purchased on November 27 are all applied to the ending inventory. The ending inventory in this LIFO perpetual illustration is \$5,700, and cost of goods sold is \$6,300, as compared to the LIFO periodic Illustration 6.8 where the ending inventory is \$5,000 and cost of goods sold is \$7,000.

Average-Cost

The average-cost method in a perpetual inventory system is called the **moving-average method**. Under this method, the company computes a new average **after each purchase**, by dividing the cost of goods available for sale by the units on hand. The average cost is then applied to (1) the units sold, to determine the cost of goods sold, and (2) the remaining units on hand, to determine the ending inventory amount. **Illustration 6A.4** shows the application of the moving-average cost method by Houston Electronics (computations of the moving-average unit cost are shown after Illustration 6A.4).

ILLUSTRATION 6A.4

Perpetual system—
moving-average method

Date	Purchases	Cost of Goods Sold	Balance (in units and cost)
January 1			(100 @ \$10) \$1,000
April 15	(200 @ \$11) \$2,200		(300 @ \$10.667) \$3,200
August 24	(300 @ \$12) \$3,600		(600 @ \$11.333) \$6,800
September 10		(550 @ \$11.333) \$6,233	(50 @ \$11.333) \$ 567
November 27	(400 @ \$13) \$5,200		(450 @ \$12.816) \$5,767

Cost of goods sold: \$6,233

Ending inventory: \$5,767

As indicated, Houston Electronics computes **a new average each time it makes a purchase**.

1. On April 15, after Houston buys 200 units for \$2,200, a total of 300 units costing \$3,200 ($\$1,000 + \$2,200$) are on hand. The average unit cost is \$10.667 ($\$3,200 \div 300$).
2. On August 24, after Houston buys 300 units for \$3,600, a total of 600 units costing \$6,800 ($\$1,000 + \$2,200 + \$3,600$) are on hand. The average cost per unit is \$11.333 ($\$6,800 \div 600$).
3. On September 10, to compute cost of goods sold, Houston uses this unit cost of \$11.333 until it makes another purchase, when the company computes a new unit cost. Accordingly, the unit cost of the 550 units sold on September 10 is \$11.333, and the total cost of goods sold is \$6,233.
4. On November 27, following the purchase of 400 units for \$5,200, there are 450 units on hand costing \$5,767 ($\$567 + \$5,200$) with a new average cost of \$12.816 ($\$5,767 \div 450$).

Compare this moving-average cost under the perpetual inventory system to Illustration 6.11, which shows the average-cost method under a periodic inventory system.

Appendix 6B

Estimating Inventories

LEARNING OBJECTIVE *6

Describe the two methods of estimating inventories.

In the chapter, we assumed that a company would be able to physically count its inventory. What if it cannot? What if the inventory were destroyed by fire or flood, for example? In that case, the company would use an estimate.

Two circumstances explain why companies sometimes estimate inventories. First, a casualty such as fire, flood, or earthquake may make it impossible to take a physical inventory. Second, managers may want monthly or quarterly financial statements, but a physical inventory is taken only annually. The need for estimating inventories occurs primarily with a periodic inventory system because of the absence of perpetual inventory records.

There are two widely used methods of estimating inventories: (1) the gross profit method, and (2) the retail inventory method.

Gross Profit Method

The **gross profit method** estimates the cost of ending inventory by applying a gross profit rate to net sales. This method is relatively simple but effective. Accountants, auditors, and managers frequently use the gross profit method to test the reasonableness of the ending inventory amount. It will detect large errors.

To use this method, a company needs to know its net sales, cost of goods available for sale, and gross profit rate. The company then can estimate its gross profit for the period.

Illustration 6B.1 shows the formulas for using the gross profit method.

Step 1:	Net Sales	—	Estimated Gross Profit	=	Estimated Cost of Goods Sold
Step 2:	Cost of Goods Available for Sale	—	Estimated Cost of Goods Sold	=	Estimated Cost of Ending Inventory

ILLUSTRATION 6B.1

Gross profit method formulas

To illustrate, assume that Kishwaukee Company wishes to prepare an income statement for the month of January. Its records show net sales of \$200,000, beginning inventory \$40,000, and cost of goods purchased \$120,000. In the preceding year, the company realized a 30% gross profit rate. It expects to earn the same rate this year. Given these facts and assumptions, Kishwaukee can compute the estimated cost of the ending inventory at January 31 under the gross profit method as shown in **Illustration 6B.2**.

Step 1:	
Net sales	\$200,000
Less: Estimated gross profit (30% × \$200,000)	60,000
Estimated cost of goods sold	\$140,000
Step 2:	
Beginning inventory	\$ 40,000
Cost of goods purchased	120,000
Cost of goods available for sale	160,000
Less: Estimated cost of goods sold	140,000
Estimated cost of ending inventory	\$ 20,000

ILLUSTRATION 6B.2

Example of gross profit method

The gross profit method is based on the assumption that the gross profit rate will remain constant. But, it may not remain constant, due to a change in merchandising policies or in market conditions. In such cases, the company should adjust the rate to reflect current operating conditions. In some cases, companies can obtain a more accurate estimate by applying this method on a department or product-line basis.

Note that companies should not use the gross profit method to prepare financial statements at the end of the year. These statements should be based on a physical inventory count.

Retail Inventory Method

A retail store such as **Home Depot**, **Ace Hardware**, or **Walmart** has thousands of different types of merchandise at low unit costs. In such cases, it is difficult and time-consuming to apply unit costs to inventory quantities. An alternative is to use the **retail inventory method** to estimate the cost of inventory. Most retail companies can establish a relationship between cost and sales price. The company then applies the cost-to-retail percentage to the ending inventory at retail prices to determine inventory at cost.

Under the retail inventory method, a company's records must show both the cost and retail value of the goods available for sale. **Illustration 6B.3** presents the formulas for using the retail inventory method.

ILLUSTRATION 6B.3

Retail inventory method formulas

Step 1:	Goods Available for Sale at Retail	—	Net Sales	=	Ending Inventory at Retail
Step 2:	Goods Available for Sale at Cost	÷	Goods Available for Sale at Retail	=	Cost-to-Retail Ratio
Step 3:	Ending Inventory at Retail	×	Cost-to-Retail Ratio	=	Estimated Cost of Ending Inventory

We can demonstrate the logic of the retail method by using unit-cost data. Assume that Ortiz Inc. has marked 10 units purchased at \$7 to sell for \$10 per unit. Thus, the cost-to-retail ratio is 70% ($\$70 \div \100). If four units remain unsold, their retail value is \$40 ($4 \times \10), and their cost is \$28 ($\$40 \times 70\%$). This amount agrees with the total cost of goods on hand on a per unit basis ($4 \times \$7$).

Illustration 6B.4 shows application of the retail method for Valley West. Note that it is not necessary to take a physical inventory to determine the estimated cost of goods on hand at any given time.

ILLUSTRATION 6B.4

Application of retail inventory method

	<u>At Cost</u>	<u>At Retail</u>
Beginning inventory	\$14,000	\$ 21,500
Goods purchased	61,000	78,500
Goods available for sale	<u>\$75,000</u>	<u>100,000</u>
Less: Net sales		<u>70,000</u>
Step (1) Ending inventory at retail =		\$ 30,000
Step (2) Cost-to-retail ratio = $\\$75,000 \div \\$100,000 = 75\%$		
Step (3) Estimated cost of ending inventory = $\\$30,000 \times 75\% = \\$22,500$		

HELPFUL HINT

In determining inventory at retail, companies use selling prices of the units.

The retail inventory method also facilitates taking a physical inventory at the end of the year. Valley West can value the goods on hand at the prices marked on the merchandise, and then apply the cost-to-retail ratio to the goods on hand at retail to determine the ending inventory at cost (see **Helpful Hint**).

The major disadvantage of the retail method is that it is an averaging technique. Thus, it may produce an incorrect inventory valuation if the mix of the ending inventory is not representative of the mix in the goods available for sale. Assume, for example, that the cost-to-retail ratio of 75% for Valley West consists of equal proportions of inventory items that have cost-to-retail ratios of 70%, 75%, and 80%. If the ending inventory contains only items with a 70% ratio, an incorrect inventory cost will result. Companies can minimize this problem by applying the retail method on a department or product-line basis.

Review and Practice

Learning Objectives Review

1 Discuss how to classify and determine inventory.

Merchandisers need only one inventory classification, merchandise inventory, to describe the different items that make up total inventory. Manufacturers, on the other hand, usually classify inventory into three categories: finished goods, work in process, and raw materials. To determine inventory quantities, manufacturers (1) take a physical inventory of goods on hand and (2) determine the ownership of goods in transit or on consignment.

2 Apply inventory cost flow methods and discuss their financial effects.

The primary basis of accounting for inventories is cost. Cost of goods available for sale includes (a) cost of beginning inventory and (b) cost of goods purchased. The inventory cost flow methods are specific identification and three assumed cost flow methods—FIFO, LIFO, and average-cost.

When prices are rising, the first-in, first-out (FIFO) method results in lower cost of goods sold and higher net income than the other methods. The last-in, first-out (LIFO) method results in the lowest income taxes. The reverse is true when prices are falling. In the balance sheet, FIFO results in an ending inventory that is closest to current value. Inventory under LIFO is the farthest from current value.

3 Indicate the effects of inventory errors on the financial statements.

In the income statement of the current year: (a) If beginning inventory is understated, net income is overstated. The reverse occurs if beginning inventory is overstated. (b) If ending inventory is overstated, net income is overstated. If ending inventory is understated, net income is understated. In the following period, its effect on net income for that period is reversed, and total net income for the two years will be correct.

In the balance sheet: Ending inventory errors will have the same effect on total assets and total owner's equity and no effect on liabilities.

4 Explain the statement presentation and analysis of inventory.

Inventory is classified in the balance sheet as a current asset immediately below receivables. There also should be disclosure of (1) the major inventory classifications, (2) the basis of accounting, and (3) the cost method.

Companies use the lower-of-cost-or-net realizable value (LCNRV) basis when the net realizable value is less than cost. Under LCNRV, companies recognize the loss in the period in which the price decline occurs.

The inventory turnover is cost of goods sold divided by average inventory. To convert it to average days in inventory, divide 365 days by the inventory turnover.

*5 Apply the inventory cost flow methods to perpetual inventory records.

Under FIFO and a perpetual inventory system, companies charge to cost of goods sold the cost of the earliest goods on hand prior to each sale. Under LIFO and a perpetual system, companies charge to cost of goods sold the cost of the most recent purchase prior to sale. Under the moving-average (average-cost) method and a perpetual system, companies compute a new average cost after each purchase.

*6 Describe the two methods of estimating inventories.

The two methods of estimating inventories are the gross profit method and the retail inventory method. Under the gross profit method, companies apply a gross profit rate to net sales to determine estimated gross profit and cost of goods sold. They then subtract estimated cost of goods sold from cost of goods available for sale to determine the estimated cost of the ending inventory.

Under the retail inventory method, companies compute a cost-to-retail ratio by dividing the cost of goods available for sale by the retail value of the goods available for sale. They then apply this ratio to the ending inventory at retail to determine the estimated cost of the ending inventory.

Glossary Review

Average-cost method Inventory costing method that uses the weighted-average unit cost to allocate to ending inventory and cost of goods sold the cost of goods available for sale. (p. 6-11).

Consigned goods Goods held for sale by one party although ownership of the goods is retained by another party. (p. 6-6).

Consistency concept Dictates that a company use the same accounting principles and methods from year to year. (p. 6-14).

Days in inventory Measure of the average number of days inventory is held; calculated as 365 divided by inventory turnover. (p. 6-18).

Finished goods inventory Manufactured items that are completed and ready for sale. (p. 6-3).

First-in, first-out (FIFO) method Inventory costing method that assumes that the costs of the earliest goods purchased are the first to be recognized as cost of goods sold. (p. 6-9).

FOB (free on board) destination Freight terms indicating that ownership of the goods remains with the seller until the goods reach the buyer. (p. 6-5).

FOB (free on board) shipping point Freight terms indicating that ownership of the goods passes to the buyer when the public carrier accepts the goods from the seller. (p. 6-5).

***Gross profit method** A method for estimating the cost of the ending inventory by applying a gross profit rate to net sales and subtracting estimated cost of goods sold from cost of goods available for sale. (p. 6-23).

Inventory turnover A ratio that measures the number of times on average the inventory sold during the period; computed by dividing cost of goods sold by the average inventory during the period. (p. 6-18).

Just-in-time (JIT) inventory Inventory system in which companies manufacture or purchase goods only when needed for use. (p. 6-3).

Last-in, first-out (LIFO) method Inventory costing method that assumes the costs of the latest units purchased are the first to be allocated to cost of goods sold. (p. 6-10).

Lower-of-cost-or-net realizable value (LCNRV) basis A basis whereby inventory is stated at the lower of either its cost or its net realizable value. (p. 6-17).

***Moving-average method** A new average is computed after each purchase, by dividing the cost of goods available for sale by the units on hand. (p. 6-22).

Net realizable value Net amount that a company expects to realize (receive) from the sale of inventory. Specifically, it is the estimated selling price in the normal course of business, less estimated costs to complete and sell. (p. 6-18)

Raw materials Basic goods that will be used in production but have not yet been placed into production. (p. 6-3).

***Retail inventory method** A method for estimating the cost of the ending inventory by applying a cost-to-retail ratio to the ending inventory at retail. (p. 6-24).

Specific identification method An actual physical flow costing method in which items still in inventory are specifically costed to arrive at the total cost of the ending inventory. (p. 6-7).

Weighted-average unit cost Average cost that is weighted by the number of units purchased at each unit cost. (p. 6-11).

Work in process That portion of manufactured inventory that has been placed into the production process but is not yet complete. (p. 6-3).

Practice Multiple-Choice Questions

- (LO 1)** When is a physical inventory usually taken?
 - When the company has its greatest amount of inventory.
 - When a limited number of goods are being sold or received.
 - At the end of the company's fiscal year.
 - Both (b) and (c).
- (LO 1)** Which of the following should **not** be included in the physical inventory of a company?
 - Goods held on consignment from another company.
 - Goods shipped on consignment to another company.
 - Goods in transit from another company shipped FOB shipping point.
 - None of the above.
- (LO 1)** As a result of a thorough physical inventory, Railway Company determined that it had inventory worth \$180,000 at December 31, 2020. This count did not take into consideration the following facts: Rogers Consignment store currently has goods worth \$35,000 on its sales floor that belong to Railway but are being sold on consignment by Rogers. The selling price of these goods is \$50,000. Railway purchased \$13,000 of goods that were shipped on December 27, FOB destination, that will be received by Railway on January 3. Determine the correct amount of inventory that Railway should report.
 - \$230,000.
 - \$215,000.
 - \$228,000.
 - \$193,000.
- (LO 2)** Cost of goods available for sale consists of two elements: beginning inventory and
 - ending inventory.
 - cost of goods purchased.
 - cost of goods sold.
 - All of the answer choices are correct.
- (LO 2)** Poppins Company has the following:

	Units	Unit Cost
Inventory, Jan. 1	8,000	\$11
Purchase, June 19	13,000	12
Purchase, Nov. 8	5,000	13

If 9,000 units are on hand at December 31, the cost of the ending inventory under FIFO is:

- \$99,000.
- \$108,000.
- \$113,000.
- \$117,000.

6. (LO 2) Using the data in Question 5, the cost of the ending inventory under LIFO is:

- \$113,000.
- \$108,000.
- \$99,000.
- \$100,000.

7. (LO 2) Hansel Electronics has the following:

	Units	Unit Cost
Inventory, Jan. 1	5,000	\$ 8
Purchase, April 2	15,000	\$10
Purchase, Aug. 28	20,000	\$12

If Hansel has 7,000 units on hand at December 31, the cost of ending inventory under the average-cost method is:

- \$84,000.
- \$70,000.
- \$56,000.
- \$75,250.

8. (LO 2) In periods of rising prices, LIFO will produce:

- higher net income than FIFO.
- the same net income as FIFO.
- lower net income than FIFO.
- higher net income than average-cost.

9. (LO 2) Considerations that affect the selection of an inventory costing method do **not** include:

- tax effects.
- balance sheet effects.
- income statement effects.
- perpetual vs. periodic inventory system.

10. (LO 3) Falk Company's ending inventory is understated \$4,000. The effects of this error on the current year's cost of goods sold and net income, respectively, are:

- understated, overstated.
- overstated, understated.

- c. overstated, overstated.
d. understated, understated.
- 11. (LO 3)** Pauline Company overstated its inventory by \$15,000 at December 31, 2019. It did not correct the error in 2019 or 2020. As a result, Pauline's owner's equity was:
- overstated at December 31, 2019, and understated at December 31, 2020.
 - overstated at December 31, 2019, and properly stated at December 31, 2020.
 - understated at December 31, 2019, and understated at December 31, 2020.
 - overstated at December 31, 2019, and overstated at December 31, 2020.
- 12. (LO 4)** The lower-of-cost-or-net realizable value rule for inventory is an example of the application of:
- the conservatism convention.
 - the historical cost principle.
 - the materiality concept.
 - the economics entity assumption.
- 13. (LO 4)** Norton Company purchased 1,000 widgets and has 200 widgets in its ending inventory at a cost of \$91 each and a net realizable value of \$80 each. The ending inventory under lower-of-cost-or-net realizable value is:
- \$91,000.
 - \$80,000.
 - \$18,200.
 - \$16,000.
- 14. (LO 4)** Santana Company had beginning inventory of \$80,000, ending inventory of \$110,000, cost of goods sold of \$285,000, and sales of \$475,000. Santana's days in inventory is:
- 73 days.
 - 121.7 days.
 - 102.5 days.
 - 84.5 days.
- 15. (LO 4)** Which of these would cause the inventory turnover to increase the most?
- Increasing the amount of inventory on hand.
 - Keeping the amount of inventory on hand constant but increasing sales.
 - Keeping the amount of inventory on hand constant but decreasing sales.
 - Decreasing the amount of inventory on hand and increasing sales.
- *16. (LO 5)** In a perpetual inventory system:
- LIFO cost of goods sold will be the same as in a periodic inventory system.
 - average costs are a simple average of unit costs incurred.
 - a new average is computed under the average-cost method after each sale.
 - FIFO cost of goods sold will be the same as in a periodic inventory system.
- *17. (LO 6)** King Company has sales of \$150,000 and cost of goods available for sale of \$135,000. If the gross profit rate is 30%, the estimated cost of the ending inventory under the gross profit method is:
- \$15,000.
 - \$30,000.
 - \$45,000.
 - \$75,000.

Solutions

- 1. d.** A physical inventory is usually taken when a limited number of goods are being sold or received, and at the end of the company's fiscal year. Choice (a) is incorrect because a physical inventory count is usually taken when the company has the least, not greatest, amount of inventory. Choices (b) and (c) are correct, but (d) is the better answer.
- 2. a.** Goods held on consignment should not be included because another company has title (ownership) to the goods. The other choices are incorrect because (b) goods shipped on consignment to another company and (c) goods in transit from another company shipped FOB shipping point should be included in a company's ending inventory. Choice (d) is incorrect because (a) is not included in the physical inventory.
- 3. b.** The inventory held on consignment by Rogers should be included in Railway's inventory balance at cost (\$35,000). The purchased goods of \$13,000 should not be included in inventory until January 3 because the goods are shipped FOB destination. Therefore, the correct amount of inventory is \$215,000 (\$180,000 + \$35,000), not (a) \$230,000, (c) \$228,000, or (d) \$193,000.
- 4. b.** Cost of goods available for sale consists of beginning inventory and cost of goods purchased, not (a) ending inventory or (c) cost of goods sold. Therefore, choice (d) All of the above is also incorrect.
- 5. c.** Under FIFO, ending inventory will consist of 5,000 units from the Nov. 8 purchase and 4,000 units from the June 19 purchase. Therefore, ending inventory is $(5,000 \times \$13) + (4,000 \times \$12) = \$113,000$, not (a) \$99,000, (b) \$108,000, or (d) \$117,000.
- 6. d.** Under LIFO, ending inventory will consist of 8,000 units from the inventory at Jan. 1 and 1,000 units from the June 19 purchase. Therefore, ending inventory is $(8,000 \times \$11) + (1,000 \times \$12) = \$100,000$, not (a) \$113,000, (b) \$108,000, or (c) \$99,000.
- 7. d.** Under the average-cost method, total cost of goods available for sale needs to be calculated in order to determine average cost per unit. The total cost of goods available is $\$430,000 = (5,000 \times \$8) + (15,000 \times \$10) + (20,000 \times \$12)$. The average cost per unit = $(\$430,000 / 40,000 \text{ total units available for sale}) = \10.75 . Therefore, ending inventory is $(\$10.75 \times 7,000) = \$75,250$, not (a) \$84,000, (b) \$70,000, or (c) \$56,000.
- 8. c.** In periods of rising prices, LIFO will produce lower net income than FIFO, not (a) higher than FIFO or (b) the same as FIFO. Choice (d) is incorrect because in periods of rising prices, LIFO will produce lower net income than average-cost. LIFO therefore charges the highest inventory cost against revenues in a period of rising prices.
- 9. d.** Perpetual vs. periodic inventory system is not one of the factors that affect the selection of an inventory costing method. The other choices are incorrect because (a) tax effects, (b) balance sheet effects, and (c) income statement effects all affect the selection of an inventory costing method.
- 10. b.** Because ending inventory is too low, cost of goods sold will be too high (overstated) and since cost of goods sold (an expense) is too high, net income will be too low (understated). Therefore, the other choices are incorrect.

11. b. Owner's equity is overstated by \$15,000 at December 31, 2019, and is properly stated at December 31, 2020. An ending inventory error in one period will have an equal and opposite effect on cost of goods sold and net income in the next period; after two years, the errors have offset each other. The other choices are incorrect because owner's equity (a) is properly stated, not understated, at December 31, 2020; (c) is overstated, not understated, by \$15,000 at December 31, 2019, and is properly stated, not understated, at December 31, 2020; and (d) is properly stated at December 31, 2020, not overstated.

12. a. Conservatism means to use the lowest value for assets and revenues when in doubt. The other choices are incorrect because (b) historical cost means that companies value assets at the original cost, (c) materiality means that an amount is large enough to affect a decision-maker, and (d) economic entity means to keep the company's transactions separate from the transactions of other entities.

13. d. Under the LCNRV basis, net realizable value is defined as the estimated selling price in the normal course of business, less estimated costs to complete and sell. Therefore, ending inventory would be valued at $200 \text{ widgets} \times \$80 \text{ each} = \$16,000$ not (a) \$91,000, (b) \$80,000, or (c) \$18,200.

14. b. Santana's days in inventory = $365 / \text{Inventory turnover} = 365 / [\$285,000 / (\$80,000 + \$110,000) / 2] = 121.7 \text{ days}$, not (a) 73 days, (c) 102.5 days, or (d) 84.5 days.

15. d. Decreasing the amount of inventory on hand will cause the denominator to decrease, causing inventory turnover to increase.

Increasing sales will cause the numerator of the ratio to increase (higher sales means higher COGS), thus causing inventory turnover to increase even more. The other choices are incorrect because (a) increasing the amount of inventory on hand causes the denominator of the ratio to increase while the numerator stays the same, causing inventory turnover to decrease; (b) keeping the amount of inventory on hand constant but increasing sales will cause inventory turnover to increase because the numerator of the ratio will increase (higher sales means higher COGS) while the denominator stays the same, which will result in a lesser inventory increase than decreasing amount of inventory on hand and increasing sales; and (c) keeping the amount of inventory on hand constant but decreasing sales will cause inventory turnover to decrease because the numerator of the ratio will decrease (lower sales means lower COGS) while the denominator stays the same.

***16. d.** FIFO cost of goods sold is the same under both a periodic and a perpetual inventory system. The other choices are incorrect because (a) LIFO cost of goods sold is not the same under a periodic and a perpetual inventory system; (b) average costs are based on a moving average of unit costs, not an average of unit costs; and (c) a new average is computed under the average-cost method after each purchase, not sale.

***17. b.** $\text{COGS} = \text{Sales } (\$150,000) - \text{Gross profit } (\$150,000 \times 30\%) = \$105,000$. Ending inventory = Cost of goods available for sale (\$135,000) – COGS (\$105,000) = \$30,000, not (a) \$15,000, (c) \$45,000, or (d) \$75,000.

Practice Brief Exercises

Compute ending inventory using FIFO and LIFO.

1. (LO 2) In its first month of operations, Moncada Company made three purchases of merchandise in the following sequence: (1) 200 units at \$7, (2) 300 units at \$8, and (3) 150 units at \$9. Assuming there are 220 units on hand, compute the cost of the ending inventory under the (a) FIFO method and (b) LIFO method. Moncada uses a periodic inventory system.

Solution

- 1. a.** The ending inventory under FIFO consists of (150 units at \$9) + (70 units at \$8) for a total allocation of \$1,910 (\$1,350 + \$560).
- b.** The ending inventory under LIFO consists of (200 units at \$7) + (20 units at \$8) for a total allocation of \$1,560 (\$1,400 + \$160).

Determine correct income statement amounts.

2. (LO 3) Financial Statement Avisail Company reports net income of \$80,000 in 2020. However, ending inventory was overstated \$9,000. What is the correct net income for 2020? What effect, if any, will this error have on total assets as reported in the balance sheet at December 31, 2020?

Solution

- 2.** The overstatement of ending inventory caused cost of goods sold to be understated \$9,000 and net income to be overstated \$9,000. The correct net income for 2020 is \$71,000 (\$80,000 – \$9,000). Total assets in the balance sheet will be overstated by the amount that ending inventory is overstated, \$9,000.

Compute inventory turnover and days in inventory.

3. (LO 4) At December 31, 2020, the following information was available for Garcia Company: ending inventory \$30,000, beginning inventory \$42,000, cost of goods sold \$240,000, and sales revenue \$400,000. Calculate inventory turnover and days in inventory for Garcia Company.

Solution

$$3. \text{ Inventory turnover: } \frac{\$240,000}{(\$30,000 + \$42,000)/2} = \frac{\$240,000}{\$36,000} = 6.67$$

$$\text{Days in inventory: } \frac{365}{6.67} = 54.7 \text{ days}$$

Practice Exercises

1. (LO 1) Matt Clark, an auditor with Grant CPAs, is performing a review of Parson Company's inventory account. Parson did not have a good year and top management is under pressure to boost reported income. According to its records, the inventory balance at year-end was \$600,000. However, the following information was not considered when determining that amount.

Determine the correct inventory amount.

- The physical count did not include goods purchased by Parson with a cost of \$30,000 that were shipped FOB destination on December 28 and did not arrive at Parson's warehouse until January 3.
- Included in the company's count were goods with a cost of \$150,000 that the company is holding on consignment. The goods belong to Alvarez Corporation.
- Included in the inventory account was \$21,000 of office supplies that were stored in the warehouse and were to be used by the company's supervisors and managers during the coming year.
- The company received an order on December 28 that was boxed and was sitting on the loading dock awaiting pick-up on December 31. The shipper picked up the goods on January 1 and delivered them on January 6. The shipping terms were FOB shipping point. The goods had a selling price of \$29,000 and a cost of \$19,000. The goods were not included in the count because they were sitting on the dock.
- On December 29, Parson shipped goods with a selling price of \$56,000 and a cost of \$40,000 to Decco Corporation FOB shipping point. The goods arrived on January 3. Decco had only ordered goods with a selling price of \$10,000 and a cost of \$6,000. However, a Parson's sales manager had authorized the shipment and said that if Decco wanted to ship the goods back next week, it could.
- Included in the count was \$27,000 of goods that were parts for a machine that the company no longer made. Given the high-tech nature of Parson's products, it was unlikely that these obsolete parts had any other use. However, management would prefer to keep them on the books at cost, "since that is what we paid for them, after all."

Instructions

Prepare a schedule to determine the correct inventory amount. Provide explanations for each item above, saying why you did or did not make an adjustment for each item.

Solution

1. Ending inventory—as reported	\$600,000
1. No effect—title does not pass to Parson until goods are received (Jan. 3).	0
2. Subtract from inventory: The goods belong to Alvarez Corporation. Parson is merely holding them for Alvarez.	(150,000)
3. Subtract from inventory: Office supplies should be carried in a separate account. They are not considered inventory held for resale.	(21,000)
4. Add to inventory: The goods belong to Parson until they are shipped (Jan. 1).	19,000
5. Add to inventory: Decco ordered goods with a cost of \$6,000. Parson should record the corresponding sales revenue of \$10,000. Parson's decision to ship extra "unordered" goods does not constitute a sale. The manager's statement that Decco could ship the goods back indicates that Parson knows this overshipment is not a legitimate sale. The manager acted unethically in an attempt to improve Parson's reported income by overshipping.	34,000
6. Subtract from inventory: GAAP require that inventory be valued at the lower-of-cost-or-net realizable value. Obsolete parts should be adjusted from cost to zero if they have no other use.	(27,000)
Correct inventory	<u><u>\$455,000</u></u>

Determine effects of inventory errors.

2. (LO 3) Rhode Software reported cost of goods sold as follows.

	2019	2020
Beginning inventory	\$ 27,000	\$ 40,000
Cost of goods purchased	200,000	235,000
Cost of goods available for sale	227,000	275,000
Ending inventory	40,000	45,000
Cost of goods sold	<u>\$187,000</u>	<u>\$230,000</u>

Rhode made two errors: (1) 2019 ending inventory was overstated \$4,000, and (2) 2020 ending inventory was understated \$9,000.

Instructions

Compute the correct cost of goods sold for each year.

Solution

2.		2019	2020
	Beginning inventory	\$ 27,000	\$ 36,000
	Cost of goods purchased	200,000	235,000
	Cost of goods available for sale	227,000	271,000
	Corrected ending inventory	(36,000) ^a	(54,000) ^b
	Cost of goods sold	<u>\$191,000</u>	<u>\$217,000</u>

^a\$40,000 – \$4,000 = \$36,000; ^b\$45,000 + \$9,000 = \$54,000

Determine LCMRV valuation.

3. (LO 4) Creve Couer Camera uses the lower-of-cost-or-net realizable value basis for its inventory. The following data are available at December 31.

	Units	Cost per Unit	Net Realizable Value per Unit
Cameras:			
Minolta	5	\$160	\$156
Canon	7	145	153
Light Meters:			
Vivitar	12	120	114
Kodak	10	130	142

Instructions

What amount should be reported on Creve Couer Camera's financial statements, assuming the lower-of-cost-or-net realizable value rule is applied?

Solution

	Cost per Unit	Net Realizable Value per Unit	Lower-of-Cost or-Net Realizable Value	Units	Inventory at Lower-of- Cost-or-Net Realizable Value
Cameras:					
Minolta	\$160	\$156	\$156	5	\$ 780
Canon	145	153	145	7	1,015
Light Meters:					
Vivitar	120	114	114	12	1,368
Kodak	130	142	130	10	1,300
Total					<u>\$4,463</u>

Practice Problems

1. (LO 2) Englehart Company has the following inventory, purchases, and sales data for the month of March.

Compute inventory and cost of goods sold using three cost flow methods in a periodic inventory system.

Inventory:	March 1	200 units @ \$4.00	\$ 800
Purchases:	March 10	500 units @ \$4.50	2,250
	March 20	400 units @ \$4.75	1,900
	March 30	300 units @ \$5.00	1,500
Sales:	March 15	500 units	
	March 25	400 units	

The physical inventory count on March 31 shows 500 units on hand.

Instructions

Under a **periodic inventory system**, determine the cost of inventory on hand at March 31 and the cost of goods sold for March under (a) FIFO, (b) LIFO, and (c) average-cost. (For average-cost, carry cost per unit to three decimal places.)

Solution

1. The cost of goods available for sale is \$6,450, as follows.

Inventory:		200 units @ \$4.00	\$ 800
Purchases:	March 10	500 units @ \$4.50	2,250
	March 20	400 units @ \$4.75	1,900
	March 30	300 units @ \$5.00	1,500
Total goods available for sale			<u>\$6,450</u>

a. FIFO Method

Ending inventory:

<u>Date</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>	
March 30	300	\$5.00	\$1,500	
March 20	200	4.75	950	\$2,450
Cost of goods sold: \$6,450 – \$2,450 =				<u>\$4,000</u>

b. LIFO Method

Ending inventory:

<u>Date</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>	
March 1	200	\$4.00	\$ 800	
March 10	300	4.50	1,350	\$2,150
Cost of goods sold: \$6,450 – \$2,150 =				<u>\$4,300</u>

c. Average-Cost Method

Average unit cost: $\$6,450 \div 1,400 = \4.607

Ending inventory: $500 \times \$4.607 =$ \$2,303.50

Cost of goods sold: $\$6,450 - \$2,303.50 =$ \$4,146.50

***2. (LO 5) Practice Problem 1** showed cost of goods sold computations under a periodic inventory system. Now let's assume that Englehart Company uses a perpetual inventory system. The company has the same inventory, purchases, and sales data for the month of March as shown earlier:

Compute inventory and cost of goods sold using three cost flow methods in a perpetual inventory system.

Inventory:	March 1	200 units @ \$4.00	\$ 800
Purchases:	March 10	500 units @ \$4.50	2,250
	March 20	400 units @ \$4.75	1,900
	March 30	300 units @ \$5.00	1,500
Sales:	March 15	500 units	
	March 25	400 units	

The physical inventory count on March 31 shows 500 units on hand.

Instructions

Under a **perpetual inventory system**, determine the cost of inventory on hand at March 31 and the cost of goods sold for March under (a) FIFO, (b) LIFO, and (c) moving-average cost.

Solution

2. The cost of goods available for sale is \$6,450, as follows.

Inventory:		200 units @ \$4.00	\$ 800
Purchases:	March 10	500 units @ \$4.50	2,250
	March 20	400 units @ \$4.75	1,900
	March 30	300 units @ \$5.00	1,500
Total:			<u>\$6,450</u>

Under a **perpetual inventory system**, the cost of goods sold under each cost flow method is as follows.

		FIFO Method	
Date	Purchases	Cost of Goods Sold	Balance
March 1			(200 @ \$4.00) \$ 800
March 10	(500 @ \$4.50) \$2,250		(200 @ \$4.00) } \$3,050 (500 @ \$4.50) }
March 15		(200 @ \$4.00) (300 @ \$4.50) \$2,150	(200 @ \$4.50) \$ 900
March 20	(400 @ \$4.75) \$1,900		(200 @ \$4.50) } \$2,800 (400 @ \$4.75) }
March 25		(200 @ \$4.50) (200 @ \$4.75) \$1,850	(200 @ \$4.75) \$ 950
March 30	(300 @ \$5.00) \$1,500		(200 @ \$4.75) } \$2,450 (300 @ \$5.00) }
	Ending inventory <u>\$2,450</u>	Cost of goods sold: \$2,150 + \$1,850 = <u>\$4,000</u>	

		LIFO Method	
Date	Purchases	Cost of Goods Sold	Balance
March 1			(200 @ \$4.00) \$ 800
March 10	(500 @ \$4.50) \$2,250		(200 @ \$4.00) } \$3,050 (500 @ \$4.50) }
March 15		(500 @ \$4.50) \$2,250	(200 @ \$4.00) \$ 800
March 20	(400 @ \$4.75) \$1,900		(200 @ \$4.00) } \$2,700 (400 @ \$4.75) }
March 25		(400 @ \$4.75) \$1,900	(200 @ \$4.00) \$ 800
March 30	(300 @ \$5.00) \$1,500		(200 @ \$4.00) } \$2,300 (300 @ \$5.00) }
	Ending inventory <u>\$2,300</u>	Cost of goods sold: \$2,250 + \$1,900 = <u>\$4,150</u>	

		Moving-Average Cost Method	
Date	Purchases	Cost of Goods Sold	Balance
March 1			(200 @ \$4.00) \$ 800
March 10	(500 @ \$4.50) \$2,250		(700 @ \$4.357) \$3,050
March 15		(500 @ \$4.357) \$2,179	(200 @ \$4.357) \$ 871
March 20	(400 @ \$4.75) \$1,900		(600 @ \$4.618) \$2,771
March 25		(400 @ \$4.618) \$1,847	(200 @ \$4.618) \$ 924
March 30	(300 @ \$5.00) \$1,500		(500 @ \$4.848) \$2,424
	Ending inventory <u>\$2,424</u>	Cost of goods sold: \$2,179 + \$1,847 = <u>\$4,026</u>	

WileyPLUS

Brief Exercises, DO IT! Exercises, Exercises, Problems, and many additional resources are available for practice in WileyPLUS.

Note: All asterisked Questions, Exercises, and Problems relate to material in the appendices to the chapter.

Questions

1. “The key to successful business operations is effective inventory management.” Do you agree? Explain.
2. An item must possess two characteristics to be classified as inventory by a merchandiser. What are these two characteristics?
3. Your friend Ben Johnson has been hired to help take the physical inventory in Pearson Hardware Store. Explain to Ben what this job will entail.
4. **a.** Jovad Company ships merchandise to Martin Company on December 30. The merchandise reaches the buyer on January 6. Indicate the terms of sale that will result in the goods being included in (1) Jovad’s December 31 inventory, and (2) Martin’s December 31 inventory.
b. Under what circumstances should Jovad Company include consigned goods in its inventory?
5. Topp Hat Shop received a shipment of hats for which it paid the wholesaler \$2,970. The price of the hats was \$3,000 but Topp was given a \$30 cash discount and required to pay freight charges of \$50. What amount will Topp record for inventory? Why?
6. Explain the difference between the terms FOB shipping point and FOB destination.
7. Leah Clement believes that the allocation of cost of goods available for sale should be based on the actual physical flow of the goods. Explain to Leah why this may be both impractical and inappropriate.
8. What is the major advantage and the major disadvantage of the specific identification method of inventory costing?
9. “The selection of an inventory cost flow method is a decision made by accountants.” Do you agree? Explain. Once a method has been selected, what accounting requirement applies?
10. Which assumed inventory cost flow method:
 - a.** usually parallels the actual physical flow of merchandise?
 - b.** divides cost of goods available by total units available for sale to determine a unit cost?
 - c.** assumes that the latest units purchased are the first to be sold?
11. In a period of rising prices, the inventory reported in Bert Company’s balance sheet is close to the current cost of the inventory, whereas Ernie Company’s inventory is considerably below its current cost. Identify the inventory cost flow method being used by each company. Which company probably has been reporting the higher gross profit?
12. Oscar Company has been using the FIFO cost flow method during a prolonged period of inflation. During the same time period, Oscar has been paying out all of its net income as dividends. What adverse effects may result from this policy?
13. Kyle Adams is studying for the next accounting mid-term examination. What should Kyle know about (a) departing from the cost basis of accounting for inventories and (b) the meaning of “net realizable value” in the lower-of-cost-or-net realizable value method?
14. Hendrix Entertainment Center has 5 TVs on hand at the balance sheet date that cost \$400 each. The net realizable value is \$380 per unit. Under the lower-of-cost-or-net realizable value basis of accounting for inventories, what value should Hendrix report for the TVs on the balance sheet? Why?
15. Warnke Stores has 20 toasters on hand at the balance sheet date. Each costs \$27. The net realizable value is \$30 per unit. Under the lower-of-cost-or-net realizable value basis of accounting for inventories, what value should Warnke report for the toasters on the balance sheet? Why?
16. Sayaovang Company discovers in 2020 that its ending inventory at December 31, 2019, was \$7,000 understated. What effect will this error have on (a) 2019 net income, (b) 2020 net income, and (c) the combined net income for the 2 years?
17. Dreher Company’s balance sheet shows Inventory \$162,800. What additional disclosures should be made?
18. Under what circumstances might inventory turnover be too high? That is, what possible negative consequences might occur?
19. What inventory cost flow does **Apple** use for its inventories? (*Hint:* You will need to examine the notes for Apple’s financial statements.)
- *20. “When perpetual inventory records are kept, the results under the FIFO and LIFO methods are the same as they would be in a periodic inventory system.” Do you agree? Explain.
- *21. How does the average-cost method of inventory costing differ between a perpetual inventory system and a periodic inventory system?
- *22. When is it necessary to estimate inventories?
- *23. Both the gross profit method and the retail inventory method are based on averages. For each method, indicate the average used, how it is determined, and how it is applied.
- *24. Pawlowski Company has net sales of \$400,000 and cost of goods available for sale of \$300,000. If the gross profit rate is 35%, what is the estimated cost of the ending inventory? Show computations.
- *25. Cinderella Shoe Shop had goods available for sale in 2020 with a retail price of \$120,000. The cost of these goods was \$84,000. If sales during the period were \$80,000, what is the ending inventory at cost using the retail inventory method?

Brief Exercises

Identify items to be included in taking a physical inventory.

BE6.1 (LO 1) Peosta Company identifies the following items for possible inclusion in the taking of a physical inventory. Indicate whether each item should be included or excluded from the inventory taking.

- Goods shipped on consignment by Peosta to another company.
- Goods in transit from a supplier shipped FOB destination.
- Goods held on consignment from another company.

Determine ending inventory amount.

BE6.2 (LO 1) Stallman Company took a physical inventory on December 31 and determined that goods costing \$200,000 were on hand. Not included in the physical count were \$25,000 of goods purchased from Pelzer Corporation, FOB shipping point, and \$22,000 of goods sold to Alvarez Company for \$30,000, FOB destination. Both the Pelzer purchase and the Alvarez sale were in transit at year-end. What amount should Stallman report as its December 31 inventory?

Compute ending inventory using FIFO and LIFO.

BE6.3 (LO 2) In its first month of operations, Weatherall Company made three purchases of merchandise in the following sequence: (1) 300 units at \$6, (2) 400 units at \$7, and (3) 200 units at \$8. Assuming there are 380 units on hand, compute the cost of the ending inventory under the (a) FIFO method and (b) LIFO method. Weatherall uses a periodic inventory system.

Compute the ending inventory using average-cost.

BE6.4 (LO 2) Data for Weatherall Company are presented in BE6.3. Compute the cost of the ending inventory under the average-cost method, assuming there are 380 units on hand.

Explain the financial statement effect of inventory cost flow assumptions.

BE6.5 (LO 2) Financial Statement The management of Mastronardo Corp. is considering the effects of inventory-costing methods on its financial statements and its income tax expense. Assuming that the price the company pays for inventory is increasing, which method will:

- Provide the highest net income?
- Provide the highest ending inventory?
- Result in the lowest income tax expense?
- Result in the most stable earnings over a number of years?

Explain the financial statement effect of inventory cost flow assumptions.

BE6.6 (LO 2) Financial Statement In its first month of operation, Hoffman Company purchased 100 units of inventory for \$6, then 200 units for \$7, and finally 140 units for \$8. At the end of the month, 180 units remained. Compute the amount of phantom profit that would result if the company used FIFO rather than LIFO. Explain why this amount is referred to as phantom profit. The company uses the periodic method.

Determine correct income statement amounts.

BE6.7 (LO 3) Larkin Company reports net income of \$90,000 in 2020. However, ending inventory was understated \$7,000. What is the correct net income for 2020? What effect, if any, will this error have on total assets as reported in the balance sheet at December 31, 2020?

Determine the LCNRV valuation using inventory categories.

BE6.8 (LO 4) Cruz Video Center accumulates the following cost and net realizable data at December 31.

Inventory Categories	Cost Data	Net Realizable Data
Cameras	\$12,000	\$12,300
Camcorders	9,500	9,700
Blu-ray players	14,000	12,900

Compute the lower-of-cost-or-net realizable value valuation for the company's total inventory.

Compute inventory turnover and days in inventory.

BE6.9 (LO 4) At December 31, 2020, the following information was available for E. Hetzel Company: ending inventory \$40,000, beginning inventory \$56,000, cost of goods sold \$270,000, and sales revenue \$380,000. Calculate inventory turnover and days in inventory for E. Hetzel Company.

Apply cost flow methods to perpetual inventory records.

***BE6.10 (LO 5)** Rosario Department Store uses a perpetual inventory system. Data for product E2-D2 include the following purchases.

<u>Date</u>	<u>Number of Units</u>	<u>Unit Price</u>
May 7	50	\$10
July 28	30	13

On June 1, Rosario sold 26 units, and on August 27, 40 more units. Prepare the perpetual inventory schedule for the above transactions using (a) FIFO, (b) LIFO, and (c) moving-average cost. (Round average cost per unit to nearest cent.)

***BE6.11 (LO 6)** At May 31, Brunet Company has net sales of \$340,000 and cost of goods available for sale of \$230,000. Compute the estimated cost of the ending inventory, assuming the gross profit rate is 35%.

Apply the gross profit method.

***BE6.12 (LO 6)** On June 30, Joanna Fabrics has the following data pertaining to the retail inventory method. Goods available for sale: at cost \$38,000; at retail \$50,000; net sales \$40,000; and ending inventory at retail \$10,000. Compute the estimated cost of the ending inventory using the retail inventory method.

Apply the retail inventory method.

DO IT! Exercises

DO IT! 6.1 (LO 1) Gresa Company just took its physical inventory. The count of inventory items on hand at the company's business locations resulted in a total inventory cost of \$300,000. In reviewing the details of the count and related inventory transactions, you have discovered the following.

Apply rules of ownership to determine inventory cost.

1. Gresa has sent inventory costing \$26,000 on consignment to Alissa Company. All of this inventory was at Alissa's showrooms on December 31.
2. The company did not include in the count inventory (cost, \$20,000) that was sold on December 28, terms FOB shipping point. The goods were in transit on December 31.
3. The company did not include in the count inventory (cost, \$14,000) that was purchased with terms of FOB shipping point. The goods were in transit on December 31.

Compute the correct December 31 inventory.

DO IT! 6.2 (LO 2) The accounting records of Americo Electronics show the following data.

Compute cost of goods sold under different cost flow methods.

Beginning inventory	3,000 units at \$5
Purchases	8,000 units at \$7
Sales	9,400 units at \$10

Determine cost of goods sold during the period under a periodic inventory system using (a) the FIFO method, (b) the LIFO method, and (c) the average-cost method. (Round unit cost to nearest tenth of a cent.)

DO IT! 6.3 (LO 3) Vanida Company understated its 2019 ending inventory by \$27,000. Determine the impact this error has on ending inventory, cost of goods sold, and owner's equity in 2019 and 2020.

Determine effect of inventory error.

DO IT! 6.4a (LO 4) Cody Company sells three different categories of tools (small, medium, and large). The cost and net realizable value of its inventory of tools are as follows.

Compute inventory value under LCNRV.

	<u>Cost</u>	<u>Net Realizable Value</u>
Small	\$ 64,000	\$ 73,000
Medium	290,000	260,000
Large	152,000	171,000

Determine the value of the company's inventory under the lower-of-cost-or-net realizable value approach.

DO IT! 6.4b (LO 4) Early in 2020, Yeng Company switched to a just-in-time inventory system. Its sales revenue, cost of goods sold, and inventory amounts for 2019 and 2020 are shown below.

Compute inventory turnover and assess inventory level.

	<u>2019</u>	<u>2020</u>
Sales	\$3,120,000	\$3,713,000
Cost of goods sold	1,200,000	1,425,000
Beginning inventory	180,000	220,000
Ending inventory	220,000	100,000

Determine the inventory turnover and days in inventory for 2019 and 2020. Discuss the changes in the amount of inventory, the inventory turnover and days in inventory, and the amount of sales across the two years.

Exercises

Determine the correct inventory amount.

E6.1 (LO 1) Tri-State Bank and Trust is considering giving Wilfred Company a loan. Before doing so, management decides that further discussions with Wilfred's accountant may be desirable. One area of particular concern is the inventory account, which has a year-end balance of \$297,000. Discussions with the accountant reveal the following.

1. Wilfred shipped goods costing \$38,000 to Lilja Company, FOB shipping point, on December 28. The goods are not expected to arrive at Lilja until January 12. The goods were not included in the physical inventory because they were not in the warehouse.
2. The physical count of the inventory did not include goods costing \$95,000 that were shipped to Wilfred FOB destination on December 27 and were still in transit at year-end.
3. Wilfred received goods costing \$22,000 on January 2. The goods were shipped FOB shipping point on December 26 by Brent Co. The goods were not included in the physical count.
4. Wilfred shipped goods costing \$35,000 to Jesse Co., FOB destination, on December 30. The goods were received at Jesse on January 8. They were not included in Wilfred's physical inventory.
5. Wilfred received goods costing \$44,000 on January 2 that were shipped FOB destination on December 29. The shipment was a rush order that was supposed to arrive December 31. This purchase was included in the ending inventory of \$297,000.

Instructions

Determine the correct inventory amount on December 31.

Determine the correct inventory amount.

E6.2 (LO 1) Kari Downs, an auditor with Wheeler CPAs, is performing a review of Depue Company's inventory account. Depue did not have a good year, and top management is under pressure to boost reported income. According to its records, the inventory balance at year-end was \$740,000. However, the following information was not considered when determining that amount.

1. Included in the company's count were goods with a cost of \$250,000 that the company is holding on consignment. The goods belong to Kroeger Corporation.
2. The physical count did not include goods purchased by Depue with a cost of \$40,000 that were shipped FOB destination on December 28 and did not arrive at Depue warehouse until January 3.
3. Included in the inventory account was \$14,000 of office supplies that were stored in the warehouse and were to be used by the company's supervisors and managers during the coming year.
4. The company received an order on December 29 that was boxed and sitting on the loading dock awaiting pick-up on December 31. The shipper picked up the goods on January 1 and delivered them on January 6. The shipping terms were FOB shipping point. The goods had a selling price of \$40,000 and a cost of \$28,000. The goods were not included in the count because they were sitting on the dock.
5. On December 29, Depue shipped goods with a selling price of \$80,000 and a cost of \$60,000 to Macchia Sales Corporation FOB shipping point. The goods arrived on January 3. Macchia had only ordered goods with a selling price of \$10,000 and a cost of \$8,000. However, a sales manager at Depue had authorized the shipment and said that if Machia wanted to ship the goods back next week, it could.
6. Included in the count was \$40,000 of goods that were parts for a machine that the company no longer made. Given the high-tech nature of Depue's products, it was unlikely that these obsolete parts had any other use. However, management would prefer to keep them on the books at cost, "since that is what we paid for them, after all."

Instructions

Prepare a schedule to determine the correct inventory amount. Provide explanations for each item above, saying why you did or did not make an adjustment for each item.

Identify items in inventory.

E6.3 (LO 1) Gato Inc. had the following inventory situations to consider at January 31, its year-end.

- a. Goods held on consignment for Steele Corp. since December 12.
- b. Goods shipped on consignment to Logan Holdings Inc. on January 5.
- c. Goods shipped to a customer, FOB destination, on January 29 that are still in transit.
- d. Goods shipped to a customer, FOB shipping point, on January 29 that are still in transit.
- e. Goods purchased FOB destination from a supplier on January 25 that are still in transit.

- f. Goods purchased FOB shipping point from a supplier on January 25 that are still in transit.
- g. Office supplies on hand at January 31.

Instructions

Identify which of the preceding items should be included in inventory. If the item should not be included in inventory, state in what account, if any, it should have been recorded.

E6.4 (LO 2) On December 1, Kiyak Electronics Ltd. has three DVD players left in stock. All are identical, all are priced to sell at \$150. One of the three DVD players left in stock, with serial #1012, was purchased on June 1 at a cost of \$100. Another, with serial #1045, was purchased on November 1 for \$88. The last player, serial #1056, was purchased on November 30 for \$80.

Calculate cost of goods sold using specific identification and FIFO.

Instructions

- a. Calculate the cost of goods sold using the FIFO periodic inventory method assuming that two of the three players were sold by the end of December, Kiyak Electronics' year-end.
- b. If Kiyak Electronics used the specific identification method instead of the FIFO method, how might it alter its earnings by "selectively choosing" which particular players to sell to the two customers? What would Kiyak's cost of goods sold be if the company wished to minimize earnings? Maximize earnings?
- c. Which of the two inventory methods do you recommend that Kiyak use? Explain why.

E6.5 (LO 2) Elsa's Boards sells a snowboard, Xpert, that is popular with snowboard enthusiasts. Information relating to Elsa's purchases of Xpert snowboards during September is shown below. During the same month, 121 Xpert snowboards were sold. Elsa's uses a periodic inventory system.

Compute inventory and cost of goods sold using FIFO and LIFO.

Date	Explanation	Units	Unit Cost	Total Cost
Sept. 1	Inventory	26	\$ 97	\$ 2,522
Sept. 12	Purchases	45	102	4,590
Sept. 19	Purchases	20	104	2,080
Sept. 26	Purchases	50	105	5,250
	Totals	141		\$14,442

Instructions

- a. Compute the ending inventory at September 30 and cost of goods sold using the FIFO and LIFO methods. Prove the amount allocated to cost of goods sold under each method.
- b. For both FIFO and LIFO, calculate the sum of ending inventory and cost of goods sold. What do you notice about the answers you found for each method?

E6.6 (LO 2) Ballas Co. uses a periodic inventory system. Its records show the following for the month of May, in which 68 units were sold.

Compute inventory and cost of goods sold using FIFO and LIFO.

		Units	Unit Cost	Total Cost
May 1	Inventory	30	\$ 8	\$240
15	Purchases	25	11	275
24	Purchases	35	12	420
	Totals	90		\$935

Instructions

Compute the ending inventory at May 31 and cost of goods sold using the FIFO and LIFO methods. Prove the amount allocated to cost of goods sold under each method.

E6.7 (LO 2) Moath Company reports the following for the month of June.

Compute inventory and cost of goods sold using FIFO and LIFO.

		Units	Unit Cost	Total Cost
June 1	Inventory	200	\$5	\$1,000
12	Purchase	400	6	2,400
23	Purchase	300	7	2,100
30	Inventory	100		

Instructions

- a. Compute the cost of the ending inventory and the cost of goods sold under (1) FIFO and (2) LIFO.
- b. Which costing method gives the higher ending inventory? Why?
- c. Which method results in the higher cost of goods sold? Why?

Compute inventory under FIFO, LIFO, and average-cost.

E6.8 (LO 2) Shawn Company had 100 units in beginning inventory at a total cost of \$10,000. The company purchased 200 units at a total cost of \$26,000. At the end of the year, Shawn had 75 units in ending inventory.

Instructions

- Compute the cost of the ending inventory and the cost of goods sold under (1) FIFO, (2) LIFO, and (3) average-cost.
- Which cost flow method would result in the highest net income?
- Which cost flow method would result in inventories approximating current cost in the balance sheet?
- Which cost flow method would result in Shawn paying the least taxes in the first year?

Compute inventory and cost of goods sold using average-cost.

E6.9 (LO 2) Inventory data for Moath Company are presented in E6.7.

Instructions

- Compute the cost of the ending inventory and the cost of goods sold using the average-cost method.
- Will the results in (a) be higher or lower than the results under (1) FIFO and (2) LIFO?
- Why is the average unit cost not \$6?

Determine effects of inventory errors.

E6.10 (LO 3) Elliott's Hardware reported cost of goods sold as follows.

	2019	2020
Beginning inventory	\$ 20,000	\$ 30,000
Cost of goods purchased	150,000	175,000
Cost of goods available for sale	170,000	205,000
Ending inventory	30,000	35,000
Cost of goods sold	<u>\$140,000</u>	<u>\$170,000</u>

Elliott's made two errors: (1) 2019 ending inventory was overstated \$3,000, and (2) 2020 ending inventory was understated \$5,000.

Instructions

Compute the correct cost of goods sold for each year.

Prepare correct income statements.

E6.11 (LO 3) **Financial Statement Writing** Smart Watch Company reported the following income statement data for a 2-year period.

	2019	2020
Sales revenue	<u>\$220,000</u>	<u>\$250,000</u>
Cost of goods sold		
Beginning inventory	32,000	44,000
Cost of goods purchased	<u>173,000</u>	<u>202,000</u>
Cost of goods available for sale	205,000	246,000
Ending inventory	<u>44,000</u>	<u>52,000</u>
Cost of goods sold	<u>161,000</u>	<u>194,000</u>
Gross profit	<u>\$ 59,000</u>	<u>\$ 56,000</u>

Smart uses a periodic inventory system. The inventories at January 1, 2019, and December 31, 2020, are correct. However, the ending inventory at December 31, 2019, was overstated \$6,000.

Instructions

- Prepare correct income statement data for the 2 years.
- What is the cumulative effect of the inventory error on total gross profit for the 2 years?
- Explain in a letter to the president of Smart Watch Company what has happened, i.e., the nature of the error and its effect on the financial statements.

Determine ending inventory under LCNRV.

E6.12 (LO 4) Freeze Frame Camera Shop uses the lower-of-cost-or-net realizable value basis for its inventory. The following data are available at December 31.

Item	Units	Unit Cost	Net Realizable Value
Cameras:			
Minolta	5	\$170	\$156
Canon	6	150	152
Light meters:			
Vivitar	10	125	115
Kodak	14	120	135

Instructions

Determine the amount of the ending inventory by applying the lower-of-cost-or-net realizable value basis.

E6.13 (LO 4) Charapata Company applied FIFO to its inventory and got the following results for its ending inventory.

Compute lower-of-cost-or-net realizable value.

Cameras	100 units at a cost per unit of \$65
Blu-ray players	150 units at a cost per unit of \$75
iPods	125 units at a cost per unit of \$80

The net realizable value at year-end was cameras \$71, Blu-ray players \$67, and iPods \$78.

Instructions

Determine the amount of ending inventory at lower-of-cost-or-net realizable value.

E6.14 (LO 4) This information is available for Abdullah's Photo Corporation for 2018, 2019, and 2020.

Compute inventory turnover, days in inventory, and gross profit rate.

	2018	2019	2020
Beginning inventory	\$ 100,000	\$ 300,000	\$ 400,000
Ending inventory	300,000	400,000	480,000
Cost of goods sold	900,000	1,152,000	1,300,000
Sales revenue	1,200,000	1,600,000	1,900,000

Instructions

Calculate inventory turnover (round to two decimal places), days in inventory (round to one decimal place), and gross profit rate (from Chapter 5, round to nearest percentage) for Abdullah's Photo Corporation for 2018, 2019, and 2020. Comment on any trends.

E6.15 (LO 4) The cost of goods sold computations for Sooner Company and Later Company are shown below.

Compute inventory turnover and days in inventory.

	Sooner Company	Later Company
Beginning inventory	\$ 45,000	\$ 71,000
Cost of goods purchased	200,000	290,000
Cost of goods available for sale	245,000	361,000
Ending inventory	55,000	69,000
Cost of goods sold	<u>\$190,000</u>	<u>\$292,000</u>

Instructions

- Compute inventory turnover (round to two decimal places) and days in inventory (round to nearest day) for each company.
- Which company moves its inventory more quickly?

***E6.16 (LO 5)** Ehrhart Appliance uses a perpetual inventory system. For its flat-screen television sets, the January 1 inventory was 3 sets at \$600 each. On January 10, Ehrhart purchased 6 units at \$660 each. The company sold 2 units on January 8 and 5 units on January 15.

Apply cost flow methods to perpetual records.

Instructions

Compute the ending inventory under (a) FIFO, (b) LIFO, and (c) moving-average cost.

***E6.17 (LO 5)** Moath Company reports the following for the month of June.

Calculate inventory and cost of goods sold using three cost flow methods in a perpetual inventory system.

Date	Explanation	Units	Unit Cost	Total Cost
June 1	Inventory	200	\$5	\$1,000
12	Purchase	400	6	2,400
23	Purchase	300	7	2,100
30	Inventory	100		

Instructions

- Calculate the cost of the ending inventory and the cost of goods sold for each cost flow assumption, using a perpetual inventory system. Assume a sale of 440 units occurred on June 15 for a selling price of \$8 and a sale of 360 units on June 27 for \$9.
- How do the results differ from the answers to E6.7 and E6.9?
- Why is the average unit cost not \$6 $[(\$5 + \$6 + \$7) \div 3 = \$6]$?

Apply cost flow methods to perpetual records.

***E6.18 (LO 5)** Information about Elsa's Boards is presented in E6.5. Additional data regarding Elsa's sales of Xpert snowboards are provided below. Assume that Elsa's uses a perpetual inventory system.

Date		Units	Unit Price	Total Revenue
Sept. 5	Sale	12	\$199	\$ 2,388
Sept. 16	Sale	50	199	9,950
Sept. 29	Sale	59	209	12,331
	Totals	121		\$24,669

Instructions

- Compute ending inventory at September 30 using FIFO, LIFO, and moving-average cost.
- Compare ending inventory using a perpetual inventory system to ending inventory using a periodic inventory system (from E6.5).
- Which inventory cost flow method (FIFO, LIFO) gives the same ending inventory value under both periodic and perpetual? Which method gives different ending inventory values?

Use the gross profit method to estimate inventory.

***E6.19 (LO 6)** Shereen Company reported the following information for November and December 2020.

	November	December
Cost of goods purchased	\$536,000	\$ 610,000
Inventory, beginning-of-month	130,000	120,000
Inventory, end-of-month	120,000	?
Sales revenue	840,000	1,000,000

Shereen's ending inventory at December 31 was destroyed in a fire.

Instructions

- Compute the gross profit rate for November.
- Using the gross profit rate for November, determine the estimated cost of inventory lost in the fire.

Determine merchandise lost using the gross profit method of estimating inventory.

***E6.20 (LO 6)** The inventory of Hang Company was destroyed by fire on March 1. From an examination of the accounting records, the following data for the first 2 months of the year are obtained: Sales Revenue \$51,000, Sales Returns and Allowances \$1,000, Purchases \$31,200, Freight-In \$1,200, and Purchase Returns and Allowances \$1,400.

Instructions

Determine the merchandise lost by fire, assuming:

- A beginning inventory of \$20,000 and a gross profit rate of 30% on net sales.
- A beginning inventory of \$30,000 and a gross profit rate of 40% on net sales.

Determine ending inventory at cost using retail method.

***E6.21 (LO 6)** Kicks Shoe Store uses the retail inventory method for its two departments, Women's Shoes and Men's Shoes. The following information for each department is obtained.



Item	Women's Shoes	Men's Shoes
Beginning inventory at cost	\$ 25,000	\$ 45,000
Cost of goods purchased at cost	110,000	136,300
Net sales	178,000	185,000
Beginning inventory at retail	46,000	60,000
Cost of goods purchased at retail	179,000	185,000

Instructions

Compute the estimated cost of the ending inventory for each department under the retail inventory method.

Problems: Set A

P6.1A (LO 1) Houghton Limited is trying to determine the value of its ending inventory as of February 28, 2020, the company's year-end. The following transactions occurred, and the accountant asked your help in determining whether they should be recorded or not.

Determine items and amounts to be recorded in inventory.

- On February 26, Houghton shipped goods costing \$800 to a customer and charged the customer \$1,000. The goods were shipped with terms FOB shipping point and the receiving report indicates that the customer received the goods on March 2.
- On February 26, Crain Inc. shipped goods to Houghton under terms FOB shipping point. The invoice price was \$450 plus \$30 for freight. The receiving report indicates that the goods were received by Houghton on March 2.
- Houghton had \$720 of inventory isolated in the warehouse. The inventory is designated for a customer who has requested that the goods be shipped on March 10.
- Also included in Houghton's warehouse is \$700 of inventory that Korenic Producers shipped to Houghton on consignment.
- On February 26, Houghton issued a purchase order to acquire goods costing \$900. The goods were shipped with terms FOB destination on February 27. Houghton received the goods on March 2.
- On February 26, Houghton shipped goods to a customer under terms FOB destination. The invoice price was \$390; the cost of the items was \$240. The receiving report indicates that the goods were received by the customer on March 2.
- Houghton had damaged goods set aside in the warehouse because they are no longer saleable. These goods originally cost \$400, and Houghton had expected to sell these items for \$600.

Instructions

For each of the preceding transactions, specify whether the item in question should be included in ending inventory, and if so, at what amount. For each item that is not included in ending inventory, indicate who owns it and in what account, if any, it should have been recorded.

P6.2A (LO 2) Glee Distribution markets CDs of the performing artist Unique. At the beginning of October, Glee had in beginning inventory 2,000 of Unique's CDs with a unit cost of \$7. During October, Glee made the following purchases of Unique's CDs.

Oct. 3	2,500 @ \$8	Oct. 19	3,000 @ \$10
Oct. 9	3,500 @ \$9	Oct. 25	4,000 @ \$11

Determine cost of goods sold and ending inventory using FIFO, LIFO, and average-cost with analysis.



During October, 10,900 units were sold. Glee uses a periodic inventory system.

Instructions

- Determine the cost of goods available for sale.
- Determine (1) the ending inventory and (2) the cost of goods sold under each of the assumed cost flow methods (FIFO, LIFO, and average-cost). Prove the accuracy of the cost of goods sold under the FIFO and LIFO methods.
- Which cost flow method results in (1) the highest inventory amount for the balance sheet and (2) the highest cost of goods sold for the income statement?

b. (2) Cost of goods sold:

FIFO	\$ 94,500
LIFO	\$108,700
Average	\$101,370

P6.3A (LO 2) Financial Statement Sekhon Company had a beginning inventory on January 1 of 160 units of Product 4-18-15 at a cost of \$20 per unit. During the year, the following purchases were made.

Mar. 15	400 units at \$23	Sept. 4	330 units at \$26
July 20	250 units at \$24	Dec. 2	100 units at \$29

Determine cost of goods sold and ending inventory, using FIFO, LIFO, and average-cost with analysis.

1,000 units were sold. Sekhon Company uses a periodic inventory system.

Instructions

- Determine the cost of goods available for sale.
- Determine (1) the ending inventory, and (2) the cost of goods sold under each of the assumed cost flow methods (FIFO, LIFO, and average-cost). Prove the accuracy of the cost of goods sold under the FIFO and LIFO methods.
- Which cost flow method results in (1) the highest inventory amount for the balance sheet, and (2) the highest cost of goods sold for the income statement?

b. (2) Cost of goods sold:

FIFO	\$23,340
LIFO	\$24,840
Average	\$24,097

Compute ending inventory, prepare income statements, and answer questions using FIFO and LIFO.

P6.4A (LO 2) Financial Statement Writing The management of Gresa Inc. is reevaluating the appropriateness of using its present inventory cost flow method, which is average-cost. The company requests your help in determining the results of operations for 2020 if either the FIFO or the LIFO method had been used. For 2020, the accounting records show these data:

Inventories		Purchases and Sales	
Beginning (7,000 units)	\$14,000	Total net sales (180,000 units)	\$747,000
Ending (17,000 units)		Total cost of goods purchased (190,000 units)	466,000

Purchases were made quarterly as follows.

Quarter	Units	Unit Cost	Total Cost
1	50,000	\$2.20	\$110,000
2	40,000	2.35	94,000
3	40,000	2.50	100,000
4	60,000	2.70	162,000
	190,000		\$466,000

Operating expenses were \$130,000, and the company's income tax rate is 40%.

Instructions

a. Gross profit:

FIFO \$312,900

LIFO \$303,000

- Prepare comparative condensed income statements for 2020 under FIFO and LIFO. (Show computations of ending inventory.)
- Answer the following questions for management.
 - Which cost flow method (FIFO or LIFO) produces the more meaningful inventory amount for the balance sheet? Why?
 - Which cost flow method (FIFO or LIFO) produces the more meaningful net income? Why?
 - Which cost flow method (FIFO or LIFO) is more likely to approximate the actual physical flow of goods? Why?
 - How much more cash will be available for management under LIFO than under FIFO? Why?
 - Will gross profit under the average-cost method be higher or lower than FIFO? Than LIFO? (Note: It is not necessary to quantify your answer.)

Calculate ending inventory, cost of goods sold, gross profit, and gross profit rate under periodic method; compare results.

P6.5A (LO 2) You are provided with the following information for Koetteritz Inc. for the month ended June 30, 2020. Koetteritz uses the periodic method for inventory.

Date	Description	Quantity	Unit Cost or Selling Price
June 1	Beginning inventory	40	\$40
June 4	Purchase	135	43
June 10	Sale	110	70
June 11	Sale return	15	70
June 18	Purchase	55	46
June 18	Purchase return	10	46
June 25	Sale	65	76
June 28	Purchase	35	50

Instructions

a. (iii) Gross profit:

LIFO \$4,330

FIFO \$4,830

Average \$4,546.90

- Calculate (i) ending inventory, (ii) cost of goods sold, (iii) gross profit, and (iv) gross profit rate under each of the following methods.
 - (1) LIFO. (2) FIFO. (3) Average-cost.
- Compare results for the three cost flow assumptions.

Compare specific identification, FIFO, and LIFO under periodic method; use cost flow assumption to justify price increase.

P6.6A (LO 2) Financial Statement You are provided with the following information for Gobler Inc. Gobler Inc. uses the periodic method of accounting for its inventory transactions.

March 1	Beginning inventory 2,000 liters at a cost of 60¢ per liter.
March 3	Purchased 2,500 liters at a cost of 65¢ per liter.
March 5	Sold 2,300 liters for \$1.05 per liter.
March 10	Purchased 4,000 liters at a cost of 72¢ per liter.
March 20	Purchased 2,500 liters at a cost of 80¢ per liter.
March 30	Sold 5,200 liters for \$1.25 per liter.

Instructions

- a. Prepare partial income statements for 2020 through gross profit, and calculate the value of ending inventory that would be reported on the balance sheet, under each of the following cost flow assumptions. (Round ending inventory and cost of goods sold to the nearest dollar.)
- Specific identification method assuming:
 - The March 5 sale consisted of 1,000 liters from the March 1 beginning inventory and 1,300 liters from the March 3 purchase; and
 - The March 30 sale consisted of the following number of units sold from beginning inventory and each purchase: 450 liters from March 1; 550 liters from March 3; 2,900 liters from March 10; 1,300 liters from March 20.
 - FIFO.
 - LIFO.
- b. How can companies use a cost flow method to justify price increases? Which cost flow method would best support an argument to increase prices?

a. Gross profit:
1. Specific identification
\$3,715

2. FIFO \$3,930

3. LIFO \$3,385

P6.7A (LO 2) Financial Statement Writing The management of Danica Co. asks your help in determining the comparative effects of the FIFO and LIFO inventory cost flow methods. For 2020, the accounting records provide the following data.

Inventory, January 1 (10,000 units)	\$ 47,000
Cost of 100,000 units purchased	532,000
Selling price of 84,000 units sold	735,000
Operating expenses	140,000

Units purchased consisted of 35,000 units at \$5.10 on May 10; 35,000 units at \$5.30 on August 15; and 30,000 units at \$5.60 on November 20. Income taxes are 30%.

Compute ending inventory, prepare income statements, and answer questions using FIFO and LIFO.

Instructions

- a. Prepare comparative condensed income statements for 2020 under FIFO and LIFO. (Show computations of ending inventory.)
- b. Answer the following questions for management.
- Which inventory cost flow method produces the most meaningful inventory amount for the balance sheet? Why?
 - Which inventory cost flow method produces the most meaningful net income? Why?
 - Which inventory cost flow method is most likely to approximate actual physical flow of the goods? Why?
 - How much additional cash will be available for management under LIFO than under FIFO? Why?
 - How much of the gross profit under FIFO is illusory in comparison with the gross profit under LIFO?

a. Net income
FIFO \$113,120
LIFO \$101,220

***P6.8A (LO 5)** Dempsey Inc. is a retailer operating in British Columbia. Dempsey uses the perpetual inventory method. All sales returns from customers result in the goods being returned to inventory; the inventory is not damaged. Assume that there are no credit transactions; all amounts are settled in cash. You are provided with the following information for Dempsey Inc. for the month of January 2020.

Calculate cost of goods sold and ending inventory under LIFO, FIFO, and moving-average cost under the perpetual system; compare gross profit under each assumption.

Date	Description	Quantity	Unit Cost or Selling Price
January 1	Beginning inventory	100	\$15
January 5	Purchase	140	18
January 8	Sale	110	28
January 10	Sale return	10	28
January 15	Purchase	55	20
January 16	Purchase return	5	20
January 20	Sale	90	32
January 25	Purchase	20	22

Instructions

- a. For each of the following cost flow assumptions, calculate (i) cost of goods sold, (ii) ending inventory, and (iii) gross profit.
- LIFO.
 - FIFO.
 - Moving-average cost. (Round cost per unit to three decimal places.)
- b. Compare results for the three cost flow assumptions.

a. (iii) Gross profit:
LIFO \$2,160
FIFO \$2,560
Average \$2,421

Determine ending inventory under a perpetual inventory system.

***P6.9A (LO 5)** Wittmann Co. began operations on July 1. It uses a perpetual inventory system. During July, the company had the following purchases and sales.

Date	Purchases		Sales Units
	Units	Unit Cost	
July 1	5	\$122	
July 6			3
July 11	7	\$136	
July 14			5
July 21	8	\$147	
July 27			5

Instructions

- a. Ending inventory
 FIFO \$1,029
 Avg. \$996
 LIFO \$957

- a. Determine the ending inventory under a perpetual inventory system using (1) FIFO, (2) moving-average cost, and (3) LIFO.
 b. Which costing method produces the highest ending inventory valuation?

Compute gross profit rate and inventory loss using gross profit method.

***P6.10A (LO 6)** Bao Company lost all of its inventory in a fire on December 26, 2020. The accounting records showed the following gross profit data for November and December.



	November	December (to 12/26)
Net sales	\$600,000	\$700,000
Beginning inventory	32,000	36,000
Purchases	389,000	420,000
Purchase returns and allowances	13,300	14,900
Purchase discounts	8,500	9,500
Freight-in	8,800	9,900
Ending inventory	36,000	?

Bao is fully insured for fire losses but must prepare a report for the insurance company.

Instructions

- a. Gross profit rate 38%

- a. Compute the gross profit rate for November.
 b. Using the gross profit rate for November, determine the estimated cost of the inventory lost in the fire.

Compute ending inventory using retail method.

***P6.11A (LO 6)** Rayre Books uses the retail inventory method to estimate its monthly ending inventories. The following information is available for two of its departments at October 31, 2020.

	Hardcovers		Paperbacks	
	Cost	Retail	Cost	Retail
Beginning inventory	\$ 420,000	\$ 640,000	\$ 280,000	\$ 360,000
Purchases	2,135,000	3,200,000	1,155,000	1,540,000
Freight-in	24,000		12,000	
Purchase discounts	44,000		22,000	
Net sales		3,100,000		1,570,000

At December 31, Rayre Books takes a physical inventory at retail. The actual retail values of the inventories in each department are Hardcovers \$744,000 and Paperbacks \$335,000.

Instructions

- a. Hardcovers: End. Inv.
 \$488,400

- a. Determine the estimated cost of the ending inventory for each department at **October 31**, 2020, using the retail inventory method.
 b. Compute the ending inventory at cost for each department at **December 31**, assuming the cost-to-retail ratios for the year are 65% for Hardcovers and 75% for Paperbacks.



Terra Images/Age Fotostock America, Inc.

Accounting Information Systems

Chapter Preview

As the following Feature Story demonstrates, a reliable information system is a necessity for any company. Whether companies use pen, pencil, or computers in maintaining accounting records, certain principles and procedures apply. The purpose of this chapter is to explain and illustrate these features.

Feature Story

QuickBooks® Helps This Retailer Sell Guitars

Starting a small business requires many decisions. For example, you have to decide where to locate, how much space you need, how much inventory to have, how many employees to hire, and where to advertise. Small business owners are typically so concerned about the product and sales side of their business that they often do not give enough thought to something that is also critical to their success—how to keep track of financial results.

Small business owners today can choose either manual or computerized accounting systems. For example, Paul and Laura West were the owners of the first independent dealership of

Carvin guitars and professional audio equipment. When they founded their company in Sacramento, California, they decided to purchase a computerized accounting system that would integrate many aspects of their retail operations. They wanted to use their accounting software to manage their inventory of guitars and amplifiers, enter sales, record and report financial data, and process credit card and debit card transactions. They evaluated a number of options and chose QuickBooks® by **Intuit Inc.**

QuickBooks®, like most other popular software packages, has programs designed for the needs of a specific type of business, which in this case is retailing. This QuickBooks® retailing package automatically collects sales information from its point-of-sale scanning devices. It also keeps track of inventory levels and automatically generates purchase orders for popular

items when re-order points are reached. It even supports specialized advertising campaigns.

For example, QuickBooks® compiled a customer database from which the Wests sent out targeted direct mailings to potential customers. The computerized system also enabled data files to be emailed to the company’s accountant. This kept down costs and made it easier and more efficient for the

Wests to generate financial reports as needed. The Wests believed that the investment in the computerized system saved them time and money, and allowed them to spend more time on other aspects of their business.

Source: Intuit Inc., “QuickBooks® and ProAdvisor® Help Make Guitar Store a Hit,” *Journal of Accountancy* (May 2006), p. 101.

Chapter Outline

LEARNING OBJECTIVES

LO 1 Explain the basic concepts of an accounting information system.	• Computerized accounting systems • Manual accounting systems	DO IT! 1 Basic AIS Concepts
LO 2 Describe the nature and purpose of a subsidiary ledger.	• Subsidiary ledger example • Advantages of subsidiary ledgers	DO IT! 2 Subsidiary Ledgers
LO 3 Record transactions in special journals.	• Sales journal • Cash receipts journal • Purchases journal • Cash payments journal • Effects of special journals • Cyber security	DO IT! 3 Special Journals
Go to the Review and Practice section at the end of the chapter for a review of key concepts and practice applications with solutions. Visit WileyPLUS with ORION for additional tutorials and practice opportunities.		

Overview of Accounting Information Systems

LEARNING OBJECTIVE 1

Explain the basic concepts of an accounting information system.

The **accounting information system** collects and processes transaction data and communicates financial information to decision-makers. It includes each of the steps in the accounting cycle that you studied in earlier chapters. It also includes the documents that provide evidence of the transactions, and the records, trial balances, worksheets, and financial statements that result. An **accounting system** may be either manual or computerized. Most businesses use some sort of computerized accounting system, whether it is an off-the-shelf system for small businesses, like QuickBooks® or Sage 50, or a more complex custom-made system.

Efficient and effective accounting information systems are based on certain basic principles. These principles, as described in **Illustration 7.1**, are (1) cost-effectiveness, (2) usefulness, and (3) flexibility. If the accounting system is cost-effective, provides useful output, and

has the flexibility to meet future needs, it can contribute to both individual and organizational goals.

Cost-Effectiveness

The accounting system must be cost-effective. Benefits of information must outweigh the costs of providing it.



Useful Output

To be useful, information must be understandable, relevant, reliable, timely, and accurate. Designers of accounting systems must consider the needs and knowledge of various users.



Flexibility

The accounting system should accommodate a variety of users and changing information needs. The system should be sufficiently flexible to meet the resulting changes in the demands made upon it.

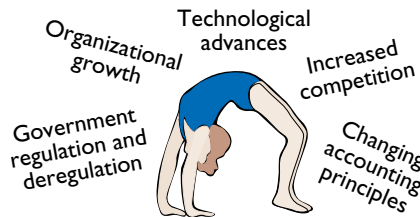


ILLUSTRATION 7.1

Principles of an efficient and effective accounting information system

Computerized Accounting Systems

Many small businesses use a computerized general ledger accounting system. **General ledger accounting systems** are software programs that integrate the various accounting functions related to sales, purchases, receivables, payables, cash receipts and disbursements, and payroll. They also generate financial statements. Computerized systems have a number of advantages over manual systems. First, the company typically enters data only once in a computerized system. Second, because the computer does most steps automatically, it eliminates many errors resulting from human intervention in a manual system, such as errors in posting or preparation of financial statements. Computerized systems also provide up-to-the-minute information. More timely information often results in better business decisions. Many different general ledger software packages are available.

Choosing a Software Package

To identify the right software for your business, you must understand your company's operations. For example, consider its needs with regard to inventory, billing, payroll, and cash management. In addition, the company might have specific needs that are not supported by all software systems. For example, you might want to track employees' hours on individual jobs or to extract information for determining sales commissions. Choosing the right system is critical because installation of even a basic system is time-consuming, and learning a new system will require many hours of employee time.

Entry-Level Software

Software publishers tend to classify businesses into groups based on revenue and the number of employees. Companies with revenues of less than \$5 million and up to 20 employees generally

use **entry-level programs**. The two leading entry-level programs are **Intuit's** QuickBooks® and **The Sage Group's** Sage 50. These programs control more than 90% of the market. Each of these entry-level programs comes in many different industry-specific versions. For example, some are designed for very specific industry applications such as restaurants, retailing, construction, manufacturing, or nonprofit.

Quality entry-level packages typically involve more than recording transactions and preparing financial statements. Here are some common features and benefits:

- **Easy data access and report preparation.** Users can easily access information related to specific customers or suppliers. For example, you can view all transactions, invoices, payments, as well as contact information for a specific client.
- **Audit trail.** As a result of the Sarbanes-Oxley Act, companies are now far more concerned that their accounting system minimizes opportunities for fraud. Many programs provide an “audit trail” that enables the tracking of all transactions.
- **Internal controls.** Some systems have an internal accounting review that identifies suspicious transactions or likely mistakes such as wrong account numbers or duplicate transactions (see **Ethics Note**).
- **Customization.** This feature enables the company to create data fields specific to the needs of its business.
- **Network-compatibility.** Multiple users in the company can access the system at the same time.

ETHICS NOTE

Entire books and movies have used cyber attacks as a major theme. Most programmers would agree that ensuring cyber security is the most difficult and time-consuming phase of their jobs.

Enterprise Resource Planning Systems

Enterprise resource planning (ERP) systems are typically used by manufacturing companies with more than 500 employees and \$500 million in sales. The best-known of these systems are **SAP AG's** SAP ERP (the most widely used) and **Oracle's** ERP. ERP systems go far beyond the functions of an entry-level general ledger package. They integrate all aspects of the organization, including accounting, sales, human resource management, and manufacturing. Because of the complexity of an ERP system, implementation can take three years and cost five times as much as the purchase price of the system. Purchase and implementation of ERP systems can cost from \$250,000 to as much as \$50 million for the largest multinational corporations.

Ethics Insight



© Sean Locke/iStockphoto

Curbing Fraudulent Activity with Software

The Sarbanes-Oxley Act (SOX) requires that companies demonstrate that they have adequate controls in place to detect significant fraudulent behavior by employees. The SOX requirements have created a huge market for software that can monitor and trace every recorded transaction and adjusting entry. This enables companies to pinpoint who used the accounting system and when they used it.

These systems also require “electronic signatures” by employees for all significant transactions. Such signatures verify that employees

have followed all required procedures, and that all actions are properly authorized. One firm that specializes in compliance software had 10 clients prior to SOX and 250 after SOX.

Note that small businesses have no standards like SOX and often do not have the resources to implement a fraud-prevention system. As a result, small businesses lose nearly \$630 billion to fraud each year. To address this problem, more sophisticated software is being designed for small business fraud prevention.

Sources: W. M. Bulkeley and C. Forelle, “Anti-Crime Program: How Corporate Scandals Gave Tech Firms a New Business Line,” *Wall Street Journal* (December 9, 2005), p. A1; and “New Software Fights Small Business Fraud,” *FOX Business* (August 9, 2013).

Why might this software help reduce fraudulent activity by employees? (Go to WileyPLUS for this answer and additional questions.)

Manual Accounting Systems

In **manual accounting systems**, someone performs each of the steps in the accounting cycle by hand. For example, someone manually enters each accounting transaction in the journal and manually posts each to the ledger. Other manual computations must be made to obtain ledger account balances and to prepare a trial balance and financial statements. In the remainder of this chapter, we illustrate the use of a manual system.

You might be wondering, “Why cover manual accounting systems if the real world uses computerized systems?” First, small businesses still abound. Most of them begin operations with manual accounting systems and convert to computerized systems as the business grows. You may work in a small business or start your own someday, so it is useful to know how a manual system works. Second, to understand what computerized accounting systems do, you also need to understand manual accounting systems.

The manual accounting system represented in the first six chapters of this textbook is satisfactory in a company with a low volume of transactions. However, in most companies, it is necessary to add additional ledgers and journals to the accounting system to record transaction data efficiently.

DO IT! 1 | Basic AIS Concepts

Indicate whether the following statements are true or false. If false, indicate how to correct the statement.

1. An accounting information system collects and processes transaction data and communicates financial information to decision-makers.
2. A company typically enters data only once in a manual accounting system.
3. Enterprise resource planning (ERP) systems are typically used by companies with revenues of less than \$5 million and up to 20 employees.

Solution

1. True. 2. False. A company typically enters data only once in a computerized accounting system. 3. False. Enterprise resource planning (ERP) systems are typically used by manufacturing companies with more than 500 employees and \$500 million in sales.

Related exercise material: **BE7.1, BE7.2, BE7.3, and DO IT! 7.1.**

ACTION PLAN

- Understand the principles of an effective accounting information system.
- Recognize the types of available general ledger software packages.

Subsidiary Ledgers

LEARNING OBJECTIVE 2

Describe the nature and purpose of a subsidiary ledger.

Imagine a business that has several thousand charge (credit) customers and shows the transactions with these customers in only one general ledger account—Accounts Receivable. It would be nearly impossible to determine the balance owed by an individual customer at any specific time. Similarly, the amount payable to one creditor would be difficult to locate quickly from a single Accounts Payable account in the general ledger.

Instead, companies use subsidiary ledgers to keep track of individual balances. A **subsidiary ledger** is a group of accounts with a common characteristic (for example, all accounts receivable). It is an addition to and an expansion of the general ledger. The subsidiary ledger frees the general ledger from the details of individual balances.

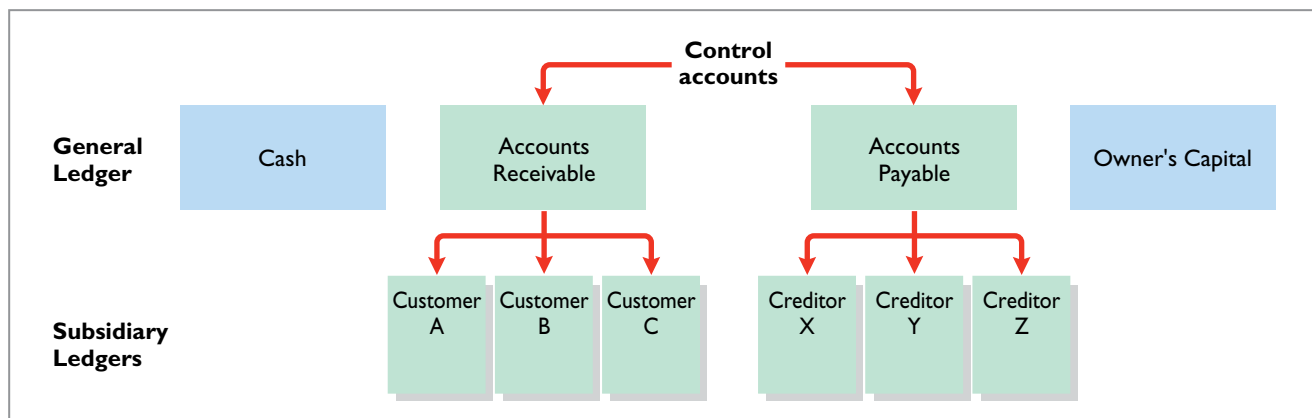
Two common subsidiary ledgers are as follows.

1. The **accounts receivable** (or **customers'**) **subsidiary ledger**, which collects transaction data of individual customers.
2. The **accounts payable** (or **creditors'**) **subsidiary ledger**, which collects transaction data of individual creditors.

In each of these subsidiary ledgers, companies usually arrange individual accounts in alphabetical order.

A general ledger account summarizes the detailed data from a subsidiary ledger. For example, the detailed data from the accounts receivable subsidiary ledger are summarized in Accounts Receivable in the general ledger. The general ledger account that summarizes subsidiary ledger data is called a **control account**. **Illustration 7.2** presents an overview of the relationship of subsidiary ledgers to the general ledger. There, the general ledger control accounts and subsidiary ledger accounts are in green. Note that Cash and Owner's Capital in this illustration are not control accounts because there are no subsidiary ledger accounts related to these accounts.

ILLUSTRATION 7.2 Relationship of general ledger and subsidiary ledgers



At the end of an accounting period, each general ledger control account balance must equal the composite balance of the individual accounts in the related subsidiary ledger. For example, the balance in Accounts Payable in Illustration 7.2 must equal the total of the subsidiary balances of Creditors X + Y + Z.

Subsidiary Ledger Example

Illustration 7.3 lists credit sales and collections on account for Pujols Company.

ILLUSTRATION 7.3

Sales and collection transactions

Credit Sales			Collections on Account		
Jan. 10	Aaron Co.	\$ 6,000	Jan. 19	Aaron Co.	\$4,000
12	Branden Inc.	3,000	21	Branden Inc.	3,000
20	Caron Co.	3,000	29	Caron Co.	1,000
		<u>\$12,000</u>			<u>\$8,000</u>

Illustration 7.4 provides an example of a control account and subsidiary ledger for Pujols Company. (Due to space considerations, the explanation column in these accounts is not

ILLUSTRATION 7.4 Relationship between general and subsidiary ledgers

General Ledger Software for Accounting

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ACCOUNTS RECEIVABLE SUBSIDIARY LEDGER

Aaron Co.

Date	Ref.	Debit	Credit	Balance
2020				
Jan 10		6,000		6,000
19			4,000	2,000

Branden Inc.

Date	Ref.	Debit	Credit	Balance
2020				
Jan 12		3,000		3,000
21			3,000	-----

Caron Co.

Date	Ref.	Debit	Credit	Balance
2020				
Jan 20		3,000		3,000
29			1,000	2,000

GENERAL LEDGER

Accounts Receivable No. 112

Date	Ref.	Debit	Credit	Balance
2020				
Jan 31		12,000		12,000
31			8,000	4,000

The subsidiary ledger is separate from the general ledger.

Accounts Receivable is a control account.

start General Ledger General Jmnl Sales Jmnl Cash Receipts Jmnl Purchases Jmnl

shown in this and subsequent illustrations.) Illustration 7.4 is based on the transactions listed in Illustration 7.3.

Pujols can reconcile the total debits (\$12,000) and credits (\$8,000) in Accounts Receivable in the general ledger to the detailed debits and credits in the subsidiary accounts. Also, the balance of \$4,000 in the control account agrees with the total of the balances in the individual accounts (Aaron Co. \$2,000 + Branden Inc. \$0 + Caron Co. \$2,000) in the subsidiary ledger.

As Illustration 7.4 shows, companies make monthly postings to the control accounts in the general ledger. This practice allows them to prepare monthly financial statements. Companies post to the individual accounts in the subsidiary ledger daily. Daily posting ensures that account information is current. This enables the company to monitor credit limits, bill customers, and answer inquiries from customers about their account balances.

Advantages of Subsidiary Ledgers

Subsidiary ledgers have several advantages:

1. **They show in a single account transactions affecting one customer or one creditor**, thus providing up-to-date information on specific account balances.
2. **They free the general ledger of excessive details.** As a result, a trial balance of the general ledger does not contain vast numbers of individual account balances.
3. **They help locate errors in individual accounts** by reducing the number of accounts in one ledger and by using control accounts.
4. **They make possible a division of labor** in posting. One employee can post to the general ledger while someone else posts to the subsidiary ledgers.

Accounting Across the Organization



© Niels Laan/iStockphoto

“I’m John Smith, a.k.a. 13695071642”

Rather than relying on customer or creditor names in a subsidiary ledger, a computerized system expands the account number of the control account in a prespecified

manner. For example, if the control account Accounts Receivable was numbered 10010, the first account in the accounts

receivable subsidiary ledger might be numbered 10010–0001. Most systems allow inquiries about specific accounts in the subsidiary ledger (by account number) or about the control account. With the latter, the system would automatically total all the subsidiary accounts whenever an inquiry to the control account was made.

Why use numbers to identify names in a computerized system? (Go to WileyPLUS for this answer and additional questions.)

DO IT! 2 | Subsidiary Ledgers

Presented below is information related to Sims Company for its first month of operations. Determine the balances that appear in the accounts payable subsidiary ledger. What Accounts Payable balance appears in the general ledger at the end of January?

Credit Purchases			Cash Paid		
Jan. 5	Devon Co.	\$11,000	Jan. 9	Devon Co.	\$7,000
11	Shelby Co.	7,000	14	Shelby Co.	2,000
22	Taylor Co.	14,000	27	Taylor Co.	9,000

ACTION PLAN

- Subtract cash paid from credit purchases to determine the balances in the accounts payable subsidiary ledger.
- Sum the individual balances to determine the Accounts Payable balance.

Solution

Subsidiary ledger balances:

Devon Co.: \$4,000 (\$11,000 – \$7,000)

Shelby Co.: \$5,000 (\$7,000 – \$2,000)

Taylor Co.: \$5,000 (\$14,000 – \$9,000)

General ledger Accounts Payable balance: \$14,000 (\$4,000 + \$5,000 + \$5,000)

Related exercise material: **BE7.4, BE7.5, DO IT! 7.2, E7.1, E7.2, E7.3, E7.4, and E7.5.**

Special Journals

LEARNING OBJECTIVE 3

Record transactions in special journals.

So far, you have learned to journalize transactions in a two-column general journal and post each entry to the general ledger. This procedure is satisfactory in only very small companies. To expedite journalizing and posting, most companies use special journals **in addition to the general journal**.

Companies use **special journals** to record similar types of transactions. Examples are all sales of merchandise on account or all cash receipts. The types of transactions

that occur frequently in a company determine what special journals the company uses. Most merchandising companies record daily transactions using the journals shown in **Illustration 7.5**.

ILLUSTRATION 7.5 Use of special journals and the general journal

Sales Journal	Cash Receipts Journal	Purchases Journal	Cash Payments Journal	General Journal
Used for: All sales of merchandise on account	Used for: All cash received (including cash sales)	Used for: All purchases of merchandise on account	Used for: All cash paid (including cash purchases)	Used for: Transactions that cannot be entered in a special journal, including correcting, adjusting, and closing entries

If a transaction cannot be recorded in a special journal, the company records it in the general journal. For example, if a company had special journals for only the four types of transactions listed above, it would record purchase returns and allowances that do not affect cash in the general journal. Similarly, **correcting, adjusting, and closing entries are recorded in the general journal.** In some situations, companies might use special journals other than those listed above. For example, when sales returns and allowances that do not affect cash are frequent, a company might use a special journal to record these transactions.

Special journals **permit greater division of labor** because several people can record entries in different journals at the same time. For example, one employee may journalize all cash receipts, and another may journalize all credit sales. Also, the use of special journals **reduces the time needed to complete the posting process.** With special journals, companies may post some accounts monthly instead of daily, as we will illustrate later in the chapter (see **Helpful Hint**). On the following pages, we discuss the four special journals shown in Illustration 7.5.

HELPFUL HINT

Postings are also made daily to individual ledger accounts in the inventory subsidiary ledger to maintain a perpetual inventory.

Sales Journal

In the **sales journal**, companies record **sales of merchandise on account**. Cash sales of merchandise go in the cash receipts journal. Credit sales of assets other than merchandise go in the general journal.

Journalizing Credit Sales

To demonstrate use of a sales journal, we will use data for Karns Wholesale Supply, which uses a **perpetual inventory system**. Under this system, each entry in the sales journal results in one entry **at selling price** and another entry **at cost**. The entry at selling price is a debit to Accounts Receivable (a control account) and a credit of equal amount to Sales Revenue. The entry at cost is a debit to Cost of Goods Sold and a credit of equal amount to Inventory (a control account). Using a sales journal with two amount columns, the company can show on only one line a sales transaction at both selling price and cost. **Illustration 7.6** shows this two-column sales journal of Karns Wholesale Supply, using assumed credit sales transactions (for sales invoices 101–107).

ILLUSTRATION 7.6 Journalizing the sales journal—perpetual inventory system

Date	Account Debited	Invoice No.	Ref.	Accts. Receivable Dr. Sales Revenue Cr.	Cost of Goods Sold Dr. Inventory Cr.
2020					
May 3	Abbot Sisters	101		10,600	6,360
7	Babson Co.	102		11,350	7,370
14	Carson Bros.	103		7,800	5,070
19	Deli Co.	104		9,300	6,510
21	Abbot Sisters	105		15,400	10,780
24	Deli Co.	106		21,210	15,900
27	Babson Co.	107		14,570	10,200
				<u>90,230</u>	<u>62,190</u>

Note that, unlike the general journal, an explanation is not required for each entry in a special journal. Also, the use of prenumbered invoices ensures that all invoices are journalized and no invoices are duplicated. Finally, the reference (Ref.) column is not used in journalizing. It is used in posting the sales journal, as explained in the next section.

Posting the Sales Journal

Companies make daily postings from the sales journal **to the individual accounts receivable** in the subsidiary ledger. Posting **to the general ledger** is done **monthly**. **Illustration 7.7** shows both the daily and monthly postings.

A check mark (✓) is inserted in the reference column to indicate that the daily posting to the customer's account has been made. If the subsidiary ledger accounts were numbered, the account number would be entered in place of the check mark. At the end of the month, Karns posts the column totals of the sales journal to the general ledger. Here, the column totals are as follows: From the selling-price column, a debit of \$90,230 to Accounts Receivable (account No. 112) and a credit of \$90,230 to Sales Revenue (account No. 401); from the cost column, a debit of \$62,190 to Cost of Goods Sold (account No. 505) and a credit of \$62,190 to Inventory (account No. 120). Karns inserts the account numbers below the column totals to indicate that the postings have been made. In both the general ledger and subsidiary ledger accounts, the reference **S1** indicates that the posting came from page 1 of the sales journal.

Proving the Ledgers

The next step is to “prove” the ledgers. To do so, Karns must determine two things: (1) The total of the general ledger debit balances must equal the total of the general ledger credit balances. (2) The sum of the subsidiary ledger balances must equal the balance in the control account. **Illustration 7.8** shows the proof of the postings from the sales journal to the general and subsidiary ledgers.

Advantages of the Sales Journal

The use of a special journal to record sales on account has a number of advantages. First, the one-line entry for each sales transaction saves time. In the sales journal, it is not necessary to write out the four account titles for each transaction. Second, only totals, rather than individual entries, are posted to the general ledger. This saves posting time and reduces the possibilities

ILLUSTRATION 7.7 Posting the sales journal

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SALES JOURNAL SI

Date	Account Debited	Invoice No.	Ref.	Accts. Receivable Dr. Sales Revenue Cr.	Cost of Goods Sold Dr. Inventory Cr.
2020					
May 3	Abbot Sisters	101	✓	10,600	6,360
7	Babson Co.	102	✓	11,350	7,370
14	Carson Bros.	103	✓	7,800	5,070
19	Deli Co.	104	✓	9,300	6,510
21	Abbot Sisters	105	✓	15,400	10,780
24	Deli Co.	106	✓	21,210	15,900
27	Babson Co.	107	✓	14,570	10,200
				<u>90,230</u>	<u>62,190</u>
				(112) / (401)	(505) / (120)

At the end of the accounting period, the company posts totals to the general ledger.

The company posts individual amounts to the subsidiary ledger daily.

ACCOUNTS RECEIVABLE SUBSIDIARY LEDGER

Abbot Sisters

Date	Ref.	Debit	Credit	Balance
2020				
May 3	SI	10,600		10,600
21	SI	15,400		26,000

Babson Co.

Date	Ref.	Debit	Credit	Balance
2020				
May 7	SI	11,350		11,350
27	SI	14,570		25,920

Carson Bros.

Date	Ref.	Debit	Credit	Balance
2020				
May 14	SI	7,800		7,800

Deli Co.

Date	Ref.	Debit	Credit	Balance
2020				
May 19	SI	9,300		9,300
24	SI	21,210		30,510

GENERAL LEDGER

Accounts Receivable No. 112

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI	90,230		90,230

Inventory No. 120

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI		62,190	62,190 ¹

Sales Revenue No. 401

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI		90,230	90,230

Cost of Goods Sold No. 505

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI	62,190		62,190

The subsidiary ledger is separate from the general ledger.

Accounts Receivable is a control account.

¹The normal balance for Inventory is a debit. But because of the sequence in which we have posted the special journals, with the sales journal first, the credits to Inventory are posted before the debits. This posting sequence explains the credit balance in Inventory, which exists only until the other journals are posted.

start General Ledger General Jnl Sales Jnl Cash Receipts Jnl Purchases Jnl

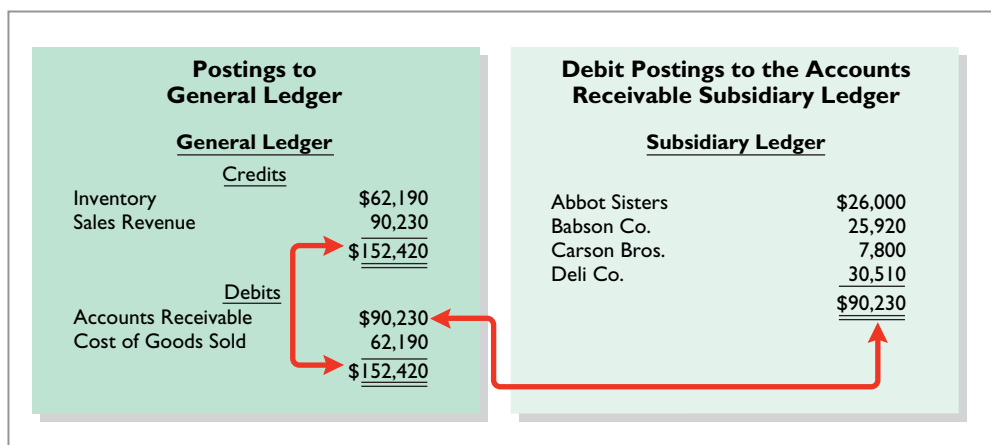
of errors in posting. Finally, a division of labor results because one individual can take responsibility for the sales journal.

Cash Receipts Journal

In the **cash receipts journal**, companies record all receipts of cash. The most common types of cash receipts are cash sales of merchandise and collections of accounts receivable. Many

ILLUSTRATION 7.8

Proving the equality of the postings from the sales journal



other possibilities exist, such as receipt of money from bank loans and cash proceeds from disposal of equipment. A one- or two-column cash receipts journal would not have space enough for all possible cash receipt transactions. Therefore, companies use a multiple-column cash receipts journal.

Generally, a cash receipts journal includes the following columns: debit columns for Cash and Sales Discounts, and credit columns for Accounts Receivable, Sales Revenue, and “Other Accounts.” Companies use the Other Accounts category when the cash receipt does not involve a cash sale or a collection of accounts receivable. Under a perpetual inventory system, each sales entry also is accompanied by an entry that debits Cost of Goods Sold and credits Inventory for the cost of the merchandise sold. **Illustration 7.9** shows a six-column cash receipts journal.

Companies may use additional credit columns if these columns significantly reduce postings to a specific account. For example, a loan company such as **Household International** receives thousands of cash collections from customers. Using separate credit columns for Loans Receivable and Interest Revenue, rather than the Other Accounts credit column, would reduce postings.

Journalizing Cash Receipts Transactions

To illustrate the journalizing of cash receipts transactions, we will continue with the May transactions of Karns Wholesale Supply. Collections from customers relate to the entries recorded in the sales journal in Illustration 7.6. The entries in the cash receipts journal are based on the following cash receipts.

- May 1 D. A. Karns makes an investment of \$5,000 in the business.
- 7 Cash sales of merchandise total \$1,900 (cost, \$1,240).
- 10 Received a check for \$10,388 from Abbot Sisters in payment of invoice No. 101 for \$10,600 less a 2% discount.
- 12 Cash sales of merchandise total \$2,600 (cost, \$1,690).
- 17 Received a check for \$11,123 from Babson Co. in payment of invoice No. 102 for \$11,350 less a 2% discount.
- 22 Received cash by signing a note for \$6,000.
- 23 Received a check for \$7,644 from Carson Bros. in full for invoice No. 103 for \$7,800 less a 2% discount.
- 28 Received a check for \$9,114 from Deli Co. in full for invoice No. 104 for \$9,300 less a 2% discount.

The following discussion provides further information about the columns in the cash receipts journal.

Debit Columns:

- Cash.** Karns enters in this column the amount of cash actually received in each transaction. The column total indicates the total cash receipts for the month.

ILLUSTRATION 7.9 Journalizing and posting the cash receipts journal

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CASH RECEIPTS JOURNAL

Date	Account Credited	Ref.	Cash Dr.	Sales Discounts Dr.	Accounts Receivable Cr.	Sales Revenue Cr.	Other Accounts Cr.	Cost of Goods Sold Dr. Inventory Cr.
2020								
May 1	Owner's Capital	301	5,000				5,000	
7			1,900			1,900		1,240
10	Abbot Sisters	✓	10,388	212	10,600			
12			2,600			2,600		1,690
17	Babson Co.	✓	11,123	227	11,350			
22	Notes Payable	200	6,000				6,000	
23	Carson Bros.	✓	7,644	156	7,800			
28	Deli Co.	✓	9,114	186	9,300			
			53,769	781	39,050	4,500	11,000	2,930
			(101)	(414)	(112)	(401)	(x)	(505)/(120)

The company posts individual amounts to the subsidiary ledger daily.

ACCOUNTS RECEIVABLE SUBSIDIARY LEDGER

Abbot Sisters

Date	Ref.	Debit	Credit	Balance
2020				
May 3	SI	10,600		10,600
10	CRI		10,600	-----
21	SI	15,400		15,400

Babson Co.

Date	Ref.	Debit	Credit	Balance
2020				
May 7	SI	11,350		11,350
17	CRI		11,350	-----
27	SI	14,570		14,570

Carson Bros.

Date	Ref.	Debit	Credit	Balance
2020				
May 14	SI	7,800		7,800
23	CRI		7,800	-----

Deli Co.

Date	Ref.	Debit	Credit	Balance
2020				
May 19	SI	9,300		9,300
24	SI	21,210		30,510
28	CRI		9,300	21,210

The subsidiary ledger is separate from the general ledger.

Accounts Receivable is a control account.

GENERAL LEDGER

Cash No. 101

Date	Ref.	Debit	Credit	Balance
2020				
May 31	CRI	53,769		53,769

Accounts Receivable No. 112

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI	90,230		90,230
31	CRI		39,050	51,180

Inventory No. 120

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI		62,190	(62,190)
31	CRI		2,930	(65,120)

Notes Payable No. 200

Date	Ref.	Debit	Credit	Balance
2020				
May 22	CRI		6,000	6,000

Owner's Capital No. 301

Date	Ref.	Debit	Credit	Balance
2020				
May 1	CRI		5,000	5,000

Sales Revenue No. 401

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI		90,230	90,230
31	CRI		4,500	94,730

Sales Discounts No. 414

Date	Ref.	Debit	Credit	Balance
2020				
May 31	CRI	781		781

Cost of Goods Sold No. 505

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI	62,190		62,190
31	CRI	2,930		65,120

start General Ledger General Jnl Sales Jnl Cash Receipts Jnl Purchases Jnl

HELPFUL HINT

A *subsidiary ledger* account is credited when the entry involves a collection of accounts receivable. A *general ledger* account is credited when the account is not shown in a special column (and an amount must be entered in the Other Accounts column). Otherwise, no account is shown in the “Account Credited” column.

2. **Sales Discounts.** Karns includes a Sales Discounts column in its cash receipts journal. By doing so, it does not need to enter sales discount items in the general journal. As a result, the cash receipts journal shows on one line the collection of an account receivable within the discount period.

Credit Columns:

3. **Accounts Receivable.** Karns uses the Accounts Receivable column to record cash collections on account (see **Helpful Hint**). The amount entered here is the amount to be credited to the individual customer’s account.
4. **Sales Revenue.** The Sales Revenue column records all cash sales of merchandise. Cash sales of other assets (plant assets, for example) are not reported in this column.
5. **Other Accounts.** Karns uses the Other Accounts column whenever the credit is other than to Accounts Receivable or Sales Revenue. For example, in the first entry, Karns enters \$5,000 as a credit to Owner’s Capital. This column is often referred to as the sundry accounts column.

Debit and Credit Column:

6. **Cost of Goods Sold and Inventory.** This column records debits to Cost of Goods Sold and credits to Inventory.

In a multi-column journal, generally only one line is needed for each entry. Debit and credit amounts for each line must be equal. When Karns journalizes the collection from Abbot Sisters on May 10, for example, three amounts are indicated. Note also that the Account Credited column identifies both general ledger and subsidiary ledger account titles. General ledger accounts are illustrated in the May 1 and May 22 entries. A subsidiary account is illustrated in the May 10 entry for the collection from Abbot Sisters.

When Karns has finished journalizing a multi-column journal, it totals the amount columns and compares the totals to prove the equality of debits and credits. **Illustration 7.10** shows the proof of the equality of Karns’s cash receipts journal.

ILLUSTRATION 7.10**Proving the equality of the cash receipts journal**

Debits		Credits	
Cash	\$53,769	Accounts Receivable	\$39,050
Sales Discounts	781	Sales Revenue	4,500
Cost of Goods Sold	2,930	Other Accounts	11,000
	<u>\$57,480</u>	Inventory	2,930
			<u>\$57,480</u>

Totaling the columns of a journal and proving the equality of the totals is called **footing** and **cross-footing** a journal.

Posting the Cash Receipts Journal

Posting a multi-column journal (Illustration 7.9) involves the following steps.

1. **At the end of the month**, the company posts all column totals, except for the Other Accounts total, to the account title(s) specified in the column heading (such as Cash or Accounts Receivable). The company then enters account numbers below the column totals to show that they have been posted. For example, Karns has posted Cash to account No. 101, Accounts Receivable to account No. 112, Inventory to account No. 120, Sales Revenue to account No. 401, Sales Discounts to account No. 414, and Cost of Goods Sold to account No. 505.
2. The company **separately posts the individual amounts comprising the Other Accounts total** to the general ledger accounts specified in the Account Credited column. See, for

example, the credit posting to Owner's Capital. The total amount of this column has not been posted. The symbol (X) is inserted below the total to this column to indicate that the amount has not been posted.

3. The individual amounts in a column, posted in total to a control account (Accounts Receivable, in this case), are posted **daily to the subsidiary ledger** account specified in the Account Credited column. See, for example, the credit posting of \$10,600 to Abbot Sisters.

The symbol **CR**, used in both the subsidiary and general ledgers, identifies postings from the cash receipts journal.

Proving the Ledgers

After posting of the cash receipts journal is completed, Karns proves the ledgers. As shown in **Illustration 7.11**, the general ledger totals agree. Also, the sum of the subsidiary ledger balances equals the control account balance.

Schedule of Accounts Receivable (from accounts receivable subsidiary ledger)		General Ledger	
		Debits	
Abbot Sisters	\$15,400	Cash	\$ 53,769
Babson Co.	14,570	Accounts Receivable	51,180
Deli Co.	21,210	Sales Discounts	781
	<u>\$51,180</u>	Cost of Goods Sold	65,120
			<u>\$170,850</u>
		Credits	
		Notes Payable	\$ 6,000
		Owner's Capital	5,000
		Sales Revenue	94,730
		Inventory	65,120
			<u>\$170,850</u>

ILLUSTRATION 7.11

Proving the ledgers after posting the sales and the cash receipts journals

Purchases Journal

In the **purchases journal**, companies record all purchases of merchandise on account. Each entry in this journal results in a debit to Inventory and a credit to Accounts Payable (see **Helpful Hint**). For example, consider the following credit purchase transactions for Karns Wholesale Supply in **Illustration 7.12**.

HELPFUL HINT

A single-column purchases journal needs only to be footed to prove the equality of debits and credits.

Date	Supplier	Amount
5/6	Jasper Manufacturing Inc.	\$11,000
5/10	Eaton and Howe Inc.	7,200
5/14	Fabor and Son	6,900
5/19	Jasper Manufacturing Inc.	17,500
5/26	Fabor and Son	8,700
5/29	Eaton and Howe Inc.	12,600

ILLUSTRATION 7.12

Credit purchase transactions

Illustration 7.13 shows the purchases journal for Karns Wholesale Supply. When using a one-column purchases journal (as in **Illustration 7.13**), a company cannot journalize other types of purchases on account or cash purchases in it. For example, if the company used the purchases journal in **Illustration 7.13**, Karns would have to record credit purchases of equipment or supplies in the general journal. Likewise, all cash purchases would be entered in the cash payments journal.

ILLUSTRATION 7.13 Journalizing and posting the purchases journal

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PURCHASES JOURNAL PI

Date	Account Credited	Terms	Ref.	Inventory Dr. Accounts Payable Cr.
2020				
May 6	Jasper Manufacturing Inc.	2/10, n/30	✓	11,000
10	Eaton and Howe Inc.	3/10, n/30	✓	7,200
14	Fabor and Son	1/10, n/30	✓	6,900
19	Jasper Manufacturing Inc.	2/10, n/30	✓	17,500
26	Fabor and Son	1/10, n/30	✓	8,700
29	Eaton and Howe Inc.	3/10, n/30	✓	12,600
				<u>63,900</u>
				(120)/(201)

The company posts individual amounts to the subsidiary ledger daily.

At the end of the accounting period, the company posts totals to the general ledger.

ACCOUNTS PAYABLE SUBSIDIARY LEDGER

Eaton and Howe Inc.

Date	Ref.	Debit	Credit	Balance
2020				
May 10	PI		7,200	7,200
29	PI		12,600	19,800

Fabor and Son

Date	Ref.	Debit	Credit	Balance
2020				
May 14	PI		6,900	6,900
26	PI		8,700	15,600

Jasper Manufacturing Inc.

Date	Ref.	Debit	Credit	Balance
2020				
May 6	PI		11,000	11,000
19	PI		17,500	28,500

GENERAL LEDGER

Inventory No. 120

Date	Ref.	Debit	Credit	Balance
2020				
May 31	SI		62,190	(62,190)
31	CR I		2,930	(65,120)
31	PI	63,900		(1,220)

Accounts Payable No. 201

Date	Ref.	Debit	Credit	Balance
2020				
May 31	PI		63,900	63,900

The subsidiary ledger is separate from the general ledger.

Accounts Payable is a control account.

start General Ledger General Jnl Sales Jnl Cash Receipts Jnl Purchases Jnl

Journalizing Credit Purchases of Merchandise

The journalizing procedure is similar to that for a sales journal. Companies make entries in the purchases journal from purchase invoices. In contrast to the sales journal, the purchases journal may not have an invoice number column because invoices received from different suppliers will not be in numerical sequence. To ensure that they record all purchase invoices, some companies consecutively number each invoice upon receipt and then use an internal document number column in the purchases journal. The entries for Karns Wholesale Supply are based on the assumed credit purchases listed in Illustration 7.12.

HELPFUL HINT

Postings to subsidiary ledger accounts are done daily because it is often necessary to know a current balance for the subsidiary accounts.

Posting the Purchases Journal

The procedures for posting the purchases journal are similar to those for the sales journal. In this case, Karns makes **daily** postings to the **accounts payable ledger** (see **Helpful Hint**).

It makes **monthly** postings to Inventory and Accounts Payable in the general ledger. In both ledgers, Karns uses **P1** in the reference column to show that the postings are from page 1 of the purchases journal.

Proof of the equality of the postings from the purchases journal to both ledgers is shown in **Illustration 7.14**.

Postings to General Ledger		Credit Postings to Accounts Payable Ledger	
Inventory (debit)	<u>\$63,900</u>	Eaton and Howe Inc.	\$19,800
		Fabor and Son	15,600
		Jasper Manufacturing Inc.	28,500
Accounts Payable (credit)	<u>\$63,900</u>		<u>\$63,900</u>

ILLUSTRATION 7.14

Proving the equality of the purchases journal

Expanding the Purchases Journal

Some companies expand the purchases journal to include all types of purchases on account, not just merchandise. Instead of one column for inventory and accounts payable, they use a multiple-column format. This format usually includes a credit column for Accounts Payable and debit columns for purchases of Inventory, Supplies, and Other Accounts. **Illustration 7.15** shows a multi-column purchases journal for Hanover Co. The posting procedures are similar to those shown earlier for posting the cash receipts journal. *For homework problems, assume the use of a single-column purchases journal unless instructed otherwise.*

ILLUSTRATION 7.15 Multi-column purchases journal

PURCHASES JOURNAL								
Date	Account Credited	Ref.	Accounts Payable Cr.	Inventory Dr.	Supplies Dr.	Other Accounts Dr.		
						Account	Ref.	Amount
2020								
June 1	Signe Audio	✓	2,000		2,000			
3	Wight Co.	✓	1,500	1,500				
5	Orange Tree Co.	✓	2,600			Equipment	157	2,600
30	Sue's Business Forms	✓	800		800			
			56,600	43,000	8,700			4,900

Cash Payments Journal

In a **cash payments (cash disbursements) journal**, companies record all disbursements of cash. Entries are made from prenumbered checks. Because companies make cash payments for various purposes, the cash payments journal has multiple columns. **Illustration 7.16** shows a four-column journal.

ILLUSTRATION 7.16 Journalizing and posting the cash payments journal

General Ledger Software for Accounting

File Edit View Go Bookmarks Tools Help

Problem Entries Post Closing Reports Tools Help

CASH PAYMENTS JOURNAL

Date	Ck. No.	Account Debited	Ref.	Other Accounts Dr.	Accounts Payable Dr.	Inventory Cr.	Cash Cr.
2020							
May 1	101	Prepaid Insurance	130	1,200			1,200
3	102	Inventory	120	100			100
8	103	Inventory	120	4,400			4,400
10	104	Jasper Manuf. Inc.	✓		11,000	220	10,780
19	105	Eaton & Howe Inc.	✓		7,200	216	6,984
23	106	Fabor and Son	✓		6,900	69	6,831
28	107	Jasper Manuf. Inc.	✓		17,500	350	17,150
30	108	Owner's Drawings	306	500			500
				6,200	42,600	855	47,945
				(x)	(201)	(120)	(101)

The company posts individual amounts to the subsidiary ledger daily.

At the end of the accounting period, the company posts totals to the general ledger.

ACCOUNTS PAYABLE SUBSIDIARY LEDGER

Eaton and Howe Inc.

Date	Ref.	Debit	Credit	Balance
2020				
May 10	PI		7,200	7,200
19	CPI	7,200		-----
29	PI		12,600	12,600

Fabor and Son

Date	Ref.	Debit	Credit	Balance
2020				
May 14	PI		6,900	6,900
23	CPI	6,900		-----
26	PI		8,700	8,700

Jasper Manufacturing Inc.

Date	Ref.	Debit	Credit	Balance
2020				
May 6	PI		11,000	11,000
10	CPI	11,000		-----
19	PI		17,500	17,500
28	CPI	17,500		-----

The subsidiary ledger is separate from the general ledger.

Accounts Payable is a control account.

GENERAL LEDGER

Cash No. 101

Date	Ref.	Debit	Credit	Balance
2020				
May 31	CR1	53,769		53,769
31	CPI		47,945	5,824

Inventory No. 120

Date	Ref.	Debit	Credit	Balance
2020				
May 3	CPI	100		100
8	CPI	4,400		4,500
31	SI		62,190	57,690
31	CR1		2,930	60,620
31	PI	63,900		3,280
31	CPI		855	2,425

Prepaid Insurance No. 130

Date	Ref.	Debit	Credit	Balance
2020				
May 1	CPI	1,200		1,200

Accounts Payable No. 201

Date	Ref.	Debit	Credit	Balance
2020				
May 31	PI		63,900	63,900
31	CPI	42,600		21,300

Owner's Drawings No. 306

Date	Ref.	Debit	Credit	Balance
2020				
May 30	CPI	500		500

start General Ledger General Jnl Sales Jnl Cash Receipts Jnl Purchases Jnl

Journalizing Cash Payments Transactions

The procedures for journalizing transactions in this journal are similar to those for the cash receipts journal. Karns records each transaction on one line, and for each line there must be equal debit and credit amounts. The entries in the cash payments journal in Illustration 7.16 are based on the following transactions for Karns Wholesale Supply.

- May 1 Issued check No. 101 for \$1,200 for the annual premium on a fire insurance policy.
 3 Issued check No. 102 for \$100 in payment of freight when terms were FOB shipping point.
 8 Issued check No. 103 for \$4,400 for the purchase of merchandise.
 10 Sent check No. 104 for \$10,780 to Jasper Manufacturing Inc. in payment of May 6 invoice for \$11,000 less a 2% discount.
 19 Mailed check No. 105 for \$6,984 to Eaton and Howe Inc. in payment of May 10 invoice for \$7,200 less a 3% discount.
 23 Sent check No. 106 for \$6,831 to Fabor and Son in payment of May 14 invoice for \$6,900 less a 1% discount.
 28 Sent check No. 107 for \$17,150 to Jasper Manufacturing Inc. in payment of May 19 invoice for \$17,500 less a 2% discount.
 30 Issued check No. 108 for \$500 to D. A. Karns as a cash withdrawal for personal use.

Note that whenever Karns enters an amount in the Other Accounts column, it must identify a specific general ledger account in the Account Debited column. The entries for checks No. 101, 102, 103, and 108 illustrate this situation. Similarly, Karns must identify a subsidiary account in the Account Debited column whenever it enters an amount in the Accounts Payable column. See, for example, the entry for check No. 104.

After Karns journalizes the cash payments journal, it totals the columns. The totals are then balanced to prove the equality of debits and credits.

Posting the Cash Payments Journal

The procedures for posting the cash payments journal are similar to those for the cash receipts journal. Karns posts the amounts recorded in the Accounts Payable column individually to the subsidiary ledger and in total to the control account. It posts Inventory and Cash only in total at the end of the month. Transactions in the Other Accounts column are posted individually to the appropriate account(s) affected. The company does not post totals for the Other Accounts column.

Illustration 7.16 shows the posting of the cash payments journal. Note that Karns uses the symbol **CP** as the posting reference. After postings are completed, the company proves the equality of the debit and credit balances in the general ledger. In addition, the control account balances should agree with the subsidiary ledger total balance. **Illustration 7.17** shows the agreement of these balances.

Schedule of Accounts Payable (from accounts payable subsidiary ledger)		General Ledger	
		Debits	
Eaton and Howe Inc.	\$12,600	Cash	\$ 5,824
Fabor and Son	8,700	Accounts Receivable	51,180
	<u>\$21,300</u>	Inventory	2,425
		Prepaid Insurance	1,200
		Owner's Drawings	500
		Sales Discounts	781
		Cost of Goods Sold	65,120
			<u>\$127,030</u>
		Credits	
		Notes Payable	\$ 6,000
		Accounts Payable	21,300
		Owner's Capital	5,000
		Sales Revenue	94,730
			<u>\$127,030</u>

ILLUSTRATION 7.17

Proving the ledgers after postings from the sales, cash receipts, purchases, and cash payments journals

Effects of Special Journals on the General Journal

Special journals for sales, purchases, and cash substantially reduce the number of entries that companies make in the general journal. **Only transactions that cannot be entered in a special journal are recorded in the general journal.** For example, a company may use the general journal to record such transactions as granting of credit to a customer for a sales return or allowance, granting of credit from a supplier for purchases returned, acceptance of a note receivable from a customer, and purchase of equipment by issuing a note payable. Also, **correcting, adjusting, and closing entries are made in the general journal.**

The general journal has columns for date, account title and explanation, reference, and debit and credit amounts. When control and subsidiary accounts are not involved, the procedures for journalizing and posting of transactions are the same as those described in earlier chapters. When control and subsidiary accounts are involved, companies make two changes from the earlier procedures:

1. In **journalizing**, they identify both the control and the subsidiary accounts.
2. In **posting**, there must be a **dual posting**: once to the control account and once to the subsidiary account.

To illustrate, assume that on May 31, Karns Wholesale Supply returns \$500 of merchandise for credit to Fabor and Son. **Illustration 7.18** shows the entry in the general journal and the posting of the entry.

ILLUSTRATION 7.18 Journalizing and posting the general journal

The screenshot displays the 'General Ledger Software for Accounting' interface. The main window shows the 'GENERAL JOURNAL' (GI) with the following entry for May 31, 2020:

Date	Account Title and Explanation	Ref.	Debit	Credit
2020				
May 31	Accounts Payable—Fabor and Son	201/✓	500	
	Inventory	120		500
	(Received credit for returned goods)			

Red arrows indicate the posting process. One arrow points from the debit amount of 500 in the general journal to the 'Accounts Payable' subsidiary ledger. The other arrow points from the credit amount of 500 in the general journal to the 'Inventory' subsidiary ledger.

The 'ACCOUNTS PAYABLE SUBSIDIARY LEDGER' for Fabor and Son shows the following entries:

Date	Ref.	Debit	Credit	Balance
2020				
May 14	PI		6,900	6,900
23	CPI	6,900		-----
26	PI		8,700	8,700
31	GI	500		8,200

The 'GENERAL LEDGER' shows the following entries for the 'Inventory' (No. 120) and 'Accounts Payable' (No. 201) accounts:

Inventory No. 120				
Date	Ref.	Debit	Credit	Balance
2020				
May 31	GI		500	500

Accounts Payable No. 201				
Date	Ref.	Debit	Credit	Balance
2020				
May 31	PI		63,900	63,900
31	CPI	42,600		21,300
31	GI	500		20,800

The bottom status bar shows the following tabs: start, General Ledger, General Jnl, Sales Jnl, Cash Receipts Jnl, Purchases Jnl.

Note that the general journal indicates two accounts (Accounts Payable, and Fabor and Son) for the debit, and two postings ("201/✓") in the reference column. One debit is posted to the control account and another debit is posted to the creditor's account in the subsidiary

ledger. If Karns receives cash instead of credit on this return, then it would record the transaction in the cash receipts journal.

Cyber Security: A Final Comment

Have you ever been hacked? With the increasing use of cell phones, tablets, and other social media outlets, a real risk exists that your confidential information may be stolen and used illegally. Companies, individuals, and even nations have all been victims of **cybercrime**—a crime that involves the Internet, a computer system, or computer technology.

For companies, cybercrime is clearly a major threat as the hacking of employees' or customers' records related to cybercrime can cost millions of dollars. Unfortunately, the numbers of security breaches are increasing. A security breach at **Target**, for example, cost the company a minimum of \$20 million, the CEO lost his job, and sales plummeted.

Here are three reasons for the rise in the successful hacks of corporate computer records.

1. Companies and their employees continue to increase their activity on the Internet, primarily due to the use of mobile devices and cloud computing.
2. Companies today collect and store unprecedented amounts of personal data on customers and employees.
3. Companies often take measures to protect themselves from cyber security attacks but then fail to check if employees are carrying out the proper security guidelines.

Note that cyber security risks extend far beyond company operations and compliance. Many hackers target highly sensitive intellectual information or other strategic assets.

Illustration 7.19 highlights the type of hackers and their motives, targets and impacts.

ILLUSTRATION 7.19 Profiles of hackers

Malicious Actors	Motives	Targets	Impacts
Nation-state	Economic, political, and/or military advantage	- Trade secrets - Sensitive business information - Emerging technologies - Critical infrastructure	- Loss of competitive advantage - Disruption to critical infrastructure
Organized crime	- Immediate financial gain - Collect information for future financial gains	- Financial/payment systems - Personally identifiable information - Payment card information - Protected health information	- Costly regulatory inquiries and penalties - Consumer and shareholder lawsuits - Loss of consumer confidence
Hacktivists	- Influence political and/or social change - Pressure businesses to change their practices	- Corporate secrets - Sensitive business information - Information related to key executives, employees, customers, and business partners	- Disruption of business activities - Harm to brand and reputation - Loss of consumer confidence
Insiders	- Personal advantage, monetary gain - Professional revenge - Patriotism	- Sales, deals, market strategies - Corporate secrets, intellectual property - Business operations - Personnel information	- Trade secret disclosure - Operational disruption - Harm to brand and reputation - National security impact

Source: PriceWaterhouseCoopers, "Answering Your Cybersecurity Questions" (January 2014).

Companies now recognize that cyber security systems that protect confidential data must be implemented. It follows that companies (and nations and individuals) must continually verify that their cyber security defenses are sound and uncompromised.

DO IT! 3
Special Journals

Swisher Company had the following transactions during March.

- Collected cash on account from Oakland Company.
- Purchased equipment by signing a note payable.
- Sold merchandise on account.
- Purchased merchandise on account.
- Paid \$2,400 for a 2-year insurance policy.

Identify the journal in which each of the transactions above is recorded. The company uses the special journals described in the chapter plus a general journal.

Solution

- | | |
|--|-----------------------|
| 1. Collected cash on account from Oakland Company. | Cash receipts journal |
| 2. Purchased equipment by signing a note payable. | General journal |
| 3. Sold merchandise on account. | Sales journal |
| 4. Purchased merchandise on account. | Purchases journal |
| 5. Paid \$2,400 for a 2-year insurance policy. | Cash payments journal |

Related exercise material: BE7.6, BE7.7, BE7.8, BE7.9, BE7.10, DO IT! 7.3, E7.6, E7.7, E7.8, and E7.10.

ACTION PLAN

- Determine if the transaction involves the receipt of cash (cash receipts journal) or the payment of cash (cash payments journal).
- Determine if the transaction is a sale of merchandise on account (sales journal) or a purchase of merchandise on account (purchases journal).
- All other transactions are recorded in the general journal.

Review and Practice

Learning Objectives Review

1
Explain the basic concepts of an accounting information system.

The basic principles in developing an accounting information system are cost-effectiveness, useful output, and flexibility. Most companies use a computerized accounting system. Smaller companies use entry-level software such as QuickBooks® or Sage 50. Larger companies use custom-made software packages, which often integrate all aspects of the organization.

2
Describe the nature and purpose of a subsidiary ledger.

A subsidiary ledger is a group of accounts with a common characteristic. It facilitates the recording process by freeing the general ledger from details of individual balances.

3
Record transactions in special journals.

Companies use special journals to group similar types of transactions. In a special journal, generally only one line is used to record a complete transaction.

Glossary Review

Accounting information system A system that collects and processes transaction data and communicates financial information to decision-makers. (p. 7-2).

Accounts payable (creditors’) subsidiary ledger A subsidiary ledger that collects transaction data of individual creditors. (p. 7-6).

Accounts receivable (customers’) subsidiary ledger A subsidiary ledger that collects transaction data of individual customers. (p. 7-6).

Cash payments (cash disbursements) journal A special journal that records all disbursements of cash. (p. 7-17).

Cash receipts journal A special journal that records all cash received. (p. 7-11).

Control account An account in the general ledger that summarizes subsidiary ledger data. (p. 7-6).

Cybercrime A crime that involves the Internet, a computer system, or computer technology. (p. 7-21).

Manual accounting system A system in which someone performs each of the steps in the accounting cycle by hand. (p. 7-5).

Purchases journal A special journal that records all purchases of merchandise on account. (p. 7-15).

Sales journal A special journal that records all sales of merchandise on account. (p. 7-9).

Special journals Journals that record similar types of transactions, such as all credit sales. (p. 7-8).

Subsidiary ledger A group of accounts with a common characteristic. (p. 7-5).

Practice Multiple-Choice Questions

1. (LO 1) The basic principles of an accounting information system include all of the following **except**:
 - a. cost-effectiveness.
 - b. flexibility.
 - c. useful output.
 - d. periodicity.
2. (LO 1) Which of the following is **not** an advantage of computerized accounting systems?
 - a. Data is entered only once in computerized accounting systems.
 - b. Computerized accounting systems provide up-to-date information.
 - c. Computerized accounting systems eliminate entering of transaction information.
 - d. Computerized accounting systems eliminate many errors resulting from human intervention.
3. (LO 2) Which of the following is **incorrect** concerning subsidiary ledgers?
 - a. The purchases ledger is a common subsidiary ledger for creditor accounts.
 - b. The accounts receivable ledger is a subsidiary ledger.
 - c. A subsidiary ledger is a group of accounts with a common characteristic.
 - d. An advantage of the subsidiary ledger is that it permits a division of labor in posting.
4. (LO 2) Two common subsidiary ledgers are:
 - a. accounts receivable and cash receipts.
 - b. accounts payable and cash payments.
 - c. accounts receivable and accounts payable.
 - d. sales and cost of goods sold.
5. (LO 2) At the beginning of the month, the accounts receivable subsidiary ledger showed balances for Apple Company \$5,000 and Berry Company \$7,000. During the month, credit sales were made to Apple \$6,000, Berry \$4,500, and Cantaloupe \$8,500. Cash was collected on account from Berry \$11,500 and Cantaloupe \$3,000. At the end of the month, the control account Accounts Receivable in the general ledger should have a balance of:
 - a. \$11,000.
 - b. \$12,000.
 - c. \$16,500.
 - d. \$31,000.
6. (LO 3) A sales journal will be used for:

	Credit Sales	Cash Sales	Sales Discounts
a.	no	yes	yes
b.	yes	no	yes
c.	yes	no	no
d.	yes	yes	no
7. (LO 3) A purchase of equipment on account is recorded in the:
 - a. cash receipts journal.
 - b. purchases journal.
 - c. cash payments journal.
 - d. general journal.
8. (LO 3) A purchase of equipment using cash is recorded in the:
 - a. cash receipts journal.
 - b. purchases journal.
 - c. cash payments journal.
 - d. general journal.
9. (LO 3) Which of the following statements is **correct**?
 - a. The sales discount column is included in the cash receipts journal.
 - b. The purchases journal records all purchases of merchandise whether for cash or on account.
 - c. The cash receipts journal records sales on account.
 - d. Merchandise returned by the buyer is recorded by the seller in the purchases journal.
10. (LO 3) Dotel Company's cash receipts journal includes an Accounts Receivable column and an Other Accounts column. At the end of the month, these columns are posted to the general ledger as:

Accounts Receivable	Other Accounts
a. a column total	a column total
b. individual amounts	a column total
c. individual amounts	individual amounts
d. a column total	individual amounts
11. (LO 3) Which of the following is **incorrect** concerning the posting of the cash receipts journal?
 - a. The total of the Other Accounts column is not posted.
 - b. All column totals except the total for the Other Accounts column are posted once at the end of the month to the account title(s) specified in the column heading.
 - c. The totals of all columns are posted daily to the accounts specified in the column heading.
 - d. The individual amounts in a column posted in total to a control account are posted daily to the subsidiary ledger account specified in the Account Credited column.
12. (LO 3) Postings from the purchases journal to the subsidiary ledger are generally made:
 - a. yearly.
 - b. monthly.
 - c. weekly.
 - d. daily.
13. (LO 3) Which statement is **incorrect** regarding the general journal?
 - a. Only transactions that cannot be entered in a special journal are recorded in the general journal.

- b. Dual postings are always required in the general journal.
- c. The general journal may be used to record acceptance of a note receivable in payment of an account receivable.
- d. Correcting, adjusting, and closing entries are made in the general journal.

14. (LO 3) When companies use special journals:

- a. they record all purchase transactions in the purchases journal.
- b. they record all cash received, except from cash sales, in the cash receipts journal.

- c. they record all cash disbursements in the cash payments journal.
- d. a general journal is not necessary.

15. (LO 3) If a customer returns goods for credit, the selling company normally makes an entry in the:

- a. cash payments journal.
- b. sales journal.
- c. general journal.
- d. cash receipts journal.

Solutions

1. d. Periodicity is not one of the basic principles of accounting information systems. The other choices are true statements.

2. c. Computerized accounting systems do not eliminate the entering of transaction information. The other choices are advantages of computerized accounting systems.

3. a. The accounts payable ledger, not the purchases ledger, is a common subsidiary ledger for creditor accounts. The other choices are true statements.

4. c. Accounts receivable and accounts payable are two common subsidiary ledgers. The other choices are incorrect because (a) cash receipts, (b) cash payments, and (d) sales revenue and cost of goods sold are not subsidiary ledgers.

5. c. The accounts receivable subsidiary ledger balances are the following: Apple Company \$11,000 (beginning balance \$5,000 + credit sales \$6,000), Berry Company \$0 (beginning balance \$7,000 + credit sales \$4,500 – cash collected on account \$11,500), and Cantaloupe \$5,500 (credit sales \$8,500 – cash collected on account \$3,000). The Accounts Receivable control account in the general ledger therefore has a balance of \$16,500 (\$11,000 + \$0 + \$5,500), not (a) \$11,000, (b) \$12,000, or (d) \$31,000.

6. c. The sales journal is used for credit sales. The cash receipts journal is used for cash sales and sales discounts. Therefore, choices (a), (b), and (d) are incorrect.

7. d. Unless the company uses a multi-column purchases journal, the general journal is used to record the purchase of equipment **on account**. The other choices are incorrect because (a) the cash receipts journal would be used only if cash is received, (b) a one-column purchases journal cannot be used to record a purchase of equipment on account, and (c) the cash payments journal would be used only if this is a cash purchase.

8. c. The cash payments journal includes all cash paid, including all **cash** purchases. The other choices are incorrect because (a) the cash receipts journal includes all cash received, including cash sales, not cash purchases; (b) the purchases journal includes only purchases of

inventory on account, not cash purchases; and (d) the general journal is only used if a transaction cannot be entered in a special journal. In this case, this transaction can be entered into the cash payments journal.

9. a. The sales discount column is included in the cash receipts journal. The other choices are incorrect because (b) the purchases journal only records purchases of inventory on account, not for cash; (c) the sales journal, not the cash receipts journal, records sales on account; and (d) when merchandise is returned by a buyer, the seller records this sales return in the general journal.

10. d. The Accounts Receivable column would be posted to the general ledger as a column total, and the Other Accounts column would be posted as individual amounts. Therefore, the other choices are incorrect.

11. c. The totals of all columns are not posted daily to the accounts specified in the column heading. Instead, all column totals except the total for the Other Accounts column (which is never posted) are posted once at the end of the month to the account title(s) specified in the column heading. The other choices are true statements.

12. d. Postings from the purchases journal to the subsidiary ledger are usually made daily, not (a) yearly, (b) monthly, or (c) weekly.

13. b. Dual postings are not always required in the general journal. Only when control and subsidiary accounts are involved are companies required to dual post: once to the control account and once to the subsidiary account. The other choices are true statements.

14. c. When special journals are used, companies record all cash disbursements in the cash payments journal. The other choices are incorrect because when special journals are used, (a) companies record only purchases of inventory on account in the purchases journal; (b) companies record all cash receipts, including cash sales, in the cash receipts journal; and (d) a general journal is still needed.

15. c. When a customer returns goods for **credit** (i.e., no cash is involved), the selling company records the transaction in the general journal, not the (a) cash payments journal, (b) sales journal, or (d) cash receipts journal.

Practice Brief Exercises

Identify subsidiary ledger balances.

1. (LO 2) The following information is related to Garcia Company for its first month of operations. Identify the balances that appear in the accounts receivable subsidiary ledger and the accounts receivable balance that appears in the general ledger at the end of January.

Credit Sales			Cash Collections		
Jan. 7	Rodan Co.	\$11,000	Jan. 17	Rodan Co.	\$6,000
15	Lawry Co.	7,000	24	Lawry Co.	5,000
23	Anderson Co.	10,000	29	Anderson Co.	8,000

Solution

1.

Accounts Receivable Subsidiary Ledger

Rodan Co.				
Date	Ref.	Debit	Credit	Balance
Jan. 7		11,000		11,000
17			6,000	5,000

Lawry Co.				
Date	Ref.	Debit	Credit	Balance
Jan. 15		7,000		7,000
24			5,000	2,000

Anderson Co.				
Date	Ref.	Debit	Credit	Balance
Jan. 23		10,000		10,000
29			8,000	2,000

General Ledger

Accounts Receivable				
Date	Ref.	Debit	Credit	Balance
Jan. 31		28,000		28,000
31			19,000	9,000

2. (LO 2) Presented below is information related to Garcia Company for its first month of operations. Identify the balances that appear in the accounts payable subsidiary ledger and the accounts payable balance that appears in the general ledger at the end of January.

Identify subsidiary ledger balances.

Credit Purchases			Cash Payments		
Jan. 4	Jones Co.	\$22,000	Jan. 13	Jones Co.	\$13,000
12	Holland Co.	14,000	20	Holland Co.	11,000
19	Geovany Co.	20,000	25	Geovany Co.	17,000

Solution

2.

Accounts Payable Subsidiary Ledger

Jones Co.				
Date	Ref.	Debit	Credit	Balance
Jan. 4			22,000	22,000
13		13,000		9,000

Holland Co.				
Date	Ref.	Debit	Credit	Balance
Jan. 12			14,000	14,000
20		11,000		3,000

Geovany Co.				
Date	Ref.	Debit	Credit	Balance
Jan. 19			20,000	20,000
25		17,000		3,000

General Ledger

Accounts Payable				
Date	Ref.	Debit	Credit	Balance
Jan. 31			56,000	56,000
31		41,000		15,000

3. (LO 3) Soto Company has the following selected transactions during March.

- Mar. 2 Purchased equipment costing \$9,400 from Jennings Company on account.
- 5 Received credit of \$630 from May Company for merchandise damaged in shipment to Soto.
- 7 Issued credit of \$500 to Morneau Company for merchandise the customer returned. The returned merchandise had a cost of \$300.

Soto Company uses a one-column purchases journal, a sales journal, the columnar cash journals used in the text, and a general journal. Journalize the transactions in the general journal.

Journalize transactions in general journal.

Solution

3.

Mar. 2	Equipment	9,400	
	Accounts Payable		9,400
5	Accounts Payable	630	
	Inventory		630
7	Sales Returns and Allowances	500	
	Accounts Receivable		500
	Inventory	300	
	Cost of Goods Sold		300

Practice Exercises

Post various journals to control and subsidiary accounts.

1. (LO 2, 3) On June 1, the balance of the Accounts Receivable control account in the general ledger of Rath Company was \$13,620. The customers' subsidiary ledger contained account balances as follows: Wilson \$2,000, Sanchez \$3,140, Roberts \$2,560, and Marks \$5,920. At the end of June, the various journals contained the following information.

Sales journal: Sales to Roberts \$900, to Wilson \$1,400, to Hardy \$1,500, and to Marks \$1,200.

Cash receipts journal: Cash received from Roberts \$1,610, from Marks \$2,600, from Hardy \$580, from Sanchez \$2,100, and from Wilson \$1,540.

General journal: An allowance is granted to Marks \$325.

Instructions

- Set up control and subsidiary accounts and enter the beginning balances. Do not construct the journals.
- Post the various journals. Post the items as individual items or as totals, whichever would be the appropriate procedure. (No sales discounts given.)
- Prepare a schedule of accounts receivable and prove the agreement of the control account with the subsidiary ledger at June 30, 2020.

Solution

1. a. and b.

GENERAL LEDGER

Accounts Receivable					
Date	Explanation	Ref.	Debit	Credit	Balance
June 1	Balance	✓			13,620
		S	5,000		18,620
		CR		8,430	10,190
		G		325	9,865

ACCOUNTS RECEIVABLE SUBSIDIARY LEDGER

Hardy					
Date	Explanation	Ref.	Debit	Credit	Balance
June 1					0
		S	1,500		1,500
		CR		580	920

Marks					
Date	Explanation	Ref.	Debit	Credit	Balance
June 1	Balance	✓			5,920
		S	1,200		7,120
		CR		2,600	4,520
		G		325	4,195

Roberts					
Date	Explanation	Ref.	Debit	Credit	Balance
June 1	Balance	✓			2,560
		S	900		3,460
		CR		1,610	1,850

Sanchez					
Date	Explanation	Ref.	Debit	Credit	Balance
June 1	Balance	✓			3,140
		CR		2,100	1,040

Wilson					
Date	Explanation	Ref.	Debit	Credit	Balance
June 1	Balance	✓			2,000
		S	1,400		3,400
		CR		1,540	1,860

c.

Rath Company
Schedule of Accounts Receivable
As of June 30, 2020

Hardy	\$ 920
Marks	4,195
Roberts	1,850
Sanchez	1,040
Wilson	1,860
Total	<u>\$9,865</u>
Accounts Receivable	<u>\$9,865</u>

2. (LO 3) Below are some typical transactions incurred by Brimmer Company.

Indicate use of special journals.

1. Received credit for merchandise purchased on credit.
2. Payment of employee wages.
3. Sales discount taken on goods sold.
4. Income summary closed to owner's capital.
5. Purchase of office supplies for cash.
6. Depreciation on building.
7. Purchase of merchandise on account.
8. Return of merchandise sold for credit.
9. Payment of creditors on account.
10. Collection on account from customers.
11. Sale of merchandise on account.
12. Sale of land for cash.
13. Sale of merchandise for cash.

Instructions

For each transaction, indicate whether it would normally be recorded in a cash receipts journal, cash payments journal, sales journal, single-column purchases journal, or general journal.

Solution

- | | |
|---|--|
| <ol style="list-style-type: none"> 2. 1. General journal 2. Cash payments journal 3. Cash receipts journal 4. General journal 5. Cash payments journal 6. General journal 7. Purchases journal | <ol style="list-style-type: none"> 8. General journal 9. Cash payments journal 10. Cash receipts journal 11. Sales journal 12. Cash receipts journal 13. Cash receipts journal |
|---|--|

Practice Problem

(LO 3) Cassandra Wilson Company uses a six-column cash receipts journal with the following columns.

Cash (Dr.)	Other Accounts (Cr.)
Sales Discounts (Dr.)	Cost of Goods Sold (Dr.) and
Accounts Receivable (Cr.)	Inventory (Cr.)
Sales Revenue (Cr.)	

Journalize transactions in cash receipts journal and explain posting procedure.

Cash receipts transactions for the month of July 2020 are as follows.

- July 3 Cash sales total \$5,800 (cost, \$3,480).
 5 Received a check for \$6,370 from Jeltz Company in payment of an invoice dated June 26 for \$6,500, terms 2/10, n/30.

- 9 Cassandra Wilson, the proprietor, made an additional investment of \$5,000 in cash in the business.
- 10 Cash sales total \$12,519 (cost, \$7,511).
- 12 Received a check for \$7,275 from R. Eliot & Co. in payment of a \$7,500 invoice dated July 3, terms 3/10, n/30.
- 15 Received an advance of \$700 cash for future services.
- 20 Cash sales total \$15,472 (cost, \$9,283).
- 22 Received a check for \$5,880 from Beck Company in payment of \$6,000 invoice dated July 13, terms 2/10, n/30.
- 29 Cash sales total \$17,660 (cost, \$10,596).
- 31 Received cash of \$200 for interest earned for July.

Instructions

- a. Journalize the transactions in the cash receipts journal.
- b. Contrast the posting of the Accounts Receivable and Other Accounts columns.

Solution

a.

CASSANDRA WILSON COMPANY								
Cash Receipts Journal								CR1
Date	Account Credited	Ref.	Cash Dr.	Sales Discounts Dr.	Accounts Receivable Cr.	Sales Revenue Cr.	Other Accounts Cr.	Cost of Goods Sold Dr. Inventory Cr.
2020								
7/3			5,800			5,800		3,480
5	Jeltz Company		6,370	130	6,500			
9	Owner's Capital		5,000				5,000	
10			12,519			12,519		7,511
12	R. Eliot & Co.		7,275	225	7,500			
15	Unearned Service Revenue		700				700	
20			15,472			15,472		9,283
22	Beck Company		5,880	120	6,000			
29			17,660			17,660		10,596
31	Interest Revenue		200				200	
			<u>76,876</u>	<u>475</u>	<u>20,000</u>	<u>51,451</u>	<u>5,900</u>	<u>30,870</u>

- b. The Accounts Receivable column total is posted as a credit to Accounts Receivable. The individual amounts are credited to the customers' accounts identified in the Account Credited column, which are maintained in the accounts receivable subsidiary ledger. The amounts in the Other Accounts column are posted individually. They are credited to the account titles identified in the Account Credited column.

WileyPLUS

Brief Exercises, DO IT! Exercises, Exercises, Problems, and many additional resources are available for practice in WileyPLUS.

Questions

- (a) What is an accounting information system? (b) "An accounting information system applies only to a manual system." Do you agree? Explain.
- Certain principles should be followed in the development of an accounting information system. Identify and explain each of the principles.
- What are common features of computerized accounting packages beyond recording transactions and preparing financial statements?
- How does an enterprise resource planning (ERP) system differ from an entry-level computerized accounting system?
- What are the advantages of using subsidiary ledgers?

6. (a) When do companies normally post to (1) the subsidiary accounts and (2) the general ledger control accounts? (b) Describe the relationship between a control account and a subsidiary ledger.
7. Identify and explain the four special journals discussed in the chapter. List an advantage of using each of these journals rather than using only a general journal.
8. Kensington Company uses special journals. It recorded in a sales journal a sale made on account to R. Stiner for \$435. A few days later, R. Stiner returns \$70 worth of merchandise for credit. Where should Kensington Company record the sales return? Why?
9. A \$500 purchase of merchandise on account from Lore Company was properly recorded in the purchases journal. When posted, however, the amount recorded in the subsidiary ledger was \$50. How might this error be discovered?
10. Why would special journals used in different businesses not be identical in format? What type of business would maintain a cash receipts journal but not include a column for accounts receivable?
11. The cash and the accounts receivable columns in the cash receipts journal were mistakenly over-added by \$4,000 at the end of the month. (a) Will the customers' ledger agree with the Accounts Receivable control account? (b) Assuming no other errors, will the trial balance totals be equal?
12. One column total of a special journal is posted at month-end to only two general ledger accounts. One of these two accounts is Accounts Receivable. What is the name of this special journal? What is the other general ledger account to which that same month-end total is posted?
13. In what journal would the following transactions be recorded? (Assume that a two-column sales journal and a single-column purchases journal are used.)
- Recording of depreciation expense for the year.
 - Credit given to a customer for merchandise purchased on credit and returned.
 - Sales of merchandise for cash.
 - Sales of merchandise on account.
 - Collection of cash on account from a customer.
 - Purchase of office supplies on account.
14. In what journal would the following transactions be recorded? (Assume that a two-column sales journal and a single-column purchases journal are used.)
- Cash received from signing a note payable.
 - Investment of cash by the owner of the business.
 - Closing of the expense accounts at the end of the year.
 - Purchase of merchandise on account.
 - Credit received for merchandise purchased and returned to supplier.
 - Payment of cash on account due a supplier.
15. What transactions might be included in a multiple-column purchases journal that would not be included in a single-column purchases journal?
16. Give an example of a transaction in the general journal that causes an entry to be posted twice (i.e., to two accounts), one in the general ledger, the other in the subsidiary ledger. Does this affect the debit/credit equality of the general ledger?
17. Give some examples of appropriate general journal transactions for an organization using special journals.

Brief Exercises

BE7.1 (LO 1) Indicate whether each of the following statements is true or false.

- When designing an accounting system, we need to think about the needs and knowledge of both the top managers and various other users.
- When the environment changes as a result of technological advances, increased competition, or government regulation, an accounting system does not have to be sufficiently flexible to meet the changes in order to save money.
- In developing an accounting system, cost is relevant. The benefits obtained from the information disseminated must outweigh the cost of providing it.

Explain basic concepts of an accounting information system.

BE7.2 (LO 1) Here is a list of words or phrases related to computerized accounting systems.

- Entry-level software.
- Enterprise resource planning systems.
- Network-compatible.
- Audit trail.
- Internal control.

Explain basic concepts of an accounting information system.

Instructions

Match each word or phrase with the best description of it.

- Allows multiple users to access the system at the same time.
- Enables the tracking of all transactions.
- Identifies suspicious transactions or likely mistakes such as wrong account numbers or duplicate transactions.
- Large-scale computer systems that integrate all aspects of the organization including accounting, sales, human resource management, and manufacturing.
- System for companies with revenues of less than \$5 million and up to 20 employees.

Explain basic concepts of an accounting information system.

BE7.3 (LO 1) Benji Borke has prepared the following list of statements about accounting information systems.

1. The accounting information system includes each of the steps of the accounting cycle, the documents that provide evidence of transactions that have occurred, and the accounting records.
2. The benefits obtained from information provided by the accounting information system need not outweigh the cost of providing that information.
3. Designers of accounting systems must consider the needs and knowledge of various users.
4. If an accounting information system is cost-effective and provides useful output, it does not need to be flexible.

Instructions

Identify each statement as true or false. If false, indicate how to correct the statement.

Identify subsidiary ledger balances.

BE7.4 (LO 2) Presented below is information related to Gantner Company for its first month of operations. Identify the balances that appear in the accounts receivable subsidiary ledger and the accounts receivable balance that appears in the general ledger at the end of January.

Credit Sales			Cash Collections		
Jan. 7	Austin Co.	\$10,000	Jan. 17	Austin Co.	\$7,000
15	Diaz Co.	8,000	24	Diaz Co.	4,000
23	Noble Co.	9,000	29	Noble Co.	9,000

Identify subsidiary ledger accounts.

BE7.5 (LO 2) Identify in what ledger (general or subsidiary) each of the following accounts is shown.

- | | |
|---------------------------------|------------------------------|
| a. Rent Expense. | c. Notes Payable. |
| b. Accounts Receivable—Cabrera. | d. Accounts Payable—Pacheco. |

Identify special journals.

BE7.6 (LO 3) Identify the journal in which each of the following transactions is recorded.

- | | |
|------------------------------|--|
| a. Cash sales. | d. Credit sales. |
| b. Owner withdrawal of cash. | e. Purchase of merchandise on account. |
| c. Cash purchase of land. | f. Receipt of cash for services performed. |

Identify entries to cash receipts journal.

BE7.7 (LO 3) Indicate whether each of the following debits and credits is included in the cash receipts journal. (Use “Yes” or “No” to answer this question.)

- | | |
|----------------------------|-----------------------------------|
| a. Debit to Sales Revenue. | c. Credit to Accounts Receivable. |
| b. Credit to Inventory. | d. Debit to Accounts Payable. |

Identify transactions for special journals.

BE7.8 (LO 3) Villar Co. uses special journals and a general journal. Identify the journal in which each of the following transactions is recorded.

- Purchased equipment on account.
- Purchased merchandise on account.
- Paid utility expense in cash.
- Sold merchandise on account.

Identify transactions for special journals.

BE7.9 (LO 3) Identify the special journal(s) in which the following column headings appear.

- | | |
|----------------------------|----------------------|
| a. Sales Discounts Dr. | d. Sales Revenue Cr. |
| b. Accounts Receivable Cr. | e. Inventory Dr. |
| c. Cash Dr. | |

Indicate postings for cash receipts journal.

BE7.10 (LO 3) Rauch Computer Components Inc. uses a multi-column cash receipts journal. Indicate which column(s) is/are posted only in total, only daily, or both in total and daily.

- | | |
|-------------------------|--------------------|
| a. Accounts Receivable. | c. Cash. |
| b. Sales Discounts. | d. Other Accounts. |

DO IT! Exercises

DO IT! 7.1 (LO 1) Indicate whether the following statements are true or false. If false, indicate how to correct the statement.

Explain basic concepts of an accounting information system.

1. A computerized accounting system must be customized to meet a company's needs.
2. Companies with revenues of less than \$5 million and up to 20 employees generally use entry-level programs rather than ERP systems.
3. A manual accounting system provides more timely financial information than a computerized system.

DO IT! 7.2 (LO 2) Presented below is information related to Rizzo Company for its first month of operations. Determine the balances that appear in the accounts payable subsidiary ledger. What Accounts Payable balance appears in the general ledger at the end of January?

Determine subsidiary and general ledger balances.

Credit Purchases			Cash Paid		
Jan. 6	Gorst Company	\$11,000	Jan. 11	Gorst Company	\$ 6,500
Jan. 10	Tian Company	12,000	Jan. 16	Tian Company	12,000
Jan. 23	Maddox Company	10,000	Jan. 29	Maddox Company	7,700

DO IT! 7.3 (LO 3) Hinske Company had the following transactions during April.

Identify special journals.

1. Sold merchandise on account.
2. Purchased merchandise on account.
3. Collected cash from a sale to Renfro Company.
4. Recorded accrued interest on a note payable.
5. Paid \$2,000 for supplies.

Identify the journal in which each of the transactions above is recorded.

Exercises

E7.1 (LO 2, 3) Nex Company uses both special journals and a general journal as described in this chapter. On June 30, after all monthly postings had been completed, the Accounts Receivable control account in the general ledger had a debit balance of \$340,000; the Accounts Payable control account had a credit balance of \$77,000.

Determine control account balances, and explain posting of special journals.

The July transactions recorded in the special journals are summarized below. No entries affecting accounts receivable and accounts payable were recorded in the general journal for July.

Sales journal	Total sales \$161,400
Purchases journal	Total purchases \$66,400
Cash receipts journal	Accounts receivable column total \$131,000
Cash payments journal	Accounts payable column total \$47,500

Instructions

- a. What is the balance of the Accounts Receivable control account after the monthly postings on July 31?
- b. What is the balance of the Accounts Payable control account after the monthly postings on July 31?
- c. To what account(s) is the column total of \$161,400 in the sales journal posted?
- d. To what account(s) is the accounts receivable column total of \$131,000 in the cash receipts journal posted?

E7.2 (LO 2) Writing Presented below is the subsidiary accounts receivable account of Jill Longley.

Explain postings to subsidiary ledger.

Date	Ref.	Debit	Credit	Balance
2020				
Sept. 2	S31	61,000		61,000
9	G4		14,000	47,000
27	CR8		47,000	—

Instructions

Write a memo to Sara Fogelman, chief financial officer, that explains each transaction.

Post various journals to control and subsidiary accounts.

E7.3 (LO 2, 3) On September 1, the balance of the Accounts Receivable control account in the general ledger of Montgomery Company was \$10,960. The customers' subsidiary ledger contained account balances as follows: Hurley \$1,440, Andino \$2,640, Fowler \$2,060, and Sogard \$4,820. At the end of September, the various journals contained the following information.

Sales journal: Sales to Sogard \$800, to Hurley \$1,260, to Giambi \$1,330, and to Fowler \$1,600.

Cash receipts journal: Cash received from Fowler \$1,310, from Sogard \$3,300, from Giambi \$380, from Andino \$1,800, and from Hurley \$1,240.

General journal: An allowance is granted to Sogard \$220.

Instructions

- Set up control and subsidiary accounts and enter the beginning balances. Do not construct the journals.
- Post the various journals. Post the items as individual items or as totals, whichever would be the appropriate procedure. (No sales discounts given.)
- Prepare a schedule of accounts receivable and prove the agreement of the controlling account with the subsidiary ledger at September 30, 2020.

Determine control and subsidiary ledger balances for accounts receivable.

E7.4 (LO 2) Kieschnick Company has a balance in its Accounts Receivable control account of \$10,000 on January 1, 2020. The subsidiary ledger contains three accounts: Bixler Company, balance \$4,000; Cuddyer Company, balance \$2,500; and Freeze Company. During January, the following receivable-related transactions occurred.

	<u>Credit Sales</u>	<u>Collections</u>	<u>Returns</u>
Bixler Company	\$9,000	\$8,000	\$ -0-
Cuddyer Company	7,000	2,500	3,000
Freeze Company	8,500	9,000	-0-

Instructions

- What is the January 1 balance in the Freeze Company subsidiary account?
- What is the January 31 balance in the control account?
- Compute the balances in the subsidiary accounts at the end of the month.
- Which January transaction would not be recorded in a special journal?

Determine control and subsidiary ledger balances for accounts payable.

E7.5 (LO 2) Pennington Company has a balance in its Accounts Payable control account of \$9,250 on January 1, 2020. The subsidiary ledger contains three accounts: Hale Company, balance \$3,000; Janish Company, balance \$1,875; and Valdez Company. During January, the following payable-related transactions occurred.

	<u>Purchases</u>	<u>Payments</u>	<u>Returns</u>
Hale Company	\$6,750	\$6,000	\$ -0-
Janish Company	5,250	1,875	2,250
Valdez Company	6,375	6,750	-0-

Instructions

- What is the January 1 balance in the Valdez Company subsidiary account?
- What is the January 31 balance in the control account?
- Compute the balances in the subsidiary accounts at the end of the month.
- Which January transaction would not be recorded in a special journal?

Record transactions in sales and purchases journal.

E7.6 (LO 3) Gomes Company uses special journals and a general journal. The following transactions occurred during September 2020.

- Sept. 2 Sold merchandise on account to H. Drew, invoice no. 101, \$620, terms n/30. The cost of the merchandise sold was \$420.
- 10 Purchased merchandise on account from A. Pagan \$650, terms 2/10, n/30.
- 12 Purchased office equipment on account from R. Cairo \$6,500.
- 21 Sold merchandise on account to G. Holliday, invoice no. 102 for \$800, terms 2/10, n/30. The cost of the merchandise sold was \$480.

- 25 Purchased merchandise on account from D. Downs \$860, terms n/30.
- 27 Sold merchandise to S. Miller for \$700 cash. The cost of the merchandise sold was \$400.

Instructions

- a. Prepare a sales journal (see Illustration 7.7) and a single-column purchases journal (see Illustration 7.13). (Use page 1 for each journal.)
- b. Record the transaction(s) for September that should be journalized in the sales journal and the purchases journal.

E7.7 (LO 3) R. Santiago Co. uses special journals and a general journal. The following transactions occurred during May 2020.

Record transactions in cash receipts and cash payments journal.

- May
- 1 R. Santiago invested \$40,000 cash in the business.
 - 2 Sold merchandise to Lawrie Co. for \$6,300 cash. The cost of the merchandise sold was \$4,200.
 - 3 Purchased merchandise for \$7,700 from J. Moskos using check no. 101.
 - 14 Paid salary to H. Rivera \$700 by issuing check no. 102.
 - 16 Sold merchandise on account to K. Stanton for \$900, terms n/30. The cost of the merchandise sold was \$630.
 - 22 A check of \$9,000 is received from M. Mangini in full for invoice 101; no discount given.

Instructions

- a. Prepare a multiple-column cash receipts journal (see Illustration 7.9) and a multiple-column cash payments journal (see Illustration 7.16). (Use page 1 for each journal.)
- b. Record the transaction(s) for May that should be journalized in the cash receipts journal and cash payments journal.

E7.8 (LO 3) Francisco Company uses the columnar cash journals illustrated in the textbook. In April, the following selected cash transactions occurred.

Explain journalizing in cash journals.

1. Made a refund to a customer as an allowance for damaged goods.
2. Received collection from customer within the 3% discount period.
3. Purchased merchandise for cash.
4. Paid a creditor within the 3% discount period.
5. Received collection from customer after the 3% discount period had expired.
6. Paid freight on merchandise purchased.
7. Paid cash for office equipment.
8. Received cash refund from supplier for merchandise returned.
9. Withdrew cash for personal use of owner.
10. Made cash sales.

Instructions

Indicate (a) the journal, and (b) the columns in the journal that should be used in recording each transaction.

E7.9 (LO 3) Writing Hasselback Company has the following selected transactions during March.

Journalize transactions in general journal and explain postings.

- Mar.
- 2 Purchased equipment costing \$7,400 from Bole Company on account.
 - 5 Received credit of \$410 from Carwell Company for merchandise damaged in shipment to Hasselback.
 - 7 Issued credit of \$400 to Dempsey Company for merchandise the customer returned. The returned merchandise had a cost of \$260.

Hasselback Company uses a one-column purchases journal, a sales journal, the columnar cash journals used in the text, and a general journal.

Instructions

- a. Journalize the transactions in the general journal.
- b. In a brief memo to the president of Hasselback Company, explain the postings to the control and subsidiary accounts from each type of journal.

E7.10 (LO 3) The following are some typical transactions incurred by Ricketts Company.

Indicate journalizing in special journals.

1. Payment of creditors on account.
2. Return of merchandise sold for credit.

3. Collection on account from customers.
4. Sale of land for cash.
5. Sale of merchandise on account.
6. Sale of merchandise for cash.
7. Received credit for merchandise purchased on credit.
8. Sales discount taken on goods sold.
9. Payment of employee wages.
10. Income summary closed to owner's capital.
11. Depreciation on building.
12. Purchase of office supplies for cash.
13. Purchase of merchandise on account.

Instructions

For each transaction, indicate whether it would normally be recorded in a cash receipts journal, cash payments journal, sales journal, single-column purchases journal, or general journal.

Explain posting to control account and subsidiary ledger.

E7.11 (LO 2, 3) The general ledger of Hensley Company contained the following Accounts Payable control account (in T-account form). Also shown is the related subsidiary ledger.

GENERAL LEDGER

Accounts Payable

Feb. 15	General journal	1,400	Feb. 1	Balance	26,025
28	?	?	5	General journal	265
			11	General journal	550
			28	Purchases	13,400
			Feb. 28	Balance	10,500

ACCOUNTS PAYABLE LEDGER

Benton			Parks		
	Feb. 28	Bal. 4,600		Feb. 28	Bal. ?
Dooley					
	Feb. 28	Bal. 2,300			

Instructions

- a. Indicate the missing posting reference and amount in the control account, and the missing ending balance in the subsidiary ledger.
- b. Indicate the amounts in the control account that were dual-posted (i.e., posted to the control account and the subsidiary accounts).

Prepare purchases and general journals.

E7.12 (LO 2, 3) Selected accounts from the ledgers of Youngblood Company at July 31 showed the following.

GENERAL LEDGER

Equipment						Inventory					
No. 157						No. 120					
Date	Explanation	Ref.	Debit	Credit	Balance	Date	Explanation	Ref.	Debit	Credit	Balance
July 1		G1	3,900		3,900	July 15		G1	400		400
						18		G1		100	300
						25		G1		200	100
						31		P1	9,300		9,400
Accounts Payable											
No. 201											
Date	Explanation	Ref.	Debit	Credit	Balance						
July 1		G1		3,900	3,900						
15		G1		400	4,300						
18		G1	100		4,200						
25		G1	200		4,000						
31		P1		9,300	13,300						

ACCOUNTS PAYABLE LEDGER**Flaherty Equipment Co.**

Date	Explanation	Ref.	Debit	Credit	Balance
July 1		G1		3,900	3,900

Marsh Co.

Date	Explanation	Ref.	Debit	Credit	Balance
July 3		P1		2,400	2,400
20		P1		1,700	4,100

Lange Corp

Date	Explanation	Ref.	Debit	Credit	Balance
July 17		P1		1,400	1,400
18		G1	100		1,300
29		P1		1,600	2,900

Weller Co.

Date	Explanation	Ref.	Debit	Credit	Balance
July 14		P1		1,100	1,100
25		G1	200		900

Yates Co.

Date	Explanation	Ref.	Debit	Credit	Balance
July 12		P1		500	500
21		P1		600	1,100

Bernardo Inc.

Date	Explanation	Ref.	Debit	Credit	Balance
July 15		G1		400	400

Instructions

From the data prepare:

- The single-column purchases journal for July.
- The general journal entries for July.

E7.13 (LO 2, 3) Tresh Products uses both special journals and a general journal as described in this chapter. Tresh also posts customers' accounts in the accounts receivable subsidiary ledger. The postings for the most recent month are included in the subsidiary T-accounts below.

Determine correct posting amount to control account.

Estes			Gehrke		
Bal.	340	250	Bal.	150	150
	200			290	
Truong			Weiser		
Bal.	–0–	145	Bal.	120	120
	145			190	
				150	

Instructions

Determine the correct amount of the end-of-month posting from the sales journal to the Accounts Receivable control account.

E7.14 (LO 3) Selected account balances for Hulse Company at January 1, 2020, are presented below.

Compute balances in various accounts.

Accounts Payable	\$14,000
Accounts Receivable	22,000
Cash	17,000
Inventory	13,500

Hulse's sales journal for January shows a total of \$110,000 in the selling-price column, and its one-column purchases journal for January shows a total of \$77,000.

The column totals in Hulse's cash receipts journal are Cash Dr. \$61,000, Sales Discounts Dr. \$1,100, Accounts Receivable Cr. \$45,000, Sales Revenue Cr. \$6,000, and Other Accounts Cr. \$11,100.

The column totals in Hulse's cash payments journal for January are Cash Cr. \$55,000, Inventory Cr. \$1,000, Accounts Payable Dr. \$46,000, and Other Accounts Dr. \$10,000. Hulse's total cost of goods sold for January is \$63,600.

Accounts Payable, Accounts Receivable, Cash, Inventory, and Sales Revenue are not involved in the Other Accounts column in either the cash receipts or cash payments journal, and are not involved in any general journal entries.

Instructions

Compute the January 31 balance for Hulse in the following accounts.

- Accounts Payable.
- Accounts Receivable.
- Cash.
- Inventory.
- Sales Revenue.

Problems: Set A

Journalize transactions in cash receipts journal; post to control account and subsidiary ledger.



P7.1A (LO 2, 3) Kozma Company's chart of accounts includes the following selected accounts.

101	Cash	401	Sales Revenue
112	Accounts Receivable	414	Sales Discounts
120	Inventory	505	Cost of Goods Sold
301	Owner's Capital		

On April 1, the accounts receivable ledger of Kozma Company showed the following balances: Morrow \$1,550, Rose \$1,200, Jennings Co. \$2,900, and Dent \$2,200. The April transactions involving the receipt of cash were as follows.

- Apr. 1 The owner, T. Kozma, invested additional cash in the business \$7,200.
 4 Received check for payment of account from Dent less 2% cash discount.
 5 Received check for \$920 in payment of invoice no. 307 from Jennings Co.
 8 Made cash sales of merchandise totaling \$7,245. The cost of the merchandise sold was \$4,347.
 10 Received check for \$600 in payment of invoice no. 309 from Morrow.
 11 Received cash refund from a supplier for damaged merchandise \$740.
 23 Received check for \$1,000 in payment of invoice no. 310 from Jennings Co.
 29 Received check for payment of account from Rose (no cash discount allowed).

Instructions

a. Balancing totals \$25,452

- a. Journalize the transactions above in a six-column cash receipts journal with columns for Cash Dr., Sales Discounts Dr., Accounts Receivable Cr., Sales Revenue Cr., Other Accounts Cr., and Cost of Goods Sold Dr./Inventory Cr. Foot and cross-foot the journal.
 b. Insert the beginning balances in the Accounts Receivable control and subsidiary accounts, and post the April transactions to these accounts.
 c. Prove the agreement of the control account and subsidiary account balances.

c. Accounts Receivable \$1,930

Journalize transactions in cash payments journal; post to control account and subsidiary ledgers.



P7.2A (LO 2, 3) Reineke Company's chart of accounts includes the following selected accounts.

101	Cash	201	Accounts Payable
120	Inventory	306	Owner's Drawings
130	Prepaid Insurance	505	Cost of Goods Sold
157	Equipment		

On October 1, the accounts payable ledger of Reineke Company showed the following balances: Ugla Company \$2,700, Orr Co. \$2,500, Rosenthal Co. \$1,800, and Clevenger Company \$3,700. The October transactions involving the payment of cash were as follows.

- Oct. 1 Purchased merchandise, check no. 63, \$300.
 3 Purchased equipment, check no. 64, \$800.
 5 Paid Ugla Company balance due of \$2,700, less 2% discount, check no. 65, \$2,646.
 10 Purchased merchandise, check no. 66, \$2,550.
 15 Paid Rosenthal Co. balance due of \$1,800, check no. 67.
 16 C. Reineke, the owner, pays his personal insurance premium of \$400, check no. 68.
 19 Paid Orr Co. in full for invoice no. 610, \$2,000 less 2% cash discount, check no. 69, \$1,960.
 29 Paid Clevenger Company in full for invoice no. 264, \$2,500, check no. 70.

Instructions

a. Balancing totals \$13,050

- a. Journalize the transactions above in a four-column cash payments journal with columns for Other Accounts Dr., Accounts Payable Dr., Inventory Cr., and Cash Cr. Foot and cross-foot the journal.
 b. Insert the beginning balances in the Accounts Payable control and subsidiary accounts, and post the October transactions to these accounts.
 c. Prove the agreement of the control account and the subsidiary account balances.

c. Accounts Payable \$1,700

Journalize transactions in multi-column purchases journal and sales journal; post to the general and subsidiary ledgers.



P7.3A (LO 2, 3) The chart of accounts of LR Company includes the following selected accounts.

112	Accounts Receivable	401	Sales Revenue
120	Inventory	412	Sales Returns and Allowances
126	Supplies	505	Cost of Goods Sold
157	Equipment	610	Advertising Expense
201	Accounts Payable		

In July, the following transactions were completed. All purchases and sales were on account. The cost of all merchandise sold was 70% of the sales price.

- July 1 Purchased merchandise from Eby Company \$8,000.
 2 Received freight bill from Shaw Shipping on Eby purchase \$400.
 3 Made sales to Fort Company \$1,300 and to Hefner Bros. \$1,500.
 5 Purchased merchandise from Getz Company \$3,200.
 8 Received credit on merchandise returned to Getz Company \$300.
 13 Purchased store supplies from Dayne Supply \$720.
 15 Purchased merchandise from Eby Company \$3,600 and from Bosco Company \$4,300.
 16 Made sales to Aybar Company \$3,450 and to Hefner Bros. \$1,870.
 18 Received bill for advertising from Welton Advertisements \$600.
 21 Sales were made to Fort Company \$310 and to Duncan Company \$2,800.
 22 Granted allowance to Fort Company for merchandise damaged in shipment \$40.
 24 Purchased merchandise from Getz Company \$3,000.
 26 Purchased equipment from Dayne Supply \$900.
 28 Received freight bill from Shaw Shipping on Getz purchase of July 24, \$380.
 30 Sales were made to Aybar Company \$5,600.

Instructions

- Journalize the transactions above in a purchases journal, a sales journal, and a general journal. The purchases journal should have the following column headings: Date, Account Credited (Debited), Ref., Accounts Payable Cr., Inventory Dr., and Other Accounts Dr.
- Post to both the general and subsidiary ledger accounts. (Assume that all accounts have zero beginning balances.)
- Prove the agreement of the control and subsidiary accounts.

- Purchases journal—
Accounts Payable \$25,100
Sales journal—Sales
Revenue \$16,830
- Accounts Receivable \$16,790
Accounts Payable \$24,800

P7.4A (LO 2, 3) Selected accounts from the chart of accounts of Mercer Company are shown below.

101 Cash	401 Sales Revenue
112 Accounts Receivable	412 Sales Returns and Allowances
120 Inventory	414 Sales Discounts
126 Supplies	505 Cost of Goods Sold
157 Equipment	726 Salaries and Wages Expense
201 Accounts Payable	

Journalize transactions in special journals.



The cost of all merchandise sold was 60% of the sales price. During January, Mercer completed the following transactions.

- Jan. 3 Purchased merchandise on account from Gallagher Co. \$9,000.
 4 Purchased supplies for cash \$80.
 4 Sold merchandise on account to Wheeler \$5,250, invoice no. 371, terms 1/10, n/30.
 5 Returned \$300 worth of damaged goods purchased on account from Gallagher Co. on January 3.
 6 Made cash sales for the week totaling \$3,150.
 8 Purchased merchandise on account from Phegley Co. \$4,500.
 9 Sold merchandise on account to Linton Corp. \$5,400, invoice no. 372, terms 1/10, n/30.
 11 Purchased merchandise on account from Cora Co. \$3,700.
 13 Paid in full Gallagher Co. on account less a 2% discount.
 13 Made cash sales for the week totaling \$6,260.
 15 Received payment from Linton Corp. for invoice no. 372.
 15 Paid semi-monthly salaries of \$14,300 to employees.
 17 Received payment from Wheeler for invoice no. 371.
 17 Sold merchandise on account to Delaney Co. \$1,200, invoice no. 373, terms 1/10, n/30.
 19 Purchased equipment on account from Dozier Corp. \$5,500.
 20 Cash sales for the week totaled \$3,200.
 20 Paid in full Phegley Co. on account less a 2% discount.
 23 Purchased merchandise on account from Gallagher Co. \$7,800.
 24 Purchased merchandise on account from Atchison Corp. \$5,100.
 27 Made cash sales for the week totaling \$4,230.
 30 Received payment from Delaney Co. for invoice no. 373.
 31 Paid semi-monthly salaries of \$13,200 to employees.
 31 Sold merchandise on account to Wheeler \$9,330, invoice no. 374, terms 1/10, n/30.

Mercer Company uses the following journals.

- Sales journal.
- Single-column purchases journal.
- Cash receipts journal with columns for Cash Dr., Sales Discounts Dr., Accounts Receivable Cr., Sales Revenue Cr., Other Accounts Cr., and Cost of Goods Sold Dr./Inventory Cr.

4. Cash payments journal with columns for Other Accounts Dr., Accounts Payable Dr., Inventory Cr., and Cash Cr.
5. General journal.

Instructions

Using the selected accounts provided:

- a. Record the January transactions in the appropriate journal noted.
- b. Foot and cross-foot all special journals.
- c. Show how postings would be made by placing ledger account numbers and checkmarks as needed in the journals. (Actual posting to ledger accounts is not required.)

- a. Sales journal \$21,180
Purchases journal \$30,100
Cash receipts journal
balancing total \$38,794
Cash payments journal
balancing total \$40,780

Journalize in sales and cash receipts journals; post; prepare a trial balance; prove control to subsidiary; prepare adjusting entries; prepare an adjusted trial balance.



P7.5A (LO 2, 3) Presented below are the purchases and cash payments journals for Fornelli Co. for its first month of operations.

PURCHASES JOURNAL				P1
Date	Account Credited	Ref.	Inventory Dr. Accounts Payable Cr.	
July 4	N. Alvarado		6,800	
5	F. Rees		8,100	
11	J. Gallup		5,920	
13	C. Werly		15,300	
20	M. Mangus		7,900	
			<u>44,020</u>	

CASH PAYMENTS JOURNAL						CP1
Date	Account Debited	Ref.	Other Accounts Dr.	Accounts Payable Dr.	Inventory Cr.	Cash Cr.
July 4	Supplies		600			600
10	F. Rees			8,100	81	8,019
11	Prepaid Rent		6,000			6,000
15	N. Alvarado			6,800		6,800
19	Owner's Drawings		2,500			2,500
21	C. Werly			15,300	153	15,147
			<u>9,100</u>	<u>30,200</u>	<u>234</u>	<u>39,066</u>

In addition, the following transactions have not been journalized for July. The cost of all merchandise sold was 65% of the sales price.

- July 1 The founder, N. Fornelli, invests \$80,000 in cash.
- 6 Sell merchandise on account to Dow Co. \$6,200 terms 1/10, n/30.
- 7 Make cash sales totaling \$8,000.
- 8 Sell merchandise on account to S. Goebel \$4,600, terms 1/10, n/30.
- 10 Sell merchandise on account to W. Leiss \$4,900, terms 1/10, n/30.
- 13 Receive payment in full from S. Goebel.
- 16 Receive payment in full from W. Leiss.
- 20 Receive payment in full from Dow Co.
- 21 Sell merchandise on account to H. Kenney \$5,000, terms 1/10, n/30.
- 29 Returned damaged goods to N. Alvarado and received cash refund of \$420.

Instructions

- a. Open the following accounts in the general ledger.

101	Cash	306	Owner's Drawings
112	Accounts Receivable	401	Sales Revenue
120	Inventory	414	Sales Discounts
126	Supplies	505	Cost of Goods Sold
131	Prepaid Rent	631	Supplies Expense
201	Accounts Payable	729	Rent Expense
301	Owner's Capital		

- b. Sales journal total \$20,700
Cash receipts journal
balancing total \$109,320

- b. Journalize the transactions that have not been journalized in the sales journal and the cash receipts journal (see Illustration 7.9).

- c. Post to the accounts receivable and accounts payable subsidiary ledgers. Follow the sequence of transactions as shown in the problem.
- d. Post the individual entries and totals to the general ledger.
- e. Prepare a trial balance at July 31, 2020.
- f. Determine whether the subsidiary ledgers agree with the control accounts in the general ledger.
- g. The following adjustments at the end of July are necessary.
 1. A count of supplies indicates that \$140 is still on hand.
 2. Recognize rent expense for July, \$500.
 Prepare the necessary entries in the general journal. Post the entries to the general ledger.
- h. Prepare an adjusted trial balance at July 31, 2020.

e. Totals \$122,520

f. Accounts Receivable \$5,000
Accounts Payable \$13,820

h. Totals \$122,520

P7.6A (LO 2, 3) The post-closing trial balance for Horner Co. is shown below.

Horner Co.
Post-Closing Trial Balance
December 31, 2020

	Debit	Credit
Cash	\$ 41,500	
Accounts Receivable	15,000	
Notes Receivable	45,000	
Inventory	23,000	
Equipment	6,450	
Accumulated Depreciation—Equipment		\$ 1,500
Accounts Payable		43,000
Owner's Capital		86,450
	<u>\$130,950</u>	<u>\$130,950</u>



The subsidiary ledgers contain the following information: (1) accounts receivable—B. Hannigan \$2,500, I. Kirk \$7,500, and T. Hodges \$5,000; (2) accounts payable—T. Igawa \$12,000, D. Danford \$18,000, and K. Thayer \$13,000. The cost of all merchandise sold was 60% of the sales price.

The transactions for January 2021 are as follows.

- Jan. 3 Sell merchandise to M. Ziesmer \$8,000, terms 2/10, n/30.
- 5 Purchase merchandise from E. Pheatt \$2,000, terms 2/10, n/30.
- 7 Receive a check from T. Hodges \$3,500.
- 11 Pay freight on merchandise purchased \$300.
- 12 Pay rent of \$1,000 for January.
- 13 Receive payment in full from M. Ziesmer.
- 14 Post all entries to the subsidiary ledgers. Issued credit of \$300 to B. Hannigan for returned merchandise.
- 15 Send K. Thayer a check for \$12,870 in full payment of account, discount \$130.
- 17 Purchase merchandise from G. Roland \$1,600, terms 2/10, n/30.
- 18 Pay sales salaries of \$2,800 and office salaries \$2,000.
- 20 Give D. Danford a 60-day note for \$18,000 in full payment of account payable.
- 23 Total cash sales amount to \$9,100.
- 24 Post all entries to the subsidiary ledgers. Sell merchandise on account to I. Kirk \$7,400, terms 1/10, n/30.
- 27 Send E. Pheatt a check for \$950.
- 29 Receive payment on a note of \$40,000 from B. Stout.
- 30 Post all entries to the subsidiary ledgers. Return merchandise of \$300 to G. Roland for credit.

Instructions

- a. Open general and subsidiary ledger accounts for the following.

101 Cash	301 Owner's Capital
112 Accounts Receivable	401 Sales Revenue
115 Notes Receivable	412 Sales Returns and Allowances
120 Inventory	414 Sales Discounts
157 Equipment	505 Cost of Goods Sold
158 Accumulated Depreciation—Equipment	726 Salaries and Wages Expense
200 Notes Payable	729 Rent Expense
201 Accounts Payable	

Journalize in special journals; post; prepare a trial balance.

- b. Sales journal \$15,400
Purchases journal \$3,600
Cash receipts journal
(balancing) \$66,060
Cash payments journal
(balancing) \$20,050
- d. Totals \$144,800
- e. Accounts Receivable \$18,600
Accounts Payable \$14,350

- b. Record the January transactions in a sales journal, a single-column purchases journal, a cash receipts journal (see Illustration 7.9), a cash payments journal (see Illustration 7.16), and a general journal.
- c. Post the appropriate amounts to the general ledger.
- d. Prepare a trial balance at January 31, 2021.
- e. Determine whether the subsidiary ledgers agree with controlling accounts in the general ledger.

Ethics Case

EC7 Wiemers Products Company operates three divisions, each with its own manufacturing plant and marketing/sales force. The corporate headquarters and central accounting office are in Wiemers, and the plants are in Freeport, Rockport, and Bayport, all within 50 miles of Wiemers. Corporate management treats each division as an independent profit center and encourages competition among them. They each have similar but different product lines. As a competitive incentive, bonuses are awarded each year to the employees of the fastest-growing and most-profitable division.

Indy Grover is the manager of Wiemers's centralized computerized accounting operation that enters the sales transactions and maintains the accounts receivable for all three divisions. Indy came up in the accounting ranks from the Bayport division where his wife, several relatives, and many friends still work.

As sales documents are entered into the computer, the originating division is identified by code. Most sales documents (95%) are coded, but some (5%) are not coded or are coded incorrectly. As the manager, Indy has instructed the data-entry personnel to assign the Bayport code to all uncoded and incorrectly coded sales documents. This is done, he says, "in order to expedite processing and to keep the computer files current since they are updated daily." All receivables and cash collections for all three divisions are handled by Wiemers as one subsidiary accounts receivable ledger.

Instructions

- a. Who are the stakeholders in this situation?
- b. What are the ethical issues in this case?
- c. How might the system be improved to prevent this situation?

Comprehensive Accounting Cycle Review

ACR7.1 (Perpetual Method) Jeter Co. uses a perpetual inventory system and both an accounts receivable and an accounts payable subsidiary ledger. Balances related to both the general ledger and the subsidiary ledgers for Jeter are indicated in the working papers presented below. Also following are a series of transactions for Jeter Co. for the month of January. Credit sales terms are 2/10, n/30. The cost of all merchandise sold was 60% of the sales price.

GENERAL LEDGER

Account Number	Account Title	January 1 Opening Balance
101	Cash	\$35,750
112	Accounts Receivable	13,000
115	Notes Receivable	39,000
120	Inventory	18,000
126	Supplies	1,000
130	Prepaid Insurance	2,000
157	Equipment	6,450
158	Accumulated Depreciation—Equip.	1,500
201	Accounts Payable	35,000
301	Owner's Capital	78,700